

MST-001
Foundation in
Mathematics and
Statistics

Block

1

FUNDAMENTALS OF MATHEMATICS-I

UNIT 1

Introduction to Sets	7
-----------------------------	----------

UNIT 2

Functions	27
------------------	-----------

UNIT 3

Progressions	49
---------------------	-----------

UNIT 4

Techniques of Counting	65
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FOUNDATION IN MATHEMATICS AND STATISTICS

Whatever way is chosen in order to define the contents of the courses of this programme, one cannot avoid the use of some elementary concepts of mathematics. That is why first 10 units of this course are devoted to introduce some mathematical terms used in the rest of the courses of this programme, in particular course MST-003. In fact having being ‘Any graduate’ as a qualification for this programme, it becomes necessary to think about those learners who don’t have mathematical background after matriculation. Having these types of learners as a part of our target group, every care has been taken in order to define mathematical terms. Most of the mathematical terms are explained with the help of some practical/real life situations followed by a large number of examples. It is tried to avoid derivations of mathematical results unless otherwise it is necessary. The aim of this course, i.e. MST-001 (in particular first 10 units) is just to put the learners (in particular those having no mathematical background after matriculation) in a position, so that whenever these mathematical terms will be used, the basic idea can easily be grasped and feel comfortable. Last six units of this course are devoted to put a foundation stone for all other courses of the programme, i.e. elementary part of statistics such as defining statistics, development stages, very important concept of measurement of scales, methods of collection of data, classification, tabulation, diagrammatical and graphical presentation of data have been discussed in last six units of this course.

This course is divided into four blocks of four units each.

In **first block**, sets, functions and their various types are introduced. Arithmetic Progression (A.P.), Geometric Progression (G.P.), concept of summation, permutation and combination also have been discussed in this block. Brief introduction of binomial theorem is also included in this block.

The **second block** is devoted to concentrate on the four very much related and useful topics namely, limit, continuity, differentiation and integration. Concept of limit, continuity, differentiation, integration and some standard results on limit, differentiation and integration also have been discussed in this block.

The **third block** is devoted to the study of matrices and determinants, different types of matrices, and some simple properties of determinants. Origin, development, definition, scope, uses, limitations of statistics also has been briefly introduced. Measurement of scales—nominal, ordinal, interval and ratio are discussed in detail. Primary data, secondary data and their methods of collection are also discussed in detail.

Block four discusses classification, tabulation, diagrammatical presentation and graphical presentation of data. Box plot, stem and leaf plot are discussed in detail.

Although the material is self contained and self explained in nature. Even though if some learners are interested to gain more and want to study the contents in greater depth/more detail, it is a friendly advice for you to put a lot of practice to attempt all the exercises given in the relevant chapters of the below listed books.

1. Mathematics Textbook for Class XI, first addition (2006), reprinted December, 2009 (NCERT) (Chapters 1, 2, 7, 8, 9, 13)

2. Mathematics Textbook for Class XII, first edition (2006), reprinted December, 2009 (NCERT) (Chapters 1, 3, 4, 5, 6, 7)
3. SCHAUM'S OUTLINE OF Theory and Problems of Discrete Mathematics, Second Edition by Seymour Lipschutz and Marc Lars Lipson [Chapters 1, 3, 5, 6], Tata McGraw-Hill Publishing Company Limited
4. SCHAUM'S OUTLINE OF Theory and Problems of STATISTICS Third Edition by Murray R. Spiegel and Larry J. Stephens [Chapters 1, 2], Tata McGraw-Hill Publishing Company Limited
5. SCHAUM'S OUTLINE OF Theory and Problems of ELEMENTS OF STATISTICS Differential Statistics and Probability Third Edition by Stephen Bernstein and Ruth Bernstein [Chapters 6, 7], Tata McGraw-Hill Publishing Company Limited
6. Grinstead and Snell's 'Introduction to Probability, 2nd Edition', by Charles M. Grinstead and J. Laurie Snell, American Mathematical Society (2006) (Chapter 3)
7. Fundamentals of Mathematical Statistics by S.C. Gupta and V.K. Kapoor (1994), Sultan Chand & Sons (Chapter 1)
8. MARKETING RESEARCH An Applied Orientation, Sixth Edition (Chapter 10) by Naresh K. Malhotra and Satyabhusan Dash, Prentice Hall
9. Fundamentals of STATISTICS, volume one by A. M. Goon, M. K. Gupta, B. Dasgupta, Calcutta the world press private LTD. 1987 (Chapter 4, 5, 6)
10. BASIC STATISTICS, Fifth Edition, By B.L. Agarwal, New Age International (P) Limited, Publishers (Chapter 1, 2, 22)
11. Business Statistics by J. S. Chandan, Prof. Jagjit Singh and K. K. Khanna, Vikas Publishing House Pvt LTD, 1994 (Chapter 1, 3, 4)
12. Elements of Statistics (Part one) by B. N. Asthana, Chaitanya Publishing House, Allahabad, 1988 (Chapter 1, 2, 4, 5)
13. Fundamentals of Statistics by Late D. N. Elhance, Kitab Mahal, 1956, (Chapter 1, 3, 4, 5, 6)

Unit wise you may refer the books as given below:

Unit Number	Serial No of the Book	Unit Number	Serial No of the Book
1	1, 3	9	2, 3
2	1, 2, 3	10	2, 3
3	1	11	1, 7, 8, 9, 10, 11, 12, 13
4	1, 3, 6	12	8, 9, 10, 11, 13
5	1, 2	13	4, 9, 10, 11, 13
6	1, 2	14	4, 10, 13
7	2	15	4, 10, 12, 13
8	2	16	5

BLOCK 1 FUNDAMENTALS OF MATHEMATICS-I

This is the first block of the course MST-001. The aim of the block is to put a foundation stone for the next block of this course and will provide a platform to the learners (especially for non mathematical background) to understand the basic ideas of probability theory i.e. course MST-003. The flow of this block is maintained by the following four units.

Unit 1: Introduction to Sets

This unit will explain what we mean by sets, various types of sets, hierarchy of sets, different operations on sets, Venn-diagrams (i.e. pictorial representation of sets) and some simple applications of sets.

Unit 2: Functions

This unit will explain the very important term ‘function’ with the help of very good real life example of daughters and mothers in a very simple and logical way. Some particular and commonly used functions and three important and useful types one-one, onto, one-one correspondence of functions are also explained with the help of a number of examples. Geometrical interpretation of one-one, onto and one-one correspondence is also explained.

Unit 3: Progressions

This unit will throw the light on two very frequently encountered progressions known as Arithmetic Progression (A.P.) and Geometric Progression (G.P.). How, n^{th} term and sum of first n terms of an A.P. or G.P. are evaluated, are explained with a variety and large number of examples. Some simple applications of A.P. and G.P. are also discussed. Concept of summation and formulae for the sum of some special sequences are also introduced. How these formulae are applied on numerical problems, is explained with the help of some examples.

Unit 4: Techniques of Counting

Last unit of this block is devoted to two very powerful techniques of counting, which provides us how many options/possibilities are there for a real life situation. Such as how many different sequences of answers of an objective type test are possible or how many different lottery numbers for a particular lottery are possible or how many different pin code of 4 digits can be provided to the customers of a particular bank. Binomial theorem is also introduced in this unit.

Notations and Symbols

$n(A)$	: cardinality of the set A i.e. number of elements in the set A
$\subseteq$	: is a sub set of or is contained in
$A \Delta B$	: symmetric difference of two sets A and B
${}^n P_r$	: total number of permutations of n things taken r ($1 \leq r \leq n$) at a time
${}^n C_r$	: total number of combinations of n things taken r ($1 \leq r \leq n$) at a time
$\supseteq$	: is a super set or contains
$\not\subseteq$	: is not subset of or is not contained in
$\subset$	: is proper subset of
ϕ	: empty set or null set or void set
$P(A)$	: power set of the set A
$l(I)$	: length of the interval I
$f : X \rightarrow Y$	: f is a function from X to Y
$ x $	: modules of x or absolute value of x
A. P.	: arithmetic progressions
G. P.	: geometric progression
$\{a_n\}$	: a sequence whose n^{th} term is a_n
a_n or t_n	: n^{th} term of an A. P. or G. P.
S_n	: sum of first n term of an A. P. or G. P.
$n!$ or $\underline{n}$	: n factorial
A^c or A'	: complement of the set A
(a, b)	: open interval
$[a, b]$	: closed interval
$(a, b]$	: left open and right closed interval
$[a, b)$	: left closed and right open interval
$\cup$	: union
$\cap$	: intersection
$\in$	: belong to
$\notin$	: does not belong to
$=$	: is equal to
$\neq$	: is not equal to
$<$	: is less than
$>$	: is greater than
$\sim$	: is equivalents to

Greek Alphabets

α	alpha	ψ	psi	π	pi
β	beta	ξ	xi	ρ	rho
$\gamma(\Gamma)$	gamma (cap. gamma)	η	eta	$\sigma(\Sigma)$	sigma (cap. sigma)
$\delta(\Delta)$	delta (cap. delta)	ζ	zeta	τ	tau
ϵ	epsilon	λ	lambda	χ	chi
i	iota	K	kappa		
θ	theta	μ	mu	$\omega(\Omega)$	omega (cap. omega)
ϕ	phi	ν	nu		

UNIT 1 INTRODUCTION TO SETS

Structure

- 1.1 Introduction
 - Objectives
- 1.2 Sets
- 1.3 Types of Sets
- 1.4 Hierarchy of Sets
- 1.5 Venn Diagrams
- 1.6 Set Operations
- 1.7 Some Useful and Important Laws
- 1.8 Summary
- 1.9 Solutions/Answers

1.1 INTRODUCTION

Sometimes, we deal with some types of collections e.g.

- i) Collection of books in a library of a university.
- ii) Collection of natural numbers which are factors of say, 80 or any other natural number.

Set is also a collection of objects but it is a well defined collection (we will learn more about this in Sec 1.2). Consider the collection of states in India. We know that presently there are 28 states in India and this figure is exactly 28 (neither one less nor one more). Also if any number of persons (having a general knowledge of states in India) are asked to write the names of the states then final list of every body will contain the same 28 names (order in which they write the names of the states does not matter). Such type of a well defined collection is known as set.

In this unit, we will introduce the notations and terminology used for sets. The unit defines set, its various types, discusses hierarchy of sets, Venn diagrams, various operations on sets and finally the unit is closed by giving an idea of some important and commonly used laws. Concept related to sets is very elementary and it is used directly or indirectly in the rest of our courses. So you must understand the concepts discussed in this unit before you proceed further in the course.

By **well defined** collection, we mean that given any object we must be able to know as to whether it belongs to the collection or does not belong to the collection.

Objectives

After completing this unit, you should be able to:

- define a set;
- write a set in different forms;
- explain the types of sets;
- draw Venn diagrams.
- apply the operations on sets;
- get the idea of super sets and subsets; and
- get an idea of some important laws related to sets like associative, De-Morgan's laws, etc.

1.2 SETS

A set remains the same if some or all of its elements are repeated or rearranged.

For example, if a set contains the elements 0, 1, -1 and another set contains the elements -1, -1, 1, 0, 0, 0, 1, 0, 1, then these two sets are nothing but represent the same set having three elements 0, 1, -1.

A well defined collection of distinct objects is called a set. A set is generally denoted by capital letters such as A, B, C, X, Y, Z, etc. and the objects which belong to the set are known as elements or members of the set and are generally denoted by small letters a, b, c, x, y, z, etc.

If 'a' is an element of a set A then we write $a \in A$ (read it as 'a' belongs to A). If 'a' is not an element of A then we write $a \notin A$ (read as 'a' does not belong to A).

Following example illustrates the term “well defined collection” or “set”.

Example 1: Consider the following collections and state reasons whether they form set or not.

- (i) Collection of good cricketers in India.
- (ii) Collection of honest students in a particular university in India.
- (iii) Collection of natural numbers which are less than 5.
- (iv) Collection of rich persons in India.
- (v) Collection of letters of the word “ASSIGNMENT”

Solution:

- (i) This collection does not form a set, because a given player may be good according to some person but the same player may not be good according to some other person.
- (ii) This collection does not form a set, because a student may be honest according to some person but the same student may not be honest according to some other person.
- (iii) Yes, this collection forms a set and elements of this set are 1, 2, 3, 4.
- (iv) Richness is not a well defined property, because according to someone, a person may be rich while he/she may not be rich in view of some other person. So this collection does not form a set.
- (v) It is a set and elements of this set are A, S, I, G, N, M, E, T.

Now, you can try the following exercise:

E1) Give reasons whether the following collections are sets or not.

- (i) Collection of intelligent students in a particular school.
- (ii) Collection of good hockey players in India.
- (iii) Collection of good actors in India.
- (iv) Collection of vowels in the word “INDIA”.

Methods of Representing a Set

A set is generally represented by two methods as given below.

1. Roster Method

Dictionary meaning of 'Roster' is 'a list showing persons who perform their duties in turn'. As its meaning suggests, in this method each and every element is listed and put, separating by commas, in curly brackets. This method is also known as **Tabular Form** or **Listing Method**.

For example,

- (i) If A is the set of vowels of English alphabets, then $A = \{a, e, i, o, u\}$
- (ii) If N is the set of natural numbers, then $N = \{1, 2, 3, 4, 5, \dots\}$
- (iii) If W is the set of whole numbers, then $W = \{0, 1, 2, 3, 4, 5, \dots\}$
- (iv) If Z is the set of integers, then $Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- (v) If E is the set of even natural numbers, then $E = \{2, 4, 6, 8, 10, 12, \dots\}$
- (vi) If O is the set of odd natural numbers, then $O = \{1, 3, 5, \dots\}$
- (vii) If P is the set of prime numbers, then
 $P = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, \dots\}$

Remark 1: Throughout the course, we use N, W and Z for the sets of natural numbers, whole numbers, and integers, respectively.

2. Set-Builder Method

In this method, we consider one or more properties that are exclusive to the elements of a set so that no other elements can be the member of the set. This method is also known as **Property Method** or **Rule Method**.

For example,

- (i) Let $A = \{x : x \text{ is a vowel of English alphabet}\}$, then elements of A are a, e, i, o, u and having exclusive property of being a vowel no other alphabet can be considered as an element of set A.
- (ii) Let $A = \{x : x \text{ is a natural number and } x \text{ is a multiple of } 3\}$, then elements of A are 3, 6, 9, 12, ... which have the exclusive property of being multiple of 3 and no other element can be consider as an element of A.
- (iii) Let $A = \{x : x \text{ is a factor of } 10 \text{ and } x > 0\}$, then elements of A are 1, 2, 5, 10 and having exclusive property of being a factor of 10 and no other element can be consider as an element of A.
- (iv) If Q is the set of rational numbers then $Q = \left\{x : x = \frac{p}{q}, p, q \in Z, q \neq 0\right\}$

Remark 2:

- (i) In terminology of set the symbol “:” used in each part above is read as “such that”.
- (ii) Throughout the course the sets of rational numbers, irrational numbers and real numbers will be denoted by Q, I and R, respectively.

Note: Advantage of the second method, i.e. Set-Builder method lies in the fact that sometimes (or in some situations) we cannot list the elements of the set or even if we can list them, it may not be practical or feasible to do so.

For example, consider the set $\{x : x \text{ is a person who born in } 2010 \text{ in India}\}$. Obviously you will be more comfortable with property method in this example.

Consider another set $\{x : x \text{ is a real number and } 1 < x < 4\}$. This set cannot be described by listing method, because number of elements in this set is uncountable.

Thus, above two examples show that in some situations either it is too difficult to describe the set by listing method or it is impossible to describe it.

No doubt, there are some examples in which listing method has its advantage For example, consider the set $\{28, \text{Bihar, India}\}$.

Introduction to Sets

Three dots ‘...’ are read as “and so on” which means all the elements following this pattern are also included in the set.

Prime number: A natural number (> 1) is called a prime number if and only if it has only two divisors 1 and itself. For example, 2 is a prime number as its only divisors are 1 and the number itself.

But 15 is not a prime number because it has 3 and 5 as its factors other than 1 and the number itself.

Rational number: A number which can be expressed in the form

$\frac{p}{q}$ such that
 $p, q \in Z, q \neq 0$ is known as rational number. For example,

$\frac{2}{3}, \frac{-4}{13}$, etc.

Irrational number: A number which cannot be expressed

in the form $\frac{p}{q}$ where
 p, q are integers and $q \neq 0$, is called an irrational number.

For example,
 $\sqrt{2}, \sqrt{5}, \sqrt[3]{10}$, etc.

Real number: A number which is either rational or irrational is called real number.

It is a set having three elements 28, Bihar and India.

Let us now consider some examples to make the ideas of two methods discussed above more clear.

Example 2: Write the following sets by roster method:

(i) $A = \{x : x \text{ is a letter of the word "FUNCTION"}\}$

(ii) $B = \{x : 2x + 5 < 17, x \in \mathbb{N}\}$

(iii) $C = \{x : x^2 - x - 12 = 0, x \in \mathbb{N}\}$

(iv) $D = \{x : x^2 - 4x - 21 = 0, x^2 - 49 = 0, x \in \mathbb{N}\}$

Solution:

(i) $A = \{F, U, N, C, T, I, O\}$ [Repeated elements are written once only]

(ii) $B = \{x : 2x < 17 - 5, x \in \mathbb{N}\} = \{x : 2x < 12, x \in \mathbb{N}\}$
 $= \{x : x < 6, x \in \mathbb{N}\} = \{1, 2, 3, 4, 5\}$

(iii) $C = \{x : x^2 - x - 12 = 0, x \in \mathbb{N}\} = \{x : x^2 - 4x + 3x - 12 = 0, x \in \mathbb{N}\}$
 $= \{x : x(x - 4) + 3(x - 4) = 0, x \in \mathbb{N}\} = \{x : (x - 4)(x + 3) = 0, x \in \mathbb{N}\}$
 $= \{x : x = 4, -3, x \in \mathbb{N}\} = \{4\} \quad [\because -3 \notin \mathbb{N}]$

(iv) $D = \{x : x^2 - 4x - 21 = 0, x^2 - 49 = 0, x \in \mathbb{N}\}$
 $= \{x : x^2 - 7x + 3x - 21 = 0, x^2 - 7^2 = 0, x \in \mathbb{N}\}$
 $= \{x : x(x - 7) + 3(x - 7) = 0, (x - 7)(x + 7) = 0, x \in \mathbb{N}\}$
 $= \{x : (x - 7)(x + 3) = 0, x = 7, -7, x \in \mathbb{N}\}$
 $= \{x : x = 7, -3, x = 7, -7, x \in \mathbb{N}\}$

$= \{7\} \quad \left[\begin{array}{l} \because \text{here are two properties one says } x = 7, -3 \text{ and second says} \\ x = 7, -7 \text{ but in case we have more than one properties, we} \\ \text{take common element(s) between them and in our case it is 7.} \end{array} \right]$

Example 3: Express the following sets in the set-builder form:

(i) $A = \{3, 6, 9, 12, 15, 18, \dots\}$

(ii) $B = \{1, 2, 3, 5, 6, 10, 15, 30\}$

(iii) $C = \{5, 25, 125, 625, \dots\}$

(iv) $D = \{1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \frac{1}{81}, \frac{1}{243}\}$

(v) $E = \{1, 4, 9, 16, 25, 36, 49, 64, 81, 100\}$

Solution:

(i) Here we see that elements in this set are multiple of 3 so in set-builder form it can be written as $A = \{x : x \text{ is a multiple of 3, } x \in \mathbb{N}\}$.

Similarly, by observing the pattern obeyed by the elements of other parts we can write them as given below.

(ii) $B = \{x : x \text{ is a factor of 30, } x \in \mathbb{N}\}$

(iii) $C = \{x : x = 5^n, n \in \mathbb{N}\}$

(iv) $D = \{x : x = \frac{1}{3^n}, 0 \leq n \leq 5, n \in \mathbb{W}\}$

(v) $E = \{x : x = n^2, 1 \leq n \leq 10, n \in \mathbb{N}\}$

E 2) Describe the following sets by roster method:

$$(i) A = \{x: x = 2n + 3, n \in W\} \quad (ii) B = \{x: x = 7^n, 0 \leq n \leq 3, n \in N\}$$

$$(iii) C = \{x: x \in N \text{ and } x \in W\} \quad (iv) D = \{x: x \in W \text{ and } x \in Q\}$$

E 3) Express the following sets in set-builder form:

$$(i) \{5, 10, 15, 20, \dots\} \quad (ii) \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right\} \quad (iii) \{2, 4, 6, 8, 10, \dots\}$$

1.3 TYPES OF SETS

We have seen that set is a well defined collection of distinct objects. Also repetition of elements in a set is not allowed. So once a set is defined, automatically number of elements contained by it has also become fixed. In this section, we shall discuss the different names given to a set on the bases of the number of elements contained by the set. Equivalent and equal sets are also defined in this section.

Null Set

Consider a collection of those sons having their ages more than their respective fathers. Of course we will find no such son in this world. This type of collection is nothing but simply known as null set or **empty set** or **void set** in the terminology of sets.

Let us now formally define null set.

A set is said to be null (or empty or void) if it has no element in it. Null set is denoted by ϕ or $\{\}$.

For example, $A = \{x: x \text{ is a natural number, } 1 < x < 2\}$ is a null set as there is no natural number between 1 and 2.

Singleton Set

Consider the collection of mothers of a baby. Obviously a baby has only one mother. This type of collection having a single element is known as singleton set in the terminology of sets. Thus, a singleton set is defined as follow.

A set is said to be singleton set if it contains only one element.

For example, $A = \{x: x \text{ is an even prime number}\}$ is a singleton set as there is only one even prime number, i.e. 2.

Finite Set

A set is said to be finite set if either it is an empty set or it has a finite number of elements.

For example,

(i) $A = \{2, 5, 7, 15\}$ is a finite set because it contains 4 elements, i.e. finite number of elements.

(ii) $B = \{1, 2, 3, 4, 5, \dots\}$ is not a finite set as the number of elements in it are infinitely many.

(iii) $C = \{x: x + 1 = 0, x \in N\} = \{x: x = -1, x \in N\} = \{\}$ is an empty set. So, it is a finite set.

Cardinal Number of a Finite Set

The number of elements in a finite set say A is called its cardinality and is denoted by $n(A)$.

For example,

- (i) If $A = \{x, y, z\}$, then $n(A) = 3$, i.e. cardinality of A is 3.
- (ii) If $B = \{14, 2, 3, 9, 15\}$, then $n(B) = 5$, i.e. cardinality of B is 5.

Infinite Set

A set is said to be infinite if it is not finite.

For example, $A = \{1, 4, 9, 16, 25, 36, \dots\}$ is an infinite set.

Remark 3: Infinite sets are either countable or uncountable. We shall discuss it in Sec. 2.6 of Unit 2 of this block.

Equivalent Sets

Two finite sets A and B (say) are said to be equivalent if number of elements in both the sets are equal in numbers, i.e. $n(A) = n(B)$ and we denote it by $A \sim B$ (read as A is equivalent to B).

For example, if $A = \{a, b, c, d\}$ and $B = \{2, 3, 5, 7\}$, then
 $A \sim B$ [$\because n(A) = n(B) = 4$]

Equal Sets

Two sets A and B are said to be equal if every element of A is in B and every element of B is in A and is written as $A = B$.

For example, if $A = \{a, b, c, d\}$ and $B = \{c, b, d, a\}$, then $A = B$ as order of elements does not matter.

If two sets A and B are not equal then we write $A \neq B$.

Example 4: Give reasons whether the following statements are true or false:

- (i) If $A = \{x : x \text{ is a vowel of English alphabet}\}$ and
 $B = \{x : x \text{ is a natural number, } 7 < x < 13\}$, then $A = B$
- (ii) If $A = \{x : x^2 = 9, x \in \mathbb{Z}\}$ and $B = \{3, -3\}$, then $A = B$
- (iii) If $A = \{x, y, z, w\}$ and $B = \{d, e, 7, 9\}$, then $A \sim B$
- (iv) If $A = \{x, x, y, z\}$ and $B = \{x, y, z, w\}$, then $A \sim B$

Solution:

- (i) Here, $A = \{a, e, i, o, u\}$ and $B = \{8, 9, 10, 11, 12\}$.
Clearly, $A \neq B$ (as their elements are not same). Hence, the given statement is false.

- (ii) Here, $A = \{3, -3\}$ and $B = \{3, -3\}$, Clearly, $A = B$

$$\left[\begin{array}{l} \because \text{we know that if } x^2 = a \text{ then } x = \pm \sqrt{a} \\ \therefore x^2 = 9 \Rightarrow x = \pm \sqrt{9} = \pm 3 \\ \therefore \text{all the elements of } A \text{ are in } B \text{ and} \\ \text{all the elements of } B \text{ are in } A. \end{array} \right]$$
 $\therefore$ the statement is true.

- (iii) Here, $n(A) = 4, n(B) = 4 \Rightarrow A \sim B$, therefore, it is a true statement.

- (iv) Here, $A = \{x, x, y, z\} = \{x, y, z\}$ [As repetition in a set is not allowed]
and $B = \{x, y, z, w\}$, so $n(A) = 3, n(B) = 4$.
 $\Rightarrow A \sim B$, is false because $n(A) \neq n(B)$.

E 4) Give reasons whether the following statements are true or false:

- (i) If $A = \{2, 9, 7, 7, 5\}$, $B = \{5, 2, 2, 9, 7\}$, then $A = B$
- (ii) If $A = \{\alpha, \beta, \gamma\}$, $B = \{\lambda, \mu, \nu\}$, then $A \sim B$
- (iii) If $A = \{4, -4, 5, 5\}$ and
 $B = \{x : \text{either } x^2 = 16 \text{ or } x^2 + x - 20 = 0, x \in \mathbb{Z}\}$, then $A = B$
- (iv) If $A = \{a, a, b, b, b, c\}$ and $B = \{d, e, e, f, g, h\}$, then $A \sim B$

1.4 HIERARCHY OF SETS

For given any two real numbers a and b , you know that either $a = b$ or $a < b$ or $a > b$. This section will focus on how this type of association is setup in case of sets. Actually, here, we consider the sets contained in some other sets and define them with their appropriate designations.

Subset

Suppose A be the set of all working ladies and B be the set of all ladies then obviously all working ladies are ladies first. That is all the members of the set A are members of the set B . If it is so, then in the terminology of the sets A is known as subset of B .

Now, let us formally define the term subset.

Let A and B be two sets. Then A is said to be subset of B (or B is **super set** of A) if every element of A belongs to B and is denoted by $A \subseteq B$.

“ $A \subseteq B$ ” read as A is contained in B or A is a subset of B .

If we write it as “ $B \supseteq A$ ” then we read it as B contains A and we call B is a super set of A .

Remark 4: From above definition of subset, we see that a set A (say) will not be subset of another set B (say) if there is at least one element in A which is not in B . And it is denoted by $A \not\subseteq B$ (read as A is not a subset of B or A is not contained in B).

For example,

- (i) If $N = \{1, 2, 3, 4, \dots\}$, $W = \{0, 1, 2, 3, 4, 5, \dots\}$,
 $Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$, $Q = \{\frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0\}$,
then $N \subseteq W$, $W \subseteq Z$, $Z \subseteq Q$, i.e. $N \subseteq W \subseteq Z \subseteq Q$.
- (ii) If $A = \{1, 2, 4\}$, $B = \{1, 2, 4, 7, 9\}$, then $A \subseteq B$
- (iii) If $A = \{a, b, c\}$, $B = \{b, c, d, e\}$, then $A \not\subseteq B$ [$\because a \in A$ but $a \notin B$]

Proper Subset

Let A and B be two sets. Then A is said to be proper subset of B if all the elements of A are in B and B has at least one element other than elements of A and is denoted by $A \subset B$.

For example, if $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4, 5\}$,

then $A \subset B$.
 $\because$ all the elements of A are in B and B
has two extra elements, i.e. 4 and 5.

Remark 5:

- (i) Empty set is subset of every set, i.e. $\phi \subseteq A$ for any set A .
- (ii) Every set is a subset of itself, i.e. $A \subseteq A$ for every set A .

Power Set

Let us first consider some examples:

- (i) If $A = \{ \}$, then ϕ is only subset of A .
That is, there is only 1 ($= 2^0$) subset of A .
- (ii) If $A = \{a\}$, then possible subsets of A are $\phi, \{a\}$.
That is, there are only 2 ($= 2^1$) subsets of A .
- (iii) If $A = \{a, b\}$, then possible subsets of A are $\phi, \{a\}, \{b\}, \{a, b\}$.
That is, there are only 4 ($= 2^2$) subsets of A .
- (iv) If $A = \{a, b, c\}$, then possible subsets of A are $\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}$.
That is, there are only 8 ($= 2^3$) subsets of A .
- (v) Similarly, if A has n elements then total number of subsets of A are 2^n .

Now, let us define what we mean by power set of a set.

Let A be any set. Then set of all subsets of A is known as power set of A and is denoted by $P(A)$.

For example, in above discussed cases (i) to (iv) power set of A is given by

- (i) $P(A) = \{ \phi \}$
- (ii) $P(A) = \{ \phi, \{a\} \}$
- (iii) $P(A) = \{ \phi, \{a\}, \{b\}, \{a, b\} \}$
- (iv) $P(A) = \{ \phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\} \}$

Here is an exercise for you.

E 5) If $A = \{a, b, c, d\}$, then write $P(A)$.

Universal Set

In IGNOU there are 21 schools such as school of humanities (SOH), school of social sciences (SOSS), school of sciences (SOS), etc. (source IGNOU dairy 2011). If U is the set of all faculties of IGNOU and $A_1, A_2, A_3, \dots, A_{21}$ are sets representing the faculties of 21 schools. Then, of course, faculties of all these 21 schools are faculties of IGNOU. That is, all the members of these 21 schools are present in the set U . Here U plays the role of universal set for the sets $A_1, A_2, A_3, \dots, A_{21}$.

Now, let us formally define the universal set.

A set U is said to be **universal set** if all the sets under study are subsets of U .

For example,

- (i) If $A = \{1, 2, 3, 5, 7, 9\}$, $B = \{2, 4, 6, 8, 10\}$, $C = \{3, 5, 7\}$, $D = \{8, 9, 10\}$, then $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ can play the role of universal set.
- (ii) If in a study, only integers are involved as the elements of the sets, then Z , the set all integers, is the universal set.

You can now solve the following exercise.

E 6) If $A = \{1, 3, 5, 7, 9\}$, $B = \{2, 4, 6, 8\}$, $C = \{2, 3, 5, 7, 11, 13, 17, 19, 23\}$, and $D = \{5, 10, 15, 20, 25\}$, then what will be the smallest universal set?

1.5 VENN DIAGRAMS

As we know that examples play an important role to understand the concepts of theory/definitions. Similarly a diagram speaks more than the words that we may use and they also make the ideas simple and easy to understand even for a fresh reader. Euler (1707-1783), a Swiss mathematician was the first who took the step to represent the sets diagrammatically. Then John Venn (1834-1923), a British mathematician, moving a step ahead simplifying the ideas and made it more user friendly. That is why diagrammatical representation of sets is also known as Venn-Euler diagram. But usually they are known as Venn-diagrams. In Venn diagrams, sets are represented by enclosed areas in a plane as described below:

Notations Used in Venn Diagrams

1. Universal Set

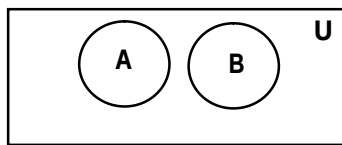
Universal set U is represented by the interior of a rectangle as shown in Fig. 1.1



Fig. 1.1

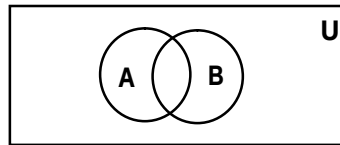
2. Subsets

Subsets of U are described by the interior of closed curves (known as circular discs) within the rectangle, representing the universal set U . Fig. 1.2 shows the case when A and B have no common element, while Fig. 1.3 shows the case when A and B have some common elements. Fig. 1.4 shows the case when $A \subseteq B$, and Fig. 1.5 shows the case when $B \subseteq A$.



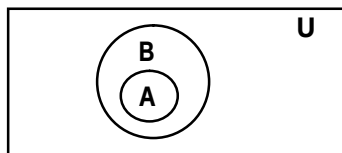
If $A \cap B = \phi$

Fig. 1.2



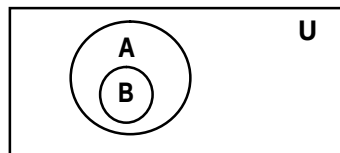
If $A \cap B \neq \phi$

Fig. 1.3



If $A \subseteq B$

Fig. 1.4



If $B \subseteq A$

Fig. 1.5

Remark 6:

- (i) In general when nothing is mentioned about the common elements of A and B , presentation of Fig. 1.3 is used.
- (ii) Sizes that we use to present A and B do not matter.

1.6 SET OPERATIONS

In school days, a child first learns counting numbers and then he/she learns how operations of addition, subtraction, multiplication and division are used on two numbers.

A similar type of approach is being used here. In sections 1.2 and 1.3 of this unit you have become familiar with the definition and types of sets respectively. In this section, we will learn about some commonly used operations on sets.

Union

Let A be a set containing the persons getting salary (in Rs) between 10000 and 100000 per month and another set B containing the persons getting salary (in Rs) between 5000 and 20000 per month. For this example, if we are interested in finding those persons who are getting the salary within the range 10000-100000 or 5000-20000 then such persons will be those having salary between 5000 and 100000. The set of such persons is nothing but the union of two sets A and B.

Now, let us formally define the union of two sets.

Let A and B be two sets then union of A and B is denoted by $A \cup B$ and is defined as

$$A \cup B = \{x: \text{either } x \in A \text{ or } x \in B\}.$$

i.e. $A \cup B$ contains all the elements of A as well as of B (see Fig. 1.6).

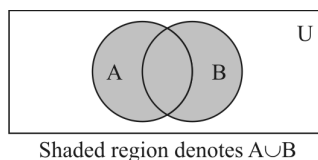


Fig. 1.6

For example,

- (i) If $A = \{2, 3, 5\}$, $B = \{3, 5, 7, 11\}$, then $A \cup B = \{2, 3, 5, 7, 11\}$.
- (ii) If $A = \{a, b, c, d\}$, $B = \{d, e, f\}$, then $A \cup B = \{a, b, c, d, e, f\}$.
- (iii) If Q = set of all rational numbers and I = set of all irrational numbers, then $Q \cup I = R$ = set of all real numbers.

That is, if all rational and irrational numbers are mixed then that mixture will be the set of real numbers.

Now, you can do the following exercises.

E 7) If $A = \{3\}$, $B = \{a, b, c\}$, then write $A \cup B$.

E 8) If $A = \{a, b\}$, $B = \{e, f\}$, then write $A \cup B$.

E 9) If $A = \{1, 2, 3\}$, $B = \{3, 4, 5, 6\}$, $C = \{3, 6, 7\}$, then write $A \cup B \cup C$.

Intersection

Let us again consider the example given above. For this example, if we are interested in finding those persons who are getting the salary within the common range then such persons will be those having salary between 10000 and 20000. The set of such persons is nothing but the intersection of two sets A and B.

Now, let us formally define the intersection of two sets.

Let A and B be two sets then intersection of A and B is denoted by $A \cap B$ and is defined as

$$A \cap B = \{x: x \in A \text{ and } x \in B\}.$$

i.e. $A \cap B$ contains common elements of A and B (see Fig. 1.7).

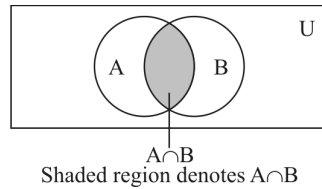


Fig. 1.7

For example,

- (i) If $A = \{a, b, c\}$, $B = \{b, c, d, e\}$, then $A \cap B = \{b, c\}$.
- (ii) If $A = \{5, 7, 9\}$, $B = \{10, 11, 18\}$, then $A \cap B = \{ \} = \phi$ = empty set, as there is no common element in the two sets.

Note: If $A \cap B = \phi$ then we say that two sets A and B are **disjoint**.

Here are some exercises for you.

E 10) If $A = \{ \}$, $B = \{1, 4, 7\}$, then write $A \cap B$.

E 11) If $A = \{2, 4, 6, 8\}$, $B = \{2, 3, 5, 7, 11, 13\}$, then write $A \cap B$.

Complement of a Set

Suppose we have set of persons of a locality having voting right. Then set of those persons of the locality who do not have voting right is its complement, if the set of all persons of that locality is considered as a universal set.

Now, let us formally define complement of a set.

Let U be the universal set, then complement of a set A (where $A \subseteq U$) is

denoted by A^c or $\bar{A}$ or A' and is defined as

$$A' = \{x \in U: x \notin A\}.$$

That is A' contains those elements of U which are not in A , i.e. A' contains all the elements of U other than A (see Fig. 1.8.).

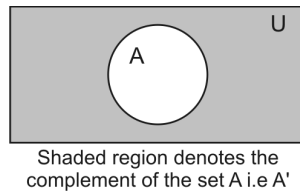


Fig. 1.8

For example, if $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and $A = \{1, 2, 4, 7, 9, 10\}$, then $A' = \{3, 5, 6, 8\}$, i.e. those elements of U which are not in A .

Here is an exercise for you.

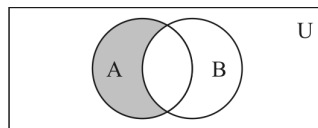
E 12) If $U = \{x: x \text{ is an English alphabet}\}$ and

$A = \{x: x \text{ is an vowel of English alphabet}\}$, then write A' .

Difference of Two Sets

Let A and B be two sets then difference of A and B is denoted by $A - B$ and is defined as

$A - B = \{x: x \in A \text{ but } x \notin B\}$. See Fig. 1.9



Shaded region denotes $A - B$

Fig. 1.9

For example,

(i) If $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7\}$, then $A - B = \{1, 2, 3\}$

(ii) If $A = \{a, b, c\}$, $B = \{a, b, c, d, e\}$, then $A - B = \{ \}$

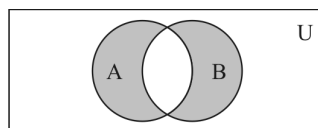
Now, you can do the following exercise.

E 13) If W = set of whole numbers and N = set of natural numbers then write $W - N$.

Symmetric Difference of Two Sets

Let A and B be two sets, then symmetric difference of A and B is denoted by $A \Delta B$ and is defined as

$A \Delta B = (A - B) \cup (B - A)$. See Fig. 1.10



Shaded region denotes $A \Delta B$

Fig. 1.10

For example, if $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6, 7\}$, then $A - B = \{1, 2\}$,

$B - A = \{5, 6, 7\}$

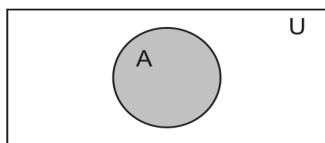
$\therefore A \Delta B = (A - B) \cup (B - A) = \{1, 2\} \cup \{5, 6, 7\} = \{1, 2, 5, 6, 7\}$.

Example 5: With the help of the Venn diagrams, show the following sets:

(i) A (ii) A' (iii) $A \cap B$ (iv) $A - B$ (v) $B - A$ (vi) $A \Delta B$ (vii) $A \subseteq B$ (viii) $B \subset A$

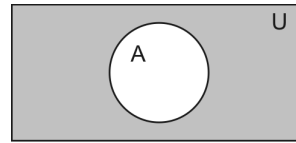
Solution:

(i)



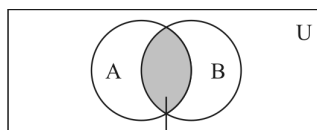
Shaded region denotes the set A

(ii)



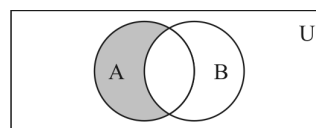
Shaded region denotes the complement of the set A i.e. A'

(iii)



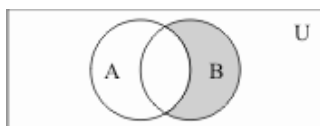
$A \cap B$
Shaded region denotes $A \cap B$

(iv)

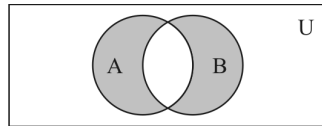


Shaded region denotes $A - B$

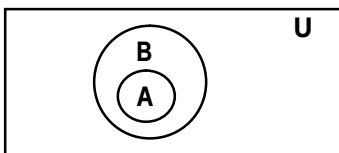
(v)

Shaded region denotes $B - A$

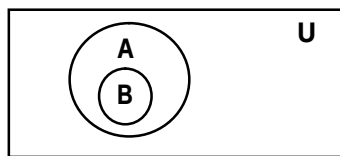
(vi)

Shaded region denotes $A \Delta B$

(vii)



(viii)



Now, you can try the following exercise.

E 14) With the help of the Venn diagram justify the following

(i) $A \subseteq A \cup B$ (ii) $B \subseteq A \cup B$ (iii) $A \cap B \subseteq A$ (iv) $A \cap B \subseteq B$

(v) $(A \cap B)' = A' \cup B'$ (De-Morgan's law) (vi) $A \subseteq B \Rightarrow B' \subseteq A'$

1.7 SOME USEFUL AND IMPORTANT LAWS

Some commonly used laws of sets are listed below.

1. Idempotent Laws

For any set A

(i) $A \cup A = A$ (ii) $A \cap A = A$

2. Identity Laws

For any subset A of the universal set U

$A \cup \phi = A$, $A \cap U = A$, where ϕ is empty set

3. Commutative Laws

For any two sets A and B

(i) $A \cup B = B \cup A$ (ii) $A \cap B = B \cap A$

4. Associative Laws

If A , B , C are any three sets then

(i) $(A \cup B) \cup C = A \cup (B \cup C)$

(ii) $(A \cap B) \cap C = A \cap (B \cap C)$

5. Distributive Laws

If A , B , C are any three sets then

(i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

6. De-Morgan's Laws

For any two sets A and B

(i) $(A \cup B)' = A' \cap B'$

(ii) $(A \cap B)' = A' \cup B'$

Example 6: If $A = \{1, 3, 5\}$, $B = \{3, 5, 7, 9\}$, $C = \{2, 6, 8, 9\}$ are subsets of the universal set $U = \{1, 2, 3, 5, 6, 7, 8, 9\}$ then verify

(i) De-Morgan's laws and

(ii) Distributive laws

Solution:

(i) De-Morgan's laws state that

(a) $(A \cup B)' = A' \cap B'$

(b) $(A \cap B)' = A' \cup B'$

To verify (a)

$$A \cup B = \{1, 3, 5, 7, 9\}$$

$$\therefore (A \cup B)' = \{2, 6, 8\}$$

$$A' = \{2, 6, 7, 8, 9\}, B' = \{1, 2, 6, 8\}$$

$$\therefore A' \cap B' = \{2, 6, 8\}$$

We see that here $(A \cup B)' = A' \cap B'$.

Hence verified.

To verify (b)

$$A \cap B = \{3, 5\}$$

$$\therefore (A \cap B)' = \{1, 2, 6, 7, 8, 9\}$$

$$A' \cup B' = \{1, 2, 6, 7, 8, 9\}$$

We see that here $(A \cap B)' = A' \cup B'$.

Hence verified.

(ii) Distributive laws state that

(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

To verify (a)

$$B \cap C = \{9\}$$

$$\therefore A \cup (B \cap C) = \{1, 3, 5, 9\}$$

$$A \cup B = \{1, 3, 5, 7, 9\}, A \cup C = \{1, 2, 3, 5, 6, 8, 9\}$$

$$\therefore (A \cup B) \cap (A \cup C) = \{1, 3, 5, 9\}$$

We see that here $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Hence verified.

To verify (b)

$$B \cup C = \{2, 3, 5, 6, 7, 8, 9\}$$

$$\therefore A \cap (B \cup C) = \{3, 5\}$$

$$A \cap B = \{3, 5\}, A \cap C = \{\}$$

$$\therefore (A \cap B) \cup (A \cap C) = \{3, 5\}$$

We see that here $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Hence verified.

Now, you can do the following exercise.

E 15) If the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and $A = \{1, 3, 5, 6, 7\}$, $B = \{5, 6, 7, 8\}$, $C = \{1, 5, 7, 8\}$ are subsets of U , then verify

(i) commutative laws and (ii) associative laws

Application of Sets

Venn diagrams are helpful in establishing many important relations between different sets, some of them are mentioned as under which are helpful in solving many practical problems too.

1. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

2. $n(A \cup B) = n(A - B) + n(A \cap B) + n(B - A)$

3. $n(A - B) = n(A) - n(A \cap B)$
4. $n(B - A) = n(B) - n(A \cap B)$
5. $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$

Let us consider an example based on above formulae.

Example 7: In a group of 500 persons, 400 can speak Hindi and 150 can speak English. Then how many can speak

- (i) both Hindi and English
- (ii) Hindi only
- (iii) English only

Solution: Let A and B denote the set of persons who can speak Hindi and English, respectively. Then in usual notations, we are given

$$n(A \cup B) = 500, n(A) = 400, n(B) = 150$$

- (i) We know that
 $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
 $\Rightarrow 500 = 400 + 150 - n(A \cap B) \Rightarrow n(A \cap B) = 550 - 500 = 50$
 $\therefore$ number of persons who can speak both Hindi and English = 50
- (ii) We know that
 $n(A - B) = n(A) - n(A \cap B) = 400 - 50 = 350$
 $\therefore$ number of persons who can speak Hindi only = 350
- (iii) We know that
 $n(B - A) = n(B) - n(A \cap B) = 150 - 50 = 100$
 $\therefore$ number of persons who can speak English only = 100

Here is an exercise for you.

E 16) Out of the 50 students in a class, 24 play cricket, 15 play hockey, 18 play football, 6 play cricket and hockey, 8 play cricket and football, 5 play hockey and football and 10 students do not play any of the three games. Then how many play (i) all the three games, (ii) hockey but not football and (iii) cricket and football but not hockey.

1.8 SUMMARY

Let us now summarise what we have covered in this unit.

- 1) Definition of a set with examples.
- 2) Two methods of writing a set.
- 3) Definition of various types of sets including empty set, singleton set, finite set, infinite set, equivalent sets, and equal sets.
- 4) Definition of subsets, proper subset, super set, universal set, power set.
- 5) Introduction of Venn diagrams.
- 6) Various operations on sets.
- 7) Idempotent, identity, commutative, associative, distributive and De-Morgan's laws.

- E 1)** (i) It is not a set, because a student may be intelligent according to someone while the same student may not be intelligent according to some other person.
- (ii) It is not a set because a hockey player may be good in someone's view while the same player may not be good in view of some other person.
- (iii) It is not a set because an actor may be good in someone's point of view while the same actor may not be good in the view of some other person.

(iv) It is a set having elements I, N, D, A $\left[\begin{array}{l} \because \text{repetition of elements} \\ \text{in a set is not allowed.} \end{array} \right]$

E 2) (i) $x = 2n + 3, n \in W = \{0, 1, 2, 3, 4, \dots\}$

$$\therefore x = 3, 5, 7, 9, 11, \dots \quad \left[\begin{array}{l} \text{Values are obtained on putting} \\ n = 0, 1, 2, \dots \text{ in } x = 2n + 3 \\ \text{e.g. when we put } n = 0, \text{ we get} \\ x = 2(0) + 3 = 3 \end{array} \right]$$

And, therefore, $A = \{3, 5, 7, 9, 11, \dots\}$

(ii) $B = \{7^0, 7^1, 7^2, 7^3\} = \{1, 7, 49, 343\}$

(iii) $N = \{1, 2, 3, 4, \dots\}$ and $W = \{0, 1, 2, 3, 4, \dots\}$
 $\therefore$ if $x \in N$ and $x \in W$ then the elements which satisfy both are 1, 2, 3, 4, ... and hence
 $C = \{1, 2, 3, 4, 5, \dots\}$.

(iv) $W = \{0, 1, 2, 3, \dots\}$ and $Q =$ set of rational numbers,
 $\therefore$ the elements which are common to both are 0, 1, 2, 3, ... as no other rational number is a whole number and hence
 $D = \{0, 1, 2, 3, \dots\}$.

E 3) (i) $\{x \in N : x = 5n, n \in N\}$

(ii) $\{x : x = \frac{1}{n}, n \in N\}$

(iii) $\{x \in N : x = 2n, n \in N\}$

E 4) (i) $A = \{2, 9, 7, 5\}, B = \{5, 2, 9, 7\}$ as repetitions in a set are not allowed.

We see that all the elements of A are in B and all the elements of B are in A.

$\therefore A = B$. Hence the statement is true.

(ii) Here, $n(A) = 3, n(B) = 3$ and so the statement $A \sim B$ is true.

(iii) $A = \{4, -4, 5\}, B = \{x : x^2 = 16, x^2 + x - 20 = 0, x \in Z\}$
 $= \{x : x = \pm 4, (x+5)(x-4) = 0, x \in Z\}$
 $= \{x : x = \pm 4, x = -5, 4, x \in Z\} = \{4\}$

Here, $-4 \in A$ but $-4 \notin B$

$\therefore A \neq B$, hence the statement is false.

(iv) Here, $A = \{a, b, c\}, B = \{d, e, f, g, h\}$. Thus, $n(A) = 3, n(B) = 5$.
 $\therefore A \sim B$ is false because $n(A) \neq n(B)$.

E 5) $P(A) = \{\phi, \{a\}, \{b\}, \{c\}, \{d\}, \{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\}, \{a, b, c\}, \{a, b, d\}, \{a, c, d\}, \{b, c, d\}, \{a, b, c, d\}\}$

Notice that it has $2^4 = 16$ elements.

E 6) Smallest universal set is $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 15, 17, 19, 20, 23, 25\}$.

E 7) $A \cup B = \{3, a, b, c\}$.

E 8) $A \cup B = \{a, b, e, f\}$.

E 9) $A \cup B \cup C = \{1, 2, 3, 4, 5, 6, 7\}$.

E 10) $A \cap B = \{\} = \phi$.

E 11) $A \cap B = \{2\}$.

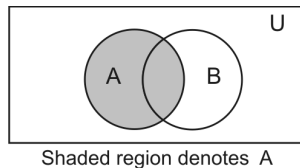
E 12) $A' = \{x: x \text{ is a consonant of English alphabet}\}$.

E 13) We know that

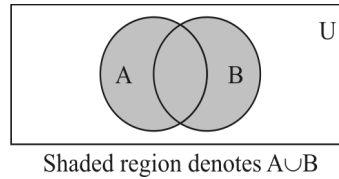
$$W = \{0, 1, 2, 3, 4, \dots\} \text{ and } N = \{1, 2, 3, 4, \dots\}$$

$$\therefore W - N = \{0, 1, 2, 3, 4, \dots\} - \{1, 2, 3, 4, \dots\} = \{0\}$$

E 14) (i)



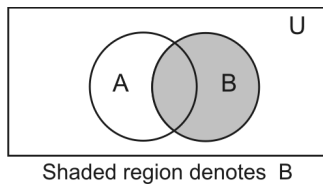
L.H.S.



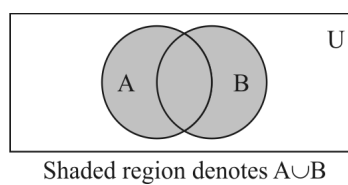
R.H.S.

Two Venn diagrams justify the relationship $A \subseteq A \cup B$.

(ii)



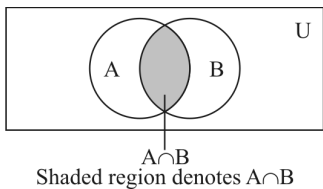
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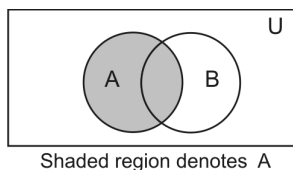
R.H.S.

Two Venn diagrams justify the relationship $B \subseteq A \cup B$.

(iii)



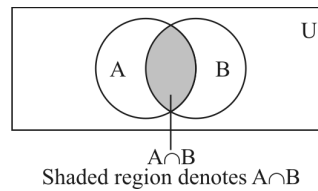
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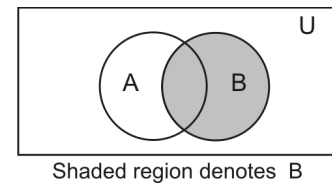
R.H.S.

Two Venn diagrams justify the relationship $A \cap B \subseteq A$.

(iv)



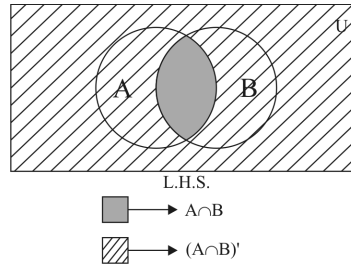
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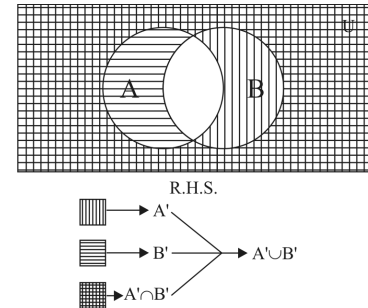
R.H.S.

Two Venn diagrams justify the relationship $A \cap B \subseteq B$.

(v)



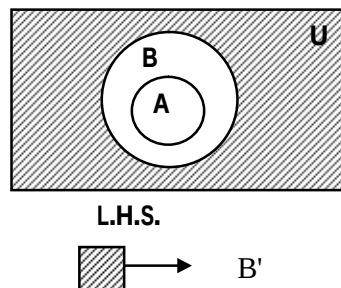
L.H.S.



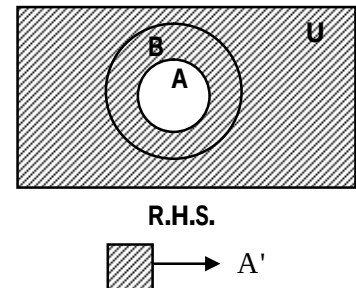
R.H.S.

Two Venn diagrams justify the relationship $(A \cap B)' = A' \cup B'$.

(vi)



L.H.S.



R.H.S.

Two Venn diagrams justify the relationship $A \subseteq B \Rightarrow B' \subseteq A'$.

E 15) (i) Commutative laws state that

(a) $A \cup B = B \cup A$

(b) $A \cap B = B \cap A$

To verify (a)

$A \cup B = \{1, 3, 5, 6, 7\} \cup \{5, 6, 7, 8\} = \{1, 3, 5, 6, 7, 8\}$
 $\therefore B \cup A = \{5, 6, 7, 8\} \cup \{1, 3, 5, 6, 7\} = \{1, 3, 5, 6, 7, 8\}$
 $\therefore$ we see that here $A \cup B = B \cup A$.
Hence verified.

To verify (b)

$A \cap B = \{1, 3, 5, 6, 7\} \cap \{5, 6, 7, 8\} = \{5, 6, 7\}$
and $B \cap A = \{5, 6, 7, 8\} \cap \{1, 3, 5, 6, 7\} = \{5, 6, 7\}$
We see that here $A \cap B = B \cap A$
Hence verified.

(ii) Associative laws state that

$$(a) (A \cup B) \cup C = A \cup (B \cup C)$$

$$(b) (A \cap B) \cap C = A \cap (B \cap C)$$

To verify (a)

$$A \cup B = \{1, 3, 5, 6, 7, 8\}$$

$$\therefore (A \cup B) \cup C = \{1, 3, 5, 6, 7, 8\}$$

$$B \cup C = \{5, 6, 7, 8\} \cup \{1, 5, 7, 8\} = \{1, 5, 6, 7, 8\}$$

$$\therefore A \cup (B \cup C) = \{1, 3, 5, 6, 7\} \cup \{1, 5, 6, 7, 8\} = \{1, 3, 5, 6, 7, 8\}$$

We see that here $(A \cup B) \cup C = A \cup (B \cup C)$.

Hence verified.

To verify (b)

$$A \cap B = \{5, 6, 7\}$$

$$\therefore (A \cap B) \cap C = \{5, 6, 7\} \cap \{1, 5, 7, 8\} = \{5, 7\}$$

$$B \cap C = \{5, 6, 7, 8\} \cap \{1, 5, 7, 8\} = \{5, 7, 8\}$$

$$\therefore A \cap (B \cap C) = \{1, 3, 5, 6, 7\} \cap \{5, 7, 8\} = \{5, 7\}$$

We see that here $(A \cap B) \cap C = A \cap (B \cap C)$.

Hence verified.

E 16) Let C, H, F, denote the set of students who play cricket, hockey and football respectively. Then in usual notations, we are given.

$$n(C) = 24, n(H) = 15, n(F) = 18, n(C \cap H) = 6, n(C \cap F) = 8,$$

$$n(H \cap F) = 5, n(C' \cap H' \cap F') = 10$$

(i) Before finding the required number of students, we are to first obtain the number of students who play at least one of the three games which is given as

$$= n(C \cup H \cup F) = 50 - n(C' \cap H' \cap F') = 50 - 10 = 40$$

Now, we know that

$$n(C \cup H \cup F) = n(C) + n(H) + n(F) - n(C \cap H) - n(C \cap F) - n(H \cap F) + n(C \cap H \cap F)$$

$$\Rightarrow 40 = 24 + 15 + 18 - 6 - 8 - 5 + n(C \cap H \cap F)$$

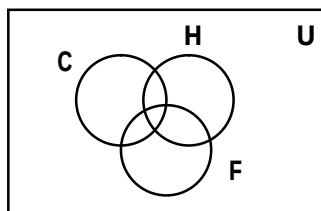
$$\Rightarrow 40 = 57 - 19 + n(C \cap H \cap F) = 38 + n(C \cap H \cap F)$$

$$\Rightarrow n(C \cap H \cap F) = 40 - 38 = 2$$

$\therefore$ number of students who play all the three games = 2

(ii) We know that

$$\begin{aligned} n(H - F) &= n(H) - n(H \cap F) \\ &= 15 - 5 = 10 \end{aligned}$$



$\therefore$ number of students who play hockey but not football = 10

$$(iii) n((C \cap F) - H) = n(C \cap F) - n((C \cap F) \cap H) = 8 - 2 = 6$$

$$\left[\begin{array}{l} \because n(A - B) = n(A) - n(A \cap B) \\ \text{Here } A = C \cap F, B = H \end{array} \right]$$

$\therefore$ number of students who play cricket and football but not hockey = 6

UNIT 2 FUNCTIONS

Structure

- 2.1 Introduction
 - Objectives
 - 2.2 Quantity
 - 2.3 Interval
 - 2.4 Function
 - 2.5 Classification of functions
 - 2.6 Types of functions
 - 2.7 Summary
 - 2.8 Solutions/Answers
-

2.1 INTRODUCTION

Many times, we observe the association of the elements of one set with the elements of another set. For example, roll numbers of students in an examination of a university are associated with their corresponding marks. Such type of associations are discussed under the heading “Function”. In this unit, we will focus on definition of function, its classification and various types.

In this unit, we will use some concepts related to sets discussed in the preceding unit.

Objectives

After completing this unit, you should be able to:

- define constant quantity, variable, interval, give some examples of each;
 - define function, some particular functions;
 - evaluate the value of some particular functions at given points;
 - get an idea of one-one, onto and one to one correspondence and their geometrical interpretation; and
 - define countable and uncountable sets.
-

2.2 QUANTITY

Before defining function, let us first explain what we mean by quantity, constant, variable and intervals. See Fig. 2.1

Quantity

Here, by quantity we mean those things on which four basic mathematical operations addition, subtraction, multiplication and division can be applied. For example, temperature, height, weight, and time all these are quantities but they are continuous in nature, where as books in a library, number of trees, number of balls is discrete in nature.

Note 1: If the nature of the quantity is such that it can take any possible value between two certain limits then such a quantity is known as continuous in nature.

Let us consider the following example:

Suppose at the time of birth, height of a baby was 1.5 ft and after 15 years, the height of the same baby is 5 ft, then we know that height of this baby took all possible values between 1.5ft. to 5ft. That is, this is not the case that at the time of birth the height was 1.5 ft and in the next moment it reached to 1.6 ft and in successive moment to 1.7 ft. In fact, there are infinitely many values between 1.5 ft and 1.6 ft and all these values were taken by the height of that baby.

Note 2: If the nature of the quantity is such that it can take at most countable values between two certain limits then such a quantity is known as discrete in nature. Countable set is defined in Sec. 2.6 of this unit.

Let us consider an example:

Number of children per family in a locality is an example of discrete quantity.

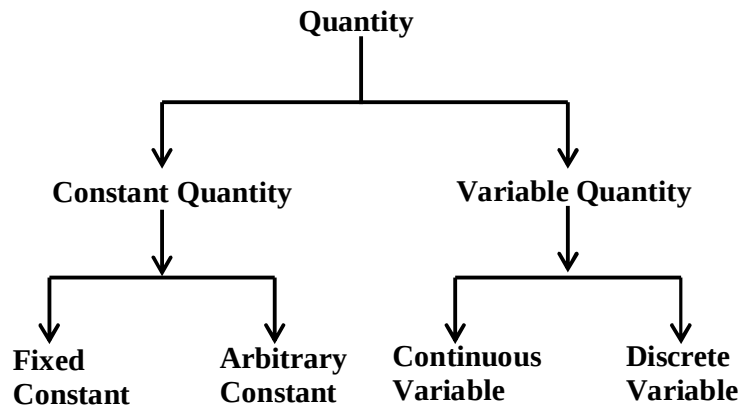


Fig. 2.1

Constant Quantity

A quantity which remains same (unchanged) throughout a particular investigation is known as a constant quantity.

Fixed Constant

Those types of constants which always remain same (unchanged) independent of the purpose of user, place and time are known as fixed constants.

For example,

(i) $2, 5, \frac{3}{2}, \sqrt{13}, -\sqrt{17}, , \text{etc.}$

(ii) quotient of circumference of a circle and its diameter which is always equal to π .

Arbitrary Constants

Those types of constants which remain same in one problem but may vary from problem to problem are known as arbitrary constants and are generally denoted by $a, b, c, l, m, n, \alpha, \beta, \gamma$, etc.

For example, heights of the houses constructed by the different owners of the plots in a locality as per their own choice, is an example of arbitrary constant because heights of the houses vary from house to house as it depends on the choice of the owners.

Variable

A quantity which may change its value even in a particular problem is known as variable.

For example, blood pressures of a person as it vary time to time.

2.3 INTERVAL

Let R be the set of all real numbers. Then a set $I \subseteq R$ is said to be an interval if whenever $a, b \in I$ and $a < x < b$ then $x \in I$

For example, the set of all real numbers satisfying $2 \leq x \leq 3$ is an interval where x can take any real value between 2 and 3 including 2 and 3.

Open Interval

An open interval $I \subseteq R$ with end points a and b ($a < b$) is denoted by (a, b) and is defined by

$$(a, b) = \{x \in R : a < x < b\}$$

i.e. an open interval contains each value between the end points but does not include the end points.

For example, open interval $(2, 5)$ contains each real number lying between 2 and 5 but does not contain 2 and 5.

Closed Interval

A closed interval $I \subseteq R$ with end points a and b ($a < b$) is denoted by $[a, b]$ and is defined by

$$[a, b] = \{x \in R : a \leq x \leq b\}$$

i.e. a closed interval contains each value between and including extreme values.

For example, closed interval $[2, 5]$ contains each real number lying between 2 and 5. It also contains its end points. i.e.

$$x \in [2, 5] \quad \forall x, \quad 2 < x < 5 \text{ and}$$

$$\text{also } 2 \in [2, 5], \quad 5 \in [2, 5]$$

Left Open and Right Closed Interval

A left open and right closed interval $I \subseteq R$ with end points a and b ($a < b$) is denoted by $(a, b]$ and is defined as

$$(a, b] = \{x \in R : a < x \leq b\}$$

In this case $x \in (a, b], \forall x, a < x \leq b$ and $a \notin (a, b]$, but $b \in (a, b]$

Left Closed and Right Open Interval

A left closed and right open interval $I \subseteq R$ with end points a and b ($a < b$) is denoted by $[a, b)$ and is defined as

$$[a, b) = \{x \in R : a \leq x < b\}$$

In this case $x \in [a, b), \forall x, a \leq x < b$ and $a \in [a, b)$, but $b \notin [a, b)$

Length of an Interval

Length of each of the intervals (a, b) , $[a, b]$, $(a, b]$, $[a, b)$ is defined as $b - a, a < b$

i.e. length of the interval = difference of the end points

For example, if $I = (2, 7)$ then $l(I) = 7 - 2 = 5$, where $l(I)$ denotes the length of the interval I .

Finite Interval

An interval is said to be finite if its length is finite.

For example, if $I = (-3, 5)$ then $l(I) = 5 - (-3) = 5 + 3 = 8$ which is finite.
 $\therefore$ interval I is finite.

Infinite Interval

An interval is said to be infinite interval if its length is not finite.

For example,

- (i) The set $\{x \in \mathbb{R} : x > a, a \in \mathbb{R}\}$ is an infinite interval and is denoted by (a, ∞)
- (ii) The set $\{x \in \mathbb{R} : x < a, a \in \mathbb{R}\}$ is an infinite interval and is denoted by $(-\infty, a)$

Similarly, infinite intervals $[a, \infty)$, $(-\infty, a]$ are defined as

$[a, \infty) = \{x \in \mathbb{R} : x \geq a\}$, where a is a fixed real number

$(-\infty, a] = \{x \in \mathbb{R} : x \leq a\}$, where a is a fixed real number

Remark 1:

- (i) Each interval contains infinitely many elements.
- (ii) Each interval is an infinite set but an infinite set may or may not be an interval. For example $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$ are infinite sets but are not intervals.
- (iii) A set may or may not be an interval.
- (iv) $\mathbb{R}$, set of real numbers, is an infinite interval given by $\mathbb{R} = (-\infty, \infty)$
- (v) Remember that $-\infty$ and ∞ are not included in the set of real numbers. Also, if extreme value is ∞ or $-\infty$, then open bracket is used on the side having extreme value.
- (vi) When we say that x is a finite number/real number it means that $-\infty < x < \infty$

Now we are in a position to define function.

2.4 FUNCTION

Definition of Function

Let X and Y be two sets. Then a rule which associates each element of X to a unique element of Y is called a function.

X is called domain of the function.

Y is called co-domain of the function and set of only those values of Y for which function is defined is called range of the function.

That is, subset $\{y \in Y : y = f(x) \text{ for some } x \in X\}$ of Y is called range of the function.

Notation:

- (i) A function is generally denoted by f , g , h , etc., in the case of above definition we write $f: X \rightarrow Y$ and read as f is a function from X to Y .
- (ii) A function $f: X \rightarrow Y$ is generally described by writing $y = f(x)$, $x \in X$, where $f(x)$ is an expression in terms of x .

There are two conditions for a rule to be a function.

- (i) Each element of X must be associated to some element of Y .
- (ii) There is unique element of Y corresponding to each element of X .

Pictorial Presentation of a Function

A function can be presented diagrammatically. As shown in the example of mothers and daughters discussed below.

To get a clear cut idea of the definition of a function without cramming and for a long time memory. Let us consider a real life example.

Let X = set of daughters and Y = set of mothers

Then consider the following situations given in (a), (b), (c) and (d) with the help of the diagrams.

(a)

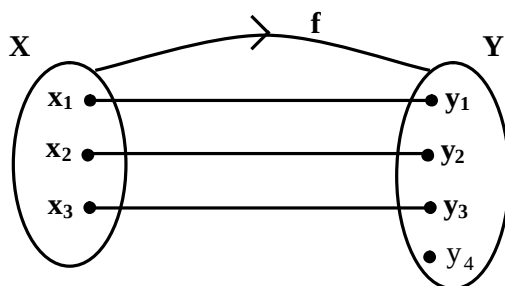


Fig. 2.2

The rule f shown in Fig. 2.2 is a function because each daughter has unique mother. [i.e. both conditions mentioned in the box are satisfied] and domain of the function f = set of all daughters = $X = \{x_1, x_2, x_3\}$

co-domain of this function = set of all mothers = $Y = \{y_1, y_2, y_3, y_4\}$

range of this function = set of those mothers who has at least one daughter
= $\{y_1, y_2, y_3\}$

Note 3: One point which may come in your mind is that if y_4 is a mother then there should be at least one daughter of y_4 . But as we know that to become a mother it is not necessary that there should be a daughter. A mother may have only one son or only two sons or more than two sons without a daughter.

(b)

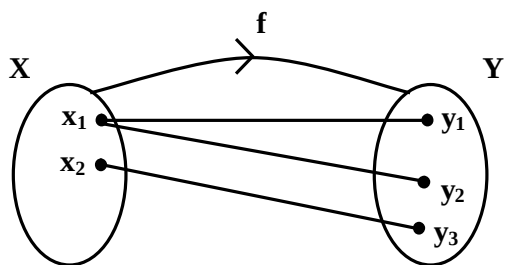


Fig. 2.3

The rule f shown in Fig. 2.3 is not a function because x_1 has two mothers y_1, y_2 which is not possible. [i.e. condition (ii) given in the box is not satisfied]

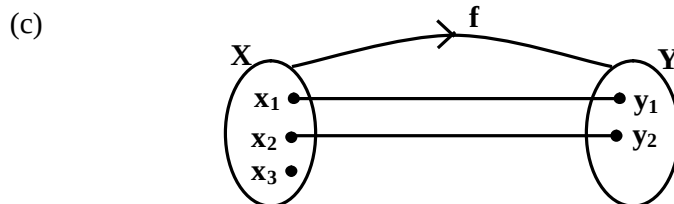


Fig. 2.4

The rule f shown in Fig. 2.4 is not a function because daughter x_3 has no mother. If x_3 came in this world then there should be some mother of x_3 . [i.e. condition (i) given in the box is not satisfied]

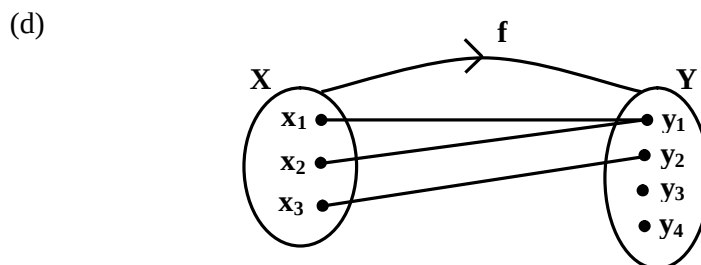


Fig. 2.5

The rule f shown in Fig. 2.5 is a function because each daughter has unique mother. We see that x_1, x_2 both have same mother, no problem it is possible. Further mothers y_3, y_4 have no daughters again no problem it is also possible. In this case:
domain of the function $f = \{x_1, x_2, x_3\} = X =$ set of all daughters,
co-domain of the function $f = \{y_1, y_2, y_3, y_4\} = Y =$ set of all mothers and
range of the function $f = \{y_1, y_2\} =$ set of only those mothers who have at least one daughter.

Some more Concepts Related to Function are given as under

- (i) If $f : X \rightarrow Y$ is a function given by $y = f(x)$ then x is known a pre image of y and y is known as image of x .
- (ii) If $y = f(x)$ is a function then values of y depend on values of x . So, y is known as dependent variable and x is known as independent variable.

Let us now take up some examples which will enable you to distinguish as to whether a rule is a function or not.

For example,

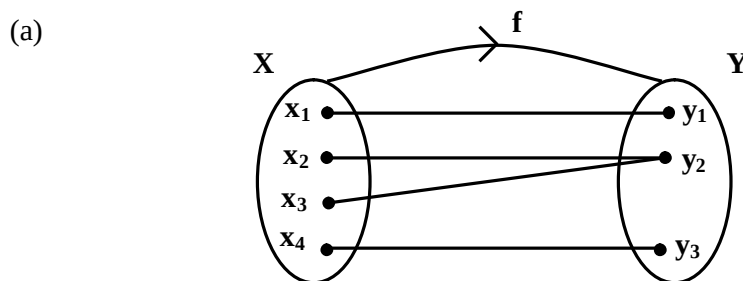


Fig. 2.6

The rule f shown in Fig. 2.6 is a function because it satisfies both conditions i.e.

- (i) Each element of X is associated to some element of Y
- (ii) There is unique element of Y corresponding to each element of X .
 i.e. y_1 is unique element of Y corresponding to $x_1 \in X$
 y_2 is unique element of Y corresponding to both $x_2, x_3 \in X$
 y_3 is unique element of Y corresponding to $x_4 \in X$

Here domain of $f = \{x_1, x_2, x_3, x_4\}$

Co-domain of $f = \{y_1, y_2, y_3\}$ and range of $f = \{y_1, y_2, y_3\}$

(b)

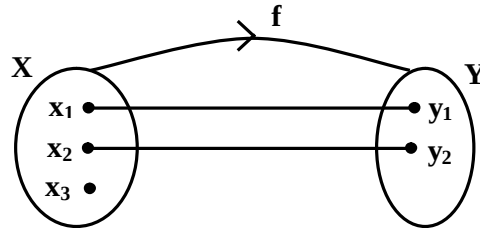


Fig. 2.7

The rule f shown in Fig. 2.7 is not a function because $x_3 \in X$, but it is not associated to the element of Y .

$\because$ out of two restriction for a rule to be a function, first is that each element of X must be associated to some element of Y .

(c)

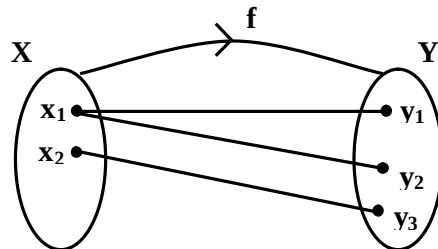


Fig. 2.8

The rule f shown in Fig. 2.8 is also not a function because $x_1 \in X$ is not associated to unique element of Y .

$\because$ for a rule to be a function it is must that each element of X is associated to a unique elemeny of Y .

(d)

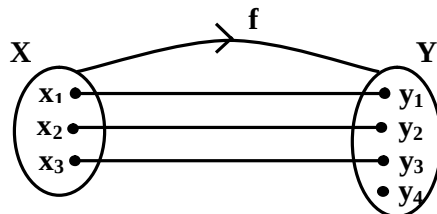


Fig. 2.9

The rule f shown in Fig. 2.9 is a function because it satisfies both the conditions for a rule to be a function, i.e.

- (i) Each element of X is associated to some element of Y .
- (ii) There is unique element of Y corresponding to each element of X .

Here domain of the function $f = \{x_1, x_2, x_3\} = X$

Co-domain of the function $f = \{y_1, y_2, y_3, y_4\} = Y$

Range of the function $f = \{y_1, y_2, y_3\} \subset Y$

Some Examples of Functions

Example 1: Let $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by

$$f(n) = 3n, \quad n \in \mathbb{N}$$

Express the function diagrammatically. Also write domain, range and co-domain of the function.

Solution: $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by

$$f(x) = 3n, \quad n \in \mathbb{N}$$

$\therefore f(1) = 3, f(2) = 6, f(3) = 9$ and so on. See Fig. 2.10

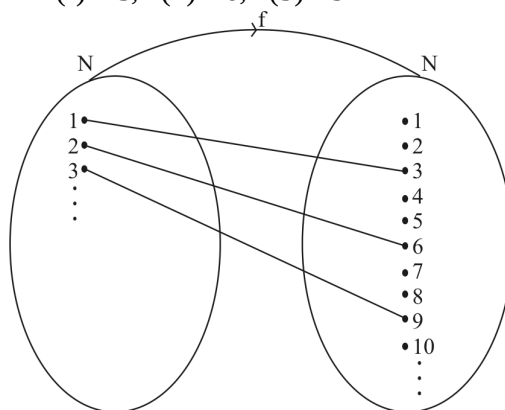


Fig. 2.10

Domain of the function $f = \{1, 2, 3, \dots\} = \mathbb{N}$

Range of the function $f =$ Set of only those values for which function is define
 $= \{3, 6, 9, \dots\}$

Co-domain = Set of all values of $Y = \{1, 2, 3, \dots\} = \mathbb{N}$

Example 2: If $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = 3^x, \quad x \in \mathbb{R}, \text{ then obtain (i) Domain of } f \text{ (ii) Range of } f$$

Solution: $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$f(x) = 3^x, \quad x \in \mathbb{R}$$

(i) Since $f(x)$ is defined for all $x \in \mathbb{R}$

$\therefore$ domain of $f = \mathbb{R} =$ set of all real numbers

(ii) We Know that

$$3^x > 0 \quad \forall x \in \mathbb{R}$$

$$\text{i.e. } f(x) > 0 \quad \forall x \in \mathbb{R}$$

$\therefore$ range of $f = (0, \infty)$

Example 3: Find the domain of the function $f : \mathbb{R} \rightarrow \mathbb{R}$, defined by

$$f(x) = \sqrt{(x-3)(5-x)}, \quad x \in \mathbb{R}$$

Also evaluate $f(3)$, $f(4)$, $f(5)$.

Solution: Given function is

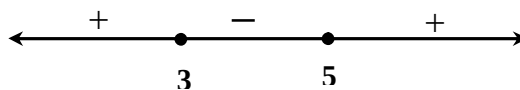
$$f(x) = \sqrt{(x-3)(5-x)}, \quad x \in \mathbb{R}$$

For $f(x)$ to be real, the quantity under the square root should be non negative and hence

$$(x-3)(5-x) \geq 0$$

$$\Rightarrow -(x-3)(x-5) \geq 0$$

$$\Rightarrow (x-3)(x-5) \leq 0$$



Now, when we take x less than 3, the L.H.S. comes out to be

$(-ve)(-ve) = +ve$ hence does not satisfy the inequality.

Also, if we take x greater than 5, the L.H.S. comes out to be

$(+ve)(+ve) = +ve$ and hence this value does not satisfy the inequality.

But, if we take $3 < x < 5$, the L.H.S. becomes $(+ve)(-ve) = -ve$ and hence satisfies the inequality.

$$\Rightarrow x \in [3, 5]$$

$$\therefore \text{domain of } f = [3, 5]$$

$$\text{Also } f(3) = \sqrt{(3-3)(5-3)} = \sqrt{0 \times 2} = \sqrt{0} = 0$$

$$f(4) = \sqrt{(4-3)(5-4)} = \sqrt{1 \times 1} = 1$$

$$f(5) = \sqrt{(5-3)(5-5)} = \sqrt{2 \times 0} = \sqrt{0} = 0$$

Now, you can try the following exercises.

E 1) Find the domain and range of the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = 4x + 5, x \in \mathbb{R}. \text{ Also, evaluate } f(0), f\left(\frac{1}{2}\right).$$

E 2) Find the domain and range of the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = \frac{1}{x-2}, x \in \mathbb{R}. \text{ Also, evaluate } f(1), f(3), f(-5).$$

E 3) Find the domain and range of the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = \frac{1}{x+3}, x \in \mathbb{R}. \text{ Also, evaluate } f(0), f(2), f(-3) \text{ if possible.}$$

2.5 CLASSIFICATION OF FUNCTIONS WITH THEIR GRAPHS

In previous Sec. we have seen that function is a rule (satisfying two conditions mentioned in the box at page number 30) which associates the elements of one set to the elements of another set. Based on the nature of classification a function may be given some particular names. In this section you will meet some of these commonly used names, their definitions followed by some examples.

Constant Function

Let $X, Y \subseteq \mathbb{R}$, then a function $f: X \rightarrow Y$ is said to be a constant function if it is defined as

$$f(x) = a, \quad \forall x \in X, \text{ where } a \text{ is a real constant.}$$

i.e. a function is constant if range is a singleton set.

i.e. all elements of the domain are associated to a single element of the co-domain of the function.

For example,

(i) $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by

$$f(x) = 3, \quad n \in \mathbb{N}$$

is a constant function because all elements of the domain are associated to the single element 3 as shown in the Fig. 2.11

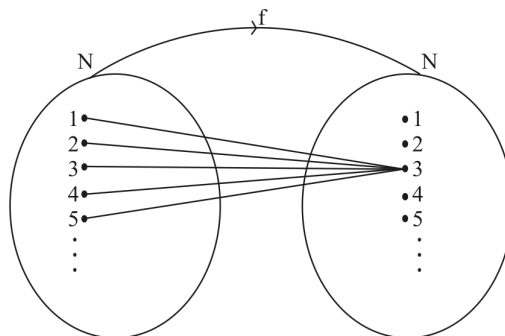


Fig. 2.11

(ii) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = 2, \quad x \in \mathbb{R}$$

is also a constant function and its graph is given below in Fig. 2.12.

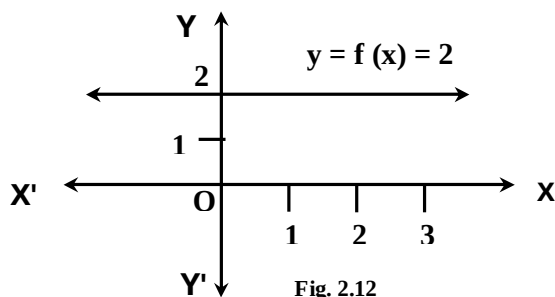


Fig. 2.12

Identity Function

Let $X \subseteq \mathbb{R}$, a function $f : X \rightarrow X$ is said to be an identity function if it is defined as

$$f(x) = x, \quad x \in X$$

i.e. a function is said to be identity function if each element is associated to itself.

For example,

(i) $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by

$$f(n) = n, \quad n \in \mathbb{N}$$

is an identity function as shown in the Fig. 2.13.

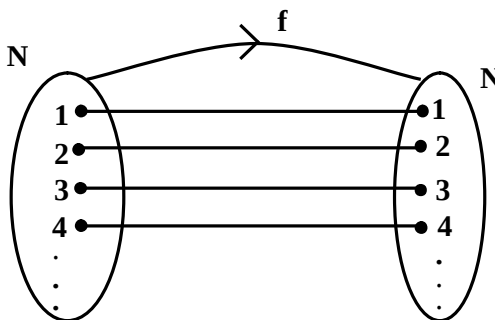


Fig. 2.13

(ii) Function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = x, \quad x \in \mathbb{R}$$

is also an identity function and its graph is given in Fig. 2.14.

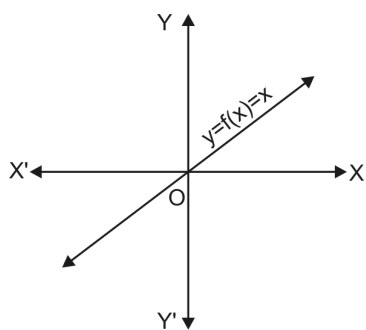


Fig. 2.14

Polynomial Function

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to be a polynomial function of degree n if it is defined as

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n, \quad x \in \mathbb{R}$$

where $a_0, a_1, a_2, \dots, a_{n-1}, a_n \in \mathbb{R}$, $a_0 \neq 0$ are constants

e.g. $f(x) = 2x^3 + x^2 - x + 5$, is a polynomial function of degree 3.

Linear Function (Polynomial Function of Degree 1)

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to be linear function if it is defined as

$$f(x) = ax + b, \quad x \in \mathbb{R}, \text{ where } a, b \in \mathbb{R}, a \neq 0 \text{ are real constants.}$$

Graph of the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = 2x + 3, \quad x \in \mathbb{R}$$

is given in Fig. 2.15.

$$y = f(x) = 2x + 3$$

$$y = 2x + 3$$

x	0	1	2
y	3	5	7

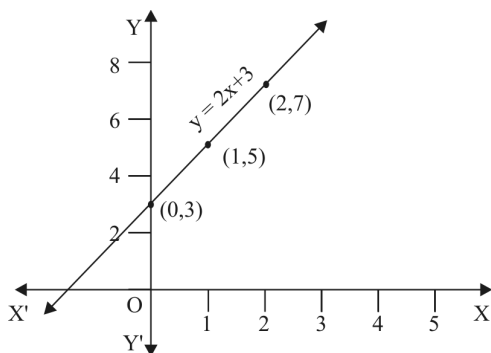


Fig. 2.15

Logarithm Function

A function $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ is said to be logarithm function if it is defined as

$$y = f(x) = \log_a x, \quad x \in \mathbb{R}^+ = \text{set of all positive real numbers.}$$

where $a > 0$ and $a \neq 1$

If $a^m = n$ then in terms of logarithm we write it as $\log_a n = m$

$$\text{i.e. } 2^3 = 8 \Rightarrow \log_2 8 = 3$$

$$2^4 = 16 \Rightarrow \log_2 16 = 4$$

$$4^3 = 64 \Rightarrow \log_4 64 = 3$$

$$16^{\frac{3}{4}} = 8 \Rightarrow \log_{16} 8 = \frac{3}{4}$$

Graph of $y = \log_a x$, $a > 0$, $a \neq 1$, $x > 0$ is given in Fig. 2.16 a, b.

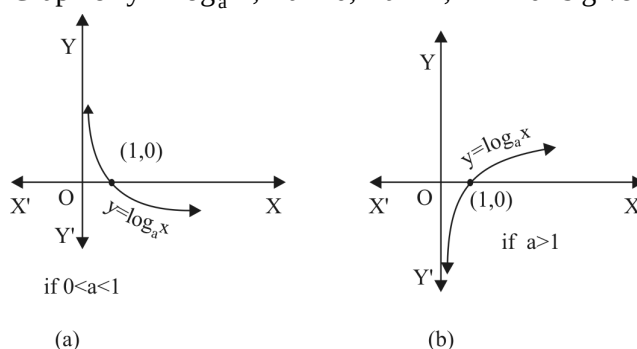


Fig. 2.16

Domain of logarithm function is $\mathbb{R}^+$ and range of logarithm function is $\mathbb{R}$.

Laws of Logarithm

- $\log_a mn = \log_a m + \log_a n$
- $\log_a \frac{m}{n} = \log_a m - \log_a n$
- $\log_a m^n = n \log_a m$
- $a^{\log_a m} = m$
- $\log_a a = 1$
- $\log_a b = \frac{1}{\log_b a}$
- $\log_a b = \frac{\log_n b}{\log_n a}$ this is known as base change formula, infact we can take any base in place of n .

Remark 2:

- If base of the logarithm is 10 then it is known as common logarithm.
- If base of the logarithm is e then it is known as natural logarithm and some time is written as $\ln x$ instead of $\log x$.
- When we write $\log x$ it means base is e . That is, in most of the cases base is mentioned only when it is other than e .

In mathematics number e is denoted by the sum of an infinite series

$$1 + \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \dots$$

In general, expansion for e^x is given by

$$1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \dots$$

where 2 read as 2 factorial, etc.

Factorial and its notations have been discussed in Sec. 4.2 in Unit 4 of this block.

Exponential Function

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = a^x, \quad x \in \mathbb{R}, \quad a > 0, a \neq 1$$

is called exponential function.

i.e. in case of exponential function there is a constant in the base and variable in the exponent.

i.e. nature of exponential function = (Constant)^{Variable}

For example, $f(x) = 2^x$ is an exponential function.

Graph of the exponential function is shown in Fig. 2.17 (a), (b).

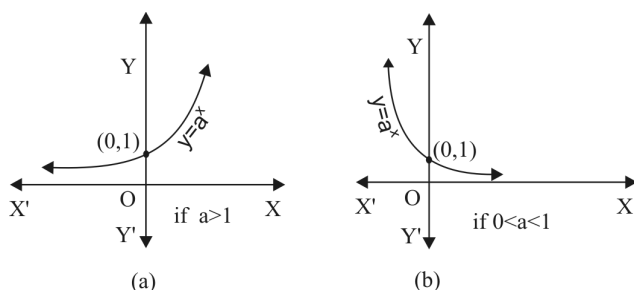


Fig. 2.17

Absolute Value Function or Modulus Function

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$y = f(x) = |x|, \quad \text{where } x \in \mathbb{R} \text{ and } |x| = \begin{cases} x, & x \geq 0 \\ -x, & x \leq 0 \end{cases}$$

is called absolute value function and graph of this function is given in Fig. 2.18

Domain of this function = $\mathbb{R}$ and range of this function = $[0, \infty)$.

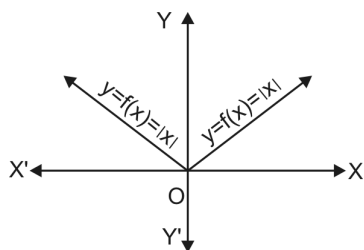


Fig. 2.18

Let us see how we calculate the value of modulus function at some particular point with the help of following example.

Example 4: If $f(x) = |x| - 3$ then evaluate $f(2)$, $f(-2)$, $f(3)$, $f(-3)$, $f(-7)$.

Solution:

$$f(2) = |2| - 3 = 2 - 3 = -1 \quad \left[\begin{array}{l} \because 2 > 0, \text{ so by definition of} \\ \text{modulus function } |2| = 2 \end{array} \right]$$

$$f(-2) = |-2| - 3 = -(-2) - 3 = 2 - 3 = -1 \quad \left[\begin{array}{l} \because -2 < 0, \text{ so by definition of} \\ \text{modulus function } |-2| = -(-2) \end{array} \right]$$

$$f(3) = |3| - 3 = 3 - 3 = 0$$

$$f(-3) = |-3| - 3 = -(-3) - 3 = 3 - 3 = 0$$

$$f(-7) = |-7| - 3 = -(-7) - 3 = 7 - 3 = 4$$

Here is an exercise for you.

E 4 If $f(x) = 5 - |x - 3|$ then evaluate $f(2), f(-2), f(6), f(-5), f(12)$.

Even Function

A function $f(x)$ is said to be even function if it satisfies

$$f(-x) = f(x), \text{ for all points } x \text{ of the domain of the function } f.$$

i.e. the value of function remains unchanged on changing x to $-x$.

For example,

(i) $f(x) = x^2 + x^4$

$$f(-x) = (-x)^2 + (-x)^4 = x^2 + x^4 = f(x)$$

$\therefore$ it is an even function.

(ii) $f(x) = |x|$

$$f(-x) = |-x| = |(-1)(x)| = |(-1)||x| = (1)|x| \quad \text{as } |-1| = -(-1) = 1$$

$$= |x|$$

$$= f(x)$$

$\therefore$ it is an even function

(iii) $f(x) = x^2 + x^3$

$$f(-x) = (-x)^2 + (-x)^3 = x^2 - x^3 \neq f(x)$$

$\therefore$ it is not an even function.

Odd Function

A function $f(x)$ is said to be odd function if it satisfies

$$f(-x) = -f(x), \text{ for all points } x \text{ of the domain of the function } f$$

i.e. the value of the function becomes $-ve$ on changing x to $-x$.

For example,

(i) $f(x) = x^3 + x$

$$f(-x) = (-x)^3 + (-x) = -x^3 - x = -(x^3 + x) = -f(x)$$

$\therefore$ it is an odd function.

(ii) $f(x) = \frac{1}{x^3}$

$$f(-x) = \frac{1}{(-x)^3} = \frac{1}{-x^3} = -\frac{1}{x^3} = -f(x)$$

$\therefore$ it is an odd function.

(iii) $f(x) = x^2 + x^3$

$$f(-x) = (-x)^2 + (-x)^3 = x^2 - x^3 = -(x^3 - x^2) \neq -f(x)$$

$\therefore$ it is not an odd function

We see that if $f(x) = x^2 + x^3$ then neither $f(-x) = f(x)$ nor $f(-x) = -f(x)$

$\therefore f(x) = x^2 + x^3$ is neither even nor odd function.

One-One Function

A function $f : X \rightarrow Y$ is said to be 1-1 or **injective** function if distinct elements of X are associated to distinct elements of Y under f .

i.e. if $x_1, x_2 \in X$ be s.t.

$$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$$

Or if $x_1, x_2 \in X$ and $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$.

If we compare this definition with example of daughters and mothers then one-one function means each daughter must have different mother, i.e. there cannot be two daughters having same mother for a function to be one-one.

For example,

(i)

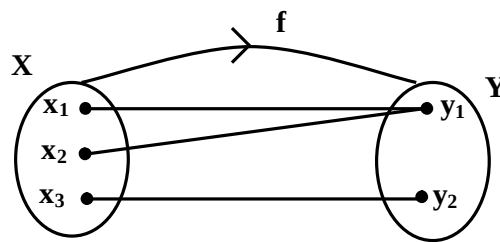


Fig. 2.19

The function f shown in Fig. 2.19 is not one-one functions because two different elements x_1, x_2 of X have the same image y_1 .

(ii)

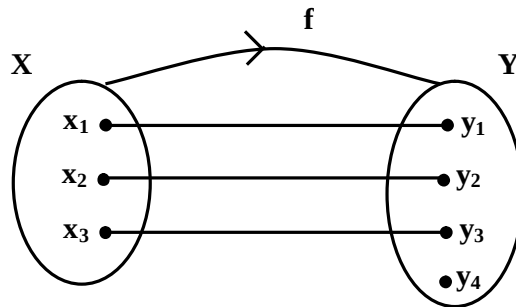


Fig. 2.20

The function f shown in Fig. 2.20 is one-one function because all the three elements of X have distinct images in Y .

Remark 3: If we want to show that a function $f(x)$ is one-one then

we take $x_1, x_2 \in X$ s.t.

$f(x_1) = f(x_2)$ and we have to show that $x_1 = x_2$.

For example,

(i) Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 7x + 5$ is 1-1 function.

Solution: Let x_1, x_2 be s.t.

$$\begin{aligned} f(x_1) &= f(x_2) \Rightarrow 7x_1 + 5 = 7x_2 + 5 \Rightarrow 7x_1 = 7x_2 \Rightarrow x_1 = x_2 \\ &\Rightarrow f \text{ is 1-1 function} \end{aligned}$$

(ii) Check whether the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$ is 1-1 or not.

Solution: $f(x) = x^2$

Let $x_1 = 2$, $x_2 = -2$ then $x_1 \neq x_2$

But $f(x_1) = f(2) = (2)^2 = 4$ and $f(x_2) = f(-2) = (-2)^2 = 4$

$\therefore f(2) = f(-2)$, i.e. $f(x_1) = f(x_2)$ but $x_1 \neq x_2$

$\Rightarrow f$ is not 1-1 function.

Onto Function

A function $f : X \rightarrow Y$ is said to be onto or **surjective** if each element of Y has at least one pre image in X .

i.e. for each $y \in Y$, there exists at least one $x \in X$ such that $f(x) = y$

If we compare this definition with example of daughters and mothers then onto function means each mother must have at least one daughter.

For example,

(i)

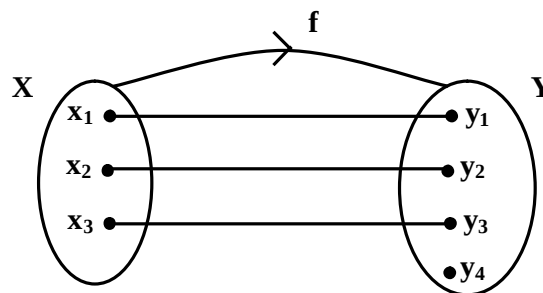


Fig. 2.21

The function f shown in Fig. 2.21 is not onto function because $y_4 \in Y$ but there is no $x \in X$ such that $y_4 = f(x)$

(ii)

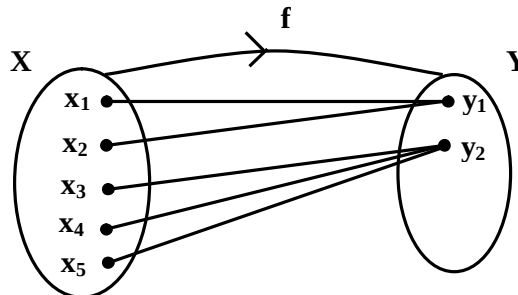


Fig. 2.22

The function f shown in Fig. 2.22 is onto function because each element of Y has at least one pre image, i.e. y_1 has two pre images and y_2 has three pre images.

Remark 4: If we want to show that a function $f(x)$ is onto then first we take an element y in Y and we have to show that there exists an element x in X such that $f(x) = y$

For example, show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 7x + 5$ is onto function.

Solution: Here $X = \mathbb{R}$, $Y = \mathbb{R}$

Let $y \in Y = \mathbb{R}$, then $\frac{y-5}{7} \in X = \mathbb{R}$ s.t.

$$f\left(\frac{y-5}{7}\right) = 7\left(\frac{y-5}{7}\right) + 5 = (y-5) + 5 = y$$

$\therefore f$ is an onto function.

One-One and Onto Function

A function $f : X \rightarrow Y$ is said to be one-one and onto or **bijective** or **one-one correspondence** if

- (i) f is one-one
- (ii) f is onto

If we compare this definition with example of daughters and mothers then one-one and onto function means each mother have exactly one daughter and there is no mother who does not have any daughter. The function f shown in Fig.2.23 represents a situation of one-one and onto function.

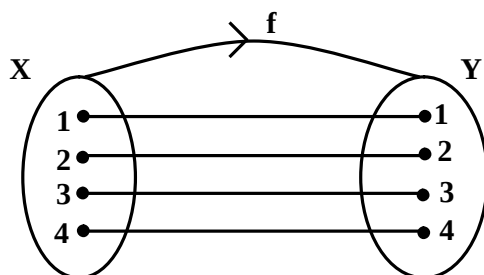


Fig. 2.23

Also the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = 7x + 5, \quad x \in \mathbb{R}$$

is one-one and onto (already shown)

$\therefore f$ is one-one and onto function.

Geometrical Meaning of Injective, Surjective and Bijective Functions

One-One Function

If a function is 1-1 then geometrically it will satisfy the following condition. Each horizontal line either does not intersect the graph of the function or if it intersects it will intersect exactly at one point.

For example,

- (i) Graph shown in Fig. 2.24 (b) is the graph of a one-one function because each horizontal line either does not intersect the graph or if it intersects, it will intersect exactly at one point, i.e. each horizontal line above x-axis will intersect the graph exactly at one point and each horizontal line below x-axis not intersect the graph at all.
- (ii) Graph shown in the Fig. 2.24 (c) is also the graph of a one-one function because each horizontal line intersects the graph exactly at one point.
- (iii) But the graph shown in the Fig. 2.24 (a) is not the graph of a one-one function because if we draw any horizontal line below the x-axis, then it will intersect the graph at two points. Similarly graph shown in Fig. 2.24 (d) is also not a one-one function.

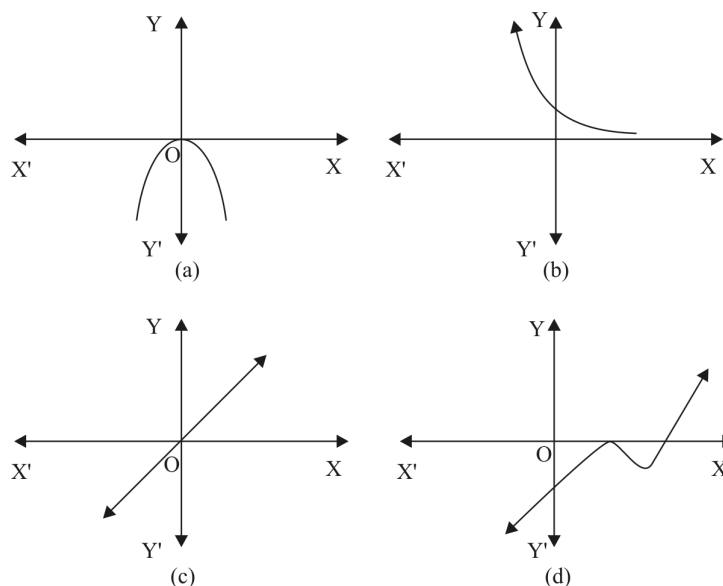


Fig. 2.24

Onto or Surjective Function

If a function is onto then geometrically it will satisfy the following condition. Each horizontal line must intersect the graph of the function at least at one point.

For example,

- (i) Graph shown in the Fig. 2.24 (d) is the graph of an onto function because each horizontal line intersect the graph of the function at least at one point. Graph shown in Fig. 2.24 (c) is also onto function because each horizontal line intersects the graph exactly at one point.
- (ii) But the graph shown in the Fig. 2.24 (b) is not the graph of a surjective function because if we draw any horizontal line below the x-axis, then it will not intersect the graph.
- (iii) Similarly the graph shown in the Fig. 2.24 (a) is not the graph of an onto function because if we draw any horizontal line above the x-axis, then it will not intersect the graph.

Bijjective Function

If a function is bijective or one-one and onto or one-one correspondence, then each horizontal line must intersect the graph of the function exactly at one point.

For example,

- (i) Graph shown in the Fig. 2.24 (c) is the graph of a bijective function because each horizontal line intersects the graph of the function exactly at one point.
- (ii) But the graphs shown in the Fig. 2.24 (a) and (b) are not the graphs of a bijective function because they are not onto.
- (iii) Similarly the graph shown in the Fig. 2.24 (d) is not the graph of a bijective function because it is not one-one.

Countable Sets

Equivalent Sets: Two sets A and B are said to be equivalent if either there exists a one- one correspondence from A to B or from B to A and is denoted by $A \sim B$.

For example, let $A = \{1, 2, 3, 4\}$ and $B = \{1, 4, 9, 16\}$, then $A \sim B$ because there exists a one-one correspondence $f : A \rightarrow B$ defined by

$f(x) = x^2$, $x \in A$ between A and B as shown in Fig. 2.25

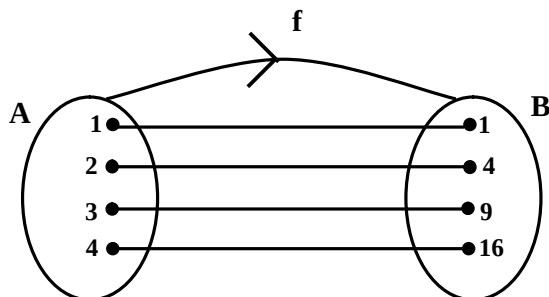


Fig. 2.25

Enumerable Set: A set E is said to be enumerable, if it is equivalent to the set of natural numbers, i.e. if $N \sim E$

i.e. if there exists a one- one correspondence between N and E.

An enumerable set is also known as denumerable set or countably infinite set.

For example,

- (i) Let $E = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$

Define a map $f : N \rightarrow E$ by

$$f(n) = \frac{1}{n}, \quad n \in N$$

Then f is both 1-1 and onto as shown in Fig. 2.26.

$\Rightarrow N \sim E \Rightarrow E$ is enumerable.

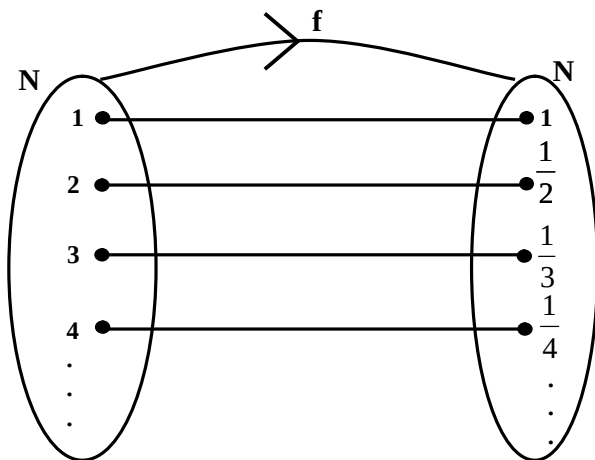


Fig. 2.26

- (ii) Let $A = \{3, 6, 9, 12, \dots\}$

Define a map $f : N \rightarrow A$ by

$$f(n) = 3n, \quad n \in N$$

Then f is both 1-1 and onto as shown in Fig. 2.27.

$\Rightarrow N \sim A \Rightarrow A$ is enumerable.

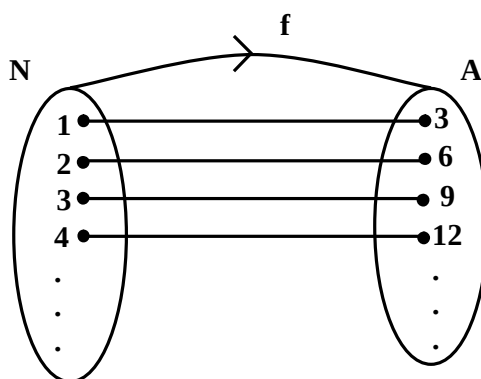


Fig. 2.27

Here is an exercise for you.

E 5 Show that

(i) $A = \{5, 25, 125, 625, \dots\}$ (ii) $B = \{1, 5, 25, 125, 625, \dots\}$

(iii) $C = \{1, 4, 7, 10, 13, \dots\}$

all are enumerable sets.

Countable Set: A set is said to be countable if either it is finite or enumerable

For example,

(i) $A = \{\} = \phi$, which is a finite set so it is countable.

(ii) $B = \{a, b, c\}$, which is a finite set and hence countable.

(iii) $C = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$, which is an enumerable set (already shown), so it is a countable set.

Remark 5: In the fifth unit of Course-3, i.e. MST-003, you will meet the word countable in the definition of the discrete random variable. So it becomes very important to understand what we mean by countable set.

We close this unit by summarising the topics that we have discussed in this unit:

2.7 SUMMARY

In this unit we have covered following topics:

- 1) Quantity, constant quantity, variable.
- 2) Interval, open interval, closed interval, semi-open and closed interval, finite and infinite intervals.
- 3) Function and its classification with examples and their graphs.
- 4) Types of functions, i.e. 1-1, onto and one-one correspondence with their geometrical interpretation.
- 5) Equivalent sets, enumerable sets and countable sets.

E 1) Since $f(x)$ takes real values for all $x \in \mathbb{R}$.

$\therefore$ domain of $f = \mathbb{R}$

Also, as x vary, over $\mathbb{R}$, then $4x + 5$ also vary over $\mathbb{R}$, so range of $f = \mathbb{R}$

Now, $f(0) = 4(0) + 5 = 5$ and $f\left(\frac{1}{2}\right) = 4\left(\frac{1}{2}\right) + 5 = 2 + 5 = 7$

E 2) Here $f(x)$, takes real values for all $x \in \mathbb{R}$, except at $x = 2$

i.e. $f(x)$ is defined for all real values of x , except at $x = 2$

$\therefore$ domain of $f = \mathbb{R} - \{2\}$ and

$f(x)$ cannot be zero at any real number,

$\therefore$ range of $f = \mathbb{R} - \{0\}$

Also, $f(1) = \frac{1}{1-2} = \frac{1}{-1} = -1$, $f(3) = \frac{1}{3-2} = \frac{1}{1} = 1$ and

$f(-5) = \frac{1}{-5-2} = \frac{1}{-7} = -\frac{1}{7}$

E 3) Here $f(x)$ takes real values for all $x \in \mathbb{R}$, except at $x = -3$

$\therefore$ domain of $f = \mathbb{R} - \{-3\}$

and $f(x)$ cannot take zero values at any real number,

$\therefore$ range of $f = \mathbb{R} - \{0\}$

Also, $f(0) = \frac{1}{0+3} = \frac{1}{3}$, $f(2) = \frac{1}{2+3} = \frac{1}{5}$

$f(-3)$ is not defined because $x = -3$ is not a point of the domain of f .

E 4) $f(x) = 5 - |x - 3|$

$f(2) = 5 - |2 - 3| = 5 - |-1| = 5 - (-(-1)) = 5 - 1 = 4$

$f(-2) = 5 - |-2 - 3| = 5 - |-5| = 5 - (-(-5)) = 5 - (5) = 5 - 5 = 0$

$f(6) = 5 - |6 - 3| = 5 - |3| = 5 - 3 = 2$

$f(-5) = 5 - |-5 - 3| = 5 - |-8| = 5 - (-(-8)) = 5 - (8) = 5 - 8 = -3$

$f(12) = 5 - |12 - 3| = 5 - |9| = 5 - 9 = -4$

E 5) (i) Define a map $f : \mathbb{N} \rightarrow A$ by

$$f(n) = 5^n, \quad n \in \mathbb{N}$$

f is both 1-1 and onto as shown in Fig. 2.28.

$\Rightarrow \mathbb{N} \sim A$

$\Rightarrow A$ is enumerable.

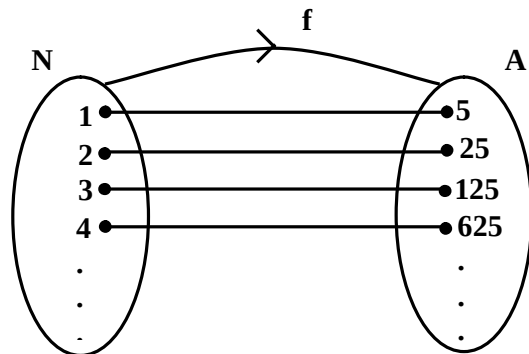


Fig. 2.28

- (ii) Define a map $f : \mathbb{N} \rightarrow \mathbb{B}$ by
 $f(x) = 5^{n-1}, \quad n \in \mathbb{N}$
 f is both 1-1 and onto as shown in Fig. 2.29.
 $\Rightarrow \mathbb{N} \sim \mathbb{B} \Rightarrow \mathbb{B}$ is enumerable.

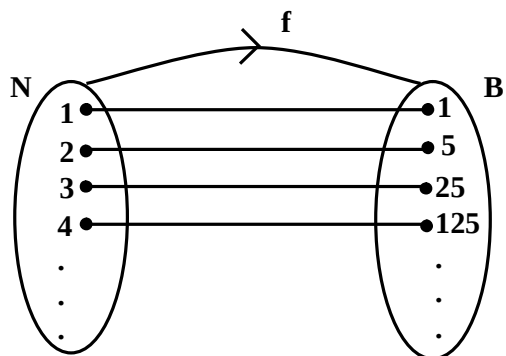


Fig. 2.29

- (iii) Define a map $f : \mathbb{N} \rightarrow \mathbb{C}$ by
 $f(x) = 3n - 2, \quad n \in \mathbb{N}$
 f is both 1-1 and onto as shown in Fig. 2.30.
 $\Rightarrow \mathbb{N} \sim \mathbb{C} \Rightarrow \mathbb{C}$ is enumerable.

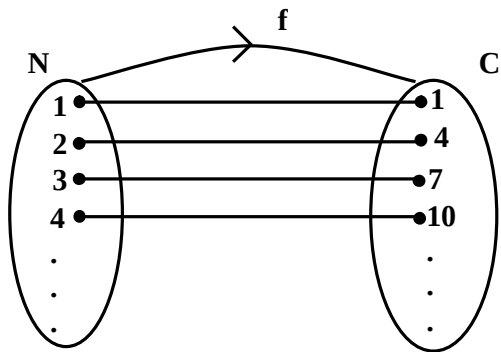


Fig. 2.30

UNIT 3 PROGRESSIONS

Structure

- 3.1 Introduction
 - Objectives
- 3.2 Sequence
- 3.3 Arithmetic Progresses (A.P.)
- 3.4 Geometric Progression (G.P.)
- 3.5 Sum of Infinite G.P.
- 3.6 Concept of Summation
- 3.7 Sum of some Special Sequences
- 3.8 Summary
- 3.9 Solutions/Answers

3.1 INTRODUCTION

In day to day life many times people use the word sequence. So we are familiar with the dictionary meaning of the word sequence, in fact to put the things in a particular order means we have put the things in a particular sequence. For example, natural numbers which are multiple of 5 can be put in the following sequence.

5, 10, 15, 20, 25, ...

In Unit 2 of this block, we have defined functions. In this unit we will define sequence mathematically and it will be interesting to know that sequences are also special types of functions. Then we will see arithmetic progression (A.P.) and geometric progression (G.P.) are special types of sequences. In fact in this unit we will focus on the n^{th} term of A.P and G.P., and sum of first n terms of A.P. and G.P. Finally, we will discuss what we mean by summation and how the initiation (origin) of summation can be changed.

Objectives

After completing this unit, you should be able to:

- define sequence;
- define and recognize arithmetic progression (A.P.);
- give formula for n^{th} term and sum of first n terms of on A.P.;
- define geometric progression (G.P.);
- give formula for n^{th} term and sum of n terms of a G.P.;
- find sum of infinite G.P.; and
- become familiar with the concept of summation.

3.2 SEQUENCE

In the introduction of this unit we have indicated that sequences are special types of functions. So let us first give a mathematical definition of sequence.

Sequence: A sequence is a function whose domain is the set of all natural numbers and range may be any set. A sequence is generally denoted by writing $a_1, a_2, a_3, \dots, a_n, \dots$

Or simply by $\{a_n\}$ or $\langle a_n \rangle$, where a_n denotes the n^{th} term of the sequence, i.e. a_1 is first term of the sequence, a_2 is second term of the sequence, a_3 is third term of the sequence, and so on.

For example, define a function $f : \mathbb{N} \rightarrow \mathbb{R}$ by

$$f(n) = \frac{1}{n}, \quad n \in \mathbb{N}$$

This function represents a sequence which can be written as $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$

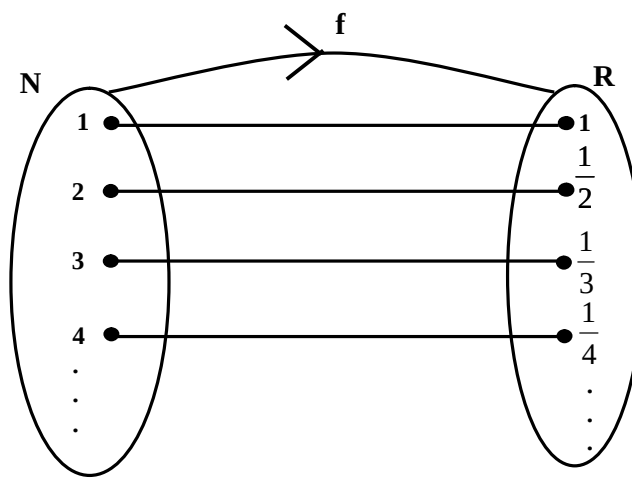


Fig. 4.1

Remark 1:

- (i) Sequences are special types of functions because here domain always remains set of natural numbers whereas in case of real functions domain may be any subset of real numbers.
- (ii) If range of a sequence is subset of $\mathbb{R}$, then we say that sequence is real.
- (iii) Here we will discuss only real sequences.
- (iv) Here geometrical representation of the sequence is given just to realize you that sequences are special types of functions. In future we will not give geometrical representation and it is neither required.

Next question which may strike your mind is that what are the commonly used methods to represent a sequence? This question is addressed in the following discussion:

Ways of Representing a Sequence

A sequence may be represented by any of the following ways:

- (a) **One of the ways of representing a sequence is writing down the first few terms of the sequence till a definite rule for writing down other terms becomes clear.**

For example,

- (i) $1, \frac{1}{4}, \frac{1}{9}, \frac{1}{16}, \dots$; is a sequence having n^{th} term $\frac{1}{n^2}$.

(ii) 5, 10, 15, 20,...; is a sequence having n^{th} term $5n$.

(b) A sequence can also be represented by giving a formula for its n^{th} term.

For example,

(i) If $a_n = n^2$, then this represents the sequence 1, 4, 9, 16, ...

(ii) If $a_n = \frac{n}{n+1}$, then this represents the sequence $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots$

(c) Recursive Relation: A sequence can also be represented by writing its first few terms and a formula to write down the other terms of the sequence. Such way of representing a sequence is known as recursive relation.

For example, if $a_1 = a_2 = 1$, and $a_{n+1} = a_n + a_{n-1}$, $n \geq 2$

then terms of this sequence are 1, 1, 2, 3, 5, 8, ...

i.e. each term (except first and second) is equal to the sum of its preceding two terms. This sequence is known as Fibonacci sequence.

(d) Sometimes, the nature of the sequence is such that it cannot be represented by giving a single formula for its n^{th} term. So, to avoid this difficulty we have another way of representing a sequence by writing more than one relation for its n^{th} term.

For example, if $a_n = \begin{cases} \frac{n-1}{2}, & \text{if } n = 1, 3, 5, \dots \\ \frac{n}{2}, & \text{if } n = 2, 4, 6, \dots \end{cases}$

then terms of this sequence are 0, 1, 1, 2, 2, 3, 3, ...

We have discussed different methods of representing a sequence, so it is the right place to provide an example to obtain some terms of the sequence given by using either of the ways.

Example 1: For the given sequences write down the given terms:

(i) $a_n = n^2 - 2n + 2$, find a_1, a_2, a_r .

(ii) $a_n = \frac{1 + (-1)^n}{2}$, find a_1, a_2, a_3, a_4 .

(iii) $a_1 = 2, a_2 = 3, a_n = 2a_{n-1} + 4a_{n-2}$, $n \geq 3$, find a_3, a_4 .

(iv) $a_n = \begin{cases} n^2, & n = 1, 3, 5, \dots \\ \frac{1}{n+1}, & n = 2, 4, 6, \dots \end{cases}$, find $a_1, a_2, a_3, a_4, a_5, a_6$.

(v) $a_n = \frac{(n-1)(n-2)}{2(n+1)}$, find a_1, a_2, a_3, a_4 .

Solution:

(i) $a_n = n^2 - 2n + 2$

For $n = 1, 2, r$, we have

$$a_1 = (1)^2 - 2 \times 1 + 2 = 1, a_2 = 2^2 - 2 \times 2 + 2 = 2, a_r = r^2 - 2r + 2$$

$$(ii) \quad a_n = \frac{1+(-1)^n}{2}$$

For $n = 1, 2, 3, 4$, we have

$$a_1 = \frac{1+(-1)^1}{2} = \frac{1-1}{2} = \frac{0}{2} = 0, \quad a_2 = \frac{1+(-1)^2}{2} = \frac{1+1}{2} = \frac{2}{2} = 1$$

$$a_3 = \frac{1+(-1)^3}{2} = \frac{1-1}{2} = \frac{0}{2} = 0, \quad a_4 = \frac{1+(-1)^4}{2} = \frac{1+1}{2} = \frac{2}{2} = 1$$

$$(iii) \quad a_1 = 2, \quad a_2 = 3$$

$$a_n = 2a_{n-1} + 4a_{n-2}, \quad n \geq 3$$

For $n = 3, 4$, we have

$$a_3 = 2a_2 + 4a_1 = 2 \times 3 + 4 \times 2 = 6 + 8 = 14$$

$$a_4 = 2a_3 + 4a_2 = 2 \times 14 + 4 \times 3 = 28 + 12 = 40$$

(iv) For $n = 1, 2, 3, 4, 5, 6$, we have

$$a_1 = (1)^2 = 1, \quad a_2 = \frac{1}{2+1} = \frac{1}{3}, \quad a_3 = (3)^2 = 9$$

$$a_4 = \frac{1}{4+1} = \frac{1}{5}, \quad a_5 = (5)^2 = 25, \quad a_6 = \frac{1}{6+1} = \frac{1}{7}$$

(v) For $n = 1, 2, 3, 4$, we have

$$a_1 = \frac{(1-1)(1-2)}{2(1+1)} = \frac{0}{4} = 0, \quad a_2 = \frac{(2-1)(2-2)}{2(2+1)} = \frac{0}{6} = 0$$

$$a_3 = \frac{(3-1)(3-2)}{2(3+1)} = \frac{2}{8} = \frac{1}{4}, \quad a_4 = \frac{(4-1)(4-2)}{2(4+1)} = \frac{6}{10} = \frac{3}{5}$$

Here is an exercise for you.

E 1 (i) If $a_n = \sqrt{2n}$ then find a_1, a_2, a_3, a_4 .

(ii) If $a_n = \frac{2}{\sqrt{n}}$ then find a_1, a_2, a_3, a_4, a_8 .

(iii) If $a_n = \frac{2^n \times 3^n}{2^n + 3^n}$ then find a_1, a_2, a_3 .

3.3 ARITHMETIC PROGRESSION (A.P.)

Some sequences follow certain pattern. Arithmetic progression (A.P.) is also a sequence which follows a particular pattern as defined below.

Arithmetic progression (A.P.): A sequence $\{a_n\}$ or $\langle a_n \rangle$ is said to be arithmetic progression (A.P.) if

$a_{n+1} - a_n = d, \quad \forall n, n = 1, 2, 3, \dots$ where d is a fixed constant known as common difference of the A.P.

i.e. difference of any term to its preceding term always remains constant.

For example, 7, 11, 15, 19, ... is an A.P. with first term = 7 and common difference = $11 - 7 = 4$.

Remark 2:

- (i) If a sequence is given by listing its first few terms and we want to know whether it is an A.P. or not, for this first of all we calculate

$$a_2 - a_1, a_3 - a_2, a_4 - a_3, \text{ etc.}$$

If $a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = \dots = d$, then we say that it is an A.P. with d as common difference, otherwise it is not an A.P.

- (ii) If a sequence is given by writing its n^{th} term a_n then we calculate

$a_{n+1} - a_n$. If this difference is independent of n , it represents A.P. and if the differences $a_{n+1} - a_n$ involve n then it is not an A.P.

Following example is based on the two points discussed in the Remark 2 given above.

Example 2: In which of the following cases given sequence is an A.P.:

(i) $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$ (ii) $1, 4, 9, 16, \dots$ (iii) $1, 4, 7, 10, \dots$

(iv) $8, 7\frac{1}{3}, 6\frac{2}{3}, 6, \dots$ (v) $a_n = 4n + 5$ (vi) $a_n = n^2 + n$

Solution:

(i) $a_2 - a_1 = \frac{1}{2} - 1 = \frac{-1}{2}$; $a_3 - a_2 = \frac{1}{3} - \frac{1}{2} = \frac{2-3}{6} = \frac{-1}{6}$

$$\therefore a_2 - a_1 \neq a_3 - a_2 \Rightarrow \text{it is not an A.P.}$$

(ii) $a_2 - a_1 = 4 - 1 = 3$; $a_3 - a_2 = 9 - 4 = 5$

$$\therefore a_2 - a_1 \neq a_3 - a_2 \Rightarrow \text{it is not an A.P.}$$

(iii) $a_2 - a_1 = 4 - 1 = 3$; $a_3 - a_2 = 7 - 4 = 3$; $a_4 - a_3 = 10 - 7 = 3$

and so on

$$\therefore a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = \dots = 3 (= \text{constant})$$

$\Rightarrow$ it is an A.P. with first term 1 and common difference 3.

(iv) $a_2 - a_1 = 7\frac{1}{3} - 8 = \frac{22}{3} - 8 = \frac{22-24}{3} = -\frac{2}{3}$

$$a_3 - a_2 = 6\frac{2}{3} - 7\frac{1}{3} = \frac{20}{3} - \frac{22}{3} = -\frac{2}{3}$$

$$a_4 - a_3 = 6 - 6\frac{2}{3} = 6 - \frac{20}{3} = \frac{18-20}{3} = -\frac{2}{3}$$

and so as

$$\therefore a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = \dots = -\frac{2}{3} (= \text{constant})$$

$\Rightarrow$ it is an A.P. with first term 8 and common difference $= -\frac{2}{3}$.

(v) $a_n = 4n + 5$

Replace n by $n+1$, we get

$$a_{n+1} = 4(n+1) + 5 = 4n + 9$$

$$a_{n+1} - a_n = 4n + 9 - (4n + 5)$$

$$= 4n + 9 - 4n - 5$$

$$= 4 \text{ which is independent of } n$$

$\therefore$ it is an A.P. with first term 9 and common difference $= 4$.

$$(vi) a_n = n^2 + n$$

Replacing n by $n+1$, we get

$$a_{n+1} = (n+1)^2 + (n+1) = n^2 + 2n + 1 + n + 1 = n^2 + 3n + 2$$

$$a_{n+1} - a_n = n^2 + 3n + 2 - (n^2 + n) = n^2 + 3n + 2 - n^2 - n = 2n + 2 \text{ which is not free from } n$$

$\therefore$ it is not an A.P.

Here is an exercise for you.

E 2) Show that sequence $\log a, \log \frac{a^2}{b}, \log \frac{a^3}{b^2}$, form an A.P.

So far in this section we have defined A.P. and also learned how to check whether a given sequence is an A.P. or not? But now question arise can we find any term of a given A.P.? and second question can we find the sum of any number of terms of an A.P.? Answers of both the questions are yes and we will discuss these questions separately in two subsections 3.3.1 and 3.3.2.

3.3.1 Standard A.P. and its General Term

A sequence defined by $a, a + d, a + 2d, a + 3d, \dots$... (1)
is known as standard A.P. with first term = a and common difference = d .

Standard A.P.: A.P. defined by (1), i.e.

$$a, a + d, a + 2d, a + 3d, \dots$$

is known as standard A.P.

General Term: From standard A.P. given by (1) we see that

$$\text{First term} = a_1 = T_1 = a = a + (1-1)d$$

$$\text{Second term} = a_2 = T_2 = a + d = a + (2-1)d$$

$$\text{Third term} = a_3 = T_3 = a + 2d = a + (3-1)d$$

$$\text{Forth term} = a_4 = T_4 = a + 3d = a + (4-1)d$$

...

...

...

$$\therefore \boxed{n^{\text{th}} \text{ term} = a_n = T_n = a + (n-1)d}$$

Remark 3: Keep this formula always in mind and is known as formula for n^{th} term or general term of an A.P. with first term 'a' and common difference 'd'.

Example 3: Find indicated term(s) in each case:

(i) If $a = 4, d = 3$, find T_n, T_{17} .

(ii) Find T_n of the A.P. $5, 5 + \sqrt{3}, 5 + 2\sqrt{3}, 5 + 3\sqrt{3}, \dots$

(iii) $20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, \dots$. Find T_{20} of this A.P.

(iv) Which term of the of the A.P. $43, 38, 33, 28, \dots$ is -457 ?

(v) Which term of the A.P. $17, 16\frac{1}{2}, 16, 15\frac{1}{2}, \dots$ is the first negative term?

(vi) If 7^{th} and 31^{st} terms of an A.P. are 29, 125 respectively, then find the A.P.

Solution:

(i) $T_n = a + (n-1)d = 4 + (n-1)3 = 3n + 1$
 For $n = 17$, $T_{17} = 3 \times 17 + 1 = 51 + 1 = 52$

(ii) Here, $a = 5$, $d = 5 + \sqrt{3} - 5 = \sqrt{3}$
 $\therefore T_n = a + (n-1)d = 5 + (n-1)\sqrt{3}$

(iii) Here, $a = 20$, $d = 19\frac{1}{4} - 20 = \frac{77}{4} - 20 = -\frac{3}{4}$
 $\therefore T_{20} = a + 19d = 20 + 19\left(-\frac{3}{4}\right) = \frac{80 - 57}{4} = \frac{23}{4} = 5\frac{3}{4}$

(iv) Here $a = 43$, $d = 38 - 43 = -5$
 Let $T_n = -457$
 $\Rightarrow a + (n-1)d = -457$
 $\Rightarrow 43 + (n-1)(-5) = -457$
 $\Rightarrow 43 - 5n + 5 = -457$
 $\Rightarrow -5n = -457 - 48 = -505$
 $\Rightarrow n = 101$

(v) Here $a = 17$, $d = 16\frac{1}{2} - 17 = \frac{33}{2} - 17 = -\frac{1}{2}$
 Let $T_n < 0$
 $\Rightarrow a + (n-1)d < 0 \Rightarrow 17 + (n-1)\left(-\frac{1}{2}\right) < 0$
 $\Rightarrow 17 - \frac{n}{2} + \frac{1}{2} < 0 \Rightarrow -\frac{n}{2} < -17 - \frac{1}{2}$
 $\Rightarrow -\frac{n}{2} < -\frac{35}{2} \Rightarrow \frac{n}{2} > \frac{35}{2} \quad [\because -x > -a \Rightarrow x < a \text{ Or } -x < -a \Rightarrow x > a]$
 $\Rightarrow n > 35 \Rightarrow n = 36$
 $\therefore 36$ is the first negative term of this A.P.

(vi) Let a , d be the first term and common difference, respectively, of the given A.P.

According to the problem,

$$\left. \begin{array}{l} T_7 = 29 \\ T_{31} = 125 \end{array} \right\} \Rightarrow \begin{cases} a + 6d = 29 & \dots (1) \\ a + 30d = 125 & \dots (2) \end{cases}$$

(2) - (1) gives

$$24d = 96 \Rightarrow d = 4$$

Putting $d = 4$ in (1), we get

$$a + 24 = 29 \Rightarrow a = 5$$

$\therefore$ given A.P. is 5, 9, 13, 17, ...

Here is an exercise for you.

E 3 (i) If $5k + 1$, $6k + 5$ and $10k + 3$ are three consecutive terms of an A.P. then find k .

(ii) Is 121 a term of the sequence 3, 9, 15, 21, ...?

(iii) How many terms are there in the A.P. $-1, \frac{-1}{4}, \frac{1}{2}, \frac{5}{4}, \dots, 14$?

3.3.2 Sum of n Terms of an A.P.

Standard A.P. is $a, a + d, a + 2d, a + 3d, \dots$

We know that $T_n = a + (n - 1)d$

$$\therefore T_{n-1} = a + (n - 2)d$$

Let S_n denotes the sum of first n terms of the above A.P., then

$$S_n = T_1 + T_2 + \dots + T_{n-1} + T_n$$

$$\text{Or } S_n = a + (a + d) + \dots + [a + (n - 2)d] + [a + (n - 1)d] \quad \dots (1)$$

Writing the terms of R.H.S. in reverse order we get

$$S_n = [a + (n - 1)d] + [a + (n - 2)d] + \dots + (a + d) + a \quad \dots (2)$$

(1) + (2) gives

$$2S_n = \underbrace{[2a + (n - 1)d] + [2a + (n - 1)d] + \dots + [2a + (n - 1)d] + [2a + (n - 1)d]}_{n \text{ times}}$$

$$\Rightarrow 2S_n = n[2a + (n - 1)d]$$

$$\Rightarrow S_n = \frac{n}{2}[2a + (n - 1)d]$$

Remark 4:

(i) This formula can also be written as

$$S_n = \frac{n}{2}[a + a + (n - 1)d] = \frac{n}{2}(a + l), \text{ where } l = a + (n - 1)d = \text{last term}$$

(ii) Keep this formula always in mind and is known as formula for sum of first n terms of an A.P. with first term 'a' and common difference 'd'.

Example 4: Find the following sums:

(i) $1 + 4 + 7 + 10 + \dots$ to 40 terms

(ii) $0.8 + 0.81 + 0.82 + \dots$ to 101 terms

(iii) $3 + 7 + 11 + \dots + 79$

Solution:

(i) Here $a = 1, d = 4 - 1 = 3, n = 40$

We know that

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$\therefore S_{40} = \frac{40}{2}[2 \times 1 + (40 - 1)3] = 20(2 + 117) = 2380$$

(ii) Here $a = 0.8, d = 0.81 - 0.8 = 0.01, n = 101$

We know that

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$\begin{aligned} \therefore S_{101} &= \frac{101}{2}[2 \times 0.8 + (101 - 1) \times 0.01] = \frac{101}{2}[2 \times 0.8 + 100 \times 0.01] \\ &= \frac{101}{2}[1.6 + 1] = \frac{101}{2} \times 2.6 = 101 \times 1.3 = 131.3 \end{aligned}$$

(iii) Here $a = 3, d = 7 - 3 = 4, T_n = 79$

We know that

$$T_n = a + (n - 1)d \Rightarrow a + (n - 1)d = 79 \Rightarrow 3 + (n - 1)4 = 79$$

$$\Rightarrow 4n - 1 = 79 \Rightarrow 4n = 80 \Rightarrow n = 20$$

Now, we also know that

$$S_n = \frac{n}{2}(a + l), \text{ where } l \text{ is last term}$$

$$\therefore S_{20} = \frac{20}{2}(3 + 79) = 10 \times 82 = 820$$

Alternatively

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$\therefore S_{20} = \frac{20}{2}[2 \times 3 + (20-1) \times 4] = 10(6 + 76) = 820$$

3.4 GEOMETRIC PROGRESSION (G.P.)

A sequence $\{a_n\}$ is said to be geometric progression (G.P.) if

$$\frac{a_{n+1}}{a_n} = r \quad \forall n \in \mathbb{N}$$

i.e. ratio of any term to its preceding term is same (remains constant).
where r is non zero fixed constant and is known as common ratio

For example, 3, 6, 12, 24, 48, ... is G.P. $\left[\because \frac{6}{3} = \frac{12}{6} = \frac{24}{12} = \frac{48}{24} = \dots = 2 \right]$

Remark 5:

(i) In case of G.P. neither a_n (for all n) nor r can be zero.

i.e. $a_n \neq 0, \quad \forall n \in \mathbb{N}$ and $r \neq 0$

(ii) If a sequence is given by listing its first few terms and we want to know whether it is a G.P. or not, for this first of all we calculate

$$\frac{a_2}{a_1}, \frac{a_3}{a_2}, \frac{a_4}{a_3}, \text{ etc.}$$

If $\frac{a_2}{a_1}, \frac{a_3}{a_2}, \frac{a_4}{a_3} = \dots = r$, then we say that it is a G.P. with common ratio r ,

otherwise it is not a G.P.

For example, we have seen just before Remark 5 that the sequence 3, 6, 12, 24, 48, ... is a G.P. by using this procedure.

So far in this section we have defined G.P. and also learned how to check whether a given sequence is a G.P. or not? But now question arise can we find any term of a given G.P.? and second question can we find the sum of any number of terms of G.P.? Answers of both the questions are yes and we will discuss these questions separately in two subsections 3.4.1 and 3.4.2.

3.4.1 Standard G.P. and its General Term

A sequence defined by $a, ar, ar^2, ar^3, \dots \dots (1)$

is known as standard G.P. with first term = a and common ratio = r .

Standard G.P.: G.P. defined by (1), i.e.

$$a, ar, ar^2, ar^3, \dots$$

is known as standard G.P.

General Term: From standard G.P. given by (1) we see that

$$\text{First term} = a_1 = T_1 = a = ar^{1-1}$$

$$\text{Second term} = a_2 = T_2 = ar = ar^{2-1}$$

$$\text{Third term} = a_3 = T_3 = ar^2 = ar^{3-1}$$

$$\text{Forth term} = a_4 = T_4 = ar^3 = ar^{4-1}$$

...

...

...

$$\therefore \boxed{n^{\text{th}} \text{ term} = a_n = T_n = ar^{n-1}}$$

Remark 6: Keep this formula always in mind and is known as formula for n^{th} term or general term of the G.P. with first term 'a' and common ratio 'r'.

Example 5: Which of the following sequences are G.P.. If a sequence is a G.P., write its first term, common ratio and n^{th} term.

(i) 2, 6, 18, 54, ...

(ii) $\frac{1}{8}, -\frac{1}{4}, \frac{1}{2}, -1, \dots$

(iii) $\sqrt{2}, 3\sqrt{2}, 6\sqrt{2}, 12\sqrt{2}, \dots$

Solution:

(i) $\frac{T_2}{T_1} = \frac{6}{2} = 3, \quad \frac{T_3}{T_2} = \frac{18}{6} = 3, \quad \frac{T_4}{T_3} = \frac{54}{18} = 3, \dots$

$$\therefore \frac{T_2}{T_1} = \frac{T_3}{T_2} = \frac{T_4}{T_3} = \dots = 3$$

$\Rightarrow$ it is a G.P with first term $a = 2$ and common ratio $r = 3$.

$$n^{\text{th}} \text{ term} = T_n = ar^{n-1} = 2(3)^{n-1}$$

(ii) $\frac{T_2}{T_1} = \frac{-1/4}{1/8} = -2, \quad \frac{T_3}{T_2} = \frac{1/2}{-1/4} = -2, \quad \frac{T_4}{T_3} = \frac{-1}{1/2} = -2, \dots$

$$\therefore \frac{T_2}{T_1} = \frac{T_3}{T_2} = \frac{T_4}{T_3} = \dots = -2$$

$\Rightarrow$ it is a G.P. with first term $a = \frac{1}{8}$ and common ratio $r = -2$.

$$\therefore n^{\text{th}} \text{ term} = T_n = ar^{n-1} = \frac{1}{8}(-2)^{n-1}$$

(iii) $\frac{T_2}{T_1} = \frac{3\sqrt{2}}{\sqrt{2}} = 3, \quad \frac{T_3}{T_2} = \frac{6\sqrt{2}}{3\sqrt{2}} = 2$

$$\therefore \frac{T_2}{T_1} \neq \frac{T_3}{T_2}$$

$\Rightarrow$ it is not a G.P.

Here is an exercise for you.

E 4) (i) Find the 10^{th} term of the G.P. 128, 32, 8, 2, ...

(ii) 4^{th} and 7^{th} terms of a G.P. are 24 and 192 respectively. Find the G.P.

3.4.2 Sum of n Terms of a G.P.

Standard G.P. is $a, ar, ar^2, ar^3, \dots$

We know that

$$T_n = ar^{n-1}$$

$$\therefore T_{n-1} = ar^{n-2}$$

Let S_n denotes the sum of first n terms of the above G.P., then

$$S_n = T_1 + T_2 + \dots + T_{n-1} + T_n$$

$$S_n = a + ar + ar^2 + \dots + ar^{n-2} + ar^{n-1} \quad \dots (1)$$

Multiply on both sides by r, we get

$$rS_n = ar + ar^2 + \dots + ar^{n-1} + ar^n \quad \dots (2)$$

(1) – (2) gives

$$(1-r)S_n = a - ar^n \quad [\text{All other terms cancel out in pairs}]$$

$$\Rightarrow S_n = \frac{a(1-r^n)}{1-r}$$

By taking negative sign common from numerator as well as denominator we can also write the above formula as

$$S_n = \frac{(-1)a(r^n - 1)}{(-1)(r - 1)} = \frac{a(r^n - 1)}{r - 1}$$

So, you can use either form of the formula for S_n result will be same, but we will use the formula depending on the value of common ratio r as given in the box below.

$S_n = \frac{a(1-r^n)}{1-r}, \quad r < 1$ $S_n = \frac{a(r^n - 1)}{r - 1}, \quad r > 1$

Remark 7: Keep this formula always in mind and is known as formula for sum of first n terms of a G.P. with first term 'a' and common ratio 'r'.

Let us evaluate the sum of a given G.P. with the help of above formula in the following example.

Example 6: Find the sum of the following, G.P.:

(i) 2, 4, 8, 16, ...to 10 terms (ii) $-1 + \frac{2}{3} - \frac{4}{9} + \dots$ to 8 terms

Solution:

(i) Here $a = 2$, $d = \frac{4}{2} = 2$, $n = 10$

We know that

$$S_n = \frac{a(r^n - 1)}{r - 1}, \quad \text{as } r = 2 > 1$$

$$\therefore S_{10} = \frac{2(2^{10} - 1)}{2 - 1} = 2(1024 - 1) = 2046$$

(ii) Here $a = -1$, $r = \frac{2/3}{-1} = -\frac{2}{3}$, $n = 8$

We know that

$$S_n = \frac{a(1-r^n)}{1-r}, \quad \text{as } |r| = \frac{2}{3} < 1$$

$$\therefore S_8 = \frac{(-1) \left[1 - \left(\frac{-2}{3} \right)^8 \right]}{1 - \left(\frac{-2}{3} \right)} = \frac{- \left[1 - \frac{256}{6561} \right]}{1 + \frac{2}{3}}$$

$$= \frac{-(6561 - 256)}{6561 \left(\frac{3+2}{3} \right)} = \frac{-6305 \times 3}{6561 \times 5} = -\frac{1261}{2187}$$

Here is an exercise for you.

E 5) Find the sum $\frac{2}{9} + \frac{2}{3} + 2 + 6 + \dots + 486$.

3.5 SUM OF INFINITE G.P.

A geometric progression (G.P.) is said to be infinite G.P. if number of terms in it are infinite. That is, G.P. given by

$$a, ar, ar^2, ar^3, \dots \text{ to } \infty \quad \dots (1)$$

is an infinite G.P.

We note that sum of an infinite G.P. will be finite if common ratio is less than 1 in magnitude. Let S denotes the sum of the infinite G.P. given by (1)

$$\text{i.e. } S = a + ar + ar^2 + ar^3 + \dots \text{ to } \infty \quad \dots (2)$$

Multiplying on both sides of (2) by r (common ratio), we get

$$rS = ar + ar^2 + ar^3 + \dots \text{ to } \infty \quad \dots (3)$$

(2) – (3) gives

$$(1-r)S = a, \quad -1 < r < 1, \text{ i.e. } |r| < 1 \quad [\text{All other terms cancel out in pairs}]$$

$$S = \frac{a}{1-r}, \quad -1 < r < 1, \text{ i.e. } |r| < 1$$

If you are interested to know the details related to the above formula, refer the remark given below.

Remark 8:

(i) If n approaches to infinity, i.e. $n \rightarrow \infty$, then behaviour of x^n is given below

$$x^n = \begin{cases} \infty, & \text{if } |x| > 1 \\ 0, & \text{if } |x| < 1 \\ 1, & \text{if } x = 1 \\ 1, & \text{if } x = -1 \text{ and } n \text{ is even} \\ -1, & \text{if } x = -1 \text{ and } n \text{ is odd} \end{cases}$$

But $(1)^\infty$ is not defined

For example, let $x = 4$, then for $n = 1, 2, 3, 4, 5, \dots$ we have

$$4^1 = 4, 4^2 = 16, 4^3 = 64, 4^4 = 256, 4^5 = 1024, \dots$$

That is, we observe that as n increases then x^n increases very fast and hence we write $x^n \rightarrow \infty$, as $n \rightarrow \infty$.

Similarly, let $x = 0.2$, then for $n = 1, 2, 3, 4, 5, \dots$ we have

$$(0.2)^1 = 0.2, (0.2)^2 = 0.04, (0.2)^3 = 0.008, (0.2)^4 = 0.0016, (0.2)^5 = 0.00032, \dots$$

That is, we observe that as n increases then x^n decreases very fast and reaches nearer and nearer to zero and hence we write $x^n \rightarrow 0$, as $n \rightarrow \infty$.

And if $x = 1$, then we define $x^n = 1$ for finite values of $n = 1, 2, 3, \dots, < \infty$.

whereas x^n is not defined in the case $n = \infty$, and we handle this type of situation by using some results from limit, etc.

$$(ii) S_n = \frac{a(1-r^n)}{1-r}, \text{ if } -1 < r < 1, \text{ then}$$

$$S = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{a(1-r^n)}{1-r} = \frac{a}{1-r} \left[\begin{array}{l} \because \lim_{n \rightarrow \infty} r^n = 0 \text{ as discussed} \\ \text{in part (i)} \end{array} \right]$$

(iii) Concept of limit will be discussed in Unit 5 of this course, i.e. MST-001.

Example 7: Find the following sums:

$$(i) 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \text{to } \infty \quad (ii) 1 - \frac{1}{5} + \frac{1}{25} - \frac{1}{125} + \dots \text{to } \infty$$

Solution:

$$(i) \text{ Here } a = 1, r = \frac{1/2}{1} = \frac{1}{2} < 1$$

$$\therefore S = \frac{a}{1-r} = \frac{1}{1-1/2} = \frac{1}{1/2} = 2 \quad \left[\because \text{sum of infinite G.P.} = \frac{a}{1-r} \right]$$

$$(ii) \text{ Here } a = 1, r = \frac{-1/5}{1} = -\frac{1}{5}$$

$\therefore |r| < 1$, so sum of infinite G.P. is given by

$$S = \frac{a}{1-r} = \frac{1}{1-(-1/5)} = \frac{1}{1+1/5} = \frac{1}{6/5} = \frac{5}{6}$$

Here is an exercise for you.

$$\text{E 6) Prove that (i) } 4^{1/4} \cdot 4^{1/8} \cdot 4^{1/16} \cdot 4^{1/32} \dots \text{to } \infty = 2 \quad (ii) 5^{1/3} \cdot 5^{1/9} \cdot 5^{1/27} \dots \text{to } \infty = \sqrt{5}$$

3.6 CONCEPT OF SUMMATION

3.6.1 Series: If $a_1, a_2, a_3, \dots$ to ∞ is a sequence then expression

$a_1 + a_2 + a_3 + \dots$ to ∞ is known as series.

This series in the form of summation is written as $\sum_{n=1}^{\infty} a_n$

$$\text{i.e. } \sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \dots \text{to } \infty$$

In case of finite expression $x_1 + x_2 + x_3 + \dots + x_n$

$$\text{We write as } \sum_{i=1}^n x_i = x_1 + x_2 + \dots + x_n$$

Remark 9:

- (i) The symbol Σ is the Greek letter pronounced as sigma.
- (ii) The letters n and i used above are known as dummy variables. These letters have nothing special other letters like m, r, s, k, j, etc. can also be used.

3.6.2 Change of Origin of Summation

A given series can be written in different ways in terms of summation

i.e. series $x_0 + x_1 + x_2 + \dots$ to ∞

can be written in any of the following ways

$$\sum_{n=0}^{\infty} x_n \quad \text{or} \quad \sum_{n=2}^{\infty} x_{n-2} \quad \text{or} \quad \sum_{x=10}^{\infty} x_{n-10} \quad \text{or} \quad \sum_{n=r}^{\infty} x_{n-r}$$

i.e. origin of the summation $\sum_{n=0}^{\infty} x_n$ is at $n = 0$ and if we want to shift the origin

at $n = k$ then we have to subtract k from the suffixes of terms within summation.

In Case of Finite Terms: Origin of the summation $\sum_{i=0}^n x_i$ is at $i = 0$ and if we

want to shift the origin at $i = k$ then we have to subtract k from the suffixes of the terms within summation and we also have to add k in range of the summation

$$\text{i.e. } \sum_{i=0}^n x_i = \sum_{i=k}^{n+k} x_{i-k}$$

3.7 SUM OF SOME SPECIAL SEQUENCES

Following are given sum of some special sequences as they will be helpful at various occasions during study of the programme. Keep these always in mind

$$(1) \sum_{k=1}^n k = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} = \text{sum of first } n \text{ natural numbers.}$$

$$(2) \sum_{k=1}^n k^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

= sum of squares of first n natural numbers.

$$(3) \sum_{k=1}^n k^3 = 1^3 + 2^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

= sum of cubes of first n natural numbers.

3.8 SUMMARY

Let us summarise the topics that we have covered in this unit:

- 1) Definition of sequence.
- 2) Ways of representation a sequence.
- 3) Arithmetic progression (A.P.).
- 4) Geometric progression (G.P.).
- 5) Sum of infinite G.P.
- 6) Concept of summation and change of initiation of summation.
- 7) Sum of some special sequences.

3.9 SOLUTIONS/ANSWERS

E 1 (i) $a_n = \sqrt{2n}$

For $n = 1, 2, 3, 4$, we have

$$a_1 = \sqrt{2 \times 1} = \sqrt{2}, a_2 = \sqrt{2 \times 2} = \sqrt{4} = 2$$

$$a_3 = \sqrt{2 \times 3} = \sqrt{6}, a_4 = \sqrt{2 \times 4} = \sqrt{8} = \sqrt{2 \times 2 \times 2} = 2\sqrt{2}$$

(ii) $a_n = \frac{2}{\sqrt{n}}$

For $n = 1, 2, 3, 4, 8$, we have

$$a_1 = \frac{2}{\sqrt{1}} = \frac{2}{1} = 2, a_2 = \frac{2}{\sqrt{2}} = \frac{2}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{2\sqrt{2}}{2} = \sqrt{2}$$

$$a_3 = \frac{2}{\sqrt{3}} = \frac{2}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{2\sqrt{3}}{3}, a_4 = \frac{2}{\sqrt{4}} = \frac{2}{2} = 1$$

$$a_8 = \frac{2}{\sqrt{8}} = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

(iii) $a_n = \frac{2^n \times 3^n}{2^n + 3^n}$

For $n = 1, 2, 3$, we have

$$a_1 = \frac{2^1 \times 3^1}{2^1 + 3^1} = \frac{6}{5}, a_2 = \frac{2^2 \times 3^2}{2^2 + 3^2} = \frac{36}{13}, a_3 = \frac{2^3 \times 3^3}{2^3 + 3^3} = \frac{8 \times 27}{8 + 27} = \frac{216}{35}$$

E 2 $a_2 - a_1 = \log \frac{a^2}{b} - \log a = \log \frac{a^2}{b} \times \frac{1}{a} = \log \frac{a}{b} \left[\because \log m - \log n = \log \frac{m}{n} \right]$

$$a_3 - a_2 = \log \frac{a^3}{b^2} - \log \frac{a^2}{b} = \log \frac{a^3}{b^2} \times \frac{b}{a^2} = \log \frac{a}{b} \quad [\text{Same reason}]$$

and so on

$$\therefore a_2 - a_1 = a_3 - a_2 = \dots = \log \frac{a}{b}$$

$\therefore$ it is an A.P.

E 3 (i) Since $5k + 1, 6k + 5$ and $10k + 3$ are three consecutive terms of an A.P.

$$\therefore (6k + 5) - (5k + 1) = (10k + 3) - (6k + 5) \quad [\because T_2 - T_1 = T_3 - T_2]$$

$$\Rightarrow k + 4 = 4k - 2 \Rightarrow 6 = 3k \Rightarrow k = 2$$

(ii) Here $a = 3, d = 9 - 3 = 6$

$$\text{Let } T_n = 121 \Rightarrow a + (n-1)d = 121 \Rightarrow 3 + (n-1)6 = 121$$

$$\Rightarrow 3 + 6n - 6 = 121 \Rightarrow 6n - 3 = 121 \Rightarrow 6n = 124 \Rightarrow n = \frac{124}{6} = \frac{62}{3}$$

$$\Rightarrow n = 20\frac{2}{3}$$

This is not possible because value of n is always a natural number.

$\therefore 121$ cannot be a term of this A.P.

(iii) Here $a = -1, d = -\frac{1}{4} - (-1) = -\frac{1}{4} + 1 = \frac{3}{4}$

$$\text{Let } T_n = 14 \Rightarrow a + (n-1)d = 14 \Rightarrow -1 + (n-1)\frac{3}{4} = 14$$

$$\Rightarrow -1 + \frac{3}{4}n - \frac{3}{4} = 14 \Rightarrow \frac{3}{4}n = 14 + 1 + \frac{3}{4} \Rightarrow \frac{3}{4}n = \frac{63}{4} \Rightarrow n = 21$$

$\therefore$ number of terms = 21. That is 14 is the 21st term of the given A.P.

E 4) (i) Here $a = 128$, $r = \frac{32}{128} = \frac{1}{4}$

We know that

$$T_n = ar^{n-1}$$

$$\therefore T_{10} = 128 \left(\frac{1}{4} \right)^{10-1} = 2^7 \left(\frac{1}{4} \right)^9 = \frac{2^7}{2^{18}} = \frac{1}{2^{11}} = \frac{1}{2048}$$

(ii) Let a, r be the first term and common ratio of the given G.P. respectively.

According to the problem

$$\left. \begin{array}{l} T_4 = 24 \\ T_7 = 192 \end{array} \right\} \Rightarrow \begin{cases} ar^3 = 24 & \dots(1) \\ ar^6 = 192 & \dots(2) \end{cases}$$

(2) $\div$ (1) gives

$$r^3 = \frac{192}{24} = 8 \Rightarrow r^3 = 2^3 \Rightarrow r = 2$$

Putting $r = 2$ in (1), we get

$$8a = 24 \Rightarrow a = 3$$

$\therefore$ G.P. is 3, 6, 12, 24, 48, ...

E 5) Here $a = \frac{2}{9}$, $r = \frac{2/3}{2/9} = \frac{2}{3} \times \frac{9}{2} = 3$

$$T_n = 486 \Rightarrow ar^{n-1} = 486 \Rightarrow \frac{2}{9}(3)^{n-1} = 486 \Rightarrow 3^{n-1} = 486 \times \frac{9}{2}$$

$$\Rightarrow 3^{n-1} = 243 \times 9 \Rightarrow 3^{n-1} = 3^7 \Rightarrow n-1 = 7 \Rightarrow n = 8$$

We know that

$$S_n = \frac{a(r^n - 1)}{r - 1}, \text{ as } r = 3 > 1$$

$$\therefore S_8 = \frac{\frac{2}{9}[(3)^8 - 1]}{3 - 1} = \frac{\frac{2}{9}[6561 - 1]}{2} = \frac{6560}{9}$$

E 6) (i) L.H.S. = $4^{1/4} \cdot 4^{1/8} \cdot 4^{1/16} \cdot 4^{1/32} \dots \text{to } \infty = 4^{\left(\frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \text{to } \infty\right)}$

$$= 4^{\left(\frac{1/4}{1-1/2}\right)} \left[\because \text{sum of infinite G.P.} = \frac{a}{1-r} \text{ here } a = \frac{1}{4} \text{ and } r = \frac{1}{2} \right]$$

$$= 4^{\frac{1}{2}} = (2^2)^{\frac{1}{2}} = 2 = \text{R.H.S.}$$

(ii) L.H.S. = $5^{\left(\frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots \text{to } \infty\right)}$

$$= 5^{\left(\frac{1/3}{1-1/3}\right)} \left[\because \text{sum of infinite G.P.} = \frac{a}{1-r} \text{ here } a = \frac{1}{3} \text{ and } r = \frac{1}{3} \right]$$

$$= 5^{\left(\frac{1/3}{2/3}\right)} = 5^{1/2} = \sqrt{5} = \text{R.H.S.}$$

UNIT 4 TECHNIQUES OF COUNTING

Structure

- 4.1 Introduction
 - Objectives
- 4.2 Factorial and its Notations
- 4.3 Fundamental Principles of Counting
- 4.4 Permutation
- 4.5 Combination
- 4.6 Selection of Permutation or Combination
- 4.7 Some Important Results
- 4.8 Binomial Theorem
- 4.9 Summary
- 4.10 Solutions/Answers

4.1 INTRODUCTION

Suppose a boy/girl has number lock in his/her cycle having 3 wheels each containing 10 digits from 0 to 9. Suppose the boy/girl forgets his/her 3 digits lock number. Then is there any technique which helps him/her to open the lock without breaking the lock? The answer is yes. This answer is provided by the techniques of counting. In case of above situation, techniques of counting tell us how many different locking options (the three digits codes) are possible with 3 wheels each containing 10 digits from 0 to 9. Out of these options, there is only one correct option. If the boy/girl starts to try them definitely at some stage, lock will get opened (because total number of options is finite in number). In our day to day life, there are many situations, where we need to count the number of ways a particular event can take place.

In this unit, we will discuss two such techniques known as permutation and combination based on fundamental principles of counting. We will introduce concept of factorial and binomial theorem also in this unit.

Objectives

After completing this unit, you should be able to:

- get the idea of factorial and its notations;
- get the logic of fundamental principles of addition and multiplication;
- define linear permutation and solve simple problems based on it;
- define circular permutation and solve simple problems based on it;
- define combination and solve simple problems based on it; and
- get an idea of binomial theorem.

4.2 FACTORIAL AND ITS NOTATIONS

The product of first n natural numbers is denoted by $n!$ or $\underline{n}$ and read as 'n factorial'.

i.e $n! = n \times (n-1) \times \dots \times 3 \times 2 \times 1$

If $n = 0$, then we define $0! = 1$

Remark: We see that $n! = n \times (n-1) \times (n-2) \times \dots \times 3 \times 2 \times 1$
 $= n \times \underline{n-1}$
 $= n \times (n-1) \times \underline{n-2}$ and so on.

Let us now consider some examples.

Example 1: Evaluate the following

(i) $8!$ (ii) $(4!)(3!)$ (iii) $\frac{10!}{8!4!}$ (iv) $5! + 4!$ (v) $6! - 4!$

Solution:

(i) $8! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 40320$
(ii) $(4!)(3!) = (4 \times 3 \times 2 \times 1)(3 \times 2 \times 1) = 24 \times 6 = 144$
(iii) $\frac{10!}{8!4!} = \frac{10.9.8!}{8!(4 \times 3 \times 2 \times 1)} = \frac{10.9}{4.3.2.1} = \frac{15}{4}$
(iv) $5! + 4! = 120 + 24 = 144$
(v) $6! - 4! = 720 - 24 = 696$

Example 2: Express the following in terms of factorial:

(i) $5.6.7.8.9.10$ (ii) $4.8.12.16.20.24$
(iii) $1.3.5.7.9.11$ (iv) $2n.4n.6n.8n$

Solution:

(i) $5.6.7.8.9.10 = \frac{1.2.3.4.5.6.7.8.9.10}{1.2.3.4} = \frac{10!}{4!}$
(ii) $4.8.12.16.20.24 = 4^6(1.2.3.4.5.6) = 4^6(6!)$
(iii) $1.3.5.7.9.11 = \frac{1.2.3.4.5.6.7.8.9.10.11}{2.4.6.8.10} = \frac{11!}{2^5(1.2.3.4.5)} = \frac{11!}{32(5!)}$
(iv) $2n.4n.6n.8n = (2n)^4(1.2.3.4) = (2n)^4(4!)$

Example 3: Solve for n , $n \in \mathbb{N}$

(i) $(n+2)! = 42.n!$ (ii) $\frac{1}{8!} + \frac{1}{9!} = \frac{n}{10!}$ (iii) $\frac{2.(n!)}{(n-2)!} = \frac{(n+1)!}{(n-1)!}$, $n \neq 1, 2$

Solution:

(i) $(n+2)(n+1)(n!) = 42(n!) \Rightarrow (n+2)(n+1) = 42$
 $\Rightarrow n^2 + 3n + 2 - 42 = 0 \Rightarrow n^2 + 3n - 40 = 0$
 $\Rightarrow n^2 + 8n - 5n - 40 = 0 \Rightarrow n(n+8) - 5(n+8) = 0$
 $\Rightarrow (n+8)(n-5) = 0$
 $\Rightarrow n = -8, 5$
But $n \in \mathbb{N}$, therefore $n = 5$
(ii) $\frac{1}{8!} + \frac{1}{9!} = \frac{n}{10!} \Rightarrow \frac{1}{8!} + \frac{1}{9.8!} = \frac{n}{10.9.8!} \Rightarrow 1 + \frac{1}{9} = \frac{n}{90}$
 $\Rightarrow \frac{10}{9} = \frac{n}{90} \Rightarrow n = \frac{10 \times 90}{9} = 100$
 $\therefore n = 100$
(iii) $\frac{2.n.(n-1).(n-2)!}{(n-2)!} = \frac{(n+1).n.(n-1)!}{(n-1)!}$

$$\Rightarrow 2n(n-1) = (n+1)n$$

$$\Rightarrow 2n^2 - 2n = n^2 + n \Rightarrow n^2 - 3n = 0$$

$$\Rightarrow n(n-3) = 0 \Rightarrow n = 0, 3$$

But $n \in \mathbb{N}$

$$\therefore n = 3$$

Now, you can try the following exercises.

E 1) Evaluate the following

(i) $\frac{22!}{19!}$

(ii) $\frac{15!}{10! \times 5!}$

E 2) Express the following in terms of factorial.

(i) $3.6.9.12.15$

(ii) $7.8.9.10.11.12$

E 3) Solve for n , $n \in \mathbb{N}$

(i) $(n-2)! = 12(n-4)!$

(ii) $n! = 72(n-2)!$

4.3 FUNDAMENTAL PRINCIPLES OF COUNTING

There are two fundamental principles of counting. These two principles solve the problems of counting. So it becomes necessary for us first to define what is the counting problem? According to Grinstead and Snell (2006) it is defined as if you

“Consider an experiment that takes place in several stages and is such that the number of outcomes m at the n th stage is independent of the outcomes of the previous stages. The number m may be different for different stages. We want to count the number of ways that the entire experiment can be carried out.”

Let us take an example.

Example 4: Statistics discipline wanted to book the lunch in the IGNOU guest house for the experts during an expert committee meeting. The incharge of the guest house explain the lunch menu like this:

- (a) there are two choices for appetizers: soup and juice
- (b) there are two choices for main course: veg and non-veg
- (c) there are three choices for dessert: sponge rashgulla, gulab jamun and ice cream.

How many options were there for statistics discipline for complete meal?

Solution: If we compare this situation with the counting problem we note that

- (i) Entire experiment means complete meal.
- (ii) Number of stages of the entire experiment are three in numbers.
- (iii) Options (number of outcomes) for first, second, third stages are 2, 2, 3 respectively.

The above situation in the form of a tree diagram can be represented as shown on the next page.

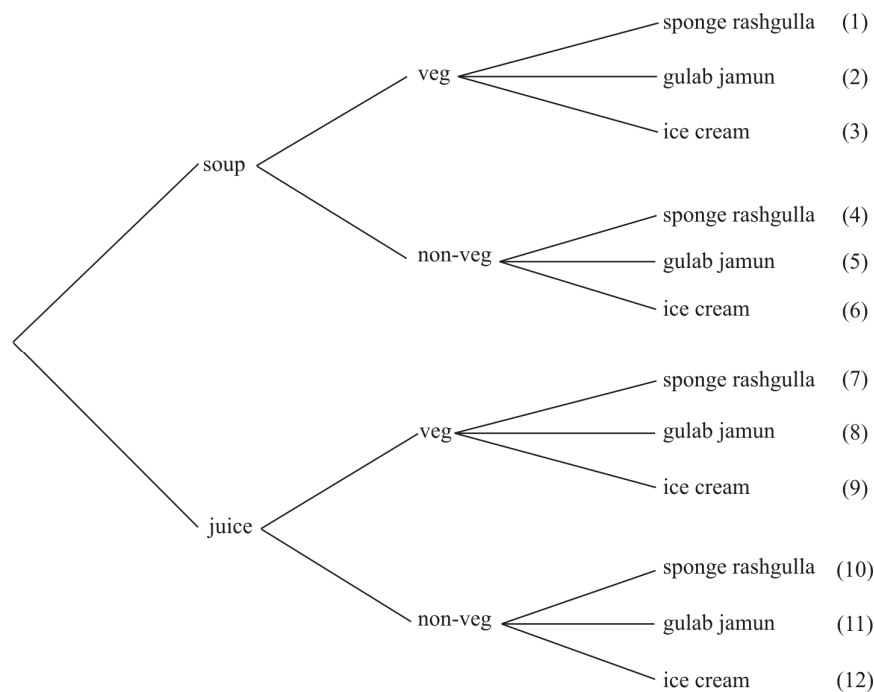


Fig 4.1 Tree Diagram of Guest House Menu

From the above tree diagram, we see that total number of options for complete meal is given by:

(total choices at first stages) $\times$ (total choices at second stage) $\times$ (total choices at third stage)

$$= 2 \times 2 \times 3$$

$$= 12$$

Hence there were 12 options for statistics discipline for complete meal. Each of these 12 options is numbered from 1 to 12 in the right margin of the tree diagram. For example option 7 is 'juice followed by veg followed by sponge rashgulla', option 12 is 'juice followed by non-veg followed by ice-cream', etc.

Now we discuss the two fundamental principles of counting in coming two sub-sections.

4.3.1 Fundamental Principle of Multiplication (FPM)

Suppose we want to complete two jobs, where first job can be done in m distinct ways, second job can be done in n distinct ways then both jobs can take place (one followed by other) in $m \times n$ distinct ways.

In general, suppose we want to complete n jobs, where
first job can be done in m_1 distinct ways,
second job can be done in m_2 distinct ways,
third job can be done in m_3 distinct ways,
and so on
 n^{th} job can be done in m_n distinct ways.

Then these n jobs can take place (in succession) in $m_1 \times m_2 \times m_3 \times \dots \times m_n$ distinct ways.

For example, suppose a teacher wants to select one boy and one girl student out of a class having 15 boys and 10 girls students, then teacher can make such selection in $15 \times 10 = 150$ distinct ways.

4.3.2 Fundamental Principle of Addition (FPA)

Suppose we want to complete one job out of two jobs, where first job can be done in m distinct ways and second independent job can be done in n distinct ways. Then one of the two jobs can take place in $m + n$ distinct ways.

In general, suppose we want to complete one job out of n jobs, where

first job can be done in m_1 distinct ways,

second job can be done in m_2 distinct ways,

third job can be done in m_3 distinct ways,

and so on

n^{th} job can be done in m_n distinct ways.

Then one of the n jobs (any two or any three... or all of these can not occur simultaneously) can take place in $m_1 + m_2 + m_3 + \dots + m_n$ distinct ways.

For example, suppose a teacher wants to select either a boy or a girl student out of a class having 15 boys and 10 girls, then teacher can select either a boy or a girl student in $15 + 10 = 25$ distinct ways.

Now we take some examples based on these two principles of counting.

Example 5: In a college, there are 40 male and 30 female faculties. The principal of that college wants to select one male and one female faculty to accompany with the students of the college going for a picnic. In how many ways can principal do this selection?

Solution: For this selection, principal has to complete two jobs:

(i) Selection of a male faculty

(ii) Selection of a female faculty

First job can be done in 40 ways and second job can be done in 30 ways.

$\therefore$ by fundamental principle of multiplication required number of ways

$$= 40 \times 30 = 1200$$

Example 6: There are 40 male and 30 female faculties in a college. The principal of the college wants to select one faculty (either male or female) for an examination duty. In how many ways this selection can be done?

Solution: For this selection, principal of the college do either of the following two jobs:

(i) Selection of a male faculty

(ii) Selection of a female faculty

First job can be done in 40 ways and second job can be done in 30 ways.

$\therefore$ by fundamental principle of addition, required number of ways

$$= 40 + 30 = 70$$

Example 7: How many different number plates of vehicles are possible using two different letters of English alphabet followed by four different digits 0 to 9?

Solution: In order to complete this job, we have to fill up six positions in succession, where

First position can be filled up in 26 ways,

Second position can be filled up in 25 ways $\left[\because \text{one letter has been used in first place} \right]$

Third position can be filled up in 10 ways $\left[\text{With one of the digits from 0 to 9} \right]$

In case of 'and' we multiply and in case of 'or' we add

Fourth position can be filled up in 9 ways $\left[\begin{array}{l} \text{With one of the 9 digits} \\ \text{leaving the one already used} \end{array} \right]$

Fifth position can be filled up in 8 ways $\left[\begin{array}{l} \text{With one of the 8 digits} \\ \text{leaving the two already used} \end{array} \right]$

Sixth position can be filled up in 7 ways $\left[\begin{array}{l} \text{With one of the 7 digits} \\ \text{leaving the three already used} \end{array} \right]$

$\therefore$ by fundamental principal of multiplication required numbers of ways

$$= 26 \times 25 \times 10 \times 9 \times 8 \times 7$$

$$= 3276000$$

Note: If in the above example repetition of digits 0 to 9 is allowed (which in practice happens) then required number of ways = $26 \times 25 \times (10 \times 10 \times 10 \times 10 - 1)$

$\left[\begin{array}{l} 1 \text{ is subtracted because we have ignore} \\ \text{the case containg all zeros, i.e. 0000} \end{array} \right]$

Now, you can try the following exercises.

-
- E 4)** In an examination there are 10 multiple choice questions. First five questions have 4 choices each and last five questions have 5 choices each. How many sequences of answers are possible?
- E 5)** How many four-letter words can be formed by using letters a, b, g, h, k, if
- (i) Repetition is not allowed (ii) Repetition is allowed
-

4.4 PERMUTATION

Permutation is related to the arrangement of things. Things arranged in a line come under the heading of **linear permutation**, while arrangement of things in a circle comes under the heading of **circular permutation**. Let us discuss these two heading one by one.

4.4.1 Linear Permutation

Possible arrangements in a line of a number of things taken some or all at a time are called the permutation. Before giving the general formula, let us consider an example, where we are to arrange say three books of different colours (Red, Green and Orange):

Permutations of three books when taken one at a time are R, G, W, i.e.

the number of permutations = $3 = \frac{3!}{(3-1)!} = {}^3P_1$ or $P(3, 1)$

Permutations of three books when taken two at a time are RG, GR, RW, WR, GW, WG, i.e.

the number of permutations = $6 = \frac{3!}{(3-2)!} = {}^3P_2$ or $P(3, 2)$

Permutations of three books when taken all at a time are RGW, RWG, GRW, GWR, WRG, WGR, i.e.

the number of permutations = $6 = \frac{3!}{(3-3)!} = {}^3P_3$ or $P(3, 3)$

In general, the total number of permutations of n things taken r ($1 \leq r \leq n$) at a time is denoted by nP_r or $P(n, r)$ and is defined as

$${}^n P_r = \frac{n!}{(n-r)!} = n(n-1)(n-2) \dots (n-(r-1))$$

i.e. ${}^n P_r = n(n-1)(n-2) \dots$ up to r factors

For example,

- (i) Total number of permutations of a, b, c taken 2 at a time are given by ab, ba, bc, cb, ca, ac.

$$\text{Also } {}^3 P_2 = 3.2 = 6$$

- (ii) Total number of permutations of a, b, c taken all at a time are given by abc, acb, bca, bac, cab, cba.

$$\text{Also } {}^3 P_3 = 3.2.1 = 6$$

1. We define ${}^n P_0 = 1$

2 Always remember the following result

$${}^n P_r = \frac{n!}{(n-r)!}, \quad 0 \leq r \leq n$$

Example 8: Evaluate the following:

- (i) ${}^8 P_2$ (ii) ${}^{20} P_5$ (iii) $P(10, 4)$ (iv) $P(8, 8)$ (v) ${}^5 P_0$

Solution:

$$(i) \quad {}^8 P_2 = \frac{8!}{(8-2)!} = \frac{8!}{6!} = \frac{8 \times 7 \times 6!}{6!} = 8 \times 7 = 56$$

$$(ii) \quad {}^{20} P_5 = \frac{20!}{(20-5)!} = \frac{20!}{15!} = \frac{20 \times 19 \times 18 \times 17 \times 16 \times 15!}{15!} \\ = 20 \times 19 \times 18 \times 17 \times 16 = 1860480$$

$$(iii) \quad P(10, 4) = \frac{10!}{(10-4)!} = \frac{10!}{6!} = \frac{10 \times 9 \times 8 \times 7 \times 6!}{6!} \\ = 10 \times 9 \times 8 \times 7 = 5040$$

$$(iv) \quad P(8, 8) = \frac{8!}{(8-8)!} = \frac{8!}{0!} = \frac{40320}{1} = 40320$$

$$(v) \quad {}^5 P_0 = \frac{5!}{(5-0)!} = \frac{5!}{5!} = 1$$

Example 9: Find n if ${}^n P_5 = 30 {}^n P_3$.

$$\text{Solution: } {}^n P_5 = 30 {}^n P_3 \Rightarrow \frac{n!}{(n-5)!} = \frac{30.n!}{(n-3)!}$$

$$\Rightarrow \frac{1}{(n-5)!} = \frac{30}{(n-3)!} \Rightarrow \frac{1}{(n-5)!} = \frac{30}{(n-3).(n-4).(n-5)!}$$

$$\Rightarrow \frac{1}{1} = \frac{30}{(n-3)(n-4)} \Rightarrow (n-3)(n-4) = 30$$

$$\Rightarrow n^2 - 7n + 12 = 30 \Rightarrow n^2 - 7n - 18 = 0$$

$$\Rightarrow n^2 - 9n + 2n - 18 = 0 \Rightarrow n(n-9) + 2(n-9) = 0$$

$$\Rightarrow (n-9)(n+2) = 0 \Rightarrow n = 9, -2$$

But n cannot be -2 .

$$\therefore n = 9$$

Here are some exercises for you.

E6) Find n if ${}^5P_n = {}^6P_{n-1}$.

E7) Evaluate the following

$$(i) {}^7P_4 \quad (ii) {}^nP_n \quad (iii) {}^nP_{n-1} \quad (iv) {}^nP_1 \quad (v) {}^nP_2 \quad (vi) {}^{16}P_3$$

Example 10: How many different words, with or without meaning, can be formed by using all the letters of the word 'EQUATION' (without repetition)?

Solution: There are 8 letters in the word 'EQUATION' which are all different.

$\therefore$ possible number of words = number of arrangement of 8 letters taken all at a time

$$= {}^8P_8 = \frac{8!}{(8-8)!} = \frac{8!}{0!} = 8! = 40320 \quad \text{as } 0! = 1$$

Example 11: How many signals are possible with 4 flags each of different colour?

Solution: Possible number of signals using one flag at a time = ${}^4P_1 = 4$

Possible number of signals using two flags at a time = ${}^4P_2 = 4.3 = 12$

Possible number of signals using three flags at a time = ${}^4P_3 = 4.3.2 = 24$

Possible number of signals using all flags at a time = ${}^4P_4 = 4! = 24$

$\therefore$ total number of signals = $4 + 12 + 24 + 24 = 64$

Here is an exercise for you.

E 8) In how many ways can 5 students stand in a queue?

Permutations of Things not all Distinct

So far we have discussed the permutations of things which were all distinct. But in usual it is not always possible that things to be permuted are all distinct.

In case of repetition of things we use following result.

If out of n things p_1 are of one kind, p_2 are of second kind, p_3 are of third kind and so on p_k are of k^{th} kind then total number of possible permutations are given by

$$\frac{n!}{p_1! \times p_2! \times p_3! \times \dots \times p_k!}, \text{ where } p_1 + p_2 + \dots + p_k = n \text{ and } p_i \geq 1, 1 \leq i \leq k$$

Example 12: How many different words, with or without meaning, can be formed by using all the letters of the word "BANANA"?

Solution: There are 6 letters in the word "BANANA"

Out of which A, N occur 3, 2 times respectively.

$$\therefore \text{total number of permutations} = \frac{6!}{3! 2!} = \frac{6.5.4.3!}{3! 2!} = \frac{120}{2} = 60$$

Here are some exercises for you.

E 9) How many different signals are possible with 3 red, 4 white and 2 green flags by using all at a time in a queue?

E10) How many words can be formed with or without meaning by using the letters of the words AMAR?

Permutation when Repetition is Allowed

Example 13: Prove that total number of permutations of n things taken r at a time any thing can repeat any number of times is given by n^r .

Proof: In order to find out total number of permutations, we have to fill up r positions, where

First position can be filled up in n ways,

Second position can also be filled up in n ways, [$\because$ repetition is allowed]

Third position can be filled up in n way, [Same reason]

And so on

r^{th} position can be filled up in n ways.

$\therefore$ required number of permutations = $\underbrace{n \times n \times n \times \dots \times n}_{r \text{ times}} = n^r$.

Example 14: In how many ways 5 letters can be posted in 3 letter boxes?

Solution: Each of the 5 letters can be posted in 3 ways, i.e. each of 5 letters can be posted by using any of the 3 letter boxes.

$\therefore$ required number of ways = $3 \times 3 \times 3 \times 3 \times 3 = 3^5 = 243$

Here is an exercise for you.

E 11) In how many ways can 3 prizes be distributed among 5 students when

- (i) No student gets more than one prize?
 - (ii) A student may get any number of prizes?
 - (iii) No student gets all the prizes?
-

4.4.2 Circular Permutation

Let us consider four letters A, B, C, D. Consider the following arrangements ABCD, BCDA, CDAB, DABC these are 4 different arrangements when arranged in a line. whereas this is a single arrangement when arranged in a circle, in clockwise direction as shown in figure.

$\therefore$ in case of 4 letters, 4 linear arrangements = 1 circular arrangement

1 linear arrangement = $\frac{1}{4}$ circular arrangement

So, 4! Linear arrangements = $\frac{4!}{4} = 3!$ circular arrangements.

In general, if anticlock wise and clock wise order of arrangements makes different permutations then number of circular permutations of n distinct things = $(n - 1)!$

And if anti-clock wise and clock wise order of arrangements does not give distinct permutations then total number of permutations of n distinct things

$$= \frac{(n - 1)!}{2}$$

For example, arrangements of flowers in a garland form the same permutation in case of anti clock wise and clockwise order.

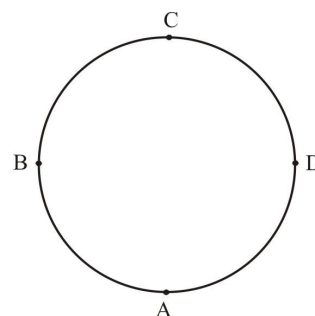
Example 15: In how many ways 10 students of a batch can be arrangements in a

- (i) Line
- (ii) Circle

Solution:

- (i) Total number of arrangements of 10 students in a line

$$= 10! = 3628800 \quad [\text{Linear permutation}]$$



- (ii) Total number of arrangements of 10 students in a circle
 $= (10 - 1)! = 9! = 362880$ [Circular permutation]

Here is an exercise for you.

-
- E 12** (i) How many different garlands are possible with 8 flowers?
 (ii) In how many ways 20 members of the management of a college can sit on a round table in a meeting, if president and vice president always sit together.
-

4.5 COMBINATION

In Sec. 4.4 of this unit we have discussed permutation. We have seen that in case of permutation we want to know the possible number of arrangements of n things taken some or all at a time. But sometimes we are interested in forming only groups or making selections or drawing items without bothering about the arrangements. These are called combinations.

Before giving the general formula, let us consider an example, where we are to form the groups of say three books of different colours (Red, Green, Orange). Combinations of three books when taken one at a time are R, G, W, i.e.

$$\text{the number of combinations} = 3 = \frac{3!}{(3-1)! \times 1!} = {}^3C_1 \text{ or } C(3, 1)$$

Combinations of three books when taken two at a time are RG, RW, GW, i.e.

$$\text{the number of combinations} = 3 = \frac{3!}{(3-2)! \times 2!} = {}^3C_2 \text{ or } C(3, 2)$$

Combination of three books when taken all at a time is RGW, i.e.

$$\text{the number of combination} = 1 = \frac{3!}{(3-3)! \times 3!} = {}^3C_3 \text{ or } C(3, 3)$$

In general, the total number of combinations of n things taken r ($1 \leq r \leq n$) at a time is denoted by nC_r or $C(n, r)$ and is defined as

$${}^nC_r = \frac{n!}{(n-r)! \times r!}$$

For example,

- (i) Total number of combinations of a, b, c taken 2 at a time are given by ab, bc, ca

$$\text{Also } {}^3C_2 = \frac{3!}{(3-2)! \times 2!} = \frac{3 \times 2 \times 1}{1 \times 2} = 3$$

- (ii) Total number of combinations of a, b, c taken all at a time are given by abc,

$$\text{Also } {}^3C_3 = \frac{3!}{(3-3)! \times 3!} = \frac{3!}{1! \times 3!} = 1$$

In this section we shall discuss another important technique of counting known as combination. Total number of groups that can be formed of n things taken

r ($0 \leq r \leq n$) at a time is called combination, denoted by nC_r or $C(n, r)$ or $\binom{n}{r}$ and is defined as

$${}^nC_r = \frac{n!}{r! \times (n-r)!}$$

For example, suppose there are six cricket teams and every team has to play one match with each other team. Then total number of matches which are to be played = combinations of six teams when taken two at a time

$$= {}^6C_2 = \frac{6!}{(6-2)! \times 2!} = \frac{6 \times 5}{2} = 15$$

4.6 SELECTION OF PERMUTATION OR COMBINATION

Following points help you in deciding which of the two permutation or combination should be used in a given situation.

- Permutation is used when the order of the selection also matters.
- Combination is used when the order does not matter, but only the selection or group formation or draw of items is taken into consideration.

Example 16: Prove the following

$$(i) \quad {}^nC_r = {}^nC_{n-r}, \quad 0 \leq r \leq n \quad (ii) \quad {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$$

$$(iii) \quad {}^nC_0 = {}^nC_n = 1 \quad (iv) \quad {}^nC_1 = {}^nC_{n-1} = n$$

Solution:

$$(i) \quad \text{R.H.S.} = {}^nC_{n-r} = \frac{n!}{(n-(n-r))! \times (n-r)!} = \frac{n!}{r! \times (n-r)!}$$

$$= \frac{n!}{(n-r)! \cdot r!} = {}^nC_r = \text{L.H.S.}$$

$$(ii) \quad \text{L.H.S.} = {}^nC_r + {}^nC_{r-1} = \frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)!}$$

$$= \frac{n!}{r.(r-1)!(n-r)!} + \frac{n!}{(r-1)!(n-r+1).(n-r)!}$$

$$= \frac{n!}{(r-1)!(n-r)!} \left[\frac{1}{r} + \frac{1}{n-r+1} \right] = \frac{n!}{(r-1)!(n-r)!} \left[\frac{n-r+1+r}{r(n-r+1)} \right]$$

$$= \frac{(n+1).n!}{r.(r-1)!(n-r+1).(n-r)!} = \frac{(n+1)!}{r!(n+1-r)!} = {}^{n+1}C_r = \text{R.H.S.}$$

$$(iii) \quad {}^nC_0 = {}^nC_n = 1$$

I II III

$$I = {}^nC_0 = \frac{n!}{0!.(n-0)!} = \frac{n!}{n!} = 1 = III \quad \text{as } 0! = 1$$

$$II = {}^nC_n = \frac{n!}{n!.(n-n)!} = \frac{n!}{n!.0!} = \frac{n!}{n!} = 1 = III$$

$$(iv) \quad {}^nC_1 = {}^nC_{n-1} = n$$

I II III

$$I = {}^nC_1 = \frac{n!}{1!.(n-1)!} = \frac{n.(n-1)!}{(n-1)!} = n = III$$

$$II = {}^nC_{n-1} = \frac{n!}{(n-1)!(n-n+1)!} = \frac{n.(n-1)!}{(n-1)! 1!} = \frac{n}{1!} = \frac{n}{1} = n = III$$

Example 17: If four cards are chosen from a pack of 52 playing cards then find the number of ways that all the four cards are

- (i) of different suit
- (ii) of same suit
- (iii) face cards
- (iv) either red or black

Solution: We know that in a pack of playing cards there are 4 suits namely, spade, diamond, heart, and club each containing 13 cards.

- (i) Required number of ways = ${}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1$
 $= 13 \times 13 \times 13 \times 13 = 28561$
- (ii) Required number of ways = ${}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4$
 $= 4 \times {}^{13}C_4 = \frac{4.13.12.11.10}{4!} = 2860$

(iii) We know that there are 12 face cards.

$$\therefore \text{required number of ways} = {}^{12}C_4 = \frac{12.11.10.9}{4!} = 495$$

(iv) We know that there are 26 red and 26 black cards in a pack of playing cards.

$$\therefore \text{required number of ways} = {}^{26}C_4 + {}^{26}C_4$$

$$= 2 \times {}^{26}C_4 = 2 \times \frac{26.25.24.23}{4!} = 29900$$

Example 18: A bag contains 4 red and 7 white balls. Find the number of ways in which 2 red and 3 white balls can be drawn.

Solution: Out of 4 red balls 2 can be drawn in 4C_2 ways = $\frac{4.3}{2!} = 6$

Out of 7 white balls 3 can be drawn in 7C_3 ways = $\frac{7.6.5}{3!} = 35$

$$\therefore \text{required number of ways} = {}^4C_2 \times {}^7C_3 = 6 \times 35 = 210$$

Here is an exercise for you.

E 13) There are 21 cricket players including 11 batsmen, 7 bowlers and 3 wicket keepers. In how many ways 11 players can be selected having 6 batsmen, 4 bowlers and 1 wicket keeper.

4.7 SOME IMPORTANT RESULTS

In this section, we will discuss some examples based on the following two important results.

Result I Total number of permutations of n distinct things taken r at a time such that

- (i) s ($0 < s < r$) particular things are always included = ${}^{n-s}P_{r-s} \times {}^rP_s$
- (ii) s ($0 < s < r$) particular things are always excluded = ${}^{n-s}P_r$

Result II Total number of combination of n distinct things taken r at a time such that

(i) s ($0 < s < r$) particular things are always included $= {}^{n-s}C_{r-s}$

(ii) s ($0 < s < r$) particular things are always excluded $= {}^{n-s}C_r$

Example 19: Find the total number of ways of selection of 15 players out of 21 players such that

(i) 3 particular players are always included

(ii) 2 particular players are always excluded.

Solution:

(i) 3 particular players are always included $\Rightarrow$ we have to select $15 - 3 = 12$ players out of $21 - 3 = 18$ players.

$$\begin{aligned}\therefore \text{required number of ways} &= {}^{18}C_{12} = \frac{18!}{12!(18-12)!} \\ &= \frac{18 \times 17 \times 16 \times 15 \times 14 \times 13 \times 12!}{12! \times 6!} \\ &= \frac{18 \times 17 \times 16 \times 15 \times 14 \times 13}{6 \times 5 \times 4 \times 3 \times 2 \times 1} = 18564\end{aligned}$$

Alternatively

Here $n = 21$, $r = 15$, $s = 3$

$$\begin{aligned}\therefore \text{required number of selections} &= {}^{n-s}C_{r-s} \quad [\text{Refer part (i) of Result II}] \\ &= {}^{21-3}C_{15-3} \\ &= {}^{18}C_{12} = 18564 \quad [\text{Already calculated}]\end{aligned}$$

(ii) 2 particular players are always excluded $\Rightarrow$ we have to select 15 players out of $21 - 2 = 19$ players

$$\begin{aligned}\therefore \text{required number of ways} &= {}^{19}C_{15} = \frac{19!}{15! \times (19-15)!} \\ &= \frac{19 \times 18 \times 17 \times 16 \times 15!}{15! \times 4!} = \frac{19 \times 18 \times 17 \times 16}{4 \times 3 \times 2 \times 1} = 3876\end{aligned}$$

Alternatively

Here $n = 21$, $r = 15$, $s = 2$

$$\begin{aligned}\therefore \text{required number of selection} &= {}^{n-s}C_r \quad [\text{Refer part (ii) of Result II}] \\ &= {}^{21-2}C_{15} \\ &= {}^{19}C_{15} = 3876 \quad [\text{Already calculated}]\end{aligned}$$

Example 20: How many 5 letters words are possible using 8 letters a, b, c, d, e, f, g, h such that

(i) Two letters a, b are always included

(ii) Three letters a, c, d are always excluded

Solution: Here concept of permutation will be used, because we have to form arrangement not groups.

(i) Here $n = 8$, $r = 5$, $s = 2$

$$\therefore \text{required number of words} = {}^{n-s}P_{r-s} \times {}^rP_s \quad [\text{Refer part (i) of Result I}]$$

$$= {}^{8-2}P_{5-2} \times {}^5P_2$$

$$= {}^6P_3 \times {}^5P_2 = 120 \times 20 = 2400$$

(ii) Here $n = 8$, $r = 5$, $s = 3$

$$\therefore \text{required number of words} = {}^{n-s}P_r \quad [\text{Refer part (ii) of Result I}]$$

$$= {}^{8-3}P_5 = {}^5P_5 = 120$$

Now, you can try the following exercises.

E 14) Find the total number of ways of selection of 11 players out of 15 players such that

- (i) captain and vice captain are always included
- (ii) one particular injured players is always excluded.

E 15) How many 4 digits numbers are possible using 9 digits 1, 2, 3, ..., 9 such that

- (i) Three digits 1, 6, 8 are always included
 - (ii) Two digits 3, 8 are always excluded.
-

4.8 BINOMIAL THEOREM

Following two sub-sections are devoted to the discussion of this theorem.

4.8.1 Binomial Theorem for Positive Integral Index

From your school days, you are familiar what we mean by monomial, binomial, trinomial and multinomial expressions. Let us recall your memory. An expression having one term, two terms, three terms, more than three terms is known as monomial, binomial, trinomial, multinomial respectively.

For example,

- (i) $7, x, 9x, 3x^2, 5y, x^2y$ all are monomial, as there is only one term in each expression.
- (ii) $a + b, a - b, 3a + 2b, a^2 + 3b, x - y, x - 4y$ all are binomial, as there are only two terms in each expression.
- (iii) $a + b + c, x - 2y + z, 3x + 2y - z$ all are trinomial, as there are only three terms in each expression.
- (iv) $a + b + c + d, x - 2y + 5z - w + 3u$ both are multinomial, as there are more than three terms in each expression.

Let us recall another memory of your school days. You have met with the identities:

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

In school days students cram these identities. But here you have become familiar with the concept of combination in Sec. 4.5, using the knowledge of Sec. 4.5 these identities can be written in a systematic manner. In fact our aim of this section is to obtain the expression for $(a + b)^n$, $n = 1, 2, 3, \dots$ known as binomial theorem (for positive integral index).

Let us see how this very interesting expression can be generated by using the knowledge what we know up to this point.

Obviously $(a+b)^1 = a+b = {}^1C_0 a^{1-0} b^0 + {}^1C_1 a^{1-1} b^1$ [$\therefore {}^1C_0 = 1, {}^1C_1 = 1$]

$$(a+b)^2 = a^2 + 2ab + b^2 = {}^2C_0 a^{2-0} b^0 + {}^2C_1 a^{2-1} b^1 + {}^2C_2 a^{2-2} b^2$$

$$[\therefore {}^2C_0 = 1, {}^2C_1 = 2, {}^2C_2 = 1]$$

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$= {}^3C_0 a^{3-0} b^0 + {}^3C_1 a^{3-1} b^1 + {}^3C_2 a^{3-2} b^2 + {}^3C_3 a^{3-3} b^3$$

$$[\therefore {}^3C_0 = 1 = {}^3C_3, {}^3C_1 = 3 = {}^3C_2]$$

$$(a+b)^4 = (a+b)(a+b)^3$$

$$= (a+b)(a^3 + 3a^2b + 3ab^2 + b^3)$$

$$= a^4 + 3a^3b + 3a^2b^2 + ab^3 + a^3b + 3a^2b^2 + 3ab^3 + b^4$$

$$= a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

$$= {}^4C_0 a^{4-0} b^0 + {}^4C_1 a^{4-1} b^1 + {}^4C_2 a^{4-2} b^2 + {}^4C_3 a^{4-3} b^3 + {}^4C_4 a^{4-4} b^4$$

$$[\therefore {}^4C_0 = {}^4C_4 = 1, {}^4C_1 = {}^4C_3 = 4, {}^4C_2 = 6]$$

...

...

...

$$(a+b)^n = {}^nC_0 a^{n-0} b^0 + {}^nC_1 a^{n-1} b^1 + {}^nC_2 a^{n-2} b^2 + {}^nC_3 a^{n-3} b^3 + \dots$$

$$+ {}^nC_{n-1} a^{n-(n-1)} b^{n-1} + {}^nC_n a^{n-n} b^n$$

$$= {}^nC_0 a^n + {}^nC_1 a^{n-1} b + {}^nC_2 a^{n-2} b^2 + {}^nC_3 a^{n-3} b^3 + \dots$$

$$+ {}^nC_{n-1} ab^{n-1} + {}^nC_n b^n$$

Some important points related to the above expression which will help you to easily remember it are given below.

1. ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ are known as binomial coefficients.
2. Exponents of a in successive terms are $n, n-1, n-2, n-3, \dots, 1, 0$, i.e. difference of super and sub subscript of C , i.e. exponent of a in the term with binomial coefficient nC_r will be $n-r$.
3. Exponents of b in successive terms are $0, 1, 2, 3, \dots, n-1, n$, i.e. equal to the sub subscript of C , i.e. exponent of b in the term with binomial coefficient nC_r will be r .
4. Sum of the exponents of a and b in each term is equal to the actual exponent of the given binomial expression.
5. If $n = 0$, then $(a+b)^0 = 1 = {}^0C_0 a^{0-0} b^0$ as ${}^0C_0 = 1$

To become user friendly with this expression, let us do some examples based on it.

Example 21: Expand $\left(x + \frac{1}{x}\right)^6$ by binomial theorem.

Solution: Comparing $\left(x + \frac{1}{x}\right)^6$ with $(a+b)^n$, we get

$$a = x, b = \frac{1}{x}, n = 6$$

$\therefore$ by binomial theorem

$$\begin{aligned}
 \left(x + \frac{1}{x}\right)^6 &= {}^6C_0 x^6 \left(\frac{1}{x}\right)^0 + {}^6C_1 x^5 \left(\frac{1}{x}\right)^1 + {}^6C_2 x^4 \left(\frac{1}{x}\right)^2 + {}^6C_3 x^3 \left(\frac{1}{x}\right)^3 \\
 &\quad + {}^6C_4 x^2 \left(\frac{1}{x}\right)^4 + {}^6C_5 x \left(\frac{1}{x}\right)^5 + {}^6C_6 \left(\frac{1}{x}\right)^6 \\
 &= x^6 + 6x^4 + 15x^2 + 20 + \frac{15}{x^2} + \frac{6}{x^4} + \frac{1}{x^6} \\
 &[\because {}^6C_0 = {}^6C_6 = 1, {}^6C_1 = {}^6C_5 = 6, {}^6C_2 = {}^6C_4 = 15, {}^6C_3 = 20]
 \end{aligned}$$

Example 22: Expand $(1-x)^{10}$ by binomial theorem.

Solution: Comparing $(1-x)^{10}$ with $(a+b)^n$, we get

$$a = 1, b = -x, n = 10$$

$\therefore$ by binomial theorem

$$\begin{aligned}
 (1-x)^{10} &= (1+(-x))^{10} \\
 &= {}^{10}C_0 (1)^{10} (-x)^0 + {}^{10}C_1 (1)^9 (-x)^1 + {}^{10}C_2 (1)^8 (-x)^2 + {}^{10}C_3 (1)^7 (-x)^3 \\
 &\quad + {}^{10}C_4 (1)^6 (-x)^4 + {}^{10}C_5 (1)^5 (-x)^5 + {}^{10}C_6 (1)^4 (-x)^6 + {}^{10}C_7 (1)^3 (-x)^7 \\
 &\quad + {}^{10}C_8 (1)^2 (-x)^8 + {}^{10}C_9 (1)^1 (-x)^9 + {}^{10}C_{10} (1)^0 (-x)^{10} \\
 &= 1 - 10x + 45x^2 - 120x^3 + 210x^4 - 252x^5 \\
 &\quad + 210x^6 - 120x^7 + 45x^8 - 10x^9 + x^{10} \\
 &\quad \left[\because {}^{10}C_0 = {}^{10}C_{10} = 1, {}^{10}C_1 = {}^{10}C_9 = 10, {}^{10}C_2 = {}^{10}C_8 = 45, \right. \\
 &\quad \left. {}^{10}C_3 = {}^{10}C_7 = 120, {}^{10}C_4 = {}^{10}C_6 = 210, {}^{10}C_5 = 252 \right]
 \end{aligned}$$

Now you can try the following exercises.

E 16) Expand $(1+\sqrt{2})^5$ by binomial theorem.

E 17) Expand $(3x-y)^7$ by binomial theorem.

4.8.2 Binomial Theorem for any Index

In subsection 4.8.1 we have discussed binomial theorem for index n , where $n = 1, 2, 3, 4, \dots$

But sometimes binomial expansion is needed for rational exponent. The aim of this section is to provide an expression which works for rational exponent.

Let us consider the expansion for positive integral index discussed in previous subsection 4.8.1

$$\begin{aligned}
 (a+b)^n &= {}^nC_0 a^n + {}^nC_1 a^{n-1} b + {}^nC_2 a^{n-2} b^2 + {}^nC_3 a^{n-3} b^3 + \dots + {}^nC_{n-1} a b^{n-1} \\
 &\quad + {}^nC_n b^n
 \end{aligned}$$

In Sec. 4.5, you have seen that ${}^nC_0, {}^nC_1, {}^nC_2, {}^nC_3$, etc. can be written as

$${}^nC_0 = {}^nC_n = 1, {}^nC_1 = {}^nC_{n-1} = n, {}^nC_2 = {}^nC_{n-2} = \frac{n(n-1)}{2!},$$

$${}^nC_3 = {}^nC_{n-3} = \frac{n(n-1)(n-2)}{3!}, \text{ etc.}$$

$$\begin{aligned}
 \therefore (a+b)^n &= a^n + na^{n-1}b + \frac{n(n-1)}{2!} a^{n-2}b^2 + \frac{n(n-1)(n-2)}{3!} a^{n-3}b^3 + \dots + nab^{n-1} \\
 &\quad + b^n
 \end{aligned}$$

Here we note that if n is not a positive integer, then this expression will never terminate. Also powers of b are increasing term by term, so in case of infinite expansion to get finite sum it becomes necessary that $|b| < 1$. In fact, in case of rational exponent n the binomial expansion of $(1+x)^n$, where $|x| < 1$ is given by

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots + \frac{n(n-1)(n-2)\dots(n-(r-1))}{r!}x^r + \dots$$

and binomial expansion of $(a+b)^n$ is given by

$$\begin{aligned}(a+b)^n &= a^n \left(1 + \frac{b}{a}\right)^n \\ &= a^n \left[1 + n\frac{b}{a} + \frac{n(n-1)}{2!}\left(\frac{b}{a}\right)^2 + \frac{n(n-1)(n-2)}{3!}\left(\frac{b}{a}\right)^3 + \dots\right], \text{ if } \left|\frac{b}{a}\right| < 1 \text{ and} \\ (a+b)^n &= b^n \left(1 + \frac{a}{b}\right)^n \\ &= b^n \left[1 + n\frac{a}{b} + \frac{n(n-1)}{2!}\left(\frac{a}{b}\right)^2 + \frac{n(n-1)(n-2)}{3!}\left(\frac{a}{b}\right)^3 + \dots\right], \text{ if } \left|\frac{a}{b}\right| < 1\end{aligned}$$

Let us do some examples based on it.

Example 23: Expand $(5+3x)^{-4}$ using binomial theorem for negative index.

$$\begin{aligned}\text{Solution: } (5+3x)^{-4} &= (5)^{-4} \left(1 + \frac{3}{5}x\right)^{-4} \\ &= \frac{1}{5^4} \left[1 + (-4)\left(\frac{3}{5}x\right) + \frac{(-4)(-4-1)}{2!}\left(\frac{3}{5}x\right)^2 + \frac{(-4)(-4-1)(-4-2)}{3!}\left(\frac{3}{5}x\right)^3 + \dots\right]\end{aligned}$$

This expansion is valid if $\left|\frac{3}{5}x\right| < 1$, i.e. $|x| < \frac{5}{3}$

$$\therefore (5+3x)^{-4} = \frac{1}{625} \left[1 - \frac{12}{5}x + \frac{18}{5}x^2 - \frac{108}{25}x^3 + \dots\right], \text{ if } |x| < \frac{5}{3}$$

Example 24: Find expansion of $(7-2x)^{2/3}$.

$$\begin{aligned}\text{Solution: } (7-2x)^{2/3} &= 7^{2/3} \left(1 - \frac{2}{7}x\right)^{2/3} \\ &= 7^{2/3} \left[1 + \left(\frac{2}{3}\right)\left(-\frac{2}{7}x\right) + \frac{\frac{2}{3}\left(\frac{2}{3}-1\right)}{2!}\left(-\frac{2}{7}x\right)^2 + \dots\right]\end{aligned}$$

This expansion is valid only if $\left|\frac{2}{7}x\right| < 1$, i.e. if $|x| < \frac{7}{2}$

$$\therefore (7-2x)^{2/3} = 7^{2/3} \left[1 - \frac{4}{21}x - \frac{4}{441}x^2 + \dots\right], \text{ if } |x| < \frac{7}{2}$$

Now, you can try the following exercise.

E 18) Expand $(1 - 3x)^{1/2}$.

4.9 SUMMARY

Let us summarise the topics that we have covered in this unit:

- 1) Concept of factorial.
- 2) Fundamental principles of multiplication and addition.
- 3) Definition and examples of permutation.
- 4) Permutation in different situations.
- 5) Circular permutation.
- 6) Definition and examples of combination.
- 7) Binomial theorem for integral index and for any index.

4.10 SOLUTIONS/ANSWERS

E 1) (i) $\frac{22!}{19!} = \frac{22 \times 21 \times 20 \times 19!}{19!} = 22 \times 21 \times 20 = 9240$

(ii) $\frac{15!}{10! \times 5!} = \frac{15 \times 14 \times 13 \times 12 \times 11 \times 10!}{10! \times 5 \times 4 \times 3 \times 2 \times 1} = \frac{15 \times 14 \times 13 \times 12 \times 11}{120} = 3003$

E 2) (i) $3.6.9.12.15 = 3^5 (1.2.3.4.5) = 3^5 (5!)$

(ii) $7.8.9.10.11.12 = \frac{1.2.3.4.5.6.7.8.9.10.11.12}{1.2.3.4.5.6} = \frac{12!}{6!}$

E 3) (i) $(n - 2)! = 12(n - 4)!$

$\Rightarrow (n - 2)(n - 3)(n - 4)! = 12(n - 4)!$

$\Rightarrow (n - 2)(n - 3) = 12 \Rightarrow n^2 - 5n + 6 - 12 = 0$

$\Rightarrow n^2 - 5n - 6 = 0 \Rightarrow n^2 - 6n + n - 6 = 0$

$\Rightarrow n(n - 6) + 1(n - 6) = 0 \Rightarrow (n - 6)(n + 1) = 0$

$\Rightarrow n = 6, -1$

But n cannot be negative

$\therefore n = 6$

(ii) $n! = 72(n - 2)!$

$\Rightarrow n(n - 1)(n - 2)! = 72(n - 2)! \Rightarrow n(n - 1) = 72$

$\Rightarrow n^2 - n - 72 = 0 \Rightarrow n^2 - 9n + 8n - 72 = 0$

$\Rightarrow n(n - 9) + 8(n - 9) = 0 \Rightarrow (n - 9)(n + 8) = 0$

$\Rightarrow n = 9, -8$

But n cannot be negative

$\therefore n = 9$

E 4) In order to solve this problem, we have to perform 10 jobs, where each of first five jobs can be done in 4 ways and each of last five jobs can be done in 5 ways.

$\therefore$ by fundamental principle of multiplication required possible sequences of answers

$= 4 \times 4 \times 4 \times 4 \times 4 \times 5 \times 5 \times 5 \times 5 \times 5 = 4^5 \times 5^5 = 3200000$

E 5) (i) When repetition is not allowed. First, second, third and fourth positions can be filled up in 5, 4, 3, 2 ways respectively.

$\therefore$ total number of 4-letter words that can be formed by using the letters a, b, g, h, k
 $= 5 \times 4 \times 3 \times 2 = 120$

(ii) When repetition is allowed. Each of first, second, third and fourth positions can be filled up in 5 ways.

$\therefore$ total number of 4-letter words that can be formed by using the letters a, b, g, h, k
 $= 5 \times 5 \times 5 \times 5 = 5^4 = 625$

$$\begin{aligned} \mathbf{E\ 6)} \quad {}^5P_n &= {}^6P_{n-1} \Rightarrow \frac{5!}{(5-n)!} = \frac{6!}{(6-n+1)!} \\ &\Rightarrow \frac{5!}{(5-n)!} = \frac{6 \cdot 5!}{(7-n)!} \Rightarrow \frac{1}{(5-n)!} = \frac{6}{(7-n) \cdot (6-n) \cdot (5-n)!} \\ &\Rightarrow (7-n)(6-n) = 6 \Rightarrow 42 - 13n + n^2 = 6 \Rightarrow n^2 - 13n + 36 = 0 \\ &\Rightarrow n^2 - 9n - 4n + 36 = 0 \Rightarrow n(n-9) - 4(n-9) = 0 \\ &\Rightarrow (n-9)(n-4) = 0 \Rightarrow n = 9, 4 \end{aligned}$$

But if $n = 9$, then ${}^5P_n = {}^5P_9$ becomes meaning less

$\left[\begin{array}{l} \because \text{selection of 9 things out of 5} \\ \text{does not make any sense.} \end{array} \right]$

$\therefore n = 4$

$$\begin{aligned} \mathbf{E\ 7) (i)} \quad {}^7P_4 &= \frac{7!}{(7-4)!} \left[\because {}^nP_r = \frac{n!}{(n-r)!} \right] \\ &= \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3!}{3!} = 7 \cdot 6 \cdot 5 \cdot 4 = 840 \end{aligned}$$

$$(ii) \quad {}^nP_n = \frac{n!}{(n-n)!} = \frac{n!}{0!} = \frac{n!}{1} = n! \quad \text{as } 0! = 1$$

$$(iii) \quad {}^nP_{n-1} = \frac{n!}{(n-(n-1))!} = \frac{n!}{1!} = \frac{n!}{1} = n!$$

$$(iv) \quad {}^nP_1 = \frac{n!}{(n-1)!} = \frac{n \cdot (n-1)!}{(n-1)!} = n$$

$$(v) \quad {}^nP_2 = \frac{n!}{(n-2)!} = \frac{n(n-1)(n-2)!}{(n-2)!} = n(n-1)$$

$$(vi) \quad {}^{16}P_3 = \frac{16!}{(16-3)!} = \frac{16!}{(16-3)!} = \frac{16!}{13!} = \frac{16 \cdot 15 \cdot 14 \cdot 13!}{13!} = 16 \cdot 15 \cdot 14 = 3360$$

E 8) Possible number of ways = Total number of arrangement of 5 things taken all at a time

$$= {}^5P_5 = \frac{5!}{(5-5)!} = \frac{5!}{0!} = \frac{5!}{1} = 120$$

E 9) Total number of flags = $3 + 4 + 2 = 9$

Out of which 3 are of one kind, 4 are of second kind and 2 are of

third kind.

$$\therefore \text{required number of signals} = \frac{9!}{4! 3! 2!} = \frac{9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4!}{4! \times 6 \times 2} = \frac{9 \cdot 8 \cdot 7 \cdot 6 \cdot 5}{12} = 1260$$

E 10) There are 4 letters in the word “AMAR”. Out of which A occur twice.

$$\therefore \text{total number of permutations} = \frac{4!}{2!} = \frac{4 \times 3 \times 2!}{2!} = 4 \times 3 = 12$$

E 11) (i) First prize can be given to any of the 5 students. Second prize can be given any of the remaining 4 students and similarly third prize

can be given in 3 ways. $\left[\begin{array}{l} \because \text{no student gets} \\ \text{more than one prize} \end{array} \right]$

$$\therefore \text{required number of ways} = 5 \times 4 \times 3 = 60$$

(ii) First prize can be given to any of the 5 students, i.e in 5 ways. Second and third each can also be given in 5 ways.

$\left[\begin{array}{l} \because \text{a student may get any} \\ \text{number of prizes} \end{array} \right]$

$$\therefore \text{required number of ways} = 5 \times 5 \times 5 = 5^3 = 125$$

(iii) There are 5 ways that all the prizes come to the same student.

$$\therefore \text{required number of ways} = 125 - 5 = 120$$

E 12) (i) Possible number of garlands with 8 flowers $= \frac{(8-1)!}{2} = \frac{7!}{2}$
 $= \frac{5040}{2} = 2520$

(ii) Let P and V denote president and vice president respectively. Therefore if we consider these two members as a single member then we are left with 19 members. These 19 members can sit in a round table in $(19-1)!$ ways. But president and vice president can change their seats in two ways (i.e. PV or VP).

$$\therefore \text{required number of ways of sitting the members in a meeting} = (19-1)! \times 2! = (18!) (2!)$$

E 13)

11 Batsmen
7 Bowlers
3 Wicket keepers.

Out of 11 batsmen 6 can be selected in ${}^{11}C_6$ ways.

Out of 7 bowlers 4 can be selected in 7C_4 ways.

Out of 3 wicket keepers 1 can be selected in 3C_1 ways.

$$\begin{aligned} \therefore \text{required number of ways} &= {}^{11}C_6 \times {}^7C_4 \times {}^3C_1 \\ &= \frac{11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6}{6!} \times \frac{7 \cdot 6 \cdot 5 \cdot 4}{4!} \times 3 \\ &= 48510 \end{aligned}$$

E 14) Here concept of combination will be used, because we have to form possible groups not arrangement.

(i) Here $n = 15, r = 11, s = 2$

$$\begin{aligned}\therefore \text{required number of ways} &= {}^{n-s}C_{r-s} \quad [\text{Refer part (i) of Result II}] \\ &= {}^{15-2}C_{11-2} \\ &= {}^{13}C_9 = \frac{13 \times 12 \times 11 \times 10}{4!} = 715\end{aligned}$$

(ii) Here $n = 15, r = 11, s = 1$

$$\begin{aligned}\therefore \text{required number of ways} &= {}^{n-s}C_r \quad [\text{Refer (ii) of Result II}] \\ &= {}^{15-1}C_{11} = {}^{14}C_{11} = \frac{14 \times 13 \times 12}{3!} = 364\end{aligned}$$

E 15) (i) Here $n = 9, r = 4, s = 3$

$$\begin{aligned}\therefore \text{possible 4 digits numbers that can be formed using 9 digits 1 to 9} \\ \text{subject to the condition that three digits 1, 6, 8 are always included} \\ = {}^{n-s}P_{r-s} \times {}^rP_s = {}^{9-3}P_{4-3} \times {}^4P_3 = {}^6P_1 \times {}^4P_3 = 6 \times 4 = 24\end{aligned}$$

(ii) Here $n = 9, r = 4, s = 2$

$$\begin{aligned}\therefore \text{required possible numbers} &= {}^{n-s}P_r = {}^{9-2}P_4 = {}^7P_4 = \frac{7!}{(7-4)!} \\ &= \frac{7 \times 6 \times 5 \times 4 \times 3!}{3!} = 840\end{aligned}$$

E 16) Comparing $(1 + \sqrt{2})^5$ with $(a + b)^n$, we get

$$a = 1, b = \sqrt{2}, n = 5$$

$\therefore$ by binomial theorem

$$\begin{aligned}(1 + \sqrt{2})^5 &= {}^5C_0 (1)^{5-0} (\sqrt{2})^0 + {}^5C_1 (1)^{5-1} (\sqrt{2})^1 + {}^5C_2 (1)^{5-2} (\sqrt{2})^2 \\ &\quad + {}^5C_3 (1)^{5-3} (\sqrt{2})^3 + {}^5C_4 (1)^{5-4} (\sqrt{2})^4 + {}^5C_5 (1)^{5-5} (\sqrt{2})^5 \\ &= 1 + 5(\sqrt{2}) + 10(2) + 10(2\sqrt{2}) + 5(4) + 1(4\sqrt{2}) \\ &\quad \left[\because {}^5C_0 = {}^5C_5 = 1, {}^5C_1 = {}^5C_4 = 5, {}^5C_2 = {}^5C_3 = 10 \right] \\ &= 1 + 5\sqrt{2} + 20 + 20\sqrt{2} + 20 + 4\sqrt{2} \\ &= 41 + 29\sqrt{2}\end{aligned}$$

E 17) Comparing $(3x - y)^7$ with $(a + b)^n$, we get

$$a = 3x, b = -y, n = 7$$

$\therefore$ by binomial theorem

$$\begin{aligned}(3x - y)^7 &= {}^7C_0 (3x)^{7-0} (-y)^0 + {}^7C_1 (3x)^{7-1} (-y)^1 + {}^7C_2 (3x)^{7-2} (-y)^2 \\ &\quad + {}^7C_3 (3x)^{7-3} (-y)^3 + {}^7C_4 (3x)^{7-4} (-y)^4 + {}^7C_5 (3x)^{7-5} (-y)^5 \\ &\quad + {}^7C_6 (3x)^{7-6} (-y)^6 + {}^7C_7 (3x)^{7-7} (-y)^7 \\ &= 2187x^7 + 7(729x^6)(-y) + 21(243x^5)(y^2) + 35(81x^4)(-y^3) \\ &\quad + 35(27x^3)(y^4) + 21(9x^2)(-y^5) + 7(3x)(y^6) - y^7 \\ &\quad \left[\because {}^7C_0 = {}^7C_7 = 1, {}^7C_1 = {}^7C_6 = 7, {}^7C_2 = {}^7C_5 = 21, {}^7C_3 = {}^7C_4 = 35 \right] \\ &= 2187x^7 - 5103x^6y + 5103x^5y^2 - 2835x^4y^3 + 945x^3y^4 - 189x^2y^5 \\ &\quad + 21xy^6 - y^7\end{aligned}$$

$$\begin{aligned} \mathbf{E\ 18)} \quad (1-3x)^{1/2} = & 1 + \left(\frac{1}{2}\right)(-3x) + \frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)}{2!}(-3x)^2 \\ & + \frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)}{3!}(-3x)^3 + \dots \end{aligned}$$

This expansion is valid only if $|3x| < 1$, i.e. if $|x| < \frac{1}{3}$

$$\therefore (1-3x)^{1/2} = 1 - \frac{3}{2}x - \frac{9}{8}x^2 - \frac{27}{16}x^3 + \dots, \text{ if } |x| < \frac{1}{3}$$

MST-001
Foundation in
Mathematics and
Statistics

Block

2**FUNDAMENTALS OF MATHEMATICS-II**

UNIT 5

Limit and Continuity	5
-----------------------------	----------

UNIT 6

Differentiation	29
------------------------	-----------

UNIT 7

Indefinite Integration	59
-------------------------------	-----------

UNIT 8

Definite Integration	81
-----------------------------	-----------

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BLOCK 2 FUNDAMENTALS OF MATHEMATICS-II

This is the second block of the course MST-001. The aim of this block is to provide sufficient material which will be needed in order to study course MST-003 and some sections of other courses of the programme.

Using the knowledge provided by the previous block of this course. The follow of the block is maintained by the following four units.

Unit 5: Limit and Continuity

In this unit concept of limit, evaluation of certain limits using factorisation, L.C.M., rationalisation and some standard rules have been discussed. Concept of left hand, right hand limits and infinite limit have been also introduced. The unit ends with the brief introduction of continuity.

Unit 6: Differentiation

This unit discusses a very important branch of calculus known as differentiation. In this unit, you will learn how differentiations of some commonly used functions are evaluated. Differentiations of functions using product rule, quotient rule and chain rule have been also discussed in this unit. Differentiation of parametric and implicit functions also takes place in the unit. Unit ends by giving a brief induction of higher order derivatives and maxima and minimum of functions.

Unit 7: Indefinite Integration

Another important branch of calculus known as integration is discussed in this unit. It discusses indefinite integral of some commonly used functions. It also discusses how we can evaluate an integral by using substitution method, partial fractions and integration by parts.

Unit 8: Definite Integration

This unit starts with the geometrical interpretation of the definite integral. Definite integral of some commonly used functions and properties of definite integral also have been discussed. Some examples based on first kind of improper integral also have been evaluated.

Notations and Symbols

$x \rightarrow a$: x approaches to a

L. H.S. : left hand limit

R.H.S. : right hand limit

∞ : infinity

$|x|$: modules of x or absolute value of x

+ ve : positive

– ve : negative

$\int$: sign of integration

$\int_a^b$: definite integral within limits a to b

UNIT 5 LIMIT AND CONTINUITY

Structure

- 5.1 Introduction
 - Objectives
- 5.2 Concept of Limit
- 5.3 Direct Substitution Method
- 5.4 Failure of Direct Substitution Method
- 5.5 Concept of Infinite Limit
- 5.6 Concept of Left Hand and Right Hand Limits
- 5.7 Continuity of a Function at a Point
- 5.8 Continuous Function
- 5.9 Summary
- 5.10 Solutions/Answers

5.1 INTRODUCTION

In Unit 2 of this course, i.e. MST-001 we have discussed, in detail, the concept of functions and various types of functions. In that unit we have also obtained the value of the function at certain points. That is, the value of a function $f(x)$ has been obtained at certain value of x in its domain.

Here, in this unit, we are going to introduce the concept of limit as well as continuity. That is, we are going to find the limiting value of the function $f(x)$ when x approaches to certain value. That is, we are interested in finding that value to which $f(x)$ approaches to as x approaches to the certain value. Also, this limiting value and the value of the function at certain value of x are compared to define continuity.

Objectives

After completing this unit, you should be able to:

- get an idea of limit;
- evaluate the limits of different functions;
- evaluate the infinite limit of some functions;
- check the continuity of a function at a point; and
- check the continuity of a function at a general point.

5.2 CONCEPT OF LIMIT

In Unit 2 of this course, we have discussed concept of function, consider a function

$$y = f(x) = 3x + 2$$

The following table shows the values of y for different values of x which are very close to 2.

x	1.9	1.98	1.998	1.9998	...	1.99999998	...
$y = f(x)$	7.7	7.94	7.994	7.9994	...	7.99999994	...

x	2.1	2.01	2.001	2.0001	...	2.00000001	...
y = f(x)	8.3	8.03	8.003	8.0003	...	8.00000003	...

We note that as x approaches to 2 either from left (means x comes nearer and nearer to 2 but remains < 2) or from right (means x comes nearer and nearer to 2 but remains > 2), then $y = f(x)$ approaches to 8 in the same manner.

i.e. as $x \rightarrow 2$ then $f(x) \rightarrow 8$ and we write it as $\lim_{x \rightarrow 2} f(x) = 8$.

In general if $f(x) \rightarrow l$ as $x \rightarrow a$ then we write it as $\lim_{x \rightarrow a} f(x) = l$.

In this unit, we discuss how to evaluate $\lim_{x \rightarrow a} f(x)$ in different situations. In this unit we shall also discuss the concept of infinite limit, some standard limits, left hand limit (L.H.L.) and right hand limit (R.H.L.). Finally we shall conclude the unit after introducing the concept of continuity.

5.3 DIRECT SUBSTITUTION METHOD

Suppose we want to evaluate $\lim_{x \rightarrow a} f(x)$. This method is applied when limiting

value of $f(x)$ remains same irrespective of this fact whether x approaches to a from left hand side (L.H.S.) or right hand side (R.H.S.). As the name of this method itself suggests, in this method we directly substitute a in place of x.

Before we take some examples based on the direct substitution method we list some results (algebra of limits) without proof.

If f and g are real valued functions (real valued function means range of the function is subset of R, set of real numbers) defined on the domain D such that $\lim_{x \rightarrow a} f(x)$, $\lim_{x \rightarrow a} g(x)$ both exist, then the following results hold good.

- $\lim_{x \rightarrow a} (f(x) + g(x)) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$
- $\lim_{x \rightarrow a} (f(x) - g(x)) = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$
- $\lim_{x \rightarrow a} f(x)g(x) = \left(\lim_{x \rightarrow a} f(x) \right) \left(\lim_{x \rightarrow a} g(x) \right)$
- $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)},$ provided $\lim_{x \rightarrow a} g(x) \neq 0$
- $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}$
- $\lim_{x \rightarrow a} \frac{1}{g(x)} = \frac{1}{\lim_{x \rightarrow a} g(x)}$
- $\lim_{x \rightarrow a} \log f(x) = \log \left(\lim_{x \rightarrow a} f(x) \right)$
- $\lim_{x \rightarrow a} e^{f(x)} = e^{\lim_{x \rightarrow a} f(x)}$
- $\lim_{x \rightarrow a} f(x)^{g(x)} = \left(\lim_{x \rightarrow a} f(x) \right)^{\lim_{x \rightarrow a} g(x)}$
- $\lim_{x \rightarrow a} kf(x) = k \lim_{x \rightarrow a} f(x),$ where k is a constant
- $\lim_{x \rightarrow a} k = k,$ where k is a constant

Remark 1:

(i) These results are used so frequently; that we have no need to mention these results each time.

(ii) Hereafter, we will use D.S.M. for Direct Substitution Method.

Now we are in position to discuss some examples based on (D.S.M.).

Example 1: Evaluate the following limits:

$$(i) \lim_{x \rightarrow 3} (x^2 - 2x + 4) \quad (ii) \lim_{x \rightarrow 2} x(x^2 - 4) \quad (iii) \lim_{x \rightarrow -1} (1 + x + x^2 + x^3 + \dots + x^{100})$$

$$(iv) \lim_{x \rightarrow \sqrt{2}} \frac{x^2 + 3}{3 - x^4} \quad (v) \lim_{x \rightarrow 3} \sqrt{25 - x^2}$$

$$(vi) \lim_{x \rightarrow 0} f(x), \text{ where } f(x) = \begin{cases} 2 - x^2, & x \neq 0 \\ 2, & x = 0 \end{cases}$$

Solution:

$$(i) \lim_{x \rightarrow 3} (x^2 - 2x + 4) = (3)^2 - 2 \times 3 + 4 \quad [\text{By D. S. M.}]$$

$$= 9 - 6 + 4 = 7$$

$$(ii) \lim_{x \rightarrow 2} x(x^2 - 4) = 2[(2)^2 - 4] = 2(4 - 4) = 0$$

$$(iii) \lim_{x \rightarrow -1} (1 + x + x^2 + x^3 + \dots + x^{100}) = 1 + (-1)^1 + (-1)^2 + (-1)^3 + \dots + (-1)^{100}$$

$$= 1 + (-1) + (1) + (-1) + (1) + \dots + (-1) + (1)$$

$$= 1 + (50 \text{ terms each containing } 1)$$

$$+ (50 \text{ terms each containing } (-1))$$

$$= 1 + 50 + (-50) = 1$$

$$(iv) \lim_{x \rightarrow \sqrt{2}} \frac{x^3 + 3}{3 - x^4} = \frac{\lim_{x \rightarrow \sqrt{2}} (x^3 + 3)}{\lim_{x \rightarrow \sqrt{2}} (3 - x^4)} = \frac{(\sqrt{2})^3 + 3}{3 - (\sqrt{2})^4} = \frac{2\sqrt{2} + 3}{3 - 4} = -(2\sqrt{2} + 3)$$

$$(v) \lim_{x \rightarrow 3} \sqrt{25 - x^2} = \sqrt{\lim_{x \rightarrow 3} (25 - x^2)} = \sqrt{25 - (3)^2} = \sqrt{25 - 9} = \sqrt{16} = 4$$

$$(vi) f(x) = \begin{cases} 2 - x^2, & x \neq 0 \\ 2, & x = 0 \end{cases}$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} (2 - x^2) \quad [\because x \rightarrow 0 \Rightarrow x \neq 0, \text{ so } f(x) = 2 - x^2]$$

$$= 2 - (0)^2 = 2 - 0 = 2$$

Remark 2: Limit of polynomial functions is always evaluated by D.S.M.

Now, you can try the following exercise.

E 1) Evaluate the following limits:

$$(i) \lim_{x \rightarrow 2} (x^2 - 2x + 3)^{x^2+1} \quad (ii) \lim_{x \rightarrow 1} \log(x^4 + x^2 + 1) \quad (iii) \lim_{x \rightarrow 5} 3$$

$$(iv) \lim_{x \rightarrow 3} 4f(x), \quad \text{where } f(x) = (x - 5)^2$$

D.S.M. discussed above does not always work, in some situations it may fail. In next section we shall see when it fails and what are the alternate methods in such situations?

5.4 FAILURE OF DIRECT SUBSTITUTION METHOD

In mathematics following seven forms are known as indeterminate form, i.e. as such these forms are not defined.

$$(i) \frac{0}{0} \quad (ii) \frac{\infty}{\infty} \quad (iii) 0 \times \infty \quad (iv) \infty - \infty \quad (v) 0^0 \quad (vi) 1^\infty \quad (vii) \infty^0$$

So, if by direct substitution any of the above mentioned forms take place then D.S.M. fails and we need some alternate methods. Some of them are listed below:

I Factorisation Method

II Least Common Multiplier Method

III Rationalisation Method

IV Use of some Standard Results

Let us discuss these methods one by one:

5.4.1 Factorisation Method

This method is useful, when we get $\frac{0}{0}$ form by direct substitution in the given

expression of the type $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$. This will happen if $f(x)$ and $g(x)$ both becomes

zero on direct substitution.

$\Rightarrow$ both have at least one common factor $(x - a)$.

In this case express $f(x) = (x - a)$ (some factor) and $g(x) = (x - a)$ (some factor) either by long division method or by any other method known to you. Then cancel out the common factor and again try D.S.M. If D.S.M. works, we get the required limit.

If D.S.M. fails again, repeat the same procedure. Ultimately, after a finite number of steps, you will get the result as the numerator and dominator both are of finite degrees.

Let us explain the method with the help of the following example.

Example 2: Evaluate the following limits:

$$(i) \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

$$(ii) \lim_{x \rightarrow -1} \frac{x^3 + 1}{x + 1}$$

$$(iii) \lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - 5x + 4}$$

$$(iv) \lim_{x \rightarrow 3} \frac{x^3 - 3x^2 + x}{x^3 - 10x^2 + 27x - 18}$$

Remember, we cannot cancel 0 by 0. But here $x - 2$ is not equal to zero because x is approaching to 2 and not equal to 2 and hence $x - 2$ is approaching to zero and not equal to zero.

Solution:

$$(i) \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

$$\left[\frac{0}{0} \text{ form, so D.S.M. fails} \right]$$

Using factorisation method, we have

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{x - 2} \quad [\because a^2 - b^2 = (a - b)(a + b)]$$

$$= \lim_{x \rightarrow 2} (x + 2)$$

$$= 2 + 2 = 4$$

$$\left[\because x \rightarrow 2 \Rightarrow x - 2 \neq 0, \text{ so dividing numerator and denominator by } x - 2. \right]$$

[By D.S.M.]

$$(ii) \lim_{x \rightarrow -1} \frac{x^3 + 1}{x + 1} \quad \left[\frac{0}{0} \text{ form, so D.S.M. fails} \right]$$

Using factorisation method, we have

$$\begin{aligned} \lim_{x \rightarrow -1} \frac{x^3 + 1}{x + 1} &= \lim_{x \rightarrow -1} \frac{x^3 + 1^3}{x + 1} \\ &= \lim_{x \rightarrow -1} \frac{(x + 1)(x^2 - x + 1)}{x + 1} \quad [\because a^3 + b^3 = (a + b)(a^2 - ab + b^2)] \\ &= \lim_{x \rightarrow -1} (x^2 - x + 1) \quad \left[\because x \rightarrow -1 \Rightarrow x + 1 \neq 0, \right. \\ &\quad \left. \text{so dividing numerator and denominator by } x + 1. \right] \\ &= (-1)^2 - (-1) + 1 = 1 + 1 + 1 = 3 \end{aligned}$$

$$(iii) \lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - 5x + 4} \quad \left[\frac{0}{0} \text{ form, so D.S.M. fails} \right]$$

Using factorisation method, we have

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - 5x + 4} &= \lim_{x \rightarrow 1} \frac{x^2 + 2x - x - 2}{x^2 - 4x - x + 4} \\ &= \lim_{x \rightarrow 1} \frac{x(x + 2) - 1(x + 2)}{x(x - 4) - 1(x - 4)} = \lim_{x \rightarrow 1} \frac{(x + 2)(x - 1)}{(x - 4)(x - 1)} \\ &= \lim_{x \rightarrow 1} \frac{x + 2}{x - 4} \quad \left[\because x \rightarrow 1 \Rightarrow x - 1 \neq 0, \right. \\ &\quad \left. \text{so dividing numerator and denominator by } x - 1. \right] \\ &= \frac{1 + 2}{1 - 4} = \frac{3}{-3} = -1 \quad [\text{By D.S.M.}] \end{aligned}$$

$$(iv) \text{ Let } I = \lim_{x \rightarrow 3} \frac{x^3 - 3x^2 + x}{x^3 - 10x^2 + 27x - 18} \quad \left[\frac{0}{0} \text{ form, so D.S.M. fails} \right]$$

As on putting $x = 3$, the numerator and as well denominator both becomes zero, therefore $x - 3$ is a factor of $x^3 - 3x^2 + x$ as well as of $x^3 - 10x^2 + 27x - 18$. Dividing $x^3 - 3x^2 + x$ by $x - 3$, we get $x^2 + 1$ as the quotient and 0 as the remainder and on dividing $x^3 - 10x^2 + 27x - 18$, we get $x^2 - 7x + 6$ as the quotient and 0 as the remainder.

$$\begin{aligned} \therefore I &= \lim_{x \rightarrow 3} \frac{(x^2 + 1)(x - 3)}{(x^2 - 7x + 6)(x - 3)} = \lim_{x \rightarrow 3} \frac{x^2 + 1}{x^2 - 7x + 6} \quad \left[\text{Cancelling out the} \right. \\ &\quad \left. \text{factor } x - 3 \right] \\ &= \frac{(3)^2 + 1}{(3)^2 - 7 \times 3 + 6} \quad [\text{By D.S.M.}] \\ &= \frac{9 + 1}{9 - 21 + 6} = \frac{10}{-6} = -\frac{5}{3} \end{aligned}$$

Here is an exercise for you.

E 2) Evaluate the following limits:

$$(i) \lim_{x \rightarrow 2} \frac{x^3 - 7x^2 + 16x - 12}{x^4 - 6x^3 - 3x^2 + 52x - 60} \quad (ii) \lim_{x \rightarrow 2} \frac{x^3 - 4x^2 + 5x - 2}{x^3 - 2x - 4}$$

5.4.2 Least Common Multiplier Method

This method is useful in $\infty - \infty$ form.

Procedure: Take L.C.M. of the given expression and simplify it. Most of the times after simplification it reduces to $\frac{0}{0}$ form then solve it as explained in factorisation method.

Let us take an example based on this method.

Example 3: Evaluate $\lim_{x \rightarrow 3} \left(\frac{1}{x-3} - \frac{3}{x^2-3x} \right)$

Solution: $\lim_{x \rightarrow 3} \left(\frac{1}{x-3} - \frac{3}{x^2-3x} \right)$ [$\infty - \infty$ form, so D.S.M. fails]

Using LCM method, we have

$$\lim_{x \rightarrow 3} \left(\frac{1}{x-3} - \frac{3}{x^2-3x} \right) = \lim_{x \rightarrow 3} \left(\frac{1}{x-3} - \frac{3}{x(x-3)} \right) = \lim_{x \rightarrow 3} \left(\frac{x-3}{x(x-3)} \right) = \lim_{x \rightarrow 3} \frac{1}{x} = \frac{1}{3}$$

Here is an exercise for you.

E 3) Evaluate $\lim_{x \rightarrow -2} \left(\frac{1}{x+2} - \frac{6}{x^3+x^2-2x} \right)$

5.4.3 Rationalisation Method

This method is explained in the following example.

Example 4: Evaluate the following limits:

$$(i) \lim_{x \rightarrow 0} \frac{\sqrt{4+x}-2}{x} \quad (ii) \lim_{x \rightarrow 3} \frac{\sqrt{5x-6}-\sqrt{x+6}}{x^2-9}$$

Solution:

$$(i) \lim_{x \rightarrow 0} \frac{\sqrt{4+x}-2}{x} \quad \left[\frac{0}{0} \text{ form, so D.S.M. fails} \right]$$

Rationalising the numerator, we have

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{4+x}-2}{x} &= \lim_{x \rightarrow 0} \frac{\sqrt{4+x}-2}{x} \times \frac{\sqrt{4+x}+2}{\sqrt{4+x}+2} \\ &= \lim_{x \rightarrow 0} \frac{(\sqrt{4+x})^2 - (2)^2}{x(\sqrt{4+x}+2)} \quad [\because a^2 - b^2 = (a-b)(a+b)] \\ &= \lim_{x \rightarrow 0} \frac{4+x-4}{x(\sqrt{4+x}+2)} = \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{4+x}+2)} \\ &= \lim_{x \rightarrow 0} \frac{1}{\sqrt{4+x}+2} = \frac{1}{\sqrt{4+0}+2} = \frac{1}{2+2} = \frac{1}{4} \end{aligned}$$

$$(ii) \lim_{x \rightarrow 3} \frac{\sqrt{5x-6}-\sqrt{x+6}}{x^2-9} \quad \left[\frac{0}{0} \text{ form, so D.S.M. fails} \right]$$

Rationalising the numerator, we have

$$\begin{aligned}
\lim_{x \rightarrow 3} \frac{\sqrt{5x-6} - \sqrt{x+6}}{x^2 - 9} &= \lim_{x \rightarrow 3} \frac{\sqrt{5x-6} - \sqrt{x+6}}{x^2 - 9} \times \frac{\sqrt{5x-6} + \sqrt{x+6}}{\sqrt{5x-6} + \sqrt{x+6}} \\
&= \lim_{x \rightarrow 3} \frac{(\sqrt{5x-6})^2 - (\sqrt{x+6})^2}{(x^2 - 9)(\sqrt{5x-6} + \sqrt{x+6})} \\
&= \lim_{x \rightarrow 3} \frac{5x - 6 - (x + 6)}{(x^2 - 9)(\sqrt{5x-6} + \sqrt{x+6})} \\
&= \lim_{x \rightarrow 3} \frac{4x - 12}{(x^2 - 3^2)(\sqrt{5x-6} + \sqrt{x+6})} \\
&= \lim_{x \rightarrow 3} \frac{4(x-3)}{(x-3)(x+3)(\sqrt{5x-6} + \sqrt{x+6})} \\
&= \lim_{x \rightarrow 3} \frac{4}{(x+3)(\sqrt{5x-6} + \sqrt{x+6})} \\
&= \frac{4}{(3+3)(\sqrt{15-6} + \sqrt{3+6})} \\
&= \frac{4}{6(\sqrt{9} + \sqrt{9})} = \frac{4}{6(3+3)} = \frac{4}{36} = \frac{1}{9}
\end{aligned}$$

Here is an exercise for you.

E 4) Evaluate $\lim_{x \rightarrow 2} \frac{\sqrt{3+x} - \sqrt{5}}{x-2}$.

5.4.4 Use of some Standard Results

Here, we list without proof some very useful standard results which hold in limits.

1. $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$ [a and n are any real numbers, provided a^n, a^{n-1} exist]

2. $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = \lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} = 1$

3. $\lim_{\theta \rightarrow 0} \cos \theta = 1$

4. $\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = \lim_{\theta \rightarrow 0} \frac{\theta}{\tan \theta} = 1$

5. $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e |a|$, in particular, $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \log_e e = 1$

6. $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$

7. $\lim_{x \rightarrow 0} (1+x)^{1/x} = e$

8. $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$

Let us consider an example based on these standard results.

Example 5: Evaluate the following limits:

$$\begin{aligned} & \text{(i) } \lim_{x \rightarrow 3} \frac{x^5 - 243}{x^4 - 81} \quad \text{(ii) } \lim_{x \rightarrow 2} \frac{x^{10/3} - 2^{10/3}}{x^{4/3} - 2^{4/3}} \quad \text{(iii) } \lim_{x \rightarrow 0} \frac{\sin 4x}{3x} \quad \text{(iv) } \lim_{x \rightarrow 0} \cos 5x \\ & \text{(v) } \lim_{x \rightarrow 0} \frac{\tan 3x}{\sin 2x} \quad \text{(vi) } \lim_{x \rightarrow 0} \frac{2^{5x} - 1}{x} \quad \text{(vii) } \lim_{x \rightarrow 0} \frac{e^{ax} - 1}{x} \quad \text{(viii) } \lim_{x \rightarrow 0} \frac{\log(1 + 5x)}{x} \\ & \text{(ix) } \lim_{x \rightarrow 0} \frac{\log(1 + 3x)}{e^{2x} - 1} \quad \text{(x) } \lim_{x \rightarrow 0} (1 + 8x)^{1/x} \end{aligned}$$

Solution:

$$\begin{aligned} \text{(i) Let } I &= \lim_{x \rightarrow 3} \frac{x^5 - 243}{x^4 - 81} = \lim_{x \rightarrow 3} \frac{x^5 - 3^5}{x^4 - 3^4} \\ &\text{Dividing numerator and denominator by } x - 3, \text{ we get} \\ I &= \lim_{x \rightarrow 3} \frac{\frac{x^5 - 3^5}{x - 3}}{\frac{x^4 - 3^4}{x - 3}} = \frac{\lim_{x \rightarrow 3} \frac{x^5 - 3^5}{x - 3}}{\lim_{x \rightarrow 3} \frac{x^4 - 3^4}{x - 3}} \\ &= \frac{5(3)^{5-1}}{4(3)^{4-1}} = \frac{5}{4} \times \frac{3^4}{3^3} = \frac{5}{4} \times 3 = \frac{15}{4} \quad \left[\text{Using } \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right] \end{aligned}$$

$$\begin{aligned} \text{(ii) } \lim_{x \rightarrow 2} \frac{x^{10/3} - 2^{10/3}}{x^{4/3} - 2^{4/3}} \\ &\text{Dividing numerator and denominator by } x - 2, \text{ we get} \\ \lim_{x \rightarrow 2} \frac{x^{10/3} - 2^{10/3}}{x^{4/3} - 2^{4/3}} &= \lim_{x \rightarrow 2} \frac{\frac{x^{10/3} - 2^{10/3}}{x - 2}}{\frac{x^{4/3} - 2^{4/3}}{x - 2}} = \frac{\lim_{x \rightarrow 2} \frac{x^{10/3} - 2^{10/3}}{x - 2}}{\lim_{x \rightarrow 2} \frac{x^{4/3} - 2^{4/3}}{x - 2}} \\ &= \frac{\frac{10}{3}(2)^{\frac{10}{3}-1}}{\frac{4}{3}(2)^{\frac{4}{3}-1}} = \frac{10}{3} \times \frac{3}{4} \times \frac{2^{\frac{7}{3}}}{2^{\frac{1}{3}}} = \frac{5}{2} \times 2^{\frac{7}{3}-\frac{1}{3}} = \frac{5}{2} \cdot 2^2 = 10 \end{aligned}$$

$$\begin{aligned} \text{(iii) } \lim_{x \rightarrow 0} \frac{\sin 4x}{3x} &= \lim_{x \rightarrow 0} \frac{\sin 4x}{4x} \times \frac{4x}{3x} \quad \left[\begin{array}{l} \text{Dividing and multiplying} \\ \text{by } 4x \end{array} \right] \\ &= \lim_{x \rightarrow 0} \frac{\sin 4x}{4x} \cdot \frac{4}{3} = \frac{4}{3} \lim_{x \rightarrow 0} \frac{\sin 4x}{4x} \\ &= \frac{4}{3} \lim_{4x \rightarrow 0} \frac{\sin 4x}{4x} \quad \left[\text{As } x \rightarrow 0 \Rightarrow 4x \rightarrow 0 \right] \\ &= \frac{4}{3} \times 1 = \frac{4}{3} \quad \left[\because \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1 \right] \end{aligned}$$

$$\begin{aligned} \text{(iv) } \lim_{x \rightarrow 0} \cos 5x &= \lim_{5x \rightarrow 0} \cos 5x = 1 \quad \left[\begin{array}{l} \text{As } x \rightarrow 0 \Rightarrow 5x \rightarrow 0 \text{ and} \\ \lim_{\theta \rightarrow 0} \cos \theta = 1 \end{array} \right] \end{aligned}$$

$$\text{(v) } \lim_{x \rightarrow 0} \frac{\tan 3x}{\sin 2x} = \lim_{x \rightarrow 0} \frac{\tan 3x}{3x} \times \frac{2x}{\sin 2x} \times \frac{3}{2}$$

$$= \frac{3}{2} \left(\lim_{3x \rightarrow 0} \frac{\tan 3x}{3x} \right) \left(\lim_{2x \rightarrow 0} \frac{2x}{\sin 2x} \right) = \frac{3}{2} (1)(1) = \frac{3}{2}$$

$$\left[\because \lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = 1 \text{ and } \lim_{x \rightarrow 0} \frac{\theta}{\sin \theta} = 1 \right]$$

$$(vi) \lim_{x \rightarrow 0} \frac{2^{5x} - 1}{x} = \lim_{x \rightarrow 0} \frac{2^{5x} - 1}{5x} \times 5 = 5 \lim_{5x \rightarrow 0} \frac{2^{5x} - 1}{5x} \left[\because x \rightarrow 0 \Rightarrow 5x \rightarrow 0 \right]$$

$$= 5 \log_e 2 \quad \left[\because \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e |a| \right]$$

$$(vii) \lim_{x \rightarrow 0} \frac{e^{ax} - 1}{x} = \lim_{x \rightarrow 0} \frac{e^{ax} - 1}{ax} \times a = a \lim_{ax \rightarrow 0} \frac{e^{ax} - 1}{ax} \left[\because x \rightarrow 0 \Rightarrow ax \rightarrow 0 \right]$$

$$= a(1) \quad \left[\because \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \right]$$

$$= a$$

$$(viii) \lim_{x \rightarrow 0} \frac{\log(1+5x)}{x} = \lim_{x \rightarrow 0} 5 \frac{\log(1+5x)}{5x}$$

$$= 5 \lim_{5x \rightarrow 0} \frac{\log(1+5x)}{5x} \quad \left[\because x \rightarrow 0 \Rightarrow 5x \rightarrow 0 \right]$$

$$= 5(1) \quad \left[\because \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \right]$$

$$= 5$$

$$(ix) \lim_{x \rightarrow 0} \frac{\log(1+3x)}{e^{2x} - 1} = \lim_{x \rightarrow 0} \frac{\log(1+3x)}{3x} \times \frac{2x}{e^{2x} - 1} \times \frac{3}{2}$$

$$= \frac{3}{2} \left(\lim_{3x \rightarrow 0} \frac{\log(1+3x)}{3x} \right) \left(\lim_{2x \rightarrow 0} \frac{2x}{e^{2x} - 1} \right)$$

$$= \frac{3}{2} (1)(1) = \frac{3}{2} \quad \text{as } \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \text{ and } \lim_{x \rightarrow 0} \frac{x}{e^x - 1} = 1$$

$$(x) \lim_{x \rightarrow 0} (1+8x)^{1/x} = \lim_{x \rightarrow 0} \left[(1+8x)^{\frac{1}{8x}} \right]^8 = \left[\lim_{8x \rightarrow 0} (1+8x)^{\frac{1}{8x}} \right]^8 \text{ as } x \rightarrow 0 \Rightarrow 8x \rightarrow 0$$

$$= (e)^8 = e^8 \quad \left[\because \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e \right]$$

Here, is an exercise for you.

E 5) Evaluate the following limits:

$$(i) \lim_{x \rightarrow \sqrt{2}} \frac{x^{10} - 32}{x - \sqrt{2}} \quad (ii) \lim_{x \rightarrow 0} \frac{(ab)^{3x} - 1}{x} \quad (iii) \lim_{x \rightarrow 0} \frac{e^{\sin x} - 1}{\tan x}$$

$$(iv) \lim_{x \rightarrow 0} \frac{\log(1+8x^2)}{e^{x^2} - 1} \quad (v) \lim_{x \rightarrow 0} \frac{a^x - e^x}{x} \quad (vi) \lim_{x \rightarrow 0} \frac{e^x - (1+2x)}{2^x - 1}$$

$$(vii) \lim_{x \rightarrow 0} \frac{x(1+2x)^{1/x} - (e^{2x} - 1)}{x}$$

5.5 CONCEPT OF INFINITE LIMIT

Consider the following cases

$$\frac{1}{10} = 0.1$$

$$\frac{1}{100} = 0.01$$

$$\frac{1}{1000} = 0.001$$

$$\frac{1}{10000} = 0.0001$$

...

...

...

$$\frac{1}{10^n} = \underbrace{0.000\dots 1}_{n \text{ times}}$$

We see that as x (denominator) becomes larger and larger than $\frac{1}{x}$ becomes smaller and smaller and approaches to zero.

$$\therefore \text{we write } \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

$$\text{Or } \lim_{x \rightarrow \infty} \frac{1}{x^n} = 0, \text{ where } n > 0$$

Remark 3: $x \rightarrow \infty$ means that whatever large real number K (say) we take then $x > K$, i.e. no real number can be greater than x .

Let us consider an example, which involve infinite limit.

Example 6: Evaluate the following limits:

$$(i) \lim_{x \rightarrow \infty} \frac{3x^2 + 5x - 1}{4x^2 - 3x + 9} \quad (ii) \lim_{x \rightarrow \infty} \frac{5x^5 + x + 1}{x^3 + 5} \quad (iii) \lim_{x \rightarrow \infty} \frac{x^5 + 1}{4x^7 + 3x^2 + 7}$$

Solution:

$$(i) \lim_{x \rightarrow \infty} \frac{3x^2 + 5x - 1}{4x^2 - 3x + 9}$$

Here degree of numerator = Degree of denominator = 2

$\therefore$ dividing numerator and denominator by x^2 , we get

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 5x - 1}{4x^2 - 3x + 9} = \lim_{x \rightarrow \infty} \frac{3 + \frac{5}{x} - \frac{1}{x^2}}{4 - \frac{3}{x} + \frac{9}{x^2}} = \frac{3 + 0 - 0}{4 - 0 + 0} = \frac{3}{4} \left[\because \lim_{x \rightarrow \infty} \frac{1}{x^n} = 0 \right. \\ \left. \text{for } n > 0 \right]$$

$$(ii) \lim_{x \rightarrow \infty} \frac{5x^5 + x + 1}{x^3 + 5}$$

Here degree of numerator > degree of denominator.

$\therefore$ dividing numerator and denominator by x^3 [Least of degrees], we get

$$\lim_{x \rightarrow \infty} \frac{5x^5 + x + 1}{x^3 + 5} = \lim_{x \rightarrow \infty} \frac{5x^2 + \frac{1}{x^2} + \frac{1}{x^3}}{1 + \frac{5}{x^3}} = \frac{\infty + 0 + 0}{1 + 0} = \infty$$

$$(iii) \lim_{x \rightarrow \infty} \frac{x^5 + 1}{4x^7 + 3x^2 + 7}$$

Here degree of numerator < degree of denominator.

∴ dividing numerator and denominator by x^5 [Least of degrees], we get

$$\lim_{x \rightarrow \infty} \frac{x^5 + 1}{4x^7 + 3x^2 + 7} = \lim_{x \rightarrow \infty} \frac{1 + \frac{1}{x^5}}{4x^2 + \frac{3}{x^3} + \frac{7}{x^5}} = \frac{1 + 0}{\infty + 0 + 0} = \frac{1}{\infty} = 0$$

In general, without calculating actual limit we can know the answer in advance of rational functions, in the cases when as $x \rightarrow \infty$ see the following result without proof.

$$\lim_{x \rightarrow \infty} \frac{a_0 x^m + a_1 x^{m-1} + a_2 x^{m-2} + \dots + a_{m-1} x + a_m}{b_0 x^n + b_1 x^{n-1} + b_2 x^{n-2} + \dots + b_{n-1} x + b_n} = \begin{cases} \frac{a_0}{b_0}, & \text{if } m = n \\ 0, & \text{if } m < n \\ \infty, & \text{if } m > n \end{cases}$$

Now, you can try the following exercise.

E 6) Evaluate $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 3} + \sqrt[4]{x^4 + 2}}{\sqrt[7]{x^7 + x^2} - \sqrt[3]{x^2 + 5}}$.

5.6 CONCEPT OF LEFT HAND AND RIGHT HAND LIMITS

We note that on the real line, we can approach any real number 2(say) either from left or from right. Consider the example $y = f(x) = 3x + 2$. We see that as x takes the values 1.9, 1.98, 1.998, 1.9998, ... then corresponding values taken by y are 7.7, 7.94, 7.994, 7.9994, ... respectively as shown below.

x	1.9	1.98	1.998	1.9998	...	1.99999998	...
$y = f(x)$	7.7	7.94	7.994	7.9994	...	7.99999994	...

x	2.1	2.01	2.001	2.0001	...	2.0000001	...
$y = f(x)$	8.3	8.03	8.003	8.0003	...	8.0000003	...

i.e. as x is coming nearer and nearer to 2 from left then y is also coming nearer and nearer to 8 from left. If x approaches like this from left (see Fig. 5.1), then we say that x is approaching from left to 2 and is denoted by putting a -ve sign as a right superscript of 2 i.e. 2^-

i.e. we write the limit of the function as

$$\lim_{x \rightarrow 2^-} f(x) \quad \dots (1)$$

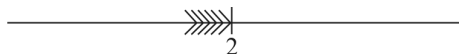


Fig. 5.1

If limit (1) exists, then we call it left hand limit (L.H.L.) of the function $f(x)$ as x tends to 2.

Similarly we see that as x takes the values 2.1, 2.01, 2.001, 2.0001, ... then corresponding values taken by y are 8.3, 8.03, 8.003, 8.0003, ... respectively. i.e. as x is coming nearer and nearer to 2 from right then y is also coming nearer and nearer to 8 from right. If x approaches like this from right (see Fig. 5.2) then we say that x is approaching from right to 2 and is denoted by putting +ve sign as a superscript of 2 i.e. 2^+ and the limit of the function as

$$\lim_{x \rightarrow 2^+} f(x) \quad \dots (2)$$

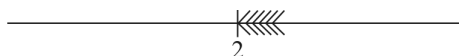


Fig. 5.2

If limit (2) exists, then we call it right hand limit (R.H.L.) of the function $f(x)$ as x tends to 2.

Remark 4:

- (i) L.H. and R.H. limits are used when functions have different values for $x \rightarrow 2^-$ and $x \rightarrow 2^+$.

For example, in case of

- (a) modules functions,
- (b) functions having different values just below or above the value to which x is tending, i.e. there is break in function.

- (ii) Limit exists if L.H.L. and R.H.L. both exist and are equal.

Following example illustrates the idea of L.H.L. and R.H.L.

Example 7: Evaluate the following limits:

$$(i) \lim_{x \rightarrow 0} |x| \quad (ii) \lim_{x \rightarrow 3} |x - 3| \quad (iii) \lim_{x \rightarrow 1} f(x), \text{ where } f(x) = \begin{cases} x^2 + 1, & x \leq 1 \\ 1 - x^2, & x > 1 \end{cases}$$

$$(iv) \lim_{x \rightarrow 4} f(x), \text{ where } f(x) = \begin{cases} \frac{|x - 4|}{x - 4}, & x \neq 4 \\ 0, & x = 4 \end{cases}$$

Solution:

$$(i) \lim_{x \rightarrow 0} |x|$$

Here we have to use the concept of L.H.L. and R.H.L., because of the presence of the modulus function.

$$\text{L.H.L.} = \lim_{x \rightarrow 0^-} |x|$$

Here, as x is approaching to zero from its left and hence x is having little bit lesser value than 0.

Let us put $x = 0 - h$, where h is +ve real and is very small.

$$\text{As } x \rightarrow 0^- \Rightarrow h \rightarrow 0^+$$

$$\begin{aligned}
\text{L.H.L.} &= \lim_{h \rightarrow 0^+} |0 - h| = \lim_{h \rightarrow 0^+} |-h| = \lim_{h \rightarrow 0^+} |-1| \times |h| = \lim_{h \rightarrow 0^+} |h| \text{ as } |-1| = -(-1) = 1 \\
&= \lim_{h \rightarrow 0^+} h \quad [\because h \rightarrow 0^+ \Rightarrow h > 0 \Rightarrow |h| = h] \\
&= 0 \quad \dots (1)
\end{aligned}$$

$$\text{R.H.L.} = \lim_{x \rightarrow 0^+} |x|$$

Here, as x is approaching to zero from its right and hence x is having slightly greater value than 0.

Let us put $x = 0 + h$, where h is +ve real and is very small.

$$\text{As } x \rightarrow 0^+ \Rightarrow h \rightarrow 0^+$$

$$\therefore \text{R.H.L.} = \lim_{h \rightarrow 0^+} |0 + h| = \lim_{h \rightarrow 0^+} |h| = \lim_{h \rightarrow 0^+} h = 0 \quad \dots (2)$$

From (1) and (2)

$$\text{L.H.L.} = \text{R.H.L.}$$

$$\therefore \lim_{x \rightarrow 0} |x| \text{ exists and equal to 0.}$$

$$(ii) \lim_{x \rightarrow 3} |x - 3|$$

$$\text{L.H.L.} = \lim_{x \rightarrow 3^-} |x - 3|$$

Putting $x = 3 - h$, where h is +ve real and very small.

$$\text{As } x \rightarrow 3^- \Rightarrow h \rightarrow 0^+$$

$$\text{L.H.L.} = \lim_{h \rightarrow 0^+} |3 - h - 3| = \lim_{h \rightarrow 0^+} |-h| = \lim_{h \rightarrow 0^+} |h| = \lim_{h \rightarrow 0^+} h = 0 \quad \dots (1)$$

$$\text{R.H.L.} = \lim_{x \rightarrow 3^+} |x - 3|$$

Putting $x = 3 + h$ as $x \rightarrow 3^+ \Rightarrow h \rightarrow 0^+$

$$\text{R.H.L.} = \lim_{h \rightarrow 0^+} |3 + h - 3| = \lim_{h \rightarrow 0^+} |h| = \lim_{h \rightarrow 0^+} h = 0 \quad \dots (2)$$

From (1) and (2)

$$\text{L.H.L.} = \text{R.H.L.}$$

$$\therefore \lim_{x \rightarrow 3} |x - 3| \text{ exists and equal to 0.}$$

$$(iii) \lim_{x \rightarrow 1} f(x), \text{ where } f(x) = \begin{cases} x^2 + 1, & x \leq 1 \\ 1 - x^2, & x > 1 \end{cases}$$

$$\begin{aligned}
\text{L.H.L.} &= \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x^2 + 1) \left[\because x \rightarrow 1^- \text{ means } x \text{ is slightly less than } 1 \right. \\
&\quad \left. \text{and hence in this case } f(x) = x^2 + 1 \right] \\
&= (1)^2 + 1 = 1 + 1 = 2 \quad \dots (1)
\end{aligned}$$

$$\begin{aligned}
\text{R.H.L.} &= \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (1 - x^2) \left[\because x \rightarrow 1^+ \text{ means } x \text{ is slightly greater than } 1 \right. \\
&\quad \left. \text{and hence in this case } f(x) = 1 - x^2 \right] \\
&= 1 - (1)^2 = 1 - 1 = 0 \quad \dots (2)
\end{aligned}$$

From (1) and (2)

$$\text{L.H.L.} \neq \text{R.H.L.}$$

$$\therefore \lim_{x \rightarrow 1} f(x) \text{ does not exist.}$$

$$(iv) \lim_{x \rightarrow 4} f(x), \text{ where } f(x) = \begin{cases} \frac{|x - 4|}{x - 4}, & x \neq 4 \\ 0, & x = 4 \end{cases}$$

$$\text{L.H.L.} = \lim_{x \rightarrow 4^-} f(x) = \lim_{x \rightarrow 4^-} \frac{|x-4|}{x-4} \left[\begin{array}{l} \because x \rightarrow 4^- \Rightarrow x \text{ is slightly less than } 4 \\ \text{i.e } x \neq 4, \text{ so in this case } f(x) = \frac{|x-4|}{x-4} \end{array} \right]$$

Putting $x = 4 - h$, where h is +ve real and very small.

As $x \rightarrow 4^- \Rightarrow h \rightarrow 0^+$

$$\begin{aligned} \text{L.H.L.} &= \lim_{h \rightarrow 0^+} \frac{|4-h-4|}{4-h-4} = \lim_{h \rightarrow 0^+} \frac{|-h|}{-h} = \lim_{h \rightarrow 0^+} \frac{-1 \times |h|}{-h} \\ &= \lim_{h \rightarrow 0^+} \frac{(1)(h)}{-h} = \lim_{h \rightarrow 0^+} (-1) = -1 \end{aligned} \quad \dots (1)$$

$$\text{R.H.L.} = \lim_{x \rightarrow 4^+} f(x) = \lim_{x \rightarrow 4^+} \frac{|x-4|}{x-4}$$

Putting $x = 4 + h$ as $x \rightarrow 4^+ \Rightarrow h \rightarrow 0^+$

$$\therefore \text{R.H.L.} = \lim_{h \rightarrow 0^+} \frac{|4+h-4|}{4+h-4} = \lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = \lim_{h \rightarrow 0^+} 1 = 1 \quad \dots (2)$$

From (1) and (2)

L.H.L. $\neq$ R.H.L.

$\therefore \lim_{x \rightarrow 4} f(x)$ does not exist.

Example 8: If $\lim_{x \rightarrow 0} f(x)$ exists, then find the value of k for

$$f(x) = \begin{cases} x - |x|, & x \leq 0 \\ k, & x > 0 \end{cases}$$

$$\text{Solution: } f(x) = \begin{cases} x - |x|, & x \leq 0 \\ k, & x > 0 \end{cases}$$

$$\text{L.H.L.} = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x - |x|)$$

Putting $x = 0 - h$ as $x \rightarrow 0^- \Rightarrow h \rightarrow 0^+$

$$\begin{aligned} \text{L.H.L.} &= \lim_{h \rightarrow 0^+} (0 - h - |0 - h|) = \lim_{h \rightarrow 0^+} (-h - |-h|) \\ &= \lim_{h \rightarrow 0^+} (-h - |-1 \times |h||) = \lim_{h \rightarrow 0^+} (-h - |h|) = \lim_{h \rightarrow 0^+} (-h - h) \\ &= \lim_{h \rightarrow 0^+} (-2h) = -2(0) = 0 \end{aligned} \quad \dots (1)$$

$$\text{R.H.L.} = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} k = k \quad \dots (2)$$

Since, it is given that $\lim_{x \rightarrow 0} f(x)$ exists.

$\therefore$ we must have

$$\text{L.H.L.} = \text{R.H.L.} \Rightarrow 0 = k \text{ or } k = 0$$

Here are some exercises for you.

E 7) Evaluate the following limits:

$$(i) \lim_{x \rightarrow 0^-} \frac{5x + |x|}{3|x| - 7x} \quad (ii) \lim_{x \rightarrow 5^+} (3 - |x|) \quad (iii) \lim_{x \rightarrow 0} \frac{x}{|x|}$$

E 8) If $\lim_{x \rightarrow 3} f(x)$ exists then find a , for $f(x) = \begin{cases} ax + 3, & x \leq 3 \\ 2(x + 1), & x > 3 \end{cases}$

5.7 CONTINUITY OF A FUNCTION AT A POINT

In Sec. 5.6, we have discussed the concept of L.H.L. and R.H.L. Adding one more-step, we can define continuity at a point.

A function $f(x)$ is said to be continuous at $x = a$ if

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a), \text{ i.e. for continuity at a point } x = a, \text{ we must have}$$

i.e. $L.H.L._{at\ x=a} = R.H.L._{at\ x=a} = \text{value of the function at } x = a$

Diagrammatically, continuity at $x = a$ means graph of the function $f(x)$ from a value slightly less than 'a' to a value slightly greater than 'a' has no gap, i.e. if we draw the graph with pencil then we don't have to pick up the pencil as we cross the point where $x = a$. Look at the Fig. 5.3 to 5.5.

In Fig. 5.3 $f(x)$ is not continuous at $x = a$.

In Fig. 5.4 $f(x)$ is not continuous at $x = a$.

In Fig. 5.5 $f(x)$ is continuous at $x = a$.

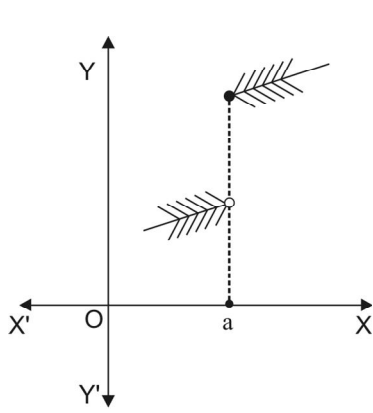


Fig. 5.3

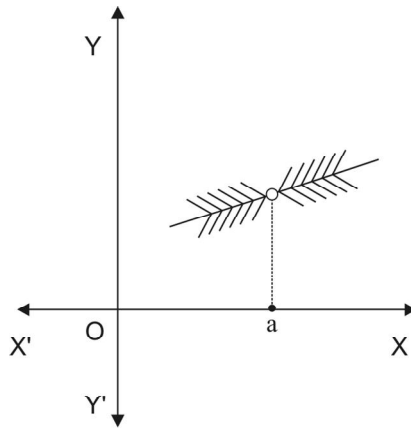


Fig. 5.4

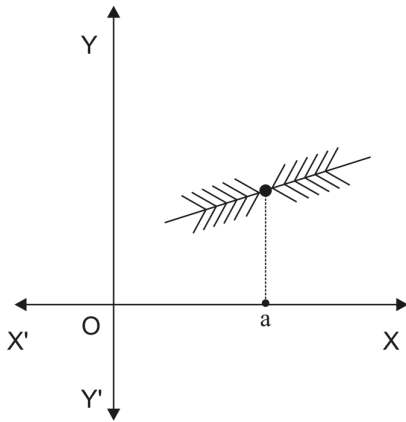


Fig. 5.5

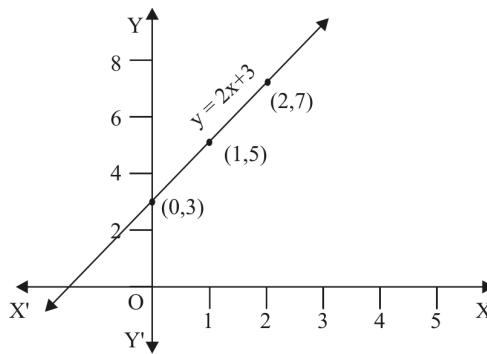


Fig. 5.6

Functions whose graphs are given in Fig. 5.6 and Fig. 5.7 are discussed below.

- (i) Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x + 3$

x	0	1	2
y	3	5	7

See the graph of this function in Fig. 5.6. We note that this function is continuous at all points of its domain as there is no gap at any point in its graph.

(ii) Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1, & x \geq 1 \\ 2, & x < 1 \end{cases}$$

See the graph of this function in Fig. 5.7.

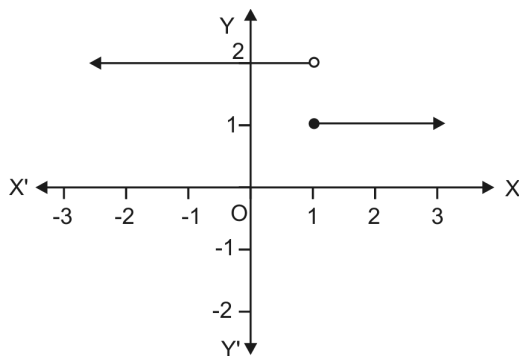


Fig .5.7

We note that, if we draw the graph of this function with pencil, then we will have to pick up the pencil as we cross the point where $x = 1$. Therefore this function is not continuous at $x = 1$. Also this function is continuous at all points of its domain except at $x = 1$.

Now, let us consider some examples on continuity at a point.

Example 9: Discuss the continuity of the following functions at given point:

(i) $f(x) = |x|$ at $x = 0$

(ii) $f(x) = |x - 3|$ at $x = 3$

(iii) $f(x) = \begin{cases} x^2 + 1, & x \leq 1 \\ 1 - x^2, & x > 1 \end{cases}$ at $x = 1$

(iv) $f(x) = \begin{cases} \frac{|x - 4|}{x - 4}, & x \neq 4 \\ 0, & x = 4 \end{cases}$ at $x = 4$

(v) $f(x) = \frac{x}{|x|}$ at $x = 0$

Solution:

(i) $f(x) = |x|$ at $x = 0$

L.H.L. = 0, R.H.L. = 0 [Already calculated in
Example 7 of this unit]

Also, at $x = 0$, $f(x) = |0| = 0$

$\therefore$ L.H.L. at $x = 0$ = R.H.L. at $x = 0$ = $f(0)$

$\Rightarrow f(x)$ is continuous at $x = 0$

$$(ii) f(x) = |x - 3|, \quad \text{at } x = 3$$

$$\text{L.H.L.} = 0, \quad \text{R.H.L.} = 0$$

[Already calculated in
Example 7 of this unit]

$$\text{Also, } f(3) = |3 - 3| = |0| = 0$$

$$\therefore \text{L.H.L.}_{\text{at } x=3} = \text{R.H.L.}_{\text{at } x=3} = f(3)$$

$\Rightarrow f(x)$ is continuous at $x = 3$

$$(iii) f(x) = \begin{cases} x^2 + 1, & x \leq 1 \\ 1 - x^2, & x > 1 \end{cases} \quad \text{at } x = 1$$

$$\text{L.H.L.} = 2, \text{ but } \text{R.H.L.} = 0$$

[Already calculated in
Example 7 of this unit]

$$\text{As } \text{L.H.L.}_{\text{at } x=1} \neq \text{R.H.L.}_{\text{at } x=1}$$

$\therefore f$ is not continuous at $x = 1$

$$(iv) f(x) = \begin{cases} |x - 4|, & x \neq 4 \\ 0, & x = 4 \end{cases} \quad \text{at } x = 4$$

$$\therefore \text{L.H.L.}_{\text{at } x=4} = -1, \text{ but } \text{R.H.L.}_{\text{at } x=4} = 1$$

[Already calculated in
Example 7 of this unit]

$$\text{As } \text{L.H.L.}_{\text{at } x=4} \neq \text{R.H.L.}_{\text{at } x=4}$$

$\therefore f(x)$ is not continuous at $x = 4$.

$$(v) f(x) = \frac{x}{|x|}, \quad \text{at } x = 0$$

$$\text{L.H.L.} = \lim_{x \rightarrow 0^-} \frac{x}{|x|}$$

Putting $x = 0 - h$ as $x \rightarrow 0^- \Rightarrow h \rightarrow 0^+$

$$\begin{aligned} \text{L.H.L.} &= \lim_{h \rightarrow 0^+} \frac{0 - h}{|0 - h|} = \lim_{h \rightarrow 0^+} \frac{-h}{|-h|} = \lim_{h \rightarrow 0^+} \frac{-h}{-1 \times |h|} \\ &= \lim_{h \rightarrow 0^+} \frac{-h}{h} = \lim_{h \rightarrow 0^+} (-1) = -1 \quad \dots (1) \end{aligned}$$

$$\text{R.H.L.} = \lim_{x \rightarrow 0^+} \frac{x}{|x|} = \lim_{x \rightarrow 0^+} \frac{x}{x} = \lim_{x \rightarrow 0^+} 1 = 1 \quad \dots (2)$$

From (1) and (2)

$$\text{L.H.L.} \neq \text{R.H.L.}$$

$$\therefore \lim_{x \rightarrow 0} \frac{x}{|x|} \text{ does not exist.}$$

Hence f is not continuous at $x = 0$.

Example 10: Find the values of a and b , if the function f given below is continuous at $x = 2$

$$f(x) = \begin{cases} 7, & x < 2 \\ ax + b, & x > 2 \\ a + 5, & x = 2 \end{cases}$$

Solution: L.H.L. = $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} 7 = 7$

R.H.L. = $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (ax + b) = 2a + b$

Also $f(2) = a + 5$

Since, f is given to be continuous at $x = 2$, therefore we must have

L.H.L._{at $x = 2$} = R.H.L._{at $x = 2$} = $f(2)$

$\Rightarrow 7 = 2a + b = a + 5$

I II III

I & III $\Rightarrow 7 = a + 5 \Rightarrow a = 2$

I & II $\Rightarrow 7 - 2a = b \Rightarrow b = 7 - 4 = 3$

$\therefore a = 2, b = 3$

Here is an exercise for you.

E 9) Find the relation between a and b if the function f is given to be continuous at $x = 0$, where

$$f(x) = \begin{cases} 2x - a, & x \geq 0 \\ ax + b + 3, & x < 0 \end{cases}$$

5.8 CONTINUOUS FUNCTION

In section 5.7, we have discussed the continuity of a function at a point. In this section, we define what we mean by continuous function.

Continuous Function: A function f is said to be continuous if it is continuous at each point of its domain.

For example, function $y = f(x) = 2x + 3$ whose graph is given in Fig. 5.6 is a continuous function as we have already discussed that it is continuous at all points of its domain.

Algebra of continuous functions:

If f and g are two continuous functions on a common domain then

- (1) $f + g$ is continuous
- (2) $f - g$ is continuous
- (3) fg is continuous
- (4) f/g is continuous, provided $g(x) \neq 0 \forall$ points x of its domain.

5.9 SUMMARY

In this unit, we have:

- 1) Given the concept of limit.
- 2) Discussed direct substitution method of evaluation of limit.
- 3) Explained factorisation, L.C.M. rationalisation, and some standard methods to evaluate a given limit.
- 4) Given the concept of infinite limit.
- 5) Given the concept of L.H.L. and R.H.L.
- 6) Discussed the continuity of a function at a point.
- 7) Discussed what we mean by continuous function.

5.10 SOLUTIONS/ANSWERS

$$\text{E 1 (i)} \quad \lim_{x \rightarrow 2} (x^2 - 2x + 3)^{x^2+1} = (2^2 - 2 \times 2 + 3)^{2^2+1} = 3^5 = 243$$

$$\begin{aligned} \text{(ii)} \quad \lim_{x \rightarrow 1} \log(x^4 + x^2 + 1) &= \log\left(\lim_{x \rightarrow 1} (x^4 + x^2 + 1)\right) \\ &= \log(1^4 + 1^2 + 1) = \log 3 \end{aligned}$$

$$\text{(iii)} \quad \lim_{x \rightarrow 5} 3 = 3 \quad \text{as } f(x) = 3 \text{ is a constant function}$$

$$\begin{aligned} \text{(iv)} \quad \lim_{x \rightarrow 3} 4f(x) &= 4 \lim_{x \rightarrow 3} f(x) = 4 \lim_{x \rightarrow 3} (x-5)^2 \\ &= 4(3-5)^2 = 4(-2)^2 = 4 \times 4 = 16 \end{aligned}$$

$$\text{E 2 (i)} \quad \text{Let } I = \lim_{x \rightarrow 2} \frac{x^3 - 7x^2 + 16x - 12}{x^4 - 6x^3 - 3x^2 + 52x - 60} \quad \left[\frac{0}{0} \text{ form, so D.S.M. fails} \right]$$

Using factorisation method, we have

$$I = \lim_{x \rightarrow 2} \frac{(x-2)(x^2 - 5x + 6)}{(x-2)(x^3 - 4x^2 - 11x + 30)}$$

Cancelling the common factor $x-2 \neq 0$, we get

$$I = \lim_{x \rightarrow 2} \frac{x^2 - 5x + 6}{x^3 - 4x^2 - 11x + 30} \quad \left[\text{Again } \frac{0}{0} \text{ form} \right]$$

Again using factorisation method, we have

$$\begin{aligned} I &= \lim_{x \rightarrow 2} \frac{(x-2)(x-3)}{(x-2)(x^2 - 2x - 15)} = \lim_{x \rightarrow 2} \frac{x-3}{x^2 - 2x - 15} \\ &= \frac{2-3}{4-4-15} \quad [\text{By D.S.M.}] \\ &= \frac{-1}{-15} = \frac{1}{15} \end{aligned}$$

$$\text{(ii)} \quad \lim_{x \rightarrow 2} \frac{x^3 - 4x^2 + 5x - 2}{x^3 - 2x - 4} \quad \left[\frac{0}{0} \text{ form, so D.S.M. fails} \right]$$

Using factorisation method, we have

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{x^3 - 4x^2 + 5x - 2}{x^3 - 2x - 4} &= \lim_{x \rightarrow 2} \frac{(x-2)(x^2 - 2x + 1)}{(x-2)(x^2 + 2x + 2)} \\ &= \lim_{x \rightarrow 2} \frac{x^2 - 2x + 1}{x^2 + 2x + 2} = \frac{4-4+1}{4+4+2} = \frac{1}{10} \end{aligned}$$

$$\text{E 3} \quad \lim_{x \rightarrow -2} \left(\frac{1}{x+2} - \frac{6}{x^3 + x^2 - 2x} \right) \quad [\infty - \infty \text{ form, so D.S.M. fails}]$$

Using LCM method, we have

$$\begin{aligned} \lim_{x \rightarrow -2} \left(\frac{1}{x+2} - \frac{6}{x^3 + x^2 - 2x} \right) &= \lim_{x \rightarrow -2} \left(\frac{1}{x+2} - \frac{6}{x(x^2 + x - 2)} \right) \\ &= \lim_{x \rightarrow -2} \left(\frac{1}{x+2} - \frac{6}{x(x+2)(x-1)} \right) \end{aligned}$$

$$\begin{aligned}
 &= \lim_{x \rightarrow -2} \left(\frac{x(x-1)-6}{x(x+2)(x-1)} \right) = \lim_{x \rightarrow -2} \frac{x^2 - x - 6}{x(x+2)(x-1)} \\
 &= \lim_{x \rightarrow -2} \frac{(x-3)(x+2)}{x(x+2)(x-1)} \\
 &= \lim_{x \rightarrow -2} \frac{x-3}{x(x-1)} \left[\begin{array}{l} \because x \rightarrow -2 \Rightarrow x \neq -2 \\ \Rightarrow x+2 \neq 0 \text{ so, cancelling} \\ \text{out } x+2 \end{array} \right] \\
 &= \frac{-2-3}{-2(-2-1)} = \frac{-5}{6} = -\frac{5}{6}
 \end{aligned}$$

E 4) $\lim_{x \rightarrow 2} \frac{\sqrt{3+x} - \sqrt{5}}{x-2}$

Rationalising the numerator, we have

$$\begin{aligned}
 \lim_{x \rightarrow 2} \frac{\sqrt{3+x} - \sqrt{5}}{x-2} &= \lim_{x \rightarrow 2} \frac{\sqrt{3+x} - \sqrt{5}}{x-2} \times \frac{\sqrt{3+x} + \sqrt{5}}{\sqrt{3+x} + \sqrt{5}} \\
 &= \lim_{x \rightarrow 2} \frac{(\sqrt{3+x})^2 - (\sqrt{5})^2}{(x-2)(\sqrt{3+x} + \sqrt{5})} \quad [\because (a-b)(a+b) = a^2 - b^2] \\
 &= \lim_{x \rightarrow 2} \frac{3+x-5}{(x-2)(\sqrt{3+x} + \sqrt{5})} = \lim_{x \rightarrow 2} \frac{x-2}{(x-2)(\sqrt{3+x} + \sqrt{5})} \\
 &= \lim_{x \rightarrow 2} \frac{1}{\sqrt{3+x} + \sqrt{5}} \quad \left[\begin{array}{l} \text{Canceling out the common} \\ \text{factor } x-2 \neq 0 \end{array} \right] \\
 &= \frac{1}{\sqrt{3+2} + \sqrt{5}} = \frac{1}{\sqrt{5} + \sqrt{5}} = \frac{1}{2\sqrt{5}} \\
 &= \frac{1}{2\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{\sqrt{5}}{10} \quad \left[\begin{array}{l} \text{Rationalising the} \\ \text{denominator} \end{array} \right]
 \end{aligned}$$

E 5) (i) $\lim_{x \rightarrow \sqrt{2}} \frac{x^{10} - 32}{x - \sqrt{2}} = \lim_{x \rightarrow \sqrt{2}} \frac{x^{10} - (\sqrt{2})^{10}}{x - \sqrt{2}}$

$$\begin{aligned}
 &= 10 \left((\sqrt{2})^{10-1} \right) \quad \left[\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right] \\
 &= 10(\sqrt{2})^9 = 10(\sqrt{2})^8 (\sqrt{2}) = 10 \times 16 \times \sqrt{2} = 160\sqrt{2}
 \end{aligned}$$

(ii) $\lim_{x \rightarrow 0} \frac{(ab)^{3x} - 1}{x} = \lim_{x \rightarrow 0} \frac{(ab)^{3x} - 1}{3x} \times 3$

$$\begin{aligned}
 &= 3 \lim_{3x \rightarrow 0} \frac{(ab)^{3x} - 1}{3x} \quad \text{as } x \rightarrow 0 \Rightarrow 3x \rightarrow 0 \\
 &= 3 \log|ab| \quad \left[\because \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log|a| \right]
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad \lim_{x \rightarrow 0} \frac{e^{\sin x} - 1}{\tan x} &= \lim_{x \rightarrow 0} \frac{e^{\sin x} - 1}{\sin x} \times \frac{\sin x}{x} \times \frac{x}{\tan x} \\
 &= \left(\lim_{\sin x \rightarrow 0} \frac{e^{\sin x} - 1}{\sin x} \right) \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right) \left(\lim_{x \rightarrow 0} \frac{x}{\tan x} \right) \\
 &\quad \text{as } x \rightarrow 0 \Rightarrow \sin x \rightarrow 0 \\
 &= (1)(1)(1) = 1 \quad \left[\begin{array}{l} \because \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1, \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1 \\ \text{and } \lim_{\theta \rightarrow 0} \frac{\theta}{\tan \theta} = 1 \end{array} \right]
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv)} \quad \lim_{x \rightarrow 0} \frac{\log(1+8x^2)}{e^{x^2} - 1} &= \lim_{x \rightarrow 0} \frac{\log(1+8x^2)}{8x^2} \times \frac{x^2}{e^{x^2} - 1} \times 8 \\
 &= 8 \left(\lim_{8x^2 \rightarrow 0} \frac{\log(1+8x^2)}{8x^2} \right) \left(\lim_{x^2 \rightarrow 0} \frac{x^2}{e^{x^2} - 1} \right) \\
 &\quad \left[\because \text{as } x \rightarrow 0 \Rightarrow 8x^2 \rightarrow 0 \text{ and } x^2 \rightarrow 0 \right] \\
 &= 8(1)(1) = 8 \quad \left[\begin{array}{l} \because \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \text{ and } \\ \lim_{x \rightarrow 0} \frac{x}{e^x - 1} = 1 \end{array} \right]
 \end{aligned}$$

$$\begin{aligned}
 \text{(v)} \quad \lim_{x \rightarrow 0} \frac{a^x - e^x}{x} &= \lim_{x \rightarrow 0} \frac{a^x - 1 - (e^x - 1)}{x} = \lim_{x \rightarrow 0} \left(\frac{a^x - 1}{x} - \frac{e^x - 1}{x} \right) \\
 &= \left(\lim_{x \rightarrow 0} \frac{a^x - 1}{x} \right) - \left(\lim_{x \rightarrow 0} \frac{e^x - 1}{x} \right) \\
 &= \log|a| - \log|e| \quad \left[\begin{array}{l} \because \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log|a| \text{ and } \\ \lim_{x \rightarrow 0} \frac{e^x - 1}{1} = \log|e| \end{array} \right] \\
 &= \log \left| \frac{a}{e} \right| \quad \text{as } \log \frac{m}{n} = \log m - \log n
 \end{aligned}$$

$$\text{or } \lim_{x \rightarrow 0} \frac{a^x - e^x}{x} = \log|a| - 1 \quad \text{as } \log e = 1$$

$$\text{(vi)} \quad \text{Let } I = \lim_{x \rightarrow 0} \frac{e^x - (1 + 2x)}{2^x - 1} = \lim_{x \rightarrow 0} \frac{(e^x - 1) - 2x}{2^x - 1}$$

Dividing numerator and denominator by x , we get

$$I = \lim_{x \rightarrow 0} \frac{\frac{e^x - 1}{x} - 2}{\frac{2^x - 1}{x}} = \frac{\lim_{x \rightarrow 0} \frac{e^x - 1}{x} - 2}{\lim_{x \rightarrow 0} \frac{2^x - 1}{x}} = \frac{1 - 2}{\log 2} = \frac{-1}{\log 2}$$

$$\text{(vii)} \quad \lim_{x \rightarrow 0} \frac{x(1 + 2x)^{1/x} - (e^{2x} - 1)}{x} = \lim_{x \rightarrow 0} \left((1 + 2x)^{1/x} - \frac{e^{2x} - 1}{x} \right)$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} (1 + 2x)^{1/x} - \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x} \\
 &= \lim_{x \rightarrow 0} \left[(1 + 2x)^{\frac{1}{2x}} \right]^2 - \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{2x} \times 2 \\
 &= \left[\lim_{2x \rightarrow 0} (1 + 2x)^{\frac{1}{2x}} \right]^2 - 2 \lim_{2x \rightarrow 0} \frac{e^{2x} - 1}{2x} \\
 &\quad \text{as } x \rightarrow 0 \Rightarrow 2x \rightarrow 0 \\
 &= (e)^2 - 2 \log e \left[\begin{array}{l} \because \lim_{x \rightarrow 0} (1 + x)^{1/x} = e \\ \text{and } \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \log e \end{array} \right] \\
 &= e^2 - 2 \quad \text{as } \log e = 1
 \end{aligned}$$

**Rule to be
Remembered:**

$$\sqrt[n]{f(x)} = [f(x)]^{1/n}$$

$$\begin{aligned}
 \text{E 6) } \lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 3} + \sqrt[4]{x^4 + 2}}{\sqrt[7]{x^7 + x^2} - \sqrt[3]{x^2 + 5}} &= \lim_{x \rightarrow \infty} \frac{x \left(\sqrt{1 + \frac{3}{x^2}} \right) + x \left(\sqrt[4]{1 + \frac{2}{x^4}} \right)}{x \left(\sqrt[7]{1 + \frac{1}{x^5}} \right) - x \left(\sqrt[3]{\frac{1}{x} + \frac{5}{x^3}} \right)} \\
 &= \lim_{x \rightarrow \infty} \frac{\sqrt{1 + \frac{3}{x^2}} + \sqrt[4]{1 + \frac{2}{x^4}}}{\sqrt[7]{1 + \frac{1}{x^5}} - \sqrt[3]{\frac{1}{x} + \frac{5}{x^3}}} \\
 &= \frac{\sqrt{1+0} + \sqrt[4]{1+0}}{\sqrt[7]{1+0} - \sqrt[3]{0+0}} = \frac{1+1}{1-0} = \frac{2}{1} = 2
 \end{aligned}$$

$$\text{E 7) (i) } \lim_{x \rightarrow 0^-} \frac{5x + |x|}{3|x| - 7x}$$

Putting $x = 0 - h$ as $x \rightarrow 0^- \Rightarrow h \rightarrow 0^+$

$$\begin{aligned}
 \lim_{x \rightarrow 0^-} \frac{5x + |x|}{3|x| - 7x} &= \lim_{h \rightarrow 0^+} \frac{5(0 - h) + |0 - h|}{3|0 - h| - 7(0 - h)} = \lim_{h \rightarrow 0^+} \frac{-5h + |-h|}{3|-h| + 7h} \\
 &= \lim_{h \rightarrow 0^+} \frac{-5h + |-1||h|}{3|-1||h| + 7h} = \lim_{h \rightarrow 0^+} \frac{-5h + h}{3h + 7h} = \lim_{h \rightarrow 0^+} \frac{-4h}{10h} \\
 &= \lim_{h \rightarrow 0^+} \frac{-2}{5} = -\frac{2}{5}
 \end{aligned}$$

$$\text{(ii) } \lim_{x \rightarrow 5^+} (3 - |x|)$$

Putting $x = 5 + h$ as $x \rightarrow 5^+ \Rightarrow h \rightarrow 0^+$

$$\begin{aligned}
 \therefore \lim_{x \rightarrow 5^+} (3 - |x|) &= \lim_{h \rightarrow 0^+} (3 - |5 + h|) \\
 &= \lim_{h \rightarrow 0^+} (3 - (5 + h)) \left[\begin{array}{l} \because \text{as } h \rightarrow 0^+ \Rightarrow 5 + h > 5 > 0 \\ \Rightarrow |5 + h| = 5 + h \end{array} \right] \\
 &= \lim_{h \rightarrow 0^+} (-2 - h) = -2 - 0 = -2
 \end{aligned}$$

$$(iii) \lim_{x \rightarrow 0} \frac{x}{|x|}$$

$$\text{L.H.L.} = \lim_{x \rightarrow 0^-} \frac{x}{|x|}$$

Putting $x = 0 - h$ as $x \rightarrow 0^- \Rightarrow h \rightarrow 0^+$

$$\begin{aligned} \text{L.H.L.} &= \lim_{h \rightarrow 0^+} \frac{0 - h}{|0 - h|} = \lim_{h \rightarrow 0^+} \frac{-h}{|-h|} = \lim_{h \rightarrow 0^+} \frac{-h}{-1|h|} \\ &= \lim_{h \rightarrow 0^+} \frac{-h}{h} = \lim_{h \rightarrow 0^+} (-1) = -1 \quad \dots (1) \end{aligned}$$

$$\begin{aligned} \text{R.H.L.} &= \lim_{x \rightarrow 0^+} \frac{x}{|x|} = \lim_{x \rightarrow 0^+} \frac{x}{x} = \lim_{x \rightarrow 0^+} 1 = 1 \quad \dots (2) \\ &\left[\because x \rightarrow 0^+ \Rightarrow x \text{ is slightly greater} \right. \\ &\quad \left. \text{than } 0, \text{ so } |x| = x \right] \end{aligned}$$

From (1) and (2)

L.H.L. $\neq$ R.H.L.

$\therefore \lim_{x \rightarrow 0} \frac{x}{|x|}$ does not exist.

$$\text{E 8) } f(x) = \begin{cases} ax + 3, & x \leq 3 \\ 2(x + 1), & x > 3 \end{cases}$$

$$\text{L.H.L.} = \lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} (ax + 3) = 3a + 3 \quad \dots (1)$$

$$\text{R.H.L.} = \lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} 2(x + 1) = 2(3 + 1) = 8 \quad \dots (2)$$

Since, it is given that $\lim_{x \rightarrow 3} f(x)$ exists.

$\therefore$ we must have

$$\text{L.H.L.} = \text{R.H.L.}$$

$$\Rightarrow 3a + 3 = 8 \Rightarrow 3a = 5 \Rightarrow a = 5/3$$

$$\text{E 9) } f(x) = \begin{cases} 2x - a, & x \geq 0 \\ ax + b + 3, & x < 0 \end{cases} \quad \text{at } x = 0$$

$$\text{L.H.L.} = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (ax + b + 3)$$

Putting $x = 0 - h$ as $x \rightarrow 0^- \Rightarrow h \rightarrow 0^+$

$$\text{L.H.L.} = \lim_{h \rightarrow 0^+} (a(0 - h) + b + 3) = \lim_{h \rightarrow 0^+} (-ah + b + 3) = b + 3$$

$$\text{R.H.L.} = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (2x - a) = 2(0) - a = -a$$

$$\text{Also } f(0) = 2(0) - a = 0 - a = -a$$

Since f is given to be continuous at $x = 0$, so we must have

$$\text{L.H.L.} = \text{R.H.L.} = f(0)$$

$$\Rightarrow b + 3 = -a \Rightarrow a + b + 3 = 0$$

Which is the required relation between a and b.

UNIT 6 DIFFERENTIATION

Structure

- 6.1 Introduction
 - Objectives
- 6.2 Definition of Derivative, its Meaning and Geometrical Interpretation
- 6.3 Derivative at a Point
- 6.4 Derivative by First Principle
- 6.5 Chain Rule
- 6.6 Derivatives of Exponential, Logarithmic, Parametric and Implicit Functions
- 6.7 Derivatives of Higher Orders
- 6.8 Concept of Maxima and Minima
- 6.9 Summary
- 6.10 Solutions/Answers

6.1 INTRODUCTION

In the preceding unit, we have discussed concept of limit and continuity. In fact, the definition of derivative involves these concepts. So, learner must go through the previous unit before starting this unit. Derivatives have large number of applications in the fields of mathematics, statistics, economics, insurance, industrial, health sector, etc.

In this unit, we will present this concept from a very simple and elementary point of view, keeping in mind that learner knows nothing about derivatives. In this unit, we have discussed some examples basically based on the formulae for derivatives of a constant, polynomial, exponential, logarithmic, parametric and implicit functions. Product rule, quotient rule, chain rule have also been discussed. Finally, we close this unit by giving a touch to higher order derivatives and maxima and minima of functions.

Objectives

After completing this unit, you should be able to:

- find derivative of a function at a particular point and at a general point;
- find derivative by first principle;
- find derivative of some commonly used functions;
- apply the chain rule;
- find derivative of exponential, logarithmic, parametric and implicit functions;
- find higher order derivatives; and
- find maxima and minima of a function.

6.2 DEFINITION OF DERIVATIVE, ITS MEANING AND GEOMETRICAL INTERPRETATION

Definition

Let $f : D \rightarrow \mathbb{R}$ be a function, where $D \subseteq \mathbb{R}$,

i.e. f is a real valued function defined on D .

Let $a \in D$ then derivative of f at $x = a$ is denoted by $f'(a)$ and is defined as

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}, \text{ provided limit exists} \quad \dots (1)$$

From definition (1), we see that $f'(a)$ measures the rate at which the function $f(x)$ changes at $x = a$. This is clear from the figure 6.1 given below.

Geometrical Interpretation

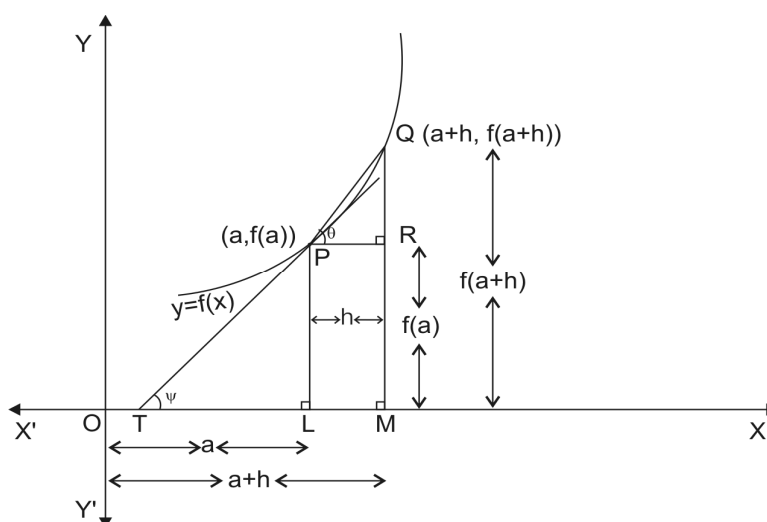


Fig. 6.1

Let PT be the tangent at point P of the curve of the function $y = f(x)$.

Draw $PL \perp OX$, $QM \perp OX$, $PR \perp QM$

Let $OL = a$, $OM = a + h$

$$\therefore PR = LM = OM - OL = a + h - a = h$$

$$\text{and } RQ = MQ - MR = MQ - LP = f(a + h) - f(a)$$

$\therefore (1) \Rightarrow$

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{RQ}{PR} \\ &= \lim_{h \rightarrow 0} \tan \theta \quad \left[\because \text{in } \triangle PQR, \tan \theta = \frac{\text{perpendicular}}{\text{Base}} = \frac{RQ}{PR} \right] \end{aligned}$$

Now as $h \rightarrow 0$, chord PQ tends to coincide with the tangent at point P,

i.e. as $h \rightarrow 0 \Rightarrow \theta \rightarrow \psi$

$$\therefore f'(a) = \lim_{h \rightarrow 0} \tan \theta = \tan \psi$$

i.e. $f'(a) = \tan \psi$

i.e. (derivative at point $x = a$) = (tangent of the angle which the tangent line at $x = a$ makes with +ve direction of x-axis)

In fact, if a line makes an angle θ with position direction of x-axis, then value of $\tan \theta$ is known as slope of the line.

Thus in mathematical language we can say

(Derivative at a point) $_{x=a}$ is the slope of the tan gent at that po int. ... (2)

i.e. we can say that derivative measures the rate at which the tangent to the curve at point $x = a$ is changing

Meaning

Rewriting (1)

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \quad \dots (3)$$

From the knowledge of previous unit, we know that limit in R.H.S. of (1) or (3) exists if

$$\lim_{h \rightarrow 0^-} \frac{f(a+h) - f(a)}{h} \text{ and } \lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h} \text{ both exist and are equal.}$$

In view of (2), we have, limit in (1) exists if

$$\left(\begin{array}{l} \text{Slope of the tan gent to the left} \\ \text{of the point } x = a \end{array} \right) = \left(\begin{array}{l} \text{Slope of the tan gent to the right} \\ \text{of the point } x = a \end{array} \right)$$

i.e. limit in (1) exists if $x = a$ is not a corner point.

i.e. $f'(a)$ does not exists at corner points. ... (4)

For example, consider the function

$$f(x) = |x|$$

See the graph of this function in Fig. 6.2 .We observe that $x = 0$ is a corner point in its graph.

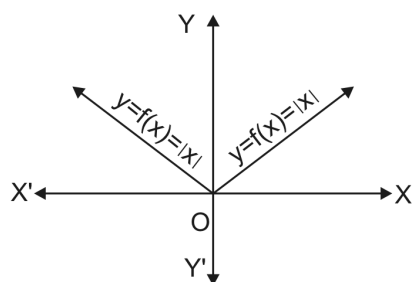


Fig . 6.2

So $f(x) = |x|$ is not differentiable at $x = 0$. but there is no other corner point in its graph, so it is differentiable at all points of the domain except $x = 0$.

Remark1:

- (i) The last paragraph is very useful to understand the concept of derivative for those learners, who do not have mathematical background.
- (ii) However, the units have been written keeping in mind that the learner has no mathematical background after 10th standard.
- (iii) In the definition of derivative of a function at a point given by (1), we see that in order to find the derivative at the said point, we have to evaluate the limit in R.H.S. But sometimes functions may have different values for $h \rightarrow 0^-$ and $h \rightarrow 0^+$. In such cases like modulus function or where there is break for function in order to evaluate the limit we have to

find out the L.H.L. and R.H.L. as we have done in the previous unit. But in the definition of derivative these L.H.L. and R.H.L. are known as left hand derivative (L.H.D.) and right hand derivative (R.H.D.) respectively.

A function is said to have derivative at a point if L.H.D. and R.H.D. both exist and are equal at that point, i.e. L.H.D. = R.H.D.

We denote L.H.D. of the function $f(x)$ at $x = a$ by $L(f'(a))$ and R.H.D. of the function $f(x)$ at $x = a$ by $R(f'(a))$. See Example 2 of this unit for more clarity.

6.3 DERIVATIVE AT A POINT

Here, we give some examples which will illustrate the idea as to how we calculate derivative of a function at a point.

Example 1: Find the derivative of the following functions at the indicated points:

(i) $f(x) = a$, at $x = 5$, where a is a real constant

(ii) $f(x) = ax + b$, at $x = 2$, $a \neq 0$

(iii) $f(x) = ax^2 + bx + c$, at $x = 3$, $a \neq 0$

(iv) $f(x) = \frac{1}{x}$, at $x = 1$

Solution:

(i) $f(x) = a$, where a is real constant

By definition

$$f'(5) = \lim_{h \rightarrow 0} \frac{f(5+h) - f(5)}{h} = \lim_{h \rightarrow 0} \frac{a - a}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = \lim_{h \rightarrow 0} 0 = 0$$

(ii) $f(x) = ax + b$, $a \neq 0$

By definition

$$\begin{aligned} f'(2) &= \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{a(2+h) + b - (2a + b)}{h} = \lim_{h \rightarrow 0} \frac{ah}{h} = \lim_{h \rightarrow 0} a = a \end{aligned}$$

(iii) $f(x) = ax^2 + bx + c$, $a \neq 0$

By definition

$$\begin{aligned} f'(3) &= \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} = \lim_{h \rightarrow 0} \frac{a(3+h)^2 + b(3+h) + c - (9a + 3b + c)}{h} \\ &= \lim_{h \rightarrow 0} \frac{ah^2 + 6ah + bh}{h} = \lim_{h \rightarrow 0} (ah + 6a + b) = 6a + b \end{aligned}$$

(iv) $f(x) = \frac{1}{x}$

By definition

$$\begin{aligned} f'(1) &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{1+h} - \frac{1}{1}}{h} \\ &= \lim_{h \rightarrow 0} \frac{1 - (1+h)}{h(1+h)} = \lim_{h \rightarrow 0} \frac{-h}{h(1+h)} = -\lim_{h \rightarrow 0} \frac{1}{1+h} = -\frac{1}{1+0} = -1 \end{aligned}$$

Here are some exercises for you.

Diffrenciation

E 1) Find the derivative of the following functions at the indicated points

(i) $f(x) = x^3 + x + 1$, at $x = -1$

(ii) $f(x) = 2 - 3x^2$, at $x = 1/2$

E 2) Find the value of a , if $f'(-2) = 3$, where $f(x) = 2x^2 - 3ax + 5$

Example 2: Find the derivative (if exists) of the following functions at the indicated points.

(i) $f(x) = |x|$ at $x = 0$

(ii) $f(x) = \begin{cases} 5 + 2x, & x \geq 1 \\ 9 - 2x, & x < 1 \end{cases}$ at $x = 1$

Solution:

(i) By definition

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{|0+h| - |0|}{h} = \lim_{h \rightarrow 0} \frac{|h| - 0}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h}$$

We note that to deal with $|h|$ we must know the sign of h in advance. So We must have to calculate L.H.D. and R.H.D. separately.

$$\text{L.H.D.} = \lim_{h \rightarrow 0^-} \frac{|h|}{h}$$

Putting $h = 0 - k$ as $h \rightarrow 0^- \Rightarrow k \rightarrow 0^+$

$$\begin{aligned} \text{L.H.D.} &= \lim_{k \rightarrow 0^+} \frac{|0-k|}{0-k} = \lim_{k \rightarrow 0^+} \frac{|-k|}{-k} = \lim_{k \rightarrow 0^+} \frac{|-1| \times |k|}{-k} = \lim_{k \rightarrow 0^+} \frac{k}{-k} \\ &= \lim_{k \rightarrow 0^+} (-1) = -1 \end{aligned} \quad \dots (1)$$

$$\text{R.H.D.} = \lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = \lim_{h \rightarrow 0^+} (1) = 1 \quad \dots (2)$$

From (1) and (2), we have

L.H.D. $\neq$ R.H.D.

$\Rightarrow f'(0)$ does not exists.

(ii) By definition

$$f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{f(1+h) - (5 + 2 \times 1)}{h} = \lim_{h \rightarrow 0} \frac{f(1+h) - 7}{h}$$

We note that function have different values for $x < 1$ and $x > 1$, so we must have to calculate L.H.D. and R.H.D. separately.

$$\begin{aligned} L(f'(1)) &= \text{L.H.D.} = \lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0^-} \frac{9 - 2(1+h) - 7}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{-2h}{h} = \lim_{h \rightarrow 0^-} (-2) = -2 \end{aligned} \quad \dots (1)$$

$$\begin{aligned} R(f'(1)) &= \text{R.H.D.} = \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0^+} \frac{5 + 2(1+h) - 7}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{2h}{h} = \lim_{h \rightarrow 0^+} (2) = 2 \end{aligned} \quad \dots (2)$$

From (1) and (2)

$L(f'(1)) \neq R(f'(1))$

$\Rightarrow f'(1)$ does not exists.

Remark 2: In part (i) $x=0$ is a corner point (see Fig. 6.2) that is why its derivative did not exist at this point, which was indicated in equation (4) in Sec. 6.2. Same is the case in part (ii).

6.4 DERIVATIVE BY FIRST PRINCIPLE

In section 6.3 of this unit, we have discussed as to how we calculate the derivative of a function at a given point $x = a$ (say). Suppose we want to calculate the derivative at 10 points, then using the definition 10 times is a very time consuming and lengthy procedure. To get rid of this difficulty, we will introduce a procedure in this section which will provide us the derivative of the function at a general point. After calculating the derivative at the general point we can replace this point by any number of points very quickly (provided derivative at the required point exists). Let us first describe the procedure as to how we calculate the derivative at a general point. After this we shall give some results to get a good understanding of the procedure. This process of finding derivative is known as derivative by **first principle** or by **definition** or by **delta** method or **ab-initio**.

Let us explain the procedure of first principle for the function

$$y = f(x) \quad \dots (1)$$

in the following steps.

Step I Let δx be the small increment (+ve or -ve) in the value of x and δy be the corresponding increment in the value of y .

$\therefore$ (1) becomes

$$y + \delta y = f(x + \delta x) \quad \dots (2)$$

Step II (2) – (1) gives

$$\delta y = f(x + \delta x) - f(x) \quad \dots (3)$$

Step III First we simplify the expression in the R.H.S. of (3). After simplifying the expression, we divide both sides by δx and get

$$\frac{\delta y}{\delta x} = \frac{f(x + \delta x) - f(x)}{\delta x}$$

Step IV Proceeding limit as $\delta x \rightarrow 0$ on both sides

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x} \quad \dots (4)$$

Step V The term in L.H.S. of (4) is denoted by $\frac{dy}{dx}$ and limit in R.H.S. of (4)

is evaluated using suitable formula discussed in the previous unit

$$\text{i.e. } \frac{dy}{dx} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x} \quad \dots (5)$$

The expression obtained after simplification of the R.H.S. of (5), is derivative of y w.r.t. x at a general point x .

If we want the derivative of the function $y = f(x)$ at a particular point $x = a$ (say), then replace x by a in the result.

Some Results

Result 1: Find the derivative of the constant function given by $f(x) = k$, where k is a real constant by using first principle.

Solution: Let $y = f(x) = k \quad \dots (1)$

Step I Let δx be the small increment in the value of x and δy be the corresponding increment in the value of y .

$\therefore$ (1) becomes

$$y + \delta y = k \quad \dots (2)$$

Step II (2) – (1) gives

$$y + \delta y - y = k - k$$

$$\text{Or } \delta y = 0 \quad \dots (3)$$

Step III Dividing on both sides of (3) by δx

$$\frac{\delta y}{\delta x} = \frac{0}{\delta x} = 0 \Rightarrow \frac{\delta y}{\delta x} = 0$$

Step IV Proceeding limit as $\delta x \rightarrow 0$, we get

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} 0$$

$$\text{Step V } \frac{dy}{dx} = 0$$

$$\text{i.e. } \frac{d}{dx}(k) = 0.$$

So, derivative of a constant function is zero.

Result 2: Find the derivatives of the functions given by

$$(i) f(x) = x^2 \quad (ii) f(x) = x^3$$

by using first principle.

$$\text{Solution: (i) Let } y = f(x) = x^2 \quad \dots (1)$$

Step I Let δx be the small increment in the value of x and δy be the corresponding increment in the value of y .

$\therefore$ (1) becomes

$$y + \delta y = (x + \delta x)^2 = x^2 + (\delta x)^2 + 2x \times \delta x \quad \dots (2)$$

Step II (2) – (1) gives

$$\delta y = x^2 + (\delta x)^2 + 2x \times \delta x - x^2 = (\delta x)^2 + 2x \times \delta x$$

Step III Dividing on both sides by δx , we get

$$\frac{\delta y}{\delta x} = \frac{(\delta x)^2 + 2x \times \delta x}{\delta x} = \delta x + 2x$$

Step IV Proceeding limit as $\delta x \rightarrow 0$, we get

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} (\delta x + 2x)$$

$$\text{Step V } \frac{dy}{dx} = \lim_{\delta x \rightarrow 0} \delta x + \lim_{\delta x \rightarrow 0} 2x = 0 + 2x = 2x$$

$$\text{i.e. } \frac{d}{dx}(x^2) = 2x = 2x^{2-1}$$

Second Method

$$\text{Let } y = f(x) = x^2 \quad \dots (1)$$

Step I Let δx be the small increment in the value of x and δy be the corresponding increment in the value of y .

$\therefore$ (1) becomes

$$y + \delta y = (x + \delta x)^2 \quad \dots (2)$$

Step II (2) – (1) gives

$$\delta y = (x + \delta x)^2 - x^2$$

Step III Dividing on both sides by δx , we get

$$\frac{\delta y}{\delta x} = \frac{(x + \delta x)^2 - x^2}{\delta x} = \frac{(x + \delta x)^2 - x^2}{(x + \delta x) - x}$$

Step IV Proceeding limit as $\delta x \rightarrow 0$, we get

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{(x + \delta x)^2 - x^2}{(x + \delta x) - x}$$

Step V $\frac{dy}{dx} = \lim_{x+\delta x \rightarrow x} \frac{(x + \delta x)^2 - x^2}{(x + \delta x) - x}$ as $\delta x \rightarrow 0 \Rightarrow x + \delta x \rightarrow x$

$$\text{i.e. } \frac{d}{dx}(x^2) = 2x^{2-1} = 2x \quad \left[\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right]$$

(ii) Let $y = f(x) = x^3 \quad \dots (1)$

Step I Let δx be the small increment in the value of x and δy be the corresponding increment in the value of y .

$\therefore$ (1) becomes

$$y + \delta y = (x + \delta x)^3 = x^3 + 3x^2 \times \delta x + 3x \times (\delta x)^2 + (\delta x)^3 \quad \dots (2)$$

Step II (2) – (1) gives

$$\begin{aligned} \delta y &= x^3 + 3x^2 \times \delta x + 3x \times (\delta x)^2 + (\delta x)^3 - x^3 \\ &= 3x^2 \times \delta x + 3x \times (\delta x)^2 + (\delta x)^3 \end{aligned}$$

Step III Dividing on both sides by δx , we get

$$\frac{\delta y}{\delta x} = \frac{3x^2 \times \delta x + 3x \times (\delta x)^2 + (\delta x)^3}{\delta x} = 3x^2 + 3x \times \delta x + (\delta x)^2$$

Step IV Proceeding limit as $\delta x \rightarrow 0$, we get

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} (3x^2 + 3x \times \delta x + (\delta x)^2)$$

Step V $\frac{dy}{dx} = \lim_{\delta x \rightarrow 0} 3x^2 + \lim_{\delta x \rightarrow 0} 3x \times \delta x + \lim_{\delta x \rightarrow 0} (\delta x)^2 = 3x^2 + 0 + 0 = 3x^2$

$$\text{i.e. } \frac{d}{dx}(x^3) = 3x^2 = 3x^{3-1}$$

Second Method

Let $y = f(x) = x^3 \quad \dots (1)$

Step I Let δx be the small increment in the value of x and δy be the corresponding increment in the value of y .

$\therefore$ (1) becomes

$$y + \delta y = (x + \delta x)^3 \quad \dots (2)$$

Step II (2) – (1) gives

$$\delta y = (x + \delta x)^3 - x^3$$

Step III Dividing on both sides by δx , we get

$$\frac{\delta y}{\delta x} = \frac{(x + \delta x)^3 - x^3}{\delta x} = \frac{(x + \delta x)^3 - x^3}{(x + \delta x) - x}$$

Step IV Proceeding limit as $\delta x \rightarrow 0$, we get

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{(x + \delta x)^3 - x^3}{(x + \delta x) - x}$$

Step V $\frac{dy}{dx} = \lim_{x+\delta x \rightarrow x} \frac{(x + \delta x)^3 - x^3}{(x + \delta x) - x}$ as $\delta x \rightarrow 0 \Rightarrow x + \delta x \rightarrow x$

$$\text{i.e. } \frac{d}{dx}(x^3) = 3x^{3-1} \quad \left[\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right]$$

Similarly, $\frac{d}{dx}(x^n) = nx^{n-1}$

Result 3: Find the derivative of the function given by

$$f(x) = (ax + b)^2, \quad \text{where } a, b \text{ are real constants and } a \neq 0$$

by using first principle.

Solution: Let us use second method here.

$$\text{Let } y = f(x) = (ax + b)^2 \quad \dots (1)$$

Step I Let δx be the small increment in the value of x and δy be the corresponding increment in the value of y .

$\therefore (1)$ becomes

$$y + \delta y = [a(x + \delta x) + b]^2 = (ax + a\delta x + b)^2 \quad \dots (2)$$

Step II $(2) - (1)$ gives

$$\delta y = (ax + a\delta x + b)^2 - (ax + b)^2$$

Step III Dividing on both sides by δx , we get

$$\begin{aligned} \frac{\delta y}{\delta x} &= \frac{(ax + a\delta x + b)^2 - (ax + b)^2}{\delta x} = a \left[\frac{(ax + a\delta x + b)^2 - (ax + b)^2}{a\delta x} \right] \\ &= a \left[\frac{(ax + a\delta x + b)^2 - (ax + b)^2}{(ax + a\delta x + b) - (ax + b)} \right] \end{aligned}$$

Step IV Proceeding limit as $\delta x \rightarrow 0$, we get

$$\begin{aligned} \lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} &= a \lim_{\delta x \rightarrow 0} \left[\frac{(ax + a\delta x + b)^2 - (ax + b)^2}{(ax + a\delta x + b) - (ax + b)} \right] \\ &= a \lim_{a\delta x \rightarrow 0} \left[\frac{(ax + a\delta x + b)^2 - (ax + b)^2}{(ax + a\delta x + b) - (ax + b)} \right] \quad \text{as } \delta x \rightarrow 0 \Rightarrow a\delta x \rightarrow 0 \end{aligned}$$

$$\text{Step V } \frac{dy}{dx} = a \lim_{ax+a\delta x+b \rightarrow ax+b} \left[\frac{(ax + a\delta x + b)^2 - (ax + b)^2}{(ax + a\delta x + b) - (ax + b)} \right]$$

$$= a \times 2(ax + b)^{2-1} \quad \left[\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right]$$

$$\text{i.e. } \frac{d}{dx}(ax + b)^2 = 2a(ax + b)^{2-1}$$

Similarly, we can easily obtain

$$\frac{d}{dx}(ax + b)^n = na(ax + b)^{n-1}$$

Result 4: Find the derivative of the exponential function $f(x) = e^{ax}$ by using first principle.

Solution: Let $y = f(x) = e^{ax}$... (1)

Step I Let δx be the small increment in the value of x and δy be the corresponding increment in the value of y .

$\therefore$ (1) becomes

$$y + \delta y = e^{a(x+\delta x)} = e^{ax+a\delta x} \quad \dots (2)$$

Step II (2) – (1) gives

$$\begin{aligned} \delta y &= e^{ax+a\delta x} - e^{ax} = e^{ax} e^{a\delta x} - e^{ax} \quad [\because a^{m+n} = a^m a^n] \\ &= e^{ax} (e^{a\delta x} - 1) \end{aligned}$$

Step III Dividing on both sides by δx , we get

$$\frac{\delta y}{\delta x} = e^{ax} \frac{(e^{a\delta x} - 1)}{\delta x} = ae^{ax} \left(\frac{e^{a\delta x} - 1}{a\delta x} \right)$$

Step IV Proceeding limit as $\delta x \rightarrow 0$, we get

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} ae^{ax} \left(\frac{e^{a\delta x} - 1}{a\delta x} \right)$$

Step V $\frac{dy}{dx} = ae^{ax} \lim_{a\delta x \rightarrow 0} \frac{e^{a\delta x} - 1}{a\delta x}$ as $\delta x \rightarrow 0 \Rightarrow a\delta x \rightarrow 0$

$$= ae^{ax} (1) \quad \left[\because \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \right]$$

$$\text{i.e. } \frac{d}{dx}(e^{ax}) = ae^{ax}$$

Result 5: Find the derivative of the logarithm function $f(x) = \log_a x$ by using first principle.

Solution: Let $y = f(x) = \log_a x$... (1)

Step I Let δx be the small increment in the values of x and δy be the corresponding increment in the value of y

$\therefore$ (1) becomes

$$y + \delta y = \log_a (x + \delta x) \quad \dots (2)$$

Step II (2) – (1) gives

$$\begin{aligned} \delta y &= \log_a (x + \delta x) - \log_a x = \log_a \left(\frac{x + \delta x}{x} \right) \quad \left[\because \log m - \log n = \log \frac{m}{n} \right] \\ &= \log_a \left(1 + \frac{\delta x}{x} \right) \end{aligned}$$

Step III Dividing on both sides by δx , we get

Diffrenciation

$$\begin{aligned}\frac{\delta y}{\delta x} &= \frac{\log_a \left(1 + \frac{\delta x}{x}\right)}{\delta x} = \frac{1}{x} \frac{x}{\delta x} \log_a \left(1 + \frac{\delta x}{x}\right) \\ &= \frac{1}{x} \log_a \left(1 + \frac{\delta x}{x}\right)^{\frac{x}{\delta x}} \quad [\because n \log m = \log m^n]\end{aligned}$$

Step IV Proceeding limit as $\delta x \rightarrow 0$, we get

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{1}{x} \log_a \left(1 + \frac{\delta x}{x}\right)^{\frac{x}{\delta x}}$$

Step V $\frac{dy}{dx} = \frac{1}{x} \lim_{\frac{\delta x}{x} \rightarrow 0} \log_a \left(1 + \frac{\delta x}{x}\right)^{\frac{x}{\delta x}} \quad \text{as } \delta x \rightarrow 0 \Rightarrow \frac{\delta x}{x} \rightarrow 0$

$$\begin{aligned}&= \frac{1}{x} \log_a \left[\lim_{\frac{\delta x}{x} \rightarrow 0} \left(1 + \frac{\delta x}{x}\right)^{\frac{x}{\delta x}} \right] \quad \left[\because \log \text{arithm is a continuous function on its domain} \right] \\ &= \frac{1}{x} \log_a e \quad \left[\because \lim_{x \rightarrow 0} (1 + x)^{1/x} = e \right]\end{aligned}$$

If a function is **continuous** then it **respects limit** i.e. if a function f is continuous and a is point of its domain, then
 $\lim_{x \rightarrow a} f(x) = f(\lim_{x \rightarrow a} x) = f(a)$
 i.e. limit can be taken inside the function.
 i.e. Role of limit and function can be interchanged.

i.e. $\frac{d}{dx}(\log_a x) = \frac{1}{x} \log_a e$

In particular, if base of the logarithmic is e in place of a , then

$$\frac{d}{dx}(\log_e x) = \frac{1}{x} \log_e e = \frac{1}{x} \quad \text{as } \log_e e = 1$$

Remark 3: Keep all these formulae put in the rectangular boxes always in mind, as we will use these formulae later on as standard results.

Some more Formulae of Finding Derivatives:

If u and v are functions of x , then

(i) $\frac{d}{dx}(cu) = c \frac{du}{dx}$, where c is a real constant

(ii) $\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$

(iii) $\frac{d}{dx}(u.v) = u \frac{dv}{dx} + v \frac{du}{dx}$ (Known as **Product Rule**)

(iv) $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$ (Known as **Quotient Rule**)

Remark 4: Aim of this unit from learners point of view is not to focus on the derivations of the formulae. But main aim of this unit is able to make the learners user friendly as to how these results can be used whenever we encounter a situation where derivative is involved. That is why we will not provide the derivations of more formulae.

Various formulae which are used in many practical situations are listed below:

S. No	Function	Derivative of the function
1	k (constant)	$\frac{d}{dx}(k) = 0$
2	x^n	$\frac{d}{dx}(x^n) = nx^{n-1}$
3	$(ax + b)^n$	$\frac{d}{dx}(ax + b)^n = na(ax + b)^{n-1}$
4	Exponential function (i) a^{bx} (ii) e^{bx}	(i) $\frac{d}{dx}(a^{bx}) = ba^{bx} \log a$ (ii) $\frac{d}{dx}(e^{bx}) = be^{bx}$
5	Logarithmic function (i) $\log_a x$ (ii) $\log_e x$	(i) $\frac{d}{dx}(\log_a x) = \frac{1}{x} \log_a e$ (ii) $\frac{d}{dx}(\log_e x) = \frac{1}{x}$
6	cu, where c is constant and u is a function of x	$\frac{d}{dx}(cu) = c \frac{d}{dx}(u)$
7	(i) $u \pm v$ (ii) uv (iii) $\frac{u}{v}$ where, u, v are functions of x.	(i) $\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$ (ii) $\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$ (Product Rule) (iii) $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$ (Quotient Rule)
8	$[f(x)]^n$, n is +ve or -ve real number	$\frac{d}{dx}[f(x)]^n = n[f(x)]^{n-1} \frac{d}{dx}(f(x))$
9	$\frac{1}{f(x)}$	$\frac{d}{dx}\left(\frac{1}{f(x)}\right) = \frac{d}{dx}(f(x))^{-1} = \frac{-1}{[f(x)]^2} \frac{d}{dx}(f(x))$
10	$y = f(u)$ $u = g(w)$ $w = h(x)$	$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dw} \frac{dw}{dx}$ (Chain Rule)
11	Parametric functions $x = f(t)$ $y = g(t)$	$\frac{dy}{dx} = \frac{dy}{dt} \bigg/ \frac{dx}{dt}$
12	Polynomial function $f(x) = a_0x^n + a_1x^{n-1} + \dots$ $+ a_{n-1}x + a_n$	$\frac{d}{dx}(f(x)) = na_0x^{n-1} + (n-1)a_1x^{n-2} + \dots + a_{n-1}$

Now we take some examples. We will write “Diff. w.r.t x” in place of “differentiating with respect to x”.

Example 3: Find derivative of the following functions:

- (i) 5 (ii) $\alpha + \beta$ (iii) $\frac{1}{\sqrt{11}}$ (iv) $\frac{-\pi}{\sqrt{17}}$ (v) x^{11}
- (vi) $x^{5/2}$ (vii) $\frac{1}{\sqrt[3]{x}}$ (viii) $\frac{1}{x^7}$ (ix) $\frac{1}{x}$ (x) $\sqrt{x}$
- (xi) $(2x+5)^3$ (xii) $(4-3x)^8$ (xiii) $\left(5-\frac{3}{2}x\right)^{4/9}$ (xiv) $\sqrt{2+3x}$
- (xv) $x^5 + x^2 + 1$ (xvi) $\frac{x^3 + x^2 + 1}{x}$ (xvii) $\left(x^2 + \frac{1}{x^2}\right)^2$ (xviii) $\left(x + \frac{1}{x}\right)^{100}$
- (xix) $(x^2 + 1)(x - 1)$ (xx) $x^{-3}(1 + x^2 + x^5 + x^8)$
- (xxi) $(x^2 + 1)(x^3 + x^2 + 1)$ (xxii) $(4x + 1)^3(7x + 1)^4$
- (xxiii) $(x + 2)^2(x + 3)^4(x + 1)^5$

Solution:

- (i) Let $y = 5$

Diff. w.r.t. x

$$\frac{dy}{dx} = 0 \quad \left[\begin{array}{l} \because 5 \text{ is a constant and derivative} \\ \text{of a constant function is zero.} \end{array} \right]$$

- (ii) Let $y = \alpha + \beta$

Diff. w.r.t. x

$$\frac{dy}{dx} = 0 \quad \left[\begin{array}{l} \because \alpha \text{ and } \beta \text{ both are constants } \Rightarrow \alpha + \beta \text{ is constant} \\ \text{and derivative of a constant function is zero.} \end{array} \right]$$

- (iii) Let $y = \frac{1}{\sqrt{11}}$

Diff. w.r.t. x

$$\frac{dy}{dx} = 0 \quad \left[\begin{array}{l} \because \frac{1}{\sqrt{11}} \text{ is a constant and derivative} \\ \text{of a constant function is zero.} \end{array} \right]$$

- (iv) Let $y = \frac{-\pi}{\sqrt{17}}$

Diff. w.r.t. x

$$\frac{dy}{dx} = 0 \quad \left[\begin{array}{l} \because \frac{-\pi}{\sqrt{17}} \text{ is a constant and derivative} \\ \text{of a constant function is zero.} \end{array} \right]$$

- (v) Let $y = x^{11}$

Diff. w.r.t. x

$$\frac{dy}{dx} = 11x^{11-1} = 11x^{10} \quad \left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

- (vi) Let $y = x^{5/2}$

Diff. w.r.t. x

$$\frac{dy}{dx} = \frac{5}{2}x^{\frac{5}{2}-1} = \frac{5}{2}x^{\frac{3}{2}} \quad \left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

(vii) Let $y = \frac{1}{\sqrt[3]{x}} = \frac{1}{x^{1/3}} = x^{-1/3}$

Diff. w.r.t. x

$$\frac{dy}{dx} = -\frac{1}{3}x^{-4/3} \quad \left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

(viii) Let $y = \frac{1}{x^7} = x^{-7}$

Diff. w.r.t. x

$$\frac{dy}{dx} = -7x^{-8} = -\frac{7}{x^8} \quad \left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

(ix) Let $y = \frac{1}{x} = x^{-1}$

Diff. w.r.t. x

$$\frac{dy}{dx} = (-1)x^{-2} = -\frac{1}{x^2} \quad \left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

(x) Let $y = \sqrt{x} = x^{1/2}$

Diff. w.r.t. x

$$\frac{dy}{dx} = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}} \quad \left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

(xi) Let $y = (2x+5)^3$

Diff. w.r.t. x

$$\frac{dy}{dx} = 3(2x+5)^2(2) = 6(2x+5)^2 \quad \left[\because \frac{d}{dx}((ax+b)^n) = n(ax+b)^{n-1}a \right]$$

(xii) Let $y = (4-3x)^8$

Diff. w.r.t. x

$$\frac{dy}{dx} = 8(4-3x)^7(-3) = -24(4-3x)^7 \quad \left[\because \frac{d}{dx}((ax+b)^n) = n(ax+b)^{n-1}a \right]$$

(xiii) Let $y = \left(5 - \frac{3}{2}x\right)^{4/9}$

Diff. w.r.t. x

$$\begin{aligned} \frac{dy}{dx} &= \frac{4}{9} \left(5 - \frac{3}{2}x\right)^{-5/9} \left(-\frac{3}{2}\right) \quad \left[\because \frac{d}{dx}((ax+b)^n) = n(ax+b)^{n-1}a \right] \\ &= -\frac{2}{3} \left(5 - \frac{3}{2}x\right)^{-5/9} \end{aligned}$$

(xiv) Let $y = \sqrt{2+3x} = (2+3x)^{1/2}$

Diff. w.r.t. x

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2}(2+3x)^{-1/2}(3) \quad \left[\because \frac{d}{dx}((ax+b)^n) = n(ax+b)^{n-1}a \right] \\ &= \frac{3}{2\sqrt{2+3x}} \end{aligned}$$

(xv) Let $y = x^5 + x^2 + 1$

Diff. w.r.t. x

$$\frac{dy}{dx} = 5x^4 + 2x + 0 = 5x^4 + 2x \quad \left[\text{Using formula written at serial number 12 of the table of formulae} \right]$$

(xvi) Let $y = \frac{x^3 + x^2 + 1}{x} = \frac{x^3}{x} + \frac{x^2}{x} + \frac{1}{x} = x^2 + x + \frac{1}{x}$

$$y = x^2 + x + \frac{1}{x}$$

Diff. w.r.t. x

$$\frac{dy}{dx} = \frac{d}{dx}(x^2) + \frac{d}{dx}(x) + \frac{d}{dx}\left(\frac{1}{x}\right) = 2x + 1 - \frac{1}{x^2} \quad \left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

(xvii) Let $y = \left(x^2 + \frac{1}{x^2}\right)^2$

Diff. w.r.t. x

$$\begin{aligned} \frac{dy}{dx} &= 2\left(x^2 + \frac{1}{x^2}\right) \frac{d}{dx}\left(x^2 + \frac{1}{x^2}\right) \quad \left[\because \frac{d}{dx}\left((f(x))^n\right) = n(f(x))^{n-1} \frac{d}{dx}(f(x)) \right] \\ &= 2\left(x^2 + \frac{1}{x^2}\right) \left(2x - \frac{2}{x^3}\right) = 4\left(x^3 - \frac{1}{x} + \frac{1}{x} - \frac{1}{x^5}\right) = 4\left(x^3 - \frac{1}{x^5}\right) \end{aligned}$$

(xviii) Let $y = \left(x + \frac{1}{x}\right)^{100}$

Diff. w.r.t. x

$$\begin{aligned} \frac{dy}{dx} &= 100\left(x + \frac{1}{x}\right)^{99} \frac{d}{dx}\left(x + \frac{1}{x}\right) \quad [\text{Same reason as given in (xvii)}] \\ &= 100\left(x + \frac{1}{x}\right)^{99} \left(1 - \frac{1}{x^2}\right) \end{aligned}$$

(xix) Let $y = (x^2 + 1)(x - 1) = x^3 - x^2 + x - 1$

Diff. w.r.t. x

$$\begin{aligned} \frac{dy}{dx} &= 3x^2 - 2x + 1 - 0 \quad \left[\text{Using formula written at serial number 12 of the table of formulae.} \right] \\ &= 3x^2 - 2x + 1 \end{aligned}$$

Alternatively: Using Product Rule

$$\begin{aligned} \frac{dy}{dx} &= (x^2 + 1) \frac{d}{dx}(x - 1) + (x - 1) \frac{d}{dx}(x^2 + 1) \\ &= (x^2 + 1)(1 - 0) + (x - 1)(2x + 0) \\ &= x^2 + 1 + 2x^2 - 2x = 3x^2 - 2x + 1 \end{aligned}$$

(xx) Let $y = x^{-3}(1 + x^2 + x^5 + x^8)$

$$y = x^{-3} + x^{-1} + x^2 + x^5$$

Diff. w.r.t. x

$$\begin{aligned} \frac{dy}{dx} &= -3x^{-4} - 1x^{-2} + 2x + 5x^4 \quad \left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right] \\ &= -3x^{-4} - x^{-2} + 2x + 5x^4 \end{aligned}$$

(xxi) Let $y = (x^2 + 1)(x^3 + x^2 + 1)$

Diff. w.r.t. x [Using Product Rule]

$$\begin{aligned}\frac{dy}{dx} &= (x^2 + 1) \frac{d}{dx}(x^3 + x^2 + 1) + (x^3 + x^2 + 1) \frac{d}{dx}(x^2 + 1) \\ &= (x^2 + 1)(3x^2 + 2x) + (x^3 + x^2 + 1)(2x) \\ &= 3x^4 + 2x^3 + 3x^2 + 2x + 2x^4 + 2x^3 + 2x \\ &= 5x^4 + 4x^3 + 3x^2 + 4x\end{aligned}$$

(xxii) Let $y = (4x + 1)^3(7x + 1)^4$

Diff. w.r.t. x [Using Product rule]

$$\begin{aligned}\frac{dy}{dx} &= (4x + 1)^3 \frac{d}{dx}(7x + 1)^4 + (7x + 1)^4 \frac{d}{dx}(4x + 1)^3 \\ &= (4x + 1)^3 4(7x + 1)^3 + (7x + 1)^4 3(4x + 1)^2 \\ &= 4(4x + 1)^2(7x + 1)^3[7(4x + 1) + 3(7x + 1)] \\ &= 4(4x + 1)^2(7x + 1)^3(49x + 10)\end{aligned}$$

(xxiii) Let $y = (x + 2)^2(x + 3)^4(x + 1)^5$

Diff. w.r.t. x

$$\begin{aligned}\frac{dy}{dx} &= (x + 2)^2(x + 3)^4 \frac{d}{dx}(x + 1)^5 + (x + 2)^2(x + 1)^5 \frac{d}{dx}(x + 3)^4 \\ &\quad + (x + 3)^4(x + 1)^5 \frac{d}{dx}(x + 2)^2\end{aligned}$$

$$\left[\because \text{if } u, v, w, \text{ are functions of } x, \text{ then} \right. \\ \left. \frac{d}{dx}(uvw) = uv \frac{d}{dx}(w) + uw \frac{d}{dx}(v) + vw \frac{d}{dx}(u) \right]$$

$$\begin{aligned}&= (x + 2)^2(x + 3)^4 5(x + 1)^4 + (x + 2)^2(x + 1)^5 4(x + 3)^3 \\ &\quad + (x + 3)^4(x + 1)^5 2(x + 2) \\ &= (x + 2)(x + 3)^3(x + 1)^4 [5(x + 2)(x + 3) + 4(x + 2)(x + 1) + 2(x + 3)(x + 1)] \\ &= (x + 2)(x + 3)^3(x + 1)^4 [5(x^2 + 5x + 6) + 4(x^2 + 3x + 2) + 2(x^2 + 4x + 3)] \\ &= (x + 2)(x + 3)^2(x + 1)^4 (11x^2 + 45x + 44)\end{aligned}$$

Example 4: Find the derivative of the following functions:

(i) $\frac{x+1}{x-1}$ (ii) $\frac{8x+3}{6-5x}$ (iii) $\frac{a^2}{x^2+a^2}$ (iv) $\sqrt{\frac{x^2+1}{x+1}}$

Solution:

(i) Let $y = \frac{x+1}{x-1}$

Diff. w.r.t. x

$$\begin{aligned}\frac{dy}{dx} &= \frac{(x-1) \frac{d}{dx}(x+1) - (x+1) \frac{d}{dx}(x-1)}{(x-1)^2} \quad [\text{Using Quotient Rule}] \\ &= \frac{(x-1) \cdot 1 - (x+1) \cdot 1}{(x-1)^2} = \frac{x-1-x-1}{(x-1)^2} = \frac{-2}{(x-1)^2}\end{aligned}$$

(ii) Let $y = \frac{8x+3}{6-5x}$

Diff. w.r.t. x

$$\begin{aligned}\frac{dy}{dx} &= \frac{(6-5x)\frac{d}{dx}(8x+3) - (8x+3)\frac{d}{dx}(6-5x)}{(6-5x)^2} \quad [\text{Using Quotient Rule}] \\ &= \frac{(6-5x).8 - (8x+3)(-5)}{(6-5x)^2} = \frac{63}{(6-5x)^2}\end{aligned}$$

(iii) Let $y = \frac{a^2}{x^2 + a^2} = a^2(x^2 + a^2)^{-1}$

Diff. w.r.t. x

$$\frac{dy}{dx} = a^2(-1)(x^2 + a^2)^{-2} \frac{d}{dx}(x^2 + a^2)$$

[Don't use quoient rule here because
there is no function of x in numerator.]

$$= -\frac{a^2}{(x^2 + a^2)^2} \cdot 2x = \frac{-2a^2x}{(x^2 + a^2)^2}$$

(iv) Let $y = \sqrt{\frac{x^2+1}{x+1}}$

Diff. w.r.t. x

$$\frac{dy}{dx} = \frac{1}{2\sqrt{\frac{x^2+1}{x+1}}} \frac{d}{dx}\left(\frac{x^2+1}{x+1}\right) \quad \left[\because \frac{d}{dx}((f(x))^n) = n(f(x))^{n-1} \frac{d}{dx}(f(x)) \right]$$

$$= \frac{1}{2\sqrt{\frac{x^2+1}{x+1}}} \left[\frac{(x+1)\frac{d}{dx}(x^2+1) - (x^2+1)\frac{d}{dx}(x+1)}{(x+1)^2} \right] \quad [\text{Using Quotient Rule}]$$

$$= \frac{(x+1)(2x-1) - (x^2+1)(1)}{2\sqrt{x^2+1}(x+1)^{3/2}} = \frac{2x^2+2x-x^2-1}{2\sqrt{x^2+1}(x+1)^{3/2}}$$

$$= \frac{x^2+2x-1}{2(x+1)^{3/2} \times \sqrt{x^2+1}} = \frac{(x-1)^2}{2(x+1)^{3/2} \sqrt{x^2+1}}$$

Here, are some exercises for you.

E 3) Differentiate the following functions w.r.t. x

(i) πe (ii) $\frac{1}{x^7}$ (iii) $\sqrt{x}$ (iv) $(4-3x)^8$ (v) $\left(x^3 + \frac{1}{x^3}\right)^2$

E 4) Find the derivative of the following functions:

(i) $\left(x + \frac{1}{x}\right)\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)$ (ii) $\left(x^3 + \frac{1}{x^3}\right)\left(x + \frac{1}{x}\right)$ (iii) $\frac{x^2}{x^3+1}$ (iv) $\frac{x^2+x}{a}$

6.5 CHAIN RULE

Sometimes variables y and x are connected by the relations of the form

$y = f(u)$, $u = g(w)$, $w = h(x)$

and we want to differentiate y w.r.t. x. then chain rule is used, which gives

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dw} \frac{dw}{dx}$$

Following example will illustrate the rule more clearly.

Example 5: Find $\frac{dy}{dx}$ in the following cases.

(i) $y = 3u$, $u = v^2$, $v = 4x^2 + 5$

(ii) $y = u^2$, $u = 3v$, $v = \frac{x}{x+1}$

Solution:

(i) $y = 3u$,	$u = v^2$,	$v = 4x^2 + 5$
Diff. w.r.t. u	Diff. w.r.t. v	Diff. w.r.t. x
$\frac{dy}{du} = 3$	$\frac{du}{dv} = 2v$	$\frac{dv}{dx} = 8x$

$\therefore$ by chain rule

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dv} \frac{dv}{dx} = 3(2v)(8x)$$

$$= 48xv = 48x(4x^2 + 5) \text{ [Replacing the value of } v \text{ in terms of } x]$$

(ii) $y = u^2$,	$u = 3v$,	$v = \frac{x}{x+1}$
Diff. w.r.t. u	Diff. w.r.t. v	Diff. w.r.t. x
$\frac{dy}{du} = 2u$	$\frac{du}{dv} = 3$	$\frac{dv}{dx} = \frac{(x+1) \cdot 1 - x \cdot (1)}{(x+1)^2} = \frac{1}{(x+1)^2}$

$\therefore$ by chain rule

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dv} \frac{dv}{dx} = (2u)(3) \left(\frac{1}{(1+x)^2} \right) \\ &= \frac{6u}{(1+x)^2} = \frac{6(3v)}{(1+x)^2} = \frac{18v}{(1+x)^2} = \frac{18x}{(1+x)^3} \end{aligned}$$

6.6 DERIVATIVES OF EXPONENTIAL, LOGRITHMIC, PARAMETRIC AND IMPLICIT FUNCTIONS

Let us first take up some examples on derivatives of exponential and logarithmic functions as given in example 12.

Example 6: Find the derivative of the following functions:

(i) 2^x (ii) 5^{ax} (iii) e^{3x} (iv) e^{9x^2} (v) $\log(1+x^2)$

(vi) $\log_x x^2$ (vii) $\log_2 x$ (viii) $\log_x 2$ (ix) $\frac{1}{\log(1+x)}$

Solution:

(i) Let $y = 2^x$

Diff. w.r.t. x

$$\frac{dy}{dx} = 2^x \log 2 \quad \left[\because \frac{d}{dx}(a^x) = a^x \log a \right]$$

(ii) Let $y = 5^{ax}$

Diff. w.r.t. x

$$\begin{aligned}\frac{dy}{dx} &= 5^{ax} \log 5 \frac{d}{dx}(ax) & \left[\because \frac{d}{dx}(a^{bx}) = ba^{bx} \log a \right] \\ &= 5^{ax} a \log 5\end{aligned}$$

(iii) Let $y = e^{3x}$

Diff. w.r.t. x

$$\frac{dy}{dx} = e^{3x} (3) = 3e^{3x} \quad \left[\because \frac{d}{dx}(e^{ax}) = ae^{ax} \right]$$

(iv) Let $y = e^{9x^2}$

Diff. w.r.t. x

$$\begin{aligned}\frac{dy}{dx} &= e^{9x^2} \frac{d}{dx}(9x^2) = e^{9x^2} (18x) & \left[\text{Using } \frac{d}{dx}(e^{f(x)}) = e^{f(x)} \frac{d}{dx}(f(x)) \right] \\ &= 18x e^{9x^2}\end{aligned}$$

(v) Let $y = \log(1+x^2)$

Diff. w.r.t. x

$$\frac{dy}{dx} = \frac{1}{1+x^2} \frac{d}{dx}(1+x^2) = \frac{2x}{1+x^2} \quad \left[\because \frac{d}{dx}(\log f(x)) = \frac{1}{f(x)} \frac{d}{dx}(f(x)) \right]$$

(vi) Let $y = \log_x x^2 = \frac{\log x^2}{\log x}$ [Using base change formula]

$$= \frac{2 \log_x x}{\log_x x} = 2 \quad \left[\because \log m^n = n \log m \right]$$

Diff. w.r.t. x

$$\frac{dy}{dx} = 0$$

(vii) Let $y = \log_2 x$

Diff. w.r.t. x

$$\frac{dy}{dx} = \frac{1}{x} \log_2 e \quad \left[\because \frac{d}{dx}(\log_a x) = \frac{1}{x} \log_a e \right]$$

(viii) Let $y = \log_x 2 = \frac{\log 2}{\log x}$ [Using base change formula]

$$y = (\log 2)(\log x)^{-1}$$

Diff. w.r.t. x

$$\frac{dy}{dx} = -(\log 2)(\log x)^{-2} \frac{d}{dx}(\log x) = -\frac{(\log 2)}{(\log x)^2} \times \frac{1}{x} = -\frac{\log 2}{x(\log x)^2}$$

(ix) Let $y = \frac{1}{\log(1+x)} y = [\log(1+x)]^{-1}$

Diff. w.r.t. x

$$\begin{aligned}\frac{dy}{dx} &= -1[\log(1+x)]^{-2} \frac{d}{dx}(\log(1+x)) \\ &= \frac{-1}{[\log(1+x)]^2} \times \frac{1}{1+x} = -\frac{1}{(1+x)[\log(1+x)]^2}\end{aligned}$$

Here is an exercise for you.

E 5) Find the derivative of the following functions:

(i) $a^{\log_a 2^x}$ (ii) $3^{\log_3 x^2}$

Following is an example on derivatives of parametric functions.

Example 7: Find $\frac{dy}{dx}$ for the following parametric functions:

(i) $x = 1 + t, y = 2 + t^2$ (ii) $x = \frac{1+3t^2}{1-t}, y = \frac{1+t^2}{1-t}$

Solution:

(i) $x = 1 + t, \quad y = 2 + t^2$
 Diff. w.r.t. t Diff. w.r.t. t
 $\frac{dx}{dt} = 1, \quad \frac{dy}{dt} = 2t$

$$\therefore \frac{dy}{dx} = \frac{dy}{dt} \bigg/ \frac{dx}{dt} = \frac{2t}{1} = 2t$$

(ii) $x = \frac{1+3t^2}{1-t}$

Diff. w.r.t. t

$$\begin{aligned} \frac{dx}{dt} &= \frac{(1-t) \frac{d}{dt}(1+3t^2) - (1+3t^2) \frac{d}{dt}(1-t)}{(1-t)^2} \\ &= \frac{(1-t)(6t) - (1+3t^2)(-1)}{(1-t)^2} = \frac{6t - 6t^2 + 1 + 3t^2}{(1-t)^2} = \frac{1+6t-3t^2}{(1-t)^2} \end{aligned}$$

Now, $y = \frac{1+t^2}{1-t}$

Diff. w.r.t. t

$$\begin{aligned} \frac{dy}{dt} &= \frac{(1-t) \frac{d}{dt}(1+t^2) - (1+t^2) \frac{d}{dt}(1-t)}{(1-t)^2} = \frac{(1-t)(2t) - (1+t^2)(-1)}{(1-t)^2} \\ &= \frac{2t - 2t^2 + 1 + t^2}{(1-t)^2} = \frac{1+2t-t^2}{(1-t)^2} \end{aligned}$$

$$\therefore \frac{dy}{dx} = \frac{dy}{dt} \bigg/ \frac{dx}{dt} = \frac{1+2t-t^2}{1+6t-3t^2}$$

Now, you can try the following the exercise.

E 6) Find $\frac{dy}{dx}$ if $x = 2 + 4t^2, y = 9t^2 + 3t + 1$.

Implicit Function

A function defined by $y = f(x)$ is known as explicit function. But sometimes y cannot be easily expressed in terms of x . A function of the form $f(x, y) = c$, where c is a constant is known as implicit function.

Procedure

In case of implicit function, differentiate the given relation w.r.t. x and collect all the terms of $\frac{dy}{dx}$ to the left hand side and finally dividing both sides by a term attached with $\frac{dy}{dx}$, we get the value of $\frac{dy}{dx}$.

Following example will explain the procedure more clearly:

Example 8: Find $\frac{dy}{dx}$ in the following cases:

(i) $x^2 + y^2 = c^2$ (ii) $(x - a)^2 + (y - b)^2 = r^2$ (iii) $x^3 + y^3 + xy = 5$

Solution:

(i) $x^2 + y^2 = c^2$

Diff. w.r.t. x

$$2x + 2y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{y}$$

(ii) $(x - a)^2 + (y - b)^2 = r^2$

Diff. w.r.t. x

$$2(x - a) + 2(y - b) \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x - a}{y - b}$$

(iii) $x^3 + y^3 + xy = 5$

Diff. w.r.t. x

$$3x^2 + 3y^2 \frac{dy}{dx} + x \frac{dy}{dx} + y \cdot 1 = 0 \quad [\text{Using Product Rule in the term } xy]$$

$$\Rightarrow (3y^2 + x) \frac{dy}{dx} = -(y + 3x^2)$$

$$\Rightarrow \frac{dy}{dx} = -\frac{3x^2 + y}{x + 3y^2}$$

Now, you can try the following exercise.

E 7) Find $\frac{dy}{dx}$, if $xy^3 + xe^x + xe^{-y} = 3$

6.7 DERIVATIVE OF HIGHER ORDERS

Sometimes we are to differentiate the function more than once.

Derivative of y w.r.t x twice is denoted by $\frac{d^2y}{dx^2}$,

Derivative of y w.r.t x thrice is denoted by $\frac{d^3y}{dx^3}$,

...
...
...

n times differentiation of y w.r.t x is denoted by $\frac{d^ny}{dx^n}$.

Following example illustrate the idea of higher order derivatives.

Example 9: Find the indicated derivatives for the following functions:

- (i) $\frac{d^4y}{dx^4}$ for $y = \frac{1}{x}$ (ii) $\frac{d^3y}{dx^3}$ for $y = e^{ax}$
 (iii) $\frac{d^3y}{dx^3}$ for $y = (ax + b)^7$ (iv) $\frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}, \frac{d^4y}{dx^4}$ for $y = x^2 + x + 1$

Solution:

(i) $y = \frac{1}{x}$
 Diff. w.r.t. x
 $\frac{dy}{dx} = \frac{-1}{x^2}$
 Again diff. w.r.t. x
 $\frac{d^2y}{dx^2} = \frac{(-1)(-2)}{x^3} = \frac{2}{x^3}$
 Again diff. w.r.t. x
 $\frac{d^3y}{dx^3} = \frac{2(-3)}{x^4} = -\frac{6}{x^4}$
 Again diff. w.r.t. x
 $\frac{d^4y}{dx^4} = \frac{-6(-4)}{x^5} = \frac{24}{x^5}$

(ii) $y = e^{ax}$
 Diff. w.r.t. x
 $\frac{dy}{dx} = e^{ax} \frac{d}{dx}(ax) \quad \left[\because \frac{d}{dx}(e^{f(x)}) = e^{f(x)} \frac{d}{dx}(f(x)) \right]$
 $= e^{ax}(a) = ae^{ax}$
 Again diff. w.r.t. x
 $\frac{d^2y}{dx^2} = a^2 e^{ax}$
 Again diff. w.r.t. x
 $\frac{d^3y}{dx^3} = a^3 e^{ax}$

(iii) $y = (ax + b)^7$
 Diff. w.r.t. x
 $\frac{dy}{dx} = 7(ax + b)^6 a \quad \left[\because \frac{d}{dx}((ax + b)^n) = n(ax + b)^{n-1} a \right]$
 Again diff. w.r.t. x
 $\frac{d^2y}{dx^2} = 42(ax + b)^5 a^2$
 Again diff. w.r.t. x
 $\frac{d^3y}{dx^3} = 210(ax + b)^4 a^3$

(iv) $y = x^2 + x + 1$

Diff. w.r.t. x

$$\frac{dy}{dx} = 2x + 1$$

[Using formula written at serial
number 12 of the table of formulae]

Again diff. w.r.t. x

$$\frac{d^2y}{dx^2} = 2$$

Again diff. w.r.t. x

$$\frac{d^3y}{dx^3} = 0$$

Again diff. w.r.t. x

$$\frac{d^4y}{dx^4} = 0$$

6.8 CONCEPT OF MAXIMA AND MINIMA

Differentiation has a large number of applications in different fields such as mathematics, statistics, economics, actuarial science, etc. Concept of differentiation is also useful to obtain maxima or minima point(s) and their corresponding value(s) of a given function. Actually sometimes we are interested only to find maximum or minimum value(s) of a function. The aim of this section is to meet this interest. Without going into theoretical details, let us discuss the concept geometrically.

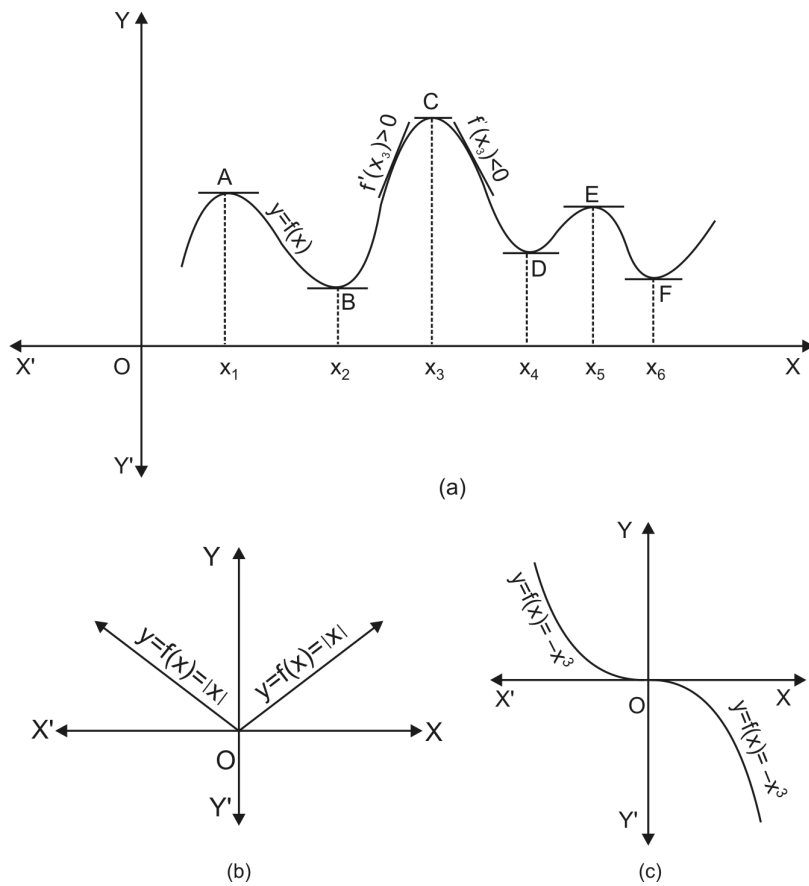


Fig. 6.3

In equation (2) of Sec. 6.2 of this unit we have seen that

(Derivative at a point) $_{x=a}$ is the slope of the tangent at that point. ... (1)

In Fig 6.3 a we see that x_1 is a point where the function $y = f(x)$ takes maximum value (local) compare to all points which are very close to x_1 i.e. $f(x_1) > f(x)$... (2)

for all points x which are very close to x_1

The point x_1 is known as local maxima point of the function $y = f(x)$ (local maxima means that it satisfy the equation (2) as given above, i.e. there may be points in the domain of the function where value of the function f is greater than $f(x_1)$)

Similarly, points x_3 and x_5 in the same figure are points of maxima (local). On the other hand in the same figure x_2 is a point where the function $y = f(x)$ takes minimum value (local) compare to all points which are very close to x_2 i.e. $f(x_2) < f(x)$

for all points x which are very close to x_2

The point x_2 is known as local minima point of the function $y = f(x)$.

Similarly x_4 and x_6 are points of local minima.

But one interesting point to be noted here is that tangent lines at all the points whether it is point of maxima or minima is parallel to x -axis.

i.e. slope of the tangent at point of maxima or minima = $\tan 0 = 0$... (3)

$\because$ if a line makes an angle θ with +ve direction of x - axis then slope of the line is defined as $\tan \theta$, in this case $\theta = 0$ as tangent line is parallel to x - axis.

In view of equation (1) and (3), we have

Derivative of the function at a point of maxima or minima = 0 ... (4)
, provided that derivative at that point exists

Also note that converse of (4) does not hold.

For example, take graph of the function $y = f(x) = -x^3$ shown in Fig. 6.3 c. we see that

$$\left. \frac{dy}{dx} \right|_{x=0} = f'(0) = -3(0)^2 = 0 \quad \left[\because \frac{dy}{dx} = f'(x) = -3x^2 \right]$$

But $x = 0$ is neither point of maxima nor minima.

Now consider the function $y = f(x) = |x|$, whose graph is shown in Fig. 6.3 b.

We see that $x = 0$ is a point of minima, in fact absolute minima or global minima (absolute minima or global minima means that function assumes smallest value at $x = 0$ in whole domain of the function not only at those points which are very close to $x = 0$)

But we have seen in part (i) of Example 2 of this unit that derivative of the function $y = f(x) = |x|$ does not exists at $x = 0$.

Equation (4) implies points obtained by putting first derivative equal to zero may be points of maxima or minima. Second derivative test differentiate between points of maxima and minima which is stated below:

Second Derivative Test: It states that if the function f is twice differentiable at a point ' c ', where c is point of the domain of the function f , then

- (i) c is point of local minima if $f'(c) = 0$ and $f''(c) > 0$.
- (ii) c is point of local maxima if $f'(c) = 0$ and $f''(c) < 0$.
- (iii) test fails if $f''(c) = 0$. In this case we use first order derivative test, which can be concluded as:
 - if $f'(c)$ changes its sign from positive to negative as we cross the point $x = c$, then $x = c$ is point of maxima (see Fig. 6.3 a at point x_3)
 - if $f'(c)$ changes its sign from negative to positive as we cross the point $x = c$, then $x = c$ is point of minima. This can be noted at points x_2, x_4, x_6 in Fig. 6.3 a.
 - if $f'(c)$ does not change its sign as we cross the point $x = c$, then $x = c$ is neither point of minima nor maxima. See Fig. 6.3 c in which $x = 0$ is such a point. Point of this nature is called point of inflection. Normal curve has two such points at $x = \mu + \sigma$ and $x = \mu - \sigma$. You can observe it by differentiating normal density twice and putting double derivative equal to zero. Normal distribution is discussed in Unit 13 and Unit 14 of MST-003.

With the following two examples followed by an exercise, let us close this Sec.

Example 10: Find local maximum and minimum values of the function

$$f(x) = 2x^3 - 15x^2 + 36x + 9.$$

Solution: Given function is

$$f(x) = 2x^3 - 15x^2 + 36x + 9$$

Dif. w.r.t. x

$$f'(x) = 6x^2 - 30x + 36 \quad \dots (1)$$

For maxima or minima

$$f'(x) = 0$$

$$\Rightarrow 6x^2 - 30x + 36 = 0$$

$$\Rightarrow x^2 - 5x + 6 = 0$$

$$\Rightarrow (x - 2)(x - 3) = 0$$

$$\Rightarrow x = 2, 3$$

Diff. (1) w.r.t. x

$$f''(x) = 12x - 30$$

$$\text{At } x = 2, f''(2) = 24 - 30 = -6 < 0$$

$\therefore$ by second order derivative test, $x = 2$ is point of maxima and maximum value is given by

$$f(2) = 2(2)^3 - 15(2)^2 + 36(2) + 9 = 16 - 60 + 72 + 9 = 27$$

$$\text{At } x = 3, f''(3) = 36 - 30 = 6 > 0$$

$\therefore$ by second order derivative test, $x = 3$ is point of minima and minimum value of the function is given by

$$f(3) = 2(3)^3 - 15(3)^2 + 36(3) + 9 = 54 - 135 + 108 + 9 = 36$$

Example 11: Find local maximum and minimum values of the function

$$f(x) = \frac{x^4}{4} + \frac{x^3}{3} - 2x^2 - 4x + 5.$$

Solution: Given function is

$$f(x) = \frac{x^4}{4} + \frac{x^3}{3} - 2x^2 - 4x + 5$$

Diff. w.r.t. x

$$f'(x) = \frac{4x^3}{4} + \frac{3x^2}{3} - 4x - 4 = x^3 + x^2 - 4x - 4 \quad \dots (1)$$

For maxima or minima

$$f'(x) = 0$$

$$\Rightarrow x^3 + x^2 - 4x - 4 = 0 \quad \dots (2)$$

By inspection $x = -1$ is a root of equation (2)

$$\Rightarrow (x+1) \text{ is a factor of } x^3 + x^2 - 4x - 4$$

$\therefore$ (2) can be written as

$$(x+1)(x^2 - 4) = 0$$

$$\Rightarrow (x+1)(x-2)(x+2) = 0$$

$$\Rightarrow x = -1, 2, -2$$

Diff. (1) w.r.t. x

$$f''(x) = 3x^2 + 2x - 4$$

$$\text{At } x = -1, f''(-1) = 3(-1)^2 + 2(-1) - 4 = 3 - 2 - 4 = -3 < 0$$

$$\text{At } x = 2, f''(2) = 3(2)^2 + 2(2) - 4 = 12 > 0 \text{ and}$$

$$\text{At } x = -2, f''(-2) = 3(-2)^2 + 2(-2) - 4 = 4 > 0$$

$\therefore$ by second order derivative test $x = 2, -2$ are points of minima and $x = -1$ is point of maxima.

$$\text{Local minimum value at } x = 2 \text{ is } f(2) = \frac{(2)^4}{4} + \frac{(2)^3}{3} - 2(2)^2 - 4(2) + 5 = -\frac{13}{3} \text{ and}$$

Local minimum value at $x = -2$ is given by

$$f(-2) = \frac{(-2)^4}{4} + \frac{(-2)^3}{3} - 2(-2)^2 - 4(-2) + 5 = \frac{19}{3}$$

Local maximum value at $x = -1$ is given by

$$f(-1) = \frac{(-1)^4}{4} + \frac{(-1)^3}{3} - 2(-1)^2 - 4(-1) + 5 = \frac{83}{12}$$

Now, you can try the following exercise.

E 8) Find local maximum and minimum values of the function

$$f(x) = 4x^3 - 21x^2 + 18x + 9.$$

6.9 SUMMARY

Let us summarise the topics that we have covered in this unit:

- 1) Derivative at a point.
- 2) Derivatives of constant, polynomials and some others commonly used functions such as x^n , $(ax + b)^n$, etc. Product Rule and Quotient Rule.
- 3) Chain rule.
- 4) Derivatives of exponential, logarithmic, parametric and implicit functions.
- 5) Derivatives of higher orders.
- 6) Concept of maxima and minima.

6.10 SOLUTIONS/ANSWERS

E 1) (i) $f(x) = x^3 + x + 1$, at $x = -1$

By definition

$$\begin{aligned} f'(-1) &= \lim_{h \rightarrow 0} \frac{f(-1+h) - f(-1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(-1+h)^3 + (-1+h) + 1 - [(-1)^3 + (-1) + 1]}{h} \\ &= \lim_{h \rightarrow 0} \frac{-1 + h^3 + 3h - 3h^2 - 1 + h + 1 - (-1 - 1 + 1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^3 - 3h^2 + 4h}{h} = \lim_{h \rightarrow 0} (h^2 - 3h + 4) = 0 - 0 + 4 = 4 \end{aligned}$$

(ii) By definition

$$\begin{aligned} f'\left(\frac{1}{2}\right) &= \lim_{h \rightarrow 0} \frac{f\left(\frac{1}{2} + h\right) - f\left(\frac{1}{2}\right)}{h} \\ &= \lim_{h \rightarrow 0} \frac{2 - 3\left(\frac{1}{2} + h\right)^2 - \left[2 - 3\left(\frac{1}{2}\right)^2\right]}{h} \\ &= \lim_{h \rightarrow 0} \frac{2 - 3\left(\frac{1}{4} + h + h^2\right) - \left(2 - \frac{3}{4}\right)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-3h - 3h^2}{h} = \lim_{h \rightarrow 0} (-3 - 3h) = -3 - 0 = -3 \end{aligned}$$

E 2) $f(x) = 2x^2 - 3ax + 5$

By definition

$$\begin{aligned} f'(-2) &= \lim_{h \rightarrow 0} \frac{f(-2+h) - f(-2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{2(-2+h)^2 - 3a(-2+h) + 5 - [2(-2)^2 - 3a(-2) + 5]}{h} \\ &= \lim_{h \rightarrow 0} \frac{2(4 + h^2 - 4h) + 6a - 3ah + 5 - (8 + 6a + 5)}{h} \end{aligned}$$

$$= \lim_{h \rightarrow 0} \frac{2h^2 - 8h - 3ah}{h} = \lim_{h \rightarrow 0} (2h - 8 - 3a) = -8 - 3a$$

But according to problem

$$f'(-2) = 3 \Rightarrow -8 - 3a = 3 \Rightarrow -3a = 11 \Rightarrow a = -\frac{11}{3}$$

E 3) (i) Let $y = \pi e$

Diff. w.r.t. x

$$\frac{dy}{dx} = 0 \quad \left[\because \pi \text{ and } e \text{ both are constant} \Rightarrow \pi e \text{ is a constant} \right. \\ \left. \text{and derivative of a constant function is zero.} \right]$$

(ii) Let $y = \frac{1}{x^7} = x^{-7}$

Diff. w.r.t. x

$$\frac{dy}{dx} = -7x^{-8} = -\frac{7}{x^8} \quad \left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

(iii) Let $y = \sqrt{x} = x^{1/2}$

Diff. w.r.t. x

$$\frac{dy}{dx} = \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x}} \quad \left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

(iv) Let $y = (4 - 3x)^8$

Diff. w.r.t. x

$$\frac{dy}{dx} = 8(4 - 3x)^7(-3) \quad \left[\because \frac{d}{dx}((ax + b)^n) = n(ax + b)^{n-1}a \right] \\ = -24(4 - 3x)^7$$

(v) Let $y = \left(x^3 + \frac{1}{x^3}\right)^2$

Diff. w.r.t. x

$$\frac{dy}{dx} = 2\left(x^3 + \frac{1}{x^3}\right) \frac{d}{dx}\left(x^3 + \frac{1}{x^3}\right) \\ \left[\because \frac{d}{dx}((f(x))^n) = n(f(x))^{n-1} \frac{d}{dx}(f(x)) \right] \\ = 2\left(x^3 + \frac{1}{x^3}\right) \left(3x^2 - \frac{3}{x^4}\right) \\ = 6\left(x^5 - \frac{1}{x} + \frac{1}{x} - \frac{1}{x^7}\right) = 4\left(x^5 - \frac{1}{x^7}\right)$$

E 4) (i) $y = \left(x + \frac{1}{x}\right) \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right) = x^{3/2} + x^{1/2} + \frac{1}{x^{1/2}} + \frac{1}{x^{3/2}}$

$$y = x^{3/2} + x^{1/2} + x^{-1/2} + x^{-3/2}$$

Diff. w.r.t. x

$$\begin{aligned}\frac{dy}{dx} &= \frac{3}{2}x^{1/2} + \frac{1}{2}x^{-1/2} - \frac{1}{2}x^{-3/2} - \frac{3}{2}x^{-5/2} \\ &= \frac{3}{2}\sqrt{x} + \frac{1}{2\sqrt{x}} - \frac{1}{2x^{3/2}} - \frac{3}{2x^{5/2}}\end{aligned}$$

(ii) Let $y = \left(x^3 + \frac{1}{x^3}\right)\left(x + \frac{1}{x}\right) = x^4 + x^2 + \frac{1}{x^2} + \frac{1}{x^4}$

Diff. w.r.t. x

$$\frac{dy}{dx} = 4x^3 + 2x - \frac{2}{x^3} - \frac{4}{x^5}$$

(iii) Let $y = \frac{x^2}{x^3 + 1}$

Diff. w.r.t. x

$$\begin{aligned}\frac{dy}{dx} &= \frac{(x^3 + 1)\frac{d}{dx}(x^2) - x^2\frac{d}{dx}(x^3 + 1)}{(x^3 + 1)^2} \quad [\text{Using Quotient Rule}] \\ &= \frac{(x^3 + 1)(2x) - x^2(3x^2)}{(x^3 + 1)^2} \\ &= \frac{2x^4 + 2x - 3x^4}{(x^3 + 1)^2} = \frac{2x - x^4}{(x^3 + 1)^2} = \frac{x(2 - x^3)}{(x^3 + 1)^2}\end{aligned}$$

(iv) Let $y = \frac{x^2 + x}{a} = \frac{1}{a}(x^2 + x)$

Diff. w.r.t. x

$$\frac{dy}{dx} = \frac{1}{a} \frac{d}{dx}(x^2 + x) = \frac{1}{a}(2x + 1) \left[\begin{array}{l} \text{Do not use quoient rule} \\ \text{because in the denominator} \\ \text{there is no function of } x. \end{array} \right]$$

E5) (i) Let $y = a^{\log_a 2^x} y = 2^x \quad [\because a^{\log_a f(x)} = f(x)]$

Diff. w.r.t. x

$$\frac{dy}{dx} = 2^x \log 2 \quad \left[\because \frac{d}{dx}(a^x) = a^x \log a \right]$$

(ii) Let $y = 3^{\log_3 x^2} = x^2 \quad [\because a^{\log_a f(x)} = f(x)]$

Diff. w.r.t. x

$$\frac{dy}{dx} = 2x \quad \left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

E 6) $x = 2 + 4t^2, \quad y = 9t^2 + 3t + 1$

Diff. w.r.t. t

Diff. w.r.t. t

$$\frac{dx}{dt} = 8t$$

$$\frac{dy}{dt} = 18t + 3$$

$$\therefore \frac{dy}{dx} = \frac{dy}{dt} \bigg/ \frac{dx}{dt} = \frac{18t + 3}{8t}$$

E 7) $xy^3 + xe^x + xe^{-y} = 3$

Diff. w.r.t. x

$$x\left(3y^2 \frac{dy}{dx}\right) + (1)y^3 + xe^x + (1)e^x + xe^{-y}\left(-\frac{dy}{dx}\right) + (1)e^{-y} = 0$$

$$\Rightarrow (3xy^2 - xe^{-y})\frac{dy}{dx} = -y^3 - xe^x - e^x - e^{-y}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{y^3 + xe^x + e^x + e^{-y}}{3xy^2 - xe^{-y}} = \frac{y^3 + xe^x + e^x + e^{-y}}{xe^{-y} - 3xy^2}$$

E 8) Given function is

$$f(x) = 4x^3 - 21x^2 + 18x + 9$$

Diff. w.r.t. x

$$f'(x) = 12x^2 - 42x + 18 \quad \dots (1)$$

For maxima or minima

$$f'(x) = 0$$

$$\Rightarrow 12x^2 - 42x + 18 = 0$$

$$\Rightarrow 2x^2 - 7x + 3 = 0$$

$$\Rightarrow 2x^2 - 6x - x + 3 = 0$$

$$\Rightarrow 2x(x - 3) - 1(x - 3) = 0$$

$$\Rightarrow (x - 3)(2x - 1) = 0$$

$$\Rightarrow x = 3, 1/2$$

Diff. (1) w.r.t. x

$$f''(x) = 24x - 42$$

$$\text{At } x = 3, f''(3) = 24 \times 3 - 42 = 72 - 42 = 30 > 0$$

$$\text{At } x = 1/2, f''(1/2) = 24 \times \frac{1}{2} - 42 = 12 - 42 = -30 < 0$$

$\therefore$ by second order derivative test $x = 3$ is point of minima and $x = 1/2$ is point of maxima.

Local minimum value at $x = 3$ is given by

$$f(3) = 4(3)^3 - 21(3)^2 + 18(3) + 9 = 108 - 189 + 54 + 9 = -18$$

Local maximum value at $x = 1/2$ is given by

$$\begin{aligned} f\left(\frac{1}{2}\right) &= 4\left(\frac{1}{2}\right)^3 - 21\left(\frac{1}{2}\right)^2 + 18\left(\frac{1}{2}\right) + 9 \\ &= \frac{1}{2} - \frac{21}{4} + 9 + 9 = \frac{21 - 21 + 72}{4} = \frac{53}{4} \end{aligned}$$

UNIT 7 INDEFINITE INTEGRATION

Structure

- 7.1 Introduction
 - Objectives
- 7.2 Meaning and Terminology used.
- 7.3 Integration of some Particular Functions
- 7.4 Integration by Substitution
- 7.5 Integration using Partial Fractions
- 7.6 Integration by Parts
- 7.7 Summary
- 7.8 Solutions/Answers

7.1 INTRODUCTION

In the previous unit, we have studied the differentiation of some functions. Here, in this unit we are going to discuss the reverse process of differentiation known as integration.

In this unit, we will study the integration of some commonly used functions in section 7.3, integration by substitutions in section 7.4, integration by using partial fractions in section 7.5 and integration by parts in section 7.6.

Objectives

After completing this unit, you should be able to:

- evaluate the integration of some commonly used functions;
- evaluate the integration by substitution method;
- evaluate the integration using partial fractions; and
- evaluate the integration by parts.

7.2 MEANING AND TERMINOLOGY USED

Notations

You have become familiar with the concept of summation discussed in Unit 3 of this course, i.e. MST-001. In fact, summation is convenient way to represent the sum of discrete values only. If the variable is continuous, then the summation cannot be used in the way it is used for discrete values. Summation is obtained by the process of integration in case of continuous variable.

Origin of integration lies in the process of summation. In mathematics, the words “Summation” and “Integration” are used for the words “to unite”.

In previous unit we have studied differentiation. The integration is just the reverse process of differentiation. Actually, it is an antiderivative of a function. That is, if $f'(x)$ is derivative of $f(x)$ and hence $f'(x)$ is derivative of $f(x) + c$ ($\because$ derivative of constant is zero). And therefore, $f(x) + c$ is the integration of $f'(x)$.

Integral of a function $f(x)$ w.r.t. x is denoted by $\int f(x)dx$

,where $f(x)$ is known as integrand, dx reflects the message that integrand is to be integrated w.r.t. to the variable x and the entire process of finding the integral of integrand is known as integration. The symbol $\int$ has its origin from the letter S, which was used for summation.

Let us consider a simple example first and then give a list of the formulae.

We know that the function x is the differentiation of $\frac{x^2}{2} + c$ w.r.t. x .

$\therefore \frac{x^2}{2} + c$ is integration of x .

i.e. $\int x dx = \frac{x^2}{2} + c$, where c is known as constant of integration.

Similarly, integration of other functions can be obtained. Integrations of some commonly used functions are listed in the following table.

List of Formulae of Integration

S. No.	Function $f(x)$	$\int f(x)dx$
1	k (constant function)	$kx + c$, where c is constant of integration
2	x^n	$\frac{x^{n+1}}{n+1} + c$, $n \neq -1$
3	$\frac{1}{x}$	$\log x + c$
4	$(ax + b)^n$	$\frac{(ax + b)^{n+1}}{a(n+1)} + c$, $n \neq -1$
5	$\frac{1}{ax + b}$	$\frac{1}{a} \log ax + b + c$
6	Exponential functions (i) a^{mx} (ii) a^{mx+n} (iii) e^{ax} (iv) e^{ax+b}	(i) $\frac{a^{mx}}{m \log_e a } + c$ (ii) $\frac{a^{mx+n}}{m \log a } + c$ (iii) $\frac{e^{ax}}{a} + c$ (iv) $\frac{e^{ax+b}}{a} + c$

Remark 1:

If f, g are integral functions such that $f + g, f - g$, are defined and a, b are real constants, then

$$(i) \int a(f(x))dx = a \int f(x)dx$$

$$(ii) \int (af(x) \pm bg(x))dx = a \int f(x)dx \pm b \int g(x)dx$$

7.3 INTEGRATION OF SOME PARTICULAR FUNCTIONS

In this section, we learn how the formulae mentioned in the table on previous page are used.

Example 1: Evaluate the following integrals:

- (i) $\int \sqrt{5} dx$ (ii) $\int 0 dx$ (iii) $\int \pi dx$ (iv) $\int x^3 dx$
 (v) $\int x^{7/2} dx$ (vi) $\int \frac{1}{x^5} dx$ (vii) $\int \frac{1}{x^{7/2}} dx$ (viii) $\int \frac{1}{\sqrt[3]{x}} dx$
 (ix) $\int (x^3 + x + 5) dx$ (x) $\int (x^2 + 1)(x - 1) dx$ (xi) $\int \frac{x^6 - x^4 + 1}{x^2} dx$
 (xii) $\int \frac{x^4 + x^3 + 3}{\sqrt{x}} dx$ (xiii) $\int \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) dx$ (xiv) $\int \left(x + \frac{1}{x} \right)^2 dx$
 (xv) $\int \left(8x^3 + 2x - 3 + \frac{1}{x^2} + \frac{2}{x^3} \right) dx$

Solution:

$$(i) \quad \int \sqrt{5} dx = \sqrt{5}x + c \quad \left[\begin{array}{l} \because \sqrt{5} \text{ is a constant and if } k \text{ is} \\ \text{constant then } \int k dx = kx + c \end{array} \right]$$

where c is constant of integration.

Note: Constant of integration c is added everywhere, so in future we will not write 'where c is constant of integration'.

$$(ii) \quad \int 0 dx = 0x + c = c \quad \text{as } 0 \text{ is constant}$$

$$(iii) \quad \int \pi dx = \pi x + c \quad \text{as } \pi \text{ is constant}$$

$$(iv) \quad \int x^3 dx = \frac{x^{3+1}}{3+1} + c = \frac{x^4}{4} + c \quad \left[\because \int x^n dx = \frac{x^{n+1}}{n+1} + c \right]$$

$$(v) \quad \int x^{7/2} dx = \frac{x^{\frac{7}{2}+1}}{\frac{7}{2}+1} + c = \frac{2}{9} x^{\frac{9}{2}} + c \quad \left[\because \int x^n dx = \frac{x^{n+1}}{n+1} + c \right]$$

$$(vi) \quad \int \frac{1}{x^5} dx = \int x^{-5} dx = \frac{x^{-5+1}}{-5+1} + c = -\frac{1}{4} x^{-4} + c$$

$$\left[\because \int x^n dx = \frac{x^{n+1}}{n+1} + c \right]$$

$$(vii) \quad \int \frac{1}{x^{7/2}} dx = \int x^{-7/2} dx = \frac{x^{-5/2}}{-5/2} + c = -\frac{2}{5} x^{-5/2} + c$$

$$\left[\because \int x^n dx = \frac{x^{n+1}}{n+1} + c \right]$$

$$(viii) \quad \int \frac{1}{\sqrt[3]{x}} dx = \int \frac{1}{x^{1/3}} dx = \int x^{-1/3} dx = \frac{x^{2/3}}{2/3} + c = \frac{3}{2} x^{2/3} + c$$

$$\left[\because \int x^n dx = \frac{x^{n+1}}{n+1} + c \right]$$

$$(ix) \int (x^3 + x + 5)dx = \int x^3 dx + \int x dx + \int 5 dx = \frac{x^4}{4} + \frac{x^2}{2} + 5x + c$$

$$(x) \int (x^2 + 1)(x - 1)dx = \int (x^3 - x^2 + x - 1)dx$$

$$= \frac{x^4}{4} - \frac{x^3}{3} + \frac{x^2}{2} - x + c \left[\because \int x^n dx = \frac{x^{n+1}}{n+1} + c \right]$$

$$\text{and } \int k dx = kx + c$$

$$(xi) \int \frac{x^6 - x^4 + 1}{x^2} dx = \int \left(\frac{x^6}{x^2} - \frac{x^4}{x^2} + \frac{1}{x^2} \right) dx = \int (x^4 - x^2 + x^{-2}) dx$$

$$= \frac{x^5}{5} - \frac{x^3}{3} + \frac{x^{-1}}{-1} + c = \frac{x^5}{5} - \frac{x^3}{3} - \frac{1}{x} + c$$

$$(xii) \int \frac{x^4 + x^3 + 3}{\sqrt{x}} dx = \int \left(\frac{x^4}{\sqrt{x}} + \frac{x^3}{\sqrt{x}} + \frac{3}{\sqrt{x}} \right) dx = \int (x^{7/2} + x^{5/2} + 3x^{-1/2}) dx$$

$$= \frac{x^{9/2}}{9/2} + \frac{x^{7/2}}{7/2} + 3 \frac{x^{1/2}}{1/2} + c = \frac{2}{9} x^{9/2} + \frac{2}{7} x^{7/2} + 6\sqrt{x} + c$$

$$(xiii) \int \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) dx = \int (x^{1/2} + x^{-1/2}) dx = \frac{x^{3/2}}{3/2} + \frac{x^{1/2}}{1/2} + c$$

$$= \frac{2}{3} x^{3/2} + 2\sqrt{x} + c$$

$$(xiv) \int \left(x + \frac{1}{x} \right)^2 dx = \int \left(x^2 + \frac{1}{x^2} + 2x \cdot \frac{1}{x} \right) dx$$

$$= \int (x^2 + x^{-2} + 2) dx = \frac{x^3}{3} + \frac{x^{-1}}{-1} + 2x + c = \frac{x^3}{3} - \frac{1}{x} + 2x + c$$

$$(xv) \int \left(8x^3 + 2x - 3 + \frac{1}{x^2} + \frac{2}{x^3} \right) dx = \int (8x^3 + 2x - 3 + x^{-2} + 2x^{-3}) dx$$

$$= \frac{8x^4}{4} + \frac{2x^2}{2} - 3x + \frac{x^{-1}}{-1} + \frac{2x^{-2}}{-2} + c$$

$$= 2x^4 + x^2 - 3x - \frac{1}{x} - \frac{1}{x^2} + c$$

Now, you can try the following exercise.

E 1) Evaluate the following integrals:

$$(i) \int \left(x^2 + \frac{1}{x^2} \right)^2 dx \quad (ii) \int \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^3 dx \quad (iii) \int (\alpha - 3) dx$$

$$(iv) \int \sqrt{x} \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) dx \quad (v) \int (x^a + 1)(x^b + 1) dx \quad (vi) \int \frac{x^m}{x^n} dx$$

$$(vii) \int \left(x + \frac{1}{x} \right) \left(x^3 + \frac{1}{x^3} \right) dx$$

Example 2: Evaluate the following integrals:

- (i) $\int (2x+3)^5 dx$ (ii) $\int (5-9x)^6 dx$ (iii) $\int (9x+5)^{3/2} dx$
 (iv) $\int \sqrt[5]{8-3x} dx$ (v) $\int \frac{(3+2x)^{7/2}}{\sqrt{3+2x}} dx$ (vi) $\int \frac{1}{(7x+2)^3} dx$
 (vii) $\int \frac{1}{\sqrt{3x+5}} dx$ (viii) $\int e^{5 \log \sqrt{3x-5}} dx$ (ix) $\int a^{\log_a \sqrt{4x+5}} dx$
 (x) $\int a^{3x} dx$ (xi) $\int e^{3x} dx$ (xii) $\int e^{5x+7} dx$
 (xiii) $\int a^{3-2x} dx$ (xiv) $\int (5e^{7x} + x^2) dx$ (xv) $\int (a^x + e^x + a^a + e^a + a^e) dx$
 (xvi) $\int 5^x 2^x dx$ (xvii) $\int \frac{2^x}{3^x} dx$ (xviii) $\int \frac{(a^x - b^x)^2}{a^x b^x} dx$
 (xix) $\int (e^{a \log x} + a^{x \log_a a} + a^{m \log_a a}) dx$
 (xx) $\int \left(\frac{x}{3} + (5x-3)^3 + x\sqrt{x} + \frac{1}{\sqrt{5+2x}} + a^a \right) dx$

Solution:

$$\begin{aligned} \text{(i)} \quad \int (2x+3)^5 dx &= \frac{(2x+3)^{5+1}}{2(5+1)} + c \quad \left[\begin{array}{l} \because \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c \\ \text{Here } a=2, n=5 \end{array} \right] \\ &= \frac{(2x+3)^6}{12} + c \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \int (5-9x)^6 dx &= \frac{(5-9x)^7}{7(-9)} + c \quad \left[\begin{array}{l} \because \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c \\ \text{Here } a=-9, n=6 \end{array} \right] \\ &= -\frac{1}{63} (5-9x)^7 + c \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \int (9x+5)^{3/2} dx &= \frac{(9x+5)^{5/2}}{\frac{5}{2} \times 9} + c \quad \left[\begin{array}{l} \because \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c \\ \text{Here } a=9, n=3/2 \end{array} \right] \\ &= \frac{2}{45} (9x+5)^{5/2} + c \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad \int \sqrt[5]{8-3x} dx &= \int (8-3x)^{\frac{1}{5}} dx = \frac{(8-3x)^{\frac{6}{5}}}{\frac{6}{5} \times (-3)} + c = \frac{-5}{18} (8-3x)^{\frac{6}{5}} + c \\ &\quad \left[\begin{array}{l} \because \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c \\ \text{Here } a=-3, n=1/5 \end{array} \right] \end{aligned}$$

$$\begin{aligned} \text{(v)} \quad \int \frac{(3+2x)^{\frac{7}{2}}}{\sqrt{3+2x}} dx &= \int (3+2x)^{\frac{7}{2} - \frac{1}{2}} dx \quad \left[\because \frac{a^m}{a^n} = a^{m-n} \right] \\ &= \int (3+2x)^3 dx \end{aligned}$$

$$= \frac{(3+2x)^4}{4 \times 2} + c \quad \left[\begin{array}{l} \because \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c \\ \text{Here } a = 2, n = 3 \end{array} \right]$$

$$= \frac{1}{8}(3+2x)^4 + c$$

$$(vi) \int \frac{1}{(7x+2)^3} dx = \int (7x+2)^{-3} dx$$

$$= \frac{(7x+2)^{-2}}{-2 \times 7} + c \quad \left[\begin{array}{l} \because \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c \\ \text{Here } a = 7, n = -3 \end{array} \right]$$

$$= -\frac{1}{14}(7x+2)^{-2} + c$$

$$(vii) \int \frac{1}{\sqrt{3x+5}} dx = \int (3x+5)^{-1/2} dx$$

$$= \frac{(3x+5)^{1/2}}{\frac{1}{2} \times 3} + c \quad \left[\begin{array}{l} \because \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c \\ \text{Here } a = 3, n = -1/2 \end{array} \right]$$

$$= \frac{2}{3}\sqrt{3x+5} + c$$

$$(viii) \int e^{5 \log \sqrt{3x-5}} dx = \int e^{\log(3x-5)^{5/2}} dx = \int (3x-5)^{5/2} dx \quad \left[\because a^{\log_a f(x)} = f(x) \right]$$

$$= \frac{(3x-5)^{7/2}}{(7/2)(3)} + c = \frac{2}{21}(3x-5)^{7/2} + c$$

$$(ix) \int a^{\log_a \sqrt{4x+5}} dx = \int \sqrt{4x+5} dx \quad \left[\because a^{\log_a f(x)} = f(x) \right]$$

$$= \frac{(4x+5)^{3/2}}{(3/2) \times 4} + c = \frac{1}{6}(4x+5)^{3/2} + c$$

$$(x) \int a^{3x} dx = \frac{a^{3x}}{3 \log |a|} + c \quad \left[\begin{array}{l} \because \int a^{mx} dx = \frac{a^{mx}}{m \log |a|} + c \\ \text{Here } a = a, m = 3 \end{array} \right]$$

$$(xi) \int e^{3x} dx = \frac{e^{3x}}{3} + c \quad \left[\begin{array}{l} \because \int e^{ax} dx = \frac{e^{ax}}{a} + c \\ \text{Here } a = 3 \end{array} \right]$$

$$(xii) \int e^{5x+7} dx = \frac{e^{5x+7}}{5} + c \quad \left[\begin{array}{l} \because \int e^{ax+b} dx = \frac{e^{ax+b}}{a} + c \\ \text{Here } a = 5, b = 7 \end{array} \right]$$

$$(xiii) \int a^{3-2x} dx = \frac{a^{3-2x}}{-2 \log |a|} + c \quad \left[\begin{array}{l} \because \int a^{mx+n} dx = \frac{a^{mx+n}}{m \log |a|} + c \\ \text{Here } m = -2, n = 3 \end{array} \right]$$

$$(xiv) \int (5e^{7x} + x^2) dx = \frac{5}{7}e^{7x} + \frac{x^3}{3} + c$$

$$(xv) \int (a^x + e^x + a^a + e^a + a^e) dx = \frac{a^x}{\log|a|} + e^x + a^a x + e^a x + a^e x + c$$

$$\left[\begin{array}{l} \text{Here } a^a, e^a, a^e \text{ all are constants} \\ \text{and if } k \text{ is constant then } \int k dx = kx + c \end{array} \right]$$

$$(xvi) \int 5^x 2^x dx = \int (5 \cdot 2)^x dx = \int 10^x dx = \frac{10^x}{\log 10} + c \left[\begin{array}{l} \because \int a^x dx = \frac{a^x}{\log|a|} + c \\ \text{Here } a = 10 \end{array} \right]$$

$$(xvii) \int \frac{2^x}{3^x} dx = \int (2/3)^x dx = \frac{(2/3)^x}{\log 2/3} + c \left[\begin{array}{l} \because \int a^x dx = \frac{a^x}{\log|a|} + c \\ \text{Here } a = 2/3 \end{array} \right]$$

$$= \frac{(2/3)^x}{\log 2 - \log 3} + c$$

$$\begin{aligned} (xviii) \int \frac{(a^x - b^x)^2}{a^x b^x} dx &= \int \frac{a^{2x} + b^{2x} - 2a^x b^x}{a^x b^x} dx \quad [\because (a-b)^2 = a^2 + b^2 - 2ab] \\ &= \int \left(\frac{a^{2x}}{a^x b^x} + \frac{b^{2x}}{a^x b^x} - \frac{2a^x b^x}{a^x b^x} \right) dx = \int \left(\frac{a^x}{b^x} + \frac{b^x}{a^x} - 2 \right) dx \\ &= \int \left((a/b)^x + (b/a)^x - 2 \right) dx \\ &= \frac{(a/b)^x}{\log|a/b|} + \frac{(b/a)^x}{\log|b/a|} - 2x + c \quad \left[\because \int m^x dx = \frac{m^x}{\log|m|} + c \right] \end{aligned}$$

$$\begin{aligned} (xix) \int (e^{a \log x} + a^{x \log_a a} + a^{m \log_a a}) dx &= \int e^{\log x^a} + a^{\log_a a^x} + a^{\log_a a^m} dx \\ &= \int (x^a + a^x + a^m) dx \quad [\because a^{\log_a f(x)} = f(x)] \\ &= \frac{x^{a+1}}{a+1} + \frac{a^x}{\log|a|} + a^m x + c \\ &\quad [\because a^m \text{ is a constant quantity}] \end{aligned}$$

$$\begin{aligned} (xx) \int \left[\frac{x}{3} + (5x-3)^3 + x\sqrt{x} + \frac{1}{\sqrt{5+2x}} + a^a \right] dx \\ &= \int \left[\frac{x}{3} + (5x-3)^3 + x^{3/2} + (5+2x)^{-1/2} + a^a \right] dx \\ &= \frac{x^2}{3 \times 2} + \frac{(5x-3)^4}{4 \times 5} + \frac{x^{5/2}}{5/2} + \frac{(5+2x)^{1/2}}{2 \times 1/2} + a^a x + c \\ &= \frac{x^2}{6} + \frac{(5x-3)^4}{20} + \frac{2}{5} x^{5/2} + (5+2x)^{1/2} + a^a x + c \end{aligned}$$

Now, you can try the following exercise.

E 2) Evaluate the following integral:

$$(i) \int \left(a^x + e^x a^x + \frac{x}{a} \right) dx \quad (ii) \int (3^{2\log_3 x} + 3^{x \log_3 a} + a^{a \log_a x} + a^{a \log_a a}) dx$$

Example 3: Evaluate the following integrals:

$$(i) \int \frac{1}{x} dx \quad (ii) \int \frac{3}{x} dx \quad (iii) \int \frac{5}{x+1} dx \quad (iv) \int \frac{7}{5x+2} dx \quad (v) \int \frac{3}{9-2x} dx$$

Solution:

$$(i) \int \frac{1}{x} dx = \log|x| + c \quad [\text{Using formula 3 of the table}]$$

$$(ii) \int \frac{3}{x} dx = 3 \int \frac{1}{x} dx = 3 \log|x| + c \quad [\text{Using formula 3 of the table}]$$

$$(iii) \int \frac{5}{x+1} dx = 5 \log|x+1| + c \quad [\text{Using formula 5 of the table}]$$

$$(iv) \int \frac{7}{5x+2} dx = \frac{7 \log|5x+2|}{5} + c \quad [\text{Using formula 5 of the table}]$$

$$= \frac{7}{5} \log|5x+2| + c$$

$$(v) \int \frac{3}{9-2x} dx = \frac{3 \log|9-2x|}{(-2)} + c \quad [\text{Using formula 5 of the table}]$$

$$= -\frac{3}{2} \log|9-2x| + c$$

Remark 2: In solving these examples you have noted that integration is in fact anti derivative of a function.

For example, consider (ix) part of Example 1

$$\text{Let } f(x) = x^3 + x + 5 \text{ then } \int f(x) dx = \frac{x^4}{4} + \frac{x^2}{2} + 5x + c \text{ (already calculated)}$$

$$\text{Now, let } F(x) = \frac{x^4}{4} + \frac{x^2}{2} + 5x + c$$

Diff. w.r.t.x

$$\frac{d}{dx}(F(x)) = \frac{4x^3}{4} + \frac{2x}{2} + 5 + 0 = x^3 + x + 5$$

$$\text{Thus, we note that if } \int f(x) dx = F(x) \text{ then } \frac{d}{dx}(F(x)) = f(x)$$

i.e. integral $F(x)$ of $f(x)$ is indefinite because of the presence of arbitrary constant c .

In the next unit you will meet definite integral, where c will be cancel out. (Refer section 8.2 of Unit 8 of this course, i.e. MST-001).

7.4 INTEGRATION BY SUBSTITUTION

In section 7.3, we have taken into consideration the integrations for which a formula can directly be used. But sometimes integrand cannot be directly

integrated using standard formula. In order to convert it into a form for which standard formula can be applied, we substitute some function in place of some other function and this technique of obtaining the integration is known as integration by substitution method.

Substitution Method

If integral is of the type $\int (f(x))^n f'(x) dx$ or $\int \frac{f'(x)}{(f(x))^n} dx$, then

Step I We substitute $f(x) = t$... (1)

Step II Differentiate on both sides of (1)

Step III Change the given integral in terms of t

Step IV After simplification, if necessary, we get one of the standard forms discussed in Sec. 7.3. Using appropriate formula we can obtain the integral of given integrand in terms of t .

Step V Replace t in terms of x , we get the desired result after simplification, if required.

Following example will explain the substitution method and the steps involved in it:

Example 4: Evaluate the following integrals:

- (i) $\int \frac{x^9}{x^{10} + 1} dx$ (ii) $\int \frac{x^{n-1}}{x^n + a} dx$ (iii) $\int \frac{e^x}{e^x + 5} dx$
 (iv) $\int \frac{1}{x \log x} dx$ (v) $\int \frac{2ax + b}{ax^2 + bx + c} dx$ (vi) $\int \frac{8x^3 + 4x}{(x^4 + x^2 + 1)^6} dx$
 (vii) $\int \frac{e^{2x} + e^{-2x}}{e^{2x} - e^{-2x}} dx$ (viii) $\int \frac{1}{x + \sqrt{x}} dx$ (ix) $\int (2ax + b) \sqrt{ax^2 + bx + c} dx$
 (x) $\int \frac{2x}{(1 + x^2) \log(1 + x^2)} dx$

Solution:

(i) Let $I = \int \frac{x^9}{x^{10} + 1} dx$... (1)

Putting $x^{10} + 1 = t$

Differentiating

$$10x^9 dx = dt \Rightarrow x^9 dx = \frac{dt}{10}$$

$\therefore$ (1) becomes

$$\begin{aligned} I &= \int \frac{1}{t} \cdot \frac{dt}{10} = \frac{1}{10} \int \frac{dt}{t} = \frac{1}{10} \log|t| + c \quad \left[\because \int \frac{1}{y} dy = \log|y| + c \right] \\ &= \frac{1}{10} \log|x^{10} + 1| + c \quad [\text{Replacing } t \text{ in terms of } x] \\ &= \frac{1}{10} \log(x^{10} + 1) + c \quad \left[\because x^{10} + 1 \text{ cannot be -ve for real } x \right] \end{aligned}$$

Alternatively: We can also put

$$x^{10} = t$$

Differentiating

$$10x^9 dx = dt \Rightarrow x^9 dx = \frac{dt}{10}$$

$\therefore$ (1) becomes

$$\begin{aligned} I &= \int \frac{1}{t+1} \times \frac{dt}{10} = \frac{1}{10} \int \frac{1}{t+1} dt \\ &= \frac{1}{10} \log|t+1| + c && \left[\because \int \frac{1}{x+a} dx = \log|x+a| + c \right] \\ &= \frac{1}{10} \log|x^{10}+1| + c && [\text{Replacing } t \text{ in terms of } x] \\ &= \frac{1}{10} \log(x^{10}+1) + c && \left[\because x^{10}+1 \text{ is always +ve for real } x \right] \end{aligned}$$

(ii) Let $I = \int \frac{x^{n-1}}{x^n + a} dx \quad \dots (1)$

Putting $x^n + a = t$

Differentiating

$$nx^{n-1} dx = dt \Rightarrow x^{n-1} dx = \frac{dt}{n}$$

$\therefore$ (1) becomes

$$\begin{aligned} I &= \frac{1}{n} \int \frac{dt}{t} = \frac{1}{n} \log|t| + c && \left[\because \int \frac{1}{x} dx = \log|x| + c \right] \\ &= \frac{1}{n} \log|x^n + a| + c && [\text{Replacing } t \text{ in terms of } x] \end{aligned}$$

(iii) Let $I = \int \frac{e^x}{e^x + 5} dx \quad \dots (1)$

Putting $e^x + 5 = t$

Differentiating

$$e^x dx = dt$$

$\therefore$ (1) becomes

$$I = \int \frac{dt}{t} = \log|t| + c = \log|e^x + 5| + c \quad \left[\because \int \frac{1}{x} dx = \log|x| + c \right]$$

(iv) Let $I = \int \frac{1}{x \log x} dx \quad \dots (1)$

Putting $\log x = t$

Differentiating

$$\frac{1}{x} dx = dt$$

$\therefore$ (1) becomes

$$\begin{aligned} I &= \int \frac{dt}{t} = \log|t| + c && \left[\because \int \frac{1}{x} dx = \log|x| + c \right] \\ &= \log|\log x| + c && [\text{Replacing } t \text{ in terms of } x] \end{aligned}$$

(v) Let $I = \int \frac{2ax + b}{ax^2 + bx + c} dx \quad \dots (1)$

Putting $ax^2 + bx + c = t$

Differentiating

$$(2ax + b)dx = dt$$

$\therefore$ (1) becomes

$$I = \int \frac{dt}{t} = \log|t| + k = \log|ax^2 + bx + c| + k$$

where k is constant of integration

$$(vi) \text{ Let } I = \int \frac{8x^3 + 4x}{(x^4 + x^2 + 1)^6} dx = 2 \int \frac{4x^3 + 2x}{(x^4 + x^2 + 1)^6} dx \quad \dots (1)$$

$$\text{Putting } x^4 + x^2 + 1 = t$$

Differentiating

$$(4x^3 + 2x)dx = dt$$

$\therefore$ (1) becomes

$$\begin{aligned} I &= 2 \int \frac{dt}{t^6} = 2 \int t^{-6} dt = 2 \times \frac{t^{-5}}{-5} + c \\ &= \frac{-2}{5} (x^4 + x^2 + 1)^{-5} + c \quad [\text{Replacing } t \text{ in terms of } x] \end{aligned}$$

$$(vii) \text{ Let } I = \int \frac{e^{2x} + e^{-2x}}{e^{2x} - e^{-2x}} dx \quad \dots (1)$$

$$\text{Putting } e^{2x} - e^{-2x} = t$$

Differentiating

$$(2e^{2x} + 2e^{-2x})dx = dt \Rightarrow (e^{2x} + e^{-2x})dx = \frac{dt}{2}$$

$\therefore$ (1) becomes

$$\begin{aligned} I &= \frac{1}{2} \int \frac{dt}{t} = \frac{1}{2} \log|t| + c \quad \left[\because \int \frac{1}{x} dx = \log|x| + c \right] \\ &= \frac{1}{2} \log|e^{2x} - e^{-2x}| + c \quad [\text{Replacing } t \text{ in terms of } x] \end{aligned}$$

$$(viii) \text{ Let } I = \int \frac{1}{x + \sqrt{x}} dx = \int \frac{1}{\sqrt{x}(\sqrt{x} + 1)} dx \quad \dots (1)$$

$$\text{Putting } \sqrt{x} + 1 = t$$

Differentiating

$$\frac{1}{2\sqrt{x}} dx = dt \Rightarrow \frac{dx}{\sqrt{x}} = 2dt$$

$\therefore$ (1) becomes

$$I = 2 \int \frac{dt}{t} = 2 \log|t| + c = 2 \log|\sqrt{x} + 1| + c$$

$$(ix) \text{ Let } I = \int (2ax + b) \sqrt{ax^2 + bx + c} dx \quad \dots (1)$$

$$\text{Putting } ax^2 + bx + c = t$$

Differentiating

$$(2ax + b)dx = dt$$

$\therefore$ (1) becomes

$$I = \int \sqrt{t} dt = \frac{t^{3/2}}{3/2} + k = \frac{2}{3} (ax^2 + bx + c)^{3/2} + k,$$

where k is constant of integration

$$(x) \text{ Let } I = \int \frac{2x}{(1+x^2)\log(1+x^2)} dx \quad \dots (1)$$

Putting $\log(1+x^2) = t$

Differentiating

$$\frac{1}{1+x^2} 2x dx = dt$$

$\therefore (1)$ becomes

$$I = \int \frac{dt}{t} = \log|t| + c = \log|\log(1+x^2)| + c \text{ [Replacing } t \text{ in terms of } x]$$

Now, you can try the following exercise.

E 3) Evaluate the following integrals:

$$(i) \int \frac{2x+1}{(x^2+x+7)^5} dx \quad (ii) \int x\sqrt{x+a} dx$$

$$(iii) \int \frac{x}{\sqrt{x+a}} dx \quad (iv) \int \frac{1}{(1+x)\log(1+x)} dx$$

7.5 INTEGRATION USING PARTIAL FRACTIONS

The integrand may be in the form that it can be integrated only after resolving it into partial fractions. Here, in this section, we are going to deal with integration of such functions:

First of all we discuss the process of resolving such functions into partial fractions:

Important steps for resolving into partial fractions are:

1. Check degree of numerator, if it is less than that of denominator, go to step 2 and if it is greater than or equal to that of denominator, then first divide the numerator by the denominator and then go to step 2.
2. We may have one of the following main types of functions which we will dealt as discussed below:

Type 1 Denominator involve all linear factors with exponent as unity.

$$\text{e.g. } \frac{x+5}{(x-1)(x-2)(x-3)}.$$

$$\text{Step I} \text{ Let } \frac{x+5}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3} \quad \dots (1)$$

Step II Equate each of the factors of denominator to zero.

$$\text{i.e. } x-1=0 \Rightarrow x=1, \quad x-2=0 \Rightarrow x=2, \quad x-3=0 \Rightarrow x=3$$

Step III Put $x = 1, 2, 3$ every where (in the given expression) but not in the factor from which it has come out,

$$A = \frac{1+5}{(1-2)(1-3)} = \frac{6}{2} = 3, \quad [\text{By putting } x=1 \text{ in L.H.S. of (1)}]$$

$$\text{Step IV } B = \frac{2+5}{(2-1)(2-3)} = \frac{7}{-1} = -7, \quad [\text{By putting } x=2 \text{ in L.H.S. of (1)}]$$

Step V and $C = \frac{3+5}{(3-1)(3-2)} = \frac{8}{2 \times 1} = 4$ [By putting $x = 3$ in L.H.S. of (1)]

Thus, we may write $\frac{x+5}{(x-1)(x-2)(x-3)} = \frac{3}{x-1} + \frac{-7}{x-2} + \frac{4}{x-3}$

R.H.S. is nothing but the partial fractions of the given expression. Here we note that integration of R.H.S. is directly available, as we will see in the Example 5 of this unit.

Type 2 Denominator involves all linear factors but some have 2, 3, etc. as exponents

e.g. $\frac{x^2 + x + 5}{(x+5)(x+2)^3}$

Step I Let $\frac{x^2 + x + 5}{(x+5)(x+2)^3} = \frac{A}{x+5} + \frac{B}{x+2} + \frac{C}{(x+2)^2} + \frac{D}{(x+2)^3}$

Multiply on both sides by denominator of L.H.S. in this case

by $(x+5)(x+2)^3$, we get

$$x^2 + x + 5 = A(x+2)^3 + B(x+5)(x+2)^2 + C(x+5)(x+2) + D(x+5) \dots (1)$$

Step II Equate each of the factors to zero.

$$\text{i.e. } x+5=0 \Rightarrow x=-5, x+2=0 \Rightarrow x=-2$$

Step III Put $x = -5$ in (1) we get value of A, as given below

$$(-5)^2 + (-5) + 5 = A(-5+2)^3 + B(0) + C(0) + D(0)$$

$$\Rightarrow 25 = -27A \Rightarrow \boxed{A = -\frac{25}{27}}$$

Step IV Put $x = -2$ in (1) we get value of D, as given below

$$(-2)^2 + (-2) + 5 = A(0) + B(0) + C(0) + D(-2+5)$$

$$\Rightarrow 7 = 3D \Rightarrow \boxed{D = \frac{7}{3}}$$

Step V In order to find the values of B, C we have to equate the coefficients of different powers of x on both sides of (1).

In present case equating coefficients of x^3 and constant terms, we get

$$0 = A + B \dots (2)$$

$$5 = 8A + 20B + 10C + 5D \dots (3)$$

By putting value of A from Step III and value of D from step IV in equations (2) and (3), we get.

$$0 = -\frac{25}{27} + B = 0 \Rightarrow \boxed{B = \frac{25}{27}}$$

$$5 = 8\left(-\frac{25}{27}\right) + 20B + 10C + 5\left(\frac{7}{3}\right)$$

$$\Rightarrow 10C = 5 + \frac{200}{27} - 20\left(\frac{25}{27}\right) - \frac{35}{3} \Rightarrow \boxed{C = -48}$$

Thus, we may write

$$\frac{x^2 + x + 5}{(x+5)(x+2)^3} = \frac{-25/27}{x+5} + \frac{25/27}{x+2} + \frac{-48}{(x+2)^2} + \frac{7/3}{(x+2)^3}$$

R.H.S. is nothing but the partial fractions of the given expression. Here we note that integration of R.H.S. is directly available, as we will see in Example 5 of this unit.

Type 3 Denominator involves quadratic expressions. We will not discuss the problems based on this type, because it will involve the integral formulae which are beyond our contents.

Type 4 Denominator involves higher powers of quadratic expressions. This type is also not discussed here because it will involve the integral formulae which are beyond our contents.

Following example will illustrate how these types are used in evaluating integrals.

Example 5: Evaluate the following integrals:

$$\begin{aligned} \text{(i)} \quad & \int \frac{4x+1}{(x-1)(x-2)} dx & \text{(ii)} \quad & \int \frac{3x+4}{x^2-x-12} dx & \text{(iii)} \quad & \int \frac{8x}{(x+1)(x-3)^2} dx \\ \text{(iv)} \quad & \int \frac{x^2+x+2}{(x+2)(x+1)^3} dx & \text{(v)} \quad & \int \frac{x^2+1}{x^2-1} dx \end{aligned}$$

Solution:

$$\begin{aligned} \text{(i)} \quad \text{Let } I &= \int \frac{4x+1}{(x-1)(x-2)} dx \\ &= \int \left[\frac{-5}{x-1} + \frac{9}{x-2} \right] dx & \left[\begin{array}{l} \text{Using type 1 procedure as already} \\ \text{discussed. Put } x=1 \text{ every where} \\ \text{except in } x-1 \text{ and } x=2 \text{ every} \\ \text{where except in } x-2, \text{ we have} \\ A = \frac{4.1+1}{1-2} = -5, B = \frac{4.2+1}{2-1} = 9 \end{array} \right] \\ &= -5 \log|x-1| + 9 \log|x-2| + c \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \text{Let } I &= \int \frac{3x+4}{x^2-x-12} dx = \int \frac{3x+4}{(x-4)(x+3)} dx \\ &= \int \left(\frac{16/7}{x-4} + \frac{5/7}{x+3} \right) dx & \left[\begin{array}{l} \text{Using partial fractions} \\ \text{as discussed in type 1} \end{array} \right] \\ &= \frac{16}{7} \log|x-4| + \frac{5}{7} \log|x+3| + c \end{aligned}$$

$$\text{(iii)} \quad \text{Let } I = \int \frac{8x}{(x+1)(x-3)^2} dx \quad \dots (1)$$

Let us first resolve into partial fractions

$$\text{Let } \frac{8x}{(x+1)(x-3)^2} = \frac{A}{x+1} + \frac{B}{x-3} + \frac{C}{(x-3)^2}$$

Multiplying on both sides by $(x+1)(x-3)^2$, we get

$$8x = A(x-3)^2 + B(x+1)(x-3) + C(x+1) \quad \dots (2)$$

Putting $x = -1$ in (2), we get $[\because x+1=0 \text{ gives } x=-1]$

$$-8 = A(-1-3)^2 + B(0) + C(0) \Rightarrow -8 = 16A \Rightarrow \boxed{A = -\frac{1}{2}}$$

Putting $x = 3$ in (2), we get $[\because x-3=0 \text{ gives } x=3]$

$$24 = A(0) + B(0) + C(3+1) \Rightarrow 24 = 4C \Rightarrow \boxed{C=6}$$

Comparing coefficient of x^2 on both sides of (2), we get

$$0 = A + B \Rightarrow B = -A \Rightarrow \boxed{B = \frac{1}{2}}$$

$$\begin{aligned} \therefore I &= \int \left(\frac{-1/2}{x+1} + \frac{1/2}{x-3} + \frac{6}{(x-3)^2} \right) dx = \int \left[\frac{-1/2}{x+1} + \frac{1/2}{x-3} + 6(x-3)^{-2} \right] dx \\ &= \frac{-1}{2} \log|x+1| + \frac{1}{2} \log|x-3| + 6 \frac{(x-3)^{-1}}{-1} + c \\ &= -\frac{1}{2} \log|x+1| + \frac{1}{2} \log|x-3| - \frac{6}{x-3} + c \end{aligned}$$

(iv) Let $I = \int \frac{x^2 + x + 2}{(x+2)(x+1)^3} dx$

Let us first resolve into partial fractions

$$\text{Let } \frac{x^2 + x + 2}{(x+2)(x+1)^3} = \frac{A}{x+2} + \frac{B}{x+1} + \frac{C}{(x+1)^2} + \frac{D}{(x+1)^3}$$

Multiplying on both sides by $(x+2)(x+1)^3$, we get

$$x^2 + x + 2 = A(x+1)^3 + B(x+2)(x+1)^2 + C(x+2)(x+1) + D(x+2) \dots (2)$$

Putting $x = -2$ in (2), we get $[\because x+2=0 \text{ gives } x=-2]$

$$(-2)^2 + (-2) + 2 = A(-2+1)^2 + B(0) + C(0) + D(0)$$

$$\Rightarrow 4 = A \Rightarrow \boxed{A=4}$$

Putting $x = -1$ in (2), we get $[\because x+1=0 \text{ gives } x=-1]$

$$(-1)^2 + (-1) + 2 = A(0) + B(0) + C(0) + D(-1+2)$$

$$\Rightarrow 2 = D \Rightarrow \boxed{D=2}$$

Comparing coefficients of x^3 and constant terms on both sides of (2), we get

$$0 = A + B \Rightarrow B = -A \Rightarrow \boxed{B = -4}$$

$$2 = A + 2B + 2C + 2D \Rightarrow 2C = 2 - A - 2B - 2D$$

$$\Rightarrow 2C = 2 - 4 + 8 - 4$$

$$\Rightarrow 2C = 2$$

$$\Rightarrow \boxed{C=1}$$

$$\begin{aligned} \therefore I &= \int \left[\frac{4}{x+2} + \frac{-4}{x+1} + \frac{1}{(x+1)^2} + \frac{2}{(x+1)^3} \right] dx \\ &= 4 \log|x+2| - 4 \log|x+1| + \frac{(x+1)^{-1}}{-1} + \frac{2(x+1)^{-2}}{-2} + c \\ &= 4 \log|x+2| - 4 \log|x+1| - \frac{1}{x+1} - \frac{1}{(x+1)^2} + c \end{aligned}$$

$$\begin{aligned}
 \text{(v) Let } I &= \int \frac{x^2+1}{x^2-1} dx = \int \left[1 + \frac{2}{x^2-1} \right] dx & x^2-1 &= \frac{x^2+1}{x^2-1} \\
 & & & - \frac{+}{2} \\
 &= \int \left[1 + \frac{2}{(x-1)(x+1)} \right] dx \\
 &= \int \left[1 + 2 \left(\frac{1/2}{x-1} + \frac{-1/2}{x+1} \right) \right] dx & \left[\begin{array}{l} \text{Using partial fractions} \\ \text{as discussed in type 1} \end{array} \right] \\
 &= x + \log|x-1| - \log|x+1| + c \\
 &= x + \log \left| \frac{x-1}{x+1} \right| + c & [\because \log m - \log n = \log \frac{m}{n}]
 \end{aligned}$$

Now, you can try following exercise.

E 4) Evaluate the following integrals:

$$\text{(i) } \int \frac{3x+2}{(x-1)(x-2)(x-3)} dx \quad \text{(ii) } \int \frac{x^3+5x+1}{x^2-4} dx$$

7.6 INTEGRATION BY PARTS

If u and v are any two functions of a single variable x such that first derivatives of u and v w.r.t. x exist, then by product rule, we have

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

Integrating on both sides, we have

$$uv = \int u \frac{dv}{dx} dx + \int v \frac{du}{dx} dx$$

$$\Rightarrow \int u \frac{dv}{dx} dx = uv - \int \frac{du}{dx} v dx \quad \dots (1)$$

$$\text{Let } u = f(x) \text{ and } \frac{dv}{dx} = g(x) \quad \dots (2)$$

$$\Rightarrow \frac{du}{dx} = f'(x) \text{ and } v = \int g(x) dx \quad \dots (3)$$

Using (2) and (3) in (1), we get

$$\int f(x) g(x) dx = f(x) \int g(x) dx - \int \left[f'(x) \int g(x) dx \right] dx$$

$$\text{Or } \int I \text{ II } dx = I \int \text{II } dx - \int \left[\frac{d}{dx}(I) \int \text{II } dx \right] dx \quad \dots (4)$$

where I = first function = $f(x)$
 II = second function = $g(x)$

R.H.S. of equation (4) is known as integration by parts of L.H.S. of equation (4), where I , and II just indicate our choice between the product of two functions taking as first and second functions.

Remark 3:

- (i) In case of integration by parts, choice of first and second function is important as explained in part (i) of Example 6 given below.
- (ii) If in the product of two functions, one is polynomial function then we take polynomial function as first function.
- (iii) Integration by parts is one of the methods (techniques) of integration. It does not mean that integration of product of any two functions exists.

Example 6: Evaluate the following integrals:

(i) $\int xe^x dx$ (ii) $\int x^2 e^{3x} dx$ (iii) $\int x^3 a^x dx$ (iv) $\int \log x dx$

Solution:

(i) Let $I = \int xe^x dx$

I II

Integrating by parts (taking x as first and e^x as second function)

$$I = x \int e^x dx - \int \left[\left(\frac{d}{dx}(x) \right) \left(\int e^x dx \right) \right] dx + c_1$$

where c_1 is constant of integration

$$= xe^x - \int (1)(e^x) dx + c_1 = xe^x - \int e^x dx + c_1 = xe^x - (e^x + c_2) + c_1$$

where c_2 is constant of integration

$$= xe^x - e^x + c, \text{ where } c = c_1 - c_2$$

Let us see what happens if we integrate by parts by taking x as second and e^x as first function:

$$I = e^x \int x dx - \int \left[\left(\frac{d}{dx}(e^x) \right) \left(\int x dx \right) \right] dx + c_1$$

$$= e^x \frac{x^2}{2} - \int \left[e^x \frac{x^2}{2} \right] dx + c_1 = \frac{x^2 e^x}{2} - \frac{1}{2} \int x^2 e^x dx + c_1$$

We see that integration becomes more complicated. So choice of first and second function is important.

Note: In future we will add c as constant of integration only once.

(ii) Let $I = \int x^2 e^{3x} dx$

I II

Integrating by parts (taking x^2 as first and e^{3x} as second function)

$$I = x^2 \left(\frac{e^{3x}}{3} \right) - \int (2x) \left(\frac{e^{3x}}{3} \right) dx + c$$

where c is constant of integration

$$= \frac{x^2 e^{3x}}{3} - \frac{2}{3} \int x e^{3x} dx + c$$

I II

Again integrating by parts (taking x as first and e^{3x} as second function)

$$\begin{aligned} I &= \frac{x^2 e^{3x}}{3} - \frac{2}{3} \left[x \left(\frac{e^{3x}}{3} \right) - \int (1) \left(\frac{e^{3x}}{3} \right) dx \right] + c \\ &= \frac{x^2 e^{3x}}{3} - \frac{2}{3} \left[\frac{x e^{3x}}{3} - \frac{1}{3} \int e^{3x} dx \right] + c \\ &= \frac{x^2 e^{3x}}{3} - \frac{2}{3} \left(\frac{x e^{3x}}{3} - \frac{e^{3x}}{9} \right) + c = \frac{1}{3} x^2 e^{3x} - \frac{2}{9} x e^{3x} + \frac{2}{27} e^{3x} + c \end{aligned}$$

(iii) Let $I = \int x^3 a^x dx$

I II

Integrating by parts (taking x^3 as first and a^x as second function)

$$I = (x^3) \left(\frac{a^x}{\log a} \right) - \int (3x^2) \left(\frac{a^x}{\log a} \right) dx + c = \frac{x^3 a^x}{\log a} - \frac{3}{\log a} \int x^2 a^x dx + c$$

Again integrating by parts (taking x^2 as first and a^x as second function)

$$\begin{aligned} I &= \frac{x^3 a^x}{\log a} - \frac{3}{\log a} \left[(x^2) \left(\frac{a^x}{\log a} \right) - \int (2x) \left(\frac{a^x}{\log a} \right) dx \right] + c \\ &= \frac{x^3 a^x}{\log a} - \frac{3x^2 a^x}{(\log a)^2} + \frac{6}{(\log a)^2} \int x a^x dx + c \end{aligned}$$

I II

Again integrating by parts (taking x as first and a^x as second function)

$$\begin{aligned} &= \frac{x^3 a^x}{\log a} - \frac{3x^2 a^x}{(\log a)^2} + \frac{6}{(\log a)^2} \left[(x) \left(\frac{a^x}{\log a} \right) - \int (1) \left(\frac{a^x}{\log a} \right) dx \right] + c \\ &= \frac{x^3 a^x}{\log a} - \frac{3x^2 a^x}{(\log a)^2} + \frac{6}{(\log a)^2} \left[\frac{x a^x}{\log a} - \frac{a^x}{(\log a)^2} \right] + c \\ &= \frac{x^3 a^x}{\log a} - \frac{3x^2 a^x}{(\log a)^2} + \frac{6x a^x}{(\log a)^3} - \frac{6a^x}{(\log a)^4} + c \end{aligned}$$

(iv) Let $I = \int \log x dx = \int 1 \times \log x dx$

II I

Integrating by parts (taking $\log x$ as first and 1 as second function)

$$I = (\log x)(x) - \int \left(\frac{1}{x} \right) (x) dx + c = x \log x - \int 1 dx + c = x \log x - x + c$$

Here, is an exercise for you.

E 5) Evaluate the following integrals:

(i) $\int x^2 e^{-x} dx$ (ii) $\int x^3 e^{x^2} dx$

7.7 SUMMARY

Let us summarise the topics that we have covered in this unit.

1) Integral of some functions like constant, x^n , $\frac{1}{x^n}$, polynomial,

$(ax + b)^n$, $\frac{1}{ax + b}$, exponential whose integral are directly available.

2) Integration by method of substitution.

3) Integration by use of partial fractions.

4) Integration by parts.

7.8 SOLUTIONS/ANSWERS

$$\begin{aligned}\text{E 1) (i)} \quad \int \left(x^2 + \frac{1}{x^2} \right)^2 dx &= \int \left(x^4 + \frac{1}{x^4} + 2x^2 \cdot \frac{1}{x^2} \right) dx = \int \left(x^4 + \frac{1}{x^4} + 2 \right) dx \\ &= \int (x^4 + x^{-4} + 2) dx \\ &= \frac{x^5}{5} + \frac{x^{-3}}{-3} + 2x + c \quad \left[\begin{array}{l} \because \int x^n dx = \frac{x^{n+1}}{n+1} + c \\ \text{and } \int k dx = kx + c \end{array} \right]\end{aligned}$$

$$= \frac{x^5}{5} - \frac{1}{3x^3} + 2x + c$$

$$\begin{aligned}\text{(ii)} \quad \int \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^3 dx &= \int \left[x^{3/2} + \frac{1}{x^{3/2}} + 3\sqrt{x} \cdot \frac{1}{\sqrt{x}} \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) \right] dx \\ &= \int (x^{3/2} + x^{-3/2} + 3x^{1/2} + 3x^{-1/2}) dx \\ &= \frac{x^{5/2}}{5/2} + \frac{x^{-1/2}}{-1/2} + \frac{3x^{3/2}}{3/2} + \frac{3x^{1/2}}{1/2} + c \\ &\quad \left[\because \int x^n dx = \frac{x^{n+1}}{n+1} + c \right] \\ &= \frac{2}{5} x^{5/2} - \frac{2}{\sqrt{x}} + 2x^{3/2} + 6\sqrt{x} + c\end{aligned}$$

$$\text{(iii)} \quad \int (\alpha - 3) dx = (\alpha - 3)x + c \quad [\because \alpha - 3 \text{ is a constant}]$$

$$\text{(iv)} \quad \int \sqrt{x} \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) dx = \int (x + 1) dx = \frac{x^2}{2} + x + c$$

$$\begin{aligned}\text{(v)} \quad \int (x^a + 1)(x^b + 1) dx &= \int (x^{a+b} + x^a + x^b + 1) dx + c \\ &= \frac{x^{a+b+1}}{a+b+1} + \frac{x^{a+1}}{a+1} + \frac{x^{b+1}}{b+1} + x + c, \\ &\quad \left[\because \int x^n dx = \frac{x^{n+1}}{n+1} \text{ and } \int k dx = kx, \text{ where } k \text{ is constant} \right] \\ &\quad \text{where } a \neq -1, b \neq -1, a + b \neq -1\end{aligned}$$

$$(vi) \int \frac{x^m}{x^n} dx = \int x^{m-n} dx = \frac{x^{m-n+1}}{m-n+1} + c, m-n \neq -1 \quad \left[\because \int x^n dx = \frac{x^{n+1}}{n+1} \right]$$

$$\begin{aligned} (vii) \int \left(x + \frac{1}{x} \right) \left(x^3 + \frac{1}{x^3} \right) dx &= \int \left(x^4 + \frac{1}{x^2} + x^2 + \frac{1}{x^4} \right) dx \\ &= \int (x^4 + x^{-2} + x^2 + x^{-4}) dx \\ &= \frac{x^5}{5} + \frac{x^{-1}}{-1} + \frac{x^3}{3} + \frac{x^{-3}}{-3} + c \quad \left[\because \int x^n dx = \frac{x^{n+1}}{n+1} \right] \\ &= \frac{x^5}{5} - \frac{1}{x} + \frac{x^3}{3} - \frac{1}{3x^3} + c \end{aligned}$$

$$\text{E 2) (i) } \int \left(a^x + e^x a^x + \frac{x}{a} \right) dx = \int \left(a^x + (ea)^x + \frac{x}{a} \right) dx = \frac{a^x}{\log a} + \frac{(ea)^x}{\log ea} + \frac{x^2}{2a} + c$$

$$\begin{aligned} (ii) \int &\left(3^{2 \log_3 x} + 3^{x \log_3 a} + a^{a \log_a x} + a^{x \log_a a} \right) dx \\ &= \int \left(3^{\log_3 x^2} + 3^{\log_3 a^x} + a^{\log_a x^a} + a^{\log_a a^x} \right) dx \\ &= \int \left(x^2 + a^x + x^a + a^a \right) dx \quad [\because a^{\log_a f(x)} = f(x)] \\ &= \frac{x^3}{3} + \frac{a^x}{\log|a|} + \frac{x^{a+1}}{a+1} + a^a x + c \end{aligned}$$

$$\text{E 3) (i) Let } I = \int \frac{2x+1}{(x^2+x+7)^5} dx \quad \dots (1)$$

Putting $x^2 + x + 7 = t$

Differentiating

$$(2x+1)dx = dt$$

$\therefore$ (1) becomes

$$I = \int \frac{dt}{t^5} = \int t^{-5} dt = \frac{t^{-4}}{-4} + c = \frac{-1}{4t^4} + c = -\frac{1}{4(x^2+x+7)^4} + c$$

$$(ii) \text{ Let } I = \int x\sqrt{x+a} dx \quad \dots (1)$$

Putting $\sqrt{x+a} = t \Rightarrow x+a = t^2$

Differentiating

$$dx = 2t dt$$

$\therefore$ (1) becomes

$$\begin{aligned} I &= \int (t^2 - a)t(2t)dt = 2 \int (t^4 - at^2)dt = 2 \left(\frac{t^5}{5} - \frac{at^3}{3} \right) + c \\ &= 2 \left[\frac{(x+a)^{5/2}}{5} - \frac{a}{3}(x+a)^{3/2} \right] + c \end{aligned}$$

$$(iii) \text{ Let } I = \int \frac{x}{\sqrt{x+a}} dx \quad \dots (1)$$

Putting $\sqrt{x+a} = t \Rightarrow x+a = t^2$

Differentiating

$$dx = 2t dt$$

$\therefore$ (1) becomes

$$I = \int \frac{dt}{t} = \log|t| + c = \log|\log(1+x)| + c$$

Using partial fractions as discussed in type1, we get

$$A = \frac{3.1+2}{(1-2)(1-3)} = \frac{5}{2}, B = \frac{3.2+2}{(2-1)(2-3)} = -8, C = \frac{3.3+2}{(3-1)(3-2)} = \frac{11}{2}$$

$$= \frac{5}{2} \log|x-1| - 8 \log|x-2| + \frac{11}{2} \log|x-3| + c$$

$$I = \int \left[x + \frac{9x+1}{x^2-4} \right] dx = \int \left[x + \frac{9x+1}{(x-2)(x+2)} \right] dx$$

$$\begin{aligned} I &= \int \left[x + \frac{19/4}{x-2} + \frac{17/4}{x+2} \right] dx && \left[\begin{array}{l} \text{Using partial fractions} \\ \text{as discussed in type 1} \end{array} \right] \\ &= \frac{x^2}{2} + \frac{19}{4} \log|x-2| + \frac{17}{4} \log|x+2| + c && \left[\because \int \frac{1}{x+a} dx = \log|x+a| \right] \\ &= \frac{x^2}{2} + \frac{1}{4} (19 \log|x-2| + 17 \log|x+2|) + c \\ &= \frac{x^2}{2} + \frac{1}{4} (\log|x-2|^{19} + \log|x+2|^{17}) + c && [\because n \log m = \log m^n] \\ &= \frac{x^2}{2} + \frac{1}{4} \log|(x-2)^{19} (x+2)^{17}| + c && [\because \log m + \log n = \log mn] \end{aligned}$$

E 5) (i) Let $I = \int x^2 e^{-x} dx$

I II

Integrating by parts (taking x^2 as first and e^{-x} as second function)

$$I = (x^2) \left(\frac{e^{-x}}{-1} \right) - \int (2x) \left(\frac{e^{-x}}{-1} \right) dx + c$$

where c is constant of integration

$$= -x^2 e^{-x} + 2 \int x e^{-x} dx + c$$

I II

Integrating by parts (taking x as first and e^{-x} as second function)

$$I = -x^2 e^{-x} + 2 \left[(x) \left(\frac{e^{-x}}{-1} \right) - \int (1) \left(\frac{e^{-x}}{-1} \right) dx \right] + c$$

$$= -x^2 e^{-x} + 2 \left[-x e^{-x} + \int e^{-x} dx \right] + c$$

$$= -x^2 e^{-x} + 2 \left[-x e^{-x} + \left(\frac{e^{-x}}{-1} \right) \right] + c$$

$$= -x^2 e^{-x} - 2x e^{-x} - 2e^{-x} + c$$

(ii) Let $I = \int x x^2 e^{x^2} dx \dots (1)$

Putting $x^2 = t$

Differentiating

$$2x dx = dt \Rightarrow x dx = \frac{dt}{2}$$

$\therefore (1)$ becomes

$$I = \frac{1}{2} \int t e^t dt$$

I II

Integrating by parts (taking t as first and e^t as second function)

$$I = \frac{1}{2} \left[(t)(e^t) - \int (1)(e^t) dt \right] + c = \frac{1}{2} (te^t - e^t) + c = \frac{1}{2} (x^2 e^{x^2} - e^{x^2}) + c$$

$$= \frac{1}{2} (x^2 - 1) e^{x^2} + c$$

UNIT 8 DEFINITE INTEGRATION

Structure

- 8.1 Introduction
 - Objectives
- 8.2 Meaning and Geometrical Interpretation
- 8.3 Definite Integral of some commonly used Functions
- 8.4 Elementary Properties of Definite Integral
- 8.5 Examples based on Properties of Definite Integral
- 8.6 Summary
- 8.7 Solutions/Answers

8.1 INTRODUCTION

In Units 6 and 7 we have discussed concept of differentiation and concept of indefinite integral. But on many occasions, we are interested in finding out the probability of a continuous random variable in certain limits, this job is done by using the concept of definite integral.

This unit discusses about definite integral, evaluation of definite integral of some commonly used functions with the help of large number of examples. Properties of definite integral and how these are used have been also discussed in this unit with the help of number of examples.

Objectives

After completing this unit, you should be able to:

- define definite integration and give its geometrical meaning;
- evaluate the integration of some commonly used functions;
- explain the properties of the definite integrations; and
- evaluate the integrations using properties of definite integrations.

8.2 MEANING AND GEOMETRICAL INTERPRETATION

Notation and Definition

You have already studied that if

$$\frac{d}{dx}(F(x)) = f(x), \text{ then } \int f(x)dx = F(x) + c$$

where c is arbitrary constant and hence value of $\int f(x)dx$ is indefinite.

Here, we are going to discuss definite integrals.

Definite integral of a function $f(x)$ within the limits

$$a < x < b \text{ or } a \leq x < b \text{ or } a < x \leq b \text{ or } a \leq x \leq b \quad \dots (1)$$

is denoted by $\int_a^b f(x)dx$

and is defined as

**Fundamental theorem
of Integral Calculus:**

If f is integrable on $[a, b]$
and F is such that

$$\frac{d}{dx}(F(x)) = f(x),$$

$x \in [a, b]$

then

$$\int_a^b f(x) dx = F(b) - F(a)$$

$$\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a) \quad \dots (2) \quad \left[\begin{array}{l} \text{By fundamental theorem} \\ \text{of integral calculus} \end{array} \right]$$

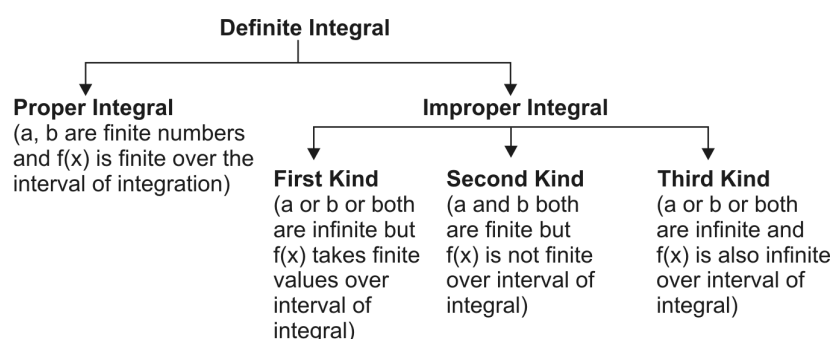
, where a is called lower limit, b is called upper limit and either of the intervals in (1) to be used is known as interval of integration.

For example,
$$\int_2^5 (x+3) dx = \left[\frac{x^2}{2} + 3x + c \right]_2^5 = \frac{(5)^2}{2} + 3(5) + c - \left[\frac{2^2}{2} + 3 \times 2 + c \right]$$

$$= 25/2 + 15 - 2 - 6 = 39/2$$

The integral $\int_a^b f(x) dx$ being definite integral, so it will have a definite value,

but equation (2) suggests that value of L.H.S. of (2) depends on a , b and $f(x)$. Following diagram shows that definite integral have different names based on the role of a , b and $f(x)$.



Here we shall discuss proper integrals and only first kind of improper integrals as these will be applicable later on in the subsequent courses.

Note: The notation $[F(x)]_a^b$ means that function $F(x)$ is to be evaluated at top and bottom limits and then subtract. Some authors use American text book notation $|F(x)|_a^b$ instead of $[F(x)]_a^b$. But here we will use square bracket notation.

Geometrical Interpretation

The definite integral $\int_a^b f(x) dx$ represents the area bounded by the function

$y = f(x)$, x -axis and between the lines $x = a$, $x = b$ as shown by the shaded region in the following Fig. 8.1

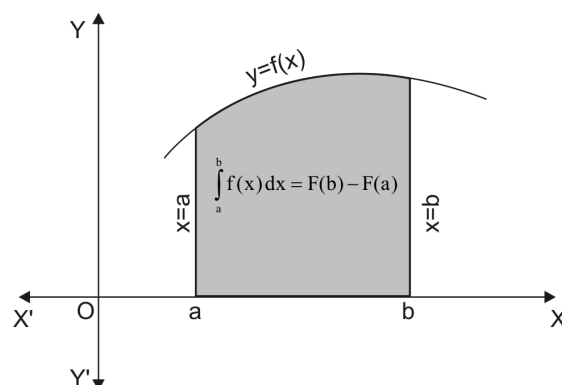


Fig. 8.1

Remark 1: In solving numerical problems, generally c is not used.

For example, if $f(x) = x^2 + x + 6$ then $\int f(x)dx = \frac{x^3}{3} + \frac{x^2}{2} + 6x + c = F(x)$

We will not write it as

$$\begin{aligned}\int_1^2 f(x)dx &= [F(x)]_1^2 = \left[\frac{x^3}{3} + \frac{x^2}{2} + 6x + c \right]_1^2 \\ &= \frac{2^3}{3} + \frac{2^2}{2} + 6 \times 2 + c - \left(\frac{1^3}{3} + \frac{1^2}{2} + 6 \times 1 + c \right) \\ &= \frac{8}{3} + 2 + 12 + c - \frac{1}{3} - \frac{1}{2} - 6 - c = \frac{8}{3} + 8 - \frac{1}{3} - \frac{1}{2} \\ &= \frac{16 + 48 - 2 - 3}{6} = \frac{59}{6}\end{aligned}$$

But throughout the unit, we will write it as

$$\begin{aligned}\int_1^2 f(x)dx &= \int_1^2 (x^2 + x + 6)dx = \left[\frac{x^3}{3} + \frac{x^2}{2} + 6x \right]_1^2 \\ &= \frac{8}{3} + \frac{4}{2} + 12 - \left(\frac{1}{3} + \frac{1}{2} + 6 \right) = \frac{59}{6}\end{aligned}$$

8.3 DEFINITE INTEGRAL OF SOME COMMONLY USED FUNCTIONS

Let us here consider some examples of definite integrals based on the formulae of indefinite integral already discussed in Unit 7 of this course.

Example 1: Evaluate the following integrals:

- | | | |
|-------------------------------------|--|-------------------------------------|
| (i) $\int_2^6 8dx$ | (ii) $\int_0^2 (x^2 + 1)dx$ | (iii) $\int_2^3 (3x + 4)dx$ |
| (iv) $\int_2^5 4x^3 dx$ | (v) $\int_2^3 x^a dx$ | (vi) $\int_2^4 \sqrt{18x - 24} dx$ |
| (vii) $\int_1^2 (x + 1)(x^2 - 1)dx$ | (viii) $\int_2^3 \frac{x^3 + 5}{x^2} dx$ | (ix) $\int_2^5 \frac{1}{2x - 3} dx$ |
| (x) $\int_2^3 \frac{4}{x} dx$ | (xi) $\int_1^4 2^x dx$ | (xii) $\int_0^2 e^{3x} dx$ |
| (xiii) $\int_0^2 3^{4x+1} dx$ | (xiv) $\int_1^3 e^{2x+5} dx$ | |

Solution:

- (i) $\int_2^6 8dx = [8x]_2^6 = 8 \times 6 - 8 \times 2 = 48 - 16 = 32$
- (ii) $\int_0^2 (x^2 + 1)dx = \left[\frac{x^3}{3} + x \right]_0^2 = \frac{8}{3} + 2 - 0 - 0 = \frac{14}{3}$

$$(iii) \int_2^3 (3x+4)dx = \left[\frac{3x^2}{2} + 4x \right]_2^3 = \frac{27}{2} + 12 - 6 - 8 = \frac{27}{2} - 2 = \frac{23}{2}$$

$$(iv) \int_2^5 4x^3 dx = \left[\frac{4x^4}{4} \right]_2^5 = \left[x^4 \right]_2^5 = 625 - 16 = 609$$

$$(v) \int_2^3 x^a dx = \left[\frac{x^{a+1}}{a+1} \right]_2^3 = \frac{1}{a+1} (3^{a+1} - 2^{a+1})$$

$$(vi) \int_2^4 \sqrt{18x-24} dx = \left[\frac{(18x-24)^{3/2}}{\frac{3}{2} \times 18} \right]_2^4 \quad \left[\because \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a \times (n+1)} \right]$$

$$= \frac{1}{27} [(72-24)^{3/2} - (36-24)^{3/2}] = \frac{1}{27} [48^{3/2} - (12)^{3/2}]$$

$$= \frac{1}{27} [48\sqrt{48} - 12\sqrt{12}] = \frac{1}{27} [48 \times 4\sqrt{3} - 12 \times 2\sqrt{3}]$$

$$= \frac{24}{27} [8\sqrt{3} - \sqrt{3}] = \frac{8}{9} (7\sqrt{3}) = \frac{56\sqrt{3}}{9}$$

$$(vii) \int_1^2 (x+1)(x^2-1)dx = \int_1^2 (x^3 - x + x^2 - 1)dx$$

$$= \left[\frac{x^4}{4} - \frac{x^2}{2} + \frac{x^3}{3} - x \right]_1^2$$

$$= \frac{16}{4} - \frac{4}{2} + \frac{8}{3} - 2 - \left(\frac{1}{4} - \frac{1}{2} + \frac{1}{3} - 1 \right)$$

$$= 4 - 2 + \frac{8}{3} - 2 - \left(\frac{3-6+4-12}{12} \right)$$

$$= \frac{8}{3} + \frac{11}{12} = \frac{32+11}{12} = \frac{43}{12}$$

$$(viii) \int_2^3 \frac{x^3+5}{x^2} dx = \int_2^3 \left(\frac{x^3}{x^2} + \frac{5}{x^2} \right) dx = \int_2^3 (x + 5x^{-2}) dx = \left[\frac{x^2}{2} + \frac{5x^{-1}}{-1} \right]_2^3$$

$$= \left[\frac{x^2}{2} - \frac{5}{x} \right]_2^3 = \frac{9}{2} - \frac{5}{3} - \left(\frac{4}{2} - \frac{5}{2} \right)$$

$$= \frac{27-10}{6} - \frac{4-5}{2} = \frac{17}{6} + \frac{1}{2} = \frac{17+3}{6} = \frac{20}{6} = \frac{10}{3}$$

$$(ix) \int_2^5 \frac{1}{2x-3} dx = \left[\frac{1}{2} \log|2x-3| \right]_2^5 = \frac{1}{2} [\log 7 - \log 1] = \frac{1}{2} \log 7 \quad \text{as } \log 1 = 0$$

$$(x) \int_2^3 \frac{4}{x} dx = 4 \int_2^3 \frac{1}{x} dx = 4 [\log|x|]_2^3 = 4(\log 3 - \log 2) = 4 \log \frac{3}{2}$$

$$(xi) \int_1^4 2^x dx = \left[\frac{2^x}{\log 2} \right]_1^4 = \frac{1}{\log 2} [2^4 - 2^1] = \frac{14}{\log 2}$$

$$(xii) \int_0^2 e^{3x} dx = \left[\frac{e^{3x}}{3} \right]_0^2 = \frac{1}{3}(e^6 - e^0) = \frac{e^6 - 1}{3}$$

$$(xiii) \int_0^2 3^{4x+1} dx = \left[\frac{3^{4x+1}}{4 \log 3} \right]_0^2 = \frac{1}{4 \log 3} [3^9 - 3^1] = \frac{1}{4 \log 3} (19683 - 3) = \frac{19680}{4 \log 3}$$

$$(xiv) \int_1^3 e^{2x+5} dx = \left[\frac{e^{2x+5}}{2} \right]_1^3 = \frac{1}{2}(e^{11} - e^7)$$

Example 2: Evaluate the following integrals:

$$(i) \int_1^6 x\sqrt{x+3} dx \quad (ii) \int_0^1 \frac{2x+3}{x^2+3x+5} dx$$

$$(iii) \int_0^2 \frac{2x+7}{(x-3)(x+1)(x-4)} dx \quad (iv) \int_0^3 \frac{x-5}{(x+1)(x+2)^2} dx$$

Solution:

$$(i) \text{ Let } I = \int_1^6 x\sqrt{x+3} dx \quad \dots (1)$$

$$\text{Putting } \sqrt{x+3} = t \Rightarrow x+3 = t^2$$

Differentiating

$$dx = 2t dt$$

Also when $x = 1$, $t = 2$ and when $x = 6$, $t = 3$

$\therefore (1)$ becomes

$$\begin{aligned} I &= \int_2^3 (t^2 - 3)t(2t) dt = 2 \int_2^3 (t^4 - 3t^2) dt = 2 \left[\frac{t^5}{5} - t^3 \right]_2^3 \\ &= 2 \left[\frac{243}{5} - 27 - \left(\frac{32}{5} - 8 \right) \right] = 2 \left(\frac{243}{5} - \frac{32}{5} - 19 \right) \\ &= 2 \left(\frac{243 - 32 - 95}{5} \right) = 2 \left(\frac{116}{5} \right) = \frac{232}{5} \end{aligned}$$

$$(ii) \text{ Let } I = \int_0^1 \frac{2x+3}{x^2+3x+5} dx \quad \dots (1)$$

$$\text{Putting } x^2 + 3x + 5 = t$$

Differentiating

$$(2x+3)dx = dt$$

Also when $x = 0$, $t = 5$ and when $x = 1$, $t = 9$

$\therefore (1)$ becomes

$$I = \int_5^9 \frac{dt}{t} = [\log|t|]_5^9 = \log 9 - \log 5 = \log \frac{9}{5}$$

$$(iii) \text{ Let } I = \int_0^2 \frac{2x+7}{(x-3)(x+1)(x-4)} dx = \int_0^2 \left(\frac{-13/4}{x-3} + \frac{1/4}{x+1} + \frac{3}{x-4} \right) dx$$

$$\left[\begin{array}{l} \text{Using partial fractions as discussed in type 1, we get} \\ A = \frac{2 \times 3 + 7}{(3+1)(3-4)} = -\frac{13}{4}, B = \frac{2(-1)+7}{(-1-3)(-1-4)} = \frac{1}{4}, C = \frac{2 \times 4 + 7}{(4-3)(4+1)} = 3 \end{array} \right]$$

$$\begin{aligned}
 I &= -\frac{13}{4}[\log|x-3|]_0^2 + \frac{1}{4}[\log|x+1|]_0^2 + 3[\log|x-4|]_0^2 \\
 &= -\frac{13}{4}(\log 1 - \log 3) + \frac{1}{4}(\log 3 - \log 1) + 3(\log 2 - \log 4) \\
 &= -\frac{13}{4}(0 - \log 3) + \frac{1}{4}(\log 3 - 0) + 3(\log 2 - \log 2^2) \quad \text{as } \log 1 = 0 \\
 &= \frac{13}{4}\log 3 + \frac{1}{4}\log 3 + 3(\log 2 - 2\log 2) \quad [\because \log m^n = n \log m] \\
 &= \frac{1}{4}(13\log 3 + \log 3) + 3(-\log 2) \\
 &= \frac{14\log 3}{4} - 3\log 2 = \frac{7}{2}\log 3 - 3\log 2
 \end{aligned}$$

$$(iv) \text{ Let } I = \int_0^3 \frac{x-5}{(x+1)(x+2)^2} dx \quad \dots (1)$$

First we resolve into partial fractions

$$\text{Let } \frac{x-5}{(x+1)(x+2)^2} = \frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{(x+2)^2}$$

Multiply on both sides by $(x+1)(x+2)^2$, we get

$$x-5 = A(x+2)^2 + B(x+1)(x+2) + C(x+1) \quad \dots (2)$$

Putting $x = -1$ in (2), we get $[\because x+1 = 0 \text{ gives } x = -1]$

$$-6 = A(-1+2)^2 + B(0) + C(0) \Rightarrow -6 = A \Rightarrow \boxed{A = -6}$$

Putting $x = -2$ in (2), we get $[\because x+2 = 0 \text{ gives } x = -2]$

$$-7 = A(0) + B(0) + C(-2+1) \Rightarrow -7 = -C \Rightarrow \boxed{C = 7}$$

Comparing coefficient of x^2 on both sides of (2), we get

$$0 = A + B \Rightarrow B = -A \Rightarrow \boxed{B = 6}$$

$$\begin{aligned}
 \therefore I &= \int_0^3 \left[\frac{-6}{x+1} + \frac{6}{x+2} + \frac{7}{(x+2)^2} \right] dx \\
 &= -6[\log|x+1|]_0^3 + 6[\log|x+2|]_0^3 + 7 \left[\frac{(x+2)^{-1}}{-1} \right]_0^3 \\
 &= -6(\log 4 - \log 1) + 6(\log 5 - \log 2) - 7 \left(\frac{1}{5} - \frac{1}{2} \right) \\
 &= -6\log 2^2 + 6\log 5 - 6\log 2 - 7 \left(\frac{2-5}{10} \right) \quad \text{as } \log 1 = 0 \\
 &= -12\log 2 + 6\log 5 - 6\log 2 + \frac{21}{10} \quad [\because \log m^n = n \log m] \\
 &= 6\log 5 - 18\log 2 + \frac{21}{10}
 \end{aligned}$$

Now, you can try the following exercises.

E 1) Evaluate the following integrals:

$$(i) \int_2^4 x^2 dx \quad (ii) \int_0^2 \left(x - \frac{5}{2}\right) dx$$

E 2) Evaluate the following integrals:

$$(i) \int_2^5 \frac{x}{x^2 + 3} dx \quad (ii) \int_0^2 \frac{e^x}{3 + e^{2x}} dx \quad (iii) \int_0^1 \frac{2x^2 + 1}{(x - 4)(x - 2)^3} dx$$

8.4 ELEMENTARY PROPERTIES OF DEFINITE INTEGRAL

Here, first we list some properties and then we will use these properties to evaluate some integrals.

$$\mathbf{P\ 1} \quad \int_a^b f(x) dx = \int_a^b f(t) dt \quad (\text{Change of variable property})$$

$$\mathbf{P\ 2} \quad \int_a^b f(x) dx = - \int_b^a f(x) dx \quad (\text{Interchange of limits property})$$

$$\mathbf{P\ 3} \quad \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx, \quad a < c < b$$

In general

We can introduce any number of points between a and b

$$\text{e.g. } \int_a^b f(x) dx = \int_a^{c_1} f(x) dx + \int_{c_1}^{c_2} f(x) dx + \dots + \int_{c_{n-1}}^{c_n} f(x) dx + \int_{c_n}^b f(x) dx$$

$$\text{where, } a < c_1 < c_2 < \dots < c_{n-1} < c_n < b$$

$$\mathbf{P\ 4} \quad \int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(x) \text{ is an even function} \\ 0, & \text{if } f(x) \text{ is an odd function} \end{cases}$$

$$\mathbf{P\ 5} \quad \int_a^b f(x) dx = \int_a^b f(a + b - x) dx$$

$$\text{In particular, } \int_0^a f(x) dx = \int_0^a f(a - x) dx$$

$$\mathbf{P\ 6} \quad \int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a - x) dx$$

$$\mathbf{P7} \quad \int_0^{2a} f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(2a - x) = f(x) \\ 0, & \text{if } f(2a - x) = -f(x) \end{cases}$$

Proof:

P 1 Let $\int f(x)dx = F(x)$, so $\int f(t)dt = F(t)$

Now, by fundamental theorem of integral calculus

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a) \quad \dots (1)$$

$$\text{and } \int_a^b f(t)dt = [F(t)]_a^b = F(b) - F(a) \quad \dots (2)$$

From (1) and (2)

$$\int_a^b f(x)dx = \int_a^b f(t)dt$$

P 2 Let $\int f(x)dx = F(x)$

$\therefore$ by fundamental theorem of integral calculus

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a) \quad \dots (1)$$

$$\text{and } \int_b^a f(x)dx = [F(x)]_b^a = F(a) - F(b) = -[F(b) - F(a)] \quad \dots (2)$$

From (1) and (2)

$$\int_a^b f(x)dx = -\int_b^a f(x)dx$$

P 3 Let $\int f(x)dx = F(x)$

$\therefore$ by fundamental theorem of integral calculus

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a) \quad \dots (1)$$

$$\begin{aligned} \text{and } \int_a^c f(x)dx + \int_c^b f(x)dx &= [F(x)]_a^c + [F(x)]_c^b = (F(c) - F(a)) + (F(b) - F(c)) \\ &= F(b) - F(a) \quad \dots (2) \end{aligned}$$

From (1) and (2)

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

P 4 Using property 3, we have

$$\int_{-a}^a f(x)dx = \int_{-a}^0 f(x)dx + \int_0^a f(x)dx \quad \dots (1)$$

$$\text{Let } I = \int_{-a}^0 f(x)dx$$

Putting $x = -t$

Differentiating

$$dx = -dt$$

Also, when $x = -a$, $t = a$ and when $x = 0$, $t = 0$

$$\therefore I = -\int_a^0 f(-t)dt = \int_0^a f(-x)dx \quad \dots (2) \quad [\text{Using properties 1 and 2}]$$

Using (2) in (1), we get

$$\begin{aligned} \int_{-a}^a f(x)dx &= \int_0^a f(-x)dx + \int_0^a f(x)dx \quad \dots (3) \\ &= \begin{cases} \int_0^a f(x)dx + \int_0^a f(x)dx, & \text{if } f \text{ is even} \\ -\int_0^a f(x)dx + \int_0^a f(x)dx, & \text{if } f \text{ is odd} \end{cases} \\ &= \begin{cases} 2\int_0^a f(x)dx, & \text{if } f \text{ is even function} \\ 0, & \text{if } f \text{ is an odd function} \end{cases} \end{aligned}$$

P 5 Let $I = \int_a^b f(x)dx \quad \dots (1)$

R.H.S. suggests that we should put

$$x = a + b - t$$

Differentiating

$$dx = -dt$$

Also, when $x = a \Rightarrow t = b$ and when $x = b \Rightarrow t = a$

$\therefore$ (1) becomes

$$\begin{aligned} I &= \int_b^a f(a+b-t)(-dt) = -\int_b^a f(a+b-t)dt \\ &= \int_a^b f(a+b-t)dt \quad [\text{Using property 2}] \\ &= \int_a^b f(a+b-x)dx \quad [\text{Using property 1}] \end{aligned}$$

In particular

If we put $a = 0, b = a$ in this result, then

$$\int_0^a f(x)dx = \int_0^a f(a-x)dx$$

P 6 $\int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_a^{2a} f(x)dx \quad [\text{Using property 3}]$

$$= I_1 + I_2 \quad \dots (1)$$

$$I_2 = \int_a^{2a} f(x)dx$$

Putting $x = 2a - t$

Differentiating

$$dx = -dt$$

Also, when $x = a, t = a$ and when $x = 2a, t = 0$

$$\begin{aligned}\therefore I_2 &= \int_a^0 f(2a-t)(-dt) = \int_0^a f(2a-t)dt \quad [\text{Using property 2}] \\ &= \int_0^a f(2a-x)dx \quad \dots (2) \quad [\text{Using property 1}]\end{aligned}$$

Using (2) in (1), we get

$$\int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a-x)dx$$

P7 From property 6

$$\begin{aligned}\int_0^{2a} f(x)dx &= \int_0^a f(x)dx + \int_0^a f(2a-x)dx \\ &= \begin{cases} \int_0^a f(x)dx + \int_0^a f(x)dx, & \text{if } f(2a-x) = f(x) \\ \int_0^a f(x)dx - \int_0^a f(x)dx, & \text{if } f(2a-x) = -f(x) \end{cases} \\ &= \begin{cases} 2 \int_0^a f(x)dx, & \text{if } f(2a-x) = f(x) \\ 0, & \text{if } f(2a-x) = -f(x) \end{cases}\end{aligned}$$

8.5 EXAMPLES BASED ON PROPERTIES OF DEFINITE INTEGRAL

In this section, you will see how the properties of definite integral, discussed in previous Sec. are used and save lot of calculation work.

Example 3: Evaluate the following integrals:

$$(i) \int_1^4 |x-2|dx \quad (ii) \int_0^2 |2x-3|dx \quad (iii) \int_0^3 f(x)dx, \text{ where } f(x) = \begin{cases} x+1, & 0 \leq x < 1 \\ 2x+3, & 1 \leq x < 3 \end{cases}$$

$$(iv) \int_{-99}^{99} (x^3 + x + e^x - e^{-x})dx \quad (v) \int_{-2}^2 \log\left(\frac{5-x}{5+x}\right)dx \quad (vi) \int_{-3}^3 2^{|2x|}dx$$

$$(vii) \int_a^b \frac{f(x)}{f(x) + f(a+b-x)}dx \quad (viii) \int_2^3 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{5-x}}dx \quad (ix) \int_2^7 \frac{\sqrt[5]{x+1}}{\sqrt[5]{x+1} + \sqrt[5]{10-x}}dx$$

Solution:

$$(i) \text{ Let } I = \int_1^4 |x-2|dx = \int_1^2 |x-2|dx + \int_2^4 |x-2|dx \quad [\text{By P3}]$$

$$I = \int_1^2 -(x-2)dx + \int_2^4 (x-2)dx \quad \left[\begin{array}{l} \because \text{ for } 1 \leq x < 2, x-2 < 0 \text{ so } \\ |x-2| = -(x-2) \\ \text{and for } 2 \leq x \leq 4, x-2 \geq 0 \text{ so } \\ |x-2| = x-2 \end{array} \right]$$

$$\begin{aligned}
 &= -\left[\frac{x^2}{2} - 2x\right]_1^2 + \left[\frac{x^2}{2} - 2x\right]_2^4 = -\left(\frac{4}{2} - 4 - \left(\frac{1}{2} - 2\right)\right) + \left(\frac{16}{2} - 8 - \left(\frac{4}{2} - 4\right)\right) \\
 &= -\left(2 - 4 - \frac{1}{2} + 2\right) + (8 - 8 - 2 + 4) = \frac{1}{2} + 2 = \frac{5}{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \int_0^2 |2x-3| dx &= \int_0^{3/2} |2x-3| dx + \int_{3/2}^2 |2x-3| dx \quad [\text{By P 3}] \\
 &= \int_0^{3/2} -(2x-3) dx + \int_{3/2}^2 (2x-3) dx \\
 &\quad \left[\begin{array}{l} \because \text{for } 0 \leq x < 3/2 \Rightarrow 2x-3 < 0 \text{ so } |2x-3| = -(2x-3) \text{ and} \\ \text{for } 3/2 \leq x \leq 2 \Rightarrow 2x-3 \geq 0 \text{ so } |2x-3| = 2x-3 \end{array} \right] \\
 &= -\left[x^2 - 3x\right]_0^{3/2} + \left[x^2 - 3x\right]_{3/2}^2 \\
 &= -\left(\frac{9}{4} - \frac{9}{2} - 0 + 0\right) + \left(4 - 6 - \frac{9}{4} + \frac{9}{2}\right) \\
 &= -\left(-\frac{9}{4}\right) - 2 + \frac{9}{4} = \frac{9}{4} - 2 + \frac{9}{4} = \frac{9-8+9}{4} = \frac{10}{4} = \frac{5}{2}
 \end{aligned}$$

$$\text{(iii) Let } I = \int_0^3 f(x) dx, \quad \text{where } f(x) = \begin{cases} x+1, & 0 \leq x < 1 \\ 2x+3, & 1 \leq x < 3 \end{cases} \quad \dots (1)$$

$$\begin{aligned}
 \text{Now, } I &= \int_0^1 f(x) dx + \int_1^3 f(x) dx \quad [\text{Using property 3}] \\
 &= \int_0^1 (x+1) dx + \int_1^3 (2x+3) dx \quad [\text{Using (1)}] \\
 &= \left[\frac{x^2}{2} + x\right]_0^1 + \left[x^2 + 3x\right]_1^3 \\
 &= \frac{1}{2} + 1 - 0 - 0 + (9 + 9 - 1 - 3) = \frac{3}{2} + 14 = \frac{31}{2}
 \end{aligned}$$

$$\text{(iv) Let } I = \int_{-99}^{99} (x^3 + x + e^x - e^{-x}) dx$$

$$\text{Let } f(x) = x^3 + x + e^x - e^{-x}$$

$$\begin{aligned}
 \therefore f(-x) &= (-x)^3 + (-x) + e^{-x} - e^{-(-x)} = -x^3 - x + e^{-x} - e^x \\
 &= -(x^3 + x + e^x - e^{-x}) = -f(x)
 \end{aligned}$$

$\Rightarrow f(x)$ is an odd function

$$\therefore I = \int_{-99}^{99} (x^3 + x + e^x - e^{-x}) dx = 0 \quad [\text{By property 4}]$$

$$\text{(v) Let } I = \int_{-2}^2 \log\left(\frac{5-x}{5+x}\right) dx$$

$$\text{Let } f(x) = \log\left(\frac{5-x}{5+x}\right)$$

$$\begin{aligned}\therefore f(-x) &= \log\left(\frac{5-(-x)}{5+(-x)}\right) = \log\left(\frac{5+x}{5-x}\right) = \log\left(\frac{5-x}{5+x}\right)^{-1} \\ &= -\log\left(\frac{5-x}{5+x}\right) = -f(x) \quad [\because \log m^n = n \log m]\end{aligned}$$

$\Rightarrow f(x)$ is an odd function

$$\therefore I = \int_{-2}^2 \log\left(\frac{5-x}{5+x}\right) dx = 0 \quad [\text{By property 4}]$$

$$\begin{aligned}\text{(vi)} \quad \int_{-3}^3 2^{|2x|} dx &= \int_{-3}^0 e^{|2x|} dx + \int_0^3 e^{|2x|} dx \\ &= \int_{-3}^0 e^{-2x} dx + \int_0^3 e^{2x} dx \quad \left[\begin{array}{l} \because \text{for } -3 < x < 0, 2x < 0 \text{ so } |2x| = -2x \text{ and} \\ \text{for } 0 < x < 3, 2x > 0 \text{ so } |2x| = 2x \end{array} \right] \\ &= \left[\frac{e^{-2x}}{-2} \right]_{-3}^0 + \left[\frac{e^{2x}}{2} \right]_0^3 = -\frac{1}{2}(1 - e^6) + \frac{1}{2}(e^6 - 1) \\ &= \frac{1}{2}(-1 + e^6 + e^6 - 1) = \frac{2e^6 - 2}{2} = e^6 - 1\end{aligned}$$

$$\begin{aligned}\text{(vii)} \quad \text{Let } I &= \int_a^b \frac{f(x)}{f(x) + f(a+b-x)} dx \quad \dots (1) \\ &= \int_a^b \frac{f(a+b-x)}{f(a+b-x) + f(a+b-(a+b-x))} dx \quad [\text{Using property 5}]\end{aligned}$$

$$I = \int_a^b \frac{f(a+b-x)}{f(a+b-x) + f(x)} dx \quad \dots (2)$$

(1) + (2) gives

$$\begin{aligned}2I &= \int_a^b \frac{f(x) + f(a+b-x)}{f(x) + f(a+b-x)} dx = \int_a^b 1 dx = [x]_a^b = b - a \\ \Rightarrow I &= \frac{b-a}{2}\end{aligned}$$

$$\begin{aligned}\text{(viii)} \quad \text{Let } I &= \int_2^3 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{5-x}} dx \quad \dots (1) \\ &= \int_2^3 \frac{\sqrt{5-x}}{\sqrt{5-x} + \sqrt{5-(5-x)}} dx \quad [\text{Using property 5}]\end{aligned}$$

$$I = \int_2^3 \frac{\sqrt{5-x}}{\sqrt{5-x} + \sqrt{x}} dx \quad \dots (2)$$

(1) + (2) gives

$$\begin{aligned}2I &= \int_2^3 \frac{\sqrt{x} + \sqrt{5-x}}{\sqrt{x} + \sqrt{5-x}} dx = \int_2^3 1 dx = [x]_2^3 = 3 - 2 = 1 \\ \Rightarrow I &= 1/2\end{aligned}$$

$$\text{(ix)} \quad \text{Let } I = \int_2^7 \frac{\sqrt[5]{x+1}}{\sqrt[5]{x+1} + \sqrt[5]{10-x}} dx \quad \dots (1)$$

$$= \int_2^7 \frac{\sqrt[5]{9-x+1}}{\sqrt[5]{9-x+1} + \sqrt[5]{10-(9-x)}} dx \quad [\text{Using property 5}]$$

$$I = \int_2^7 \frac{\sqrt[5]{10-x}}{\sqrt[5]{10-x} + \sqrt[5]{1+x}} dx \quad \dots (2)$$

(1) + (2) gives

$$2I = \int_2^7 \frac{\sqrt[5]{x+1} + \sqrt[5]{10-x}}{\sqrt[5]{10-x} + \sqrt[5]{x+1}} dx = \int_2^7 1 dx = [x]_2^7 = 7 - 2 = 5$$

$$\Rightarrow I = 5/2$$

Now, you can try the following exercise.

E 3 Evaluate the following integrals:

$$(i) \int_0^5 |x^2 - 3x + 2| dx \quad (ii) \int_1^5 (|x-1| + |x+2| + |x-3|) dx$$

$$(iii) \int_{-1}^2 f(x) dx, \text{ where } f(x) = \begin{cases} 4-3x, & -1 \leq x < 1 \\ 2x+1, & 1 \leq x \leq 2 \end{cases}$$

$$(iv) \int_{-77}^{77} x^5 \sqrt{a^2 - x^2} dx \quad (v) \int_{-1}^1 \frac{x^5 + x^7}{4 - x^4} dx \quad (vi) \int_0^2 x(2-x)^{11/2} dx \quad (vii) \int_{-3}^3 4x|x| dx$$

$$(viii) \int_1^5 f(x) dx, \text{ where } f(x) = \begin{cases} 1, & 1 \leq x < 2 \\ 2, & 2 \leq x < 3 \\ 3, & 3 \leq x < 4 \\ 4, & 4 \leq x < 5 \\ 5, & x \geq 5 \end{cases}$$

8.6 SUMMARY

Let us summarise the topics that we have covered in this unit:

1) Integration of some particular functions like

$$k, x^n (n \neq -1), \frac{1}{x}, (ax+b)^n, \frac{1}{ax+b} \text{ polynomial and exponential functions.}$$

2) Definite integral by use of substitution and partial fraction.

3) Elementary properties of definite integral.

4) Examples based on elementary properties of the definite integral.

8.7 SOLUTIONS/ANSWERS

$$\text{E 1) (i) } \int_2^4 x^2 dx = \left[\frac{x^3}{3} \right]_2^4 = \frac{1}{3}(64-8) = \frac{56}{3}$$

$$(ii) \int_0^2 \left(x - \frac{5}{2} \right) dx = \left[\frac{x^2}{2} - \frac{5}{2}x \right]_0^2 = \frac{4}{2} - \frac{5}{2} \times 2 - 0 - 0 = 2 - 5 = -3$$

E 2) (i) Let $I = \int_2^5 \frac{x}{x^2 + 3} dx \quad \dots (1)$

Putting $x^2 = t$

Differentiating

$$2x dx = dt \Rightarrow x dx = dt / 2$$

Also, when $x = 2$, $t = 4$ and when $x = 5$, $t = 25$

$$\begin{aligned} \therefore I &= \frac{1}{2} \int_4^{25} \frac{dt}{t+3} = \frac{1}{2} [\log|t+3|]_4^{25} = \frac{1}{2} (\log 28 - \log 7) = \frac{1}{2} \log \left(\frac{28}{7} \right) \\ &= \frac{1}{2} \log 4 = \frac{1}{2} \log 2^2 = \frac{1}{2} \times 2 \log 2 = \log 2 \end{aligned}$$

(ii) Let $I = \int_0^2 \frac{e^{2x}}{3 + e^{2x}} dx \quad \dots (1)$

Putting $e^{2x} = t$

Differentiating

$$2e^{2x} dx = dt \Rightarrow e^{2x} dx = \frac{dt}{2}$$

Also, when $x = 0$, $t = 1$ and when $x = 2$, $t = e^4$

$$\therefore I = \frac{1}{2} \int_1^{e^4} \frac{dt}{3+t} = \frac{1}{2} [\log|3+t|]_1^{e^4} = \frac{1}{2} [\log(3+e^4) - \log 4] = \frac{1}{2} \log \left(\frac{3+e^4}{4} \right)$$

(iii) $\int_0^1 \frac{2x^2 + 1}{(x-4)(x-2)^3} dx$

Let us first resolve into partial fractions

$$\text{Let } \frac{2x^2 + 1}{(x-4)(x-2)^3} = \frac{A}{x-4} + \frac{B}{x-2} + \frac{C}{(x-2)^2} + \frac{D}{(x-2)^3}$$

Multiplying on both sides by $(x-4)(x-2)^3$, we get

$$\begin{aligned} 2x^2 + 1 &= A(x-2)^3 + B(x-4)(x-2)^2 \\ &\quad + C(x-4)(x-2) + D(x-4) \quad \dots (2) \end{aligned}$$

Putting $x = 4$ in (2) we get $[\because x-4=0 \text{ gives } x=4]$

$$33 = A(4-2)^2 + B(0) + C(0) + D(0) \Rightarrow 33 = 4A \Rightarrow \boxed{A = \frac{33}{4}}$$

Putting $x = 2$ in (2), we get $[\because x-2=0 \text{ gives } x=2]$

$$9 = A(0) + B(0) + C(0) + D(2-4) \Rightarrow 9 = -2D \Rightarrow \boxed{D = -\frac{9}{2}}$$

Comparing coefficients of x^3 and constant terms on both sides of (2), we get

$$0 = A + B \Rightarrow B = -A \Rightarrow \boxed{B = -\frac{33}{4}}$$

$$1 = -8A - 16B + 8C - 4D \quad \dots (3)$$

Putting values of A, B and D in (3), we get

Definite Integral

$$1 = -8 \times \frac{33}{4} - 16 \left(-\frac{33}{4} \right) + 8C - 4 \left(-\frac{9}{2} \right)$$

$$1 = -66 + 132 + 8C + 18 \Rightarrow 8C = -83 \Rightarrow \boxed{C = \frac{-83}{8}}$$

$$\begin{aligned} \therefore I &= \int_0^1 \left[\frac{33/4}{x-4} + \frac{-33/4}{x-2} + \frac{-83/8}{(x-2)^2} + \frac{-9/2}{(x-2)^3} \right] dx \\ &= \frac{33}{4} \left[\log|x-4| \right]_0^1 - \frac{33}{4} \left[\log|x-2| \right]_0^1 - \frac{83}{8} \left[\frac{(x-2)^{-1}}{-1} \right]_0^1 \\ &\quad - \frac{9}{2} \left[\frac{(x-2)^{-2}}{-2} \right]_0^1 \\ &= \frac{33}{4} (\log|1-4| - \log|0-4|) - \frac{33}{4} (\log|1-2| - \log|0-2|) \\ &\quad + \frac{83}{8} \left[\frac{1}{1-2} - \frac{1}{0-2} \right] + \frac{9}{4} \left[\frac{1}{(1-2)^2} - \frac{1}{(0-2)^2} \right] \\ &= \frac{33}{4} (\log 3 - \log 4) - \frac{33}{4} (\log 1 - \log 2) + \frac{83}{8} \left(-1 + \frac{1}{2} \right) + \frac{9}{4} \left(1 - \frac{1}{4} \right) \\ &= \frac{33}{4} (\log 3 - \log 2^2) - \frac{33}{4} (0 - \log 2) - \frac{83}{16} + \frac{27}{16} \quad \text{as } \log 1 = 0 \\ &= \frac{33}{4} \log 3 - \frac{33}{2} \log 2 + \frac{33}{4} \log 2 - \frac{56}{16} \quad [\text{as } \log m^n = n \log m] \\ &= \frac{33}{4} \log 3 - \frac{33}{4} \log 2 - \frac{56}{16} \end{aligned}$$

E 3) (i) Let $I = \int_0^5 |x^2 - 3x + 2| dx = \int_0^5 |(x-2)(x-1)| dx$

$$\begin{aligned} &= \int_0^1 |(x-2)(x-1)| dx + \int_1^2 |(x-2)(x-1)| dx + \int_2^5 |(x-2)(x-1)| dx \\ &= \int_0^1 (x-2)(x-1) dx + \int_1^2 -(x-2)(x-1) dx + \int_2^5 (x-2)(x-1) dx \end{aligned}$$

$$\left[\begin{array}{l} \because \text{for } 0 \leq x < 1, (x-2)(x-1) > 0 \text{ so } |(x-2)(x-1)| = (x-2)(x-1) \\ \text{for } 1 \leq x < 2, (x-2)(x-1) \leq 0 \text{ so } |(x-2)(x-1)| = -(x-1)(x-2) \text{ and} \\ \text{for } 2 \leq x < 5, (x-2)(x-1) \geq 0 \text{ so } |(x-2)(x-1)| = (x-2)(x-1) \end{array} \right]$$

$$\begin{aligned} &= \int_0^1 (x^2 - 3x + 2) dx - \int_1^2 (x^2 - 3x + 2) dx + \int_2^5 (x^2 - 3x + 2) dx \\ &= \left[\frac{x^3}{3} - \frac{3x^2}{2} + 2x \right]_0^1 - \left[\frac{x^3}{3} - \frac{3x^2}{2} + 2x \right]_1^2 + \left[\frac{x^3}{3} - \frac{3x^2}{2} + 2x \right]_2^5 \end{aligned}$$

$$\begin{aligned}
 &= \left[\frac{1}{3} - \frac{3}{2} + 2 - (0 - 0 + 0) \right] - \left[\left(\frac{8}{3} - 6 + 4 \right) - \left(\frac{1}{3} - \frac{3}{2} + 2 \right) \right] \\
 &\quad + \left[\left(\frac{125}{3} - \frac{75}{2} + 10 \right) - \left(\frac{8}{3} - 6 + 4 \right) \right] \\
 &= \frac{2 - 9 + 12}{6} - \left(\frac{8}{3} - 2 - \frac{1}{3} + \frac{3}{2} - 2 \right) + \left(\frac{125}{3} - \frac{75}{2} + 10 - \frac{8}{3} + 2 \right) \\
 &= \frac{5}{6} - \frac{16 - 12 - 2 + 9 - 12}{6} + \frac{250 - 225 + 60 - 16 + 12}{6} \\
 &= \frac{5}{6} - \frac{-1}{6} + \frac{81}{6} = \frac{5 + 1 + 81}{6} = \frac{87}{6} = \frac{29}{2}
 \end{aligned}$$

(ii) Let $I = \int_1^5 (|x-1| + |x+2| + |x-3|) dx$

$$\begin{aligned}
 &= \int_1^5 |x-1| dx + \int_1^5 |x+2| dx + \int_1^5 |x-3| dx \\
 &= \int_1^5 (x-1) dx + \int_1^5 (x+2) dx + \int_1^3 |x-3| dx + \int_3^5 |x-3| dx \\
 &\quad \left[\because \text{for } 1 < x < 5, x-1 > 0 \right. \\
 &\quad \left. \text{so } |x-1| = x-1 \right] \\
 &= \left[\frac{x^2}{2} - x \right]_1^5 + \left[\frac{x^2}{2} + 2x \right]_1^5 + \int_1^3 -(x-3) dx + \int_3^5 (x-3) dx \\
 &\quad \left[\because \text{for } 1 < x < 3, x-3 < 0 \text{ so } |x-3| = -(x-3) \text{ and } \right. \\
 &\quad \left. \text{for } 3 \leq x \leq 5, x-3 \geq 0 \text{ so } |x-3| = x-3 \right] \\
 &= \left(\frac{25}{2} - 5 - \frac{1}{2} + 1 \right) + \left(\frac{25}{2} + 10 - \frac{1}{2} - 2 \right) - \left[\frac{x^2}{2} - 3x \right]_1^3 + \left[\frac{x^2}{2} - 3x \right]_3^5 \\
 &= \frac{25 - 10 - 1 + 2}{2} + \frac{25 + 20 - 1 - 4}{2} - \left(\frac{9}{2} - 9 - \frac{1}{2} + 3 \right) + \left(\frac{25}{2} - 15 - \frac{9}{2} + 9 \right) \\
 &= \frac{16}{2} + \frac{40}{2} - \left(\frac{9 - 18 - 1 + 6}{2} \right) + \left(\frac{25 - 30 - 9 + 18}{2} \right) \\
 &= 8 + 20 - \frac{-4}{2} + \frac{4}{2} = 28 + 2 + 2 = 32
 \end{aligned}$$

(iii) Let $I = \int_{-1}^2 f(x) dx,$

where $f(x) = \begin{cases} 4-3x, & -1 \leq x < 1 \\ 2x+1, & 1 \leq x \leq 2 \end{cases} \quad \dots (1)$

Now,

$$I = \int_{-1}^1 f(x) dx + \int_1^2 f(x) dx \quad [\text{Using property (2)}]$$

$$\begin{aligned}
&= \int_{-1}^1 (4-3x)dx + \int_1^2 (2x+1)dx \quad [\text{Using (1)}] \\
&= \left[4x - \frac{3x^2}{2} \right]_{-1}^1 + \left[x^2 + x \right]_1^2 = 4 - \frac{3}{2} - \left(-4 - \frac{3}{2} \right) + 4 + 2 - 1 - 1 \\
&= \frac{8-3}{2} - \left(\frac{-8-3}{2} \right) + 4 = \frac{5}{2} + \frac{11}{2} + 4 = \frac{5+11+8}{2} = \frac{24}{2} = 12
\end{aligned}$$

(iv) Let $I = \int_{-77}^{77} x^5 \sqrt{a^2 - x^2} dx$

Let $f(x) = x^5 \sqrt{a^2 - x^2}$

$$\therefore f(-x) = (-x)^5 \sqrt{a^2 - (-x)^2} = -x^5 \sqrt{a^2 - x^2} = -f(x)$$

$\Rightarrow f(x)$ is an odd function.

$$\therefore I = \int_{-77}^{77} x^5 \sqrt{a^2 - x^2} dx = 0 \quad [\text{By property 4}]$$

(v) $\int_{-1}^1 \frac{x^5 + x^7}{4 - x^4} dx$

Let $f(x) = \frac{x^5 + x^7}{4 - x^4}$

$$f(-x) = \frac{(-x)^5 + (-x)^7}{4 - (-x)^4} = \frac{-x^5 - x^7}{4 - x^4} = -\frac{x^5 + x^7}{4 - x^4} = -f(x)$$

$\Rightarrow f(x)$ is an odd function.

$$\therefore \int_{-1}^1 \frac{x^5 + x^7}{4 - x^4} dx = 0 \quad [\text{By property 4}]$$

(vi) Let $I = \int_0^2 x(2-x)^{11/2} dx = \int_0^2 (2-x)(2-(2-x))^{11/2} dx$ [Using property 5]

$$= \int_0^2 (2-x)(x)^{11/2} dx = \int_0^2 (2x^{11/2} - x^{13/2}) dx$$

$$= \left[2 \frac{x^{13/2}}{13/2} - \frac{x^{15/2}}{15/2} \right]_0^2 = \frac{4}{13} \times 2^{13/2} - \frac{2}{15} \times (2)^{15/2} - 0 + 0$$

$$= \frac{4}{13} \times 2^6 \times \sqrt{2} - \frac{2}{15} \times 2^7 \times \sqrt{2} = 2^8 \left(\frac{1}{13} - \frac{1}{15} \right) \sqrt{2}$$

$$= 2^8 \left(\frac{15-13}{13 \times 15} \right) \sqrt{2} = \frac{2^8 \times 2}{195} \sqrt{2} = \frac{512}{195} \sqrt{2}$$

(vii) Let $I = \int_{-3}^3 4x|x|dx$

Let $f(x) = 4x|x|$

$f(-x) = 4(-x)|-x| = -4x|(-1)x| = -4x|-1| \times |x| = -4x|x| = -f(x)$

$\Rightarrow f(x)$ is an odd function.

$\therefore I = \int_{-3}^3 4x|x|dx = 0$ [Using property 4]

(viii) $\int_1^5 f(x)dx,$

where $f(x) = \begin{cases} 1, & 1 \leq x < 2 \\ 2, & 2 \leq x < 3 \\ 3, & 3 \leq x < 4 \\ 4, & 4 \leq x < 5 \\ 5, & x \geq 5 \end{cases} \quad \dots (1)$

Now, $\int_1^5 f(x)dx = \int_1^2 f(x)dx + \int_2^3 f(x)dx + \int_3^4 f(x)dx + \int_4^5 f(x)dx$

[Using property 2]

$= \int_1^2 1dx + \int_2^3 2dx + \int_3^4 3dx + \int_4^5 4dx$ [Using (1)]

$= [x]_1^2 + [2x]_2^3 + [3x]_3^4 + [4x]_4^5$

$= (2-1) + (6-4) + (12-9) + (20-16)$

$= 1 + 2 + 3 + 4 = 10$



Indira Gandhi National Open University
School of Sciences

MST-001
Foundation in
Mathematics and
Statistics

Block

3

MATRICES, DETERMINANTS AND COLLECTION OF DATA

UNIT 9

Matrices and Determinants	5
----------------------------------	----------

UNIT 10

Applications of Matrices and Determinants	37
--	-----------

UNIT 11

Introduction to Statistics	55
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UNIT 12

Collection and Scrutiny of Data	75
--	-----------

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BLOCK 3 MATRICES, DETERMINANTS AND COLLECTION OF DATA

This is the third block of the course MST-001. This block is devoted to matrices, determinants, and introduction to statistics, collection and scrutiny of data. The aim of units 9 and 10 is just to give you an idea of matrices and determinants as the concepts related to these terms will be used in some sections of some courses. Last two units of this block and all the four units of the next block of this course put a foundation stone to the statistics.

Unit 9: Matrices and Determinants

This unit discusses what we mean by matrices and its different types, operation on matrices and trace of a square matrix. Concept of determinants and how we evaluate the determinants of square matrices of orders 1, 2, 3 and higher orders have been also discussed. This unit ends by giving some properties of determinants and how these properties are used, is explained with the help of some examples.

Unit 10: Applications of Matrices and Determinants

In this unit adjoint and inverse of a square matrix are discussed. Applications of matrices and determinants in solving a system of linear equations by matrix method and Cramer's rule have been also discussed.

Unit 11: Introduction to Statistics

This unit throws the light on origin and development, definition, scope and uses, and limitations of statistics. Different types of measurement of scale have been discussed in detail. Time series, cross section, discrete, continuous, frequency and non frequency data have been also discussed.

Unit 12: Collection and Scrutiny of Data

In this unit main methods of collection of primary data are discussed. Sources of collection of secondary data including some government publications, scrutiny of data and preparation of different kinds of questionnaires have been also discussed.

Notations and Symbols

$[a_{ij}]_{m \times n}$: matrix of order $m \times n$ with $(i, j)^{\text{th}}$ element as a_{ij}

A' : transpose of matrix A

$\text{tr}(A)$: trace of matrix A

$|A|$: determinant of the square matrix A

M_{ij} : minor of $(i, j)^{\text{th}}$ element of the matrix $[a_{ij}]$

A_{ij} : cofactor of $(i, j)^{\text{th}}$ element of the matrix $[a_{ij}]$

$\text{adj}A$: adjoint of A

A^{-1} : inverse of matrix A

Δ : capital delta

UNIT 9 MATRICES AND DETERMINANTS

Structure

- 9.1 Introduction
 - Objectives
- 9.2 Definition of a Matrix
- 9.3 Types of Matrices
- 9.4 Operations on Matrices
- 9.5 Transpose of a Matrix
- 9.6 Trace of a Matrix
- 9.7 Determinant of Square Matrices
- 9.8 Properties of Determinants
- 9.9 Summary
- 9.10 Solutions/Answers

9.1 INTRODUCTION

The knowledge of matrices has become necessary for the individuals working in different branches of science, technology, commerce, management and social sciences. In this unit, we introduce the concept of matrices and its elementary properties. The unit also discusses the determinant, which is a number associated with a square matrix and its properties. Trace of a matrix is also defined.

Objectives

After completing this unit, you should be able to:

- define a matrix and give examples of matrices;
- explain the types of matrices;
- know how operations on matrices are done;
- find multiplication of a matrix by a scalar;
- compute transpose of a matrix;
- find the trace of a square matrix;
- evaluate determinants find minors and cofactors of square matrices of different orders; and
- apply properties of determinants.

9.2 DEFINITION OF A MATRIX

Let us consider the following example to arrive at the definition of a matrix: Suppose there are three girls “Kavita, Preksha and Tanu” Kavita has 9 hundred rupees notes, 4 fifty rupees notes and 5 ten rupees notes. Preksha has 17 hundred rupees notes, 6 fifty rupees notes and one ten rupee note. Tanu has 8 hundred rupees notes, 3 fifty rupees notes and 2 ten rupees notes. This information can be represented as:

	Column1	Column 2	Column 3
	↓	↓	↓
	Rs.100	Rs.50	Rs.10
	Notes	Notes	Notes
Row 1 → Kavita	9	4	5
Row 2 → Preksha	17	6	1
Row 3 → Tanu	8	3	2

This is an arrangement of 9 (3×3) numbers in 3 rows and 3 columns. Such an arrangement is nothing but a matrix. Let us now define a matrix as follows:

Definition of a Matrix

An arrangement of $m \times n$ elements in m rows and n columns enclosed by the brackets () or [] only, is called a matrix of order $m \times n$ and is generally denoted by

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \dots & \vdots \\ \vdots & \vdots & \vdots & \dots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix} \quad \text{or} \quad \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \dots & \vdots \\ \vdots & \vdots & \vdots & \dots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{pmatrix}$$

where a_{ij} denotes the $(i, j)^{\text{th}}$ element of the matrix, i.e. element of i^{th} row and j^{th} column is denoted by a_{ij} .

Remark 1:

- (i) A matrix is denoted by capital letters A, B, C, etc. of the English alphabets.
- (ii) First suffix of an element of the matrix indicates the position of row and second suffix of the element of the matrix indicates position of column.
e.g. a_{23} means it is an element in the second row and the third column.
- (iii) The order of a matrix is written as “number of rows $\times$ number of columns”

For example,

$$(i) \quad A = \begin{bmatrix} 2 & 5 & 7 \\ 3 & 8 & 9 \end{bmatrix} \text{ is a matrix of order } 2 \times 3$$

$$(ii) \quad B = \begin{bmatrix} 9 & 6 \\ 1 & 0 \\ 8 & 4 \end{bmatrix} \text{ is a matrix of order } 3 \times 2$$

Let us consider some examples:

Example 1: Write the order of the matrix

$$A = \begin{bmatrix} 9 & 7 & 8 & -3 & -8 \\ 4 & 3 & 6 & 1 & -10 \\ 10 & 12 & 15 & 2 & 5 \end{bmatrix}$$

Also write the elements a_{23} , a_{14} , a_{35} , a_{22} , a_{31} , a_{32} .

Solution: Order of the matrix A is 3×5 and the desired elements are:

$$a_{23} = 6, a_{14} = -3, a_{35} = 5, a_{22} = 3, a_{31} = 10, a_{32} = 12$$

Example 2: Write all the possible orders of the matrix having following elements. (i) 8 (ii) 13

Solution:

(i) All the 8 elements can be arranged in single row, i.e. 1 row and 8 columns.

Or

They can be arranged in two rows with 4 elements in each row, i.e. 2 rows and 4 columns.

Or

in four rows with 2 elements in each row, i.e. 4 rows and 2 columns.

Or

in eight rows with 1 element in each row, i.e. 8 rows and 1 column.

$\therefore$ the possible orders are $1 \times 8, 2 \times 4, 4 \times 2, 8 \times 1$.

(ii) All the 13 element can be arranged in single row, i.e. 1 row and 13 columns.

Or

in 13 rows with 1 element in each row, i.e. 13 rows and 1 column.

$\therefore$ the possible orders are $1 \times 13, 13 \times 1$.

Example 3: Construct the matrix $A = [a_{ij}]_{2 \times 3}$, where $a_{ij} = \frac{(i-j)^2}{2}$

Solution: $A = [a_{ij}]_{2 \times 3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$, where $a_{ij} = \frac{(i-j)^2}{2}$

$$a_{11} = \frac{(1-1)^2}{2} = \frac{0}{2} = 0, a_{12} = \frac{(1-2)^2}{2} = \frac{(-1)^2}{2} = \frac{1}{2},$$

$$a_{13} = \frac{(1-3)^2}{2} = \frac{(-2)^2}{2} = \frac{4}{2} = 2, a_{21} = \frac{(2-1)^2}{2} = \frac{(1)^2}{2} = \frac{1}{2},$$

$$a_{22} = \frac{(2-2)^2}{2} = \frac{0}{2} = 0, a_{23} = \frac{(2-3)^2}{2} = \frac{(-1)^2}{2} = \frac{1}{2}$$

$$\therefore A = \begin{bmatrix} 0 & 1/2 & 2 \\ 1/2 & 0 & 1/2 \end{bmatrix}$$

Here is an exercise for you.

E 1) Construct $A = [a_{ij}]_{3 \times 2}$, where $a_{ij} = |i - j|$

9.3 TYPES OF MATRICES

On the basis of number of rows and number of columns and depending on the values of elements, the type of a matrix gets changed. Various types of matrix are explained as below:

Row Matrix

A matrix having only one row is called a row matrix.

For example, $[2 \ 5 \ 7]$, $[8 \ 9]$, $[1 \ 0 \ 3 \ 2]$ all are row matrices.

Column Matrix

A matrix having only one column is called a column matrix.

For example, $\begin{bmatrix} 9 \\ 6 \\ 7 \end{bmatrix}$, $\begin{bmatrix} 9 \\ -3 \\ 2 \\ 8 \end{bmatrix}$, $\begin{bmatrix} 5 \\ -11 \end{bmatrix}$ all are column matrices.

Remark 2: If a matrix has one element, e.g. $A = [6]$, then matrix A has only one row and only one column. So, it is both row matrix as well as column matrix.

Rectangular Matrix

A matrix having m rows and n columns is called a rectangular matrix if $m \neq n$.

For example, $\begin{bmatrix} 2 & 5 & 7 \\ 3 & 8 & 9 \end{bmatrix}$ is a rectangular matrix having 2 rows and 3 columns.

Square Matrix

A matrix having equal number of rows and columns is called a square matrix.

For example,

(i) $\begin{bmatrix} 4 & 6 \\ 5 & 3 \end{bmatrix}$ is a square matrix of order 2.

(ii) $\begin{bmatrix} 2 & 1 & 3 \\ 4 & 5 & 6 \\ 3 & -4 & 8 \end{bmatrix}$ is a square matrix of order 3.

Remark 3: For a square matrix, there is no need of mentioning the number of columns, e.g. in example (i) the order has been written as 2 and not 2×2 .

Diagonal Matrix

Principal Diagonal of a Matrix

If $A = [a_{ij}]_{n \times n}$ be a square matrix of order n then the elements

$a_{11}, a_{22}, a_{33}, \dots, a_{nn}$ are called diagonal elements of the square matrix A, and the diagonal along which these elements lie is called principal diagonal or **main diagonal** or simply **diagonal** of the matrix A.

For example,

(i) Diagonal elements of the matrix $A = \begin{bmatrix} 8 & 9 \\ 5 & 6 \end{bmatrix}$ are 8, 6.

(ii) Write the diagonal elements (if possible) of the matrix $A = \begin{bmatrix} 8 & 9 & 7 \\ 6 & 5 & 2 \end{bmatrix}$

Here, A is not a square matrix, so writing diagonal elements of a rectangular matrix is impossible.

Diagonal Matrix

A square matrix $A = [a_{ij}]_{n \times n}$ is said to be diagonal matrix if

$$a_{ij} = 0, \quad \forall i \neq j$$

For example,

$$(i) \text{ If } A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix} \text{ then it is a diagonal matrix because all its non-diagonal}$$

elements are zero. Sometimes, we denote it by writing $\text{diag. } [4, 3, 6]$.

$$(ii) A = \begin{bmatrix} 2 & 9 \\ 0 & 5 \end{bmatrix} \text{ is not a diagonal matrix because non-diagonal element } a_{12} \neq 0.$$

Remark 4:

(i) For a diagonal matrix all non diagonal elements must be zero.

(ii) In a diagonal matrix some or all the diagonal elements may be zero.

Example 4: Write all the diagonal matrices of order 2×2 having its elements only 0 or 1.

Solution: For a diagonal matrix, all the non-diagonal elements are zero. Therefore, we are to write 0 and 1 in the diagonal elements in different ways, i.e. 0, 0; 0, 1; 1, 0; and 1, 1.

$\therefore$ possible diagonal matrices with elements only 0 and 1 are given below:

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Scalar Matrix

A diagonal matrix is said to be scalar matrix if all its diagonal elements are same.

$$\text{For example, } \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ all are scalar matrices.}$$

Identity Matrix

A diagonal matrix is said to be Identity or **Unit matrix** if all the diagonal elements are equal to unity.

$$\text{For example, } \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ all are identity (or Unit)}$$

matrices of order 2, 3, 4 respectively.

Upper Triangular Matrix

A square matrix $A = [a_{ij}]_{n \times n}$ is said to be upper triangular matrix if all the elements below the principal diagonal are zero.

For example, $\begin{bmatrix} 2 & 5 \\ 0 & 7 \end{bmatrix}$, $\begin{bmatrix} 9 & 0 & 6 \\ 0 & 5 & 4 \\ 0 & 0 & 7 \end{bmatrix}$, $\begin{bmatrix} 8 & 0 & 5 & 0 \\ 0 & 9 & 3 & 0 \\ 0 & 0 & 4 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ all are upper triangles matrices.

But $\begin{bmatrix} 2 & 9 & 7 \\ 0 & 5 & 8 \\ 2 & 0 & 9 \end{bmatrix}$ is not an upper triangular matrix because one element below the diagonal line, i.e. a_{31} is non zero, which is 2, in this case.

Lower Triangular Matrix

A square matrix $A = [a_{ij}]_{n \times n}$ is said to be lower triangular matrix if all the elements above the principal diagonal are zero.

For example, $\begin{bmatrix} 5 & 0 \\ 3 & 2 \end{bmatrix}$, $\begin{bmatrix} 3 & 0 & 0 \\ 2 & 6 & 0 \\ 1 & 9 & 7 \end{bmatrix}$, are lower triangular matrices of orders 2 and 3 respectively.

Null Matrix

A matrix $A = [a_{ij}]_{m \times n}$ is said to be null matrix if all its elements are equal to zero.

i.e. $a_{ij} = 0$, $\forall i, j$

$\therefore$ a null matrix is generally denoted by **O**.

For example, $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, etc. are null matrices.

Comparable Matrices

Two matrices are said to be comparable if they are of the same order.

For example,

if $A = \begin{bmatrix} 2 & 5 & 3 \\ 6 & 8 & 9 \end{bmatrix}$, $B = \begin{bmatrix} a & b & c \\ x & y & z \end{bmatrix}$ then A and B are comparable because both are of the same order, i.e. of order 2×3 .

Equal Matrices

Two matrices are said to be equal if

- (i) they are of same order, and
- (ii) the corresponding elements of the matrices are equal.

For example, if $A = \begin{bmatrix} 2 & 8 \\ 3 & x \end{bmatrix}$, $B = \begin{bmatrix} a & 8 \\ 3 & 5 \end{bmatrix}$, then $A = B$, if $a = 2$, $x = 5$.

Example 5: Write orders and types of the following matrices:

(i) $\begin{bmatrix} 2 & 9 \\ 3 & 4 \end{bmatrix}$ (ii) $\begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$ (iii) $\begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix}$ (iv) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(v) $\begin{bmatrix} 2 & 5 & 7 \\ 0 & 8 & 0 \\ 0 & 0 & 9 \end{bmatrix}$ (vi) $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 7 & 6 \end{bmatrix}$ (vii) $\begin{bmatrix} 2 \\ 9 \\ 6 \end{bmatrix}$ (viii) $[8 \ 9 \ 1 \ 5]$ (ix) $\begin{bmatrix} 2 & 9 & 3 \\ 6 & 4 & 5 \end{bmatrix}$

Solution:

	Order	Type
(i)	2×2	Square matrix [$\because$ rows and columns are equal in number.]
(ii)	2×2	Diagonal matrix [$\because$ all the non-diagonal elements are zero.]
(iii)	2×2	Scalar matrix [$\because$ all the diagonal elements are equal and non diagonal element, are zero.]
(iv)	2×2	Identify matrix [$\because$ all the diagonal elements are unity and non diagonal element are zero.]
(v)	3×3	Upper triangular matrix [$\because$ all the elements below the principal diagonal are zero.]
(vi)	3×3	Lower triangular matrix [$\because$ all the elements above the principal diagonal are zero.]
(vii)	3×1	Column matrix [$\because$ it has only one column.]
(viii)	1×4	Row matrix [$\because$ it has only one row.]
(ix)	2×3	Rectangular matrix [$\because$ number of rows $\neq$ numbers of columns.]

Example 6:

(i) If $\begin{bmatrix} 3 & x+y \\ xy & 7+z \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 8 & 4 \end{bmatrix}$, find x, y, z.

(ii) If $\begin{bmatrix} a+5 & -2a & b+6 \\ 2c & 3b+2 & x \\ y+1 & z+3 & x+2 \end{bmatrix} = \begin{bmatrix} 2 & a+9 & 11 \\ c+4 & -b+22 & 3-x \\ 2 & 2z+3 & 5-x \end{bmatrix}$ find a, b, c, x, y, z.

Solution:

(i) We know that two matrices A and B are equal if

- their orders are same, and
- the corresponding elements of A and B are equal.

$\therefore$ on comparing corresponding elements of two matrices, we have

$$3 = 3$$

$$x + y = 6 \quad \dots (1)$$

$$xy = 8 \quad \dots (2)$$

$$7 + z = 4 \Rightarrow z = -3$$

$$\text{From (1), } y = 6 - x \quad \dots (3)$$

Putting y from (3) in (2), we get

$$x(6 - x) = 8$$

$$\Rightarrow 6x - x^2 - 8 = 0 \Rightarrow x^2 - 6x + 8 = 0 \Rightarrow x^2 - 4x - 2x + 8 = 0$$

$$\Rightarrow x(x - 4) - 2(x - 4) = 0 \Rightarrow (x - 4)(x - 2) = 0 \Rightarrow x = 4, 2$$

When $x = 4$, $y = 6 - 4 = 2$ and when $x = 2$, $y = 6 - 2 = 4$

$$\therefore x = 4, y = 2, z = -3 \text{ or } x = 2, y = 4, z = -3.$$

(ii) We know that two matrices A and B are equal if

(a) their orders are same, and

(b) the corresponding elements of A and B are equal.

$\therefore$ on comparing corresponding elements of two matrices, we have

$$a + 5 = 2 \Rightarrow a = -3$$

$$-2a = a + 9 \Rightarrow -3a = 9 \Rightarrow a = -3$$

$$b + 6 = 11 \Rightarrow b = 5$$

$$2c = c + 4 \Rightarrow c = 4$$

$$3b + 2 = -b + 22 \Rightarrow 4b = 20 \Rightarrow b = 5$$

$$x = 3 - x \Rightarrow 2x = 3 \Rightarrow x = \frac{3}{2}$$

$$y + 1 = 2 \Rightarrow y = 1$$

$$z + 3 = 2z + 3 \Rightarrow -z = 0 \Rightarrow z = 0$$

$$x + 2 = 5 - x \Rightarrow 2x = 3 \Rightarrow x = \frac{3}{2}$$

$$\therefore a = -3, b = 5, c = 4, x = \frac{3}{2}, y = 1, z = 0.$$

Here is an exercise for you.

E 2) Find the values of x, y, z, w if $\begin{bmatrix} 3x - 2y & z + w \\ 3z - w & x + y \end{bmatrix} = \begin{bmatrix} -1 & 7 \\ 5 & 3 \end{bmatrix}$.

9.4 OPERATIONS ON MATRICES

In school times, a child first learns the natural numbers and then learns how these numbers are added, subtracted, multiplied and divided. Similarly, here also we now see as to how such operations (except division) are applied on matrices.

These operations are explained by first giving a general formula and then examples followed by some exercises.

Remark 5: Division of a matrix by another matrix is meaning less and hence it is not permitted in case of matrices.

9.4.1 Addition of Matrices

Addition of two matrices A and B make sense only if they are of the same order and obtained by adding their corresponding elements. It is denoted by $A + B$.

That is, if $A = [a_{ij}]_{m \times n}$, $B = [b_{ij}]_{m \times n}$ then $A + B = [a_{ij} + b_{ij}]_{m \times n}$

For example,

(i) If $A = \begin{bmatrix} 2 & 3 & 4 \\ 7 & 5 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 5 & 6 \\ 2 & 9 & 8 \end{bmatrix}$ then

$$A + B = \begin{bmatrix} 2+1 & 3+5 & 4+6 \\ 7+2 & 5+9 & 1+8 \end{bmatrix} = \begin{bmatrix} 3 & 8 & 10 \\ 9 & 14 & 9 \end{bmatrix}.$$

- (ii) If $A = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 6 & 9 \\ 2 & 4 & 5 \end{bmatrix}$ then $A + B$ does not make any sense because
A and B are of different orders.

Properties of Addition of Matrices

If A, B, C are of the same orders over R, (i.e. elements of A, B, C are real numbers) then

- (i) $A + B = B + A$ (commutative law)
- (ii) $(A + B) + C = A + (B + C)$ (associative law)
- (iii) $A + \mathbf{O} = \mathbf{O} + A = A$, where $\mathbf{O}$ is a null matrix. (existence of additive identity)
- (iv) For a given matrix A, there exists a matrix B of the same order such that $A + B = \mathbf{O} = B + A$.
Here B is called additive inverse of A. (existence of additive inverse)

9.4.2 Scalar Multiplication

Let $A = [a_{ij}]_{m \times n}$ and k is any scalar then scalar multiplication of A by k is denoted by kA and obtained by multiplying each element of A by k.

i.e. $kA = [ka_{ij}]_{m \times n}$

For example,

If $A = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$ and $k = 7$, then $kA = 7A = \begin{bmatrix} 7 \times 3 & 7 \times 4 \\ 7 \times 5 & 7 \times 6 \end{bmatrix} = \begin{bmatrix} 21 & 28 \\ 35 & 42 \end{bmatrix}$.

Properties of Scalar Multiplication

If A and B are two matrices of the same order and α, β are scalars (real numbers), then

- (i) $\alpha(A + B) = \alpha A + \alpha B$
- (ii) $\alpha(\beta A) = (\alpha\beta)A$
- (iii) $(\alpha + \beta)A = \alpha A + \beta A$
- (iv) $1A = A$

9.4.3 Subtraction of Matrices

Subtraction of two matrices A and B make sense only if they are of the same order, and is given by

$A - B = A + (-B) = A + (-1)B$, i.e. $A - B$ means addition of two matrices A and $-B$. So, if $A = [a_{ij}]_{m \times n}$, $B = [b_{ij}]_{m \times n}$, then

$$A - B = [a_{ij} + (-1)b_{ij}]_{m \times n} = [a_{ij} - b_{ij}]_{m \times n}$$

For example,

(i) If $A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$, $B = \begin{bmatrix} 6 & 2 \\ 1 & 10 \end{bmatrix}$, then $A - B = \begin{bmatrix} 2-6 & 4-2 \\ 6-1 & 8-10 \end{bmatrix} = \begin{bmatrix} -4 & 2 \\ 5 & -2 \end{bmatrix}$.

- (ii) If $A = \begin{bmatrix} 2 & 9 & 3 \\ 8 & 4 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 8 & 7 \\ 6 & 5 \end{bmatrix}$, then $A - B$ does not make any sense
because A and B are of different orders.

9.4.4 Matrix Multiplication

Let $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{n \times p}$ be two matrices, then product of A and B is denoted by AB and is defined only if number of columns in A = number of rows in B and is given by

$$AB = C = [c_{ij}]_{m \times p}$$

where $c_{ij} = (i, j)^{\text{th}}$ element of C and is equal to $(i^{\text{th}}$ row of A) $(j^{\text{th}}$ column of B)

$$= [a_{i1} \quad a_{i2} \quad \dots \quad a_{in}] \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{bmatrix} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj}$$

$$= \sum_{k=1}^n a_{ik}b_{kj}, \text{ i.e. sum of product of first, second, third, ... elements of } i^{\text{th}} \text{ row of A with first, second, third, ... , elements of } j^{\text{th}} \text{ column of B respectively.}$$

You may notice that the number of rows in AB = number of rows in A, and number of columns in AB = number of columns in B.

Let us make the above concept more clear by taking the following matrices, in particular let

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \text{ and } B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}.$$

Here A is a matrix of order 3×2 and B be a matrix of order 2×2 . As number of columns of A = 2 = number of rows of B.

$\therefore AB$ is defined and is given by

$$AB = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \\ c_{31} & c_{32} \end{bmatrix}_{3 \times 2}$$

$$\begin{aligned} \text{where } c_{11} &= \text{Product of first row of A and first column of B} \\ &= \text{Sum of product of first, second elements of first row of A with first, second elements of first column of B respectively.} \\ &= a_{11}b_{11} + a_{12}b_{21}, \\ c_{12} &= \text{Product of first row of A and second column of B} \\ &= \text{Sum of product of first, second elements of first row of A with first, second elements of second column of B respectively.} \\ &= a_{11}b_{12} + a_{12}b_{22}, \\ c_{21} &= \dots, \text{ etc.} \end{aligned}$$

Properties of Matrix Multiplication

If A, B, C are three matrices such that corresponding multiplications hold then

- (1) $A(BC) = (AB)C$ (associative law)
- (2) (i) $A(B + C) = AB + AC$ (left distributive law)

$$(ii) (A + B)C = AC + BC \quad (\text{right distributive law})$$

(3) If A is a square matrix of order n, then

$$I_n A = A I_n = A, \quad \text{where } I_n \text{ is the identity matrix of order } n.$$

Remark 6: Commutative law does not hold, in general,

i.e. $AB \neq BA$, in general. But for some cases AB may be equal to BA . This has been explained below:

(i) AB may be defined but BA may not be defined and hence $AB \neq BA$ in this case.

For example, let A be a matrix of order 3×2 and B be a matrix of order 2×4 .

Here AB is defined and is of order 3×4 .

But BA is not defined ($\because$ number of columns of B $\neq$ number of rows of A).

(ii) AB and BA both may be defined but may not be of same order and hence $AB \neq BA$.

For example, let A be a matrix of order 3×2 and B be a matrix of order 2×3 .

Here as number of columns of A = number of rows of B.

$\therefore AB$ is defined and is of order 3×3 .

Also, number of columns of B = number of rows of A.

Hence BA is defined but of order 2×2 .

$\therefore AB \neq BA$.

(iii) AB and BA both may be defined and of same order but even then they may not be equal.

$$\text{Let } A = \begin{bmatrix} 5 & 4 \\ 2 & 7 \end{bmatrix}, B = \begin{bmatrix} 6 & 8 \\ 2 & 9 \end{bmatrix}$$

Here, AB and BA both are defined and are of same order.

$$\text{But } AB = \begin{bmatrix} 5 & 4 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} 6 & 8 \\ 2 & 9 \end{bmatrix} = \begin{bmatrix} 30+8 & 40+36 \\ 12+14 & 16+63 \end{bmatrix} = \begin{bmatrix} 38 & 76 \\ 26 & 79 \end{bmatrix} \text{ and}$$

$$BA = \begin{bmatrix} 6 & 8 \\ 2 & 9 \end{bmatrix} \begin{bmatrix} 5 & 4 \\ 2 & 7 \end{bmatrix} = \begin{bmatrix} 30+16 & 24+56 \\ 10+18 & 8+63 \end{bmatrix} = \begin{bmatrix} 46 & 80 \\ 28 & 71 \end{bmatrix}.$$

So, $AB \neq BA$.

However, sometimes, we may observe that $AB = BA$.

$$\text{For example, Let } A = \begin{bmatrix} 2 & -3 \\ 4 & 5 \end{bmatrix}, \text{ and } B = \begin{bmatrix} 5 & 3 \\ -4 & 2 \end{bmatrix}.$$

$$\text{Here } AB = \begin{bmatrix} 2 & -3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 5 & 3 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} 10+12 & 6-6 \\ 20-20 & 12+10 \end{bmatrix} = \begin{bmatrix} 22 & 0 \\ 0 & 22 \end{bmatrix} \text{ and}$$

$$BA = \begin{bmatrix} 5 & 3 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 10+12 & -15+15 \\ -8+8 & 12+10 \end{bmatrix} = \begin{bmatrix} 22 & 0 \\ 0 & 22 \end{bmatrix}.$$

Here, $AB = BA$.

Example 7: If $A = \begin{bmatrix} 2 & 4 & 5 \\ -3 & 6 & 7 \\ 1 & 8 & 9 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 6 & 2 \\ 1 & 4 & 5 \\ 8 & 7 & -1 \end{bmatrix}$, then evaluate the following

(i) $3A + 2B$ (ii) $2A - 3B$ (iii) AB (iv) BA

Solution:

$$\begin{aligned} \text{(i)} \quad 3A + 2B &= 3 \begin{bmatrix} 2 & 4 & 5 \\ -3 & 6 & 7 \\ 1 & 8 & 9 \end{bmatrix} + 2 \begin{bmatrix} 3 & 6 & 2 \\ 1 & 4 & 5 \\ 8 & 7 & -1 \end{bmatrix} = \begin{bmatrix} 6 & 12 & 15 \\ -9 & 18 & 21 \\ 3 & 24 & 27 \end{bmatrix} + \begin{bmatrix} 6 & 12 & 4 \\ 2 & 8 & 10 \\ 16 & 14 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 6+6 & 12+12 & 15+4 \\ -9+2 & 18+8 & 21+10 \\ 3+16 & 24+14 & 27-2 \end{bmatrix} = \begin{bmatrix} 12 & 24 & 19 \\ -7 & 26 & 31 \\ 19 & 38 & 25 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad 2A - 3B &= 2 \begin{bmatrix} 2 & 4 & 5 \\ -3 & 6 & 7 \\ 1 & 8 & 9 \end{bmatrix} - 3 \begin{bmatrix} 3 & 6 & 2 \\ 1 & 4 & 5 \\ 8 & 7 & -1 \end{bmatrix} = \begin{bmatrix} 4 & 8 & 10 \\ -6 & 12 & 14 \\ 2 & 16 & 18 \end{bmatrix} - \begin{bmatrix} 9 & 18 & 6 \\ 3 & 12 & 15 \\ 24 & 21 & -3 \end{bmatrix} \\ &= \begin{bmatrix} 4-9 & 8-18 & 10-6 \\ -6-3 & 12-12 & 14-15 \\ 2-24 & 16-21 & 18+3 \end{bmatrix} = \begin{bmatrix} -5 & -10 & 4 \\ -9 & 0 & -1 \\ -22 & -5 & 21 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad AB &= \begin{bmatrix} 2 & 4 & 5 \\ -3 & 6 & 7 \\ 1 & 8 & 9 \end{bmatrix} \begin{bmatrix} 3 & 6 & 2 \\ 1 & 4 & 5 \\ 8 & 7 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 6+4+40 & 12+16+35 & 4+20-5 \\ -9+6+56 & -18+24+49 & -6+30-7 \\ 3+8+72 & 6+32+63 & 2+40-9 \end{bmatrix} \\ &= \begin{bmatrix} 50 & 63 & 19 \\ 53 & 55 & 17 \\ 83 & 101 & 33 \end{bmatrix} \quad \dots (1) \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad BA &= \begin{bmatrix} 3 & 6 & 2 \\ 1 & 4 & 5 \\ 8 & 7 & -1 \end{bmatrix} \begin{bmatrix} 2 & 4 & 5 \\ -3 & 6 & 7 \\ 1 & 8 & 9 \end{bmatrix} \\ &= \begin{bmatrix} 6-18+2 & 12+36+16 & 15+42+18 \\ 2-12+5 & 4+24+40 & 5+28+45 \\ 16-21-1 & 32+42-8 & 40+49-9 \end{bmatrix} \\ &= \begin{bmatrix} -10 & 64 & 75 \\ -5 & 68 & 78 \\ -6 & 66 & 80 \end{bmatrix} \quad \dots (2) \end{aligned}$$

9.4.5 Integral Powers of a Square Matrix

Here, we will learn how higher powers of A are evaluated. We define

$$A^2 = A.A$$

$$A^3 = A^2A \quad \text{or} \quad A^3 = AA^2$$

$$A^4 = A^3A \text{ or } A^4 = AA^3 \text{ or } A^4 = A^2A^2$$

and so on

$$\text{in general } A^{p+q} = A^p A^q = A^q A^p.$$

Remark 7:

- (i) We define $A^0 = I$, where I is the identity matrix of the same order as A .
 (ii) $(A + B)^2 = A^2 + AB + BA + B^2$.
 (iii) $(A + B)^2 = A^2 + 2AB + B^2$ if and only if $AB = BA$.

Example 8: If $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$ then find A^4 .

Solution: $A^2 = AA = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 1+8 & 2+6 \\ 4+12 & 8+9 \end{bmatrix} = \begin{bmatrix} 9 & 8 \\ 16 & 17 \end{bmatrix}$

$$A^4 = A^2 A^2 = \begin{bmatrix} 9 & 8 \\ 16 & 17 \end{bmatrix} \begin{bmatrix} 9 & 8 \\ 16 & 17 \end{bmatrix} = \begin{bmatrix} 81+128 & 72+136 \\ 144+272 & 128+289 \end{bmatrix} = \begin{bmatrix} 209 & 208 \\ 416 & 417 \end{bmatrix}$$

Now, you can try the following exercises.

E 3) If $3X + 2Y = \begin{bmatrix} 4 & 13 \\ 18 & 13 \end{bmatrix}$ and $2X - 3Y = \begin{bmatrix} 7 & 0 \\ -1 & -13 \end{bmatrix}$, then find matrices X and Y .

E 4) Find AB , if defined, in each of the following cases:

(i) $A = \begin{bmatrix} 5 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ (ii) $A = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 6 \end{bmatrix}$

(iii) $A = \begin{bmatrix} 3 & 4 & 1 \\ 2 & 5 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$ (iv) $A = \begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 4 \\ 3 & 2 \end{bmatrix}$

E 5) Evaluate the product $\begin{bmatrix} 2 & 3 & 5 \end{bmatrix} \begin{bmatrix} 4 & 5 \\ 0 & 1 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 2 & 4 & 1 \\ 5 & 6 & 8 \end{bmatrix}$.

E 6) If $A = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$, then find A^8 .

9.5 TRANSPOSE OF A MATRIX

Transpose of a matrix A is denoted by A' or A^T and is obtained by interchanging rows and columns of A .

For example, if $A = \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 8 \end{bmatrix}$ then $A' = \begin{bmatrix} 2 & 5 \\ 3 & 6 \\ 4 & 8 \end{bmatrix}$.

Properties of Transpose

- (i) $(A')' = A$
 (ii) $(kA)' = kA'$, where k is a scalar
 (iii) $(A + B)' = A' + B'$
 (iv) $(A - B)' = A' - B'$
 (v) $(AB)' = B'A'$

Symmetric Matrix

A square matrix A is said to be symmetric matrix if $A' = A$.

For example, let $A = \begin{bmatrix} 2 & 5 & 6 \\ 5 & 4 & 3 \\ 6 & 3 & 8 \end{bmatrix}$, then $A' = \begin{bmatrix} 2 & 5 & 6 \\ 5 & 4 & 3 \\ 6 & 3 & 8 \end{bmatrix} = A$.

$\therefore A$ is symmetric.

Skew-Symmetric Matrix

A square matrix A is said to be skew-symmetric matrix if $A' = -A$.

For example, let $A = \begin{bmatrix} 0 & 5 & -3 \\ -5 & 0 & -2 \\ 3 & 2 & 0 \end{bmatrix}$ then

$$A' = \begin{bmatrix} 0 & -5 & 3 \\ 5 & 0 & 2 \\ -3 & -2 & 0 \end{bmatrix} = -\begin{bmatrix} 0 & 5 & -3 \\ -5 & 0 & -2 \\ 3 & 2 & 0 \end{bmatrix} = -A.$$

$\therefore A$ is skew-symmetric.

Remark 8: A square matrix $A = [a_{ij}]_{m \times n}$ will be symmetric if $a_{ij} = a_{ji}$, $\forall i, j$ and will be skew-symmetric if $a_{ij} = -a_{ji}$, $\forall i, j$ and hence for a skew-symmetric matrix

$$a_{ii} = -a_{ii} \Rightarrow 2a_{ii} = 0 \Rightarrow a_{ii} = 0$$

That is, all the diagonal elements of a skew-symmetric matrix are zero.

Example 9: If $A = \begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix}$ then show that

(i) $\frac{1}{2}(A + A')$ is symmetric, and (ii) $\frac{1}{2}(A - A')$ is skew-symmetric.

Solution:

$$\begin{aligned} \text{(i) Let } P &= \frac{1}{2}(A + A') = \frac{1}{2} \left(\begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix} + \begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix}' \right) \\ &= \frac{1}{2} \left(\begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix} + \begin{bmatrix} 3 & -2 \\ 5 & 4 \end{bmatrix} \right) = \frac{1}{2} \begin{bmatrix} 6 & 3 \\ 3 & 8 \end{bmatrix} = \begin{bmatrix} 3 & 3/2 \\ 3/2 & 4 \end{bmatrix} \quad \dots (1) \end{aligned}$$

$$P' = \begin{bmatrix} 3 & 3/2 \\ 3/2 & 4 \end{bmatrix}' = \begin{bmatrix} 3 & 3/2 \\ 3/2 & 4 \end{bmatrix} \quad \dots (2)$$

From (1) and (2)

$$P' = P \Rightarrow P \text{ is symmetric, i.e. } \frac{1}{2}(A + A') \text{ is symmetric.}$$

$$\begin{aligned} \text{(ii) Let } Q &= \frac{1}{2}(A - A') = \frac{1}{2} \left(\begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix} - \begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix}' \right) = \frac{1}{2} \left(\begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix} - \begin{bmatrix} 3 & -2 \\ 5 & 4 \end{bmatrix} \right) \\ &= \frac{1}{2} \begin{bmatrix} 3-3 & 5+2 \\ -2-5 & 4-4 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & 7 \\ -7 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 7/2 \\ -7/2 & 0 \end{bmatrix} \end{aligned}$$

$$Q' = \begin{bmatrix} 0 & 7/2 \\ -7/2 & 0 \end{bmatrix}' = \begin{bmatrix} 0 & -7/2 \\ 7/2 & 0 \end{bmatrix} = -\begin{bmatrix} 0 & 7/2 \\ -7/2 & 0 \end{bmatrix} = -Q$$

$\Rightarrow Q$ is skew symmetric, i.e. $\frac{1}{2}(A - A')$ is skew symmetric.

Remark 9:

$$A = \frac{1}{2}A + \frac{1}{2}A = \frac{1}{2}A + \frac{1}{2}A' + \frac{1}{2}A - \frac{1}{2}A' = \frac{1}{2}(A + A') + \frac{1}{2}(A - A') = P + Q$$

i.e. $A = P + Q$, where P is symmetric and Q is skew symmetric.

i.e. every square matrix can be expressed as a sum of a symmetric and a skew-symmetric matrix.

9.6 TRACE OF A MATRIX

In this section we will define trace of a matrix.

Trace of a square matrix $A = [a_{ij}]_{n \times n}$ is denoted by $\text{tr}(A)$ and is defined as $\text{tr}(A)$ = sum of diagonal elements of the matrix.

i.e. $\text{tr}(A) = a_{11} + a_{22} + a_{33} + \dots + a_{nn}$

For example, if $A = \begin{bmatrix} 2 & 3 & 5 \\ 6 & 8 & 4 \\ 9 & 1 & -3 \end{bmatrix}$ then $\text{tr}(A) = 2 + 8 + (-3) = 7$.

Properties of Trace of a Matrix

If $A = [a_{ij}]_{n \times n}$ and $B = [b_{ij}]_{n \times n}$ then

- (i) $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$
- (ii) $\text{tr}(kA) = k \text{tr}(A)$, where k is a scalar
- (iii) $\text{tr}(AB) = \text{tr}(BA)$

Remark 10: $\text{tr}(AB) \neq \text{tr}(A) \text{tr}(B)$

Here is an exercise for you.

E 7) (i) Find trace of the matrix A , where $A = \begin{bmatrix} 8 & 7 \\ 5 & 6 \end{bmatrix}$

(ii) Find trace of the matrices I_2, I_3, I_n .

9.7 DETERMINANT OF SQUARE MATRICES

Determinant is a number associated with each square matrix. In this section, we will deal with determinant of square matrices of order 1, 2, 3 and 4.

Determinants of square matrices of order greater than 4 can be evaluated in a similar fashion.

9.7.1 Determinant of a Square Matrix of Order 1

If $A = [a_{11}]$ be a square matrix of order 1 then determinant of A is given by

$$|A| = |a_{11}| = a_{11}.$$

For example,

(i) If $A = [5]$ then $|A| = |5| = 5$.

(ii) If $A = [-3]$ then $|A| = |-3| = -3$.

Remark 11:

(i) $|A|$ is read as determinant of A, do not read it modulus of A, i.e.

if $A = [-8]$ then $|A| = |-8| = -8$.

But in case of modulus $|-8| = -(-8) = 8$.

(ii) The context in which we are using $| \quad |$ will clear whether it represents modulus or determinant.

9.7.2 Determinant of a Square Matrix of Order 2×2

If $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ then $|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$

Let us take an example:

Example 10: Evaluate the following determinants:

(i) $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ (ii) $\begin{vmatrix} 3 & 5 \\ 8 & 9 \end{vmatrix}$ (iii) $\begin{vmatrix} x^2 & x^2 + 1 \\ x & x + 1 \end{vmatrix}$

Solution:

(i) $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

(ii) $\begin{vmatrix} 3 & 5 \\ 8 & 9 \end{vmatrix} = 27 - 40 = -13$

(iii) $\begin{vmatrix} x^2 & x^2 + 1 \\ x & x + 1 \end{vmatrix} = x^3 + x^2 - (x^3 + x) = x^2 - x$

Now, you can try the following exercise.

E 8) Find x in each of the following cases:

(i) $\begin{vmatrix} x & 7 \\ 9 & x + 2 \end{vmatrix} = 0$ (ii) $\begin{vmatrix} x & x^2 \\ 15 & 5 \end{vmatrix} = 0$

9.7.3 Determinant of a Square Matrix of Order 3×3

Before evaluating, the determinant of order 3×3 , let us define the minors and cofactors of a square matrix as follows:

Minors and Cofactors

Minor

If $A = [a_{ij}]_{n \times n}$ be a square matrix of order n then minor of $(i, j)^{\text{th}}$ element a_{ij} is denoted by M_{ij} and is defined as

M_{ij} = determinant of sub matrix of order $n - 1$ obtained after deleting i^{th} row and j^{th} column from A.

Example 11: Find the minor of each element of the following matrices:

$$(i) \begin{bmatrix} 2 & 5 \\ 4 & -7 \end{bmatrix} \quad (ii) \begin{bmatrix} -3 & 4 & -2 \\ 6 & 5 & 7 \\ -8 & 9 & 1 \end{bmatrix}$$

Solution:

$$(i) \text{ Let } A = \begin{bmatrix} 2 & 5 \\ 4 & -7 \end{bmatrix}$$

Let M_{ij} denotes the minor of $(i, j)^{\text{th}}$ element of the matrix A, $i, j = 1, 2$.

$$\therefore M_{11} = |-7| = -7 \quad \left[\begin{array}{l} \text{Determinant obtained after deleting first} \\ \text{row and first column of matrix A} = |-7| \end{array} \right]$$

$$\text{Similarly, } M_{12} = |4| = 4, \quad M_{21} = |5| = 5, \quad M_{22} = |2| = 2$$

$$(ii) \text{ Let } A = \begin{bmatrix} -3 & 4 & -2 \\ 6 & 5 & 7 \\ -8 & 9 & 1 \end{bmatrix}$$

Let M_{ij} denotes the minor of $(i, j)^{\text{th}}$ element of the matrix A, where $i, j = 1, 2, 3$.

$$\therefore M_{11} = \begin{vmatrix} 5 & 7 \\ 9 & 1 \end{vmatrix} = 5 - 63 = -58 \quad \left[\begin{array}{l} \text{After deleting the first row and} \\ \text{first column from A.} \end{array} \right]$$

$$M_{12} = \begin{vmatrix} 6 & 7 \\ -8 & 1 \end{vmatrix} = 6 + 56 = 62 \quad \left[\begin{array}{l} \text{After deleting the first row and} \\ \text{second column from A.} \end{array} \right]$$

Similarly,

$$M_{13} = \begin{vmatrix} 6 & 5 \\ -8 & 9 \end{vmatrix} = 54 + 40 = 94$$

$$M_{21} = \begin{vmatrix} 4 & -2 \\ 9 & 1 \end{vmatrix} = 4 + 18 = 22$$

$$M_{22} = \begin{vmatrix} -3 & -2 \\ -8 & 1 \end{vmatrix} = -3 - 16 = -19$$

$$M_{23} = \begin{vmatrix} -3 & 4 \\ -8 & 9 \end{vmatrix} = -27 + 32 = 5$$

$$M_{31} = \begin{vmatrix} 4 & -2 \\ 5 & 7 \end{vmatrix} = 28 + 10 = 38$$

$$M_{32} = \begin{vmatrix} -3 & -2 \\ 6 & 7 \end{vmatrix} = -21 + 12 = -9$$

$$M_{33} = \begin{vmatrix} -3 & 4 \\ 6 & 5 \end{vmatrix} = -15 - 24 = -39$$

Cofactor

If $A = [a_{ij}]_{n \times n}$ be a square matrix of order n then cofactor of $(i, j)^{\text{th}}$ element a_{ij} of matrix A is denoted by C_{ij} and is defined by

$C_{ij} = (-1)^{i+j} M_{ij}$, where M_{ij} denotes the minor of $(i, j)^{\text{th}}$ element of the matrix A .

Example 12: Find the cofactor of each element of the following matrices:

$$(i) \begin{bmatrix} 2 & 5 \\ 4 & -7 \end{bmatrix} \quad (ii) \begin{bmatrix} -3 & 4 & -2 \\ 6 & 5 & 7 \\ -8 & 9 & 1 \end{bmatrix}$$

Solution:

$$(i) \text{ Let } A = \begin{bmatrix} 2 & 5 \\ 4 & -7 \end{bmatrix}$$

Let C_{ij} denotes the cofactor of $(i, j)^{\text{th}}$ element of the matrix A , $i, j = 1, 2$.

$$\therefore C_{11} = (-1)^{1+1} M_{11} = (-1)^2 (-7) = -7 \quad [\text{Using Example 11 (i)}]$$

Similarly,

$$C_{12} = (-1)^{1+2} M_{12} = (-1)^3 (4) = -4$$

$$C_{21} = (-1)^{2+1} M_{21} = (-1)^3 (5) = -5$$

$$C_{22} = (-1)^{2+2} M_{22} = (-1)^4 (2) = 2$$

$$(ii) \text{ Let } A = \begin{bmatrix} -3 & 4 & -2 \\ 6 & 5 & 7 \\ -8 & 9 & 1 \end{bmatrix}$$

Let C_{ij} denotes the cofactor of $(i, j)^{\text{th}}$ element of the matrix A , then

$$C_{11} = (-1)^{1+1} M_{11} = (-1)^2 (-58) = -58 \quad [\text{Using Example 11 (ii)}]$$

Similarly,

$$C_{12} = (-1)^{1+2} M_{12} = (-1)^3 (62) = -62$$

$$C_{13} = (-1)^{1+3} M_{13} = (-1)^4 (94) = 94$$

$$C_{21} = (-1)^{2+1} M_{21} = (-1)^3 (22) = -22$$

$$C_{22} = (-1)^{2+2} M_{22} = (-1)^4 (-19) = -19$$

$$C_{23} = (-1)^{2+3} M_{23} = (-1)^5 (5) = -5$$

$$C_{31} = (-1)^{3+1} M_{31} = (-1)^4 (38) = 38$$

$$C_{32} = (-1)^{3+2} M_{32} = (-1)^5 (-9) = 9$$

$$C_{33} = (-1)^{3+3} M_{33} = (-1)^6 (-39) = -39$$

Here is an exercise for you.

E 9) Find minor and cofactor of the elements $a_{12}, a_{23}, a_{31}, a_{13}$ where

$$A = [a_{ij}]_{3 \times 3} = \begin{bmatrix} 5 & 6 & -2 \\ -3 & 8 & 9 \\ 7 & 10 & -4 \end{bmatrix}$$

Now, we discuss the determinant of a square matrix of order 3×3 .

If $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ then

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \text{Sum of products of the elements of any line (row or column) with their corresponding co-factors.}$$

Let us expand along first row (R_1), we have

$$|A| = a_{11} (\text{co-factor of } a_{11}) + a_{12} (\text{co-factor of } a_{12}) + a_{13} (\text{co-factor of } a_{13})$$

$$\begin{aligned} &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11}(a_{22}a_{33} - a_{32}a_{23}) - a_{12}(a_{21}a_{33} - a_{31}a_{23}) + a_{13}(a_{21}a_{32} - a_{31}a_{22}) \end{aligned}$$

Remark 12:

- (i) We can expand the determinant along any row or column, we will get the same value.
- (ii) When we expand a determinant along any row or column we attach + or – sign with each term containing the product of elements of a row (or column) and its corresponding minor. Pattern of +, – signs is shown as under.

$$\begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$$

We put + at (1, 1) position and then alternatively– and + are placed, provided either we can move along row or column (we cannot walk diagonally).

- (iii) There is no hard and fast rule, to choose a row or column to expand a determinant. But if we choose that row or column which contains maximum number of zero, it will reduce a lot of our calculation work.

Example 13: Evaluate the following determinants:

(i) $\begin{vmatrix} 3 & 2 & -1 \\ 5 & 4 & 6 \\ -3 & 1 & 7 \end{vmatrix}$ (ii) $\begin{vmatrix} 2 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 1 \end{vmatrix}$ (iii) $\begin{vmatrix} 3 & 1 & 2 \\ 0 & 9 & 6 \\ 0 & 5 & 4 \end{vmatrix}$

Solution:

(i) Let $\Delta = \begin{vmatrix} 3 & 2 & -1 \\ 5 & 4 & 6 \\ -3 & 1 & 7 \end{vmatrix}$

Expanding along R_1 (first row)

$$\Delta = 3(28 - 6) - 2(35 + 18) - 1(5 + 12) = 66 - 106 - 17 = -57$$

$$(ii) \text{ Let } \Delta = \begin{vmatrix} 2 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 1 \end{vmatrix}$$

Expanding along R_1 (first row)

$$\Delta = 2(2 - 4) - 1(1 - 4) + 2(2 - 4) = -4 + 3 - 4 = -5$$

$$(iii) \text{ Let } \Delta = \begin{vmatrix} 3 & 1 & 2 \\ 0 & 9 & 6 \\ 0 & 5 & 4 \end{vmatrix}$$

Expanding along C_1 (first column)

$\left[\because \text{it contains maximum} \right]$
 $\left[\text{number of zeros.} \right]$

$$\Delta = 3(36 - 30) - 0 + 0 = 18$$

Here is an exercise for you.

E 10) If $A = \begin{bmatrix} 1 & 2 & 4 \\ -3 & 5 & 1 \\ 2 & 4 & 8 \end{bmatrix}$ then show that $|A| = 0$.

9.7.4 Determinant of Square Matrices of Order 4×4 and of Higher Order

The procedure of expanding the determinant of order 4 or more is the same as we discussed in case of order 3×3 .

Example 14: Evaluate $\Delta = \begin{vmatrix} 1 & 2 & 3 & 4 \\ 2 & -1 & 3 & 5 \\ 6 & 4 & -2 & 7 \\ -4 & 8 & 9 & -5 \end{vmatrix}$

Solution: Expanding along R_1 , we get

$$\Delta = 1 \begin{vmatrix} -1 & 3 & 5 \\ 4 & -2 & 7 \\ 8 & 9 & -5 \end{vmatrix} - 2 \begin{vmatrix} 2 & 3 & 5 \\ 6 & -2 & 7 \\ -4 & 9 & -5 \end{vmatrix} + 3 \begin{vmatrix} 2 & -1 & 5 \\ 6 & 4 & 7 \\ -4 & 8 & -5 \end{vmatrix} - 4 \begin{vmatrix} 2 & -1 & 3 \\ 6 & 4 & -2 \\ -4 & 8 & 9 \end{vmatrix}$$

Expanding each determinant of order 3×3 along R_1 , we get

$$\begin{aligned} \Delta &= 1[-1(10 - 63) - 3(-20 - 56) + 5(36 + 16)] - 2[2(10 - 63) \\ &\quad - 3(-30 + 28) + 5(54 - 8)] + 3[2(-20 - 56) - (-1)(-30 + 28) + 5(48 + 16)] \\ &\quad - 4[2(36 + 16) - (-1)(54 - 8) + 3(48 + 16)] \\ &= (53 + 228 + 260) - 2(-106 + 6 + 230) + 3(-152 - 2 + 320) - 4(104 + 46 + 192) \\ &= 541 - 260 + 498 - 1368 \\ &= -589 \end{aligned}$$

Remark 13:

- (i) If A is square matrix then determinant of A is unique.
- (ii) If A is not a square matrix then determinant of A does not exist.

9.8 PROPERTIES OF DETERMINANTS

In Sec. 9.7 of this unit you have become familiar about how to expand the determinants of orders 1, 2, 3, or of higher order. But as you have seen that it requires lot of calculations and is a time consuming process. To avoid such calculations and to reduce the time of evaluation, we will use properties of determinants.

In this section, we will discuss some properties of the determinants. We shall give the proofs of these properties only for determinants of order 3×3 . But remember that these properties hold good for all orders of the determinants. Let us discuss these one by one. Our way to move further is that, first we list all the properties and then some examples will be solved to get the idea how these properties are used and useful.

P 1 $|A'| = |A|$, i.e. determinants of a matrix and its transpose are equal.

Proof: Let $A = \begin{bmatrix} a & b & c \\ x & y & z \\ l & m & n \end{bmatrix}$... (1)

$$|A| = \begin{vmatrix} a & b & c \\ x & y & z \\ l & m & n \end{vmatrix}$$

Expanding along R_1

$$|A| = a(ny - mz) - b(nx - lz) + c(mx - ly) \quad \dots (2)$$

From (1), we get

$$A' = \begin{bmatrix} a & x & l \\ b & y & m \\ c & z & n \end{bmatrix}$$

$$\therefore |A'| = \begin{vmatrix} a & x & l \\ b & y & m \\ c & z & n \end{vmatrix}$$

Expanding along R_1

$$\begin{aligned} |A'| &= a(ny - mz) - x(bn - cm) + l(bz - cy) \\ &= a(ny - mz) - bnx + cmx + lbz - cly \\ &= a(ny - mz) - b(nx - lz) + c(mx - ly) \quad \dots (3) \end{aligned}$$

From (2) and (3), we get

$$|A'| = |A|$$

P 2 If any two rows (or columns) of a determinant are interchanged, then sign of determinant is multiplied by (-1) .

Proof: Let $\Delta = \begin{vmatrix} a & b & c \\ x & y & z \\ l & m & n \end{vmatrix}$... (1)

Expanding along R_1

$$\Delta = a(ny - mz) - b(nx - lz) + c(mx - ly) \quad \dots (2)$$

Let us interchange the first and second rows of the given determinant we have a new determinant Δ_1 (say) as

$$\Delta_1 = \begin{vmatrix} x & y & z \\ a & b & c \\ l & m & n \end{vmatrix}$$

Expanding along R_1

$$\begin{aligned} \Delta_1 &= x(bn - cm) - y(an - cl) + z(am - bl) \\ &= bnx - cmx - any + cly + amz - blz \\ &= -a(ny - mz) + b(nx - lz) - c(mx - ly) \\ &= -[a(ny - mz) - b(nx - lz) + c(mx - ly)] \\ &= -\Delta \quad [\text{Using (2)}] \end{aligned}$$

Remark 14: Here we interchanged R_1 and R_2 . In fact we can interchange any two rows or any two columns, result remains the same in each case.

P 3 If any two rows or columns of a determinant are identical then value of the determinant vanishes.

Proof: Let $\Delta = \begin{vmatrix} a & b & c \\ a & b & c \\ x & y & z \end{vmatrix}$, where R_1 and R_2 are identical

Expanding along R_1 , we get

$$\Delta = a(bz - cy) - b(az - cx) + c(ay - bx) = abz - acy - abz + bcx + acy - bcx = 0$$

P 4 If each element of a row (or a column) of a determinant is multiplied by a scalar k (say), then value of the new determinant is k times the original given determinant.

Proof: Let $\Delta = \begin{vmatrix} a & b & c \\ x & y & z \\ l & m & n \end{vmatrix}$

Expanding along R_1

$$\Delta = a(ny - mz) - b(nx - lz) + c(mx - ly) \quad \dots (1)$$

Let $\Delta_1 = \begin{vmatrix} ka & b & c \\ kx & y & z \\ kl & m & n \end{vmatrix}$ [Here, the elements of first column of Δ have been multiplied with k .]

Expanding along R_1

$$\begin{aligned} \Delta_1 &= ka(ny - mz) - b(knx - klz) + c(kmx - kly) \\ &= k[a(ny - mz) - b(nx - lz) + c(mx - ly)] = k\Delta \quad \dots (2) \text{ [Using (1)]} \end{aligned}$$

From (1) and (2)

$$\Delta_1 = k\Delta$$

Hence proved

Remark 15: This property implies that if there is some factor common in all elements of any line then we can write it as the factor of the whole determinant.

For example, $\begin{vmatrix} 5a & b & c \\ 5x & y & z \\ 5l & m & n \end{vmatrix} = 5 \begin{vmatrix} a & b & c \\ x & y & z \\ l & m & n \end{vmatrix}$

P 5 If each element of a row (or column) of a determinant is expressed as a sum of two (or more) terms, then the determinant can be expressed as the sum of two (or more) determinants.

Proof: Let $\Delta = \begin{vmatrix} a+\lambda & b+\mu & c+v \\ x & y & z \\ l & m & n \end{vmatrix}$, then expanding along R_1 , we get

$$\begin{aligned} \Delta &= (a+\lambda) \begin{vmatrix} y & z \\ m & n \end{vmatrix} - (b+\mu) \begin{vmatrix} x & z \\ l & n \end{vmatrix} + (c+v) \begin{vmatrix} x & y \\ l & m \end{vmatrix} \\ &= \left[a \begin{vmatrix} y & z \\ m & n \end{vmatrix} - b \begin{vmatrix} x & z \\ l & n \end{vmatrix} + c \begin{vmatrix} x & y \\ l & m \end{vmatrix} \right] + \left[\lambda \begin{vmatrix} y & z \\ m & n \end{vmatrix} - \mu \begin{vmatrix} x & z \\ l & n \end{vmatrix} + v \begin{vmatrix} x & y \\ l & m \end{vmatrix} \right] \\ &= \begin{vmatrix} a & b & c \\ x & y & z \\ l & m & n \end{vmatrix} + \begin{vmatrix} \lambda & \mu & v \\ x & y & z \\ l & m & n \end{vmatrix} \end{aligned}$$

P 6 If to each element of any row (or column), we add some scalar multiple of another row (or column) and some other scalar multiple of some other row (or column), the value of determinant remains unaltered.

Proof: Let $\Delta = \begin{vmatrix} a & b & c \\ x & y & z \\ l & m & n \end{vmatrix} \quad \dots (1)$

$$\text{and } \Delta_1 = \begin{vmatrix} a & b & c \\ x & y & z \\ l+ka & m+kb & n+kc \end{vmatrix}$$

where Δ_1 is obtained from Δ by operating $R_3 \rightarrow R_3 + kR_1$

i.e. k times R_1 has been added to R_3 .

$$\Delta_1 = \begin{vmatrix} a & b & c \\ x & y & z \\ l & m & n \end{vmatrix} + \begin{vmatrix} a & b & c \\ x & y & z \\ ka & kb & kc \end{vmatrix} \quad [\text{Using property 5}]$$

$$= \Delta + k \begin{vmatrix} a & b & c \\ x & y & z \\ a & b & c \end{vmatrix} \quad \left[\begin{array}{l} \text{Using (1) and taking } k \\ \text{common from third} \\ \text{rows of second determinant} \end{array} \right]$$

$$= \Delta + k(0) = \Delta + 0 = \Delta \quad \left[\begin{array}{l} \because R_1 \text{ and } R_2 \text{ are identical and} \\ \text{so using property 3} \end{array} \right]$$

Hence proved

Remark 16: If operations of the type $R_i \rightarrow R_i + kR_j$ are used more than one in a single step then keep it always in mind that row which has been affected in one operation cannot be used in other operation.

For example,

- (i) $R_1 \rightarrow R_1 + 2R_3$, $R_2 \rightarrow R_2 + 5R_1$ is not allowed because R_1 has been affected by first operation, so it cannot be used in second operation in the same step.
- (ii) $R_1 \rightarrow R_1 + 3R_3$, $R_2 \rightarrow R_2 + 2R_3$, etc. are allowed.

P 7 If all the elements of any line (row or column) are zero then value of the determinant vanishes.

Proof: Let $\Delta = \begin{vmatrix} 0 & 0 & 0 \\ x & y & z \\ l & m & n \end{vmatrix}$, then evaluating along R_1 , we get

$$\Delta = (0)(ny - mz) - (0)(nx - lz) + (0)(mx - ly) = 0 - 0 + 0 = 0$$

Example 15: Evaluate the following determinants:

$$(i) \begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} \quad (ii) \begin{vmatrix} -3 & 5 & -2 \\ 8 & 9 & -17 \\ 3 & -6 & 3 \end{vmatrix} \quad (iii) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ b+c & c+a & a+b \end{vmatrix}$$

$$(iv) \begin{vmatrix} 3 & x & xyz \\ 3 & y & xyz \\ 3 & z & xyz \end{vmatrix} \quad (v) \begin{vmatrix} 2 & 3 & 30 \\ 5 & 4 & 54 \\ 6 & 1 & 42 \end{vmatrix} \quad (vi) \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

$$(vii) \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \quad (viii) \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^3 & y^3 & z^3 \end{vmatrix} \quad (ix) \begin{vmatrix} 2x+3 & x & x \\ x & 2x+3 & x \\ x & x & 2x+3 \end{vmatrix}$$

Solution:

$$(i) \text{ Let } \Delta = \begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix}$$

Operating $C_1 \rightarrow C_1 + C_2 + C_3$

$$\Delta = \begin{vmatrix} 0 & b-c & c-a \\ 0 & c-a & a-b \\ 0 & a-b & b-c \end{vmatrix} = 0 \quad [\because \text{all elements of } C_1 \text{ are zero, so using P7.}]$$

$$(ii) \text{ Let } \Delta = \begin{vmatrix} -3 & 5 & -2 \\ 8 & 9 & -17 \\ 3 & -6 & 3 \end{vmatrix}$$

Operating $C_1 \rightarrow C_1 + C_2 + C_3$

$$\begin{aligned} \Delta &= \begin{vmatrix} -3+5-2 & 5 & -2 \\ 8+9-17 & 9 & -17 \\ 3-6+3 & -6 & 3 \end{vmatrix} \\ &= \begin{vmatrix} 0 & 5 & -2 \\ 0 & 9 & -17 \\ 0 & -6 & 3 \end{vmatrix} = 0 \quad [\because \text{all the element of } C_1 \text{ are zero, so using P7.}] \end{aligned}$$

$$(iii) \text{ Let } \Delta = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ b+c & c+a & a+b \end{vmatrix}$$

Operating $R_3 \rightarrow R_3 + R_2$

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a+b+c & a+b+c & a+b+c \end{vmatrix}$$

Taking $(a + b + c)$ common from R_3

$$\Delta = (a + b + c) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix} = (a + b + c)(0) = 0 [\because R_1 \text{ and } R_3 \text{ are identical.}]$$

$$(iv) \text{ Let } \Delta = \begin{vmatrix} 3 & x & xyz \\ 3 & y & xyz \\ 3 & z & xyz \end{vmatrix}$$

Taking 3, xyz common from C_1 and C_3 respectively

$$\Delta = 3xyz \begin{vmatrix} 1 & x & 1 \\ 1 & y & 1 \\ 1 & z & 1 \end{vmatrix} = 3xyz (0) = 0 \quad [\because C_1 \text{ and } C_3 \text{ are identical.}]$$

$$(v) \text{ Let } \Delta = \begin{vmatrix} 2 & 3 & 30 \\ 5 & 4 & 54 \\ 6 & 1 & 42 \end{vmatrix}$$

Taking 6 common from C_3

$$\Delta = 6 \begin{vmatrix} 2 & 3 & 5 \\ 5 & 4 & 9 \\ 6 & 1 & 7 \end{vmatrix}$$

Operating $C_3 \rightarrow C_3 - C_1 - C_2$

$$\Delta = 6 \begin{vmatrix} 2 & 3 & 0 \\ 5 & 4 & 0 \\ 6 & 1 & 0 \end{vmatrix} = 6 (0) = 0 [\because \text{all the elements of } C_3 \text{ are zero, so using P7.}]$$

$$(vi) \text{ Let } \Delta = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

Operating $R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 - R_1$

$$\Delta = \begin{vmatrix} 1 & x & x^2 \\ 0 & y-x & y^2-x^2 \\ 0 & z-x & z^2-x^2 \end{vmatrix} = \begin{vmatrix} 1 & x & x^2 \\ 0.(y-x) & (y-x).1 & (y-x)(y+x) \\ 0.(z-x) & (z-x).1 & (z-x)(z+x) \end{vmatrix}$$

Taking $y - x$, $z - x$ common from R_2, R_3 respectively

$$\Delta = (y-x)(z-x) \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y+x \\ 0 & 1 & z+x \end{vmatrix}$$

Operating $R_3 \rightarrow R_3 - R_2$

$$\Delta = (y-x)(z-x) \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y+x \\ 0 & 0 & z-y \end{vmatrix}$$

$$= (y-x)(z-x) \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y+x \\ 0.(z-y) & 0.(z-y) & 1.(z-y) \end{vmatrix}$$

Taking $(z-y)$ common from R_3

$$\Delta = (y-x)(z-x)(z-y) \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y+x \\ 0 & 0 & 1 \end{vmatrix}$$

Expanding along C_1 , we get

$$\Delta = (y-x)(z-x)(z-y)[1(1-0)-0+0]$$

$$= (x-y)(y-z)(z-x)$$

(vii) Let $\Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$

Operating $C_1 \rightarrow C_1 + C_2 + C_3$

$$\Delta = \begin{vmatrix} a+b+c & b & c \\ a+b+c & c & a \\ a+b+c & a & b \end{vmatrix}$$

Taking $(a+b+c)$ common from C_1

$$\Delta = (a+b+c) \begin{vmatrix} 1 & b & c \\ 1 & c & a \\ 1 & a & b \end{vmatrix}$$

Operating $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$

$$\Delta = (a+b+c) \begin{vmatrix} 1 & b & c \\ 0 & c-b & a-c \\ 0 & a-b & b-c \end{vmatrix}$$

Expanding along C_1

$$\Delta = (a+b+c)[1\{(c-b)(b-c)-(a-b)(a-c)\}-0+0]$$

$$= (a+b+c)[bc-c^2-b^2+bc-(a^2-ac-ab+bc)]$$

$$= (a+b+c)(ab+bc+ca-a^2-b^2-c^2)$$

(viii) Let $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^3 & y^3 & z^3 \end{vmatrix}$

Operating $C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1$

$$\Delta = \begin{vmatrix} 1 & 0 & 0 \\ x & y-x & z-x \\ x^3 & y^3-x^3 & z^3-x^3 \end{vmatrix}$$

Taking $y - x$, $z - x$ common from C_2 , C_3 respectively

$$\Delta = (y-x)(z-x) \begin{vmatrix} 1 & 0 & 0 \\ x & 1 & 1 \\ x^3 & y^2 + x^2 + xy & z^2 + x^2 + zx \end{vmatrix}$$

Operating $C_3 \rightarrow C_3 - C_2$

$$\begin{aligned} \Delta &= (y-x)(z-x) \begin{vmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ x^3 & x^2 + y^2 + xy & z^2 - y^2 + x(z-y) \end{vmatrix} \\ &= (y-x)(z-x) \begin{vmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ x^3 & x^2 + y^2 + xy & (z-y)(z+y+x) \end{vmatrix} \end{aligned}$$

Taking $(z-y)(x+y+z)$ common from C_3

$$\Delta = (y-x)(z-x)(z-y)(x+y+z) \begin{vmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ x^3 & x^2 + y^2 + xy & 1 \end{vmatrix}$$

Expanding along R_1

$$\begin{aligned} \Delta &= (y-x)(z-x)(z-y)(x+y+z)[1(1-0)-0+0] \\ &= (x-y)(y-z)(z-x)(x+y+z) \end{aligned}$$

$$(ix) \text{ Let } \Delta = \begin{vmatrix} 2x+3 & x & x \\ x & 2x+3 & x \\ x & x & 2x+3 \end{vmatrix}$$

Operating $C_1 \rightarrow C_1 + C_2 + C_3$

$$\Delta = \begin{vmatrix} 4x+3 & x & x \\ 4x+3 & 2x+3 & x \\ 4x+3 & x & 2x+3 \end{vmatrix}$$

Taking $4x+3$ common from C_1

$$\Delta = (4x+3) \begin{vmatrix} 1 & x & x \\ 1 & 2x+3 & x \\ 1 & x & 2x+3 \end{vmatrix}$$

Operating $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$

$$\Delta = (4x+3) \begin{vmatrix} 1 & x & x \\ 0 & x+3 & 0 \\ 0 & 0 & x+3 \end{vmatrix}$$

Expanding along C_1

$$\Delta = (4x+3)[1\{(x+3)^2 - 0\} - 0 + 0] = (4x+3)(x+3)^2$$

Now, you can try the following exercise.

E 11) Prove the following

$$(i) \begin{vmatrix} ab & 1 & c(a+b) \\ bc & 1 & a(b+c) \\ ca & 1 & b(c+a) \end{vmatrix} = 0 \quad [\text{Without expanding}]$$

$$(ii) \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ 1+x^3 & 1+y^3 & 1+z^3 \end{vmatrix} = (x-y)(y-z)(z-x)(1+xyz)$$

[Using properties]

$$(iii) \begin{vmatrix} a+b & c & c \\ a & b+c & a \\ b & b & c+a \end{vmatrix} = 4abc \quad [\text{Using properties}]$$

9.9 SUMMARY

In this unit we have covered following topics:

- 1) Definition with examples of a matrix.
- 2) Types of matrices with examples.
- 3) Operations on matrices.
- 4) Integral powers of a square matrix.
- 5) Trace of a matrix.
- 6) Determinant and its properties.

9.10 SOLUTIONS/ANSWERS

$$\mathbf{E\ 1)} \quad A = [a_{ij}]_{3 \times 2} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}, \text{ where } a_{ij} = |i - j|$$

$$a_{11} = |1-1| = |0| = 0, \quad a_{12} = |1-2| = |-1| = -(-1) = 1, \quad a_{21} = |2-1| = |1| = 1,$$

$$a_{22} = |2-2| = |0| = 0, \quad a_{31} = |3-1| = |2| = 2, \quad a_{32} = |3-2| = |1| = 1$$

$$\therefore A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 2 & 1 \end{bmatrix}$$

E 2) We know that two matrices A and B are equal if

- (a) their orders are same, and
- (b) the corresponding elements of A and B are equal.

$\therefore$ on comparing corresponding elements of two matrices, we have

$$3x - 2y = -1 \quad \dots (1)$$

$$z + w = 7 \quad \dots (2)$$

$$3z - w = 5 \quad \dots (3)$$

$$x + y = 3 \quad \dots (4)$$

Equation (1) + 2 × equation (4) gives

$$\begin{array}{r} 3x - 2y = -1 \\ 2x + 2y = 6 \\ \hline 5x = 5 \end{array}$$

$$\Rightarrow x = 1$$

Putting $x = 1$ in (4), we get

$$1 + y = 3 \Rightarrow y = 2$$

(2) + (3) gives.

$$4z = 12 \Rightarrow z = 3$$

Putting $z = 3$ in (2), we get

$$3 + w = 7 \Rightarrow w = 4$$

$$\therefore x = 1, y = 2, z = 3, w = 4.$$

$$\text{E 3) } 3X + 2Y = \begin{bmatrix} 4 & 13 \\ 18 & 13 \end{bmatrix} \quad \dots (1)$$

$$2X - 3Y = \begin{bmatrix} 7 & 0 \\ -1 & -13 \end{bmatrix} \quad \dots (2)$$

Equation (1) × 3 + 2 × equation (2) gives

$$9X + 6Y + 4X - 6Y = 3 \begin{bmatrix} 4 & 13 \\ 18 & 13 \end{bmatrix} + 2 \begin{bmatrix} 7 & 0 \\ -1 & -13 \end{bmatrix}$$

$$13X = \begin{bmatrix} 12 & 39 \\ 54 & 39 \end{bmatrix} + \begin{bmatrix} 14 & 0 \\ -2 & -26 \end{bmatrix} = \begin{bmatrix} 26 & 39 \\ 52 & 13 \end{bmatrix}$$

$$\Rightarrow X = \frac{1}{13} \begin{bmatrix} 26 & 39 \\ 52 & 13 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \quad [\text{By scalar multiplication property}]$$

Putting this value of X in (1), we get

$$3 \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} + 2Y = \begin{bmatrix} 4 & 13 \\ 18 & 13 \end{bmatrix}$$

$$\begin{aligned} \Rightarrow 2Y &= \begin{bmatrix} 4 & 13 \\ 18 & 13 \end{bmatrix} - 3 \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 13 \\ 18 & 13 \end{bmatrix} + \begin{bmatrix} -6 & -9 \\ -12 & -3 \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ 6 & 10 \end{bmatrix} \end{aligned}$$

$$\Rightarrow Y = \frac{1}{2} \begin{bmatrix} -2 & 4 \\ 6 & 10 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ 3 & 5 \end{bmatrix} \quad [\text{By scalar multiplication property}]$$

$$\therefore X = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \text{ and } Y = \begin{bmatrix} -1 & 2 \\ 3 & 5 \end{bmatrix}.$$

E 4) (i) Order of A is 1×2 and order of B is 3×1 .

$\therefore$ number of columns in A $\neq$ number of rows in B.

$\Rightarrow AB$ is not defined.

(ii) Number of columns in A = number of rows in B = 1.

$\Rightarrow AB$ is defined and is given by

$$AB = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \end{bmatrix} = \begin{bmatrix} 3 \times 5 & 3 \times 6 \\ 4 \times 5 & 4 \times 6 \end{bmatrix} = \begin{bmatrix} 15 & 18 \\ 20 & 24 \end{bmatrix}$$

(iii) AB is defined and is given by

$$AB = \begin{bmatrix} 3 & 4 & 1 \\ 2 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 3+12+5 & 6+16+6 \\ 2+15+30 & 4+20+36 \end{bmatrix} = \begin{bmatrix} 20 & 28 \\ 47 & 60 \end{bmatrix}$$

(iv) AB is defined and is given by

$$AB = \begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 5 & 4 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 10+9 & 8+6 \\ 5+0 & 4+0 \end{bmatrix} = \begin{bmatrix} 19 & 14 \\ 5 & 4 \end{bmatrix}$$

$$\begin{aligned} \text{E 5) } [2 \ 3 \ 5] \begin{bmatrix} 4 & 5 \\ 0 & 1 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 2 & 4 & 1 \\ 5 & 6 & 8 \end{bmatrix} &= [8+0+10 \quad 10+3+30] \begin{bmatrix} 2 & 4 & 1 \\ 5 & 6 & 8 \end{bmatrix} \\ &= [18 \quad 43] \begin{bmatrix} 2 & 4 & 1 \\ 5 & 6 & 8 \end{bmatrix} \\ &= [36+215 \quad 72+258 \quad 18+344] \\ &= [251 \quad 330 \quad 362] \end{aligned}$$

$$\begin{aligned} \text{E 6) } A^2 = AA &= \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1+0 & 0+0 \\ 2+6 & 0+9 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 8 & 9 \end{bmatrix} \\ A^4 = A^2 A^2 &= \begin{bmatrix} 1 & 0 \\ 8 & 9 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 8 & 9 \end{bmatrix} = \begin{bmatrix} 1+0 & 0+0 \\ 8+72 & 0+81 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 80 & 81 \end{bmatrix} \\ A^8 = A^4 A^4 &= \begin{bmatrix} 1 & 0 \\ 80 & 81 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 80 & 81 \end{bmatrix} \\ &= \begin{bmatrix} 1+0 & 0+0 \\ 80+6480 & 0+6561 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 6560 & 6561 \end{bmatrix} \end{aligned}$$

E 7) (i) $\text{tr}(A) = \text{sum of diagonal elements} = 8 + 6 = 14$

(ii) We know that in an identity matrix, all the diagonal elements are unity.

$\therefore \text{tr}(I_2) = 1 + 1 = 2$ [$\because I_2$ is identity matrix of order 2×2].

Similarly, $\text{tr}(I_3) = 1 + 1 + 1 = 3$, $\text{tr}(I_n) = \underbrace{1+1+1+\dots+1}_{n \text{ times}} = n$.

$$\begin{aligned} \text{E 8) (i) } \begin{vmatrix} x & 7 \\ 9 & x+2 \end{vmatrix} &= 0 \Rightarrow x(x+2) - 63 = 0 \\ &\Rightarrow x^2 + 2x - 63 = 0 \Rightarrow x^2 + 9x - 7x - 63 = 0 \\ &\Rightarrow x(x+9) - 7(x+9) = 0 \\ &\Rightarrow (x+9)(x-7) = 0 \Rightarrow x = 7, -9 \end{aligned}$$

$$\begin{aligned} \text{(ii) } \begin{vmatrix} x & x^2 \\ 15 & 5 \end{vmatrix} &= 0 \Rightarrow 5x - 15x^2 = 0 \\ &\Rightarrow 15x^2 - 5x = 0 \Rightarrow 5x(3x-1) = 0 \Rightarrow x = 0, 1/3 \end{aligned}$$

E 9) Let M_{ij} and C_{ij} denote the minor and cofactor of $(i, j)^{\text{th}}$ element in the matrix A respectively then

$$M_{12} = \begin{vmatrix} -3 & 9 \\ 7 & -4 \end{vmatrix} = 12 - 63 = -51, \quad M_{23} = \begin{vmatrix} 5 & 6 \\ 7 & 10 \end{vmatrix} = 50 - 42 = 8$$

$$M_{31} = \begin{vmatrix} 6 & -2 \\ 8 & 9 \end{vmatrix} = 54 - (-16) = 54 + 16 = 70$$

$$M_{13} = \begin{vmatrix} -3 & 8 \\ 7 & 10 \end{vmatrix} = -30 - 56 = -86$$

$$C_{12} = (-1)^{1+2} M_{12} = (-1)^3 (-51) = 51, \quad C_{23} = (-1)^{2+3} M_{23} = (-1)^5 (8) = -8$$

$$C_{31} = (-1)^{3+1} M_{31} = (-1)^4 (70) = 70$$

$$C_{13} = (-1)^{1+3} M_{13} = (-1)^4 (-86) = -86$$

$$\text{E 10) } |A| = \begin{vmatrix} 1 & 2 & 4 \\ -3 & 5 & 1 \\ 2 & 4 & 8 \end{vmatrix}$$

Expanding along R_1

$$|A| = 1(40 - 4) - 2(-24 - 2) + 4(-12 - 10) = 36 + 52 - 88 = 0$$

$$\text{E 11) (i) L.H.S.} = \begin{vmatrix} ab & 1 & c(a+b) \\ bc & 1 & a(b+c) \\ ca & 1 & b(c+a) \end{vmatrix} = \begin{vmatrix} ab & 1 & ac+bc \\ bc & 1 & ab+ac \\ ca & 1 & bc+ab \end{vmatrix}$$

Operating $C_3 \rightarrow C_3 + C_1$

$$\text{L.H.S.} = \begin{vmatrix} ab & 1 & ab+bc+ca \\ bc & 1 & ab+bc+ca \\ ca & 1 & ab+bc+ca \end{vmatrix}$$

Taking $ab + bc + ca$ common from C_3

$$\text{L.H.S.} = (ab + bc + ca) \begin{vmatrix} ab & 1 & 1 \\ bc & 1 & 1 \\ ca & 1 & 1 \end{vmatrix}$$

$$= (ab + bc + ca)(0) = 0 = \text{R.H.S.} \quad [\because C_2 \text{ and } C_3 \text{ are identical.}]$$

$$\text{(ii) Let } \Delta = \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ 1+x^3 & 1+y^3 & 1+z^3 \end{vmatrix}$$

$$= \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ 1 & 1 & 1 \end{vmatrix} + \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \end{vmatrix} \quad [\text{Using property 5}]$$

Taking x, y, z common from C_1, C_2, C_3 of the second determinant respectively.

$$\Delta = \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ 1 & 1 & 1 \end{vmatrix} + xyz \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

Operating $R_1 \leftrightarrow R_3$ on first determinant

$$\Delta = (-1) \begin{vmatrix} 1 & 1 & 1 \\ x^2 & y^2 & z^2 \\ x & y & z \end{vmatrix} + xyz \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

Operating $R_2 \leftrightarrow R_3$ on first determinant

$$\Delta = (-1)(-1) \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} + xyz \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

$$= 1 \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} + xyz \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

Taking determinant common from both terms

$$\Delta = (1 + xyz) \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

Operating $C_2 \rightarrow C_2 - C_1$; $C_3 \rightarrow C_3 - C_1$

$$\Delta = (1 + xyz) \begin{vmatrix} 1 & 0 & 0 \\ x & y - x & z - x \\ x^2 & y^2 - x^2 & z^2 - x^2 \end{vmatrix}$$

Taking $y - x$, $z - x$ common from C_2 , C_3 respectively

$$\Delta = (1 + xyz)(y - x)(z - x) \begin{vmatrix} 1 & 0 & 0 \\ x & 1 & 1 \\ x^2 & x + y & z + x \end{vmatrix}$$

Operating $C_3 \rightarrow C_3 - C_2$

$$\Delta = (1 + xyz)(y - x)(z - x) \begin{vmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ x^2 & x + y & z - y \end{vmatrix}$$

Expanding along R_1

$$\Delta = (1 + xyz)(y - x)(z - x)[1\{(z - y) - 0\} - 0 + 0]$$

$$= (1 + xyz)(y - x)(z - x)(z - y) = (x - y)(y - z)(z - x)(1 + xyz) = \text{R.H.S.}$$

$$(iii) \text{ L.H.S} = \begin{vmatrix} a + b & c & c \\ a & b + c & a \\ b & b & c + a \end{vmatrix}$$

Operating $R_1 \rightarrow R_1 - R_2 - R_3$

$$\text{L.H.S.} = \begin{vmatrix} 0 & -2b & -2a \\ a & b + c & a \\ b & b & c + a \end{vmatrix}$$

Operating $C_2 \rightarrow C_2 - C_1$, $C_3 \rightarrow C_3 - C_1$

$$\text{L.H.S.} = \begin{vmatrix} 0 & -2b & -2a \\ a & b + c - a & 0 \\ b & 0 & c + a - b \end{vmatrix}$$

Expanding along R_1

$$\text{L.H.S.} = 0 - (-2b)[a(c + a - b) - 0] + (-2a)[0 - b(b + c - a)]$$

$$= 2ab(c + a - b) + 2ab(b + c - a) = 2ab(c + a - b + b + c - a)$$

$$= 2ab(2c) = 4abc = \text{R.H.S.}$$

UNIT 10 APPLICATIONS OF MATRICES AND DETERMINANTS

Structure

- 10.1 Introduction
 - Objectives
- 10.2 Adjoint of a Matrix
- 10.3 Inverse of a Matrix
- 10.4 Application of Matrices
- 10.5 Application of determinants
- 10.6 Summary
- 10.7 Solutions/Answers

10.1 INTRODUCTION

In the previous unit, we have discussed matrices, types of matrices and determinants of square matrices of orders 1, 2, 3 and higher orders. We have also discussed minors, cofactors of square matrices and properties of determinants.

In this unit, we will learn some applications of matrices and determinants such as solutions of simultaneous linear equations by using matrix method and Cramer's rule.

Objectives

After completing unit, you should be able to:

- find the adjoint of a square matrix;
- find the inverse of a square matrix;
- solve the simultaneous linear equations with the help of matrix methods; and
- solve the simultaneous linear equations with the help of Cramer's rule.

10.2 ADJOINT OF A MATRIX

In next section, i.e. Sec.10.3, we will discuss inverse of a square matrix. But in order to define inverse of a square matrix we use the concept of adjoint of the square matrix, so in this section we are going to discuss adjoint of the square matrix.

Adjoint of a Matrix: Let $A = [a_{ij}]_{n \times n}$ be a square matrix of order $n \times n$, then adjoint of A is denoted by $\text{adj}A$ and is defined as

$$\text{adj}A = \begin{bmatrix} A_{11} & A_{12} & A_{13} & \dots & A_{1n} \\ A_{21} & A_{22} & A_{23} & \dots & A_{2n} \\ A_{31} & A_{32} & A_{33} & \dots & A_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & A_{n3} & \dots & A_{nn} \end{bmatrix},$$

where A_{ij} denotes the cofactor of $(i, j)^{\text{th}}$ element of the matrix A .

It may be verified that $A(\text{adj}A) = (\text{adj}A)A = |A|I$,
where I is an identity matrix of order n .

Example 1: Find the $\text{adj}A$, where $A = \begin{bmatrix} 1 & 2 & 3 \\ -4 & 5 & 0 \\ -1 & -2 & 1 \end{bmatrix}$.

Also verify that $A(\text{adj}A) = (\text{adj}A)A = |A|I$.

Solution: $A = \begin{bmatrix} 1 & 2 & 3 \\ -4 & 5 & 0 \\ -1 & -2 & 1 \end{bmatrix}$

Let A_{ij} denotes the cofactor of $(i, j)^{\text{th}}$ element of the matrix A .

$$\therefore A_{11} = (-1)^{1+1} \begin{vmatrix} 5 & 0 \\ -2 & 1 \end{vmatrix} = (-1)^2 (5 - 0) = 5$$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} -4 & 0 \\ -1 & 1 \end{vmatrix} = (-1)^3 (-4 - 0) = 4$$

$$A_{13} = (-1)^{1+3} \begin{vmatrix} -4 & 5 \\ -1 & -2 \end{vmatrix} = (-1)^4 (8 + 5) = 13$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} 2 & 3 \\ -2 & 1 \end{vmatrix} = (-1)^3 (2 + 6) = -8$$

$$A_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix} = (-1)^4 (1 + 3) = 4$$

$$A_{23} = (-1)^{2+3} \begin{vmatrix} 1 & 2 \\ -1 & -2 \end{vmatrix} = (-1)^5 (-2 + 2) = 0$$

$$A_{31} = (-1)^{3+1} \begin{vmatrix} 2 & 3 \\ 5 & 0 \end{vmatrix} = (-1)^4 (0 - 15) = -15$$

$$A_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 3 \\ -4 & 0 \end{vmatrix} = (-1)^5 (0 + 12) = -12$$

$$A_{33} = (-1)^{3+3} \begin{vmatrix} 1 & 2 \\ -4 & 5 \end{vmatrix} = (-1)^6 (5 + 8) = 13$$

$$\therefore \text{adj}A = \begin{bmatrix} 5 & 4 & 13 \\ -8 & 4 & 0 \\ -15 & -12 & 13 \end{bmatrix} = \begin{bmatrix} 5 & -8 & -15 \\ 4 & 4 & -12 \\ 13 & 0 & 13 \end{bmatrix}$$

To verify $A(\text{adj}A) = (\text{adj}A)A = |A|I$, we proceed as follows

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ -4 & 5 & 0 \\ -1 & -2 & 1 \end{vmatrix}$$

Expanding along C_3

$$|A| = 3(8 + 5) - 0 + 1(5 + 8) = 39 + 13 = 52 \quad \dots (1)$$

$$\begin{aligned}
 A(\text{adj}A) &= \begin{bmatrix} 1 & 2 & 3 \\ -4 & 5 & 0 \\ -1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 5 & -8 & -15 \\ 4 & 4 & -12 \\ 13 & 0 & 13 \end{bmatrix} \\
 &= \begin{bmatrix} 5+8+39 & -8+8+0 & -15-24+39 \\ -20+20+0 & 32+20+0 & 60-60+0 \\ -5-8+13 & 8-8+0 & 15+24+13 \end{bmatrix} = \begin{bmatrix} 52 & 0 & 0 \\ 0 & 52 & 0 \\ 0 & 0 & 52 \end{bmatrix} \\
 &= 52 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 52I = |A|I \quad \dots (2) \quad [\text{Using (1)}] \\
 (\text{adj}A)A &= \begin{bmatrix} 5 & -8 & -15 \\ 4 & 4 & -12 \\ 13 & 0 & 13 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ -4 & 5 & 0 \\ -1 & -2 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 5+32+15 & 10-40+30 & 15-0-15 \\ 4-16+12 & 8+20+24 & 12+0-12 \\ 13+0-13 & 26+0-26 & 39+0+13 \end{bmatrix} = \begin{bmatrix} 52 & 0 & 0 \\ 0 & 52 & 0 \\ 0 & 0 & 52 \end{bmatrix} \\
 &= 52 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 52I = |A|I \quad \dots (3)
 \end{aligned}$$

From (2) and (3), we have $A(\text{adj}A) = (\text{adj}A)A = |A|I$

Now, you can try the following exercise.

E 1) If $A = \begin{bmatrix} 2 & -3 \\ 4 & 5 \end{bmatrix}$ then verify that $A(\text{adj}A) = (\text{adj}A)A = |A|I_2$.

10.3 INVERSE OF A MATRIX

You are familiar with the concept of the multiplicative inverse of a real number.

e.g. you know the multiplicative inverse of 5, $\frac{2}{3}$, etc.

Multiplicative inverse of 5 is $\frac{1}{5}$ and of $\frac{2}{3}$ is $\frac{3}{2}$, etc. $\left[\because 5 \times \frac{1}{5} = 1, \frac{2}{3} \times \frac{3}{2} = 1 \right]$

i.e. when a number is multiplied with its multiplicative inverse, it gives 1. Similarly, in case of matrices, when a square matrix is multiplied with its multiplicative inverse, it gives identity matrix (I). That is, in case of matrices, identity matrix plays the role as 1 plays in case of real numbers.

Inverse of a Matrix

Let A be a square matrix of order n. If there exists a square matrix B of order n such that

$AB = BA = I_n$, then B is known as inverse of A and is denoted by A^{-1} .

As verified above that $A(\text{adj}A) = (\text{adj}A)A = |A|I$,

$\therefore A(\text{adj}A) = |A|I$

$$\Rightarrow A \left[\frac{1}{|A|} (\text{adj}A) \right] = I, \text{ provided } |A| \neq 0$$

Now, by definition of inverse of a matrix, $\frac{1}{|A|} (\text{adj}A)$ is inverse of A.

$$\text{i.e. } A^{-1} = \frac{1}{|A|} (\text{adj}A)$$

Properties of Inverse of a Matrix

- (i) The inverse of a square matrix, if exists is unique.
- (ii) A^{-1} exists if and only if $|A| \neq 0$.
- (iii) $(A^{-1})' = (A')^{-1}$
- (iv) $(AB)^{-1} = B^{-1}A^{-1}$

To find inverse of a square matrix use following steps

I Find $|A|$

II If $|A| = 0$ then A^{-1} does not exists.

III If $|A| \neq 0$ then find $\text{adj}A$ and $A^{-1} = \frac{1}{|A|} (\text{adj}A)$.

Example 2: Find A^{-1} , where $A = \begin{bmatrix} 4 & 2 & 3 \\ 0 & -1 & -2 \\ 5 & -3 & 6 \end{bmatrix}$

Solution: $A = \begin{bmatrix} 4 & 2 & 3 \\ 0 & -1 & -2 \\ 5 & -3 & 6 \end{bmatrix}$

$$|A| = 4(-6-6) - 2(0+10) + 3(0+5) = -48 - 20 + 15 = -53 \neq 0$$

$\therefore A^{-1}$ exists.

Let A_{ij} denotes the cofactor of $(i, j)^{\text{th}}$ element of the matrix A.

$$\therefore A_{11} = (-1)^{1+1}(-6-6) = -12, \quad A_{12} = (-1)^{1+2}(0+10) = -10$$

$$A_{13} = (-1)^{1+3}(0+5) = 5, \quad A_{21} = (-1)^{2+1}(12+9) = -21$$

$$A_{22} = (-1)^{2+2}(24-15) = 9, \quad A_{23} = (-1)^{2+3}(-12-10) = 22$$

$$A_{31} = (-1)^{3+1}(-4+3) = -1, \quad A_{32} = (-1)^{3+2}(-8-0) = 8$$

$$A_{33} = (-1)^{3+3}(-4-0) = -4$$

$$\therefore \text{adj}A = \begin{bmatrix} -12 & -10 & 5 \\ -21 & 9 & 22 \\ -1 & 8 & -4 \end{bmatrix}' = \begin{bmatrix} -12 & -21 & -1 \\ -10 & 9 & 8 \\ 5 & 22 & -4 \end{bmatrix}$$

$$\text{and } A^{-1} = \frac{1}{|A|} (\text{adj}A) = \frac{1}{-53} \begin{bmatrix} -12 & -21 & -1 \\ -10 & 9 & 8 \\ 5 & 22 & -4 \end{bmatrix}$$

Here is an exercise for you.

E 2) For $A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 4 \\ 5 & -2 \end{bmatrix}$, verify the result $(AB)^{-1} = B^{-1}A^{-1}$.

10.4 APPLICATION OF MATRICES

Matrix Method

Inverse of a matrix can be used to solve a system of linear simultaneous equations.

Consider a system of n equations in n unknowns $x_1, x_2, x_3, \dots, x_n$ given below.

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

.

.

.

$$a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n$$

This system of equations can be written in matrix form as

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \cdot & & & \\ \cdot & & & \\ \cdot & & & \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \cdot \\ \cdot \\ \cdot \\ b_n \end{bmatrix} \quad \dots (1)$$

Or $AX = B$

$$\text{, where } A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \cdot & & & \\ \cdot & & & \\ \cdot & & & \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix}, B = \begin{bmatrix} b_1 \\ b_2 \\ \cdot \\ \cdot \\ \cdot \\ b_n \end{bmatrix}$$

If $n = 3$, $x_1 = x$, $x_2 = y$, $x_3 = z$, then (1) reduces to

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Or $AX = B \quad \dots (2)$

Or $X = A^{-1}B \quad \dots (3)$

$$\text{Or } X = \frac{1}{|A|}(\text{adj}A)B \quad \left[\because A^{-1} = \frac{1}{|A|}(\text{adj}A) \right]$$

Or $|A|X = (\text{adj}A)B \quad \dots (4)$

If $|A| = 0$, then A^{-1} does not exist. In this case, the given system of equations either has no solution or infinitely many solutions. In this case, we find

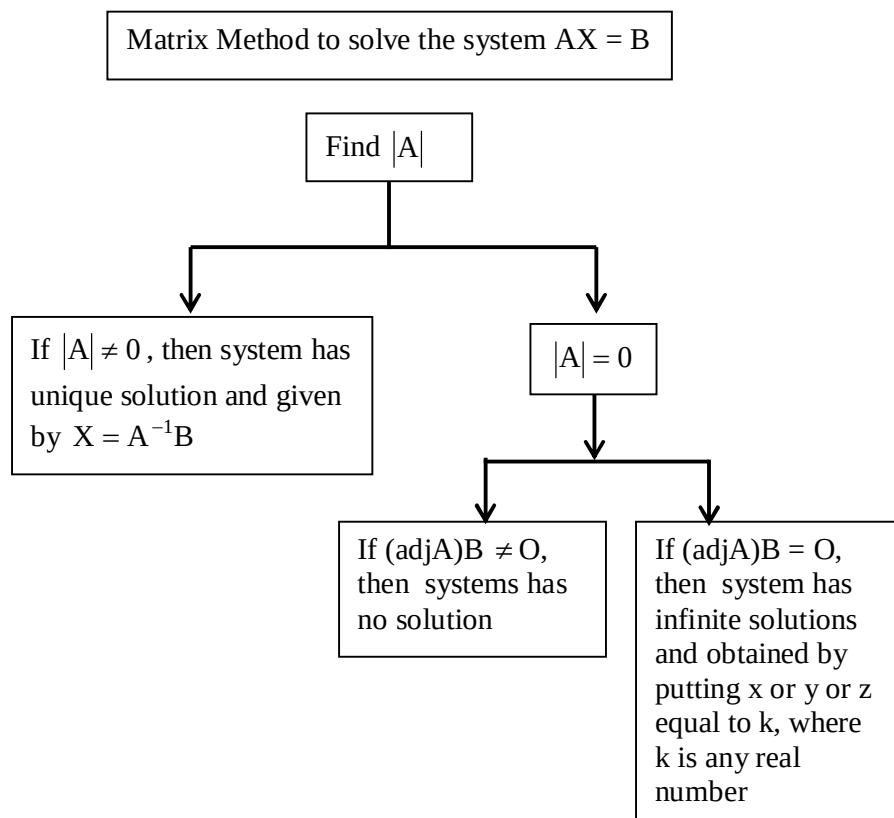
$(\text{adj}A)B$. If $(\text{adj}A)B = O$, then system has infinitely many solutions

$\left[\begin{array}{l} \because \text{in this case, (4)} \Rightarrow 0X = (\text{adj}A)B \\ \text{which holds for all matrices } X, \text{ i.e.} \\ \text{for all real values of } x, y, z. \end{array} \right]$

and if $(\text{adj}A)B \neq O$, then system has no solution.

$\left[\begin{array}{l} \because \text{in this case, L.H.S. of (4)} = O = \text{null matrix} \\ \text{but R.H.S. of (4)} = \text{non zero matrix.} \end{array} \right]$

Above discussion can be summarised in the following diagram.



Remark 1:

- (i) If $B = O$, i.e. if B is null matrix, then system given by (2) reduces to $AX = O$ and is known as linear homogeneous system of equations.
- (ii) Homogeneous system is always consistent.

Let us explain this method with the help of following example.

Example 3: The cost of 2 pens, 3 note-books, and 1 book is Rs 90. The cost of 1 pen, 4 note-books and 2 books is Rs 120. The cost of 2 pens, 4 note-books and 5 books is Rs 205. Find the cost of 1 pen, 1 note-book and 1 book by matrix method.

Solution: Let Rs x, y, z be the cost of 1 pen, 1 note-book and 1 book respectively, then according to given

$$2x + 3y + z = 90$$

$$x + 4y + 2z = 120$$

$$2x + 4y + 5z = 205$$

In matrix form this system can be written as

Consistent system: A system of equations is said to be consistent if there is either unique solution or infinite number of solutions.
Inconsistent system: If system has no solution then it is known as inconsistent system.

$$\begin{bmatrix} 2 & 3 & 1 \\ 1 & 4 & 2 \\ 2 & 4 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 90 \\ 120 \\ 205 \end{bmatrix}$$

Or $AX = B$... (1)

$$|A| = \begin{vmatrix} 2 & 3 & 1 \\ 1 & 4 & 2 \\ 2 & 4 & 5 \end{vmatrix}$$

Expanding along R_1

$$\begin{aligned} |A| &= 2(20 - 8) - 3(5 - 4) + 1(4 - 8) \\ &= 24 - 3 - 4 = 17 \neq 0 \end{aligned}$$

$\therefore A^{-1}$ exists.

Let A_{ij} denotes the cofactor of $(i, j)^{\text{th}}$ element of the matrix A .

$$\therefore A_{11} = (-1)^{1+1}(20 - 8) = 12$$

$$A_{12} = (-1)^{1+2}(5 - 4) = -1$$

$$A_{13} = (-1)^{1+3}(4 - 8) = -4$$

$$A_{21} = (-1)^{2+1}(15 - 4) = -11$$

$$A_{22} = (-1)^{2+2}(10 - 2) = 8$$

$$A_{23} = (-1)^{2+3}(8 - 6) = -2$$

$$A_{31} = (-1)^{3+1}(6 - 4) = 2$$

$$A_{32} = (-1)^{3+2}(4 - 1) = -3$$

$$A_{33} = (-1)^{3+3}(8 - 3) = 5$$

$$\text{adj}A = \begin{bmatrix} 12 & -1 & -4 \\ -11 & 8 & -2 \\ 2 & -3 & 5 \end{bmatrix}' = \begin{bmatrix} 12 & -11 & 2 \\ -1 & 8 & -3 \\ -4 & -2 & 5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}(\text{adj}A) = \frac{1}{17} \begin{bmatrix} 12 & -11 & 2 \\ -1 & 8 & -3 \\ -4 & -2 & 5 \end{bmatrix}$$

$$\begin{aligned} \text{Equation (1)} \Rightarrow X &= A^{-1}B = \frac{1}{17} \begin{bmatrix} 12 & -11 & 2 \\ -1 & 8 & -3 \\ -4 & -2 & 5 \end{bmatrix} \begin{bmatrix} 90 \\ 120 \\ 205 \end{bmatrix} \\ &= \frac{1}{17} \begin{bmatrix} 1080 - 1320 + 410 \\ -90 + 960 - 615 \\ -360 - 240 + 1025 \end{bmatrix} = \frac{1}{17} \begin{bmatrix} 170 \\ 255 \\ 425 \end{bmatrix} = \begin{bmatrix} 10 \\ 15 \\ 25 \end{bmatrix} \end{aligned}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 15 \\ 25 \end{bmatrix}$$

By definition of equality of two matrices, we have

$$x = 10, y = 15, z = 25$$

$\therefore$ costs of 1 pen, 1 note-book and one book are Rs 10, Rs 15, Rs 25, respectively.

Example 4: Solve the following system of equations:

$$(i) \begin{cases} 4x + 2y = 6 \\ 6x + 3y = 8 \end{cases} \quad (ii) \begin{cases} 3x + 6y - 4z = 3, \quad 3x - z = 0, \\ 12x - 6y - z = -3 \end{cases}$$

Solution:

(i) Given system of equations is

$$4x + 2y = 6 \quad \dots (1)$$

$$6x + 3y = 8 \quad \dots (2)$$

This system of equations can be written in matrix form as

$$\begin{bmatrix} 4 & 2 \\ 6 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

$$\text{Or } AX = B \quad \dots (3), \text{ where } A = \begin{bmatrix} 4 & 2 \\ 6 & 3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix}, B = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 4 & 2 \\ 6 & 3 \end{vmatrix} = 12 - 12 = 0$$

$\Rightarrow$ system has either no solution or infinite many solutions.

Let A_{ij} denotes the cofactor of $(i, j)^{\text{th}}$ element of the matrix A .

$$\therefore A_{11} = (-1)^{1+1} (3) = 3, \quad A_{12} = (-1)^{1+2} (6) = -6$$

$$A_{21} = (-1)^{2+1} (2) = -2, \quad A_{22} = (-1)^{2+2} (4) = 4$$

$$\text{adj}A = \begin{bmatrix} 3 & -6 \\ -2 & 4 \end{bmatrix}^T = \begin{bmatrix} 3 & -2 \\ -6 & 4 \end{bmatrix}$$

$$(\text{adj}A)B = \begin{bmatrix} 3 & -2 \\ -6 & 4 \end{bmatrix} \begin{bmatrix} 6 \\ 8 \end{bmatrix} = \begin{bmatrix} 18 - 16 \\ -36 + 32 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \end{bmatrix} \neq O = \text{null matrix}$$

Hence system has no solution.

i.e. given system of equations is inconsistent.

(ii) Given system of equations is

$$3x + 6y - 4z = 3 \quad \dots (1)$$

$$3x - z = 0 \quad \dots (2)$$

$$12x - 6y - z = -3 \quad \dots (3)$$

This system of equations can be written in matrix forms as

$$\begin{bmatrix} 3 & 6 & -4 \\ 3 & 0 & -1 \\ 12 & -6 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ -3 \end{bmatrix}$$

$$\text{Or } AX = B \quad \dots (4)$$

$$\text{where } A = \begin{bmatrix} 3 & 6 & -4 \\ 3 & 0 & -1 \\ 12 & -6 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 3 \\ 0 \\ -3 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 3 & 6 & -4 \\ 3 & 0 & -1 \\ 12 & -6 & -1 \end{vmatrix}$$

Expanding along R_1

$$|A| = 3(0 - 6) - 6(-3 + 12) - 4(-18 - 0) = -18 - 54 + 72 = 0$$

$\therefore A^{-1}$ does not exist.

$\Rightarrow$ system has either no solution or infinite many solutions.

Let A_{ij} denotes the cofactor of $(i, j)^{\text{th}}$ element of the matrix A .

$$\therefore A_{11} = (-1)^{1+1}(0 - 6) = -6$$

$$A_{12} = (-1)^{1+2}(-3 + 12) = -9$$

$$A_{13} = (-1)^{1+3}(-18 - 0) = -18$$

$$A_{21} = (-1)^{2+1}(-6 - 24) = 30$$

$$A_{22} = (-1)^{2+2}(-3 + 48) = 45$$

$$A_{23} = (-1)^{2+3}(-18 - 72) = 90$$

$$A_{31} = (-1)^{3+1}(-6 - 0) = -6$$

$$A_{32} = (-1)^{3+2}(-3 + 12) = -9$$

$$A_{33} = (-1)^{3+3}(0 - 18) = -18$$

$$\text{adj}A = \begin{bmatrix} -6 & -9 & -18 \\ 30 & 45 & 90 \\ -6 & -9 & -18 \end{bmatrix} = \begin{bmatrix} -6 & 30 & -6 \\ -9 & 45 & -9 \\ -18 & 90 & -18 \end{bmatrix}$$

$$(\text{adj}A)B = \begin{bmatrix} -6 & 30 & -6 \\ -9 & 45 & -9 \\ -18 & 90 & -18 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ -3 \end{bmatrix} = \begin{bmatrix} -18 + 0 + 18 \\ -27 + 0 + 27 \\ -54 + 0 + 54 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = O = \text{null matrix}$$

Hence system has infinite many solutions and given by putting either $x = k$ or $y = k$ or $z = k$ in any two equations given by (1), (2) and (3). Let us put $z = k$ in (1) and (2), we get, where k is any real number

$$3x + 6y = 3 + 4k \quad \dots (5)$$

$$3x = k \quad \dots (6)$$

$$(6) \Rightarrow x = \frac{k}{3}$$

Putting $x = k/3$ in (5), we get

$$3(k/3) + 6y = 3 + 4k \Rightarrow 6y = 3 + 4k - k \Rightarrow 6y = 3 + 3k \Rightarrow y = (k + 1)/2$$

$$\therefore x = \frac{k}{3}, y = \frac{k+1}{2}, z = k, \text{ where } k \text{ is any real number.}$$

Here is an exercise for you.

E 3) Solve the following system of equations by matrix method:

$$\begin{array}{lll} \text{(i) } x + y = 3 & \text{(ii) } 2x - 3y = 3 & \text{(iii) } 2x + 3y = 5 \\ 4x - 3y = 5 & 6x - 9y = 5 & 4x + 6y = 10 \end{array}$$

6.5 APPLICATION OF DETERMINANTS

There are many applications of determinants but here we shall discuss as to how the concept of determinant provides the solution of a given system of linear equations. The procedure which we will use is known as Cramer's rule.

Cramer's Rule: Cramer's rule can be used in any system of n linear equations in n variables. But here we shall discuss Cramer's rule only for system of 2 (or 3) equations in 2 (or 3) variables respectively. Suppose we are given following system of equations

$$a_{11}x + a_{12}y + a_{13}z = b_1$$

$$a_{21}x + a_{22}y + a_{23}z = b_2$$

$$a_{31}x + a_{32}y + a_{33}z = b_3$$

Cramer's rule suggests the following:

- (i) Write the determinant for the coefficients. For the above given system of

$$\text{equations, it is } \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \Delta (\text{say}).$$

- (ii) Write determinant by interchanging first column of Δ with the right side constants of the given equations. Let this determinant denoted by Δ_1 .

$$\therefore \Delta_1 = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}$$

Similarly, writing the determinants by interchanging the second and third columns of Δ with the right side constants and denoting them by Δ_2, Δ_3 respectively, we have

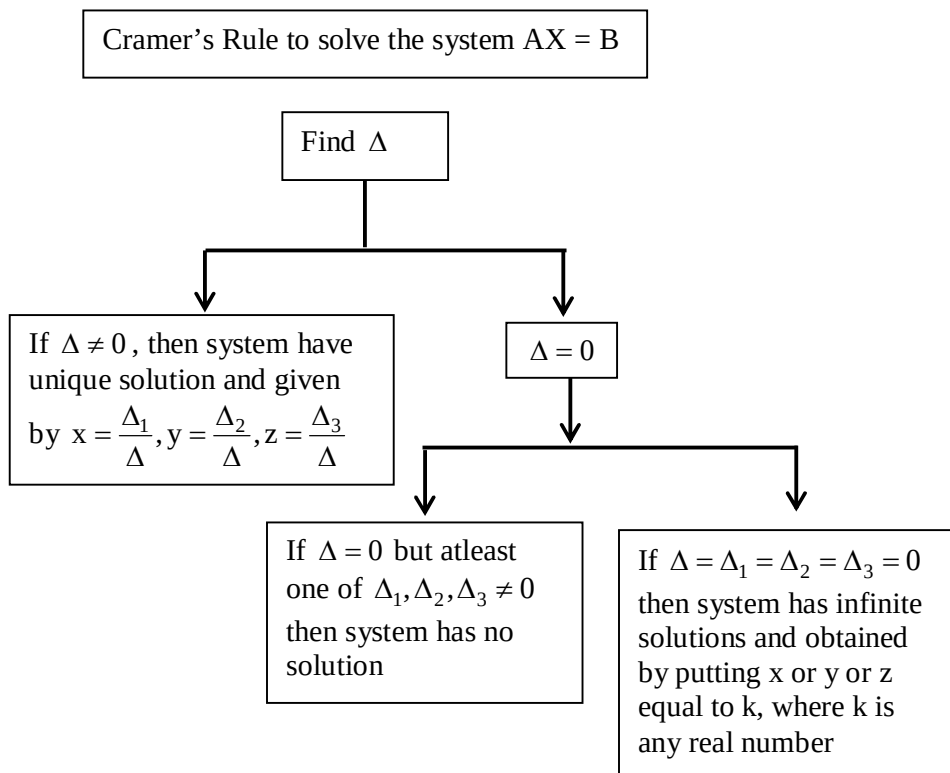
$$\Delta_2 = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}, \Delta_3 = \begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}.$$

- (iii) If $\Delta \neq 0$, system has unique solution and is given by

$$x = \frac{\Delta_1}{\Delta}, y = \frac{\Delta_2}{\Delta}, z = \frac{\Delta_3}{\Delta}.$$

If $\Delta = 0$ but at least one of the other determinants is non-zero then system has no solution. If $\Delta = 0$ and other determinants are also zero then system has infinitely many solutions given by putting x or y or z equal to k , where k is any real number.

Above discussion can be summarised in the following diagram.



Let us now consider some examples wherein Cramer's rule is applied.

Example 5: Solve the following system of equations:

$$2x + 3y + z = 4$$

$$x - y + 2z = 9$$

$$3x + 2y - z = 1$$

using Cramer's rule

Solution: Given system of equations is

$$2x + 3y + z = 4 \quad \dots (1)$$

$$x - y + 2z = 9 \quad \dots (2)$$

$$3x + 2y - z = 1 \quad \dots (3)$$

$$\text{Here, } \Delta = \begin{vmatrix} 2 & 3 & 1 \\ 1 & -1 & 2 \\ 3 & 2 & -1 \end{vmatrix}$$

Expanding along R_1 , we get

$$\Delta = 2(1 - 4) - 3(-1 - 6) + 1(2 + 3) = -6 + 21 + 5 = 20 \neq 0$$

$\therefore$ system has unique solution.

$$\Delta_1 = \begin{vmatrix} 4 & 3 & 1 \\ 9 & -1 & 2 \\ 1 & 2 & -1 \end{vmatrix}$$

Expanding along R_1

$$\Delta_1 = 4(1 - 4) - 3(-9 - 2) + 1(18 + 1) = -12 + 33 + 19 = 40$$

$$\Delta_2 = \begin{vmatrix} 2 & 4 & 1 \\ 1 & 9 & 2 \\ 3 & 1 & -1 \end{vmatrix}$$

Expanding along R_1

$$\Delta_2 = 2(-9 - 2) - 4(-1 - 6) + 1(1 - 27) = -22 + 28 - 26 = -20$$

$$\Delta_3 = \begin{vmatrix} 2 & 3 & 4 \\ 1 & -1 & 9 \\ 3 & 2 & 1 \end{vmatrix}$$

Expanding along R_1

$$\Delta_3 = 2(-1 - 18) - 3(1 - 27) + 4(2 + 3) = -38 + 78 + 20 = 60$$

$\therefore$ by Cramer's rule

$$x = \frac{\Delta_1}{\Delta} = \frac{40}{20} = 2, y = \frac{\Delta_2}{\Delta} = \frac{-20}{20} = -1, z = \frac{\Delta_3}{\Delta} = \frac{60}{20} = 3$$

$$\therefore x = 2, y = -1, z = 3$$

Example 6: Solve the following system of equations:

$$x + y + 2z = 4$$

$$x - y + 3z = 3$$

$$2x + 2y + 4z = 7$$

using Cramer's rule.

Solution: Given system of equations is

$$x + y + 2z = 4 \quad \dots(1)$$

$$x - y + 3z = 3 \quad \dots(2)$$

$$2x + 2y + 4z = 7 \quad \dots(3)$$

$$\text{Here, } \Delta = \begin{vmatrix} 1 & 1 & 2 \\ 1 & -1 & 3 \\ 2 & 2 & 4 \end{vmatrix}$$

Taking 2 common from R_3

$$\Delta = 2 \begin{vmatrix} 1 & 1 & 2 \\ 1 & -1 & 3 \\ 1 & 1 & 2 \end{vmatrix} \quad \left[\begin{array}{l} \because \text{if a row is multiplied with some} \\ \text{number, the whole determinant} \\ \text{is multiplied with that number.} \end{array} \right]$$

$$= 2(0) = 0 \quad [\because R_1 \text{ and } R_3 \text{ are identical}]$$

$\therefore$ system has either no solution or infinite many solutions.

$$\Delta_1 = \begin{vmatrix} 4 & 1 & 2 \\ 3 & -1 & 3 \\ 7 & 2 & 4 \end{vmatrix}$$

Expanding along R_1

$$\Delta_1 = 4(-4 - 6) - 1(12 - 21) + 2(6 + 7) = -40 + 9 + 26 = -5 \neq 0$$

$\therefore$ systems have no solution.

Example 7: Solve the following system of equations:

$$x + 3y + 2z = 6$$

$$-x + 4y + 5z = 8$$

$$2x + 5y + 3z = 10$$

Solution: Given system of equations is

$$x + 3y + 2z = 6 \quad \dots(1)$$

$$-x + 4y + 5z = 8 \quad \dots(2)$$

$$2x + 5y + 3z = 10 \quad \dots(3)$$

$$\text{Here, } \Delta = \begin{vmatrix} 1 & 3 & 2 \\ -1 & 4 & 5 \\ 2 & 5 & 3 \end{vmatrix}$$

Expanding along R_1

$$\Delta = 1(12 - 25) - 3(-3 - 10) + 2(-5 - 8) = -13 + 39 - 26 = -39 + 39 = 0$$

$\therefore$ system has either no solution or infinite many solutions.

$$\text{Now, } \Delta_1 = \begin{vmatrix} 6 & 3 & 2 \\ 8 & 4 & 5 \\ 10 & 5 & 3 \end{vmatrix}$$

Expanding along R_1

$$\Delta_1 = 6(12 - 25) - 3(24 - 50) + 2(40 - 40) = -78 + 78 + 0 = 0$$

$$\Delta_2 = \begin{vmatrix} 1 & 6 & 2 \\ -1 & 8 & 5 \\ 2 & 10 & 3 \end{vmatrix}$$

Expanding along R_1

$$\Delta_2 = 1(24 - 50) - 6(-3 - 10) + 2(-10 - 16) = -26 + 78 - 52 = -78 + 78 = 0$$

$$\Delta_3 = \begin{vmatrix} 1 & 3 & 6 \\ -1 & 4 & 8 \\ 2 & 5 & 10 \end{vmatrix}$$

Expanding along R_1

$$\Delta_3 = 1(40 - 40) - 3(-10 - 16) + 6(-5 - 8) = 0 + 78 - 78 = 0$$

As $\Delta = 0$ and also $\Delta_1 = \Delta_2 = \Delta_3 = 0$

$\Rightarrow$ system has infinite many solutions which are given by putting $z = k$ (where k is any real number) in any of the two equations given by (1), (2) and (3).

Let us put $z = k$ in (1) and (2), we get

$$x + 3y = 6 - 2k \quad \dots (4)$$

$$-x + 4y = 8 - 5k \quad \dots (5)$$

Again for this system of equations

$$\Delta = \begin{vmatrix} 1 & 3 \\ -1 & 4 \end{vmatrix} = 4 + 3 = 7$$

$$\Delta_1 = \begin{vmatrix} 6 - 2k & 3 \\ 8 - 5k & 4 \end{vmatrix} = 24 - 8k - 24 + 15k = 7k$$

$$\Delta_2 = \begin{vmatrix} 1 & 6 - 2k \\ -1 & 8 - 5k \end{vmatrix} = 8 - 5k + 6 - 2k = 14 - 7k$$

$\therefore$ by Cramer's rule

$$x = \frac{\Delta_1}{\Delta} = \frac{7k}{7} = k, y = \frac{\Delta_2}{\Delta} = \frac{14 - 7k}{7} = 2 - k$$

$\therefore x = k, y = 2 - k, z = k$, where k is any real number

Remark 2: Here, in the above example, we have taken $z = k$. You may also take $y = k$ or $x = k$ and then can solve by taking any two equations out of (1), (2), (3) in remaining two unknowns using Cramer's rule.

Now, you can try the following exercise.

E 4) Solve the following system of equations using Cramer's rule:

$$\begin{array}{lll} \text{(i)} \quad 3x + 5y = -11 & \text{(ii)} \quad x - 2y = 5 & \text{(iii)} \quad 2x - y = 6 \\ \quad \quad \quad 2x - 3y = 18 & \quad \quad \quad -2x + 4y = 8 & \quad \quad \quad -6x + 3y = -18 \end{array}$$

6.6 SUMMARY

Let us summarise the topics that we have covered in this unit:

- 1) Adjoint of a square matrix.
- 2) Inverse of a square matrix.
- 3) Application of matrices for solving a given system of linear equations, i.e. matrix method.
- 4) Application of determinants for solving a given system of linear equations, i.e. Cramer's rule.

6.7 SOLUTIONS/ANSWERS

E 1) $A = \begin{bmatrix} 2 & -3 \\ 4 & 5 \end{bmatrix}$

Let A_{ij} denotes the cofactor of $(i, j)^{\text{th}}$ element of the matrix A .

$$\therefore A_{11} = (-1)^{1+1}(5) = 5, \quad A_{12} = (-1)^{1+2}(4) = -4$$

$$A_{21} = (-1)^{2+1}(-3) = 3, \quad A_{22} = (-1)^{2+2}(2) = 2$$

$$\therefore \text{adj}A = \begin{bmatrix} 5 & -4 \\ 3 & 2 \end{bmatrix}' = \begin{bmatrix} 5 & 3 \\ -4 & 2 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 2 & -3 \\ 4 & 5 \end{vmatrix} = 10 - (-12) = 10 + 12 = 22$$

$$\begin{aligned} A(\text{adj}A) &= \begin{bmatrix} 2 & -3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 5 & 3 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} 10+12 & 6-6 \\ 20-20 & 12+10 \end{bmatrix} = \begin{bmatrix} 22 & 0 \\ 0 & 22 \end{bmatrix} \\ &= 22 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 22I_2 = |A|I_2 \quad \dots (1) \end{aligned}$$

$$\begin{aligned} (\text{adj}A)A &= \begin{bmatrix} 5 & 3 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 10+12 & -15+15 \\ -8+8 & 12+10 \end{bmatrix} = \begin{bmatrix} 22 & 0 \\ 0 & 22 \end{bmatrix} \\ &= 22 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 22I_2 = |A|I_2 \quad \dots (2) \end{aligned}$$

From (1) and (2), we get

$$A(\text{adj}A) = (\text{adj}A)A = |A|I_2$$

E 2) $A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 4 \\ 5 & -2 \end{bmatrix}$

$$|A| = \begin{vmatrix} 2 & 3 \\ -1 & 4 \end{vmatrix} = 8 + 3 = 11 \neq 0$$

$\therefore A^{-1}$ exists.

$$|B| = \begin{vmatrix} 1 & 4 \\ 5 & -2 \end{vmatrix} = -2 - 20 = -22 \neq 0$$

$\therefore B^{-1}$ exists.

$$AB = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 5 & -2 \end{bmatrix} = \begin{bmatrix} 2+15 & 8-6 \\ -1+20 & -4-8 \end{bmatrix} = \begin{bmatrix} 17 & 2 \\ 19 & -12 \end{bmatrix}$$

$$|AB| = \begin{vmatrix} 17 & 2 \\ 19 & -12 \end{vmatrix} = -204 - 38 = -242 \neq 0$$

$\therefore (AB)^{-1}$ exists.

Let A_{ij} , B_{ij} , C_{ij} denotes the cofactors of $(i, j)^{\text{th}}$ element of matrices A , B , AB respectively.

$$\therefore A_{11} = (-1)^{1+1}(4) = 4, \quad A_{12} = (-1)^{1+2}(-1) = 1$$

$$A_{21} = (-1)^{2+1}(3) = -3, \quad A_{22} = (-1)^{2+2}(2) = 2$$

$$B_{11} = (-1)^{1+1}(-2) = -2, \quad B_{12} = (-1)^{1+2}(5) = -5$$

$$B_{21} = (-1)^{2+1}(4) = -4, \quad B_{22} = (-1)^{2+2}(1) = 1$$

$$C_{11} = (-1)^{1+1}(-12) = -12, \quad C_{12} = (-1)^{1+2}(19) = -19$$

$$C_{21} = (-1)^{2+1}(2) = -2, \quad C_{22} = (-1)^{2+2}(17) = 17$$

$$\therefore \text{adj}A = \begin{bmatrix} 4 & 1 \\ -3 & 2 \end{bmatrix}' = \begin{bmatrix} 4 & -3 \\ 1 & 2 \end{bmatrix}, \text{adj}B = \begin{bmatrix} -2 & -5 \\ -4 & 1 \end{bmatrix}' = \begin{bmatrix} -2 & -4 \\ -5 & 1 \end{bmatrix}$$

$$\text{adj}(AB) = \begin{bmatrix} -12 & -19 \\ -2 & 17 \end{bmatrix}' = \begin{bmatrix} -12 & -2 \\ -19 & 17 \end{bmatrix}, A^{-1} = \frac{1}{|A|}(\text{adj}A) = \frac{1}{11} \begin{bmatrix} 4 & -3 \\ 1 & 2 \end{bmatrix}$$

$$B^{-1} = \frac{1}{|B|}(\text{adj}B) = \frac{1}{-22} \begin{bmatrix} -2 & -4 \\ -5 & 1 \end{bmatrix} = \frac{-1}{22} \begin{bmatrix} -2 & -5 \\ -4 & 1 \end{bmatrix}'$$

$$(AB)^{-1} = \frac{1}{|AB|}(\text{adj}AB) = \frac{-1}{242} \begin{bmatrix} -12 & -2 \\ -19 & 17 \end{bmatrix} \quad \dots (1)$$

$$\begin{aligned} B^{-1}A^{-1} &= \frac{-1}{22} \begin{bmatrix} -2 & -4 \\ -5 & 1 \end{bmatrix} \frac{1}{11} \begin{bmatrix} 4 & -3 \\ 1 & 2 \end{bmatrix} \\ &= \frac{-1}{242} \begin{bmatrix} -2 & -4 \\ -5 & 1 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 1 & 2 \end{bmatrix} = \frac{-1}{242} \begin{bmatrix} -8-4 & 6-8 \\ -20+1 & 15+2 \end{bmatrix} \\ &= \frac{-1}{242} \begin{bmatrix} -12 & -2 \\ -19 & 17 \end{bmatrix} \quad \dots (2) \end{aligned}$$

From (1) and (2), we have

$$(AB)^{-1} = B^{-1}A^{-1}$$

E3) (i) Given system of equations is

$$x + y = 3$$

$$4x - 3y = 5$$

This system of equations can be written in matrix form as

$$\begin{bmatrix} 1 & 1 \\ 4 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

Or $AX = B$ (1)

where $A = \begin{bmatrix} 1 & 1 \\ 4 & -3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$, $B = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$

$$|A| = \begin{vmatrix} 1 & 1 \\ 4 & -3 \end{vmatrix} = -3 - 4 = -7 \neq 0$$

$\therefore A^{-1}$ exists.

Let A_{ij} denotes the cofactor of $(i, j)^{\text{th}}$ element of the matrix A .

$$\therefore A_{11} = (-1)^{1+1}(-3) = -3, \quad A_{12} = (-1)^{1+2}(4) = -4$$

$$A_{21} = (-1)^{2+1}(1) = -1, \quad A_{22} = (-1)^{2+2}(1) = 1$$

$$\text{adj}A = \begin{bmatrix} -3 & -4 \\ -1 & 1 \end{bmatrix}^t = \begin{bmatrix} -3 & -1 \\ -4 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}(\text{adj}A) = -\frac{1}{7} \begin{bmatrix} -3 & -1 \\ -4 & 1 \end{bmatrix}$$

$$\begin{aligned} \text{Equation (1)} \Rightarrow X = A^{-1}B &= -\frac{1}{7} \begin{bmatrix} -3 & -1 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} \\ &= -\frac{1}{7} \begin{bmatrix} -9-5 \\ -12+5 \end{bmatrix} = -\frac{1}{7} \begin{bmatrix} -14 \\ -7 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \end{aligned}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

By definition of equality of two matrices, we have

$$x = 2, y = 1$$

(ii) Given system of equations is

$$2x - 3y = 3$$

$$6x - 9y = 5$$

This system of equations in matrix form can be written as

$$\begin{bmatrix} 2 & -3 \\ 6 & -9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

Or $AX = B$... (1), where $A = \begin{bmatrix} 2 & -3 \\ 6 & -9 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$, $B = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$

$$|A| = \begin{vmatrix} 2 & -3 \\ 6 & -9 \end{vmatrix} = -18 + 18 = 0$$

$\therefore A^{-1}$ does not exists.

$\Rightarrow$ system has either no solution or infinite many solutions.

Let A_{ij} denotes the cofactor of $(i, j)^{\text{th}}$ element of A .

$$\therefore A_{11} = (-1)^{1+1}(-9) = -9, \quad A_{12} = (-1)^{1+2}(6) = -6$$

$$A_{21} = (-1)^{2+1}(-3) = 3, \quad A_{22} = (-1)^{2+2}(2) = 2$$

$$\text{adj}A = \begin{bmatrix} -9 & -6 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} -9 & 3 \\ -6 & 2 \end{bmatrix}$$

$$(\text{adj}A)B = \begin{bmatrix} -9 & 3 \\ -6 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} -27+15 \\ -18+10 \end{bmatrix} = \begin{bmatrix} -12 \\ -8 \end{bmatrix} \neq O = \text{null matrix}$$

Hence system has no solution.

(iii) Given system of equations is

$$2x + 3y = 5 \quad \dots (1)$$

$$4x + 6y = 10 \quad \dots (2)$$

This system of equations can be written in matrix form as

$$\begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$\text{Or } AX = B, \text{ where } A = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix}, B = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = 2 \times 6 - 4 \times 3 = 12 - 12 = 0$$

$\therefore A^{-1}$ does not exist.

$\Rightarrow$ system has either no solution or infinite many solutions.

Let A_{ij} denotes the cofactor of $(i, j)^{\text{th}}$ element of the matrix A .

$$\therefore A_{11} = (-1)^{1+1}(6) = 6, A_{12} = (-1)^{1+2}(4) = -4$$

$$A_{21} = (-1)^{2+1}(3) = -3, A_{22} = (-1)^{2+2}(2) = 2$$

$$\text{adj}A = \begin{bmatrix} 6 & -4 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ -4 & 2 \end{bmatrix}$$

$$(\text{adj}A)B = \begin{bmatrix} 6 & -3 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 10 \end{bmatrix} = \begin{bmatrix} 30-30 \\ -20+20 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = O = \text{null matrix}$$

$\Rightarrow$ system has infinitely many solutions and given by putting $y = k$ in (1),

we get

$$2x + 3k = 5 \Rightarrow x = \frac{5-3k}{2}$$

$$\therefore x = \frac{5-3k}{2}, y = k, \text{ where } k \text{ is any real number.}$$

E 4) (i) Given system of equations is

$$3x + 5y = -11$$

$$2x - 3y = 18$$

$$\text{Here, } \Delta = \begin{vmatrix} 3 & 5 \\ 2 & -3 \end{vmatrix} = -9 - 10 = -19 \neq 0$$

$\Rightarrow$ system has unique solution.

$$\Delta_1 = \begin{vmatrix} -11 & 5 \\ 18 & -3 \end{vmatrix} = 33 - 90 = -57$$

$$\Delta_2 = \begin{vmatrix} 3 & -11 \\ 2 & 18 \end{vmatrix} = 54 + 22 = 76$$

$\therefore$ by Cramer's rule

$$x = \frac{\Delta_1}{\Delta} = \frac{-57}{-19} = 3, \quad y = \frac{\Delta_2}{\Delta} = \frac{76}{-19} = -4$$

$$\therefore x = 3, \quad y = -4$$

(ii) Given system of equations is

$$x - 2y = 5$$

$$-2x + 4y = 8$$

$$\text{Here, } \Delta = \begin{vmatrix} 1 & -2 \\ -2 & 4 \end{vmatrix} = 4 - 4 = 0$$

$\therefore$ system has either no solution or infinite many solutions.

$$\Delta_1 = \begin{vmatrix} 5 & -2 \\ 8 & 4 \end{vmatrix} = 20 + 16 = 36 \neq 0$$

$$\therefore \Delta = 0 \text{ but } \Delta_1 = 36 \neq 0$$

$\Rightarrow$ system has no solution.

(iii) Given system of equations is

$$2x - y = 6 \quad \dots (1)$$

$$-6x + 3y = -18 \quad \dots (2)$$

$$\text{Here, } \Delta = \begin{vmatrix} 2 & -1 \\ -6 & 3 \end{vmatrix} = 6 - 6 = 0$$

$$\Delta_1 = \begin{vmatrix} 6 & -1 \\ -18 & 3 \end{vmatrix} = 18 - 18 = 0, \quad \Delta_2 = \begin{vmatrix} 2 & 6 \\ -6 & -18 \end{vmatrix} = -36 + 36 = 0$$

$$\therefore \Delta = 0, \quad \Delta_1 = \Delta_2 = 0$$

$\Rightarrow$ system has infinite many solutions and given by putting either x or y equal to some arbitrary constant.

Let $y = k$, where k is any real number.

$$\text{Equation (1)} \Rightarrow 2x = 6 + k \Rightarrow x = \frac{6 + k}{2}$$

$$\therefore x = \frac{6 + k}{2}, \quad y = k, \quad \text{where k is any real number}$$

UNIT 11 INTRODUCTION TO STATISTICS

Structure

- 11.1 Introduction
 - Objectives
- 11.2 Origin and Development of Statistics
- 11.3 Definition of Statistics
- 11.4 Scope and Uses of Statistics
- 11.5 Limitations of Statistics
- 11.6 Measurement Scales
- 11.7 Types of Data
- 11.8 Summary
- 11.9 Solutions/Answers

11.1 INTRODUCTION

As you know every subject has its origin, development stages, scope, uses and limitations.

In this unit, we will discuss origin and development, definition, scope and uses, and limitations of statistics. Different measurement scales and different types of data also have been discussed in this unit.

Objectives

After completing this unit, you should be able to:

- know origin and development stages of statistics;
- know definition, scope, uses and limitations of statistics;
- get an idea of different types of measurement scales; and
- get an idea of different types of data.

11.2 ORIGIN AND DEVELOPMENT OF STATISTICS

As we know that every subject, race, machine, etc have its own origin and development stages. Similarly, Statistics also have its own origin and development stages. In fact, in a single sentence it can be said that human society has been using Statistics knowingly or unknowingly right from the beginning of its existence. This is because (i) most of the decisions taken by a human being are based on the past experience (i.e. based on statistical data he/she has experienced actually), and (ii) future events are also predicted by using/examining the past behavior of that particular event.

The word “Statistics” seems to have been derived from Latin word ‘Status’ or Italian word ‘Statista’ or German word ‘Statistik’. But according to the observations of great John Graunt (1620-1674), the word ‘Statistics’ is of Italian origin and it is derived from the word ‘Stato’ and statista means a person who deals with affairs of the state. That is, initially kings or monarchs or governments used it to collect the information related to the population, agricultural land, wealth, etc. of the state. Their aim behind it was just to get an

State Craft:
The art of
managing state
affairs.

idea about the men power of the state, force needed for the purpose of a war and necessary taxes to be impose to meet the financial need of the state.

So, it indicates that initially it was used by kings or monarchs or governments for administrative requirements of the state. That is why its origin lies in the state craft.

On the basis of evidences form papyrus manuscripts and ancient monuments in pharaonic temples, it is assumed that first census in the world was carried out in Egypt in 3050 BC. Yet, China's census data around 2000 BC is considered as the oldest surviving census data in the world.

Statistics in India

Though the use of Statistics, knowingly or unknowingly, has been there in India from ancient times, yet if we talk about its origin on the basis of evidences, it takes us in 3rd century BC when "Arthashastra" came into existence written by one of the greatest geniuses of political administration, Kautilya. In it, he had described the details related to conduct of population, agriculture and economic census. An efficient system of collecting official and administrative statistics was in use during the reign of Chandra Gupta Maurya (324-300 BC) under the guidance of Kautilya. Many things like taxation policy of the state, governance and administration, public finance, duties of a king, etc. had also been discussed in this celebrated Arthashastra. Another evidence that statistics was in use during Emperor Akbar's empire (1556-1605) is in the form of "Ain-I-Akbari" written by Abul Fazl, one of the nine jems of Akbar. Raja Todar Mal, Akbar's finance minister and another one of the nine jems of Akbar, used to keep very good records of land and revenue and he developed a very systematic revenue collections system in the kingdom of Akbar by using his expertise and the recorded data. Revenue collection system developed by Raja Todar Mal was so systematic that it became a model for future Mughals and later on for British.

British Government, after transfer of the power from East India Company to it, started a publication entitled 'Statistical Abstract of British India' as a regular annual feature in 1868 in which all the useful statistical information related to local administrations to all the British Provinces was provided. In between some census reports were coming on based on a particular area, but not at the national level. The first attempt to get detailed information on the whole population of India was made between 1867 and 1872. First decennial census was undertaken on 17th February 1881 by W.W. Plowden, first census commissioner of India. After that a census has been carried out over a period of 10 years in India. 2011 census was the 15th census in India.

Credit of establishing Statistics as a discipline in India goes to Prasanta Chandra Mahalanobis (P.C. Mahalanobis). He was a professor of physics in the Presidency College in Kolkata. During his study at Cambridge he got a chance to go through the work of Karl Pearson and R. A. Fisher. Continuing his interest in Statistics, he established a Statistical laboratory in the Presidency College Kolkata. On 17 December 1931, this statistical laboratory was given the name Indian Statistical Institute (ISI) mainly to promote the study, dissemination of knowledge of Statistics, research in it and to develop statistical techniques which play major role in addressing various problems of planning of national development and social welfare in the country. P.C. Mahalanobis was the founder director of ISI. Some well known personalities other than P.C. Mahalanobis associated to ISI whose research work made the

institute unique internationally are Professor C.R. Rao, Professor R.C. Bose, Professor S.N. Roy, etc.

First post graduate course in Statistics was started by Kolkata University in 1941, while first under graduate course in Statistics was started by the Presidency College Kolkata. With the passage of time some more universities/institutes came up with courses in Statistics. Some of these are University of Mumbai, University of Pune, University of Madras, University of Mysore, University of Kerala and University of Lucknow. This list of institutions went on increasing with time and at present more than 1100 institutes are there in the country, which are offering under graduate or post graduate courses in statistics.

Statistics in World

Without going into more details, we will concentrate on only some major discoveries in the area of Statistics at international level. A lot of theoretical development in different areas of statistics took place in seventeenth and eighteenth centuries in many countries of the world. John Graunt (1620-1674 born in London), being a haberdasher by profession, has the credit of producing the first life table with probabilities of survival to each age. Due to this great achievement, he is known as father of vital statistics. This was the period when some other persons also did their contribution in the same area such as Edmund Haller (1656-1742) prepared a life table on the basis of the data collected by Casper Newman in 1691, relating to death records of Breslau. Sir William Petty (1623-1687) also prepared mortality tables and calculated expectation of life at different ages. G.F. Knapp (1842-1926) and W. Lexis (1837-1914) also did valuable work on the statistics of mortality. Study of probability was also found to be very important in the area of Statistics, quantitative measure of which was given by Galileo (1564-1642), an Italian mathematician [For detail discussion on development of Probability Sec 1.1 of Unit 1 of MST-003 may be referred to]. Guass (1777-1855) gave the principal of least square and normal law of errors. J. Bernoulli (1654-1705) was the first person who states the law of large numbers in his great work *Ars conjectandi* published eight years after his death. Statistical methods in the field of biometry were first introduced of Sir Francis Galton (1822-1911). Later on Professor Karl Pearson (1857-1936) followed up the work of Galton and did significant contribution to Anthropology and correlation coefficient theories. Karl Pearson was also the founder of Statistical Research Laboratory in the university college, London in 1911. Credit of discovery of Chi-Square test also goes to Karl Pearson. Credit of discovery of 't test' or 'student t' test goes to W.S. Gosset who wrote under the pseudonym of student's 't'.

List of contributors in the area of Statistics did not end here but we conclude by throwing some light on the work done by Fisher. Credit of discovering of very powerful test known as Analysis of Variance (ANOVA) goes to Sir Ronald A. Fisher (1890-1962). Fisher also did a lot of work in the area of point estimation. Due to his remarkable contribution in the field of Statistics, he is known as Father of Statistics.

11.3 DEFINITION OF STATISTICS

In previous section of this unit, we have seen that statistics is a very old science and it has developed/grown up through ages. So, it is not surprising that through its long journey its definitions given by different authors time to time

may also vary. In fact, today statistics is quite different that of earlier times. Let us first see some definitions by different authors and then it will be clear that which definition is more broad and may be consider as a good definition of statistics.

- “Statistics is the science of counting”. – **A.L. Bowly**.
- “Statistics is the science of average.” – **A.L. Bowly**.
- Statistics is “The science of the measurement of the social organism, regarded as a whole, in all its manifestations.” – **A.L. Bowly**.
- “Statistics are the numerical statements of facts in any department of enquiry placed in relation to each other.” – **A.L. Bowly**.
- “By statistics we mean quantitative data affected to a marked extent by multiplicity of causes.” – **Yule and Kendall**.
- “Science of estimates and probabilities.” – **Boddington**.
- “The method of judging collective natural or social phenomena from the results obtained by the analysis of an enumeration or collection of estimaties.” – **W.I. King**.
- “Statistics is the science which deals with collection classification and tabulation of numerical facts as the basis for explanation description and comparison of phenomenon”. – **Lovitt**.
- “The science which deals with the collection, tabulation, analysis and interpretation of numerical data.” – **Croxton and Cowden**.

From the list of above definitions given by different authors, and various other definitions the comprehensive definition of Statistics may be given as:

“Statistics is a branch of science which deals with collection, classification, tabulation, analysis and interpretation of data.”

11.4 SCOPE AND USES OF STATISTICS

In present times, statistics is not known as only collection of data, but it is regarded as a science having sound techniques of even handling huge data and providing valuable conclusions. Today statistical methods are universally applied. That is, statistics find its application in almost every sphere of human activity such as economics, commerce, management, information technology, education, planning, banking, insurance sector, medical science, biology, industrial, agriculture, market research, etc. That is, scope of statistics is very wide and in single sentence we can say that statistics is the queen of all sciences.

In the above paragraph, we have listed many fields where statistics has its application. Let us see its uses in some of these fields.

Statistics and Industry

In industrial line, statistics plays an important role such as in quality control and production engineering various control charts are used to maintain a certain quality level and different inspection plans are used in production engineering. Even to find average life of some products such as electric bulb, sampling technique is used. Those learners who will opt Industrial Statistics specialisation will get these terms in detail in courses MSTE-001 and MSTE-002.

Professor Karl Pearson has stated that the whole doctrine of heredity rests on statistical basis. This is generally said that height of the child is associated with the height of the father. To test this type of hypothesis, statistics is the only science which provides the scientific methods. Vital statistics is totally devoted to the different aspects of human life like average life of men and women, birth and death rates, etc. Those learners who will opt Bio-Statistics specialisation will get these terms in detail in courses MSTE-003 and MSTE-004.

Statistics and Medicine

Statistics also plays an important role in the field of medicine. The hypothesis of the types:

- (i) Drug A is better than drug B.
- (ii) Smoking and cancer are associated.
- (iii) Smoking and TB are associated.

All are tested using t-test or χ^2 -test as the case may be. Statistics also find its application in clinical trials.

Statistics and Planning

Every institution/organisation plans for its future targets. Now, a days for a good planning, it has become necessary to analysis the statistical data according to the field of interest such as availability of raw material, consumption, investment, resources available, income, expenditure, quality needed, etc. In order to analysis these types of data, one has to totally depend on the statistical techniques. Thus statistics is essential for planning.

Statistics and Commerce

In present times, there is a very tough competition in almost every business. Also fashions, likings/tastes, requirements, trends, levels of qualities, technologies, etc. are changing very fast. So for the success of the business, it has become necessary for a business man to know the coming trend of market in advance or as soon as possible. This can also be achieved only with the help of market survey, which requires statistical techniques.

Statistics and Agriculture

Presently there are a number of varieties of seeds for a particular crop. Also different types of fertilizers are available in the market. For a good yield, it has become necessary to know that which one is better. This job is again done by a very popular and widely used test known as Analysis of variance (ANOVA) discovered by Professor R.A. Fisher. You will learn about ANOVA in more detail in block 2 of course MST-005. Complete Block 2 of MST-005 is totally devoted to ANOVA.

Statistics and Insurance Sector

Whole insurance sector totally depends on the statistical data and different concepts of probability theory. Life tables lies in the heart of human insurances. Curtate future life time and complete future life time, of a life are calculated using concept of random variables and their expected values. (you will learn in detail about random variables and their expected values in block 2 of course MST-003). Due to the large use of statistics in insurance sector, a new branch of statistics known as Actuarial statistics has been started in some institutes throughout the world.

Statistics and Research

Research is very important aspect in every discipline. In many disciplines such as psychology, tourism, education, M.B.A., etc. one has to collect the data on the characteristics of interest under study. Now a very important question arises, related to the measurement of scale to be used and appropriate test to be used. This requires the knowledge of different types of measurement scales and accordingly suitable statistical tool need. (Different types of measurement scales-nominal, ordinal, interval and ratio have been discussed in detail in Sec 11.6 of this unit). Also appropriate statistical tool to be needed in a given situation have been listed in Table 11.1.

Statistics and Economics

In order to know about the development of a country, it has become necessary to obtain the data related to its economical growth. Again, statistical tools are needed to collect relevant data (such as related to agricultural, industrial, literacy, etc) and for its analysis.

Statistics and Common Man

Statistics also plays an important role in the welfare of common man. Common man of any country faces lot of problems in his routine life such as food shortage, hygienic drinking water, unemployment, poverty, medical, shortage of public transport, etc. Time to time statistical figures on these issues enables the government to think and sort out these problems.

Statistics helps the common man in their day to day life in another way also, e.g. in purchasing any good he/she used his/her past experience (actually based on the data he/she faced/experienced) and take the decision to buy or not buy a particular object. Similarly, a farmer decide about the crop to be yield based on his past experience (actually based on the data he has faced) and labourer choose one of the works which gives him more wages based on his past experienced (actually based on the data he has faced).

List of fields/areas where statistics is used does not end here. We have just touch some of the areas where statistics has its application. We close this section by saying that there is hardly any field where statistics cannot be used. Infact, statistics can be used in any field of interest.

11.5 LIMITATIONS OF STATISTICS

In previous section of this unit, you have seen wide range of application of statistics. Being the queen of all sciences, statistics also have its own limitations, some of them are described as follows:

(1) Indirect Approach Towards Qualitative Characteristic

Science of statistics basically deals with numerical data. Therefore statistical tools are applicable only for quantitative measures. But many a times characteristic under study is qualitative in nature such as honesty, beauty, intelligence, boldness, drinking, smoking, etc. So any statistical tool cannot be directly applied on these types of characteristics. However, study of these types of characteristics can be made possible by first converting the characteristic under study into numerical figures based on some uniform criteria. For example, intelligence can be converted into numerical figures with the help of the marks obtained by the individuals in a common test.

(2) Dealingness with a Group

Science of statistics deals with aggregates of objects not with individuals. The individual's figures, when taken separately do not come under the category of statistical data. So, applicability of any statistical tool becomes meaningless. For example, salary of one employee of an institute does give any message related to the salaries of the employees of that particular institution.

(3) Lack of Exactness

Statistical results are not exactly true, but they are true on an average.

For example,

- (i) If a statistical report says that 70% population of India lived in rural area. It does not imply that if you visit at public place like bus stand, railway station, etc. and asked the people about their living place. Results may surprise you and may highly differ with the above figure. But you may note that as sample size increases, the result will also come nearer and nearer to exact figure 70%.
- (ii) Consider another example, suppose past data show that 90% operations of a doctor are successful. It does not imply that out of the next 100 operations, exactly 90 will be successful. It may happen that figure that will obtain in future may be 90%, 80%, 87%, 95%, etc. But there are sciences like mathematics where exactness is maintained. For example, if a book seller get 5% profit on selling a particular book. Then it is sure that if sell of that particular book is of Rs 200 he/she will get Rs 10 as profit and in case of sale of Rs 300 profit will be Rs 15 and so on.

(4) Requirement of Experts Hands for Effective Use

Requirement of experts' hands for effective and appropriate use is one of the main draw backs of the science of statistics. There are many statistical tools of similar type.

For example,

- (i) To find average in a particular situation, which of the possible tools like mean, median, mode, geometric mean, harmonic mean, etc. is appropriate needs the hands of experts.
- (ii) Similarly to test a given statistical hypothesis which of the possible tools like Z-test, t-test, χ^2 -test, F-test, ANOVA, median test, run test, sign test, etc. is appropriate again needs the hands of experts. This limitation of the statistics limits the range of its effective users.

11.6 MEASUREMENT SCALES

Two words “counting” and “measurement” are very frequently used by everybody. For example, if you want to know the number of pages in a note book, you can easily count them. Also, if you want to know the height of a man, you can easily measure it. But, in Statistics, act of counting and measurement is divided into 4 levels of measurement scales known as

- (1) Nominal Scale
- (2) Ordinal Scale

(3) Interval Scale

(4) Ratio Scale

Let us discuss these scales of measurement one by one:

(1) Nominal Scale

In Latin, 'Nomen' means name. The word nominal has come from this Latin word, i.e. 'Nomen'. Therefore, under nominal scale we divide the objects under study into two or more categories by giving them unique names. The classification of objects into atleast two or more categories is done in such a way that

- (a) Each object takes place only in one category, i.e. each object falls in a unique category, i.e. it either belongs to a category or not. Mathematically, we may use the symbol (" $=$ ", " $\neq$ ") if an object falls in a category or not.
- (b) Number of categories must be sufficient to include all objects, i.e. there should not be scope for missing even a single object which does not fall in any of the categories. That is, in statistical language categories must be mutually exclusive and exhaustive.

Generally nominal scale is used when we want to categorize the data based on the characteristic such as gender, race, region, religion, etc.

To get more familiar with the idea of nominal scale, let us consider some examples:

(i) Classification into Different Categories Based on Gender

This can be done by dividing the population into two categories male 'M' and female 'F'

Category	Name/Code
Male	M
Female	F

Here we have named male as 'M' and female as 'F'. This is not the only way, we can also code male by '0' and female by '1' or we may use any other convenient symbols. So, we note that main thing is that we have to give a unique name to each category.

(ii) Classification into Different Categories Based on Caste

Different categories	Code allotted /Name given
General	Gen
Scheduled caste	SC
Scheduled tribes	ST
Backward class	BC
Others	'O'

Here also we can give a code to general, scheduled caste, scheduled tribes, backward class and other categories by '0', '1', '2', '3', '4' respectively.

(iii) Classification into Different Categories Based on Region

28 states and 7 union territories together classified India into 35 categories which can be coded by their usual names or may be coded by using some other symbols.

(iv) Classification into Different Categories Based on Religion

Population of India can be broadly categorised based on the following different religions:

Different categories	Codes allowed /Names given
Hindu	1
Muslim	2
Sikh	3
Isaiah	4
Others	5

(v) Classification into Different Categories Based on Number Allotted

In a sport event, the numbers allotted to the participants also come under nominal scale.

Note 1: We note that in nominal scale we have just coded the objects. Sign of less than or greater than does not make any sense in nominal scale. That is here we have coded Hindu, Muslim, by '1' and '2' respectively. But Hindu > Muslim or Muslim > Hindu does not make any sense.

Similarly, male > female or female > male does not make any sense.

That is, we cannot talk about the order between two categories in case of nominal scale.

If in a measurement scale orders also make sense then, this scale comes under the heading ordinal scale discussed below.

(2) Ordinal Scale

We have seen that order does not make any sense in nominal scale. As the name ordinal itself suggests that other than the names or codes given to the different categories, it also provides the order among the categories. That is, we can place the objects in a series based on the orders or ranks given by using ordinal scale. But here we cannot find actual difference between the two categories.

Generally ordinal scale is used when we want to measure the attitude scores towards the level of liking, satisfaction, preference, etc. Different designation in an institute can also be measured by using ordinal scale.

To get more familiar with the concept of ordinal scale let us consider some examples:

- (i) Opinion of persons about proposal of introducing co-education in a college comes under this scale. Suppose we assign '1' to strongly disagree, '2' to disagree '3' to indifferent (or neutral) '4' to agree and '5' to strongly agree. Here, the order also matters and mathematically we may use the symbols $>$, $<$ in addition to those used for nominal scale, i.e. $=$, $\neq$ as here strongly agree opinion comes first in order as compared to agree and so on, i.e. $5 > 4 > 3 > 2 > 1$ or $1 < 2 < 3 < 4 < 5$. But actual difference between different categories in this Likert Scale is not possible. This is because, suppose "strongly agree" means he/she gives marks from 75% to 100% for the co-education to be introduced and suppose "agree" means the marks are given in the range say 50% to 75%. Now, the actual difference between "strongly agree" and "agree" is not feasible in this sense.

$\left[\begin{array}{l} \therefore \text{"strongly agree" may have the marks percentage as} \\ 80\% \text{ and "agree" marks } 74\%. \text{ Similarly in other} \\ \text{case these values may be } 90\% \text{ and } 70\% \text{ respectively.} \end{array} \right]$

- (ii) Suppose, a school boy is asked to list the name of three ice-cream flavours according to his preference. Suppose he lists them in the following order:

Vanilla
Straw berry
Tooty-frooty

This indicates that he likes vanilla more compared to straw berry and straw berry more as compared to tooty-frooty. But the actual difference between his liking between vanilla and straw berry cannot be measured.

- (iii) In sixth pay commission, teachers of colleges and universities are designated as Assistant Professor, Associate Professor and Professor. The rank of Professor is higher than that of Associate Professor and designation of Associate Professor is higher than Assistant Professor. But you cannot find the actual difference between Professor and Associate Professor or Professor and Assistant Professor or Associate Professor and Assistant Professor. This is because, one teacher in a designation might have served certain number of years and have done a good quality of research work, etc. and other teacher in the same designation might have served for lesser number of years have done unsatisfactory research work, etc. So, the actual difference between one designation and other designation cannot be found. So one may be very near to his next higher designation and other may be very far from it depending on their quality of teaching/research.

- (iv) Based on economic condition of a family, generally families of a society are divided into three categories:

Higher class family
Middle class family
Lower class family

Every body knows that economic condition of higher class family is better than middle class family and middle class family is in a better condition compare to lower class family. But the actual difference between the economic condition of a higher class family and middle class family or between middle class family and lower class family cannot be measured.

That is, we can only give order/rank to the three classes of the families but actual difference cannot be measured. In all the above examples, the actual difference is not possible because, all the ranks are on the ranges and not on fixed points.

Now, we are going to study the next higher level of measurement wherein the actual differences can be found. This scale is known as interval scale.

(3) Interval Scale

You have become familiar with the concept of interval and its length in Sec. 2.2 of Unit 2 of this course. If $I = [4, 9]$ then length of this interval is $9 - 4 = 5$, i.e. difference between 4 and 9 is 5, i.e. we can find the difference between any two points of the interval. For example, $7, 7.3 \in I$ and difference

between 7 and 7.3 is 0.3. Thus we see that property of difference holds in case of intervals. Similarly, third level of measurement, i.e. interval scale possesses the property of difference which was not satisfied in case of nominal and ordinal scales.

Nominal scale gives only names to the different categories, ordinal scale moving one step further also provides the concept of order between the categories and interval scale moving one step ahead to ordinal scale also provides the characteristic of the difference between any two categories.

Interval scale is used when we want to measure years/historical time/calendar time, temperature (except in the Kelvin scale), sea level, marks in the tests where there is negative marking also, etc. Mathematically, this scale includes $+$, $-$ in addition to $>$, $<$ and $=$, $\neq$.

To get more familiar with the concept of interval scale, let us consider some examples:

- (i) The measurement of time of an historical event comes under interval scale because there is no fixed origin of time (i.e. '0' year). As '0' year differ calendar to calendar or society/country to society/country e.g. Hindus, Muslim and Hebrew calendars have different origin of time, i.e. '0' year is not defined. In Indian history also, we may find BC (Before Christ).
- (ii) Measurement of temperature in degree Celsius ($^{\circ}\text{C}$) assumes 0°C when water starts freezing to ice and it becomes ice at -40°C . So, in degree Celsius origin is arbitrary that's why measurement of temperature in degree Celsius comes in interval scale. Because in degree Celsius origin is arbitrary, so we cannot say that 30°C is twice as hot as 15°C . Because if it is so then can we say that 4°C is -1 times -4°C ? No it is meaningless. Similarly, measurement of temperature in Fahrenheit comes in the interval scale.
- (iii) Mean sea level (MSL) also have arbitrary origin because it is mean of two means, mean high tide and mean low tide(and mean high tide and mean low tide vary according to high and low pressure zones). Further it also varies place to place and time to time. So measurement of sea level also comes in the interval scale.

(4) Ratio Scale

Ratio scale is the highest level of measurement because nominal scale gives only names to the different categories, ordinal scale provides orders between categories other than names, interval scale provides the facility of difference between categories other than names and orders but ratio scale other than names, orders and characteristic of difference also provides natural zero (absolute zero). In ratio measurement scale values of characteristic cannot be negative.

Ratio scale is used when we want to measure temperature in Kelvin, weight, height, length, age, mass, time, plane angle, etc. Ratio scale

includes $\times$, $\div$ in addition to $+$, $-$, $>$, $<$, $=$, $\neq$. But be careful never take '0' in the denominator while finding ratios. For example, $\frac{4}{0}$ is meaningless.

To get more familiar with the concept of ratio scale let us consider some examples, where ratio scale is used:

- (i) Measurement of temperature in Kelvin scale comes under ratio scale because it has an absolute zero which is equivalent to -273.15°C . This characteristic of origin allows us to make the statement like 50K ('50K' read as 50 degree Kelvin) is 5 time hot compare to 10K.
- (ii) Measurement of money also comes under ratio scale because it satisfies all the requirement of interval scale and has a natural zero. For example, suppose there are 60 teachers in a particular school in Delhi. If we associate a unique number to each teacher related to the cash (in rupees) he/she has with him/her at the time of investigation. Then we have a fixed whole number corresponding to each teacher. Of course two or more teachers may have same cash (in rupees). These teachers will be allotted the same whole number and will fall in one category. Here we note that, the whole numbers allotted to the teachers can be ordered, have an actual difference and also have origin (i.e. absolute zero '0'). Here natural zero indicates the absence of money in the pocket of the teacher. If a teacher has Rs 500 and another teacher has Rs 100 then we can say that the teacher having Rs 500 has 5 times amount than a teacher having Rs 100. Thus it satisfies all the requirement of ratio scale.
- (iii) Both height (in cm.) and age (in days) of students of M.Sc. Statistics of a particular university satisfy all the requirements of a ratio scale. Because height and age both cannot be negative (i.e have an absolute zero).

Permissible Statistical Tools

One of the advantages of measurement scale is that these help us to decide which statistical tool should be used in a given situation.

Table 11.1 shows the list of permissible statistical tools in case of nominal, ordinal, interval and ratio scales. Based on information provided by these scales, their levels from lowest to height are nominal, ordinal, interval and ratio (see Fig 11.1). That is why all the Statistical tools applicable on the lower scale will automatically be applicable on the next level scale. So, we will not repeat the permissible statistical tools used in lower level scale. It is understood that statistical tools which are permissible for nominal will be permissible in case of ordinal and so on.

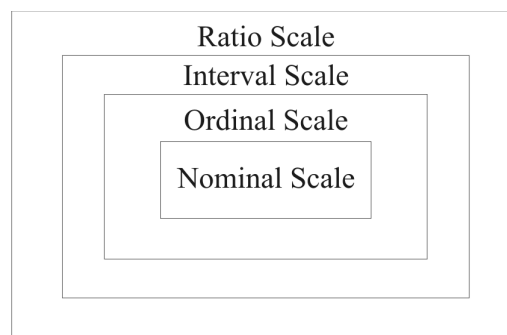


Fig. 11.1

Table 11.1

MEASUREMENT SCALE	PERMISSIBLE STATISTICAL TOOLS	LOGIC/REASON
NOMINAL SCALE	Mode, chi-square test and run test	Here counting is only permissible operation.
ORDINAL SCALE	Median all positional averages like quartile, Decile, percentile, Spearman's Rank correlation.	Here other than counting, order relation (less than or greater than) also exists.
INTERVAL SCALE	Mean, S.D., t-test, F-test, ANOVA, sample multiple and moment correlations, regression.	Here counting, order and difference operations hold.
RATIO SCALE	Geometric mean (G.M.), Harmonic mean (H.M.), Coefficient of variation.	Here counting, order, difference and natural zero exist.

Before closing this section let us consider some situations and appropriate measurement scale that can be used with the help of some examples followed by some exercises.

Example 1: If you want to collect the data based on the characteristic of literacy then which scale will be used? Explain with reasons.

Solution: Appropriate scale is nominal scale because population can be categorised in two categories literate (L) and illiterate (I). The symbols for literate and illiterate can be used according to our choice like 0, 1 or A, B or X, Y, etc.

Example 2: At a picnic spot in India, 1000 tourists visit over a period of 7 days. Each tourist is asked the name of the country of his/her birth. Then the data thus obtained come under which measurement scale.

Solution: Nominal scale, because the characteristic 'name of the country' divides the tourists into different categories each labels with the name of his/her country.

Example 3: Answer the following questions:

- Which scale is at lowest level?
- Which scale is at highest level?
- Which scale has absolute zero?
- Which scale is used to find the mean sea level (MSL)?

Solution:

- Nominal scale is at lowest level, because it has only one permissible operation counting.
- Ratio scale is at highest level, because it has all the four operations counting, order, distance and absolute zero.
- Ratio scale is only scale out of the four measurement scales nominal, ordinal, interval and ratio scales which has absolute zero.
- Because sea level has no absolute zero, so interval scale is used to find the mean sea level (MSL).

Example 4: Answer the following questions:

- (i) In which scale median is not permissible?
- (ii) In which scale(s) mean is not permissible?
- (iii) In which scale(s) geometric mean and harmonic mean are not permissible?
- (iv) In which scale geometric mean and harmonic mean are permissible?

Solution:

- (i) In order to find median, we have to arrange the data in ascending or descending order of magnitude. But in nominal scale order operation is not present. So, in case of nominal scale data, median is not permissible.
- (ii) In order to find mean, each observation of the data must be associated with a numerical quantity (which exactly measure the quantity of the characteristic). But, this requirement is not fulfilled by nominal and ordinal scales data. So, mean is not permissible in case of nominal and ordinal data.
- (iii) In order to find geometric mean (G.M.) and harmonic mean (H.M.) absolute zero must be defined so that one can talk of quotient/ratio of two numbers. But as absolute zero is defined only in ratio scale, so G.M. and H.M. are not permissible in nominal, ordinal and interval scales data.
- (iv) As discussed in (iii), G.M. and H.M. are defined only in ratio scale.

Now, here are some exercises for you.

E 1) Answer the following questions:

- (i) Which scale is considered as the best scale of measurement so far as criteria of information provide is concerned?
- (ii) Which scale is used in case of measurement of height, weight and age?
- (iii) Allotment of license plates to different car comes under which scale of measurement?
- (iv) Characteristic of equal distance between any two observations is maintained by which scale(s) of measurement?

E 2) Measurement of blood group comes under which scale of measurement?

11.7 TYPES OF DATA

Data play the role of raw material for any statistical investigation and defined in a single sentence as

“The values of different objects collected in a survey or recorded values of an experiment over a time period taken together constitute what we call data in Statistics”

Each value in the data is known as observation. Statistical data based on the characteristic, nature of the characteristic, level of measurement, time and ways of obtaining it may be classified as follows:

- Quantitative data
 - Qualitative data
- } based on the characteristic
-
- Discrete data
 - Continuous data
- } based on nature of the characteristic
-
- Nominal data
 - Ordinal data
 - Interval data
 - Ratio data
- } based on level of measurement
-
- Time Series data
 - Cross Sectional data
- } based on time component
-
- Primary data
 - Secondary data
- } based on the ways of obtaining the data

Let us discuss different types of data one by one:

Quantitative Data

As the name quantitative itself suggests that it is related to the quantity. In fact, data are said to be quantitative data if a numerical quantity (which exactly measure the characteristic under study) is associated with each observation.

Generally, interval or ratio scales are used as a measurement of scale in case of quantitative data. Data based on the following characteristics generally gives quantitative type of data. Such as weight, height, ages, length, area, volume, money, temperature, humidity, size, etc.

For example,

- (i) Weights in kilogram (say) of students of a class.
- (ii) Height in centimeter (say) of the candidates appearing in a direct recruitment of Indian army organised by a particular cantonment.
- (iii) Age of the females at the time of marriage celebrated over a period of week in Delhi.
- (iv) Length (in cm) of different tables in a showroom of furniture.

Here, is an exercise for you

E 3 Provide an example based on each of the following characteristic:

- (i) Area (ii) Volume (iii) Money (iv) Temperature (v) Humidity (vi) Size
-

Qualitative Data

As the name qualitative itself suggests that it is related to the quality of an object/thing. It is obvious that quality cannot be measured numerically in exact terms. Thus, if the characteristic/attribute under study is such that it is measured only on the bases of presence or absence then the data thus obtained is known as qualitative data.

Generally nominal and ordinal scales are used as a measurement of scale in case of qualitative data. Data based on the following characteristics generally gives qualitative data. Such as gender, marital status, qualification, colour, religion, satisfaction, types of trees, beauty, honesty, etc.

For example,

- (i) If the characteristic under study is gender then objects can be divided into two categories, male and female.
- (ii) If the characteristic under study is marital status then objects can be divided into four categories married, unmarried, divorcee, widower.
- (iii) If the characteristic under study is qualification (say) 'matriculation' then objects can be divided into two categories as 'Matriculation passed' and 'not passed'.
- (iv) If the characteristic under study is 'colour' then the objects can be divided into a number of categories Violet, Indigo, Blue, Green, Yellow, Orange and Red.

Here, is an exercise for you.

E 4 Give an example based on the following characteristic:

- (i) Religion (ii) Satisfaction
-

Discrete Data

If the nature of the characteristic under study is such that values of observations may be at most countable between two certain limits then corresponding data are known as discrete data (concept of countability have already been discussed in Sec 2.6 of Unit 2 of this course).

For example,

- (i) Number of books on the shelf of an elmira in a library form discrete data. Because number of books may be 0 or 1 or 2 or 3,... But number of books cannot take any real values such as 0.8, 1.32, 1.53245, etc.
- (ii) If there are 30 students in a class, then number of students presents in a lecture forms discrete data. Because number of present students may be 1 or 2 or 3 or 4 or...or 30. But number of present students cannot take any real values between 0 and 30 such as 1.8675, 22.56, 29.95, etc.
- (iii) Number of children in a family in a locality forms discrete data. Because number of children in a family may be 0 or 1 or 2 or 3 or 4 or.... But number of children cannot take any real values such as 2.3, 3.75, etc.
- (iv) Number of mistakes on a particular page of a book. Obviously number of mistakes may be 0 or 1 or 2 or 3.... But cannot be 6.74, 3.9832, etc.

Continuous Data

Data are said to be continuous if the measurement of the observations of a characteristic under study may be any real value between two certain limits.

For example,

- (i) Data obtained by measuring the heights of the students of a class of say 30 students form continuous data, because if minimum and maximum heights are 152cm and 175 cm then heights of the students may take any possible values between 152 cm and 175 cm. For example, it may be 152.2375 cm, 160.31326... cm, etc.
- (ii) Data obtained by measuring weights of the students of a class also form continuous data because weights of students may be 48.25796...kg, 50.275kg, 42.314314314...kg, etc.

E 5) Identify whether the data are discrete or continuous in the following cases:

- (i) Number of people present in a party.
- (ii) Length of leafs of a plant.
- (iii) Lifetime in hours of an electrical bulb.
- (iv) Number of cars standing in a showroom over a period of 7 days.
- (v) Number of patients visited to a hospital on a particular day.

Nominal Data

Data collected using nominal scale is called nominal data.

Similarly, data collected using ordinal scale, interval scale and ratio scale are called **ordinal data**, **interval data** and **ratio data** respectively. These scales of measurement have already been discussed in detail in Sec. 11.6.

Time Series Data

Collection of data is done to solve a purpose in hand. The purpose may have its connection with time, geographical location or both. If the purpose of data collection has its connection with time then it is known as time series data. That is, in time series data, time is one of the main variables and the data collected usually at regular interval of time related to the characteristic(s) under study show how characteristic(s) changes over the time.

For example, quarterly profit of a company for last eight quarters, yearly production of a crop in India for last six years, yearly expenditure of a family on different items for last five years, weekly rate of inflation for last ten weeks, etc. all form time series data.

Yearly expenditures (in Rs) for a family on different items from 2006 to 2010 are given in the following table.

Year	Food	Education	Rent	Miscellaneous	Total
2006	40000	10000	36000	20000	106000
2007	45000	12000	40000	28000	125000
2008	54000	15000	45000	32000	146000
2009	60000	20000	50000	40000	170000
2010	70000	30000	55000	45000	200000

Data given in above table is nothing but time series data.

Note 2: If the purpose of the data collection has its connection with geographical location then it is known as **Spatial Data**.

For example,

- (i) Price of petrol in Delhi, Haryana, Punjab, Chandigarh at a particular time.
- (ii) Number of runs scored by a batsman in different matches in a one day series in different stadiums.

Note 3: If the purpose of the data collection has its connection with both time and geographical location then it is known as **Spacio Temporal Data**.

For example, data related to population of different states in India in 2001 and 2011 will be Spacio Temporal Data.

Note 4: In time series data, spatial data and spacio temporal data we see that concept of frequency have no significance and hence known as **non-frequency**

data. For instance, in the example discussed in case of time series data, expenditure of Rs 40000 on food in 2006 is itself important, here its frequency say 3 (repeated three times) does not make any sense.

Note 5: Now consider the case of marks of 40 students in a class out of 10 (say). Here we note that there may be more than one student who score same marks in the test. Suppose out of 40 students 5 score 10 out of 10, it means marks 10 have frequency 5. This type of data where frequency is meaningful is known as **frequency data**.

Cross Sectional Data

Sometimes we are interested to know that how a characteristic (such as income or expenditure, population, votes in an election, etc.) under study at one point in time is distributed over different subjects (such as families, countries, political parties, etc.). This type of data which is collected at one point in time is known as cross sectional data.

For example, annual income of different families of a locality, survey of consumer's expenditure conducted by a research scholar, opinion polls conducted by an agency, salaries of all employees of an institute, etc.

Note 6:

- (i) If you are interested to know the changes in a characteristic say expenditure of a family over a period of time then you have to use time series data.
- (ii) If you are interested to know the changes in a characteristic say expenditure of different families at single point in time you have to use cross sectional data.

Primary Data

Data which are collected by an investigator or agency or institution for a specific purpose and these people are first to use these data, are called primary data. That is, these data are originally collected by these people and they are first to use these data. Primary data have been discussed in Sec. 12.2 of next unit (i.e. UNIT 12) of this course in detail.

For example, suppose a research scholar wants to know the mean age of students of M.Sc. Chemistry of a particular university. If he collects data related to the age of each student of M.Sc. Chemistry of that particular university by contacting each student personally then data so obtained by the research scholar is an example of primary data for the same research scholar.

Secondary Data

Data obtained/gathered by an investigator or agency or institution from a source which already exists, are called secondary data. That is, these data were originally collected by an investigator or agency or institution and has been used by them at least once. And now, these data are going to be used at least second time. Secondary data have been discussed in Sec. 12.3 of next unit (i.e. UNIT 12) of this course in detail.

For example, consider the same example as discussed in case of primary data. If the research scholar collects the ages of the students from the record of that particular university, then the data thus obtained is an example of secondary data. Note that, in both the cases data remain the same, only way of collecting the data differs.

11.8 SUMMARY

In this unit, we covered following topics:

- 1) Origin and development of statistics.
 - 2) Definitions of statistics by different authors.
 - 3) Scope and uses of statistics.
 - 4) Limitations of statistics.
 - 5) Different measurement scales and types of data.
 - 6) Frequency and non frequency data.
-

11.9 SOLUTIONS/ANSWERS

- E 1** (i) Ratio scale is considered as the best measurement scale so far as the criteria of information provide are concerned because all the four operations counting, order, distance and absolute zero are defined on the observations.
- (ii) Measurement of height, weight, age requires absolute zero and only ratio scale has absolute zero. So, appropriate scale of measurement for height, weight, age is ratio scale.
- (iii) Allotment of license plates to the different cars comes under nominal scale measurement, because license plates categories the cars or license plates only provide unique names to the cars. Further, the car remains the same if some other registration number is provided to it.
- (iv) Characteristic of equal distance between any two observations is maintained by two scales of measurements interval and ratio scales. For example, distance between temperatures of 18K and 13K is same as distance between 100K and 105K.
- E 2**) Blood group just divides the objects/things into four categories named as A, B, AB, O. So it comes under nominal scale.
- E 3**) Answers are not unique. There are a number of examples for each part, here one answer is provided for each part.
- (i) Area of each state (in km^2) of India.
- (ii) Volume of different buckets available at a particular shop.
- (iii) Income of each family over a period of one year in a particular locality.
- (iv) Highest or lowest temperature of a place over a period of 50 days.
- (v) Level of humidity of a particular place at each hour of a particular day.
- (vi) Size of different shoes present at a particular showroom on a specified day.
- E 4**) (i) If the characteristic under study is 'religion' then the objects can be divided into five categories Hindu, Muslim, Sikh, Isai, and others.
- (ii) If the characteristic under study is 'satisfaction' then the objects can be divided into five categories (Likert scale) as shown on the next page:

Highly satisfied	Satisfied	Neutral	Dissatisfied	Highly dissatisfied
5	4	3	2	1
OR				
2	1	0	-1	-2

- E 5** (i) Number of people present in a party may be 2 or 3 or 4 or 5 or 6 and so on, but cannot be 2.3, 4.375, 9.62875, etc.
 $\therefore$ it is an example of discrete of data.
- (ii) Lengths of leafs of a plant form continuous data because lengths of leafs may be any real number, e.g. 3.75 cm, 2.959595... cm, etc.
- (iii) It is an example of continuous data because life time of an electrical bulb may be any possible fraction of time. For example, 8 hours 8.76 hours, 100.25796 hours, 0.25 hours, etc.
- (iv) It is an example of discrete data because number of cars may be 1 or 2 or 3 or 4 or 5 or 6 or 7 and so on, but cannot be 2.87, 5.687, etc.
- (v) It is an example of discrete data because number of patients visited to a hospital on a particular day may be 0 or 1 or 2 or 3 or 4 and so on, but cannot be 2.8, 10.357, 7.856, etc.

UNIT 12 COLLECTION AND SCRUTINY OF DATA

Structure

- 12.1 Introduction
 - Objectives
- 12.2 Primary Data
- 12.3 Secondary Data
- 12.4 Scrutiny of Primary Data
- 12.5 Preparation of Questionnaire
- 12.6 Summary
- 12.7 Solutions/Answers

12.1 INTRODUCTION

Recall the definition of statistics, given in Sec. 11.3 of previous unit “Statistics is a branch of science which deals with collection, classification, tabulation, analysis and interpretation of data”.

In the above definition out of the five successive steps (i.e. collection, classification, tabulation, analysis and interpretation) used in any statistical investigation, the first step is collection of data, and last step is interpretation of data which ultimately depends on the collection of data. So, if collection of data is not done carefully and sincerely then goal of the statistical investigation is not achieved or objective(s) of the statistical investigation is/are not fulfilled or final outcomes of the investigation will not be satisfactory.

Therefore, it becomes very important to focus on the collection of data in detail. In this unit two main types of collection of data namely primary and secondary will be discussed in detail. Also we shall discuss their different methods of collection with their merits and demerits. Scrutiny of data is also discussed in this unit. Finally we conclude this unit by throwing some light on preparing a questionnaire.

Objectives

After completing this unit, you should be able to:

- define primary data and get familiar with the different methods of collection of primary data;
- define secondary data and get familiar with the different sources of collection of secondary data;
- get an idea about the scrutiny of data; and
- know some important points related to the preparation of a questionnaire.

12.2 PRIMARY DATA

In previous section of this unit, we have seen why collection of data is important in any statistical investigation. In fact there are mainly two types of data namely Primary Data and Secondary Data. Both have their own importance. In a statistical investigation which of the two is to be used, totally

depends upon many factors such as nature of problem, purpose of investigation, time period in which conclusion required and finally another important factor is availability of money/resources.

In this section we will discuss concept of primary data, different methods of collection of primary data with their merits and demerits.

Let us first formally define primary data and secondary data. Here we will just define secondary data with an example. Next section is devoted to discuss secondary data in detail.

Primary Data

Data which are collected by an investigator or agency or institution for a specific purpose and these people are first to use these data, are called primary data. That is, these data are originally collected by these people and they are first to use these data.

For example, suppose a research scholar wants to know the mean age of students of M.Sc. Chemistry of a particular university. If he collects the data related to the age of each student of M.Sc. Chemistry of that particular university by contacting each student personally. The data so obtained by the research scholar is an example of primary data for the same research scholar.

Secondary Data

The data obtained/gathered by an investigator or agency or institution from a source which already exists, are called secondary data. That is, these data were originally collected by an investigator or agency or institution and have been used by them at least once. And now, these data are going to be used at least second time.

For example, consider the same example as discussed in case of primary data. If the research scholar collects the ages of the students from the record of that particular university, then the data thus obtained is an example of secondary data. Note that, in both the cases data remain the same, only way of collecting these data differs. Our aim of just defining secondary data in this section is over, because we just want to click the idea how primary and secondary data differ from each other and are discussed into two different sections.

Now we move towards the aim of this section, which is to focus on primary data. We have already define primary data. There are a number of methods of collection of primary data depending upon many factors such as geographical area of the field, money available, time period, accuracy needed, literacy of the respondents/informants, etc.

Here we will discuss only following commonly used methods.

- (1) Direct Personal Investigation Method
- (2) Telephone Method
- (3) Indirect Oral Interviews Method
- (4) Local Correspondents Method
- (5) Mailed Questionnaires Method
- (6) Schedules Method

Let us discuss these methods one by one with some examples, merits and demerits.

(1) Direct Personal Investigation Method

In this method, the investigator personally contacts the informants and collects the related data through face to face interviews of the informants. Due to face to face meeting of investigator and informants data collected under this method has maximum degree of accuracy. But the degree of accuracy depends upon the sincerity, honesty, unbiasedness and expertness of the investigator, because it is the investigator, who ultimately gives the final shape to the information provided by the informants. This method of collection of primary data is recommended/suitable when field of enquiry is small, secrecy related to data is need, high accuracy is required and time as well as money is sufficient.

Following are some **merits** of this method:

- (i) It is simple to apply.
- (ii) It is convenient for both investigator as well as informants.
- (iii) Data have high degree of accuracy.
- (iv) Data are homogenous in nature as there is only one investigator.
- (v) Due to the presence of both investigator and informants, there is flexibility to clear any doubt or some other modification are possible according to physibility.
- (vi) Confidential information can also be obtained.

Having so many plus points, this method is not free from the demerits.

Following are some **demerits** of this method:

- (i) It is time consuming and costly.
- (ii) It is not suitable when area of investigation is large.
- (iii) It suffers from the biasness of the investigator.
- (iv) Data may be misleading if the investigator do not collect the data sincerely and honestly.
- (v) If the investigator does not have expertise, data again may be misleading.

(2) Telephone Method

In the direct personal investigation method investigator has to personally contact with the informants, but now a day's telephone is very good communication tool. If the information of the interest is collected through telephone then data so obtained come under telephone method.

Some merits and demerits of this method are listed below:

Merits

- (i) All merits of method 1 are also the merits of this method. Some additional merits are given below.
- (ii) It is easy to apply compare to method 1.
- (iii) It is time and cost saving method compare to method 1.
- (iv) It is suitable when area of investigation is large compare to method 1.

Demerits

- (i) Information related to those informants who do not have telephone and those who actually have telephone but their number are not in the list will not be included.
- (ii) This method also suffers from the demerits (iii) to (v) listed in method 1.

(3) Indirect Oral Interviews Method

In this method, investigator does not meet to the actual informants directly, but the related information is collected/obtained from other persons who are supposed to have the required type of information. The informants who provide the information about actual informants are known as 'witnesses'. The success of this method mainly depends upon the skill and experience of the investigator. Because it depends upon the way, sequence and trick of questions prepared and asked by the investigator from the informants. It also depends on the behavior and how much he/she is capable to create confidence in the informants. This method of collection of primary data is generally adopted to obtain the information related to the cases such as

- (i) murder
- (ii) theft
- (iii) in the cases where a person hesitates to provide correct information.

For example, if a researcher wants to collect the data on the smoking habit of students of a particular class of a college then it may happen that actual informants does not provide correct information. So, data can be collected with the help of class met or college met or from the neighbours.

Following are some **merits** of this method:

- (i) As far as time and money is concerned, it is economical compare to direct personal investigation.
- (ii) This method is easy to use, even if the area of investigation is large.
- (iii) Informants being a third person, so it is free from the biasness of both actual informants and investigator.

Following are some **demerits** of this method:

- (i) Here data totally depends upon the information provided by the third person. So, data suffer from the biasness of the third person.
- (ii) If the investigator is not experienced and well behaved, then data will not be reliable.
- (iii) It also depends on the honesty of the investigator.

(4) Local Correspondents Method

In this method, first of all some local correspondents or agents are appointed by the investigator or agency or institute to collect data. These local agents directly meet to the informants and collect data related to the required purpose in hand. Data collected by these local agents have high degree of accuracy, because they are familiar with the local language, traditions, general behaviour of the people, etc. of that particular area.

This method of collection of primary data is suitable if

- Area of investigation is large, e.g. news channels have their reporters throughout India.
- Time period in which information is needed is very short. For example, news related to the happening of any special incident can be easily seen on the news channels in very short time period after its happening.
- Information is needed on regular basis. For example, news are provided daily using this method.

Below some **merits** of this method are listed.

- (i) It is very economical in terms of money, time and man power.
- (ii) Time period in which information are provided by this method is very short.
- (iii) Large area and heterogeneity of the informants can easily handled by this method.
- (iv) This method can provide regular basis information.

Having so many merits, this method also has some **demerits**, listed below.

- (i) Data suffer from the biasness of the local agents.
- (ii) If local agents do not perform their duties with honesty and sincerity, then data will not be reliable.

(5) Mailed Questionnaires Method

In this method, first of all a list of questions related to the information required by the investigator is prepared. At the time of preparing the list of questions following points should keep in mind.

- Number of questions should not be too many.
- Each question should be related directly or indirectly to the objective(s) of the investigation.
- Each question should be clear.
- Generally objective type questions should be used, but if necessary multiple choice and open-ended questions can also be used.
- Language should be simple and effective.

Open ended questions are defined on page number 87 of this unit.

After preparing the 'final list' of questions known as questionnaire, it is sent through mail to the informants. With this questionnaire a covering letter in which it is requested to the informants that please sent it back after completing and a brief introduction about the objective of the investigation is also attached.

This method is suitable when the informants are literate and area of investigation is large.

Following are some **merits** of this method:

- (i) This method is very useful if area of investigation is large.
- (ii) This method is very economical as far as time, money and labour is concerned.
- (iii) This method provides very good results when informants are literate.
- (iv) Informants have enough time to think and give correct information. Thus data obtained by this method have high degree of accuracy.
- (v) Biasness of the investigator is not involved.

Having being very economical and suitable for large area, following are some **demerits** of this method.

- (i) This method fails if informants are illiterate.
- (ii) Generally percentage of responses are very less because people take less interest in filling up the questionnaires.
- (iii) If informants do not fill up the questionnaire sincerely or honestly then biasness of the informants may mislead the investigator.

(6) Schedules Method

In this method first of all a list of questions based on the information to be required is prepared like mailed questionnaires method, known as schedule. After doing this whole area of investigation is divided into sub areas. Then a number of people are appointed to collect the information directly from the informants. The appointed people are known as enumerators. The exact figure of the enumerator to be appointed depends upon the area of investigation and the amount of information to be required. These enumerators meet with the informants face to face and after giving a brief introduction about the objectives of the investigation they ask the answer of each question listed in the schedule. The answers provided by the informants are filled up in the schedule by enumerator themselves. This is one of the main differences of the two methods namely mailed questionnaires method and schedules method. In mailed questionnaires method answers are filled up by the informants themselves while in schedules method this job is done by enumerators. This characteristic of the schedules method make it superior to mailed questionnaire method in the case when informants are illiterate or semi-illiterate. Here enumerators have to meet directly with the informants, therefore it becomes important that enumerators should be well behaved, honest sincere and unbiased in nature.

This method is suitable when area of investigation is large and informants are illiterate or semi-illiterate.

Following are some **merits** of this method:

- (i) This method is suitable when informants are illiterate or semi-illiterate.
- (ii) It is application whatever large the area is.
- (iii) After every ten years census data is collected by using this method in India.
- (iv) Data are least affected by the bias of the enumerators and investigators.
- (v) Since the information is directly obtained from the informants, so data collected by this method are more reliable and have a high degree of accuracy.
- (vi) Because enumerator is present in front of the informants so if informants have any doubt, he/she can easily clear it from the enumerator.

There are so many merits of this method, even though it is not free from the demerits. Following are some **demerits** of this method:

- (i) It is very time-consuming and large amount of money is needed.
- (ii) Because a large number of enumerators have to be appointed, so it becomes too difficult to get all well trained and experienced persons.
- (iii) Training is also needed to the enumerators.
- (iv) Even after providing the training, some enumerators may not do their responsibilities sincerely, honestly and efficiently.
- (v) Accuracy of the data will suffer if enumerator is bias or not devoted.

12.3 SECONDARY DATA

Discussion in the previous section shows that collection of primary data requires lot of time, money, manpower, etc. But sometimes some or all these resources are not sufficient to go for the collection of primary data. Also, in some situations it may not be feasible to collect primary data easily. To

overcome these types of difficulties, there is another way of collecting data known as secondary data. The data obtained/gathered by an investigator or agency or institution from a source which already exists, are called secondary data. That is, these data were originally collected by an investigator or agency or institution and has been used by them at least once and now, these are going to be used at least second time. Already existed data in different sources may be in published or unpublished form. So sources of secondary data can broadly be classified under the following two heads.

- (1) Published Sources, and
- (2) Unpublished Sources.

Let us discuss these two main types of sources one by one:

(1) Published Sources

When an institution or organisation publishes its own collected data (primary data) in public domain either in printed form or in electronic form then these data are said to be secondary data in published form and the source where these data are available is known as published source of the secondary data of the corresponding institution or organisation. Some of the published sources of secondary data are given below:

(a) International Publications

There are many international organisations including the governments' organisations of different countries which collect data regarding the characteristics under their objectives and publish these data. Some of these organisations or publications are:

- (i) Annual Abstract of Statistics (United Kingdom)
- (ii) Annual Reports of International Labour Organisation (ILO)
- (iii) World Health Organisation (WHO)
- (iv) World Bank
- (v) UNESCO Institute for Statistics, etc.

(b) Government Publications in India

There are number of government organisations or bodies at national level or state level which collect and publish data on different characteristics of interest. Data published by these bodies play a significant role in the sources of secondary data. Some of these organisations or bodies or publications are:

- (i) Office of Registrar General of India
- (ii) Central Statistical Organisation (CSO)
- (iii) National Sample Survey Organisation (NSSO)
- (iv) Reserve Bank of India Bulletin
- (v) Directorate of Economics and Statistics (DES), etc.

(c) Published Reports of Commissions and Committees

From time to time, Central government and State governments constitute or appoint some commissions and committees to get a road map on some issues of interest. Some of these (with time of appointment in brackets) are listed below:

- (i) Sarkaria Commission (1983)
- (ii) Sixth Pay Commission (2006)

- (iii) Shunglu Committee (2010)
- (iv) Kalodkar Committee (2010), etc.

(d) Research Publications

Published research work of research scholars in different journals throughout the world is another major source of published data. Most of these research works are carried out in universities or other research institutes. Some of the journals related to the subject Statistics are:

- (i) International Journal of Probability and Statistics
- (ii) American Journal of Mathematics and Statistics
- (iii) Sankhya
- (iv) Journal of Statistics & Management Systems
- (v) Journal of Statistical Research, etc.

(e) Reports of Trade and Industry Associations

In India, there are number of trade and industry associations whose published reports are also secondary sources of published data. Some of these associations are given below:

- (i) All India Association of Industries
- (ii) The Indian Cotton Mill Association
- (iii) All India Resort Development Association
- (iv) All India Biotech Association, etc.

(f) Published Printed Sources

Books, directories, newspapers, magazines, etc. are published printed sources of secondary data. Some of these are:

- (i) Statistical Year Book, India
- (ii) Business Line
- (iii) The Economic Times
- (iv) Business Standard
- (v) Business Today, etc.

(g) Published Electronic Sources

Today, internet is a huge source of published data because most of the published material is available on internet. Information of interest which is available on internet can easily be obtained in very short time. Some of the electronic sources of secondary data are listed below:

- (i) Online databases
- (ii) e-books
- (iii) e-journals
- (iv) Websites of different institutes or organisations or agencies, etc.

(2) Unpublished Sources

Collected information in term of data or data observed through own experience by an individual or by an organisation which is in unpublished form is known as unpublished source of secondary data. Some of the sources of unpublished secondary data are given on the next page:

- (i) Records and statistics maintained by different institutions or organisations whether they are government or non-government
- (ii) Unpublished projects works, field works or some other research related works submitted by students in their corresponding institutes
- (iii) Records of Central Bureau of Investigation
- (iv) Personal diaries, etc.

After discussing sources of secondary data, natural questions which may arise in your mind are:

- (1) What are the precautions one should use before using secondary data?
- (2) What are the advantages of secondary data?
- (3) What are the limitations of secondary data?

Let us address these questions one by one.

(1) Precautions to be taken before using Secondary Data

Every investigation in hand has some specific objectives and data are collected keeping these objectives in view. So, secondary data which we are planning to use in our investigation may be collected for some different objectives. Therefore some precautions which are necessary before using the secondary data in our investigation are given below.

(i) Reliability of Data

Reliability of data is judged by:

- Reliability and experience of the investigator or institution for collecting data.
- Reliability of the source(s) from where data were collected.
- Whether the proper methods of collecting data were used. Whether the sample size was proper if sample technique was used in data collection?
- Whether data collected in normal times? That is, whether data were free from periods such as floods, famines, earthquakes, etc?
- Whether data were free from the biasness of the collecting investigator or institution?

If above criteria are met, we assume, generally, that reliability of data is all right.

(ii) Suitability of Data

Suitability of data is judged by:

- Comparing the nature, scope and objectives of the investigation at hand to the original one.
- Comparing definitions of different terms and units used in original investigation to the one at hand. For example, if word “large” is used as a measurement unit then what figure it represents such as 100-200 or 500-1000 or 10000 and above, etc.
- Checking uniformity of different terms or units, i.e. we have to check whether the definition of different terms and units is maintained throughout or not.

(iii) Adequacy of Data

Adequacy of data is judged by:

- Comparing geographical area covered and to be covered in original and one at hand investigations respectively. If variation between the areas of two investigations is large then data will not be adequate. For example, data collected for the purpose of estimating per person income of a state say Delhi cannot be used to estimate the per person income in India.
- Similar argument applies on time factor also. For example, if price of a commodity are available for a particular month of a year then on the basis of prices of one particular month one cannot accurately estimate the price of that commodity for whole year.

(2) Advantages of Secondary Data over Primary Data

Use of secondary data in an investigation has following advantages:

- It saves lot of time and money.
- It is easy to use.
- In some investigations primary data cannot be collected.
- The only source in case of historical documents.
- Longitudinal study can be possible.
- Secondary data complements primary data in many ways such as better understanding of the problem(s) in terms of what are gaps and deficiencies in the earlier investigation(s) which need to improve.

(3) Limitations of Secondary Data

Some of the limitations of the secondary data are given below:

- It is very difficult to get secondary data which is appropriate for all objectives of our investigation at hand.
- It is very difficult to get secondary data which meet all the requirements like reliability, suitability, adequacy and accuracy.
- Secondary data are generally not available for all types of investigations.
- Data may be beyond our reach.
- Available data may be out dated.

Example 1: Giving reason(s) in each of the following cases, specify whether data are primary or secondary?

- (i) A Television channel telecasts the published survey report of an agency XYZ (say) based on the data collected by the agency before the general election in India to know the opinion of the people about casting their votes.
- (ii) Data source in part (i) used by agency XYZ.
- (iii) An Industrial Statistics student estimate the average life of electric bulbs of a company in which he/she works by observing the lives of a random sample of 100 bulbs.
- (iv) A Bio-Statistics student collected data from the records of 10 hospitals of a state in order to conduct his/her study, whether smoking and T.B. are associated?

Solution:

- (i) Secondary because TV channel used the data which already existed in a published form.

- (ii) Primary source because agency itself collected data from the field.
- (iii) Data used by the student were primary as data observed by him/her were original in character.
- (iv) Student collected data from already existed sources (i.e. records of the hospitals) so data used by the student were secondary.

Now you can try the following exercises.

E 1) Give reasons in each case whether the data are primary or secondary?

- i) 2011 census data published by Office of Registrar General of India and to be used by itself.
- ii) 2011 census data to be used by a demographer in his study.

E 2) What are the differences between primary and secondary data?

12.4 SCRUTINY OF PRIMARY DATA

At the time of collecting primary data, information provided by the informants is recorded either in the form of questionnaire or schedule or by other means. Before tabulating primary data, it is necessary to scrutinise the questionnaires or schedules or other means used while collecting primary data to maintain

- (i) Completeness of data
- (ii) Consistency of data
- (iii) Accuracy of data, and
- (iv) Homogeneity of data

(i) Completeness of Data

We may find some questionnaires or schedules which are incomplete. Incomplete questionnaires or schedules, if possible, should be completed by revisiting to the informants otherwise we should reject them. Incomplete questionnaires or schedules should not be entertained.

(ii) Consistency of Data

Sometimes, the information provided by the informants either does not make any sense or may contradicts some other information. For example,

- If an informant puts his age 40 years while age of his son as 32 years. This information does not make any sense.
- If total age and date of birth provided by the informant do not match then these two types of information contradict each other. Again either these information should be corrected by revisiting to the informants or we should reject them.

(iii) Accuracy of Data

To maintain 100% accuracy is not an easy task. We can only do correction(s) related to certain figures by checking their sum, subtraction, etc. But we have no control if some informants provide wrong information. For example, some informants may provide wrong figure about their annual income.

(iv) Homogeneity of Data

Keeping the homogeneity of data is also an important part of editing/scrutiny of data. Here we have to check that the units of measurements used by the informants are same or different. For example,

- Some informants may put their heights in centimeters while others may put in inches.
- Some informants may put their monthly income while others may put yearly income, etc.

To maintain homogeneity of data, we have to convert all the informations provided by all the informants in the same unit(s).

12.5 PREPARATION OF QUESTIONNAIRE

We have already become familiar with the words questionnaire and schedule during our discussion of different methods of collecting primary data. It has also been mentioned there that the difference between questionnaire and schedule is that questionnaires are filled by informants themselves while schedules are filled by the enumerators/investigators. Both are formalised set of the questions for obtaining information from the informants. Questionnaire plays the role of an instrument for the investigator to get the information from the informants. So, it becomes necessary that questionnaire or schedule should be well drafted.

This whole section is devoted to discuss various aspects of the questionnaire. There are no hard and fast rules for preparing a questionnaire or schedule. Actually designing of a good questionnaire is an art not a science. Development of this art requires skill, experience, dedication and hard work. Following are some guidelines to improve the quality of the questionnaire.

(1) Identification of Objectives and Target Population

First of all, we identify objectives of the investigation at hand. Once our objectives are clear we can accordingly set the type of information needed and target population. Target population is very important because questions appropriate for one group may not be appropriate for other, e.g. questions appropriate for university students may not be appropriate for farmers.

(2) Selection of Method

After setting the information needed and the target group, we have to decide one of the methods of primary data collection from amongst those studied in Sec 12.2. The method used for the data collection also affects the questionnaire. For example, in direct personal interviews method, investigator personally contacts the informants and hence lengthy, complex and varied questions can be asked whereas in the telephone method we have to ask simple and short questions because on telephone lengthy questions may confuse the informants.

(3) Content of Individual Questions

Keep only those questions in the questionnaire which directly or indirectly contribute to the information needed or serve some specific purpose. Unnecessary questions should not be included in the questionnaire.

(4) Wording of the Questions

Whatever method of data collection is used, one should maintain the following features in the questionnaire:

- Language should be simple, i.e., use of technical terms should be avoided unless informants are technically trained.
- Questions should be short, simple and easy to understand.

- Questions should have clear meanings, i.e. words having multiple meaning should be avoided. For example, words like poor, rich, etc should be avoided because meaning of these words vary from person to person.
- Questionnaire should be self explanatory. That is, if a particular question needs some clarification, it should be provided with the help of footnote.
- Questions which are sensitive and personal in nature should be avoided.

(5) Sequence of the Questions

Arrangement of the questions in the questionnaire should be logical and all questions which are related to a particular topic should be asked before beginning of a new topic.

(6) Types of Questions

Main types of the questions which are generally used in a questionnaire are given below:

- **Dichotomous Questions:** Questions having only two alternatives like yes or no, agree or disagree, etc. are known as dichotomous questions. For example, Have you visited USA?
- **Multiple Choice Questions:** Questions having more than two alternatives are known as multiple choice questions. For example, which soap do you use for bath?
 - (i) Lux
 - (ii) Dettol
 - (iii) Dove
 - (iv) Kanti
 - (v) Other
- **Specific Information Questions:** Questions which are used to know some specific information from the informants are known as specific information questions. For example, which is your favourite soft drink?
- **Open Ended Questions:** Those questions which provide freedom to the informants to give their opinion are known as open ended questions. For example, what are your hobbies?
- **Scale Questions:** Consider following question:
Are you satisfied with the service provided by your mobile company?
Options for this type of questions are based on Likert scale as given below (Refer Unit 11):

Highly satisfied	Satisfied	Neutral	Dissatisfied	Highly dissatisfied
5	4	3	2	1
OR				
2	1	0	-1	-2

Such types of questions based on Likert Scale or some other scale are known as scale questions.

Any of the above types of questions can be used in a questionnaire depending on the suitability of the information required.

(7) Size of the Questionnaire

Generally people do not like to answer lengthy questionnaire, so we should try to make the number of questions in the questionnaire the smallest possible. Usually, the number of questions in a questionnaire lies between 15-25. But less than 15 and more than 25 questions may be used depending on the type of information needed. For example, the number of questions in the questionnaire of 2011 census data collected by Office of Registrar General of India was 21.

(8) Cross Questions

Some cross questions can also be put in a questionnaire to check the accuracy of the information provided by the informants. For example, we can put two questions:

- What is your date of birth?
- What is your age as on date?

These two questions put a cross check on each other.

(9) Format, Layout and Quality of paper of a Questionnaire

To make the questionnaire eye catching and to give it professional appearance, one has to focus on:

- Proper line spacing
- Size and colour of headings and sub-headings
- Quality of paper used.

(10) Pretesting of Questionnaire

This is the last step to give a final shape to a questionnaire. Pretesting means “testing the questionnaire on a small group of informants to get some valuable suggestion(s) from the informants to improve the questionnaire”. After incorporating valuable suggestions of the informants, questionnaire becomes ready to use.

This final questionnaire is sent to informants with a covering letter, stating briefly the objectives of the enquiry and with the request that please fill up the questionnaire and return it to the said address:

Let us prepare a questionnaire for the following example:

Example 2: A market research scholar wants to study the behaviour of Indian consumers regarding food and grocery items. Prepare a questionnaire for this purpose.

Solution: Questionnaire for this purpose is given on the next page:

Dear Informants,

I am conducting a market research study to explore the store choice behaviour of Indian consumers regarding food and grocery items. Information provided by you in the following questionnaire will be used only for academic purpose. So, you are requested to provide the feedback to the best of your knowledge and experience. Questionnaire is divided into three parts.

PART A

Put tick (✓) mark in the appropriate box.

1. **Gender:** ☐ Male ☐ Female
2. **Marital Status:** ☐ Single ☐ Married
3. **In which age (in years) group you belong?**
☐ 0 – 20 ☐ 20 – 30 ☐ 30 – 40 ☐ 40 – 50 ☐ above 50
4. **Education Level**
☐ Below 10th ☐ 10th ☐ 10 + 2
☐ Graduation ☐ PG and above
5. **Monthly Income in Rupees**
☐ Below 10000 ☐ 10000 – 20000 ☐ 20000 – 40000
☐ 40000 – 60000 ☐ Above 60000
6. **Employment Status**
☐ Employed ☐ Unemployed ☐ Business person
☐ House Wife ☐ Other (Please specify)
7. **Store where I often buy my most of food and grocery items**
☐ Kirana Store ☐ Kirana Store with self service
☐ Chain Store – Individual Outlet (Please specify)
☐ Chain Store – Located in Mall (Please specify)

PART B

8. **Please put a tick (✓) mark according to your importance level for each of the following factors while selecting a food and grocery retail store.**

Factor	Not at all Important	Not Important	Sometimes Important	Important	Extremely Important
Quality of product					
Variety of products					
Proper product display					
Price					
Spaciousness of the store					
Distance from residence					
Knowledge of sales person					
Behaviours of the workers of the store					
Location of store					
Music					
Air conditioning facility					
Cleanliness of the store					
Store lighting					

**Matrices, Determinants
and Collection of Data**

Parking facility					
Payment options (debit card, credit card, cash)					
Checkout time/billing time					
Home delivery facility					
Easy return and exchange					
Complaint handling					
Long opening hours					
Low crowd size					
Clear indication of prices					
Advance communication of discounts and offers					
Facility for membership holders					
Various offers and schemes					

9. Give your response relating to the following statements by putting a tick (✓) mark in the appropriate box. Some of the statements may or not be applicable to you.

Statement	Strongly disagree	Disagree	Neither Agree nor Disagree	Agree	Strongly Agree
I wait for special offers to buy food and grocery products					
I like to compare prices of different stores before buying the items					
I like to try the new grocery outlets					
I frequently look for new products					
I make unplanned visits to stores					
I usually shop from a nearest grocery store					
I tend to buy from a particular grocery store					
I like to shop with my family					
I like to shop alone					

Now you can try the following exercise.

-
- E 3)** Suppose you are a member of the team involved in preparing units of this block. Being a team member you want to get the feedback of the learners related to this material so that at the time of revising the material valuable suggestions of the learners can be incorporated. Prepare a questionnaire to mail your learners.
-

12.6 SUMMARY

In this unit, we have discussed the following topics:

- 1) Primary data and different methods of collection of primary data.
- 2) Secondary data and its major sources.
- 3) Scrutiny of primary data.
- 4) Preparation of questionnaires.

12.7 SOLUTIONS/ANSWERS

- E 1)** (i) 2011 census data will be considered as primary data for Office of Registrar General of India because census data were collected by this office after every 10 years.
- (ii) 2011 census data will be secondary data for the demographer because for him it will be an already existed source.

- E 2)** Difference between primary and secondary data is explained in the following table:

Factor of Difference	Primary Data	Secondary Data
Time	Long time is required for collection	Less time is required for collection
Money	Needs more money	Needs less money
Reliability	More reliable	Less reliable
Suitability	More suitability	Less suitable
Adequacy	More adequacy	Less adequate
Hand	First hand data	Second hand data
Precaution	Needs no extra precautions	Needs many precautions like reliability, suitability, adequacy and accuracy
Manpower	Needs more manpower	Needs less manpower

- E 3)** Answer of this exercise is not unique. One of the possible answers is given below:

Dear Learner,

During the study of the units of this block you may have found certain portions of the material where you faced some difficulty to grasp. We believe that there is always a scope for improvement. So we wish to know your difficulties and valuable suggestions to improve the material. Therefore, we request you to fill out and send us the following questionnaire. If you find that space provided is insufficient, kindly use a separate sheet.

QUESTIONNAIRE

1. Enrolment No

2. Mathematical Background

☐ Up to 10th

☐ Up to 10 + 2

☐ Up to Graduation

☐ Higher than Graduation

3. How many hours did you need for studying the units?

Unit Number	9	10	11	12
Number of hours				

4. Provide your feedback unit wise by putting a tick (✓) mark in the appropriate box based on your experience get at the time of studying the units of this block.

Item	Unit No	Excellent	Very Good	Good	Poor	Very Poor	Give specific example(s), in case of poor and very poor
Unit structure	9	☐	☐	☐	☐	☐	
	10	☐	☐	☐	☐	☐	
	11	☐	☐	☐	☐	☐	
	12	☐	☐	☐	☐	☐	
Way of presenting the content	9	☐	☐	☐	☐	☐	
	10	☐	☐	☐	☐	☐	
	11	☐	☐	☐	☐	☐	
	12	☐	☐	☐	☐	☐	
Conceptual clarity	9	☐	☐	☐	☐	☐	
	10	☐	☐	☐	☐	☐	
	11	☐	☐	☐	☐	☐	
	12	☐	☐	☐	☐	☐	
Language and style used	9	☐	☐	☐	☐	☐	
	10	☐	☐	☐	☐	☐	
	11	☐	☐	☐	☐	☐	
	12	☐	☐	☐	☐	☐	
Examples, exercises used	9	☐	☐	☐	☐	☐	
	10	☐	☐	☐	☐	☐	
	11	☐	☐	☐	☐	☐	
	12	☐	☐	☐	☐	☐	

5. Number of diagrams and flow charts used

Unit 9 ☐ Sufficient ☐ Insufficient

Unit 10 ☐ Sufficient ☐ Insufficient

Unit 11 ☐ Sufficient ☐ Insufficient

Unit 12 ☐ Sufficient ☐ Insufficient

6. Any other comments

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Foundation in
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Statistics

Block

4**PRESENTATION OF DATA**

UNIT 13

Classification and Tabulation of Data	5
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UNIT 14

Diagrammatic Presentation of Data	23
--	-----------

UNIT 15

Graphical Presentation of Data-I	47
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UNIT 16

Graphical Presentation of Data-II	61
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BLOCK 4 PRESENTATION OF DATA

In previous block, we have become familiar with origin, development, definition, importance of statistics and its applications in different areas. We have also discussed the collection of data and preparation of questionnaires to collect information. After collecting the information in term of data, we may like to arrange the collected data in a proper manner because the collected data may be huge in volume. So, here the need for proper arrangement of data arises. In statistical terminology, the proper arrangement of data is known as presentation of data. In this block, we shall try to learn some basic tools to represent the collected data. There are some frequently used tools available for representing data and they may be classified in three basic forms statistical table, diagrams and graphs. This block is devoted to discuss these things. The flow of the block is maintained by the following four units.

Unit 13: After collection of data next step is classification followed by tabulation of data. Unit 13 is devoted to discuss what we mean by classification and tabulation of data.

Unit 14: A pictorial presentation of the tabulated data may be done either with the helps of different kinds of diagrams or by graphs. This unit discusses about some commonly used diagrams, while **Unit 15** and **Unit 16** are devoted to discuss different types of graphical presentation of the data. That is graphs for frequency distributions, graphs for time series data, stem-and-leaf displays and box plots are discussed in Unit 15 and Unit 16.

Notations and Symbols

f : frequency

$N = \sum f$: total of all frequencies

C. I. : class interval

$^{\circ}\text{C}$: degree Celsius

UNIT 13 CLASSIFICATION OF DATA

Structure

13.1 Introduction

Objectives

13.2 Classification of Data

13.3 Tabulation of Data

13.4 Summary

13.5 Solutions/Answers

13.1 INTRODUCTION

In Unit 12 of Block-3 of this course, we have discussed some methods of data collection whether the target population from where the information collected was small or large. After collection of data, next step is to classify the data in such a manner that it becomes ready for proper presentation.

The need for proper presentation arises because of the fact that statistical data in their raw form are almost defy comprehension. When data are presented in easy-to-read form, it can help the reader to acquire knowledge in much shorter period of time and also facilitate statistical analysis.

A statistical table is a presentation of numbers in a logical arrangement, with some brief explanation to show what they are. However, before tabulating data, it is often necessary to first classify them. So, the concept of classification is described in Sec. 13.2 of the unit and that of tabulation is discussed in Sec. 13.3.

Objectives

After studying this unit, you should be able to:

- classify a data set according to the nature of the data;
- construct a discrete frequency distribution for a discrete type of data;
- construct a continuous frequency distribution for a continuous type of data;
- classify the collected data according to the class intervals; and
- arrange the data into a suitable form of a table.

13.2 CLASSIFICATION OF DATA

This unit is a combination of classification and presentation in tabular form of given data. After collection, classification is the next step in processing collected data. Classification means grouping of related facts into different classes. Information in one class differs from those of other class with respect to some characteristics. Sorting particulars according to one basis of classification and then on another basis is called cross-classification. This process can be repeated as many times as the possible sources of classification are there. Classification of data is a function very similar to that of sorting letters in a post office. Let us explain it further by considering a situation where university receives applications of candidates for filling up some posts for its various departments or disciplines. The applications received for the posts in the university are sorted according to the departments or disciplines to which they pertain. It is well known that the applications collected in an office are

sorted into different lots, department or discipline wise, i.e. in accordance with their destinations as Social Sciences, Engineering, Basic Sciences, etc. They are then put in separate belongings each containing applications with a common characteristic, viz, having the same discipline. Classification of statistical data is comparable to the categorisation process. The process of classification gives distinction to important information gathered, while dropping unnecessary facts, enables comparison and a statistical treatment of the material collected. Now the question may arise in your mind that how collected data is classified. The answer of this question is given under the heading types of classification as discussed below.

13.2.1 Types of Classification

Broadly, data can be classified under following categories:

- (i) Geographical classification
- (ii) Chronological classification
- (iii) Qualitative classification
- (iv) Quantitative classification

Let us discuss these one by one:

(i) Geographical Classification

In geographical classification, data are classified on the basis of location, region, etc. For example, if we present the data regarding production of sugarcane or wheat or rice, in view of the four main regions in India, this would be known as geographical classification as given below in Table-13.1.

Geographical classification is usually listed in alphabetical order for easy reference. Items may also be listed by size to emphasis the magnitude of the areas under consideration such as ranking the states based on population. Normally, in reference tables, the first approach (i.e. listing in alphabetical order) is followed.

Table -13.1: Classification of Production of Wheat

Region	Production of Wheat (in .000 kg.)
Eastern Region	2873
Northern Region	1646
Southern Region	2059
Western Region	986

(ii) Chronological Classification

Classification of data observed over a period of time is known as chronological classification. For example, let us consider the profit figures of a company as shown below for the year from 2001 to 2010.

Table –13.2: Profits of the Company from Year 2001 to 2010

Year	Profit (in crores of rupees)	Year	Profit (in crores of rupees)
2001	20	2006	12
2002	21	2007	25
2003	10	2008	14
2004	18	2009	19
2005	15	2010	23

Time series data are usually listed in chronological order, normally in ascending order of time, like 2001, 2002,... .When the major emphasis falls on the most recent events, a reverse time order may be used.

(iii) Quantitative Classification

Quantitative classification refers to the classification of data according to some characteristics that can be measured numerically such as height, weight, income, age, sales, etc. For example, the employees of an institute may be classified according to their pay scales as follows:

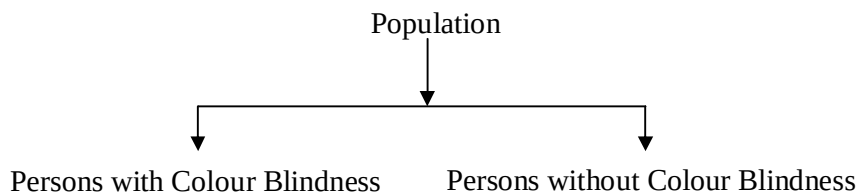
Table-13.3: Quantitative Classification of 840 Employees According to their Pay Scales

Scale of Pay	Number of Employees
9300 - 34800	467
15600 - 39100	215
37400 - 67000	158
Total	840

The quantitative classification is a combination of two elements, namely “Variable, i.e. the pay scale” and “the frequency (the number of employees in each class)” in the above example. There are 467 employees getting salary according to the pay scale 9300-34800, 215 employees are getting salary according to the pay scale 15600-39100 and so on. The quantitative classification gives birth to a frequency distribution which is discussed in subsection 13.2.2.

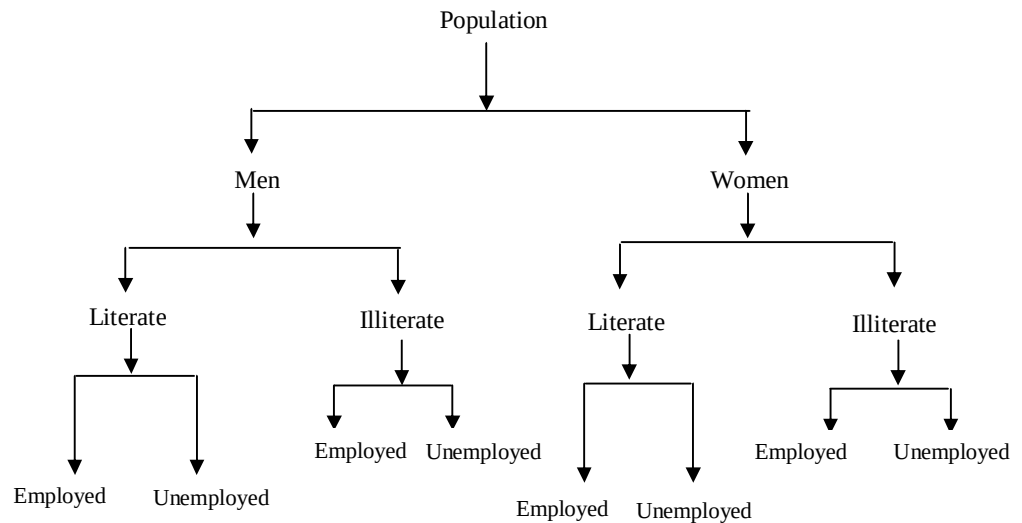
(iv) Qualitative Classification

In qualitative classification, data are classified on the basis of some attributes or qualitative characteristics such as sex, colour of hair, literacy, religion, etc. You should note that in this type of classification the attribute under study cannot be measured quantitatively. One can only count it according to its presence or absence among the individuals of the population under study. For example, in case of colour blindness, we may find out as how many persons are colour blind in a given population. It is not possible to measure the degree of colour blindness in each case. Thus, when only one attribute is studied, two classes are formed – one for possessing the attribute and the other for not possessing it. This type of classification is known as simple classification. For example, the population under study may be divided into two categories based on the characteristic ‘Colour blindness’ as follows:



In a similar manner, we may classify population of a colony on the basis of education qualification, employment, sex, etc. This type of classification where two by two classes are formed is called two fold or dichotomous classification. If, instead of forming only two classes, we further divide the data on the basis of some other attributes within those attributes is known as manifold classification. For example, we may first divide the population into ‘men’ and ‘women’ on the basis of the attribute ‘sex’. Each of these classes may be further subdivided into literate and illiterate on the basis of the attribute

'literacy'. Further classification can be done on the basis of some other attribute say employment. Such type of classification is known as manifold classification and is shown as follows:



Now, you can try the following exercises:

-
- E1)** The amount of production of wheat (in ,000 kg.) are 230, 376, 136, 583 for the cities Bhopal, Agra, Mumbai and Chandigarh respectively. Classify the data.
- E2)** If a company is manufacturing a product from 2001 to 2010 and earning the profits (in crores of rupees) as 10, 15, 13, 17, 12, 16, 17, 21, 20, 18 for the last 10 years respectively. Classify the given data.
-

13.2.2 Frequency Distribution

When observations, whether they are discrete or continuous, available on a single characteristic of a large number of individuals, it becomes necessary to condense the data as far as possible without losing any information of interest. Let us consider the ages of 30 students selected at random from among those studying in a certain class.

20, 22, 25, 22, 21, 22, 25, 24, 23, 22, 21, 20, 21, 22, 23, 25, 23, 24, 22, 24, 21, 20, 23, 21, 22, 21, 20, 21, 22, 25.

This presentation of the data is not considered as good since for large number of observations it is not easy to handle the data in this form. A better way to express the figures is shown in Table 13.4 below:

Table-13.4: Frequency Distribution of 30 Students According to their Age

Age of students	Tally Mark	Frequency
20		04
21		07
22		08
23		04
24		03
25		04
Total		30

A bar (|) called tally mark is put against the number when it occurs. After putting this mark four times against the value, a cross tally is put on these 4 tallies for the fifth mark as shown in the above table. From the sixth mark onwards, we start afresh in the similar manner. This technique facilitates easy counting of the tally marks at the end. The presentation of the data as given in Table 13.4 is known as frequency distribution.

A frequency distribution refers to the data which are classified on the basis of some variables that can be measured such as wages, age of children, etc. A variable refers to the characteristic that varies in magnitude in a frequency distribution. It may be either discrete or continuous. A discrete variable is that which generally takes integer values. For example, the number of students, the number of books, etc. A continuous variable can take integer or fractional values within the range of possibilities, such as the height or weight of individuals. Generally speaking, continuous data are obtained through measurements while discrete data are derived by counting. A series described by a continuous variable is called continuous series. Similarly, series represented by a discrete variable is called discrete series.

According to the nature of the variable, the frequency distribution may be of two types, i.e. discrete frequency distribution and continuous frequency distribution. Let us discuss them one by one.

Discrete Frequency Distribution

A frequency distribution in which the information is distributed in different classes on the basis of a discrete variable is known as discrete frequency distribution. For example, frequency distribution of number of children in 20 families is discrete frequency distribution as shown in Table–13.5.

Table –13.5: Frequency Distribution of the Number of Children in 20 Families

No. of children	Tally Mark	Frequency
0		3
1		4
2		6
3		4
4		3
Total		20

Continuous Frequency Distribution

A distribution in which the information is distributed in different classes on the basis of a continuous variable is known as continuous frequency distribution.

There may be some variables which have integer values as well as fractional values. Frequency distribution of such variables is called continuous frequency distribution. An example of a continuous frequency distribution is given below in Table-13.6.

Table 13.6: Frequency Distribution of Heights of 50 Persons

Heights (cm)	Tally Mark	Frequency
120 -130		3
130 -140		5
140- 150		10
150 -160		14
160 -170		12
170-180		4
180-190		2
Total		50

After discussing the discrete and continuous frequency distributions let us discuss the Relative and Cumulative frequency distributions which are of the similar importance as analysis point of view of data is considered.

Relative Frequency Distribution

A relative frequency corresponding to a class is the ratio of the frequency of that class to the total frequency. The corresponding frequency distribution is called relative frequency distribution. If we multiply each relative frequency by 100, we get the percentage frequency corresponding to that class and the corresponding frequency distribution is called “Percentage frequency distribution”. Let us take an example in which both relative and percentage frequency distributions are prepared.

Example 1: A frequency distribution of marks of 50 students in a subject is as given below:

Class (Marks): 0-10 10-20 20-30 30-40 40-50

Frequency: 6 10 14 18 2

Prepare relative and percentage frequency distributions.

Solution: The relative and percentage frequency distributions can be formed as given in the following table:

Class (Marks) X	Frequency (f)	Relative frequency (f/N)	Percentage Frequency (f/N) × 100
0-10	6	$6/50 = 0.12$	$0.12 \times 100 = 12\%$
10-20	10	$10/50 = 0.20$	$0.20 \times 100 = 20\%$
20-30	14	$14/50 = 0.28$	$0.28 \times 100 = 28\%$
30-40	18	$18/50 = 0.36$	$0.36 \times 100 = 36\%$
40-50	2	$2/50 = 0.04$	$0.04 \times 100 = 4\%$
Total	$\sum f = N = 50$	1.00	100

Cumulative Frequency Distribution

The cumulative frequency of a class is the total of all the frequencies up to and including that class. A cumulative frequency distribution is a frequency distribution which shows the observations ‘less than’ or ‘more than’ a specific value of the variable.

The number of observations less than the upper class limit of a given class is called the less than cumulative frequency and the corresponding cumulative frequency distribution is called less than cumulative frequency distribution.

Similarly, the number of observations corresponding to the value of more than the lower class limit of a given class is called more than cumulative frequency and the corresponding cumulative frequency distribution is called ‘more than’ cumulative frequency distribution. Following is an example, wherein ‘less than’ and ‘more than’ cumulative frequency distributions have been obtained.

Example 2: For the following frequency distribution of marks of 50 students in a subject, form both types of cumulative frequency distributions.

Class (Marks)	0-10	10-20	20-30	30-40	40-50
No. of Students	7	11	15	12	5

Solution: Cumulative frequency distributions are formed as given in the following table:

Given Frequency Distribution		Less Than Cumulative Frequency Distribution		More Than Cumulative Frequency Distribution	
Classes	No. of Students	Marks Less than	No. of students	Marks More than	No of students
0-10	07	10	07	0	50
10-20	11	20	18	10	43
20-30	15	30	33	20	32
30-40	12	40	45	30	17
40-50	05	50	50	40	05
Total	50				

Now, you can try the following exercises.

E3) Construct a discrete frequency distribution for 25 students studying in a class having the following ages (in years):

20, 21, 19, 18, 20, 20, 19, 18, 21, 19, 22, 21, 18, 19, 21, 22, 19, 18, 20, 19, 20, 22, 20, 21, 20.

E4) Construct a continuous frequency distribution for the 50 students studying in a class having the following heights (in cm): 146, 156, 152, 167, 178, 180, 172, 162, 148, 153, 161, 173, 163, 174, 147, 179, 148, 151, 168, 172, 165, 173, 172, 180, 175, 145, 153, 154, 162, 164, 170, 172, 160, 161, 158, 152, 163, 165, 170, 168, 158, 149, 155, 160, 150, 149, 167, 176, 169, 159.

After discussing the frequency distributions we now discuss how the concept of frequency distribution can be used to classify the data according to the class intervals in the next subsection.

13.2.3 Classification According to Class Intervals

To make data understandable, data are divided into number of homogeneous groups or sub groups. In classification, according to class intervals, the observations are arranged systematically into a number of groups called classes. Such classification is most popular in practice. But before this discussion we have to define some terms which will be used in the above classification.

(i) Class Limits

The class limits are the lowest and the highest values of a class. For example, let us take the class 10-20. The lowest value of this class is 10 and the highest

20. The two boundaries of a class are known as the lower limit and upper limit of the class. The lower limit of a class is the value below which there can not be any value in that class. The upper class limit of a class is the value above which no value can belong to that class.

(ii) Class Intervals

The class interval of a class is the difference between the upper class limit and the lower class limit. For example, in the class 10-20 the class interval is 10 (i.e. 20 minus 10). This is valid in the case of exclusive method discussed in this subsection later on. If the inclusive frequency distribution (discussed later on in this subsection) is given then first it is converted to exclusive form and then class interval is calculated. The size of the class interval is determined by number of classes and the total range of data.

(iii) Range of Data

The range of data may be defined as the difference between the lower class limit of the first class interval and the upper class limit of the last class interval.

(iv) Class Frequency

The number of observations corresponding to the particular class is known as the frequency of that class or the class frequency. In the given frequency distribution (Table -13.7), the frequency of the class 10-20 is 12 which implies that there are 12 persons having ages between 10-20. If we add together the frequencies of all individual classes, we obtain the total frequency.

Table-13.7: Frequency Distribution of 50 Persons having Ages between 0-50 Years.

Classes	Frequencies
0-10	08
10-20	12
20-30	15
30-40	10
40-50	05
Total	50

(v) Class Mid Value

It is the value lying half way between the lower and upper class limits of class-interval, mid-point or mid value of a class is defined as follows:

$$\text{Mid Value of a Class} = \frac{\text{Upper class limit} + \text{Lower class limit}}{2}$$

For the purpose of further calculations in statistical analysis, mid value of each class is taken to represent that class.

Now we are in position to discuss the two methods of classification according to class intervals, namely “**Exclusive Method**” and “**Inclusive Method**”. Let us discuss these two methods one by one:

Exclusive Method

Under this method, a class interval is such that each upper class limit is excluded from the class interval. Here in this method, class intervals are so fixed that the upper limit of one class is the lower limit of the next class. In the

following example there are 24 students who have secured the marks between 0 and 50. A student who secured 20 marks would be included in class 20-30, not in 10-20. This method is widely followed in practice.

Example 3: 24 students appeared in an entrance test where all questions are objective type with 25% –ve marking. The marks obtained out of 50 maximum marks are as follows:

17, 16, 7, 30, 21, 42, 44, 36, 22, 22, 25, 31, 31, 34, 30, 36, 35, 45, 25, 15,
20, 42, 40, 30

Prepare a frequency distribution by using exclusive method.

Solution: Frequency distribution of marks obtained by above 24 students is given below in table 13.8 using exclusive method as follows:

Table 13.8: Frequency Distribution of 24 Students by Exclusive Method

Classes	Tally bar	No. of Students
0-10		1
10-20		3
20-30	 	6
30-40	 	9
40-50	 	5
Total		24

Inclusive Method

Under the inclusive method of classification both lower class limit as well as the upper limit of a class is included in that class itself. Following frequency distribution is formed using inclusive method for the data of Example 3 given above.

Table 13.9: Frequency Distribution of 24 Students by Inclusive Method

Class	Tally bar	No. of Students
0-9		1
10-19		3
20-29	 	6
30-39	 	9
40-49	 	5
Total		24

That means if data are classified in such a way that the lower as well as the upper class limits are included in the same class interval, it is called inclusive class interval.

For converting data from inclusive form to exclusive form, first of all we find the half of the difference of lower limit of that class and upper limit of the preceding class. This value is then subtracted from lower limit of each class and added to the upper limit of each class. In the above example, this can be easily understood as $(10-9)/2 = 0.5$. So, the class intervals are as – 0.5- 9.5, 9.5-19.5, ... , 39.5-49.5. If all the observations of data are positive then the lower limit of first class can be taken 0. Therefore, in this case the class intervals are as 0-9.5, 9.5-19.5, ..., 39.5-49.5.

Remark

- (i) Lower limit of a class interval is always included in the class in both the method discussed above.
- (ii) In exclusive method upper limit of a class is not included in the class. That is why the name exclusive.
- (iii) In inclusive method upper limit of a class is also included in the class. That is why the name inclusive.

13.2.4 Principles of Classification

It is difficult to formulate any hard and fast rule for classifying the data. However, the following general considerations may be considered for ensuring meaningful classification of data:

- (1) The whole data should be preferably divided into number of classes between 5 and 15. However, there is no rigidity about it. The classes can be more than 15 depending upon the total number of observations and variations between them and the details required for given data, but they should not be less than 5 because in that case the classification may not reveal the essential characteristics.

To determine the approximate number of classes (K) the following formula is suggested by “Struges”:

$K = 1 + 3.322 \log N$, where K = the approximate number of classes

N = total number of observations

Log = the natural logarithm

However, the appropriate number of classes to be taken for a given data depends upon the personal judgment and other considerations such as range of data, total number of observations, etc.

- (2) One should avoid odd values of class intervals as far as possible, e.g. 3, 7, 11, 26, 39, etc. One should prefer 5 or 10 or multiple of 5 or 10 as class intervals such as 5, 10, 20, 25, 100, etc, because the human mind is accustomed more to think in terms of certain multiples of 5 or 10.
- (3) The lower class limit of the first class of a frequency distribution should either be zero or 5 or multiple of five. For example if the lowest value of the data is 26 and we have taken a class interval of 5, then the first class should be 25-30, instead of 26-31. Similarly if the lowest value of the series is 43 and the class interval is 5 then the first class should be 40-45 inspite of 43-48.
- (4) To maintain continuity and to get correct class interval, we should adopt exclusive method of classification. However, where ‘inclusive’ method has been adopted it is necessary to make an adjustment to determine the correct class interval and to maintain continuity.

How the adjustment is made when data are given by inclusive method explained in the previous sub Sec. 13.2.4. The same adjustment has been done in the frequency distribution given in Table 13.9, which is given in Table 13.10 as shown on the next page:

Table 13.10: Frequency Distribution of 24 Persons by Inclusive Method

Classes	No. of Students
– 0.5-9.5	01
9.5-19.5	03
19.5-29.5	06
29.5-39.5	09
39.5-49.5	05
Total	24

Classification and Tabulation
of Data

- (5) The intervals of all the classes should be of the same size, because if the class intervals are not of the same width, it is difficult to make meaningful comparison between classes. Sometimes the data may require the inclusion of so many class intervals that the frequency distribution will become large. Then the classification may be done as follows:

below 10
10-20
20-30
30-40
above 40

These classes are called open end classes and the distribution is known as open end frequency distribution.

It may be noted that the frequency distributions, like other types of data presentation, are always constructed to serve some specific purpose. The technical requirements outlined above must be supplemented by sound subjective judgments if proper frequency distributions are to be formed.

After learning so much about classification of data, you have got/realised the importance of classification. So before move to next section, let us just highlight/outline some of the main points related to the importance of classification:

- It is preliminary for further statistical analysis,
- It facilitates comparison and make conclusion easy,
- It facilitates tabulation.

Now, you can try the following exercises.

E5) The marks of 30 students in statistics are given below:

10, 12, 25, 32, 27, 32, 38, 43, 39, 55, 29, 38, 57, 08, 06, 13, 27, 25, 29, 53, 55, 45, 35, 48, 47, 59, 15, 19, 48, 55

Classify the above data by taking a suitable class interval.

E6) Present the following data of the profits (in crores of Rs.) of the 60 companies in the years 2009-10:

41, 17, 83, 63, 55, 92, 60, 58, 70, 06, 67, 82, 33, 44, 57, 49, 34, 73, 54, 63, 36, 52, 32, 75, 60, 33, 09, 79, 28, 30, 42, 93, 43, 80, 03, 32, 57, 67, 84, 64, 63, 11, 35, 28, 10, 23, 08, 41, 60, 32, 72, 53, 92, 88, 62, 55, 60, 33, 40, 57

Classify data by inclusive method.

E7) Use the data given in the E6 to present the same using principle of adding and subtracting the correction factor.

13.3 TABULATION OF DATA

One of the simplest and most revealing devices for summarising and presenting data in a meaningful arrangement is statistical table. We can also define a statistical table as the logical listing of quantitative data in columns and rows of numbers with sufficient explanatory statements. The statements may be given in the form of titles, headings and notes to make clear the full meaning of data and their origin.

In other words, a table is a systematic arrangement of statistical data in columns and rows. Rows are horizontal arrangements, whereas columns are vertical ones. A table can solve the purpose of the presentation and facilitate comparison. The simplification results from the clear-cut and systematic arrangement, which enables the reader to quickly locate the desired information. Comparison is facilitated by bringing related items of information close together.

13.3.1 Components of a Table

The various components of a table may vary case to case depending upon the given data. But a good table must contain at least the following components:

1. Table Number
2. Table Heading
3. Caption
4. Stub
5. Body of Table
6. Head Note
7. Foot Note

Let us throw some light on these components one by one:

1. Table Number

A statistical table should be numbered. There are different ways with regard to the place where table number is to be given. The table number may be shown either in the centre at the top above the title or in the left hand side of the table at the top. When there are many columns, it is desirable to number each column so that easy reference to it is possible.

2. Table Heading

A good table should have a suitable heading. The heading is a brief description of the contents of the table. It should be placed above the table. It should answer the following questions:

- (a) What categories of statistical data are shown?
- (b) Where the data occurred?
- (c) When the data occurred?

In other words the heading of the table should be clear, brief and self-explanatory, but some times long title may have to be used for the sake of clarity. The title should be so worded that it permits one and only one interpretation.

3. Caption

Caption refers to the column heading, and explains what information column presents. It may consist of one or more column headings, i.e. under a column

heading there may be two or more sub headings. The caption should be clearly defined and placed at the middle of the column. If the different columns are expressed in different units, the unit should be specified along with the captions.

4. Stub

The stubs are row headings. They are placed at the extreme left of the table and perform the same function for the horizontal rows in the table as the captions do for the vertical columns.

5. Body

The body of the table is the central part of table that contains the numerical information presented in table. This is the most vital part of the table.

6. Head Note

Head note is a brief explanatory statement applying to all or a major part of the material presented in the table and is placed below the title entered and enclosed in brackets. It is used to explain certain points relating to the whole table that have not been included in the title nor in the captions or stubs. For example, the unit of measurement is frequently written as the head note such as “in thousands” or “million tons” or “in crores”, etc.

7. Foot Note

Anything in a table which the reader supposed to find difficult to understand should be explained in footnotes. Footnotes may be placed directly below the body of the table. The footnotes are generally used for the following purposes:

- (a) Any special circumstances affecting the data, for example, strike, fire, etc.
- (b) To clarify any thing in the table.
- (c) To give the source in case of the secondary data. If any information in the table obtained from some journal, its name, date of publication, page number, table number, etc. should be mentioned so that if the user wishes to check the data from the original source, he could know where to look for the information.

After discussing the parts of a table, let us discuss different kinds of tables, through which we can represent or arrange the different types of informations.

13.3.2 Types of Tables

Tables may broadly be classified into following two categories.

1. Simple and Complex Tables
2. General Purpose and Special Purpose Tables

1. Simple and Complex Tables

The simple and complex tables can be differentiated on the basis of number of characteristics presented and studied. If the data based on one characteristic is presented, the table is known as simple table. The simple table is also known as one way table. On the other hand, in a complex table, two or more characteristics are presented. The complex tables are frequently used in practice because they facilitate to incorporate full information and a proper consideration of all related facts. If the data are tabulated on the basis of only two characteristics then the table is known as two way table. If three

characteristics are arranged in a table then the table is known as treble table. When four or more characteristics are simultaneously presented it is known as manifold tabulation.

The following table presenting the distribution of marks obtained by 100 students in a test is an illustration of a simple table:

Table-13.11: Distribution of Marks Obtained by 100 Students in Statistics

Marks	No. of Students
Below 10	5
10-20	8
20-30	12
30-40	10
40-50	15
50-60	18
60-70	17
70-80	13
Above 80	02
Total	100

Two Way Table

Two way table shows two characteristics and is formed when either the stub or the caption or both are divided into two categories. In the following example the nature of such a table is given and is an illustration of two-way table (a complex table):

Table -13.12: Number of Persons Living in a Colony According to Age and Sex.

Age	Persons Living in the Colony		Total
	Males	Females	
Below 15	12	6	18
15-25	20	12	32
25-35	42	27	69
35-45	25	18	43
45-55	10	8	18
55-65	8	5	13
65 and Above	5	2	07
Total	122	78	200

Higher Order Table

When three or more characteristics are represented in the table then such a table is called higher order table. The need for such a table arises when we are interested in presenting three or more characteristics simultaneously.

It should be remembered that as the number of characteristics increases, the table becomes more and more conducting. It is advised normally not more than four characteristics should be represented in the same table. When more than four characteristics are to be represented we should form more than one table depicting relationship between different attributes.

2. General Purpose and Special Purpose Tables

General purpose tables, also known as reference tables or repository tables, and provide the information for general use or reference. They usually contain detailed information and are not used for specific discussion. In other words, these tables serve as a repository of information and are arranged for easy reference such as the tables published by government agencies, the tables contained in the statistical abstract of the Indian Union, tables in the census reports, etc.

The general tables tell facts which are not for particular discussion. If general tables are used by a researcher, they are usually placed in the form of appendix at the end of the report for easy reference.

Special purpose tables, also known as summary tables or analytical tables, provide information for particular discussion. These tables are also called derivative tables since they are often derived from general tables. A special purpose table should be designed in such a way that a reader may easily refer to the table for comparison, analysis or emphasis concerning the specific discussion.

Now, you can try the following exercises.

E8) In a sample survey study about the drinking habits in two cities, it is observed that, in city X 57% are male, 22% are drinkers, and 14% are male drinkers, whereas in city Y 52% are male, 28% are drinkers and 21% are male drinkers. Tabulate the above information.

E9) Present the following information in a suitable tabular form:

In 2009 out of a total 2000 employees in a company 1550 were members of a trade union. The number of women employees was 250, out of which 200 did not belonging to any trade union. While in 2010 the number of union employees was 1725 out of which 1600 were men. The number of none union employees was 380 among which 155 were women.

13.4 SUMMARY

In this unit we have covered the concepts of classification and tabulation of data. That is we have discussed:

- 1) Classification of a data set according to the nature of data.
- 2) The methods of construction of a frequency distribution.
- 3) The methods of construction of discrete and continuous frequency distributions.
- 4) Fundamentals of classification of data according to the class intervals.
- 5) The methods of construction of relative and cumulative frequency distributions.
- 6) Parts of a table.
- 7) Types of the tables and presenting data into a suitable form of a table.

13.5 SOLUTIONS/ANSWERS

- E1)** The classification of the data for the production of wheat according to the given cities can be done in the following way:

Table 13.13: Geographical Classification of the Production of Wheat

Region	Production of Wheat (in .000 kg.)
Agra	376
Bhopal	230
Chandigarh	583
Mumbai	136

- E2)** Classification of the profits of a company from 2001 to 2010 can be done in the following way:

Table 13.14: Chronological Classification of Profits from 2001 to 2010

Year	Profits (in crores of rupees)	Year	Profits (in crores of rupees)
2001	10	2006	16
2002	15	2007	17
2003	13	2008	21
2004	17	2009	20
2005	12	2010	18

- E3)** Discrete frequency distribution for the given information can be constructed in the following way:

Table 13.15: Discrete Frequency Distribution of 25 Students According to their Age

Age of the students	Tally Mark	No. of the students
18		04
19		06
20		07
21		05
22		03
Total		25

- E4)** The continuous frequency distribution for the given information can be constructed in the following way:

Table 13.16: Continuous Frequency Distribution of 50 Students According to their Heights

Heights (cm)	Tally Mark	Frequency
145-150		07
150-155		07
155-160		05
160-165		09
165-170		07
170-175		09
175-180		04
180-185		02
Total		50

- E5)** Let us determine the suitable class interval with the help of the following formula:

$$i = \frac{\text{Range}}{1 + 3.322 \log N}$$

$$\text{Range} = 59 - 06 = 53, N = 30$$

$$i = \frac{53}{1 + 3.322 \log 30} = \frac{53}{1 + 4.91} = 8.97 \approx 9$$

Since values like 3, 7, 9 etc., should be avoided and therefore, we will take 10 as the class interval and hence let us take the first class as 5-15 and thus the following table is formed:

Table 13.17: Continuous Frequency Distribution of 30 Students According to their Heights

Heights (cm)	Tally Mark	Frequency
05-15		5
15-25		2
25-35		8
35-45		5
45-55		5
55-65		5
Total		30

- E6)** As the least value is 3 and the highest value is 93, so using

$$i = \frac{\text{Range}}{1 + 3.322 \log N} = \frac{93 - 3}{1 + 3.322 \log 60} \approx 13.03 \approx 13$$

since, values like 3, 7, 9, 11, 13 etc., should be avoided and therefore, we will take 14 as class interval and hence let us take the first class as 0-14 and thus the following table is formed.

Table 13.18: Continuous Frequency Distribution of 60 Students According to their Heights

Heights (cm)	Tally Mark	Frequency
0-14		06
15-29		04
30-44		16
45-59		10
60-74		14
75-89		07
90-104		03
Total		60

- E7)** Table 13.19 given on next page illustrates the way of classification of data according to the exclusive method and principle of correction factor in classification.

Table 13.19: Continuous Frequency Distribution of 60 Students According to their Heights

Heights (cm)	Tally Mark	Frequency
–0.5-14.5		06
14.5-29.5		04
29.5-44.5		16
44.5-59.5		10
59.5-74.5		14
74.5- 04.5		07
89.5-94.5		03
Total		60

E8) The following table is the representation of the data for the given information's regarding the drinkers in city X and city Y.

Table13.20: Presentation of Data regarding the Drinkers in City X and City Y in the form of Two Way Table

Attributes	City X		Total	City Y		Total
	Males	Females		Males	Females	
Drinkers	14	8	22	21	7	28
Non-drinkers	43	35	78	31	41	72
Total	57	43	100	52	48	100

E9) The following table is showing the trade union membership.

Table 13.21: Presentation of Data regarding the Trade Union Membership in the Year 2009 and 2010 in the form of Two Way Table

Category	2009			2010		Total
	Trade Union Members	None Union Members	Total	Trade Union Members	None Union Members	
Men	1500	250	1750	1600	225	1825
Women	50	200	250	125	155	280
Total	1550	450	2000	1725	380	2105

UNIT 14 DIAGRAMMATIC PRESENTATION OF DATA

Structure

- 14.1 Introduction
 - Objectives
- 14.2 Diagrammatic Presentation
- 14.3 One Dimensional or Bar Diagrams
- 14.4 Two Dimensional Diagrams
- 14.5 Pie Diagrams
- 14.6 Pictogram
- 14.7 Cartogram
- 14.8 Choice of a Suitable Diagram
- 14.9 Summary
- 14.10 Solutions/Answers

14.1 INTRODUCTION

In Unit 13, we have discussed about the classification and tabulation of data. Though these methods are very helpful to make easy and systematic presentation of the data, even then people are least interested in tables. A group of large number of observations always makes misperception to the reader and he/she may understand it wrongly. If data are presented in the form of diagrams, it attracts the reader and he/she tries to understand it. Diagrammatic presentation helps in quick understanding of data. Confirmation of this can be found in the financial pages of news papers, journals, advertisement, etc. There are many methods of representing the numerical figures through diagrams but sometimes, it is very difficult to decide that which is the best diagram in a specific situation?

In this unit we will discuss one-dimensional, two dimensional and pie diagrams. Pictogram and Cartogram have been also discussed in this unit. Unit ends with a note on choice of a suitable diagram in a given situation.

Objectives

After studying this unit, you should be able to:

- become familiar with the diagrammatic presentation of data;
- draw suitable bar diagrams for given data;
- draw rectangle and square diagrams for the given data;
- draw pie diagram;
- draw pictograms and cartograms; and
- select an appropriate diagram to represent data.

14.2 DIAGRAMMATIC PRESENTATION OF DATA

Before we discuss different types of diagrams, let us first see what are the significance and general rules for constructing the diagrams?

14.2.1 Significance of Diagrammatic Presentation of Data

Significance of the diagrams can be explained by the following points.

(i) Easy Understanding

A large number of observations become, easy to understand through diagrams. As the number of observations increases, their analysis tends to be more tedious, but through diagrams the presented data can be understood easily. It is saying also that a picture have explanation power of worth more than 10000 words.

(ii) Attractive Look

Diagrams look attractive to the eyes. The numbers are boring whereas the diagrams give pleasure to the eyes. Diagrams are more attractive and impressive than the numbers. That is why, the reader gives more attention to the diagrams rather than the numbers, while reading a newspaper or magazine. Therefore, the use of diagrams is increasing very fast in exhibitions, fairs, newspapers and common festivals day by day.

(iii) Greater Memorising Effect

Diagrams are long lasting than numbers. Numbers may not be remembered easily but diagrams have greater memorising effect, as the impressions created by them remains in mind for long time.

(iv) Comparison of Data

Through the diagrams, one can easily compare the data related to different areas and time. It is difficult to read and compare the numbers whereas diagrams can be compared easily by viewing the presented informations.

14.2.2 Components of Diagrams

The following components should be considered carefully while constructing diagrams:

(i) Title of the Diagram

Every diagram should have a suitable title. The title of the diagram should convey the main idea in as least words as possible, but it should not omit the necessary information. The title of the diagram may be preferably placed at the top of the diagram.

(ii) Size of the Diagram

Diagram should have a proper size. A proper proportion between the height and width of the diagram should be maintained. If either height or width is too short or too long in proportion, the diagram would give an odd impression. There are no fixed rules about the dimensions, but we may follow an important suggestion given by Lutz in the book entitled “Graphic Presentation” that the proportion between height and width should be 1:1.414. In this proportion diagram looks attractive.

(iii) Scale of the Diagram

Before constructing diagram, a proper scale should be identified. No hard and fast rules are to be followed about the scale. The concern data and the required size of diagram are the guiding factors. The diagram should neither become

too big nor too small. Similar scale is necessary for comparison of diagrams. Scale should be mentioned clearly at the top of the diagram or below it.

(iv) Footnotes

To clarify certain points about the diagram, footnotes are to be used. Footnotes may be given at the bottom of the diagram.

(v) Index of Diagram

An Index should be given to illustrate different types of lines or different types of shades or colours, so that the reader can easily make out the meaning of the diagram.

(vi) Neat and Clean Diagram

A good diagram should be absolutely neat and clean. Too many information should not be given in one diagram otherwise reader may get confused.

(vii) Simple Diagram

A good diagram should be as simple as possible so that the reader can understand its meaning clearly, otherwise the complexity can omit its main theme.

In previous two subsections we have explained the significance and general rules for construction of diagrams. In next subsection we will just list the types of the diagrams. Then in subsequent sections we will discuss each type of diagrams in detail.

14.2.3 Types of Diagrams

In practice, various types of diagrams are in use and new ones are constantly being added. For the sake of application and simplicity several types of diagrams are categorised under the following heads:

- (i) One Dimensional Diagrams or Bar Diagrams
- (ii) Two Dimensional Diagrams
- (iii) Pie Diagrams
- (iv) Pictogram
- (v) Cartogram

14.3 ONE DIMENSIONAL OR BAR DIAGRAMS

Bar diagrams are the most commonly used diagrams. Shape of a bar is like a rectangle filled with some colour (see Example 1). They are called one dimensional diagrams because only length of the bar matters and not the width. That is, width of each bar remains same in a diagram, but it may vary diagram to diagram depending on the space available and number of bars to be presented. For large number of observations lines may be drawn instead of bars to save space.

Following are the special merits of bar diagrams or one dimensional diagrams:

- (i) They are easily understood even by those who are not chart minded.
- (ii) They are the simplest and easiest in comparing two or more diagrams.
- (iii) They are the only form that can be used effectively for comparing the large number of observations.

After looking the merits of bar diagrams, you will be keen to know how bar diagrams are constructed and how many types of bar diagrams are generally used. Coming two subsections will address the above two points/questions.

14.3.1 Types of Bar Diagrams

The following are the different types of bar diagrams:

- (i) Simple Bar Diagram
- (ii) Subdivided Bar Diagram
- (iii) Multiple Bar Diagram
- (iv) Percentage Bar Diagram
- (v) Deviation Bar Diagram
- (vi) Broken Bar Diagram

Let us discuss these types of bar diagrams one by one.

(i) Simple Bar Diagram

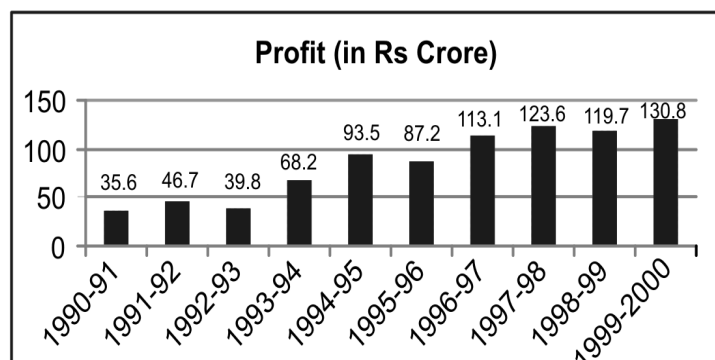
If someone has to represent the data based on one variable, then the simple bar diagram can be used. For example, the figures of productions, profits, sales, etc. for various years may be represented by the help of simple bar diagrams. From simple bar diagrams reader can easily see the variation in the characteristic under study with respect to time or some other given factor, because width of each bar is same and only lengths of the bars vary. In our representation we will take length of bars along vertical axis and other given factor along horizontal axis. They are very popular in practice. For example, while presenting the total turnover of a company for last five decades, one can only depicts the total turnover amount in the simple bar diagrams. Let us construct a simple bar diagram in the following example.

Example 1: The profit (in Rs crore) of a company from 1990-91 to 1999-2000 are given below:

Year	Profit (in Rs crore)	Year	Profit (in Rs crore)
1990-91	35.6	1995-96	87.2
1991-92	46.7	1996-97	113.1
1992-93	39.8	1997-98	123.6
1993-94	68.2	1998-99	119.7
1994-95	93.5	99-2000	130.8

Represent this data by a simple bar diagram.

Solution: The simple bar diagram of the above data is given below:



(ii) Subdivided Bar Diagram

If various components of a variable are to be represented in a single diagram then subdivided bar diagrams are made in this situation. For example, a number of members of teaching staff in various departments of an institute may be represented by a subdivided bar diagram. Each bar is divided into the number of components in this diagram. First of all the cumulative or total amount is calculated from the amounts of components. Then bar is divided with respect to the magnitude of the components. The length of the bar is equal to the total of the amounts of the components.

A bar is represented in the order of magnitude from the largest component at the base of the bar to the smallest at the end of the bar, but the order of various components in each bar is kept in the same order. Different shades or colours are used to distinguish between different components. To explain such differences, the index should be used in the bar diagram.

Subdivided bar diagrams can be represented vertically or horizontally. If the number of components are more than 10 or 12, the subdivided bar diagrams are not used because in that case, the diagram would be over loaded with information and cannot easily be compared and understood. Let us see how subdivided bar diagram is constructed with the help of the following example:

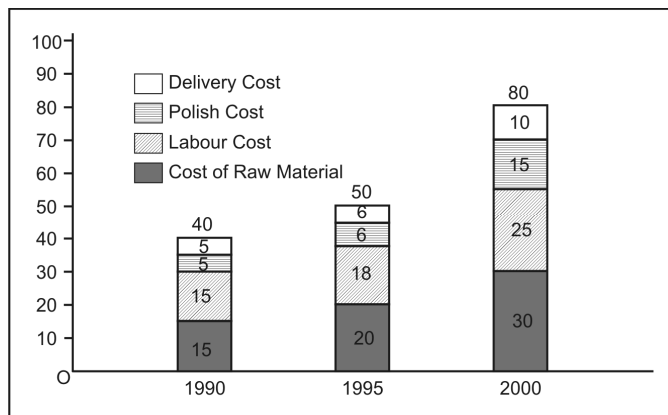
Example 2: Represent the following data by subdivided bar diagram:

Category	Cost per chair (in Rs) year wise		
	1990	1995	2000
Cost of Raw Material	15	20	30
Labour Cost	15	18	25
Polish	5	6	15
Delivery	5	6	10
Total	40	50	80

Solution: First of all we calculate the cumulative cost on the basis of the given amounts:

Category	1990		1995		2000	
	Cost (in Rs)	Cumulative Cost (in Rs)	Cost (in Rs)	Cumulative Cost (in Rs)	Cost (in Rs)	Cumulative Cost (in Rs)
Cost of RM	15	15	20	20	30	30
L Cost	15	30	18	38	25	55
Polish cost	5	35	6	44	15	70
Delivery	5	40	6	50	10	80
Total	40		50		80	

On the basis of above table required subdivided bar diagram is given below:



(iii) Multiple Bar Diagram

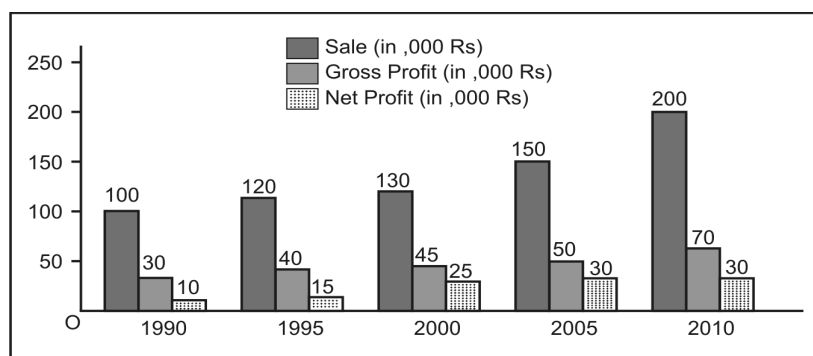
In multiple bar diagram, we construct two or more than two bars together. The multiple bars are constructed for either the different components of the total or for the magnitudes of the variables. All the bars of one group of data are made together so that the comparison of the bars of different groups can be done properly. The height of the bars will be magnitude of the component to be presented as similar as we do in simple bar diagram. In this diagram the space between the vertical axis and the first bar of the first group of bars is left but no space is left between the bars of the same group. There must also be left the space between the bars of the two different groups of bars.

In multiple bar diagrams two or more groups of interrelated data are presented. The technique of drawing such type of diagrams is the same as that of simple bar diagram. The only difference is that since more than one components are represented in each group, so different shades, colours, dots or crossing are used to distinguish between the bars of the same group, and same symbols are used for the corresponding components of the other groups. The multiple bar diagrams are very useful in situations of either the number of relative components are large or the change in the values of the components of one variable is important. Following example will illustrate how a multiple bar diagram is drawn for given data.

Example 3: Draw the multiple bar diagram for the following data.

Year	Sale (in ,000 Rs)	Gross profit (in '000 Rs)	Net profit (in, '000 Rs)
1990	100	30	10
1995	120	40	15
2000	130	45	25
2005	150	50	30
2010	200	70	30

Solution: Multiple bar diagram for the above data is given below.

**(iv) Percentage Bar Diagram**

Subdivided bar diagram drawn on the basis of the percentage of the total is known as percentage bar diagram. When such diagrams are drawn, the length of all the bars is kept equal to 100 and segments are formed in these bars to represent the components on the basis of percentage of the aggregate. First of all the total of the given variable is assumed equal to 100. Then the percentage is calculated for each and every component of the variable. After then the cumulative percentage are calculated for every component. Finally the bars are subdivided into the cumulative percentage and presented like subdivided bar

diagram. Let us explain the procedure with the help of the example given below.

Diagrammatic Presentation of Data

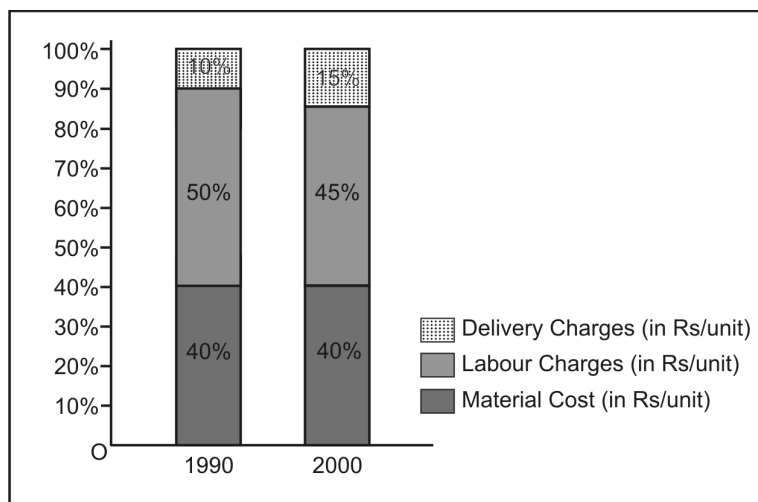
Example 4: Draw a percentage bar diagram for the following data:

Category	Cost Per Unit (1990)	Cost Per Unit (2000)
Material	20	32
Labour	25	36
Delivery	5	12
Total	50	80

Solution: First of all percentage and cumulative percentage are obtained for both the years in various category.

Category	Cost Per Unit (1990)			Cost Per Unit (2000)		
	Cost	% Cost	Cumulative % Cost	Cost	% Cost	Cumulative % Cost
Material	20	40	40	32	40	40
Labour	25	50	90	36	45	85
Delivery	5	10	100	12	15	100
Total	50	100		80	100	

On the basis of above table required percentage bar diagram is given below



(v) Deviation Bar Diagram

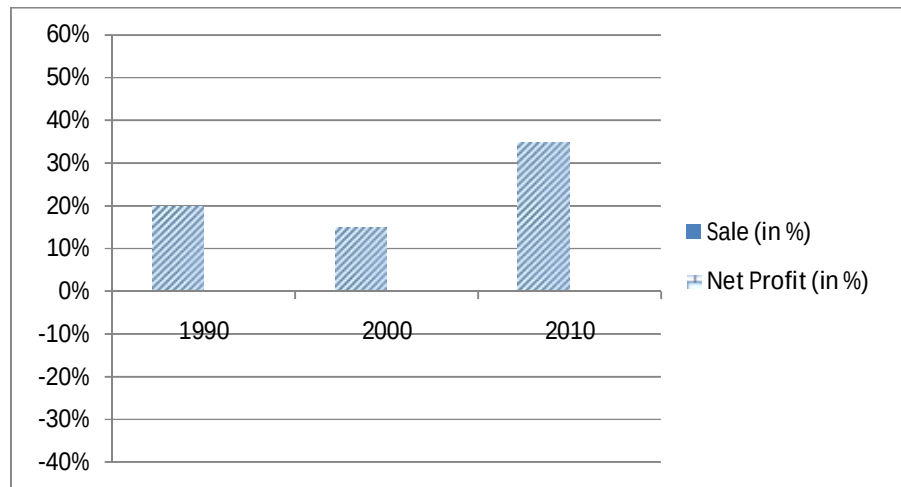
For representing net quantities excess or deficit, i.e. net profit, net loss, net exports, net imports, etc., the deviation bar diagrams are used. Through this kind of bars we can represent both positive and negative values. The values which are positive can be drawn above the base line and negative values can be drawn below it. The following example would explain this type of diagram:

Example 5: Draw a deviation diagram for the following data:

Year	Sale	Net profits
1990	20%	35%
2000	15%	50%
2010	35%	-30%

Solution: Deviation diagram for the given data is shown on the next page:

Presentation of Data



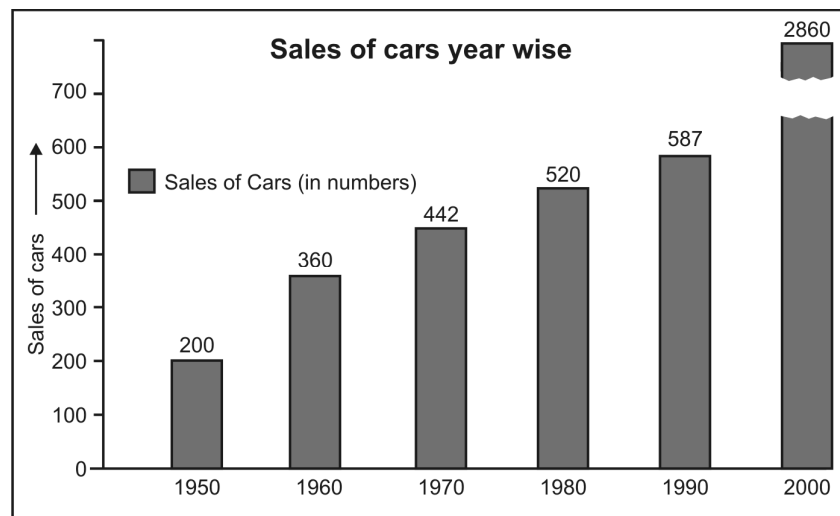
(vi) Broken Bar Diagram

If large variation exists in the values of certain type of data, i.e. some values are very small and some are very large, then in order to gain space for the smaller bars of the data, the large bar(s) may be presented as broken bars. These bars are similar to the other bars but the form of presentation is different because of having much variation from others. Let us illustrate the idea of broken bar diagram with the help of the following example:

Example 6: Represent the following data by a suitable bar diagram.

Year	Sale of cars
1950	200
1960	360
1970	442
1980	520
1990	587
2000	2860

Solution: The sale of the cars in year 2000 is almost 14 times that of in year 1950. In order to gain space for the sale figure in the year 1950, we have to use broken bar to represent the sale of cars for year 2000. Subdivided bar diagram for the given data is shown below.



14.3.1 Principles of Construction of Bar Diagrams

- (i) The width of each bar must be uniform in a diagram.
- (ii) The gap between two bars should be uniform throughout the diagram.
- (iii) Bars may be either horizontal or vertical. The vertical bars should be preferred because they give a better look than horizontal bars and also facilitate comparison. We will use vertical bars in our presentation.
- (iv) The respective figures should also be written at the top of bars so that the reader may be able to know the precise value without looking at the scale.

Now, you can try the following exercises:

E1) Represent the following data by a suitable diagram:

Years:	2005	2006	2007	2008	2009	2010
Enrollment of the students:	280	294	302	270	325	406

E2) Represent the following data by a suitable bar diagram:

Year:	2007-08	2008-09	2009-10
Gross Income:	440	480	520
Gross Expenditure:	410	440	490
Net Income:	160	180	175
Tax:	180	165	190

E3) Represent the following information by a suitable diagram:

Class	Average marks in Mathematics	Average Marks in Statistics	Average Marks in Physics
A	58	70	65
B	62	68	72

E4) Draw a suitable diagram for given expenditure data of two families.

Item	Family A	Family B
Food	300	350
Clothing	250	200
Education	280	300
Others	220	200

E5) Draw a suitable diagram to represent the following information:

Item	Company A	Company B
Selling Price	9500	8000
Raw Material	5500	6500
Direct Wages	3500	4000
Rent of Office	1500	1500

14.4 TWO DIMENSIONAL DIAGRAMS

In one dimensional diagrams only length of the bar is important and comparison of bars are done on the basis of their lengths only, while in two dimensional diagrams both length and width of the bars are considered, i.e. in two dimensional diagrams given numerical figures are represented by areas of the bars. So, two dimensional diagrams are also known as “Area Diagrams.” The following are the types of two dimensional diagrams:

- (i) Rectangles
- (ii) Squares
- (iii) Circles

Let us discuss these one by one:

(i) Rectangles Diagram

In rectangles diagram given numerical figures are represented by areas of the rectangles. We know that area of a rectangle = (length) $\times$ (breadth). So, rectangles diagram is drawn by taking one of the two variables as lengths and another variable as breadths of the rectangles along two axes. To understand this diagram, go through to the following illustration.

Example 7: Two companies A and B produce the same item. Company A produced 2000 units in January 2011 and in the same month company B produced 2400 units. The production cost per unit for company A and company B was Rs 12 and Rs 10.5 respectively. Represent these facts by using rectangles diagram.

Solution: The rectangles for both companies are to be drawn on the following basis:

Company A

Length = 2000 (total produced units)

Breadth = 12 (per unit production cost)

Area = $2000 \times 12 = 24000$

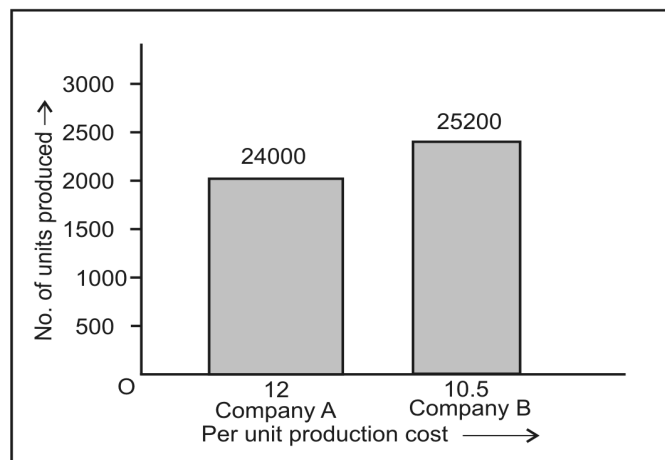
Company B

Length = 2400 (total produced units)

Breadth = 10.5 (per unit production cost)

Area = $2400 \times 10.5 = 25200$

Therefore, the length and width of rectangles of these companies will be in proportion of 2000 : 2400 and 12 : 10.5 respectively. Now, the areas calculated for both companies on the basis of their length and breadth given above, represent the total cost of the two companies. These rectangles are represented below.



(ii) Squares Diagram

When variation between given numerical figures is high then choice of squares diagram is more suitable instead of rectangles diagram. Like rectangles diagram here given numerical figures are represented by areas of squares. We know that area of a square = (side) $\times$ (side) = (side)². So, we take

(side)² = given numerical figure $\Rightarrow$ side of square = $\sqrt{\text{given numerical figure}}$

Remember that the base line would be same for all squares.

In other words, we follow the following steps for the construction of the square diagram:

Step 1 Take the given numerical observations/figures as areas of the corresponding squares.

Step 2 Take square roots of the given numerical observations/figures as sides of the corresponding squares.

Step 3 Construct the corresponding squares like rectangle diagrams.

Let us discuss the method of drawing the square diagram with the help of the following example:

Example 8: Represent the following data of the number of schools in a city A from 1970-80 to 2000-10 in a square diagram.

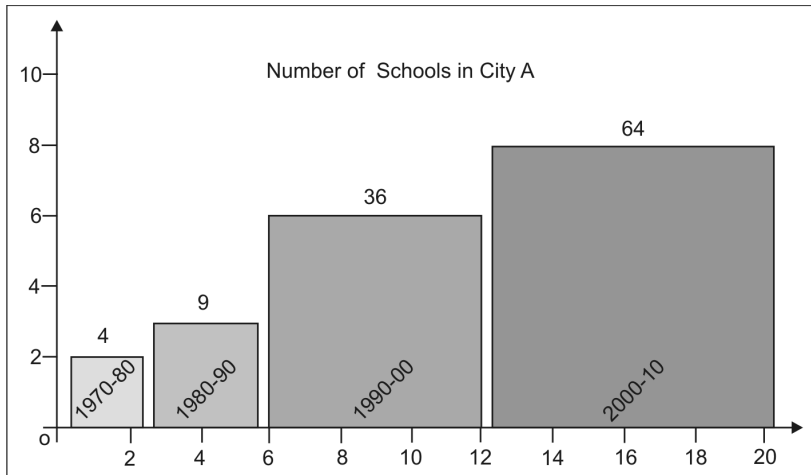
Years	1970-80	1980-90	1990-2000	2000-10
Number of schools in city A	4	9	36	64

Solution:

Step 1 Areas of the corresponding squares = 4, 9, 36, 64

Step 2 Sides of the corresponding squares = $\sqrt{4}$, $\sqrt{9}$, $\sqrt{36}$, $\sqrt{64}$ = 2, 3, 6, 8

Step 3 Square diagram for the given data is shown below.



Remark 1: If in some cases given observations are large and so their square roots, then we can adjust the scale in usual way.

For example, suppose the given observations are 256, 1600, 5184, 9216, then sides of the squares will be

$$\sqrt{256}, \sqrt{1600}, \sqrt{5184}, \sqrt{9216} = 16, 40, 72, 96.$$

Here we can adjust the scale by taking 16 units = 1 unit, after this, sides of the squares reduces to 1, 2.5, 4.5, 6. Now using sides of the squares as 1, 2.5, 4.5, 6, we can construct the square diagram as done in above example, provided we have to mention in the right top most corner the scale used (i.e. 16 units = 1 unit along both axes).

(iii) Circles Diagram

Another form of preparing the two dimensional diagram is circle diagram. As in square diagram we took given numerical figures/observations as the areas of the corresponding squares. Similarly, here we take given numerical figures/observations as areas of the corresponding circles. But as we know that

Area (A) of a circle = πr^2 , where r is radius of the circle

$$\Rightarrow A \propto r^2, \text{ read as A is proportional to } r^2 \quad \left[\begin{array}{l} \because \text{ if } y = ax, \text{ where a is} \\ \text{constant, then we say that} \\ y \text{ is proportional to x.} \end{array} \right]$$

$$\Rightarrow \pi r^2 \propto \text{Given numerical figures/observations}$$

$$\Rightarrow r^2 \propto \text{Given numerical figures/observations} \quad \text{as } \pi \text{ is constant}$$

$$\Rightarrow r \propto \text{Square roots of the given numerical figures/observations}$$

Therefore, we follow the following steps for the construction of the circle diagram:

Step 1 Take the given numerical observations/figures as areas of the corresponding circles.

Step 2 Take squares of the radii (r^2) of the corresponding circles proportional to the given numerical figures/observations as sides of the corresponding squares.

Step 3 Take radii (r) of the corresponding circles proportional to the square roots of the given numerical figures/observations.

Step 4 Construct the corresponding circles like rectangles/squares diagrams.

Circles diagram is the simplest of the two dimensional diagrams used for illustrating the totals having large differences in them like squares diagram. But circles diagram looks more attractive than squares diagram and therefore use of circle diagram is more popular compare to squares diagram. There are as many circles drawn as the totals for representation.

Let us discuss the method of drawing the square diagram with the help of the following example:

Example 9: Draw a circles diagram for the data given in Example 8.

Solution: Using the data of Example 8 for drawing a circles diagram, we have

Step 1 Areas of the corresponding circles ($\pi r_1^2, \pi r_2^2, \pi r_3^2, \pi r_4^2$) = 4, 9, 36, 64

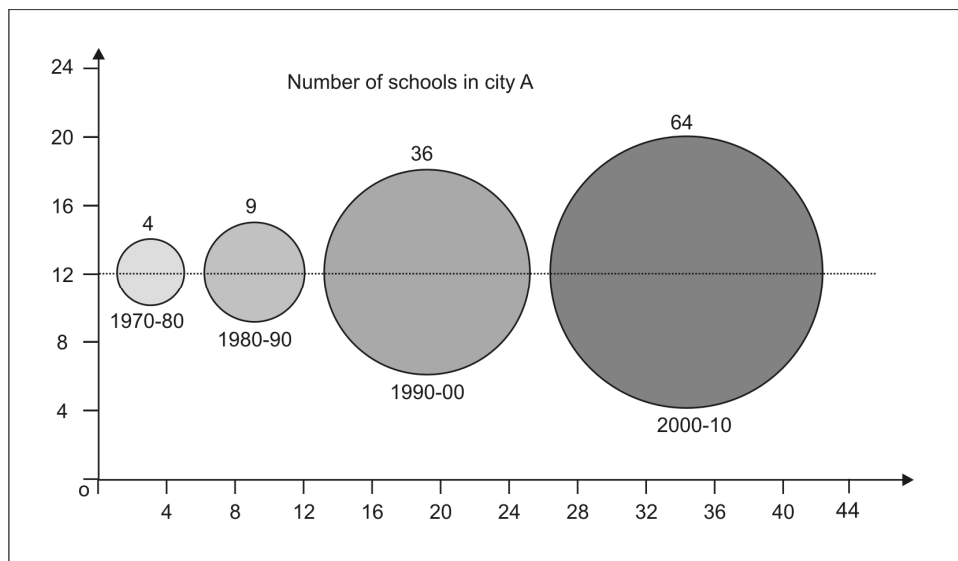
Step 2 Square of radii of corresponding circles are proportional to 4, 9, 36, 64

$$\text{i.e. } r_1^2, r_2^2, r_3^2, r_4^2 \propto 4, 9, 36, 64$$

Step 3 Radii of corresponding circles are proportional to $\sqrt{4}, \sqrt{9}, \sqrt{36}, \sqrt{64}$

$$\text{i.e. } r_1, r_2, r_3, r_4 \propto \sqrt{4}, \sqrt{9}, \sqrt{36}, \sqrt{64} = 2, 3, 6, 8$$

Step 4 Circles diagram for the given data is shown on the next page. Radii of the circles lie on the dotted line.



Remark 2: Here also we can follow similar approach as discussed in Remark1.

Now, you can try the following exercises.

E6) Draw a suitable diagram to represent the following data:

	Rate per item	Sale of item
Company P	20	400
Company Q	30	600

E7) Draw a squares diagram for the data given below:

Year	1980	1990	2000	2010
Number of colonies in city A	16	25	65	150

E 8) Draw a circle diagram for the data given in E 7)

14.5 PIE DIAGRAMS

Pie diagram/chart is used when the requirement of the situation is to know the relationship between whole of a thing and its parts, i.e. pie chart provides us the information that how the entire thing is divided up into different parts. For example, if the total monthly expenditure of a family is Rs 1000, out of which Rs 250 on food, Rs 200 on education, Rs 100 on rent, Rs 150 on transport, and Rs 300 on miscellaneous items are spent. Then this gives us the information that 25%, 20%, 10%, 15% and 30% of the total expenditure of the family are spent on food, education, rent, transport and miscellaneous items respectively. Here we note that if money spent on food (say) increased from 25% to 30% then percentages of other head(s) must shrink so that total remains 100%. Similarly, if money spent on any one of the heads decreased then percentages

Presentation of Data

of other head(s) must spread so that total remains 100%. That is why pie chart gives relationship between whole and its parts.

Steps used for constructing a pie chart.

Step 1 Find the total of different parts.

Step 2 Find the sector angles (in degrees) of each part keeping in mind that total angle around the centre of a circle is of 360° .

Step 3 Find the percentage of each part taking the total obtained in step 1 as 100 percent.

Step 4 Draw a circle and divide it into sectors, where each sector (or area of the sector) of the circle with corresponding angles obtained in step 2 will represent the size of corresponding parts. Diagram thus obtained is nothing but pie chart fitted to the given data.

Let us explain the procedure with the help of the following example:

Example 10: A company is started by the four persons A, B, C and D and they distribute the profit or loss between them in proportion of 4 : 3 : 2 : 1. In year 2010 company earned a profit of Rs 14400. Represent the shares of their profits in a pie chart.

Solution: Given ratio is 4 : 3 : 2 : 1

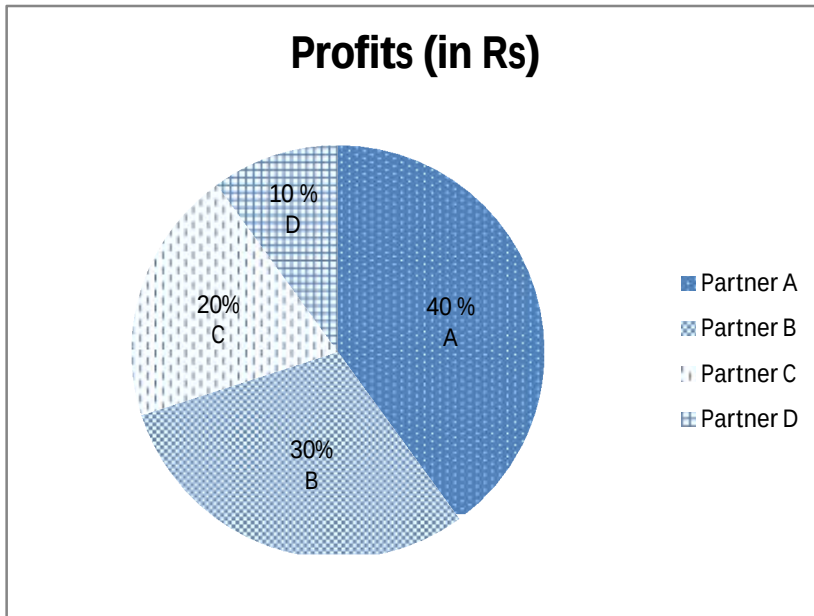
$\therefore$ sum of ratios = $4 + 3 + 2 + 1 = 10$

Calculation of Degrees and Percentages

Partners	Profits (in Rs)	Sector Angles (in degree)	Percentages
A	$14400 \times \frac{4}{10} = 5760$	$\frac{5760}{14400} \times 360 = 144$ or $\frac{4}{10} \times 360 = 144$	$\frac{5760}{14400} \times 100 = 40$
B	$14400 \times \frac{3}{10} = 4320$	$\frac{4320}{14400} \times 360 = 108$ or $\frac{3}{10} \times 360 = 108$	$\frac{4320}{14400} \times 100 = 30$
C	$14400 \times \frac{2}{10} = 2880$	$\frac{2880}{14400} \times 360 = 72$ or $\frac{2}{10} \times 360 = 72$	$\frac{2880}{14400} \times 100 = 20$
D	$14400 \times \frac{1}{10} = 1440$	$\frac{1440}{14400} \times 360 = 36$ or $\frac{1}{10} \times 360 = 36$	$\frac{1440}{14400} \times 100 = 10$
Total	14400	360	100

Solution: On the basis of above calculation, pie chart which shows the shares of profit of the four partners is shown on the next page:

Diagrammatic Presentation of Data



Note:

- (i) In drawing the components on the pie diagram it is advised to follow some logical arrangements, pattern or sequence. For example, according to size, with largest on top and others in sequence running clock wise.
- (ii) Pie chart is used only when
 - (a) total of the parts make a meaningful whole. For example, total of the expenditures of a family on different items make a meaningful whole, but if in a city there are 100 doctors, 40 engineers, 50 milkmen, 80 businessmen then total of these do not make a meaningful whole so pie chart should not be used here.
 - (b) observations in different parts are mutually exclusive. For example in the situation discussed in part (a) a businessman may also be an engineer so the observations in different parts are not mutually exclusive.
 - (c) observations of the different parts are observed at the same time.

We have discussed the method of drawing pie diagram, in this section. Let us discuss some limitations of the pie diagram.

Limitation of Pie Diagram

The following are the limitations of the pie diagram/chart:

- (i) For accurate reading and interpretation, particularly when data are divided into a large number of components or the difference among the values of components is very small, the pie diagram is less effective than the bar diagrams.
- (ii) Attractiveness of a pie chart suffers if the number of parts of the whole is more than 7 or 8. That is, pie chart should be avoided if number of parts of the whole is more than 7 or 8.

Presentation of Data

- (iii) In terms of comparison, the pie diagram appears inferior to simple bar diagram or divided bar diagram.
- (iv) Pie chart is used only when total of the parts make a meaningful whole.
- (v) Pie chart should not be used if observations of the different parts are not mutually exclusive.
- (vi) Pie chart should not be used if observations of the different parts are observed at different time.

Now, you can try the following exercises.

E9) Represent the following data of utilization of 100 paise of income by XYZ company in year 2009-10.

Item/Head	Money spent (in paise)
Manufacturing Expenses	42
Salaries of employees	14
Selling and distribution Expenses	8
Interest Charges	6
Advertisement Expenses	15
Excise duty of sales	5
Taxation	10

E10) Draw a pie diagram to represent the expenditure of Rs 100 over different budget heads as given below of a family

Item	Expenditure (in Rs.)
Food	25
Clothing	15
Education	20
Transport	10
Outing	10
Miscellaneous	5
Saving	15

14.6 PICTOGRAM

Pictograms, also known as picture grams, are very frequently used in representing statistical data. Pictograms are drawn with the help of pictures. These diagrams indicate towards the nature of the represented facts. Pictograms are attractive and easy to comprehend and as such this method is particularly useful in presenting statistics to the layman.

The picture which is used as symbols to represent the units or values of any variable or commodity selected carefully. The picture symbol must be self explanatory in nature. For example, if the increase in number of Airlines Company is to be shown over a period of time then the appropriate symbol would be an aeroplane.

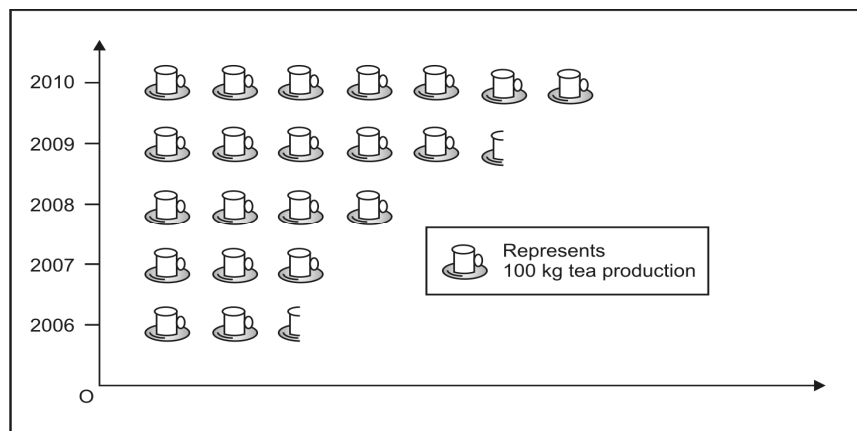
The pictograms have the following merits:

- (i) The magnitudes of the variables may be known by counting the pictures.
- (ii) An illiterate person can also get the information.
- (iii) The facts represented in a pictorial form can be remembered longer.

Example 11: Draw a pictogram for the data of production of tea (in hundred kg) in a particular area of Assam from year 2006 to 2010.

Year	2006	2007	2008	2009	2010
Production of Tea (in 100 kg.)	2.5	3.0	4.0	5.5	7.0

Solution: Pictogram for the production of tea in a particular area of Assam from year 2006 to 2010 is shown below:



Now, you can try the following exercise.

E 11) Draw a pictogram for production of mangoes in a particular area of Maharashtra from 2006 to 2010.

Year	2006	2007	2008	2009	2010
Production of Mangoes (in tons)	5.0	4.5	6.0	3.5	5.5

14.7 CARTOGRAM

Representation of the numerical facts with the help of a map is known as cartogram. By representing the facts by maps, the impact of the results on different geographical area may be shown and to be compared also. Maps are helpful in comparative study of various districts of a state or different states of a country. For example, the production of wheat in different geographical areas can also be represented by cartogram. The quantities on the map can be shown in many ways, such as through shades or colours or by dots or by placing pictograms in each geographical area or by the appropriate numerical figure in each geographical area.

Let us take an example to get a look of the cartogram.

Example 11: Density per square kilometer in different states and union territories in India according 2011 census data is given below.

State/Union Territory	Density (per sq. km.)	State/Union Territory	Density (per sq. km.)	State/Union Territory	Density (per sq. km.)
Andhra P	308	Kerala	859	Tripura	350
Arunachal P	17	Madhya P	236	Uttarakhand	189
Assam	397	Maharashtra	365	Uttar P	828

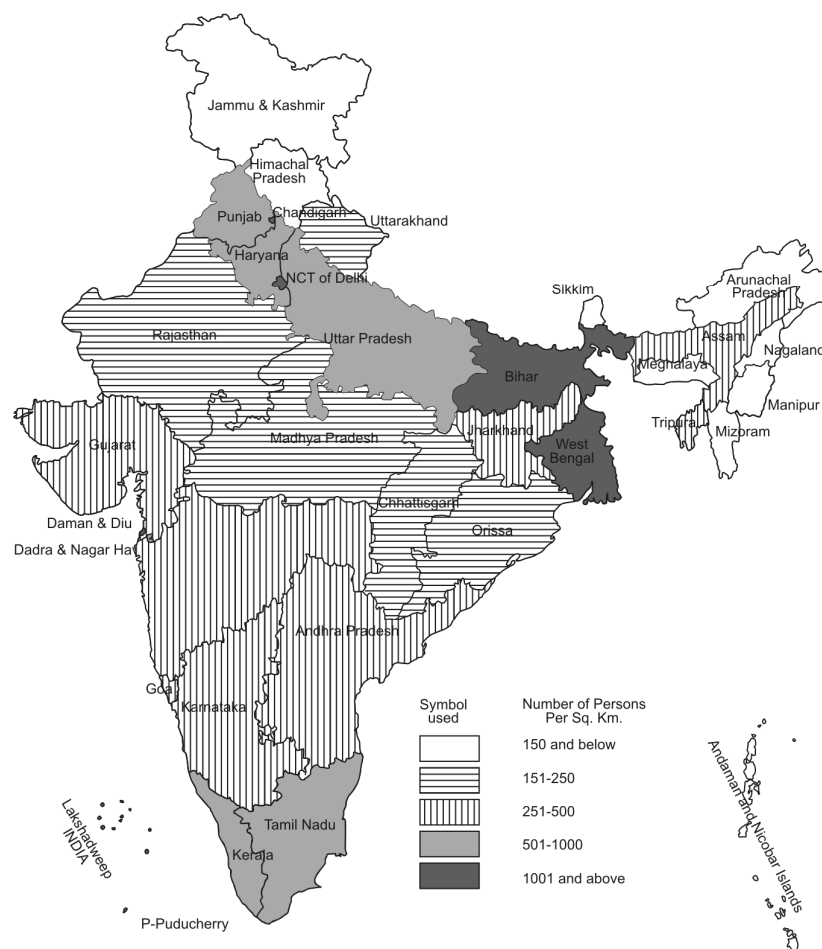
Presentation of Data

Bihar	1102	Manipur	122	West Bengal	1029
Chhattisgarh	189	Meghalaya	132	Andaman and N I	46
Goa	394	Mizoram	52	Chandigarh	9252
Gujarat	308	Nagaland	119	Dadar and N H	698
Haryana	573	Orissa	269	Daman and Diu	2169
Himachal P	123	Punjab	550	Delhi	11297
J and K	124	Rajasthan	201	Lakshadeep	2013
Jharkhand	414	Sikkim	86	Pondicherry	2598
Karnataka	319	Tamil Nadu	555		

Represent the above data with the help of cartogram.

Solution: Cartogram for the above data is given below:

Density of Population, 2011



14.8 CHOICE OF A SUITABLE DIAGRAM

In Secs. 14.3 to 14.7 we have studied many types of diagrams, so a reasonable question may arise in your mind is that how we come to know that which is the suitable diagram in a given situation? To answer this question absolutely is not an easy job because there are situations in which more than one diagram may be used, secondly this is not a complete list of the diagrams. Even though there are some suggestions which may help you to select an appropriate diagram in a given situation.

The choice would primarily depend upon two factors, namely:

- (i) **The Nature of the Data:** The nature of data would depend whether to use one dimensional, two dimensional or three dimensional diagrams and if it is one dimensional, whether it is simple, sub-divided, multiple or some other type. A cubic diagram would be preferred to a bar if the magnitudes of the figures are very wide apart.
- (ii) **The Type of People for whom the Diagram is to be made:** For drawing attention of an undedicated mass pictogram or cartograms are more effective.

Some more points which may address the question raised are given below:

- Simple bar diagrams should be used when changes in totals are required to be represented.
- Sub-divided bar diagrams are more useful when changes in totals as well as in components figures (absolute ones) are required to be represented.
- Multiple bar diagrams should be used where changes in the absolute values of the component figures are to be emphasised and the overall total is of no importance.
- The multiple and sub- divided bar diagrams are used for not more than four or five components. For more than five components, pie diagrams will be the best choice.
- Percentage bar diagrams are better choices when changes in the relative size of component figures are to be displayed.
- Pictograms and cartograms are very elementary forms of visual presentation.
- The pictogram is admirably suited to the publications of articles in newspapers and magazines or in reports.
- Cartograms or statistical maps are particularly effective in bringing out the geographical pattern that may be handled in the data.

14.9 SUMMARY

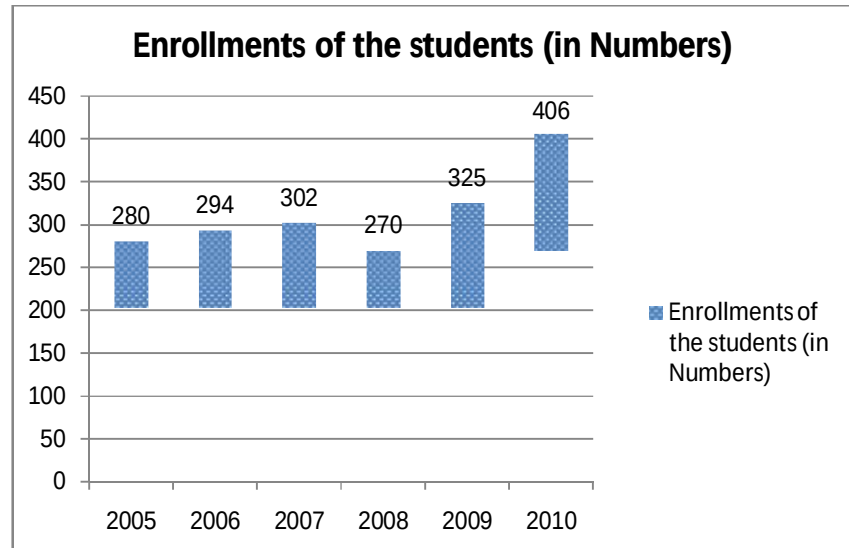
This unit covered the diagrammatic presentation of the data. In this unit, we have discussed:

- 1) One dimensional diagrammatic presentation of the data.
- 2) How to draw different types of bar diagrams.
- 3) How to draw two dimensional diagrams to represent the given data.
- 4) How to draw Pie diagram.
- 5) How to draw Pictograms and Cartograms for the pictorial representations.
- 6) The selection of an appropriate diagram to represent the data of a given situation.

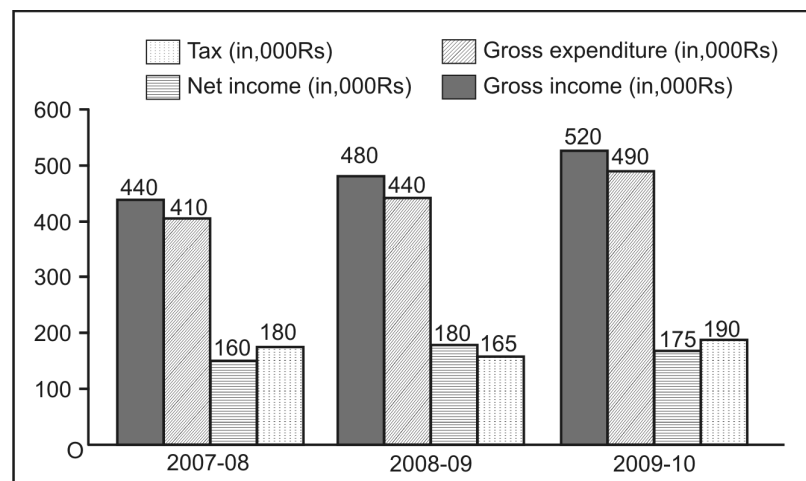
14.10 SOLUTIONS/ANSWERS

- E1) The suitable diagram in this case is simple bar diagram which is shown on the next page:

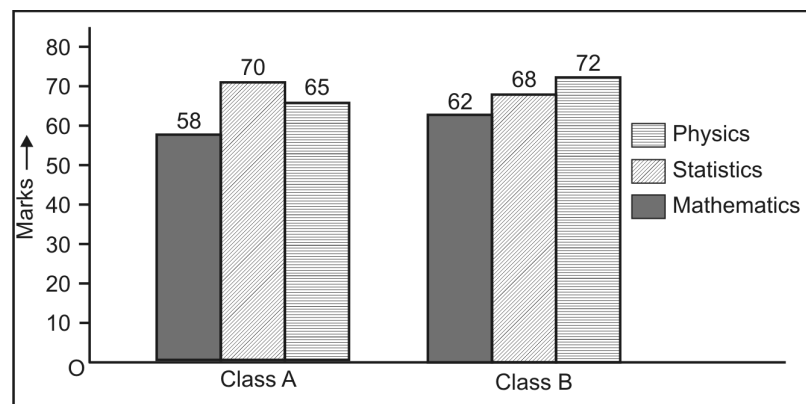
Presentation of Data



E2) The suitable diagram in this case is multiple bar diagram which is shown as follows:

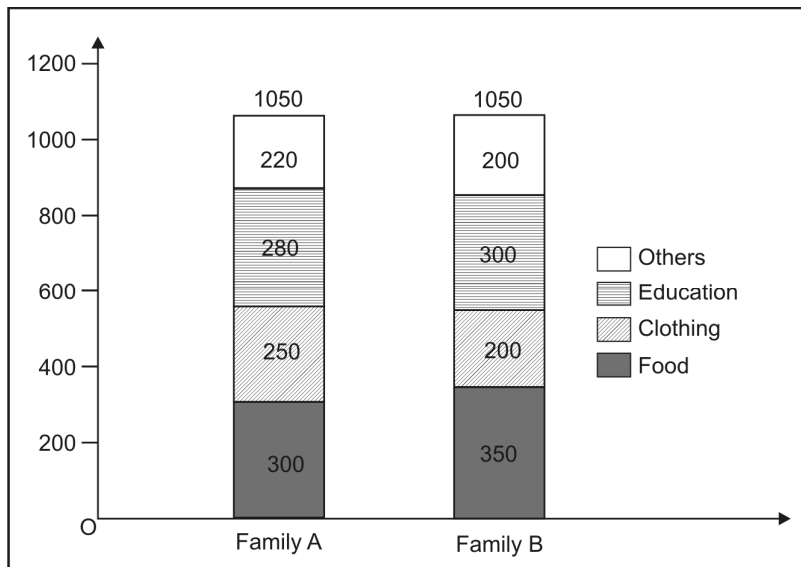


E3) The suitable diagram in this case is multiple bar diagram which is shown as follows:



E4) The suitable diagram in this case is subdivided bar diagram which is shown as follows:

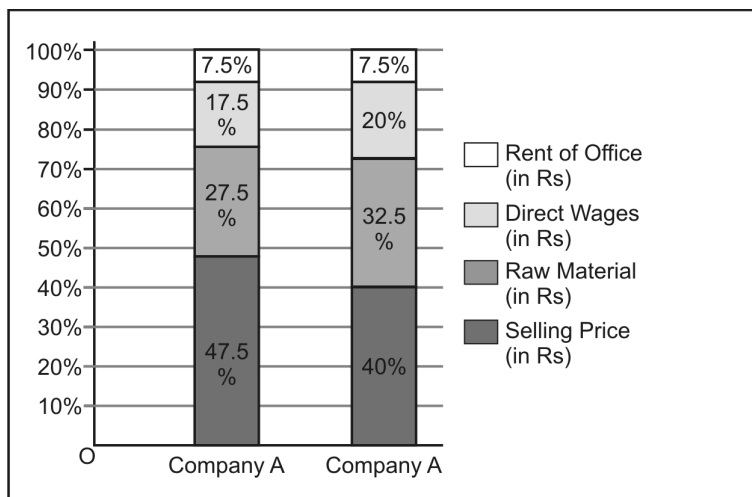
Diagrammatic Presentation of Data



E5) The suitable diagram in this case is percentage bar diagram. So first of all we have to calculate percentage and cumulative percentage for both the companies in various categories as given below:

Category	Company A			Company B		
	Cost	% Cost	Cumulative % Cost	Cost	% Cost	Cumulative % Cost
Selling price	9500	47.5	47.5	8000	40	40
RM	5500	27.5	75	6500	32.5	72.5
DW	3500	17.5	92.5	4000	20	92.5
ROO	1500	7.5	100	1500	7.5	100
Total	20000	100		20000	100	

On the basis of the above calculation subdivided bar diagram is given below:



- E6)** The suitable diagram in this case is rectangles diagram. The rectangles for both companies are to be drawn on the following basis.

Company P

Length = 400 (items sold)

Breadth = 20 (rate per item)

Area = $400 \times 20 = 8000$

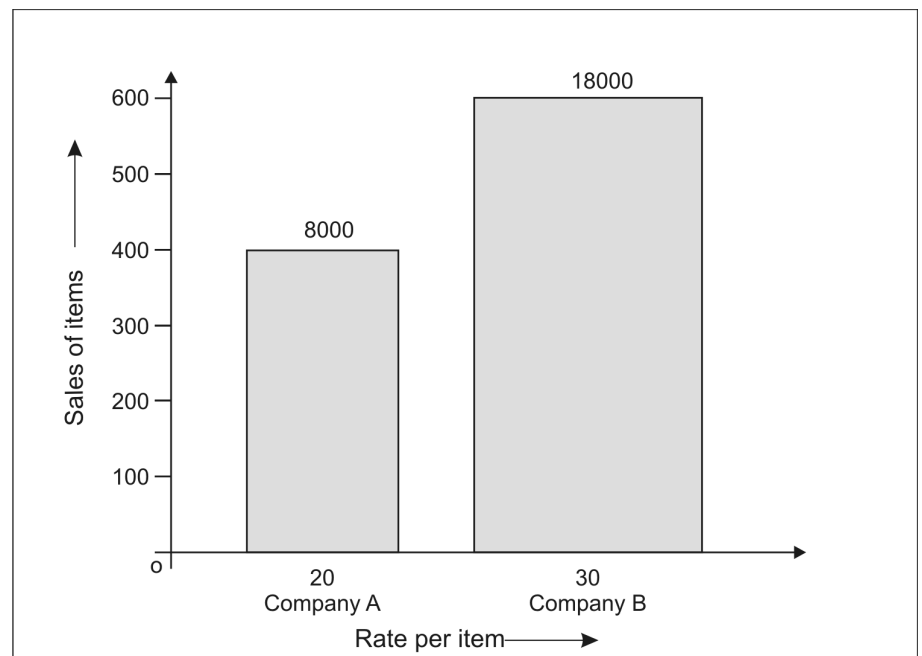
Company Q

Length = 600 (items sold)

Breadth = 30 (rate per item)

Area = $600 \times 30 = 18000$

Therefore, the length and breadth of the two rectangles will be in proportion of 400 : 600 and 20 : 30 respectively. Now, the areas calculated for both companies on the bases of their length and breadth given above, represent the total cost of the companies. These rectangles are represented below.



- E7)** Step 1 Areas of the corresponding squares = 16, 25, 65, 150

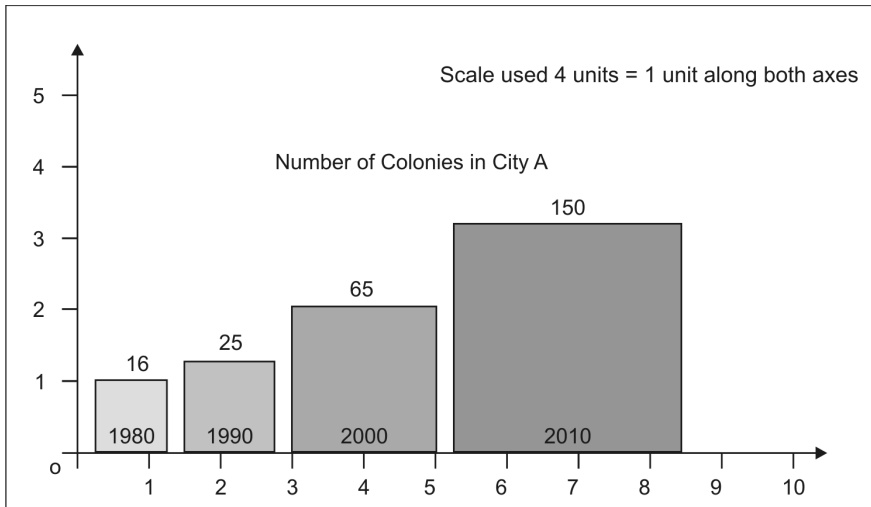
Step 2 Sides of the corresponding squares = $\sqrt{16}, \sqrt{25}, \sqrt{65}, \sqrt{150}$
 $= 4, 5, 8.06, 12.25$

Here we can adjust the scale (as discussed in Remark 1).

Let us take 4 units = 1 unit, then we have

Sides of the corresponding squares = 1, 1.25, 2.02, 3.06

Step 3 Square diagram for the given data is shown on the next page:



E 8) Following the similar steps as in Example 9 we have

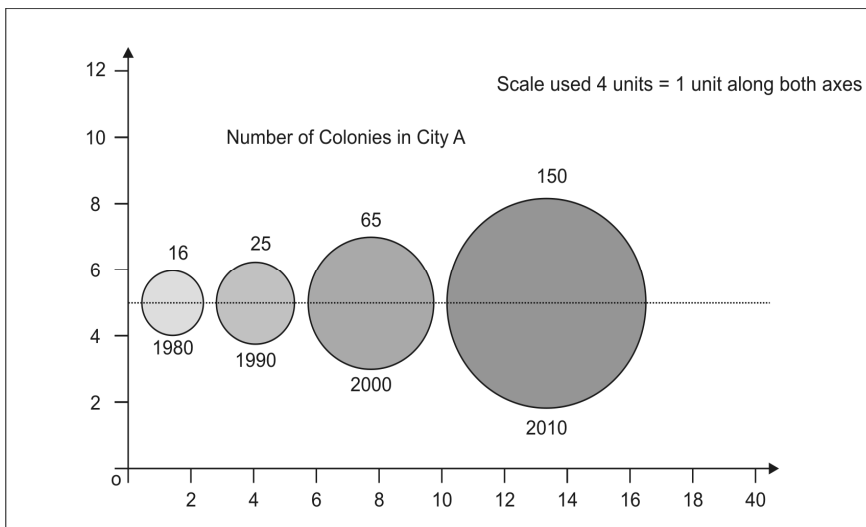
$$r_1, r_2, r_3, r_4 \propto \sqrt{16}, \sqrt{25}, \sqrt{65}, \sqrt{150} = 4, 5, 8.06, 12.25.$$

Here we can adjust the scale (as discussed in Remark 1).

Let us take 4 units = 1 unit, then we have

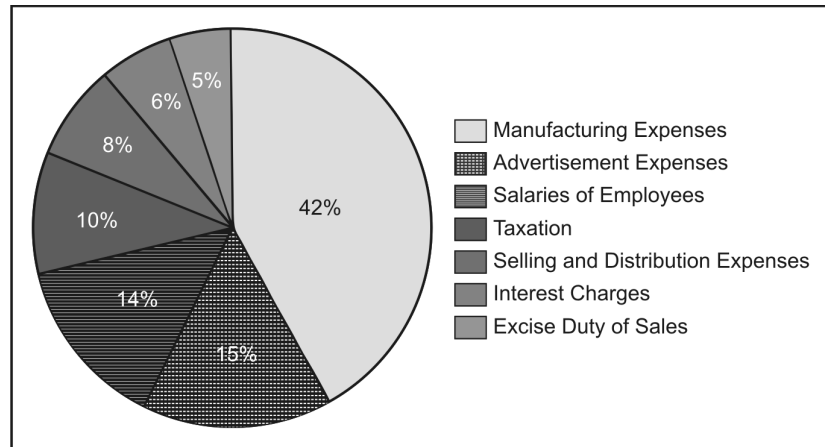
$$r_1, r_2, r_3, r_4 \propto 1, 1.25, 2.02, 3.06.$$

Circle diagram for the given data is shown below:

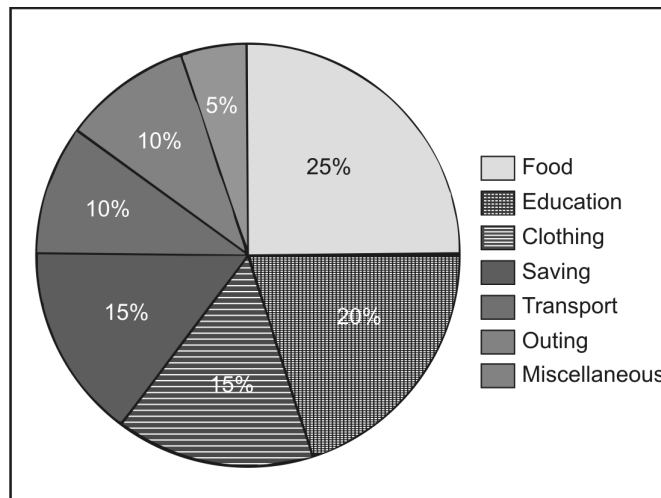


E9) The suitable diagram in this case is pie diagram. Calculation of degrees and percentages (as we did in Example 10) is an exercise for you. On the basis of calculation, pie chart which shows the utilization of 100 paise of income by XYZ company in year 2009-2010 is shown on the next page:

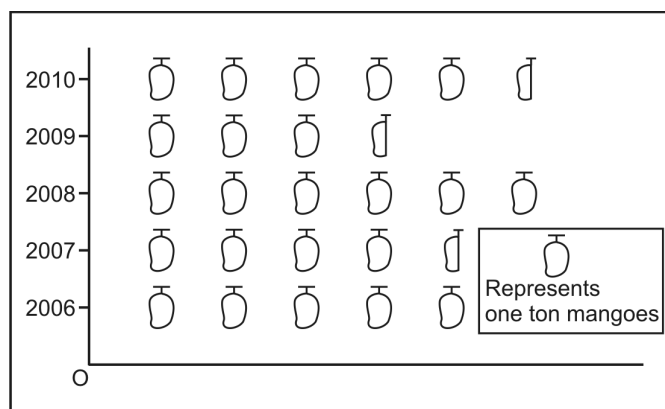
Presentation of Data



E10) The suitable diagram in this case is pie diagram. Calculation of degrees and percentages (as we did in Example 10) is an exercise for you. On the basis of calculation, pie chart which shows the expenditure of a family on different items is shown below:



E 11) We locate the production of mangoes through the picture of mango for the different years according to different magnitude of the data (taking 1 mango = 1 tons mangoes)



UNIT 15 GRAPHICAL PRESENTATION OF DATA-I

Structure

- 15.1 Introduction
 - Objectives
- 15.2 Graphical Presentation
- 15.3 Types of Graphs
 - Histogram
 - Frequency Polygon
 - Frequency Curve
 - Ogive
- 15.4 Summary
- 15.5 Solutions/Answers

15.1 INTRODUCTION

An important function of Statistics is to present the complex and huge data in such a way that they can easily understandable. In previous unit, we have discussed the diagrammatic presentation of the data where we have become familiar with some of the most commonly used diagrams. After discussing the diagrammatic presentation of data, we are now moving towards the graphical presentation of data. The graphs are plotted for frequency distributions and are used to interpolate/extrapolate items in a series including locating various partition values. In this unit, we shall discuss some of the most useful and commonly used graphs.

The graphical presentation can be divided into two categories

- (i) Graphs for frequency distributions.
- (ii) Graphs for time series.

In this unit, we will concentrate ourselves to the graphs for frequency distributions only. In this regard, we would like to discuss the most commonly used graphs for frequency distributions, i.e. Histograms, Frequency polygon, Frequency curve and Cumulative frequency curves, or Ogives.

Objectives

After studying this unit, you would be able to:

- describe the graphical presentation;
- explain the advantages of graphical presentation;
- draw the histogram for continuous frequency distribution;
- draw the frequency polygon for a frequency distribution;
- draw the frequency curves of different shapes; and
- draw the cumulative frequency curves.

15.2 GRAPHICAL PRESENTATION

A graphical presentation is a geometric image of a set of data. Graphical presentation is done for both frequency distributions and times series. Unlike diagrams, they are used to locate partition values like median, quartiles, etc, in particular, and interpolate/extrapolate items in a series, in general. They are also used to measure absolute as well as relative changes in the data. Another important feature of graphs is that if a person once sees the graphs, the figure representing the graphs is kept in his/her brain for a long time. They also help us in studying cause and affect relationship between two variables. The graph of a frequency distribution presents the huge data in an interesting and effective manner and brings to light the salient features of the data at a glance. Before closing this Sec. let us see some advantages of graphical presentation.

Advantages of Graphical Presentation

The following are some advantages of the graphical presentation:

- It simplifies the complexity of data and makes it readily understandable.
- It attracts attention of people.
- It saves time and efforts to understand the facts.
- It makes comparison easy.
- A graph describes the relationship between two or more variables.

After going through the advantages of graphical presentation of data, you were keen to know the commonly used graphs to represent the data and how these graphs are drawn. Next section will address these issues.

15.3 TYPES OF GRAPHS

Now days a large variety of graphs are in practical use. However, we shall discuss only some important graphs which are more popularly used in practice. The various types of graphs can be divided broadly under the following two heads:

- (i) Graphs of Frequency Distributions
- (ii) Graphs for Time Series Data

In this Sec. we will focus on graphs for frequency distributions and graphs for time series data will be discussed in next unit, i.e. Unit 16.

Graphs of Frequency Distributions

The graphical presentation of frequency distributions is drawn for discrete as well as continuous frequency distributions.

Let us first consider the frequency distribution of a discrete variable.

To represent a discrete frequency distribution graphically, we take two rectangular axes of co-ordinates, the horizontal axis for the variable and the vertical axis for the frequency. The different values of the variable are then located as points on the horizontal axis. At each of these points, a perpendicular bar is drawn to present the corresponding frequency.

Such a diagram is called a 'Frequency Bar Diagram'. For example, if we take the frequency distribution for the number of peas per pod for 198 pods as given in Table 15.1:

Table 15.1

No of peas per pod	1	2	3	4	5	6	7
Frequency (number of pods)	14	23	66	40	26	18	11

Graphical Presentation of Data-I

Then, the frequency bar diagram is shown in Fig. 15.1:

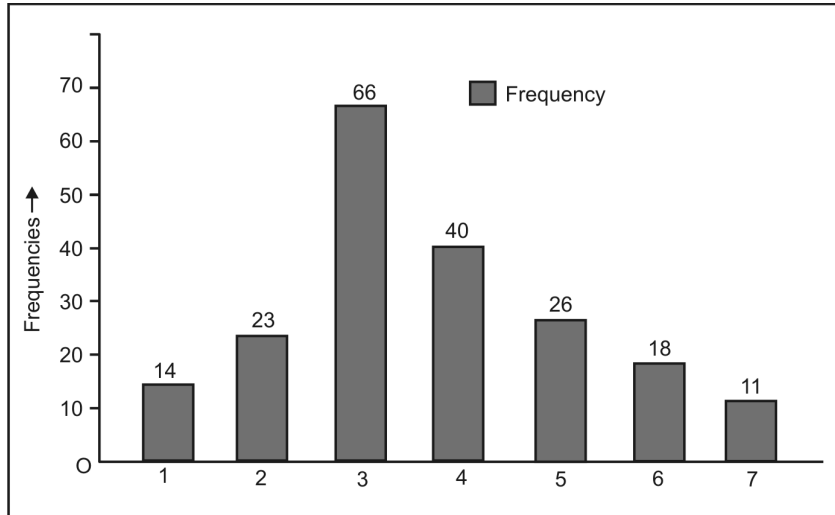


Fig. 15.1: Frequency Bar Diagram for the Frequency Distribution of Number of the Peas for 198 Pods.

Note 1: O represents origin and choice of scale used along horizontal and vertical axes depends upon given data.

Now, we take the case of frequency distribution of a continuous variable.

The following are the most commonly used graphs for continuous frequency distributions:

- (i) Histogram
- (ii) Frequency Polygon
- (iii) Frequency Curve
- (iv) Cumulative Frequency Curve or Ogives

Let us discuss these one by one:

(i) Histogram

In previous example, we have discussed how a graph is drawn for discrete frequency distribution.

For the continuous frequency distribution, a better way to represent the data graphically is to use a histogram. A histogram is drawn by constructing adjacent rectangles over the class intervals such that the length of the rectangles is proportional to the corresponding class-frequencies.

Histogram is similar to a bar diagram which represents a frequency distribution with continuous classes. The width of all bars is equal to class interval. Each rectangle is joined with the other so as to give a continuous picture.

The class-boundaries are located on the horizontal axis. If the class-intervals are of equal size, the heights of the rectangles will be proportional to the class-frequencies themselves. If the class-intervals are not of equal size, the heights of the rectangles will be proportional to the ratios of the frequencies to the

width of the corresponding classes. In other words, the frequencies of the class-intervals having the least width are written as they are and the frequencies of other class intervals are written as follows:

$$\frac{\text{Given frequency}}{\text{Width of its Class-interval}} \times (\text{The least width}) \quad \dots (15.1)$$

Let us draw a histogram to the following frequency distribution given below in the table 15.2

Table 15.2

Class Intervals	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	2	3	13	18	9	7	6	2

Histogram for the above data is given below.

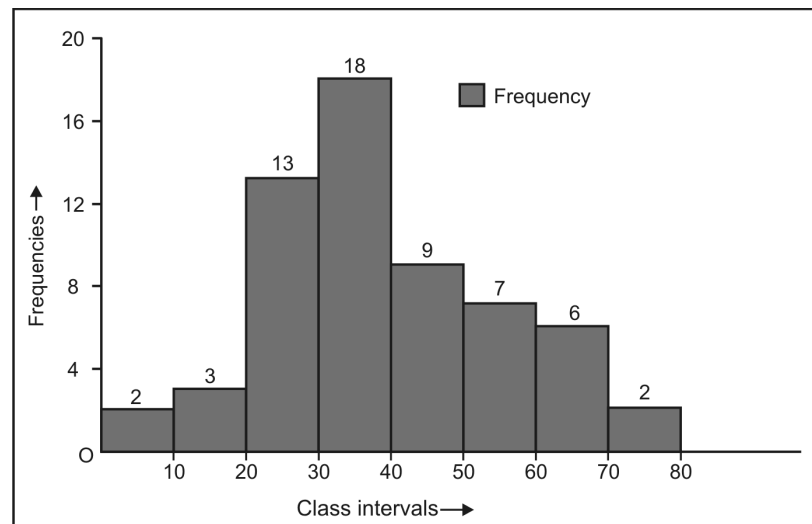


Fig. 15.2: Histogram for Frequency Distribution when Class-intervals are of Equal Width.

Now, let us consider the frequency distribution for unequal class intervals as given in the Table 15.3

Table 15.3

Class	0-10	10-20	20-30	30-40	40-70	70-80	80-100
Frequency	20	32	8	2	60	35	10

As it is a case of unequal class intervals, so we have to adjust the frequencies of the classes 40-70 and 80-100 by the formula suggested in equation 15.1. These calculations are shown in table 15.4 given below:

Table 15.4

Class Interval (CI)	Frequency	Width of (CI)	Heights of the rectangles
0-10	20	10	20
10-20	32	10	32
20-30	8	10	8
30-40	2	10	2
40-70	60	30	$(60/30) \times 10 = 20$
70-80	35	10	35
80-100	10	20	$(10/20) \times 10 = 5$

The histogram for this frequency distribution is shown in Fig. 15.3.

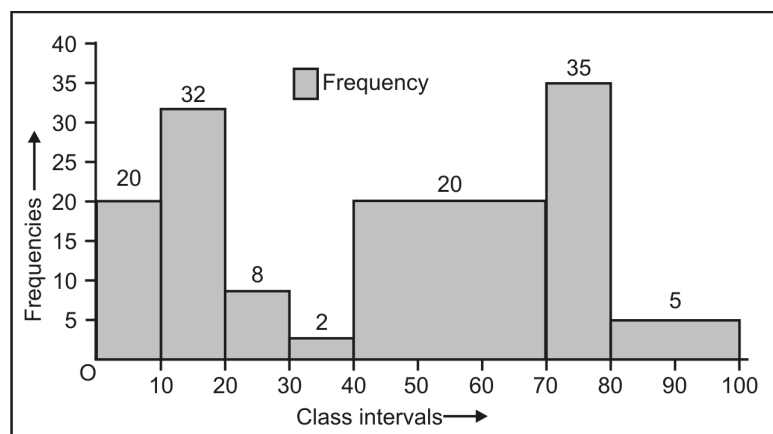


Fig. 15.3: Histogram for Frequency Distribution when Class Intervals are of Unequal Width

Note 2: Sometimes, a histogram is also used for the frequency distribution of a discrete variable. Each value of the discrete variable is regarded as the mid-point of an interval. But generally, its use is not recommended, because in discrete case each frequency actually corresponds to a single point and not to an interval.

Now, you can try the following exercises.

E1) Draw a histogram from the following data

Class Interval:	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
Frequency:	3	5	10	14	24	17	14	10	3

E2) Draw a histogram for the following frequency distribution

Wage (Rs):	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40	40-45	45-50
No. of Workers:	30	70	100	110	140	150	130	100	90	60

(ii) Frequency Polygon

Another method of presenting a frequency distribution graphically other than histogram is to use a frequency polygon. In order to draw the graph of a frequency polygon, first of all the mid values of all the class intervals and the corresponding frequencies are plotted as points with the help of the rectangular co-ordinate axes. Secondly, we join these plotted points by line segments. The graph thus obtained is known as frequency polygon, but one important point to keep in mind is that whenever a frequency polygon is required we take two imaginary class intervals each with frequency zero, one just before the first class interval and other just after the last class interval. Addition of these two class intervals facilitate the existence of the property that

Area under the polygon = Area of the histogram

For example, if we take the frequency distribution as given in Table 15.2 then, we have to first plot the points (5, 2), (15, 3), ..., (75, 2) on graph paper along with the horizontal bars. Then we join the successive points (including the mid points of two imaginary class intervals each with zero frequency) by line segments to get a frequency polygon. The frequency polygon for frequency distribution given in Table 15.2 is shown in Fig. 15.4.

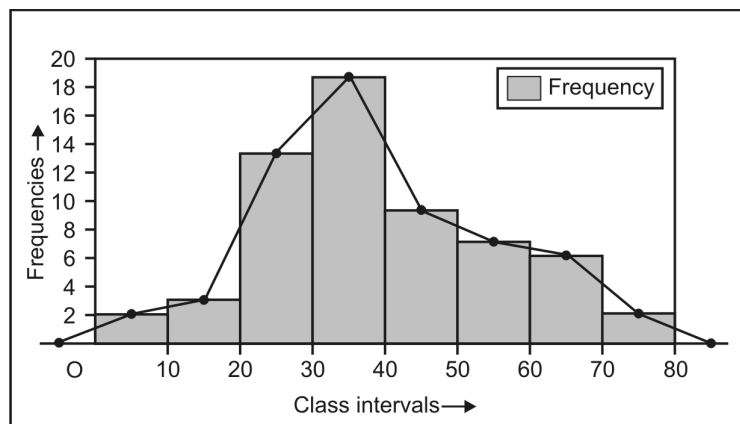


Fig. 15.4: Frequency Polygon for the Frequency Distribution given in Table 15.2.

Note 3: In some cases first class interval does not start from zero. In such situations we mark a kink on the horizontal axis, which will indicate the continuity of the scale starting from zero. Let us take an example of this type.

Example 1: Draw a frequency polygon for the following frequency distribution:

Class Interval	40-50	50-60	60-70	70-80	80-90	90-100	100-110	110-120
Frequency	4	10	11	13	18	14	11	5

Solution: Frequency polygon for the given data is shown in Fig. 15.5:

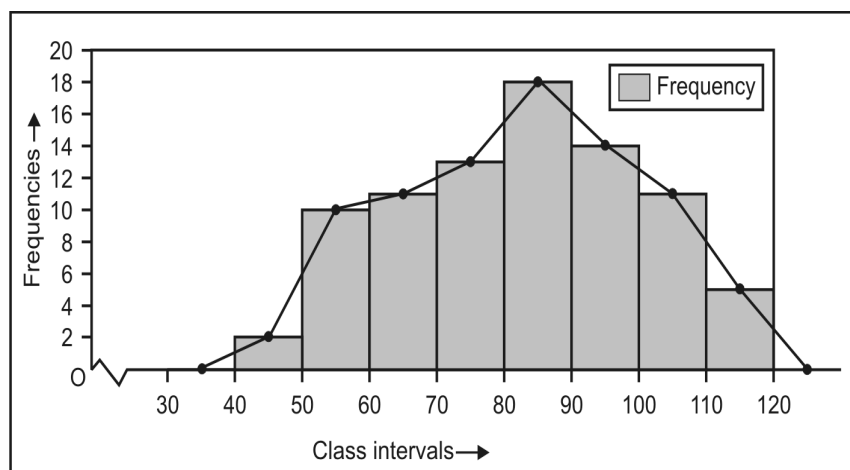


Fig. 15.5: Frequency Polygon for Continuous Frequency Distribution.

Now, you can try the following exercise.

E3) Draw a frequency polygon from the following data
Wage: 0-5 5-10 10-15 15-20 20-25 25-30 30-35 35-40 40-45 45-50
No of
Workers: 20 50 80 110 140 150 120 150 100 80

(iii) Frequency Curve

In simple words frequency curve is a smooth curve obtained by joining the

points (not necessary all points) of the frequency polygon such that

- Like frequency curve it also starts from the base line (horizontal axis) and ends at the base line.
- Area under frequency curve remains approximately equal to the area under the frequency polygon.

In other words, let us try to explain the concept theoretically. Suppose we draw a sample of size n from a large population. Frequency curve is the graph of a continuous variable. So theoretically continuity of the variable implies that whatever small class interval we take there will be some observations in that class interval. That is, in this case there will be large number of line segments and the frequency polygon tends to coincide with the smooth curve passing through these points as sample size (n) increases. This smooth curve is known as frequency curve.

In the following example we have drawn both frequency polygon and frequency curve to make the idea clear for you.

Example 2: Draw frequency polygon and frequency curve for the following frequency distribution.

Class Intervals	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
frequency	2	5	8	15	18	10	3	1

Solution: Frequency polygon and frequency curve for the above data is given below in Fig. 15.6.

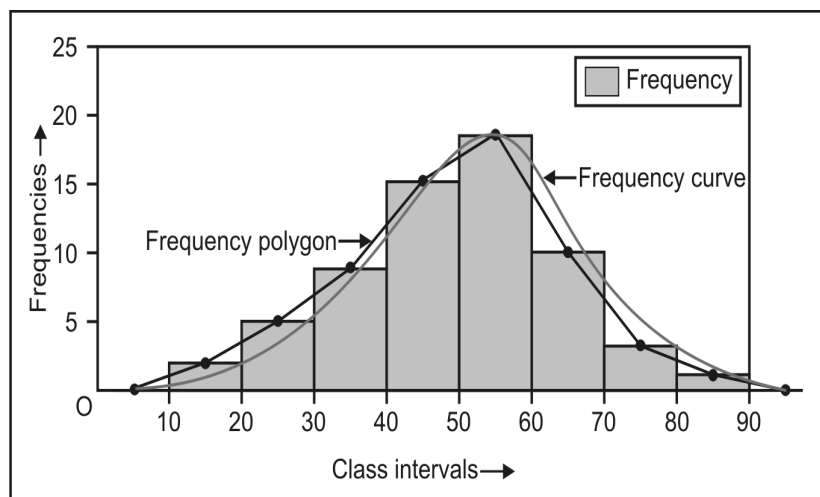


Fig. 15.6: Frequency Curve along with Frequency Polygon.

On the next page some important types of frequency curves are given which are generally obtained in the graphical presentations of frequency distributions. That is, symmetrical, positively skewed, negatively skewed, J shaped, U shaped, bimodal and multimodal frequency curves. You note that the shapes of these curves justify their names.

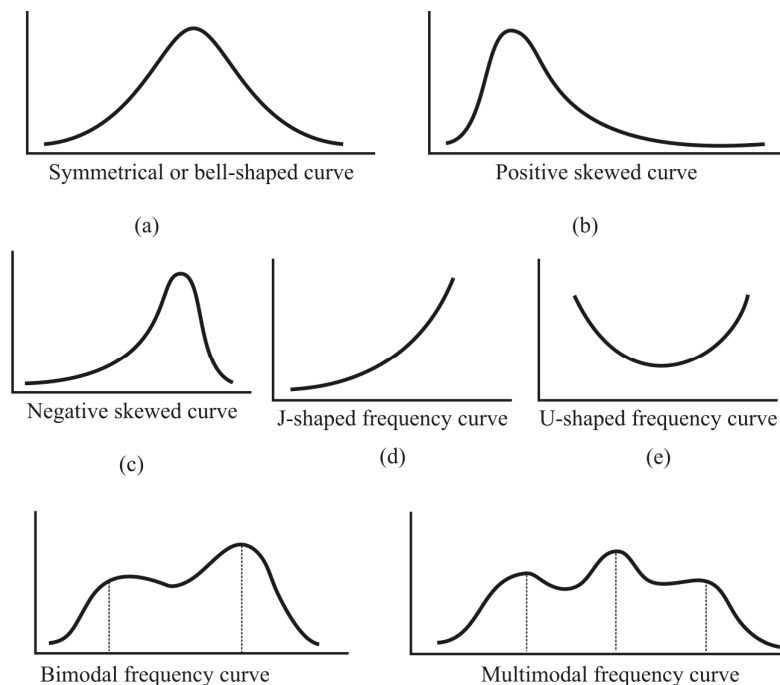


Fig. 15.7

Now, you can try the following exercise.

E4) Draw a frequency curve from the following frequency distribution								
Class Intervals:	1-2	2-3	3-4	4-5	5-6	6-7	7-8	8-9
Frequency:	3	9	20	25	18	12	6	3

(iv) Cumulative Frequency Curves

In sub Sec. 13.2.2 of the Unit 13 of this course you have already studied the concept of cumulative frequency, cumulative frequency distribution, less than cumulative frequency, more than cumulative frequency, less than cumulative frequency distribution, more than cumulative frequency distribution, etc. Here our aim is to study graphical presentation of less than and more than cumulative frequency distributions, which are known as less than frequency curve (or less than ogive) and more than frequency curve (or more than ogive) respectively.

For drawing less than cumulative frequency curve (or less than ogive), first of all the cumulative frequencies are plotted against the values (upper limits of the class intervals) up to which they correspond and then we simply join the points by line segments, curve thus obtained is known as less than ogive. Similarly, more than frequency curve (more than ogive) can be obtained by plotting more than cumulative frequencies against lower limits of the class intervals. As we have already mentioned within brackets that less than cumulative frequency curve and more than cumulative frequency curve are also called **less than ogive** and **more than ogive** respectively.

In other words we may define less than ogive and more than ogive as follow:

Less Than Ogive: If we plot the points with the upper limits of the classes as abscissae and the cumulative frequencies corresponding to the values less than the upper limits as ordinates and join the points so plotted by line segments, the curve thus obtained is nothing but known as “less than cumulative frequency curve” or “less than ogive”. It is a rising curve.

More Than Ogive: If we plot the points with the lower limits of the classes as abscissae and the cumulative frequencies corresponding to the values more than the lower limits as ordinates and join the points so plotted by line segments, the curve thus obtained is nothing but known as “more than cumulative frequency curve” or “more than ogive”. It is a falling curve.

Let us draw both the ogives (‘less than’ and ‘more than’) for the following frequency distribution of the weekly wages of number of workers given in Table 15.5.

Table 15.5

Weekly wages	0-10	10-20	20-30	30-40	40-50
No. of workers	45	55	70	40	10

Before drawing the ogives, we make a cumulative frequency distribution as given in table 15.6

Table 15.6

Weekly wages	No. of workers	Less than Cumulative frequency distribution		More than Cumulative frequency distribution	
		Wages Less than	Number of workers	Wages More than	Number of workers
0-10	45	10	45	0	220
10-20	55	20	100	10	175
20-30	70	30	170	20	120
30-40	40	40	210	30	50
40-50	10	50	220	40	10

From above data, we construct both the ogives as shown in Fig. 15.8 and Fig. 15.9:

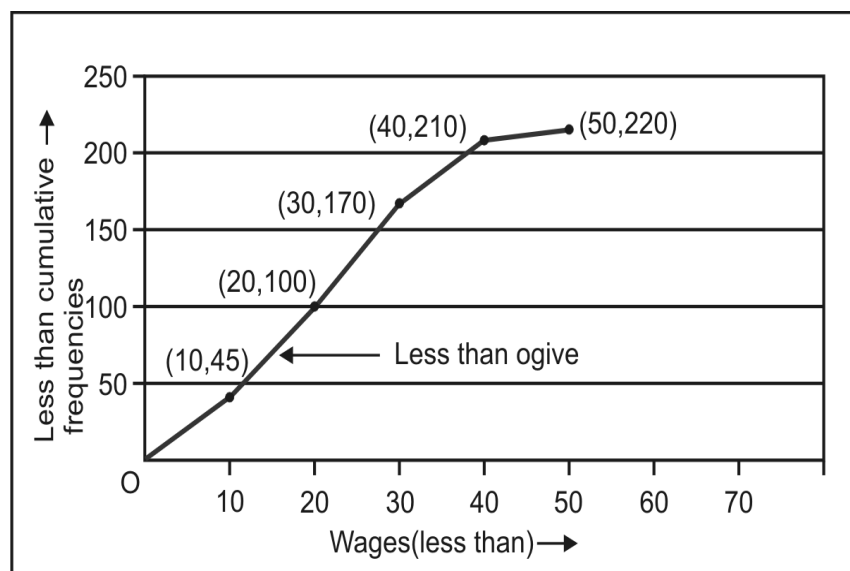


Fig. 15.8: Less Than Ogive.

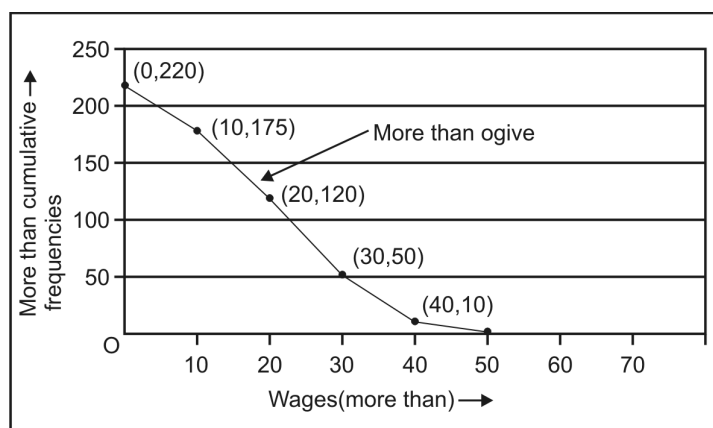


Fig. 15.9: More Than Ogive.

For “less than ogive” as shown on previous page in Fig. 15.8, we have plotted the points (10, 45), (20, 100), (30, 170), (40, 210), (50, 220) and then joined them by line segments. Similarly, for “more than ogive” as shown above in Fig. 15.9, we have plotted the points (0, 220), (10, 175), (20, 120), (30, 50), (40, 10), and then joined them by line segments.

If we want to obtain a partition value, using ogives, we draw dotted horizontal line through that value at y-axis which corresponds to the partition value and then from the point, where it meets the less then ogive, we draw a dotted vertical line and let it meets the x-axis. The abscissa of the point, where it meets the x-axis is the required partition value. For example, suppose we want to find first quartile, then we draw a dotted horizontal line starting from y-axis at a point corresponding to $N/4$ and let it meets the “less than ogive”. From that point at “less than ogive”, we draw a dotted vertical line and let it meets the x-axis. The abscissa corresponding to this point is the first quartile. Similarly, for finding median or second quartile, we start drawing dotted horizontal line from y-axis at a point corresponding to $N/2$ and then we proceed as described above. Similarly, for third quartile $3N/4$ is taken in place of $N/2$. In this way, we may find any partition value.

Note 4: Median may also be obtained by drawing dotted vertical line through the point of intersection of both the ogives, when drawn on a single figure.

Now, you can try the following exercises.

-
- E5)** Draw two ogives from the following data
- | Class: | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 | 80-90 |
|------------|------|-------|-------|-------|-------|-------|-------|-------|-------|
| Frequency: | 3 | 6 | 10 | 13 | 20 | 18 | 15 | 9 | 6 |
- Hence find median. Compare your result by calculating median by direct calculations.
- E6)** Draw less than ogive from the following frequency distribution of marks of 90 students
- | Marks: | 0-9 | 10-19 | 20-29 | 30-39 | 40-49 | 50-59 | 60-69 | 70-79 |
|------------------|-----|-------|-------|-------|-------|-------|-------|-------|
| No. of Students: | 7 | 11 | 19 | 8 | 20 | 14 | 8 | 3 |
- Hence find Q_1 , Q_2 and Q_3 .
- E7)** Draw the more than ogive for the following frequency distribution of the weekly wages of workers:
- | Weekly wages: | 0-10 | 10-20 | 20-40 | 40-50 | 50-60 | 60-70 | 70-80 | 80-90 | 90-100 |
|-----------------|------|-------|-------|-------|-------|-------|-------|-------|--------|
| No. of Workers: | 5 | 15 | 20 | 30 | 45 | 35 | 25 | 15 | 10 |
-

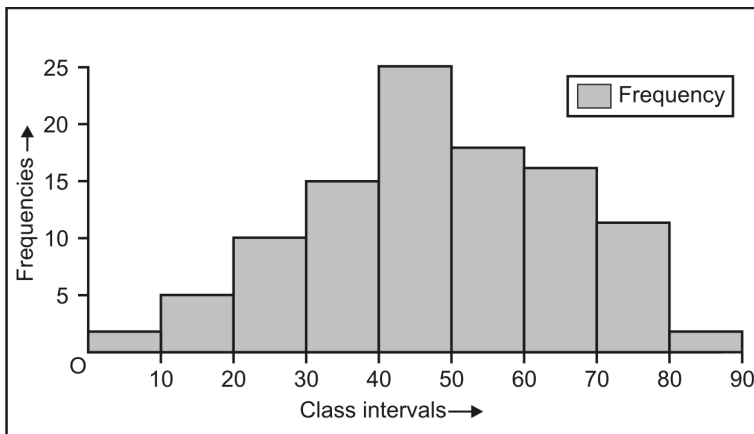
15.4 SUMMARY

In this unit we have discussed:

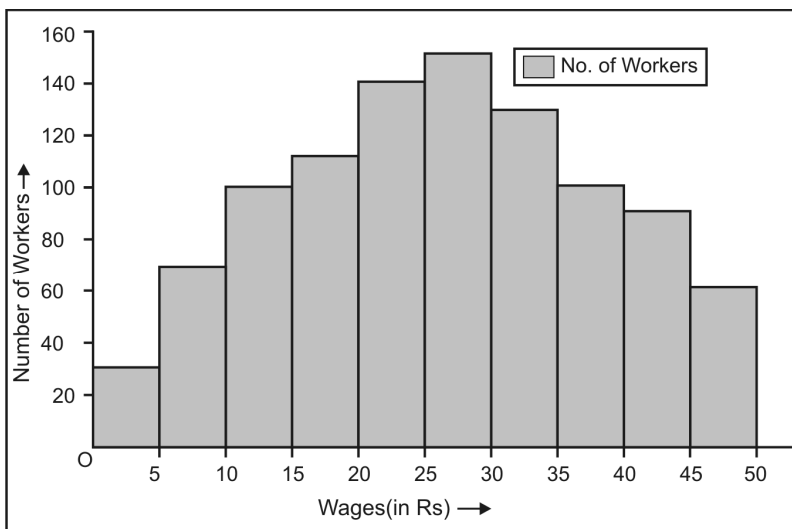
- 1) Various types of graphical presentation of data.
- 2) Way of drawing histogram for continuous frequency distributions.
- 3) Frequency polygon for a frequency distribution.
- 4) Frequency curves of different shapes, and
- 5) Way of drawing cumulative frequency graphs or ogives.

15.5 SOLUTIONS/ANSWERS

E1) Histogram of the given data is given below:

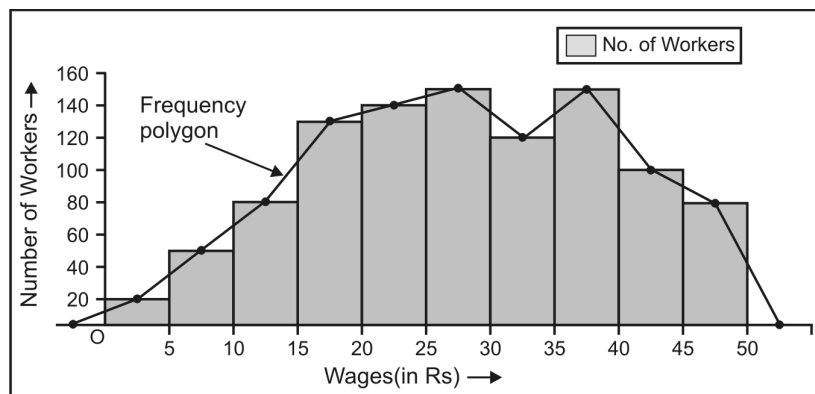


E2) Histogram of the given data is given below:

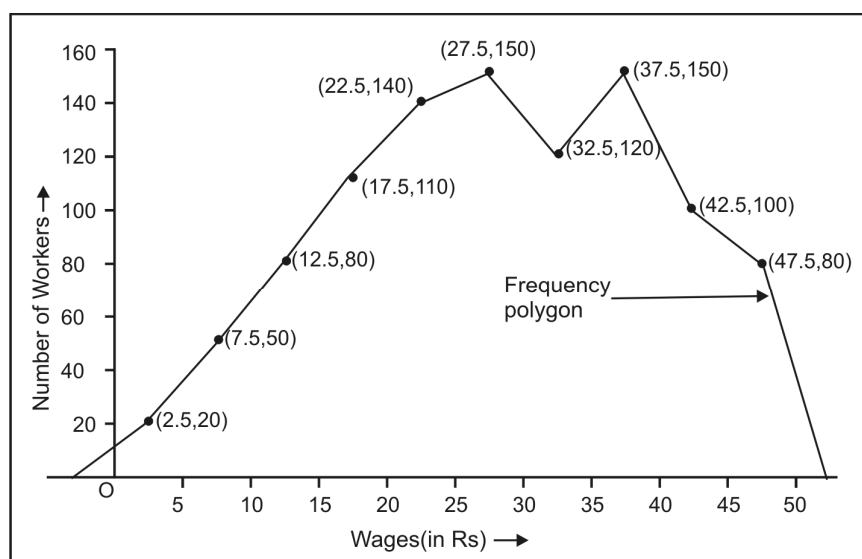


E3) Frequency polygon of the given data by first drawing histogram is given on the next page.

Presentation of Data

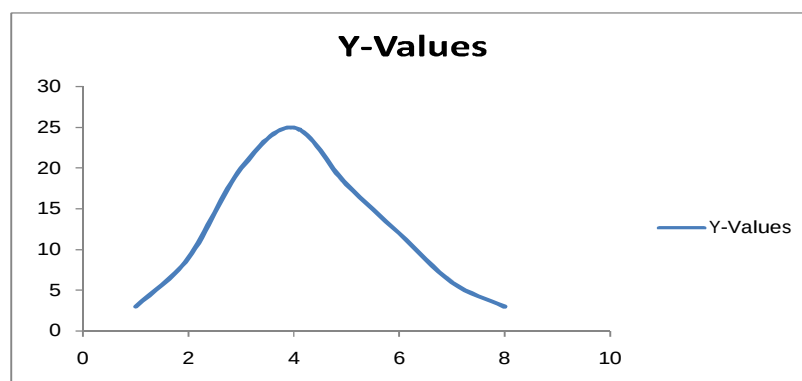


Frequency polygon can also be drawn without histogram as given below:

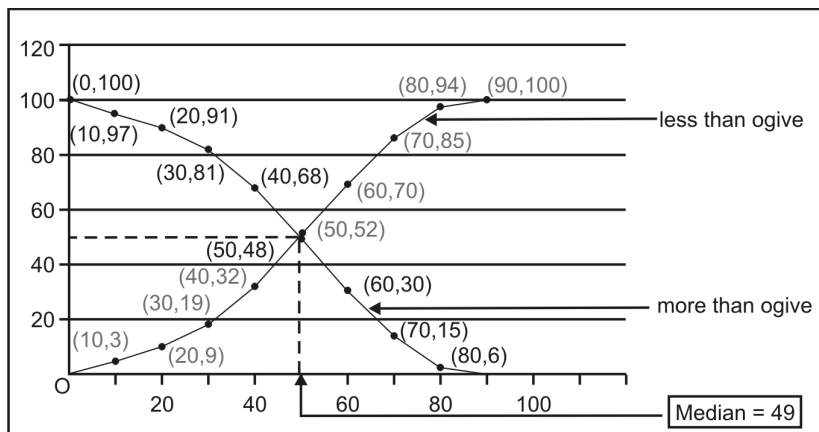


Note 5: We can draw frequency polygon by first drawing a histogram or we can draw directly.

E4) Frequency curve for the given data is given below.



E5) Two ogives for the given data are given on the next page.



Calculations of Median through direct calculation:

Given data are

Class Interval (CI)	Frequency (f)	Cumulative frequency (cf)
0-10	3	3
10-20	6	9
20-30	10	19
30-40	13	32
40-50	20	52
50-60	18	70
60-70	15	85
70-80	9	94
80-90	6	100
Total	100	

Here $N = \sum f = \text{sum of all frequencies} = 100$

$\therefore \frac{N}{2} = \frac{100}{2} = 50$ and hence median class is 40-50.

So, in usual notations we have

l = lower limit of the median class = 40,

h = width of the median class = $50 - 40 = 10$

c = cumulative frequency preceding the median class = 32

f = frequency of the median class = 20

$$\begin{aligned} \text{Now, Median} &= l + \frac{h}{f} \left(\frac{N}{2} - c \right) = 40 + \frac{10}{20} \left(\frac{100}{2} - 32 \right) = 40 + \frac{1}{2} (50 - 32) \\ &= 40 + \frac{1}{2} (18) = 40 + 9 = 49 \end{aligned}$$

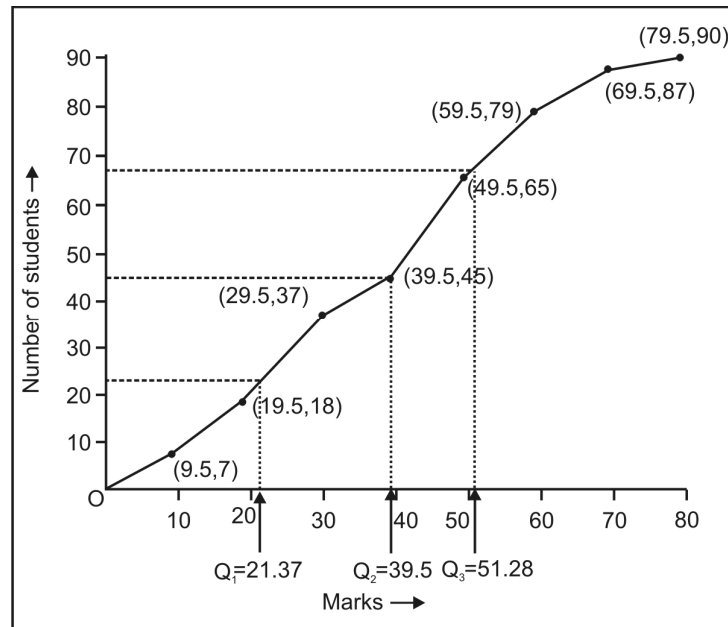
Thus we see that median obtained by using two ogives graphically and by direct calculation are same.

E6) Less than ogive for the given frequency distribution is given below.

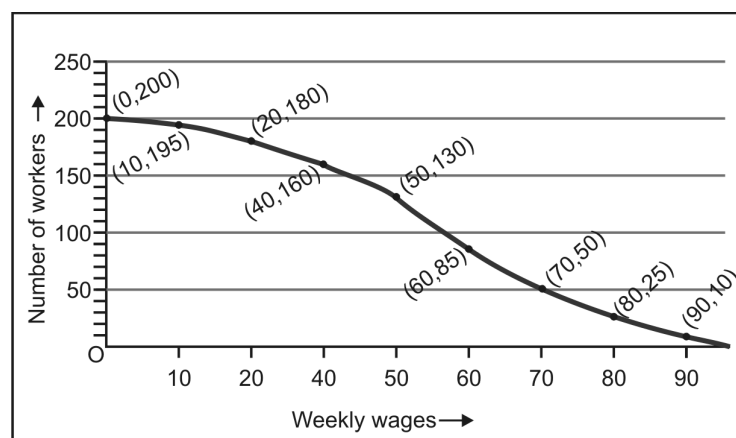
To find Q_1 : $\frac{N}{4} = \frac{90}{4} = 22.5$ first of all we have to draw a dotted

horizontal line starting from y-axis at 22.5 $\left(\text{as } \frac{N}{4} = 22.5 \right)$ and then from

the point where it meets the less ogive, we shall draw dotted vertical line, the value corresponding to the point where it meets the horizontal axis is the value of Q_1 as shown in figure. Similarly, values of Q_2 and Q_3 can be obtained as shown in the figure below:



E7) More than ogive for the given frequency distribution is given below:



UNIT 16 GRAPHICAL PRESENTATION OF DATA-II

Structure

16.1 Introduction

Objectives

16.2 Time Series Graphs

Method of Drawing a Time Series Graph

Types of Time Series Graph

16.3 Stem-and-Leaf Displays

Stem-and-Leaf Display for More than one Set of Data

Merits of Stem-and-Leaf Display

16.4 Box Plots

Method of Construction of the Box Plots

Components of a Box Plot

Box Plots with Outliers

Box Plots with + Signs

Box Plots with Whisker, + Sign and Outliers

16.5 Summary

16.6 Solutions/Answers

16.1 INTRODUCTION

In Unit 15 of this block, we have discussed some of the techniques of graphical presentation of data. In that unit, we have restricted ourselves to the graphical methods which are used for representing frequency distributions. The present unit discusses the graphical methods for time series data. A time series graph is frequently used for analysing and presenting the time series data. Range chart is a type of time series graph which is used for showing the range of variation. The Band chart is another type of time series graph which shows the total for successive time periods broken up into subtotals for each of the component parts of the total. In this unit, we shall also discuss as to how data are represented by plotting stem-and-leaf displays and box plots. Stem-and-leaf display is like histograms, but here the additional feature is that the given value of each individual is also shown in these displays. Further, in this unit we shall discuss box plots to represent the data through five-number summary.

Objectives

After studying this unit, you should be able to:

- describe a time series graph;
 - describe the method of drawing a time series graph;
 - draw a range chart and band chart;
 - describe the method of drawing a stem-and-leaf display;
 - describe the box plot and the different parts of the box plot; and
 - draw the box plots.
-

16.2 TIME SERIES GRAPHS

A time series graph is drawn to analyse the time series data. It brings out the pattern of fluctuations in time series data and facilitates in obtaining

meaningful results about its future behaviour. To draw a time series graph, time is measured along the horizontal axis and the observed data along the vertical axis. After then points are plotted against the magnitudes corresponding to each successive time period and finally these points are joined by the line segments. The resulting zigzag curve is called the time series graph. For examples, a time series graph for the production (in tons) of a commodity during the period 2001-2009 given in following table is shown in Fig. 16.1

Years	2001	2002	2003	2004	2005	2006	2007	2008	2009
Production	30	25	35	40	25	30	40	45	20

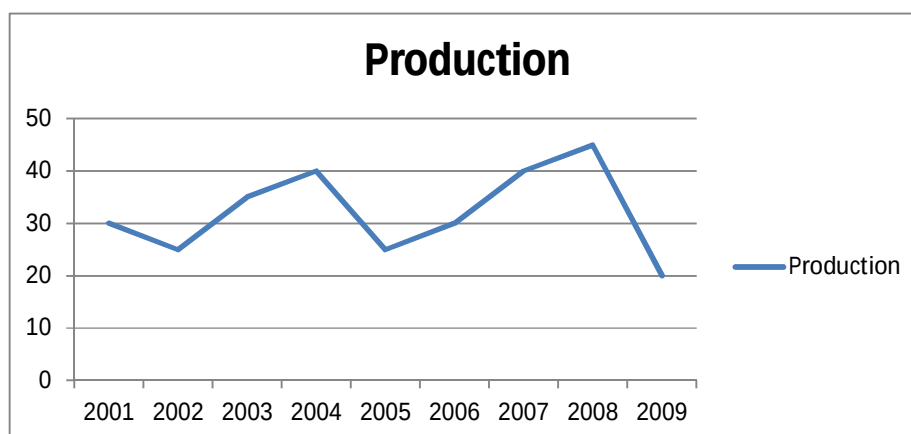


Fig. 16.1 Production of a Commodity from 2001 to 2009

In rest of this section we shall discuss on method of drawing a time series graph in subsection 16.2.1 and types of time series graph in subsection 16.2.2.

16.2.1 Method of Drawing a Time Series Graph

A time series is formed by the observations of a variable under study at different phases of time. The time series graph is extremely helpful in analysing the fluctuations in the values of the variable under study at different phases of time. We generally take time variable along x-axis and the values or magnitude of the observations of the variable under study along y-axis. After plotting the values of the variable against the corresponding values of the time variable as points, we join such points by line segments. The graph so drawn is known as a **time series graph** or **line graph**. Such type of graph is simplest to understand, easiest to draw and most widely used in practice. With the help of these graphs several variables can be shown on the same graph and a comparison can also be made. Following are some points which should be followed while constructing a time series graph:

- The scale of the y-axis should begin from zero even if the lowest y-value associated with any x-period or value is far from zero. If necessary concept of kink can be used.
- If the unit of measurement is same, we can represent two or more variables on the same graph.
- Not more than 5 or 6 number of variables is shown on the same graph, otherwise the chart/graph becomes quite confusing.
- When two or more variables are to be shown on the same graph, it is advised to use different designs of lines to distinguish between the variables.

Kink: Refer Note 3 on page number 52 of Unit 15 of this course.

16.2.2 Types of Time Series Graph

There are two types of time series graphs:

- (i) Range Chart (ii) Band Chart

Let us discuss these one by one:

(i) Range Chart

A range chart is a very useful method of showing the series of range of variation or fluctuation between the maximum and minimum values of a variable at the same point of time. For example, if we are interested in showing the minimum and maximum prices of a commodity for different periods of time or the minimum and maximum marks obtained by the students in different years, etc. the range chart would be the appropriate option.

For drawing a range chart, we take time variable along x-axis and the value of other variable on the y-axis. Then we draw two line graphs together by plotting the maximum and minimum observations in the given data. One curve representing the highest values at different point of time of the variable and the other one representing the lowest values at the same different point of time. The gap between both the curves represents the range of variation. For highlighting the difference between the lowest and highest values, the use of some colour or shade should be made. Let us take an example of drawing a range chart.

Example 1: Represent the following data by range chart.

Days	Max. Temp. (in °C)	Min. Temp. (in °C)
Monday	38	12
Tuesday	41	16
Wednesday	35	14
Thursday	42	15
Friday	44	18
Saturday	45	20
Sunday	46	21

Solution: Since there are two variables with same scales of measurement, both the variables are shown on the same graph as in Fig. 16.2.

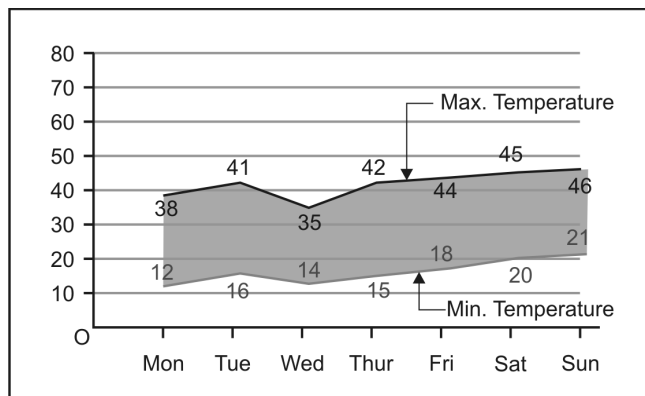


Fig. 16.2 Range Chart for the Maximum and Minimum Temperature in a Week

(i) Band Graph

Another type of a time series graph which shows the total for successive time periods broken up into sub divisions for each of the components of the total is known as band graph. In other words, the band graph represents the range of the components of total and shows as to how and what to be distributed. The

different components of the total are plotted as line graph one over the other in band graph and the gaps between the successive lines are represented and differentiated with filling up them by different shades, colours, etc., so that the graph looked like the series of bands. The band graph is especially useful in representing the components which divide such as total costs, total sales, total production, etc. into their various respective components. For example, total production may be divided into its components like nature of commodity, machines, plants, etc. Band graph is also used where the data are put in percentage form. The whole graph depicts 100% and the bands as the percentage of the various components of the total. Let us take an example of drawing a band graph.

Example 2: Present the data on the amount of production (in million tones) of various plants from 2000-01 to 2007-08 given in the following table:

Year	Plant-1	Plant-2	Plant -3	Plant -4
2000-01	42.6	38.3	26.5	31.7
2001-02	48.7	36.4	28.6	34.7
2002-03	47.2	32.4	25.3	30.8
2003-04	44.8	38.8	30.1	29.6
2004-05	46.7	37.2	27.4	32.6
2005-06	45.2	34.9	25.2	35.4
2006-07	49.1	37.8	29.1	38.2
2007-08	48.2	37.8	28.4	33.5

Solution: The above data can be most suitably presented through a band graph. We proceed for constructing such a graph as follows:

We take the time on the x-axis and the other variable on the y-axis. Then we plot the various points for different years for Plant-1 and join them by straight line segments. This is represented by curve A (see Fig. 16.3). Now we add the values of production of Plant-2 for various years to the values of production of Plant-1 and plot the values and finally join them by straight line segments. This is represented by curve B (see Fig. 16.3). The difference between the curves B and A, gives the production of Plant-2. Now we add the values of production of Plant-1 and Plant-2 to that of Plant-3 and plot the various points. This is represented by curve C (see Fig. 16.3). The difference between curve C and curve B represents production of plant -3. After that we add the values of production of Plant-1, Plant-2 and Plant-3 to the values of production of Plant -4 and draw a curve. This is represented by curve D (see Fig. 16.3). The difference between curve C and curve D represents the production in Plant-4. Using these steps required band graph is shown in Fig. 16.3.

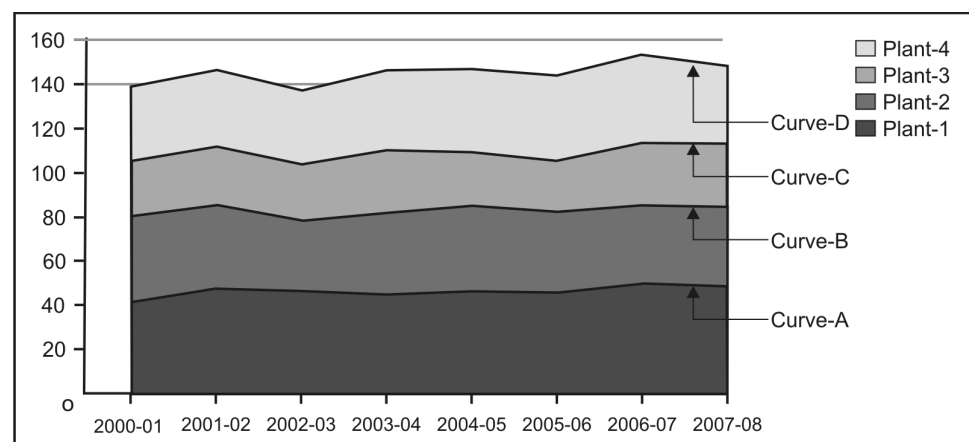


Fig. 16.3: Band Graph for the Production in four Plants

Now, you can try the following exercises.

E1) Draw a range chart for the following data:

Class:	1	2	3	4	5	6	7	8	9	10
Max: Marks:	58	65	74	61	87	65	78	92	67	84
Min. Marks:	15	21	25	32	26	16	19	22	24	17

E2) Draw a band graph for the following data of quarterly results for profit (in lakhs of rupees):

Quarters:	Plant-I	Plant-II	Plant-III
Quarter-I	34	43	46
Quarter-II	41	47	41
Quarter-III	38	39	44
Quarter-IV	51	57	53

16.3 STEM-AND-LEAF DISPLAYS

In previous units of this block as well as in previous Sec. of this unit, we have seen that data can be represented in a variety of ways including graphs, charts and tables. In this Sec. we discuss another type of graph named stem-and-leaf display. A stem-and-leaf display is very similar to a histogram but shows more information. The stem-and-leaf display summarises the shape of a set of data and provides the details regarding individual values. A stem-and-leaf display quickly summarises data while maintaining the individual data points. Now a day's use of stem-and-leaf displays is increasing, so let us formally define it in the next paragraph with some examples.

It has a vertical line of numbers obtained after removing the last digits (i.e. unit digits) from the given numbers called starting parts and for each starting part there is a horizontal line of numbers, i.e. the digits at the unit places of the given numbers called leaves. And each complete horizontal line including starting part and leaves is known as stem. The data displayed like this is nothing but known as stem-and-leaf display. The distance between the lowest values that are recorded in two consecutive stems is known as **stem width** or **category interval**, which plays very important role in stem-and-leaf displays.

Stem-and-leaf displays are used in many situations like series of scores of sports teams, series of temperature or rainfall over a period of time, series of classroom test scores, etc.

Following example will illustrate the above discussion more clearly.

Example 3: We have a set of values of the test scores of 22 students in a class as 11, 2, 28, 33, 48, 0, 42, 17, 24, 14, 0, 18, 26, 29, 35, 42, 22, 8, 28, 8, 46, 14. Draw a simple stem-and-leaf display by taking stem width 10.

Solution: Simple stem-and-leaf display for the given data can be shown as follows:

0	2 0 0 8 8	(5)
1	1 7 4 8 4	(5)
2	8 4 6 9 2 8	(6)
3	3 5	(2)
4	8 2 2 6	(4)

Total observations = 22

After arranging the leaves in ascending order of magnitude, we have

0	0 0 2 8 8
1	1 4 4 7 8
2	2 4 6 8 8 9
3	3 5
4	2 2 6 8

Here starting parts show the 'tens digits' and the leaves show the 'ones digits' in the above stem-and-leaf display. At a glance, one can see that 4 students got marks in the 40's in their test out of 50. Out of these four students two got 42 marks each, whereas the other two got 46 and 48 marks in the test. Fourth row (i.e. fourth stem) indicates that two students got marks in 30's in their test out of 50. And actual marks of these two students are 33 and 35. Similarly, we can get the information about the marks of the other students from successive rows (stems). When you count the total numbers of leaves, you may know how many students appeared in the test. The information is nicely organised when a stem-and-leaf display is used. Stem-and-leaf display provides a tool for specific information in large sets of data, otherwise one would have a long list of marks to arrange and analyse.

16.3.1 Stem-and-Leaf Display for more than one Set of Data

Stem-and-leaf display is also used to compare two sets of data. That is known as 'back to back' stem-and-leaf display. For example, if you want to compare the batting scores of two cricket players, then stem-and-leaf display is right way to represent the data.

Example 4: Draw a stem-and-leaf display for batting scores of two players given below.

Player A	102, 61, 82, 88, 90, 63, 69, 85, 105, 93, 65, 94, 107, 97, 67
Player B	104, 62, 83, 95, 106, 95, 108, 63, 108, 82, 93, 109

Solution: The scores of two players can be compared with the help of back to back stem-and-leaf display as follows:

Leaf (Player A)	Starting part	Leaf (Player B)
1 3 5 7 9	6	2 3
2 5 8	8	2 3
0 3 4 7	9	3 5 5
2 5 7	10	4 6 8 8 9

Here column of starting parts is now in the middle and the leaves columns are to the right (player B) and left (player A) of the column of starting parts. You can see that the player B has more innings with a highest score than the player A. The player B has only 2 innings with scores of 62 and 63, while the player A has 5 innings with the scores of 61, 63, 65, 67 and 69. You can also see that player B has the highest score of 109, compared to player A with highest score of 107. Thus we see that presentation of the data by stem-and-leaf display provides us lot of information in very quick time.

In above two examples stem width or category interval was 10. Now we take an example in which stem width is 5 instead of 10.

Example 5: Arrange the numbers 47, 35, 37, 20, 43, 15, 15, 26, 46, 25, 29, 12, 39, 44, 21, 24, 16, 40, 19, 46, 30, 34, 17, 39, 16, 40, 31, 21, 14, 42, 16, 43, 22, 11, 24, 25, 31, 27, 40, 33 in a stretched stem-and-leaf display that has single-digit starting parts and leaves, but has stem width of 5.

Solution: A simple stem-and-leaf display has a unique starting part for each stem with stem width 10, while the stretched stem-and-leaf display shown below has a stem width 5 which means we have stretched the stem (of stem width 10) into two stems each of width 5, and the same starting part is used for both stems (i.e. for stem 1 we used 1a and 1b, for stem 2 we used 2a and 2b, etc., it is also explained below the display). The required stretched stem-and-leaf display is given as follows:

1a	2 4 1
1b	5 5 6 9 7 6 6
2a	0 1 4 12 4
2b	6 5 9 5 7
3a	0 4 1 1 3
3b	5 7 9 9
4a	3 4 0 0 2 3 0
4b	6 5 5

In this stem-and-leaf display 'a' stands for the interval 0-4 and 'b' stands for 5-9. The values between 10-19 of stem 1 are now represented into two stems 1a and 1b which include values between 10-14 and 15-19 respectively. Similarly, values between 20-29 of stem 2 are now represented by two stems 2a and 2b, which include values between 20-24 and 25-29 respectively, and so on.

16.3.2 Merits of Stem-and-Leaf Display

Following are some merits of stem-and-leaf display:

- Stem-and-leaf display arranges the data in place values.
- Total number of observations and mode can easily be obtained from stem-and-leaf display (see Example 3).
- Summarises the shape of a set of data (the distribution) and provides the detail regarding individual values.
- Stem-and-leaf display also enables you to find quantiles such as median, quartiles (i.e. Q_1, Q_2, Q_3), deciles (i.e. $D_1, D_2, D_3, \dots, D_9$), percentiles (i.e. $P_1, P_2, P_3, \dots, P_{99}$), etc. As discussed below.

Formula for Calculating Quantiles: First of all given observations are arranged in ascending order of magnitude. Then j^{th} quantiles denoted by $Q_{j/m}$ (e.g. 7/10 of the data are below $Q_{7/10}$) is given by

$Q_{j/m} = x_i$, where x_i is that value of the variable below which j^{th} observations lie and

$$i = \left(\frac{j \times n}{m} \right) + \frac{1}{2} \quad \dots (16.1) \quad \text{, where } n = \text{total number of observations}$$

For example, let us apply this formula for finding median for the data of Example 3:

Median = Second quartile = $Q_{2/4}$:

$$i = \left(\frac{j \times n}{m} \right) + \frac{1}{2} = \left(\frac{2 \times 22}{4} \right) + \frac{1}{2} = 11.5$$

$$\begin{aligned} \therefore \text{median} &= x_{11.5} = 11^{\text{th}} \text{ observation in the array} \\ &\quad + 0.5(12^{\text{th}} \text{ observation} - 11^{\text{th}} \text{ observation}) \\ &= 22 + 0.5(24 - 22) = 23 \end{aligned}$$

Now, you can try the following exercises.

E3) Draw a stem-and-leaf display with the following marks obtained by 30 students.

77, 80, 82, 68, 65, 59, 61, 57, 50, 62, 61, 70, 69, 64, 67,
70, 62, 65, 65, 73, 76, 87, 80, 82, 83, 79, 79, 71, 80, 77

Also determine the median for the marks.

E4) Draw a stem-and-leaf display for the following data.

31, 42, 22, 27, 33, 57, 67, 58, 64, 44, 65, 59, 46, 61, 35, 26, 63

Also find seventh decile.

E5) Draw a stem-and-leaf display for the given data:

141, 137, 105, 139, 107, 144, 110, 135, 117, 125, 147, 113, 109, 120,
132, 110, 130, 112

Also find sixty seventh percentile.

16.4 BOX PLOTS

In previous section we have discussed the stem-and-leaf displays. Now let us discuss another type of plot which is known as Box plot in this section. In descriptive statistics, a box plot also known as a box-and-whisker plot is a convenient way of graphically representing numerical data. It represents the data through five-number summary, i.e. the smallest observation (sample minimum), lower quartile (Q_1), median (Q_2), upper quartile (Q_3), and the largest observation (sample maximum). Box plots are used to describe a distribution generally when it is extremely skewed or multimodal. A box plot also indicates which observation(s), if any, might be considered as outliers. A box plot is a quick graphic approach for examining one or more sets of data.

Box plots display differences between populations without making any assumptions of the distributions. The spacing between the different parts of the box helps in indicating the degree of dispersion (spread) and skewness in the data, and identifies outliers. Box plots can be drawn either horizontally or vertically. Here we will draw box plots vertically.

16.4.1 Method of Construction of the Box Plots

Box plots are useful for identifying key values while comparing two or more distributions. To understand more clearly the method of constructing the box plots, let us consider the following data of 37 students in a class who were examined by a game with a box containing some balls. Their task was to select a ball from one box placed at one corner and put it in another blank box placed at another corner of the class as quickly as possible, and their times (in seconds) were recorded. The scores were compared for 16 boys and 21 girls who participated in game. Observed data are given in the following table:

Time (in seconds) for completing the given task

Boys	18, 19, 20, 22, 24, 25, 26, 16, 17, 19, 25, 27, 28, 23, 23, 31
Girls	15, 17, 18, 19, 20, 21, 23, 14, 17, 18, 19, 20, 21, 24, 19, 16, 17, 18, 20, 22, 28

The method of construction of separate box plots for the data of boys and girls is discussed below:

There are several ways of constructing a box plot. The first relies on the quartiles, lowest and greatest values in the distribution of scores. Fig. 16.4 shows how these three statistics are used for the above example. We draw a

box plot for each gender extending from the 1st quartile to the 3rd quartile. The 2nd quartile is drawn inside the box. Therefore,

- (i) The bottom of each box is the 1st quartile,
- (ii) The top is the 3rd quartile,
- (iii) The line in the middle is the 2nd quartile.
- (iv) A line extending from the point corresponding to the smallest observation to 1st quartile is drawn and known as **lower whisker**.
- (v) A line extending from 3rd quartile to the point corresponding largest observation is drawn and is known as **upper whisker**.

Let us arrange the given data for each of the gender in ascending order as shown in following table to find out the above components, i.e. Q_1, Q_2, Q_3 , smallest observation and largest observation.

Gender	Times (in Seconds)
Boys	16, 17, 18, 19, 19, 20, 22, 22, 23, 23, 24, 25, 25, 27, 28, 31
Girls	14, 15, 16, 17, 17, 17, 18, 18, 18, 19, 19, 19, 20, 20, 20, 21, 21, 22, 23, 24, 28

For Boys

The lowest or smallest observation = $x_s = 16$.

$$\begin{aligned}\text{First quartile } (Q_1) &= \left(\frac{16+1}{4} \right)^{\text{th}} \text{ item} \\ &= 4.25^{\text{th}} \text{ item} = 4^{\text{th}} \text{ item} + 0.25 (5^{\text{th}} \text{ item} - 4^{\text{th}} \text{ item}) \\ &= 19 + 0.25 (19 - 19) = 19\end{aligned}$$

$$\begin{aligned}\text{Second quartile } (Q_2) &= 2 \left(\frac{16+1}{4} \right)^{\text{th}} \text{ item} \\ &= 8.5^{\text{th}} \text{ item} = \text{Mean of } 8^{\text{th}} \text{ and } 9^{\text{th}} \text{ items} \\ &= \frac{22 + 23}{2} = 22.5\end{aligned}$$

$$\begin{aligned}\text{Third quartile } (Q_3) &= 3 \left(\frac{16+1}{4} \right)^{\text{th}} \text{ item} \\ &= 12.75^{\text{th}} \text{ item} = 12^{\text{th}} \text{ item} + 0.75 (13^{\text{th}} \text{ item} - 12^{\text{th}} \text{ item}) \\ &= 25 + 0.75 (25 - 25) = 25\end{aligned}$$

The largest observation = $x_l = 31$

For Girls

The lowest or smallest observation = $x_s = 14$

$$\begin{aligned}\text{First quartile } (Q_1) &= \left(\frac{21+1}{4} \right)^{\text{th}} \text{ item} \\ &= 5.5^{\text{th}} \text{ item} = \text{mean of } 5^{\text{th}} \text{ and } 6^{\text{th}} \text{ items} = \frac{17+17}{2} = 17\end{aligned}$$

$$\text{Second quartile } (Q_2) = 2 \left(\frac{21+1}{4} \right)^{\text{th}} \text{ item} = 11^{\text{th}} \text{ item} = 19$$

$$\begin{aligned}\text{Third quartile } (Q_3) &= 3 \left(\frac{21+1}{4} \right)^{\text{th}} \text{ item} \\ &= 16.5^{\text{th}} \text{ item} = \text{mean of } 16^{\text{th}} \text{ and } 17^{\text{th}} \text{ item} = \frac{21+21}{2} = 21\end{aligned}$$

The largest observation = $x_l = 28$

Box plots for boys and girls on the basis of the above findings are shown below in Fig. 16.4.

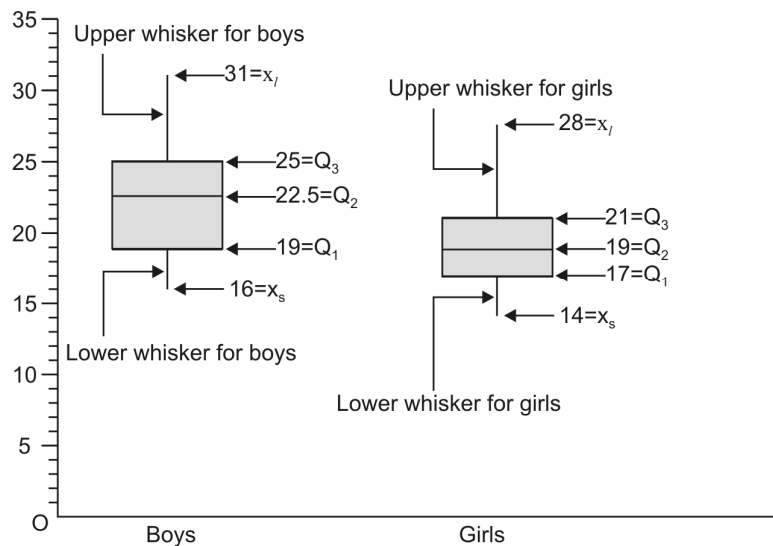


Fig. 16.4: The Box Plots with the Whiskers for Boys and Girls

16.4.2 Components of a Box Plot

Let us now discuss the various components and terminologies which are used in drawing the various types of box plots.

1. Upper and Lower Hinges

The upper and lower hinges are constructed to represent the third quartile and first quartile respectively. For the example discussed above, the values of upper and lower hinges are 21 and 17 for the girls whereas for the boys they are 25 and 19 respectively.

2. H-Spread

This is calculated by taking the difference between the upper and lower hinges. The H-Spread shows the spreadness of the elements of the data between the first quartile and third quartile. For the data related to boys in the example discussed above, the H-spread is $25 - 19 = 6$ whereas for girls it is $21 - 17 = 4$.

3. Whiskers

The lines extending above and below the box are called whiskers. Lower whisker extends from the point corresponding to the smallest observation to Q_1 and the upper whisker extends from Q_3 to the largest observation.

4. Step

The step is calculated by multiplying the H-spread by 1.5. For the data of boys in the above example, the value of 1 step is $1.5 \times 6 = 9$, whereas for girls it is $1.5 \times 4 = 6$.

5. Upper and Lower Inner Fences

The upper and lower inner fences are calculated by adding one step to the upper hinge and subtracting 1 step from the lower hinge respectively. In other words, upper inner fence is equal to the 1 step ahead from upper

hinge whereas the lower inner fence is one step before the lower hinge. For the data of boys in the example discussed above, the upper and lower inner fences are $25 + 9 = 34$ and $19 - 9 = 10$ whereas for the data of girls the upper and lower inner fences are $21 + 6 = 27$ and $17 - 6 = 11$ respectively.

6. Upper and Lower Outer Fences

The upper and lower outer fences are calculated by adding two steps to the upper hinge and subtracting 2 steps from the lower hinge respectively. For the data of boys in the example discussed above the upper and lower outer fence are $25 + 2 \times 9 = 43$ and $19 - 2 \times 9 = 1$, whereas for the data of girls they are, $21 + 2 \times 6 = 33$ and $17 - 2 \times 6 = 5$, respectively.

7. Upper and Lower Adjacent

The upper and lower adjacent are used to represent the largest and the smallest observations of the data. For the data of boys in the example discussed above upper and lower adjacent are 31 and 16 whereas for the girls data they are 28 and 14 respectively.

8. Outside Value

The outside value is a value which is beyond an inner fence but not beyond an outer fence. This is used to represent the scattered values of the data. To represent these values circles are used.

9. Far Out Value

A value which is beyond the upper outer fence or lower outer fence is known as far out value. To represent these values, the asterisks are used.

Box plot with the components discussed above for data of boys is shown in Fig 16.5.

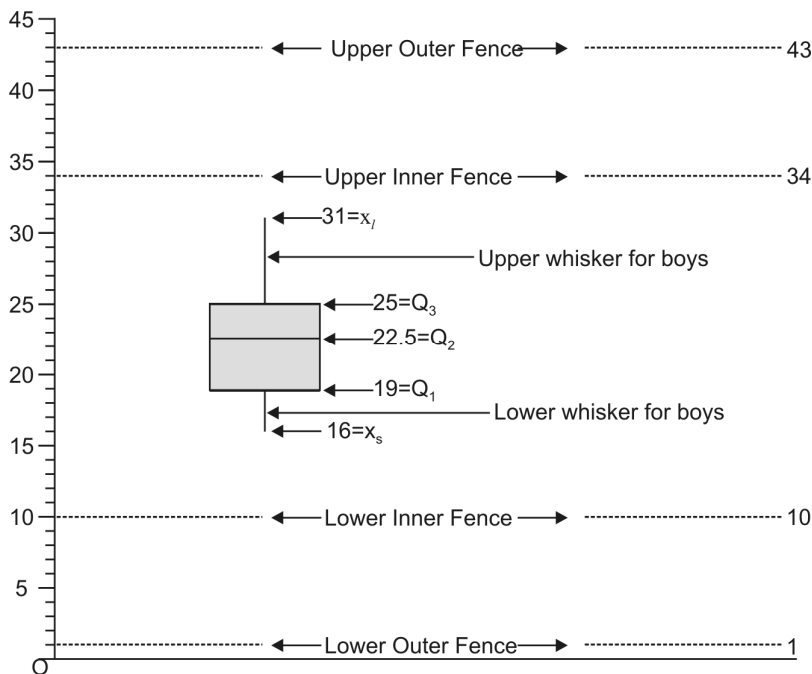


Fig. 16. 5 Box Plot for Boys with Inner and Outer Fences

Box plot with the components discussed above for data of girls is shown on next page in Fig 16.6.

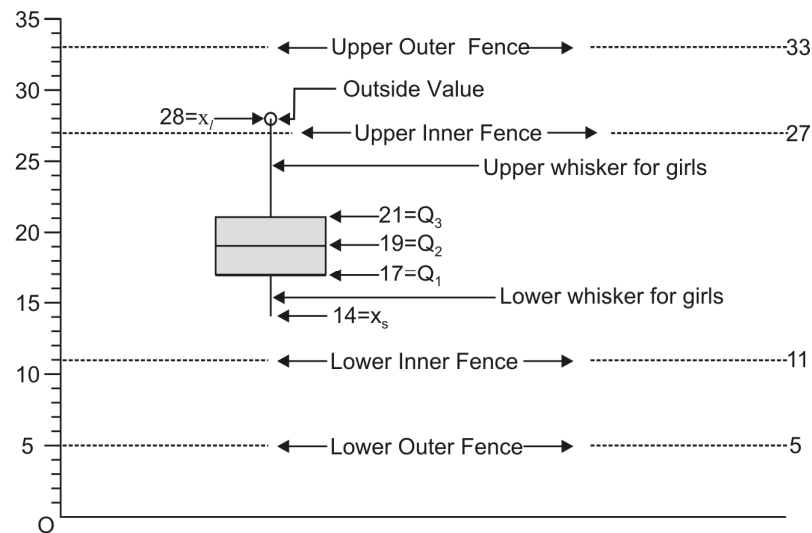


Fig. 16.6 Box Plot for Girls with Inner and Outer Fences

16.4.3 Box Plots with Outliers

Outside value(s) and far out value(s) are known as extreme observations. If the extreme observations are present in the data then those observations may be represented in box plot by an individual mark. The extreme observations are described as outliers when they are represented in box plots. The outside value is a value which is beyond an inner fence but not beyond an outer fence whereas the far out value is a value which is beyond the lower and upper outer fences. The individual marks for extreme values can be plotted above and below to the whiskers in box plot. Specially, outside values are indicated by small circles. In the data of girls in the above given example 28 is only far out value, whereas in the data of boys no value is beyond the lower or upper inner fence. The box plot for girls data indicating outside value by a circle is shown in Fig. 16.6.

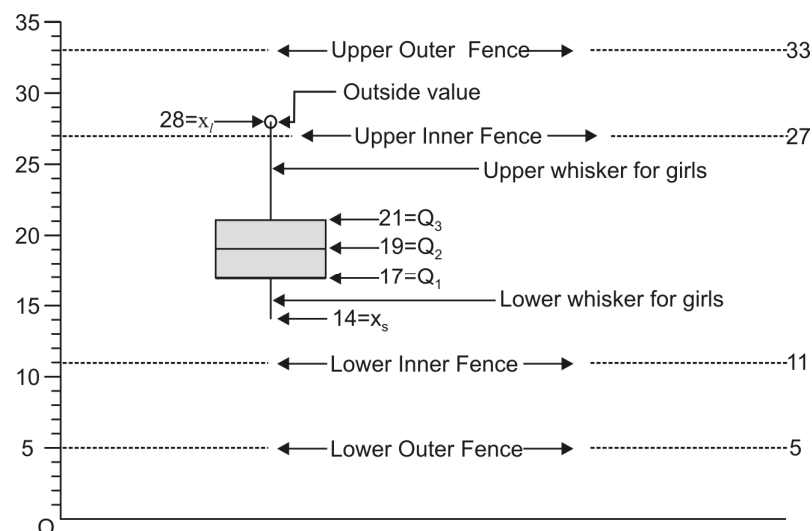


Fig. 16.6: The Box Blot for Girls with the Outlier.

16.4.4 Box Plots with + Signs

One more component which is to be included in box plots are the value of some important parameters like, mean, mode, etc. We indicate the mean score for a group of values by inserting a plus sign in box plots. For the example discussed in sub-section 16.4.1 mean in case of boys data is 20.875 and mean

in case of girls data is 19.33, which are shown by a plus sign in Fig. 16.7 as given below.

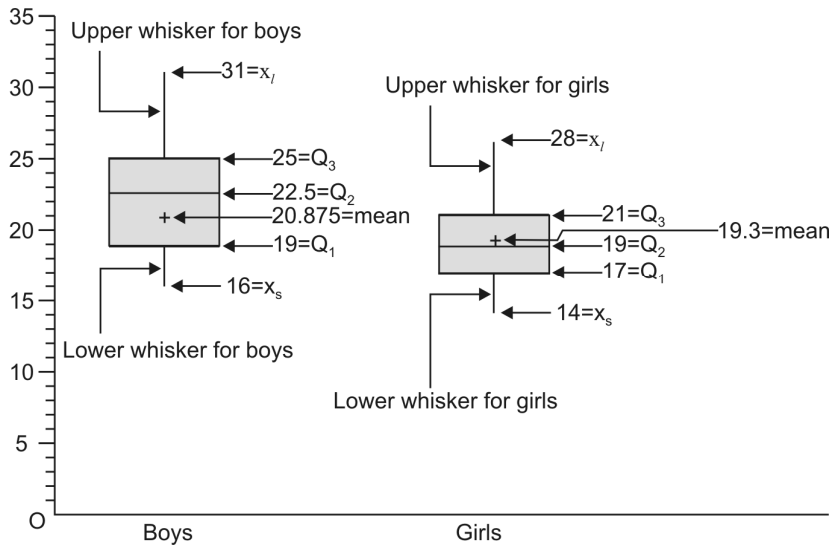


Fig. 16.7: The Box Plot with Whisker and + Signs for Boys and Girls Data.

Fig.

16.7 provides a revealing summary of the data. Since half of the scores are between the hinges (recall that the hinges are the first and third quartiles), we see that half of the girl's times are between 17 and 21, whereas half of the boy's times are between 19 and 25.

16.4.5 Box Plots with Whisker, + Sign and Outliers

On the basis of data of the example discussed in sub-section 16.4.1 we see that girls generally dropped the balls from one box to another faster than boys. We also see that one boy was slower than almost all of the women (except 3). Fig. 16.8 shows the box plot for the girl's data with whisker, + sign and outliers.

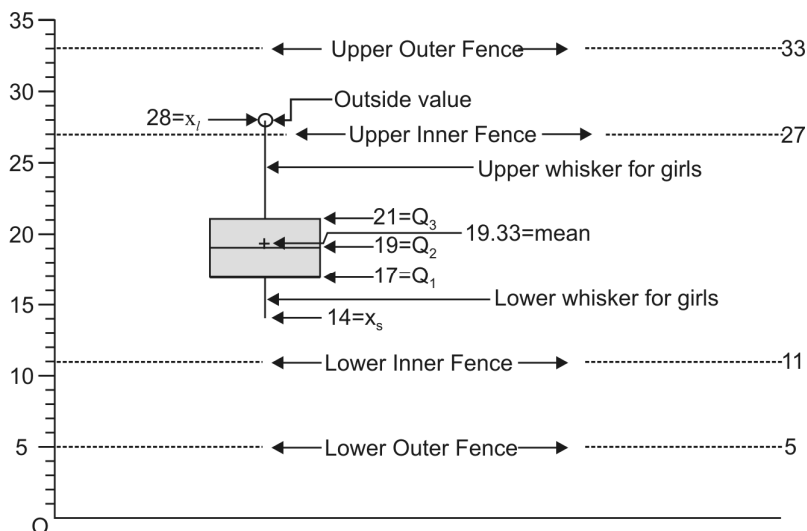


Fig. 16.8: A Box-Plot for the Girl's Data with Whisker, + Sign and Outliers

Note 2: If some learner is interested to know more about the topics discussed in Secs. 16.4 and 16.5 he/she may refer chapters 6 and 7 of the book written at serial number 5 in the reference books listed below the introduction of MST-001 on page number 4 of block 1.

Now, you can try the following exercises.

E6) Draw a box plot for the given data:

17, 15, 17, 20, 13, 15, 15, 16, 16, 15, 19, 12, 19, 14, 11, 14, 16, 10, 19, 18,
20, 14, 17, 19, 16, 22, 21, 23, 14, 12, 18, 13, 12, 25, 14, 15, 31, 17, 10, 21

E7) Draw a box plot for the given data:

31, 42, 22, 27, 33, 27, 37, 28, 34, 44, 25, 39, 26, 31, 26, 33, 46, 48, 50

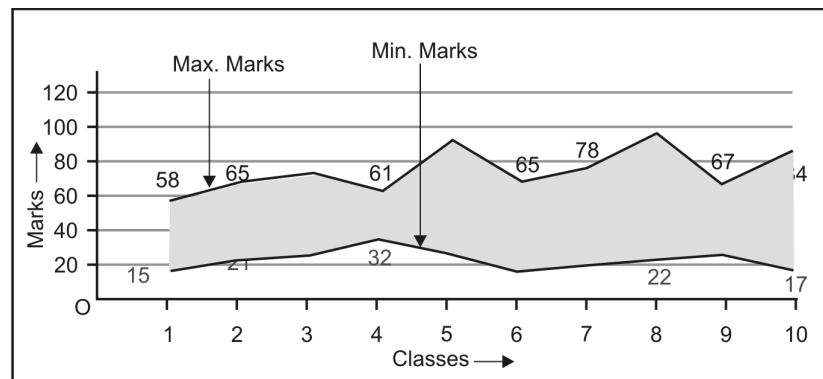
16.7 SUMMARY

In this unit, we have discussed:

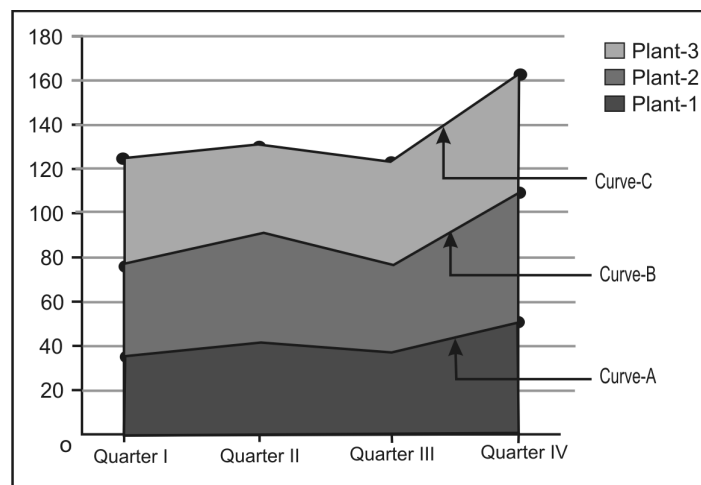
- 1) Time series data and methods of drawing a time series graph.
- 2) How to draw a range chart and band chart.
- 3) Methods of drawing a stem-and-leaf displays.
- 4) The box plots and the different components of the box plot.

16.8 SOLUTIONS / ANSWERS

E1) Range chart of the given data is given below.



E2) Required band graph of the given data is given below.



E3) Stem-and-leaf display of the given data of marks obtained by 30 students is given below.

5		9 7 0
6		8 5 1 2 1 9 4 7 2 5 5
7		7 0 0 3 6 9 9 1 7
8		0 2 7 0 2 3 0

After arranging the leaves in ascending order of magnitude, we have

5		0 7 9
6		1 1 2 2 4 5 5 5 7 8 9
7		0 0 1 3 6 7 7 9 9
8		0 0 0 2 2 3 7

Median = second quartile = $Q_{2/4}$:

$$i = \left(\frac{j \times n}{m} \right) + \frac{1}{2} = \left(\frac{2 \times 30}{4} \right) + \frac{1}{2} = 15.5$$

$$\therefore \text{median} = x_{15.5} = 15^{\text{th}} \text{ value in the array} + 0.5(16^{\text{th}} \text{ value} - 15^{\text{th}} \text{ value}) \\ = 70 + 0.5(70 - 70) = 70$$

E4) Stem-and-leaf display of the given data is given below.

2		2 7 6
3		1 3 5
4		2 4 6
5		7 8 9
6		7 4 5 1 3

After arranging the leaves in ascending order of magnitude, we have

2		2 6 7
3		1 3 5
4		2 4 6
5		7 8 9
6		1 3 4 5 7

D_7 = seventh decile = $Q_{7/10}$:

$$i = \left(\frac{j \times n}{m} \right) + \frac{1}{2} = \left(\frac{7 \times 17}{10} \right) + \frac{1}{2} = 12.4$$

$$\therefore D_7 = x_{12.4} = 12^{\text{th}} \text{ value in the array} + 0.4(13^{\text{th}} \text{ value} - 12^{\text{th}} \text{ value}) \\ = 59 + 0.4(61 - 59) = 59.8$$

E5) Stem-and-leaf display of the given data is given below.

10		5 7 9
11		0 7 3 0 2
12		5 0
13		7 9 5 2 0
14		1 4 7

After arranging the leaves in ascending order of magnitude, we have

10		5 7 9
11		0 0 2 3 7
12		0 5
13		0 2 5 7 9
14		1 4 7

Presentation of Data

P_{67} = sixty seventh percentile = $Q_{67/100}$:

$$i = \left(\frac{j \times n}{m} \right) + \frac{1}{2} = \left(\frac{67 \times 18}{100} \right) + \frac{1}{2} = 12.56$$

$$\begin{aligned} \therefore P_{67} &= x_{12.56} = 12^{\text{th}} \text{ value in the array} + 0.56(13^{\text{th}} \text{ value} - 12^{\text{th}} \text{ value}) \\ &= 132 + 0.56(135 - 132) = 133.68 \end{aligned}$$

E6) After arranging the given data in ascending order of magnitude, we have

10, 10, 11, 11, 12, 12, 12, 13, 13, 14, 14, 14, 14, 15, 15, 15, 15, 15, 16, 16, 16, 16, 17, 17, 17, 17, 18, 18, 19, 19, 19, 19, 20, 20, 21, 21, 22, 23, 25, 25

The lowest or smallest observation = $x_s = 10$

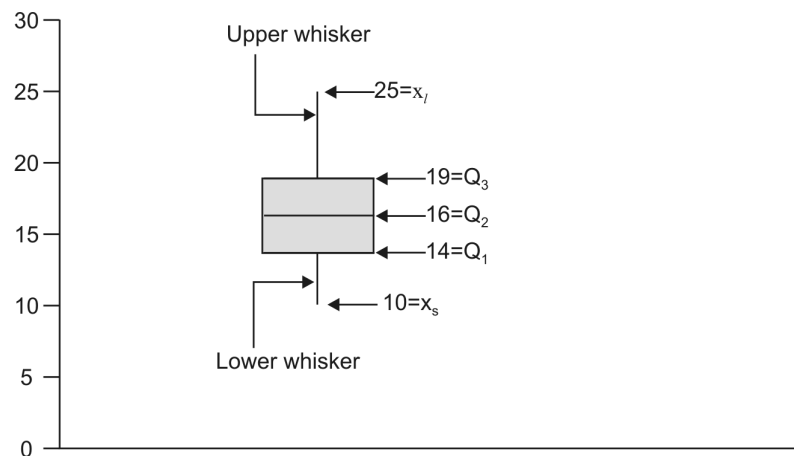
$$\begin{aligned} \text{First quartile } (Q_1) &= \left(\frac{40+1}{4} \right)^{\text{th}} \text{ item} \\ &= 10.25^{\text{th}} \text{ item} = 10^{\text{th}} \text{ item} + 0.25 (11^{\text{th}} \text{ item} - 10^{\text{th}} \text{ item}) \\ &= 14 + 0.25 (14 - 14) = 14 \end{aligned}$$

$$\begin{aligned} \text{Second quartile } (Q_2) &= 2 \left(\frac{40+1}{4} \right)^{\text{th}} \text{ item} \\ &= 20.5^{\text{th}} \text{ item} = \text{Mean of } 20^{\text{th}} \text{ and } 21^{\text{st}} \text{ items} \\ &= \frac{16+16}{2} = 16 \end{aligned}$$

$$\begin{aligned} \text{Third quartile } (Q_3) &= 3 \left(\frac{40+1}{4} \right)^{\text{th}} \text{ item} \\ &= 30.75^{\text{th}} \text{ item} = 30^{\text{th}} \text{ item} + 0.75 (31^{\text{st}} \text{ item} - 30^{\text{th}} \text{ item}) \\ &= 19 + 0.75 (19 - 19) = 19 \end{aligned}$$

The largest observation = $x_l = 25$

Using above calculations box plot based on five-number summary (i.e. smallest observation x_s , first quartile (Q_1), second quartile (Q_2), third quartile Q_3), largest observation x_l) is given below:



E7) After arranging the given data in ascending order of magnitude, we have
22, 25, 26, 26, 27, 27, 28, 31, 31, 33, 33, 34, 37, 39, 42, 44, 46, 48, 50

**Graphical Presentation of
Data - II**

The lowest or smallest observation = $x_s = 22$

First quartile (Q_1) = $\left(\frac{19+1}{4}\right)^{\text{th}}$ item = 5th item = 27

Second quartile (Q_2) = $2\left(\frac{19+1}{4}\right)^{\text{th}}$ item = 10th item = 33

Third quartile (Q_3) = $3\left(\frac{19+1}{4}\right)^{\text{th}}$ item = 15th item = 42

The largest observation = $x_l = 50$

Using above calculations box plot based on five-number summary (i.e. smallest observation x_s , first quartile (Q_1), second quartile (Q_2), third quartile Q_3), largest observation x_l) is given below:

